

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Complete Lecture Notes

Modules 1-26

جزوه کامل درس - مازول‌های ۱ تا ۲۶



Lecture Notes

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Contents

Course Description	11
Introduction	11
Structure and Content	11
Assessment and Practical Work	12
Main References	12
Course Syllabus	13
Course Information	13
Course Objectives	13
Module List (26 Modules)	14
Grading Breakdown	15
Recommended Textbooks	15
Week-by-Week Schedule	16
Lab Assignments	17
Project	17
Academic Integrity	17
Prerequisites Review	18
1 OSI Model Review (Telecom Focus)	18
2 Basic RF Concepts	18
3 Digital Communications Basics	19
4 IP Networking Fundamentals	20
5 Basic Probability	20
6 Decibel Math	21
Summary Checklist	22
1 GSM, GPRS, and EDGE	24
1 GSM Architecture	25
2 GSM Network Elements — Detailed Explanation	27
3 GSM Interfaces	30
4 GSM Mobility & Location Management	31
5 GSM Authentication & Security	32
6 WHY GPRS Was Needed	34
7 GPRS Architecture	35
8 PDP Context	37
9 GPRS Mobility Management	39
10 EDGE — Enhanced Data Rates for GSM Evolution	41
11 Evolution Path: GSM → GPRS → EDGE → 3G	43
Summary Table	44
Review Questions	44

References	45
2 2G Circuit Switching	46
1 Circuit Switching Fundamentals	47
2 SS7 Signaling Concepts	48
3 Call Setup Process	49
4 Call Routing	53
5 Location Update Procedure	54
6 Authentication Procedure	57
7 Roaming	60
8 Call Release	61
9 Why Circuit Switching?	62
Key Terms Summary	64
3 2G/3G Core Networks	65
3.1 Evolution from 2G to 3G — The Driving Forces	66
3.2 UMTS Architecture Overview	66
3.3 UTRAN: NodeB and RNC	68
3.4 3G Core Network: CS Domain + PS Domain	69
3.5 The Iu Interface	70
3.6 Release 99 vs Release 5+ (HSPA Evolution)	71
3.7 IMS (IP Multimedia Subsystem) — Introduction	73
3.8 Major Comparison Table: Mobile Network Generations	74
3.9 The CS/PS Split — And Why LTE Killed It	75
3.10 Summary: Key Takeaways	77
Review Questions	77
4 Mobility and Handover	78
1 Why Mobility Management Is Needed	79
2 Location Management: LA vs RA vs TA	80
3 Cell Reselection (Idle Mode Mobility)	81
4 Handover Types	83
5 Handover Decision Process	87
6 Handover Procedure — LTE X2 Handover	91
7 Handover Failures	96
8 Evolution of Handover: 2G → 3G → LTE → 5G	101
9 Engineering Examples with Typical Parameter Values	107
10 Summary Diagrams	112
Key Formulas Summary	114
Review Questions	114
5 4G LTE Architecture	116
5.1 Why LTE Was Developed	117

5.2	LTE Design Principles	118
5.3	LTE Architecture Overview	119
5.4	Network Elements — Detailed	120
5.5	LTE Interfaces — Reference Points	126
5.6	IMS Overview (IP Multimedia Subsystem)	132
5.7	Control Plane vs User Plane Separation (CUPS)	134
5.8	Key Differences from 3G	135
5.9	Control Plane vs User Plane Paths	137
5.10	Interface Summary Table	140
	Key Takeaways	141
	References	141
6	E-UTRAN — LTE Radio Access Network	142
6.1	E-UTRAN Architecture	143
6.2	eNodeB Functions	144
6.3	LTE Protocol Stack	145
6.4	Protocol Layers in Detail	146
6.5	LTE Channel Structure	151
6.6	S1 Interface	154
6.7	X2 Interface	156
6.8	RRC States and Transitions	157
	Summary	160
7	LTE Scheduling	161
1	Why Scheduling Matters	162
2	The LTE Resource Grid	162
3	Scheduling Granularity	164
4	Downlink Scheduling	164
5	Uplink Scheduling	166
6	Channel Quality Indicator (CQI)	167
7	Link Adaptation	168
8	HARQ (Hybrid Automatic Repeat Request)	170
9	Scheduling Algorithms	172
10	Engineering Trade-offs	173
11	Numerical Example	174
	Key Takeaways	176
	References	177
8	Random Access Channel (RACH)	178
8.1	Why Random Access is Needed	179
8.2	PRACH (Physical Random Access Channel)	179
8.3	Preamble Sequences	180

8.4	Contention-Based Random Access (4-Step CBRA)	182
8.5	Contention-Free Random Access (CFRA)	184
8.6	Timing Advance (TA)	185
8.7	Collisions and Backoff	186
8.8	RACH Overload	189
8.9	5G NR Random Access	191
8.10	Summary and Key Equations	193
	Review Questions	193
	References	194
9	Evolved Packet Core (EPC)	195
9.1	Why EPC Was Designed	196
9.2	EPC Architecture Overview	197
9.3	EPC Network Functions	198
9.4	EPC Bearers	205
9.5	Authentication in EPC	208
9.6	Session Management	211
9.7	Mobility Management	212
9.8	Failure Scenarios	215
9.9	Summary	217
10	Diameter Protocol	218
1	Why Diameter Was Introduced	219
2	Diameter vs RADIUS Comparison	220
3	Diameter Architecture	220
4	Diameter Message Structure	221
5	Diameter Commands	224
6	Important Diameter Applications in LTE	227
7	Diameter in LTE Authentication (S6a Deep Dive)	230
	Gx Policy Flow (CCR-I/CCA-I at Session Start)	234
8	Diameter vs HTTP/2 in 5G	236
9	Diameter = Control Plane Only	237
	Summary	238
11	Home Subscriber Server (HSS)	239
1	Why a Centralized Subscriber Database?	240
2	HSS Role in LTE/EPC	240
3	HSS Data Model	242
4	Authentication Vector Generation	244
5	HSS Interaction with MME (S6a / Diameter)	245
6	HSS Interaction with Other Network Elements	247
7	Authentication Scenario (Step-by-Step)	249

8	Evolution to 5G: HSS Decomposition	252
	Summary	254
	References	255
12	Mobility Management Entity (MME)	256
1	Why MME Exists	257
2	MME Functions	257
3	NAS Protocol Stack	258
4	Detailed Procedures	260
5	Bearer Management	268
6	MME Selection	269
7	S1-Flex and MME Pooling	269
8	Evolution: MME → AMF (5G)	270
13	S-GW (Serving Gateway) and P-GW (PDN Gateway)	271
1	Why Separate S-GW and P-GW?	272
2	S-GW (Serving Gateway)	272
3	P-GW (PDN Gateway)	274
4	GTP Protocol	277
5	Packet Path Trace (End-to-End with Encapsulation)	279
6	CUPS: Control and User Plane Separation	282
7	Evolution: S-GW + P-GW → UPF (5G)	283
	Summary	285
14	5G Networks Introduction	286
1	Why 5G? — Limitations of LTE/4G	287
2	5G Requirements — ITU IMT-2020 Targets	288
3	Three Pillars — 5G Use Cases	289
4	5G NR (New Radio) — Physical Layer	291
5	5G NSA vs SA — Deployment Strategies	293
6	5G Architecture Overview	295
7	gNB Architecture — CU/DU Split	297
8	Timeline — 3GPP Releases	299
	Summary — Key Takeaways	300
	References	301
15	Software Defined Networking (SDN)	302
1	Why SDN?	303
2	Traditional vs SDN Architecture	303
3	SDN Architecture Layers	304
4	SDN Controller Functions	305
5	Southbound Interfaces	305
6	Northbound Interfaces	306

7	OpenFlow Basics	308
8	SDN in Telecom	309
9	Benefits of SDN	311
10	Challenges of SDN	311
	Key Takeaways	312
16	Network Function Virtualization (NFV)	313
1	Why NFV?	314
2	Traditional vs NFV Comparison	314
3	ETSI NFV Architecture	316
4	NFV Infrastructure (NFVI)	317
5	VNF vs CNF	318
6	MANO (Management and Orchestration)	319
7	NFV in Telecom (Real Examples)	320
8	Evolution: Physical → Virtual → Cloud-Native	321
9	Benefits of NFV	322
10	Challenges of NFV	322
	Key Takeaways	323
17	3GPP Standardization for Advanced Communication Networks	324
1	What is 3GPP?	325
2	3GPP Organizational Structure	326
3	Specification Types	327
4	Release System	329
5	How to Navigate 3GPP Specifications	331
6	Study Items vs Work Items	333
7	Practical Guide: Referencing 3GPP Specs in Academic Work	335
8	Important Specification Series Table	336
	Summary	337
18	5G Service-Based Architecture (SBA)	338
1	Why 5G Moved Away from EPC Point-to-Point Interfaces	339
2	SBA Design Principles	340
3	5G Core Network Functions	342
4	NRF — The Key Enabler of SBA	346
5	Service Discovery and Communication Flow	349
6	SBI Protocol Stack	351
7	Comparison: EPC Interfaces vs 5G SBA	352
	Summary	354
	Review Questions	354
	References	354
19	5G Core Architecture	355

1	5G Core Overview	356
2	Reference Architecture	357
3	Network Functions in Detail	358
4	Control Plane vs User Plane in 5G Core	368
5	Major Comparison: 4G EPC vs 5G Core	370
6	Reference Points	372
7	Deployment Options	373
	Summary	376
20	5G Procedures	377
1	Registration (Initial Registration)	378
2	Authentication (5G-AKA)	384
3	Security Establishment	390
4	PDU Session Establishment	394
5	PDU Session Release	400
6	Service Request	404
7	Paging	407
8	Xn Handover	410
9	N2 Handover (Inter-AMF)	416
	Summary: Procedure Comparison Table	421
	Key References	422
21	Network Slicing	423
1	Why Network Slicing?	424
2	Network Slicing Concept	424
3	Slice Identification	426
4	Slice Selection (NSSF Role)	428
5	Slice Architecture	429
6	Slice Isolation	431
7	Slice Lifecycle Management	432
8	Slice Use Cases	434
9	Slicing vs. QoS vs. DNN	435
10	RAN Slicing	438
	Summary	439
	Key Takeaways	439
22	Quality of Service (QoS)	441
1	Why QoS?	442
2	QoS Evolution Across Generations	442
3	LTE Bearer-Based QoS	444
4	5G Flow-Based QoS	447
5	QoS Mapping and Enforcement	450

6	Bearer vs Flow Comparison	451
7	GBR vs Non-GBR vs Delay-Critical GBR	452
8	AMBR (Aggregate Maximum Bit Rate)	454
9	QoS Enforcement Architecture	455
	Summary: Key Takeaways	456
	QCI vs 5QI Quick Reference	456
23	User Plane	457
1	User Plane Concept	458
2	GTP-U Protocol (GPRS Tunneling Protocol — User Plane)	458
3	User Plane in LTE (4G)	460
4	User Plane in 5G	461
5	UPF Functions (3GPP TS 23.501)	463
6	Packet Trace Example: Downlink Data	464
7	User Plane Optimization	465
	Complete 5G User Plane Path Diagram	467
	Summary Table: User Plane Across Generations	468
	Key Takeaways	468
24	Control Plane	469
1	Control Plane Concept	470
2	Control Plane Protocols — MAJOR Reference Table	470
3	NAS (Non-Access Stratum)	472
4	RRC (Radio Resource Control)	474
5	NGAP (NG Application Protocol) — TS 38.413	476
6	GTP-C (GTP Control Plane)	477
7	HTTP/2 SBI (Service-Based Interface)	478
8	Signaling Protocol Evolution: SS7 → Diameter → HTTP/2	480
	Complete Control Plane Architecture Diagram (5G)	482
	Summary: Control Plane vs. User Plane	483
	Key Takeaways	483
25	Network Evolution — Grand Synthesis	484
1	Introduction	485
2	Master Comparison Table	485
3	Generation-by-Generation Analysis	487
4	Timeline Diagram	491
5	Architecture Evolution Diagram	492
6	Data Rate Evolution Graph (Exponential Growth)	494
7	Key Lessons Learned from Each Transition	495
8	Summary of Evolutionary Patterns	498
9	Exam Preparation Quick Reference	499
10	Conclusion	500

26	Future 6G Technologies for Advanced Communication Networks	502
⚠	IMPORTANT DISCLAIMER	503
	Overview: The Road to 6G	503
1	AI-Native Networks / AI-RAN	504
2	Extremely Large MIMO (XL-MIMO)	506
3	Cell-Free Massive MIMO	508
4	Integrated Sensing and Communication (ISAC / JCS)	510
5	Reconfigurable Intelligent Surfaces (RIS)	512
6	Sub-THz / THz Communications	515
7	Non-Terrestrial Networks (NTN)	517
8	Semantic Communications	519
9	Digital Twins for Networks	522
10	Advanced Positioning (cm-Level)	524
	Technology Maturity Matrix	527
	Projected Timeline: 3GPP Releases and 6G Technologies	528
	Summary: 5G vs 6G Paradigm Shifts	529
	Key Takeaways	529
	References and Further Reading	530

Introduction

Advanced Communication Networks is a 3-credit course (48 hours, 16 weeks $\times$ 3 hours) for senior undergraduate and graduate students of electrical and telecommunication engineering, taught as lectures with laboratories and a project. The course is a coherent technical journey through the evolution of mobile networks from GSM (2G) to LTE (4G), 5G NR/SA and the emerging concepts of 6G — not a collection of separate technologies, but a continuous analysis of **why** each generation appeared, **what** problem it solved and **how** traffic flows through its architecture. The course is built on a “why / what / how” logic so that students can analyse and evaluate architectural decisions instead of memorising lists of protocols.

Structure and Content

The material is organised in **26 modules**:

- **2G/3G foundations (modules 1–4):** GSM architecture, GPRS and EDGE, circuit switching and SS7 signalling, 2G/3G core networks and the basics of mobility management and handover.
- **4G access and core (modules 5–13):** LTE architecture, the E-UTRAN access network, the protocol stack, resource scheduling, the random access channel (RACH), the Evolved Packet Core (EPC), the Diameter protocol, the HSS and authentication, the MME, and the S-GW/P-GW with GTP tunnelling.
- **Software-driven architectures (modules 15–18):** software-defined networking (SDN), network function virtualisation (NFV), the 3GPP standardisation process and the 5G service-based architecture (SBA).
- **5G core and procedures (modules 14, 19–22):** the 5G core architecture, signalling procedures (registration, 5G-AKA authentication, PDU session establishment and release, paging, handover), network slicing and the evolution of the QoS framework (from QCI to 5QI).
- **Planes and outlook (modules 23–26):** the user plane and the control plane, a comprehensive synthesis of network evolution, and 6G research directions including ISAC, RIS, AI-native networks, cell-free MIMO and sub-THz.

Every module comes with comparison tables, architecture diagrams, step-by-step signalling sequence diagrams and review questions. A prerequisites review recalls the basics of OSI, RF, modulation, OFDM, IP networking, probability and decibel arithmetic.

Learning Objectives

By the end of the course, students can trace the evolution of the generations; analyse core architectures (circuit-switched, packet-switched, the all-IP EPC and the service-based 5G core); compare mobility management and handover across generations; apply network slicing, QoS, SDN and NFV in modern networks; explain 5G core procedures at the signalling level; and critically assess 6G research directions.

Assessment and Practical Work

Component	Weight
Exams (midterm 20% + final 30%)	50%
Laboratories (6 labs $\times$ 5%)	30%
Project (proposal 5% + report and demo 15%)	20%

The laboratory assignments cover the LTE link budget, EPC signalling traces, 5G NR scheduling and resource allocation, SBA service discovery, network slicing configuration and user-plane traffic tracing. The project, individual or in pairs, is defined on topics such as a comparative analysis of LTE versus 5G, an enterprise network-slicing design, implementing a scheduling algorithm with the simulator, or signalling analysis with real packet captures.

Main References

Sesia et al. (*LTE – The UMTS Long Term Evolution*), Dahlman et al. (*5G NR: The Next Generation Wireless Access Technology*), Holma and Toskala (*LTE for UMTS*), the 3GPP specifications (TS 23.501, 23.502, 24.501, 38.300) and Rappaport (*Wireless Communications*).

Course Information

Field	Details
Course Title	Advanced Communication Networks
Credits	3
Total Contact Hours	48 hours
Duration	16 weeks × 3 hours/week
Format	Lectures + Labs + Project
Target Students	Senior undergraduate / graduate EE, Telecom Engineering

Course Objectives

By the end of this course, students will be able to:

1. Trace the complete evolution of mobile networks from GSM (2G) through UMTS (3G), LTE (4G), 5G NR/SA, to emerging 6G concepts.
2. Explain WHY each generation was developed, WHAT problem it solved, and HOW traffic flows through each architecture.
3. Analyze core network architectures: circuit-switched, packet-switched, all-IP EPC, and service-based 5G Core.
4. Describe mobility management, handover mechanisms, and their evolution across generations.
5. Understand network slicing, QoS frameworks, SDN, and NFV in modern networks.
6. Explain 5G Core procedures (registration, authentication, PDU session) at the signaling level.
7. Critically assess 6G research directions (ISAC, RIS, AI-native, cell-free MIMO, sub-THz).

Module List (26 Modules)

#	Module Title	File
1	GSM, GPRS, and EDGE	01_gsm_gprs_edge.md
2	2G Circuit Switching	02_circuit_switching.md
3	2G/3G Core Networks	03_2g_3g_core.md
4	Mobility and Handover	04_mobility_handover.md
5	4G LTE Architecture	05_4g_lte.md
6	E-UTRAN (LTE Radio Access)	06_eutran.md
7	LTE Scheduling	07_lte_scheduling.md
8	Random Access Channel (RACH)	08_random_access.md
9	Evolved Packet Core (EPC)	09_epc.md
10	Diameter Protocol	10_diameter.md
11	Home Subscriber Server (HSS)	11_hss.md
12	Mobility Management Entity (MME)	12_mme.md
13	S-GW and P-GW	13_sgw_pgw.md
14	5G Networks Introduction	14_5g_networks.md
15	Software Defined Networking (SDN)	15_sdn.md
16	Network Function Virtualization (NFV)	16_nfv.md
17	3GPP Standardization	17_3gpp.md
18	5G Service-Based Architecture (SBA)	18_sba.md
19	5G Core Architecture	19_5g_core.md
20	5G Procedures	20_5g_procedures.md
21	Network Slicing	21_network_slicing.md
22	Quality of Service (QoS)	22_qos.md
23	User Plane	23_user_plane.md
24	Control Plane	24_control_plane.md
25	Network Evolution — Grand Synthesis	25_network_evolution.md
26	Future 6G Technologies	26_future_6g.md

Supporting files: [00_prerequisites.md](#) (background review), [00_syllabus.md](#) (this file)

Grading Breakdown

Component	Weight	Details
Exams	50%	Midterm (20%) + Final (30%)
Labs	30%	6 lab assignments (5% each)
Project	20%	Proposal (5%) + Report & Demo (15%)

Recommended Textbooks

1. **Sesia, S., Toufik, I., & Baker, M.** — *LTE – The UMTS Long Term Evolution: From Theory to Practice*, 2nd Ed., Wiley.
2. **Dahlman, E., Parkvall, S., & Sköld, J.** — *5G NR: The Next Generation Wireless Access Technology*, Academic Press.
3. **Holma, H. & Toskala, A.** — *LTE for UMTS*, Wiley.
4. **3GPP Specifications** — TS 23.501 (5GS), TS 23.502 (procedures), TS 24.501 (NAS), TS 38.300 (NR) — at 3gpp.org.
5. **Rappaport, T. S.** — *Wireless Communications: Principles and Practice*, 2nd Ed., Prentice Hall.

Week-by-Week Schedule

Week	Modules	Topic Focus
1	1–2	GSM architecture, GPRS/EDGE, circuit switching fundamentals
2	3	2G/3G core networks, UMTS architecture, CS/PS domains
3	4	Mobility management, handover types (2G→5G evolution)
4	5–6	LTE architecture, E-UTRAN, protocol stack; Lab 1 assigned
5	7–8	LTE scheduling, RACH procedure
6	9	Evolved Packet Core (deep dive); Lab 2 assigned
7	10–11	Diameter protocol, HSS, authentication
8	12–13	MME procedures, S-GW/P-GW, GTP; Midterm Review
9	—	MIDTERM EXAM
10	14	5G introduction, NR, NSA vs SA; Lab 3 assigned
11	15–16	SDN, NFV, cloud-native networks
12	17–18	3GPP standardization, Service-Based Architecture; Lab 4 assigned
13	19–20	5G Core architecture, 5G procedures (registration, PDU session)
14	21–22	Network slicing, QoS evolution (QCI→5QI); Lab 5 assigned
15	23–25	User plane, control plane, network evolution synthesis; Lab 6 assigned
16	26	Future 6G technologies, project presentations & final review
17	—	FINAL EXAM

Lab Assignments

Lab	Week	Topic	Related Modules
1	4	LTE Link Budget & Path Loss	5, 6
2	6	EPC Signaling (Attach/Detach trace)	9, 12
3	10	5G NR Scheduling & Resource Allocation	7, 14
4	12	5G Core SBA Service Discovery	18, 19
5	14	Network Slicing Configuration	21, 22
6	15	Traffic Flow Tracing (User Plane)	23, 24

Project

Students work individually or in pairs (max 2) on a semester project. Topics may include:

- Comparative analysis of LTE vs 5G for a specific use case
- Network slicing design for an enterprise scenario
- Implementation of a scheduling algorithm (using the wireless_lab simulator)
- 6G research topic survey and feasibility analysis
- Signaling analysis using real-world packet captures

Deliverables: Proposal (Week 13), Final Report + Demo/Presentation (Week 16)

Academic Integrity

All submitted work must be original. Collaboration on labs is encouraged for discussion but each student submits individually. Plagiarism will result in a zero for the assignment and referral to the academic integrity board.

Last updated: August 2026

This document reviews foundational concepts assumed as prior knowledge. Use it as a refresher before diving into the course modules.

1 OSI Model Review (Telecom Focus)

The OSI 7-layer model provides the reference framework. In mobile/telecom networks, the most relevant layers are:

Layer	Name	Telecom Relevance
1	Physical	RF transmission, modulation, antenna design, channel coding
2	Data Link	MAC scheduling, ARQ/HARQ, radio resource management
3	Network	IP routing, GTP tunneling, mobility management
4	Transport	TCP/UDP for end-to-end delivery, SCTP in signaling
7	Application	IMS/SIP for voice, HTTP/2 in 5G SBA interfaces

Key points to remember:

- Layers 1–2 are where most wireless-specific innovation happens (PHY/MAC).
- The “user plane” carries data (Layer 3+), while the “control plane” manages signaling.
- In 5G, the protocol stack is split: CU (PDCP and above) vs. DU (RLC/MAC) vs. RU (PHY).
- GTP-U tunnels encapsulate user IP packets between base station and core.

2 Basic RF Concepts

Frequency & Bandwidth

- Frequency (f):** Number of oscillations per second (Hz). Mobile bands range from 700 MHz to 100 GHz.
- Wavelength:** $\lambda = c/f$. Higher frequency → shorter wavelength → smaller antennas but higher path loss.
- Bandwidth (BW):** Range of frequencies occupied by a signal. More bandwidth → higher data rate (Shannon).

Modulation

- Purpose:** Map digital bits onto analog carrier signals.
- Common schemes:** BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM.
- Higher-order modulation = more bits/symbol but requires better SNR.

Signal-to-Noise Ratio (SNR)

- $\text{SNR} = P_{\text{signal}}/P_{\text{noise}}$ (linear) or $10 \log_{10}(P_{\text{signal}}/P_{\text{noise}})$ in dB.
- Shannon capacity: $C = \text{BW} \cdot \log_2(1 + \text{SNR})$ [bits/s].
- Determines which modulation and coding scheme (MCS) can be used.

Quick Reference

Modulation	Bits/symbol	Typical use
QPSK	2	Robust, used at the cell edge
64-QAM	6	High throughput, needs high SNR
256-QAM	8	Peak rates, very close to the base station

3 Digital Communications Basics

Bit Error Rate (BER)

- Probability that a received bit differs from the transmitted bit.
- Depends on SNR, modulation order, and coding.
- Typical targets: 10^{-3} (voice), 10^{-6} (data before ARQ).

Channel Coding

- **Purpose:** Add redundancy to detect/correct errors.
- **Turbo codes:** Used in 3G/4G for data channels.
- **LDPC codes:** Used in 5G NR for data (eMBB).
- **Polar codes:** Used in 5G NR for control channels.
- **Code rate (R):** Ratio of information bits to total bits. $R = k/n$ (e.g., $R = 1/2$ means 50% redundancy).

OFDM Concept

- **Problem:** Wideband channels suffer frequency-selective fading.
- **Answer** Divide wideband channel into many narrow subcarriers, each experiencing flat fading.
- **Key parameters:**
 - Subcarrier spacing (Δf): 15 kHz (LTE), 15/30/60/120/240 kHz (5G NR).
 - FFT size: determines number of subcarriers.
 - Cyclic prefix (CP): guard interval to combat inter-symbol interference (ISI).
- **Advantages:** Simple equalization (1-tap per subcarrier), flexible resource allocation.
- **OFDMA:** Multiple users share subcarriers → basis of LTE & 5G downlink.

4 IP Networking Fundamentals

Addressing

- **IPv4:** 32-bit addresses (e.g., 192.168.1.1), NAT common in mobile networks.
- **IPv6:** 128-bit addresses, increasingly used in 5G (avoids NAT).
- **Subnetting:** CIDR notation (e.g., 10.0.0.0/24 = 256 addresses).

Routing

- Mobile core networks use IP routing between network functions.
- **Static routes** for simple topologies; **BGP/OSPF** in carrier networks.
- **GTP tunneling:** Encapsulates subscriber IP packets inside GTP/UDP/IP between gNB and UPF.

Transport Protocols

Protocol	Properties	Use in Telecom
TCP	Reliable, ordered, congestion-controlled	HTTP/2 (5G SBA), web traffic
UDP	Unreliable, low-latency	VoIP/RTP, GTP-U user plane
SCTP	Multi-homing, message-oriented	S1AP/NGAP signaling (4G/5G)

DNS & NAT

- UEs obtain IP via PDN/PDU session establishment (not DHCP directly).
- DNS resolution for service discovery in 5G SBA (NRF).

5 Basic Probability

Why Probability in Telecom?

- Random access (RACH): collision probability with multiple users.
- Scheduling: modeling traffic arrivals (Poisson process).
- Fading: Rayleigh/Rician distributions model multipath.
- BER analysis: Q-function, error probability calculations.

Key Concepts

- **Random variable (X):** Outcome of a random experiment.
- **PDF / PMF:** Probability density/mass function.
- **Expected value:** $E[X] = \sum_i x_i P(x_i)$ – average outcome.
- **Variance:** $\text{Var}(X) = E[(X - \mu)^2]$ – spread around the mean.

Distributions You'll Encounter

Distribution	Use Case
Uniform	Random backoff timers
Exponential	Inter-arrival times (memoryless)
Poisson	Number of arrivals in a time window
Rayleigh	Envelope of multipath fading (NLOS)
Gaussian (Normal)	Noise modeling (AWGN)

Useful Formula

- Collision probability in slotted ALOHA with n users, 1 slot: $P(\text{collision}) = 1 - (1 - 1/K)^n \approx 1 - e^{-n/K}$ for large K .

6 Decibel Math

Why Decibels?

Wireless signals span enormous dynamic ranges (pW to W). Decibels compress this into manageable numbers and turn multiplications into additions.

Definitions

Unit	Formula	Reference
dB	$10 \log_{10}(P_1/P_2)$	Ratio (dimensionless)
dBm	$10 \log_{10}(P/1 \text{ mW})$	Absolute power ref. to 1 mW
dBW	$10 \log_{10}(P/1 \text{ W})$	Absolute power ref. to 1 W

Common Conversions

Linear	dB
$\times 2$	+3 dB
$\times 10$	+10 dB
$\times 100$	+20 dB
$\times 1000$	+30 dB
$\times 0.5$	−3 dB
$\times 0.1$	−10 dB

Quick Reference Table

Power	dBm
1 mW	0 dBm
10 mW	10 dBm
100 mW	20 dBm
1 W	30 dBm
20 W (macro BS)	43 dBm

Link Budget (Preview)

$$P_{\text{Rx}} [\text{dBm}] = P_{\text{Tx}} [\text{dBm}] + G_{\text{Tx}} [\text{dBi}] - L_{\text{path}} [\text{dB}] + G_{\text{Rx}} [\text{dBi}]$$

(received power = transmit power + transmit-antenna gain − path loss + receive-antenna gain)

Practice Problems

1. A transmitter outputs 2 W. Express in dBm. $\rightarrow 10 \log_{10}(2000) = 33 \text{ dBm}$.
2. Path loss is 130 dB, Tx = 43 dBm, antenna gains total 18 dBi. Rx power? $\rightarrow 43 + 18 - 130 = -69 \text{ dBm}$.
3. SNR = 20 dB. What is the linear ratio? $\rightarrow 10^{20/10} = 100$.

Summary Checklist

Before starting Module 2, verify you can:

- ☐ Identify which OSI layers handle PHY, MAC, and IP functions
- ☐ Convert between linear power and dBm
- ☐ Explain why OFDM combats frequency-selective fading
- ☐ Calculate Shannon capacity given BW and SNR
- ☐ Describe the difference between TCP and UDP and where each is used
- ☐ Compute simple collision probabilities for random access

If any topic feels unfamiliar, review the referenced sections or consult the recommended textbooks in the syllabus.

Last updated: August 2026

Module 1

GSM, GPRS, and EDGE

EDGE 9 GPRS 8 GSM

01



TDMA FRAME · 8 TIMESLOTS



In This Module

1	GSM Architecture	25
2	GSM Network Elements — Detailed Explanation	27
3	GSM Interfaces	30
4	GSM Mobility & Location Management	31
5	GSM Authentication & Security	32
6	WHY GPRS Was Needed	34
7	GPRS Architecture	35
8	PDP Context	37
9	GPRS Mobility Management	39
10	EDGE — Enhanced Data Rates for GSM Evolution	41
11	Evolution Path: GSM → GPRS → EDGE → 3G	43
	Summary Table	44
	Review Questions	44
	References	45

Learning Objectives

By the end of this module, you will be able to:

- Explain the complete GSM architecture and the role of each network element
- Describe how GSM handles mobility, location management, and authentication
- Articulate **why** GPRS was introduced and what limitations of GSM it addressed
- Explain the GPRS packet-switched overlay architecture
- Describe PDP context activation and GPRS mobility management
- Explain how EDGE improved data rates through modulation changes
- Trace the evolution path from GSM through EDGE to 3G

1 GSM Architecture

1.1 WHY GSM?

Before GSM (Global System for Mobile Communications), Europe had multiple incompatible analog cellular systems (NMT, TACS, C-Netz). A subscriber in France could not roam to Germany. GSM was created to solve three fundamental problems:

1. **Interoperability** — One standard across all of Europe (and eventually the world)
2. **Security** — Analog systems had no encryption; anyone with a scanner could listen
3. **Capacity** — Digital modulation serves more users per MHz of spectrum

1.2 GSM Network Architecture Diagram

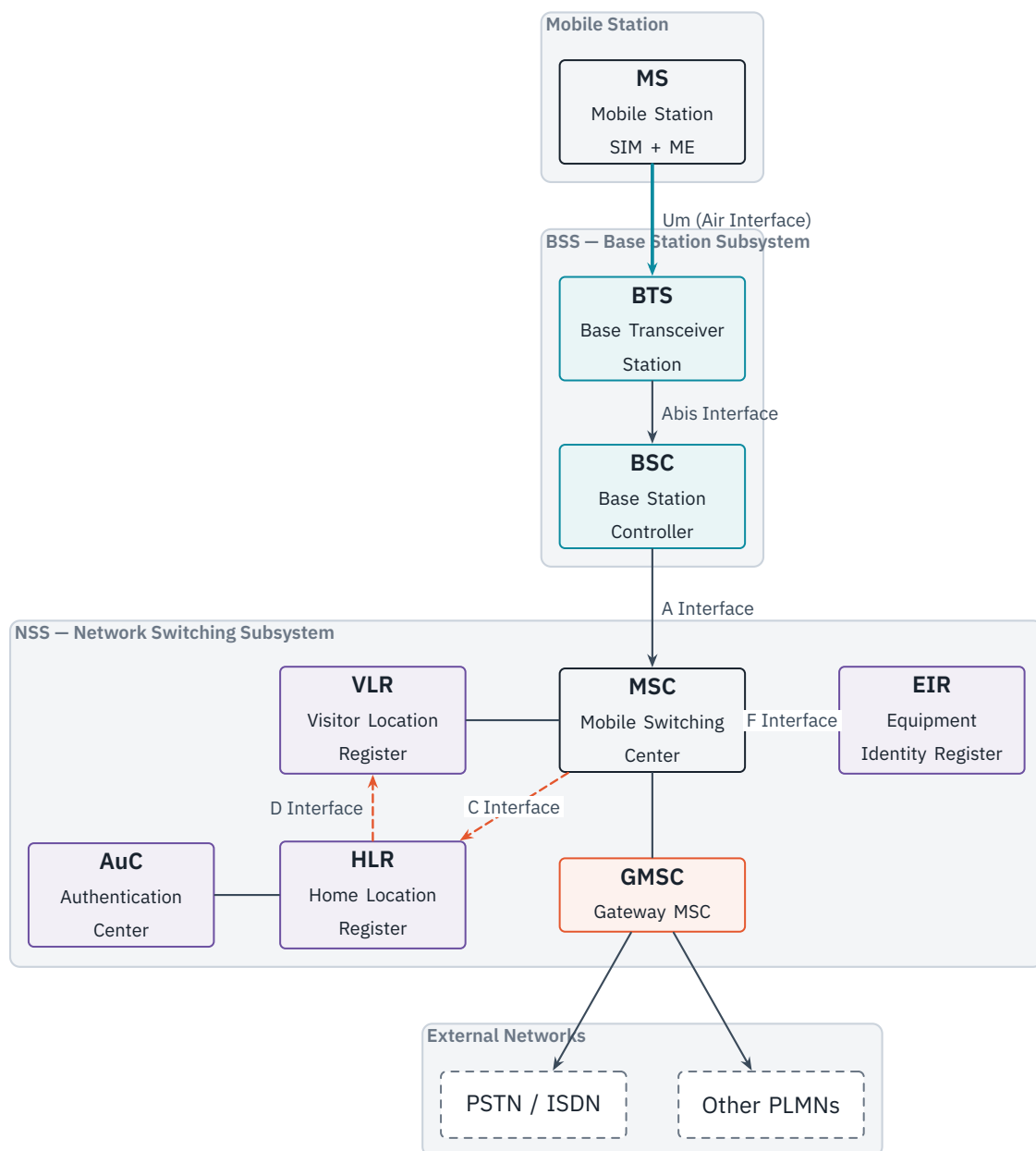


Figure 1.1 GSM network architecture: MS, BSS, NSS and external networks with their interfaces

1.3 Understanding the Architecture Layers

The GSM architecture is organized into three logical subsystems, each solving a distinct problem:

Subsystem	Problem It Solves	Elements
BSS (Base Station Subsystem)	Radio access — how does the phone talk to the network?	BTS, BSC
NSS (Network Switching Subsystem)	Call routing — how do calls get connected?	MSC, GMSC, HLR, VLR, AuC, EIR
OSS (Operation Support Subsystem)	Management — how do operators monitor and configure the network?	OMC, NMC

2 GSM Network Elements — Detailed Explanation

2.1 Mobile Station (MS)

What it is: The subscriber's device, consisting of two parts:

- **ME (Mobile Equipment):** The physical handset hardware, identified by its IMEI
- **SIM (Subscriber Identity Module):** A smart card carrying the subscriber's identity (IMSI), authentication key (Ki), and algorithms

Why it's split into ME + SIM: This separation means you can swap your SIM into any phone and keep your number, or an operator can block a stolen phone (by IMEI) without affecting the subscriber's account.

2.2 Base Transceiver Station (BTS)

What it is: The radio equipment (antennas, transceivers, amplifiers) at a cell site.

Why it exists: Someone has to convert between the wired network and radio waves. The BTS handles:

- Transmission and reception of radio signals
- Channel coding and encryption at the air interface
- Frequency hopping (if configured)
- Timing advance measurements (to compensate for propagation delay)

Key insight: The BTS is intentionally “dumb” — it has minimal intelligence. This is a deliberate design choice so that expensive radio sites don't need software upgrades for new features.

2.3 Base Station Controller (BSC)

What it is: The intelligence behind multiple BTSs. One BSC typically controls dozens to hundreds of BTSs.

Why it exists: The BSC solves the **resource management** problem:

- **Handover decisions:** When should a call be moved from one BTS to another? The BSC collects measurements from the MS and neighboring BTSs to decide.
- **Radio resource allocation:** Which frequency and timeslot should be assigned to a new call?
- **Power control:** Instructs the MS and BTS to adjust power (saves battery, reduces interference)
- **Transcoding:** Converts between the compressed speech codec on the air interface (13 kbps) and standard 64 kbps PCM used in the core network

2.4 Mobile Switching Center (MSC)

What it is: A telephone exchange specifically designed for mobile subscribers.

Why it exists: The MSC solves the **call routing for mobile users** problem. Unlike a fixed-line exchange, the MSC must handle:

- **Call setup and teardown** for users who may be moving
- **Handover between BSCs** (inter-BSC handover)
- **Interface to external networks** (PSTN, ISDN, other mobile networks)
- **Supplementary services** (call forwarding, call waiting, conference calls)
- **Coordination with VLR** for subscriber validation

2.5 Gateway MSC (GMSC)

What it is: A specialized MSC that acts as the entry point for calls coming FROM external networks.

Why it exists: When someone on a landline calls a mobile subscriber, the call arrives at the GMSC. The GMSC's job is to **find the subscriber** — it queries the HLR to determine which MSC/VLR area the subscriber is currently in, then routes the call there. Without the GMSC, external networks would need to know the internal structure of the mobile network.

2.6 Home Location Register (HLR)

What it is: The central database containing ALL permanent subscriber information.

Why it exists: The HLR solves the “**where is this subscriber?**” problem. It stores:

- Subscriber identity (IMSI, MSISDN)
- Subscription information (which services are allowed)
- **Current location** (which VLR/MSC area the subscriber is registered in)
- Call forwarding settings
- Supplementary service data

Key insight: There is logically ONE HLR per network (though physically it may be distributed). It is the single source of truth about every subscriber.

2.7 Visitor Location Register (VLR)

What it is: A temporary database associated with each MSC, holding information about subscribers currently in that MSC's area.

Why it exists: The VLR solves the **performance** problem. Without it, every single call setup would require a query to the HLR (which could be far away). Instead:

1. When a subscriber enters an MSC area, the VLR fetches their profile from the HLR
2. Subsequent operations use the local VLR copy
3. This reduces signaling load on the HLR and reduces call setup latency

The VLR also stores temporary data: the TMSI (temporary identity), location area, and authentication triplets.

2.8 Authentication Center (AuC)

What it is: A protected database storing the secret authentication keys (Ki) for every subscriber.

Why it exists: The AuC solves the “**how do we prove this subscriber is genuine?**” problem. It:

- Stores each subscriber's Ki (which never leaves the AuC or SIM)
- Generates authentication triplets (RAND, SRES, Kc) on request
- Protects against cloning and unauthorized access

Why it's separate from the HLR: Security isolation. The Ki values are extremely sensitive. By placing them in a physically secured, separate element, a compromise of the HLR doesn't expose authentication keys.

2.9 Equipment Identity Register (EIR)

What it is: A database of mobile equipment identities (IMEIs) classified into three lists:

- **White list:** Approved equipment
- **Grey list:** Equipment under observation (perhaps reported lost but not confirmed)
- **Black list:** Stolen or non-type-approved equipment — service denied

Why it exists: The EIR solves the **stolen phone** problem. Even if a thief puts a new SIM in a stolen phone, the network can identify and block the hardware by its IMEI.

3 GSM Interfaces

WHY standardized interfaces matter

GSM's genius was defining **open interfaces** between elements. This means a BTS from Ericsson can connect to a BSC from Nokia. Without standardized interfaces, operators would be locked to a single vendor for everything.

3.1 Interface Summary

Interface	Between	Protocol Stack	Purpose
Um	MS ↔ BTS	LAPDm / RR / MM / CC	Air interface — the radio link
Abis	BTS ↔ BSC	LAPD over PCM (E1/T1)	Carries traffic + signaling from radio sites to controller
A	BSC ↔ MSC	MTP / SCCP / BSSAP	Connects radio subsystem to switching subsystem
B	MSC ↔ VLR	MAP	MSC queries local VLR for subscriber data
C	GMSC ↔ HLR	MAP	GMSC asks HLR where to route incoming calls
D	VLR ↔ HLR	MAP	Location updates, subscriber data transfer
E	MSC ↔ MSC	MAP / ISUP	Inter-MSC handover
F	MSC ↔ EIR	MAP	IMEI checking
G	VLR ↔ VLR	MAP	Subscriber info transfer during inter-VLR moves
H	HLR ↔ AuC	Proprietary	Authentication vector requests

3.2 The Um Interface (Air Interface) — In Depth

The Um interface is the most complex because it must deal with the hostile radio environment:

- **Multiple access:** TDMA with 8 timeslots per 200 kHz carrier (FDMA/TDMA hybrid)
- **Modulation:** GMSK (Gaussian Minimum Shift Keying) — chosen for constant envelope (efficient power amplifiers) and spectral efficiency
- **Frame structure:** Hyperframe → Superframe → Multiframe → Frame → Timeslot → Burst

- **Logical channels:**

- **TCH (Traffic Channel):** Carries user speech/data
- **BCCH (Broadcast Control Channel):** System info broadcast to all phones
- **RACH (Random Access Channel):** MS requests access (slotted ALOHA)
- **PCH (Paging Channel):** Network pages MS for incoming call
- **SDCCH (Standalone Dedicated Control Channel):** Signaling (SMS, location update)
- **SACCH/FACCH:** Associated control for measurements and handover commands

3.3 The Abis Interface

Why it exists: BTS sites are remote (mountain tops, rooftops). The Abis interface lets them connect back to the BSC over standard 2 Mbps E1 lines (or microwave links). It carries:

- 16 kbps traffic channels (compressed speech)
- Signaling messages (LAPD protocol, similar to ISDN)
- Operation & Maintenance messages

3.4 The A Interface

Why it exists: This is where the radio world meets the telephony world. The A interface carries:

- 64 kbps PCM speech (after transcoding from 13 kbps)
- BSSAP signaling (BSS Application Part) over SS7

4 GSM Mobility & Location Management

4.1 WHY Location Management Is Needed

The fundamental challenge of mobile networks: **how do you deliver a call to someone whose location is unknown?**

Naive solution: Page every cell in the network. Problem: This doesn't scale — millions of subscribers would require constant paging on every cell.

GSM's solution: **Location Areas (LA)**

4.2 Location Area Concept

- The network is divided into **Location Areas**, each containing multiple cells (BTSs)
- Each LA has a unique **LAI** (Location Area Identity) = MCC + MNC + LAC
- The network only knows a subscriber's location to the **LA granularity**
- When a call arrives, the subscriber is paged in ALL cells of their current LA

Trade-off:

- Smaller LAs → less paging load, but more location updates (signaling)

- Larger LAs → fewer location updates, but more paging traffic
- Operators tune LA sizes based on traffic patterns

4.3 Location Update Procedure

A mobile performs a **location update** when:

1. **Normal location update:** MS detects it has entered a new LA (different LAI on BCCH)
2. **Periodic location update:** Timer expires — proves the MS is still reachable
3. **IMSI attach:** MS is powered on and registers with the network

Location Update Flow:

1. MS sends LOCATION UPDATING REQUEST to new MSC/VLR
2. New VLR contacts HLR: “Subscriber X is now here”
3. HLR sends subscriber profile to new VLR
4. HLR tells OLD VLR to delete the subscriber record
5. New VLR confirms to MS: LOCATION UPDATING ACCEPT
6. MS stores new TMSI (if reassigned)

4.4 TMSI — Temporary Mobile Subscriber Identity

Why: Broadcasting the IMSI over the air interface would let eavesdroppers track subscribers. The TMSI:

- Is assigned by the VLR and has only local significance
- Is changed frequently (at each location update or call)
- Means the permanent identity (IMSI) rarely travels over the air

5 GSM Authentication & Security

5.1 WHY Authentication Is Critical

Without authentication:

- Cloned SIMs could make calls billed to someone else
- Unauthorized users could access the network
- Eavesdroppers could intercept calls

5.2 The Authentication Triplet Mechanism

GSM uses a **challenge-response** protocol based on shared secret cryptography.

The players:

- **Ki:** A 128-bit secret key, stored ONLY on the SIM and in the AuC. It **never** travels over any interface.
- **RAND:** A 128-bit random number generated by the AuC (the challenge)

- **SRES:** A 32-bit Signed Response (the expected answer)
- **Kc:** A 64-bit cipher key for encrypting the air interface

How it works:

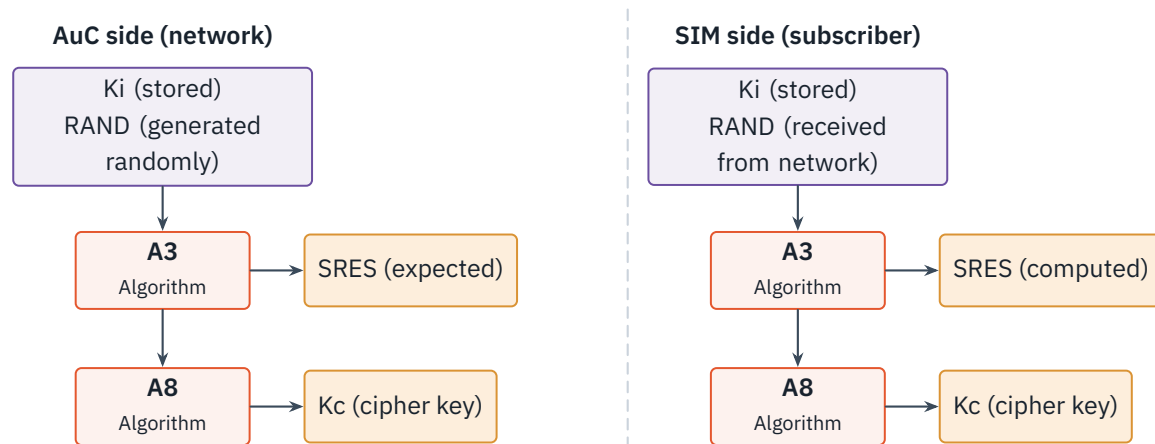


Figure 1.2 A3 and A8 run on both sides: the AuC and the SIM derive the same SRES and Kc from Ki and RAND

Step-by-step:

1. AuC generates RAND, computes $SRES = A3(Ki, RAND)$ and $Kc = A8(Ki, RAND)$
2. The triplet (RAND, SRES, Kc) is sent to the VLR (pre-computed, stored for later use)
3. When authentication is needed, VLR sends RAND to the MS
4. SIM computes $SRES = A3(Ki, RAND)$ using its stored Ki
5. MS sends SRES back to the network
6. VLR compares received SRES with stored SRES — if they match, subscriber is authentic
7. Both sides now have Kc — used with algorithm A5 to encrypt the air interface

Why triplets are pre-computed: The AuC may be geographically distant. Pre-computing multiple triplets and caching them at the VLR means authentication can happen locally without real-time AuC access.

5.3 Encryption (A5 Algorithm)

After authentication, both the MS and BTS share Kc. The A5 algorithm generates a keystream that XORs with the traffic. This provides:

- **Confidentiality** on the air interface (Um only — not end-to-end)
- Protection against casual eavesdropping

Known weakness: A5/1 has been broken. Modern networks use A5/3 (KASUMI-based) or have moved to 3G/4G with stronger security.

6 WHY GPRS Was Needed

6.1 The GSM Data Problem

GSM was designed for **voice**. Data was an afterthought:

GSM Data Service	Speed	Problem
CSD (Circuit-Switched Data)	9.6 kbps	Occupies a full timeslot for entire session
HSCSD (High-Speed CSD)	Up to 57.6 kbps	Occupies MULTIPLE timeslots — extremely expensive

The fundamental issue: Circuit switching is wrong for data.

Why? Consider web browsing:

- You request a page (1 second of activity)
- You read it (30 seconds of inactivity)
- You click a link (1 second of activity)
- You read again (30 seconds of inactivity)

With circuit switching, you're **paying for and consuming a timeslot during those 30 seconds of reading**. The channel is allocated but idle 95% of the time.

6.2 What GPRS Provides

GPRS (General Packet Radio Service) introduces **packet switching** to GSM:

- Resources are allocated **only when data is actually being sent**
- Multiple users **share** timeslots statistically
- Billing can be **per-megabyte** instead of per-minute
- “Always-on” connectivity — no dial-up delay
- Theoretical speeds: 8 timeslots × 21.4 kbps = **171.2 kbps** (theoretical max)
- Practical speeds: 30–80 kbps

6.3 GPRS Design Philosophy

GPRS was designed as an **overlay** on existing GSM infrastructure. Operators didn't need to replace their entire network — they added new packet-switching nodes alongside the existing circuit-switched core. This was critical for business adoption.

7 GPRS Architecture

7.1 New Network Elements

GPRS introduces three key new elements:

■ SGSN — Serving GPRS Support Node

What it is: The packet-switched equivalent of the MSC/VLR combination.

Why it exists: Someone needs to handle packet-switched sessions the way the MSC handles voice calls. The SGSN:

- Tracks the location of GPRS-attached mobiles (at Routing Area granularity)
- Performs authentication and ciphering for packet sessions
- Handles mobility management for packet data
- Routes packets between the MS and the GGSN
- Collects charging information

■ GGSN — Gateway GPRS Support Node

What it is: The packet-switched equivalent of the GMSC — the gateway to external data networks.

Why it exists: The mobile network must connect to the Internet (and corporate intranets). The GGSN:

- Acts as the **IP router** between the GPRS network and external packet data networks (PDNs)
- Assigns IP addresses to mobile devices (or relays from DHCP/RADIUS)
- Performs **encapsulation/de-encapsulation** of user data using GTP (GPRS Tunneling Protocol)
- Enforces policy and charging rules
- From the external network's perspective, the GGSN looks like a normal IP router

■ PCU — Packet Control Unit

What it is: An addition to the BSC that handles packet-switched radio resource management.

Why it exists: The original BSC only understood circuit-switched resource allocation. The PCU adds:

- Packet scheduling on shared radio channels (PDCH — Packet Data Channels)
- Segmentation and reassembly of LLC frames into radio blocks
- Dynamic allocation of timeslots between voice (TCH) and data (PDCH)
- Uplink state flag management and acknowledgment

7.2 GPRS Packet Data Path

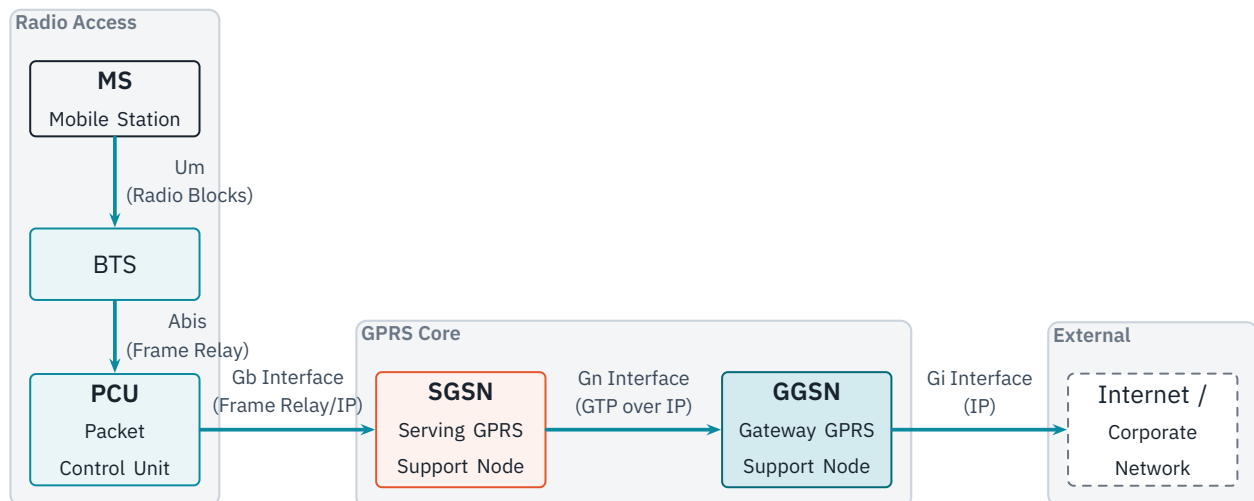


Figure 1.3 GPRS packet data path from the MS to the external packet network

7.3 GPRS Protocol Stack

The GPRS protocol stack introduces new layers:

Layer	Protocol	Purpose
Application	HTTP, SMTP, etc.	User application
Transport	TCP / UDP	End-to-end reliability
Network	IP	Addressing and routing
Tunneling	GTP (GPRS Tunneling Protocol)	Encapsulates IP packets between SGSN and GGSN
Link	LLC (Logical Link Control)	Reliable link between MS and SGSN
Link	SNDCP (SubNetwork Dependent Convergence Protocol)	Header compression, segmentation
Radio	RLC/MAC	Radio block transmission, ARQ
Physical	GMSK modulation on PDCH	Bits on the air

7.4 GPRS Interfaces

Interface	Between	Transport	Purpose
Gb	PCU/BSS ↔ SGSN	Frame Relay or IP	Packet data from radio to core
Gn	SGSN ↔ GGSN (same PLMN)	GTP/UDP/IP	User data tunneling within network
Gp	SGSN ↔ GGSN (different PLMN)	GTP + security	Roaming data
Gi	GGSN ↔ External PDN	IP	Connection to Internet
Gr	SGSN ↔ HLR	MAP	Subscriber data for GPRS
Gs	SGSN ↔ MSC/VLR	BSSAP+	Coordination of CS and PS

8 PDP Context

8.1 WHAT Is a PDP Context?

A **PDP Context (Packet Data Protocol Context)** is the “session” that must exist before a mobile can send or receive packet data. Think of it as the GPRS equivalent of “dialing in” — but without wasting resources when idle.

A PDP context defines:

- **PDP Type:** Usually IPv4 or IPv6
- **PDP Address:** The IP address assigned to the MS
- **Access Point Name (APN):** Which external network to connect to (e.g., “internet”, “corporate.vpn”)
- **QoS Profile:** Requested quality of service (priority, throughput, delay class)
- **GGSN Address:** Which GGSN serves this context

8.2 WHY PDP Context Exists

Without some form of session establishment:

- The GGSN wouldn’t know where to send downlink packets for a given IP address
- The network couldn’t enforce QoS or charging policies
- There would be no tunnel established between SGSN and GGSN

8.3 PDP Context Activation Procedure

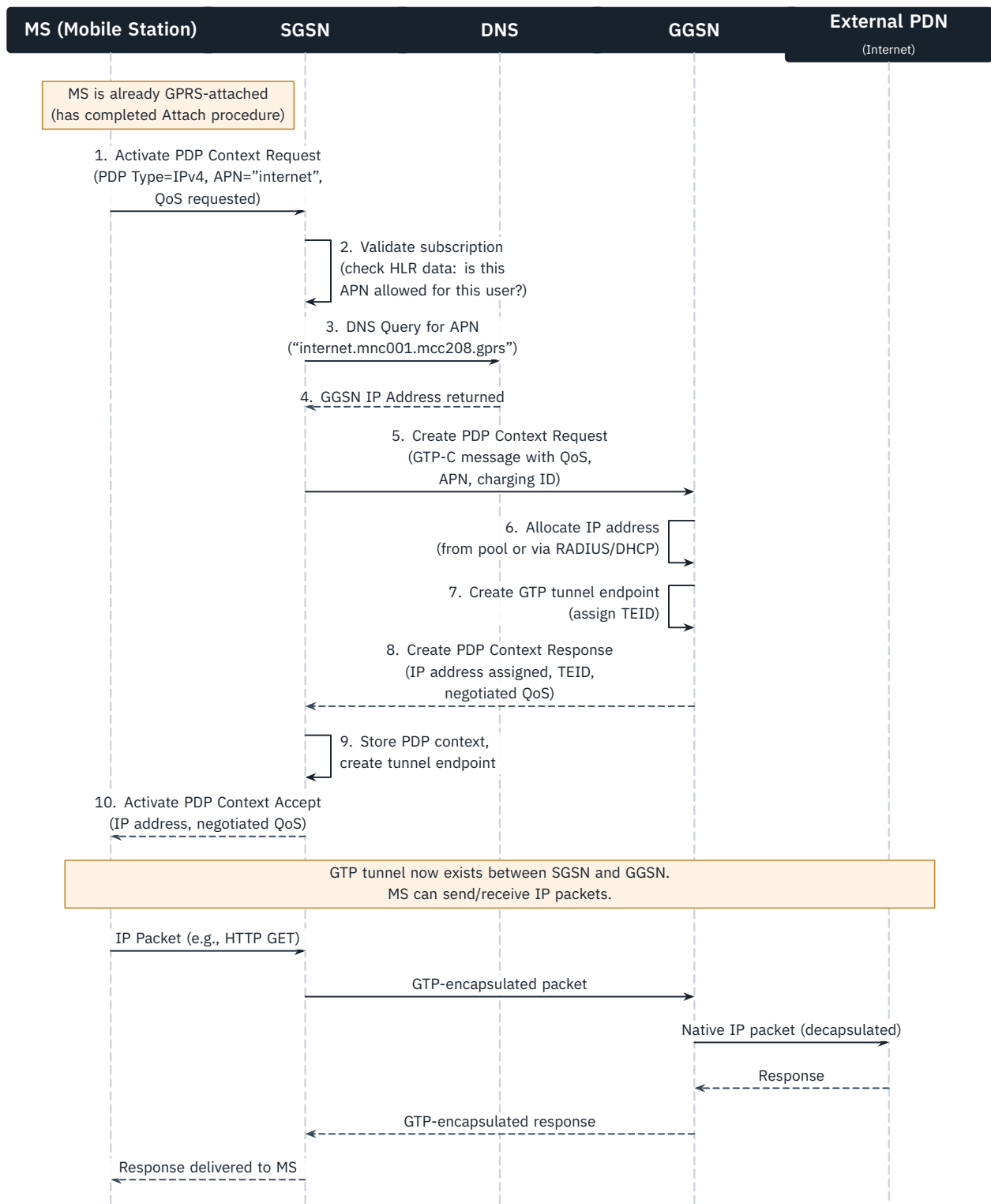


Figure 1.4 PDP context activation and the first IP packets through the GTP tunnel

8.4 PDP Context States

State	Meaning	Resources Used
INACTIVE	No PDP context — cannot send data	None
ACTIVE	Full context exists, data can flow	SGSN/GGSN tunnel, IP address allocated
STANDBY (optional)	Context exists but radio resources released	Tunnel maintained, no radio allocation

9 GPRS Mobility Management

9.1 WHY GPRS Needs Its Own Mobility Management

GSM already has location management (Location Areas, location updates). Why does GPRS need a separate system?

Because packet-switched and circuit-switched traffic have different needs:

- For a **voice call**, the network needs to find you immediately (paging must be fast)
- For **packet data**, there's more tolerance for delay — but the network needs to know which SGSN to route your downlink packets to

GPRS introduces the **Routing Area (RA)** — a subdivision of the Location Area.

9.2 Routing Area vs. Location Area

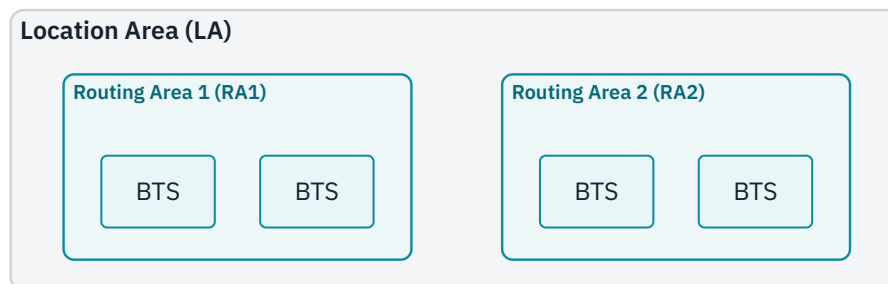


Figure 1.5 A Location Area is subdivided into Routing Areas

- A **Routing Area** is always a subset of (or equal to) a Location Area
- RA is identified by **RAI** = LAI + RAC (Routing Area Code)
- Finer granularity than LA → more precise location for packet routing

9.3 GPRS Mobility Management States

State	What the network knows	When
IDLE	Nothing — MS is not GPRS-attached	Power off or GPRS detached
STANDBY	Current Routing Area	Attached but no active data transfer
READY	Current Cell (exact)	Actively sending/receiving data

State transitions:

- IDLE → STANDBY: GPRS Attach
- STANDBY → READY: Data transfer begins (or signaling)
- READY → STANDBY: READY timer expires (no data for ~44 seconds)
- STANDBY → IDLE: GPRS Detach

9.4 Routing Area Update (RAU)

Similar to GSM's location update, but for the packet domain:

When triggered:

1. MS moves to a new Routing Area (different RAI on BCCH)
2. Periodic RAU timer expires
3. Combined RA/LA update (if CS and PS are coordinated)

Types:

- **Intra-SGSN RAU:** New RA served by same SGSN — local procedure, no GTP tunnel modification needed
- **Inter-SGSN RAU:** New RA served by different SGSN — requires:
 1. New SGSN contacts old SGSN to get PDP contexts
 2. New SGSN informs GGSN to update tunnel endpoints
 3. New SGSN updates HLR with new location
 4. Old SGSN releases resources

9.5 GPRS Attach Procedure

Before any PDP context can be activated, the MS must **attach** to the GPRS network:

1. MS sends **Attach Request** (with IMSI or P-TMSI)
2. SGSN authenticates the subscriber (same Ki-based mechanism as GSM)
3. SGSN updates HLR with subscriber's new SGSN location
4. SGSN assigns a **P-TMSI** (Packet-TMSI — equivalent of TMSI for packet domain)
5. SGSN sends **Attach Accept** to MS

After attach, the MS is “registered” for packet service but has no active data session yet — that requires PDP context activation.

10 EDGE — Enhanced Data Rates for GSM Evolution

10.1 WHY EDGE Was Needed

GPRS improved efficiency (packet switching) but was still limited by **GSM’s modulation scheme**:

- GSM/GPRS uses **GMSK** (Gaussian Minimum Shift Keying)
- GMSK encodes **1 bit per symbol**
- Maximum raw data rate per timeslot: ~21.4 kbps (after coding)
- Maximum practical throughput: ~40-60 kbps with 4 timeslots

For emerging applications (email attachments, early mobile web), this wasn’t enough. But operators couldn’t yet afford to deploy entirely new 3G networks.

EDGE’s value proposition: Significantly more speed using the **same spectrum, same cell sites, same network** — just upgraded software and transceivers.

10.2 WHAT EDGE Changes

■ Modulation: GMSK → 8PSK

Parameter	GPRS (GMSK)	EDGE (8PSK)
Modulation	GMSK	8PSK (8-Phase Shift Keying)
Bits per symbol	1	3
Raw bit rate per timeslot	22.8 kbps	69.2 kbps (3× faster)
Symbol rate	270.833 ksps	270.833 ksps (unchanged!)
Carrier bandwidth	200 kHz	200 kHz (unchanged!)
Peak data rate (8 TS)	171.2 kbps	473.6 kbps
Practical peak	~80 kbps	~200-236 kbps

Key insight: EDGE achieves 3× the raw bit rate because 8PSK encodes 3 bits in every symbol instead of 1. The symbol rate and carrier bandwidth don’t change — it’s purely a modulation efficiency gain.

■ Trade-off: Why Not Always Use 8PSK?

8PSK has **less distance between constellation points** than GMSK. This means:

- More susceptible to noise and interference
- Higher bit error rate at cell edges
- Not as robust in poor radio conditions

EDGE's solution: Link Adaptation

- The system measures channel quality in real-time
- Near the BTS (good signal): Use 8PSK with light coding (MCS-9: 59.2 kbps/slot)
- Far from BTS (weak signal): Fall back to GMSK with heavy coding (MCS-1: 8.8 kbps/slot)
- 9 different Modulation and Coding Schemes (MCS-1 through MCS-9)

10.3 EDGE Coding Schemes

MCS	Modulation	Code Rate	Data Rate per Slot	When Used
MCS-1	GMSK	~0.53	8.8 kbps	Very poor channel
MCS-2	GMSK	~0.66	11.2 kbps	Poor channel
MCS-3	GMSK	~0.80	14.8 kbps	Below average
MCS-4	GMSK	~1.0	17.6 kbps	Average (no redundancy)
MCS-5	8PSK	~0.37	22.4 kbps	Average channel
MCS-6	8PSK	~0.49	29.6 kbps	Good channel
MCS-7	8PSK	~0.76	44.8 kbps	Very good channel
MCS-8	8PSK	~0.92	54.4 kbps	Excellent channel
MCS-9	8PSK	~1.0	59.2 kbps	Best conditions

10.4 EGPRS: EDGE Applied to Packet Data

When EDGE modulation is applied to GPRS, the result is called **EGPRS (Enhanced GPRS)**:

- Same GPRS architecture (SGSN, GGSN, PDP contexts)
- Same protocol stack
- Only the physical layer and RLC layer are enhanced
- Adds **Incremental Redundancy (IR)**: If a block fails, retransmit with different puncturing; receiver combines attempts for better decoding probability

10.5 Network Changes Required for EDGE

Change	Description	Cost
New transceivers	BTS needs 8PSK-capable transmitters (linear amplifiers)	Moderate
Software upgrade	BSC, SGSN need new coding scheme support	Low
No new spectrum	Uses existing GSM frequencies	Zero
No new sites	Uses existing cell sites	Zero
No new core nodes	Same SGSN/GGSN infrastructure	Zero

This made EDGE extremely attractive as a “2.5G+” stepping stone.

11 Evolution Path: GSM → GPRS → EDGE → 3G

11.1 The Evolution Timeline

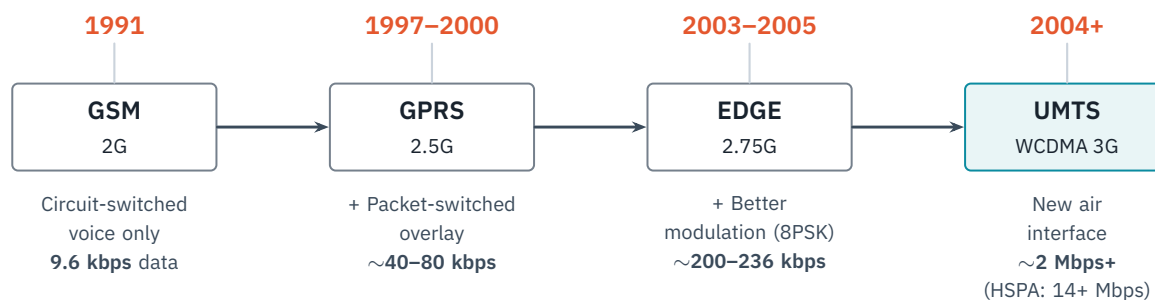


Figure 1.6 Evolution timeline from GSM to UMTS

11.2 What Each Step Added

Generation	Key Innovation	Problem Solved	What Stayed the Same
GSM	Digital cellular, roaming	Incompatible analog systems	—
GPRS	Packet switching overlay	Data was too expensive (circuit-switched)	GSM radio, BSS, spectrum
EDGE	8PSK modulation + link adaptation	GPRS was too slow for emerging apps	GPRS architecture, spectrum, sites
UMTS (3G)	New air interface (WCDMA), 5 MHz carriers	EDGE still couldn't support video/rich media	Core network concepts (SGSN → modernized)

11.3 WHY This Evolution Path Matters

The GSM → GPRS → EDGE progression demonstrates a key principle in telecom engineering:

Maximize return from existing infrastructure before deploying new generations.

Each step was designed to:

1. **Reuse** existing investments (spectrum, sites, core network)
2. **Add** capability incrementally
3. **Allow** operators to upgrade gradually (not “big bang”)

This philosophy continues today: 4G → 4G-Advanced → 5G NSA → 5G SA.

Summary Table

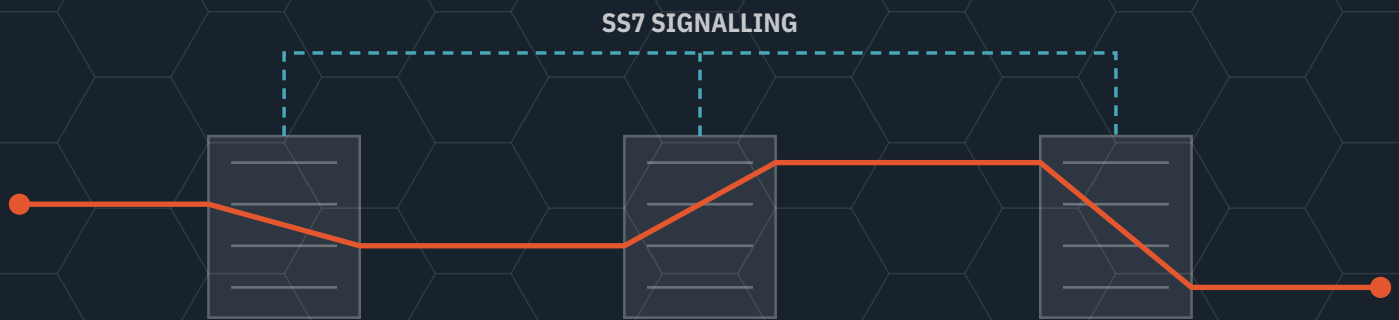
Topic	GSM	GPRS	EDGE
Switching	Circuit	Packet	Packet
Modulation	GMSK	GMSK	GMSK + 8PSK
Peak Data Rate	9.6 kbps (14.4 CSD)	171.2 kbps (theoretical)	473.6 kbps (theoretical)
Practical Data	9.6 kbps	30–80 kbps	120–236 kbps
Billing	Per minute	Per MB	Per MB
Always-on?	No (dial-up)	Yes	Yes
New Infrastructure	Complete network	+ SGSN, GGSN, PCU	+ 8PSK transceivers
New Spectrum?	Yes (900/1800 MHz)	No	No

Review Questions

1. Explain why the BSS is designed with “dumb” BTSs and an intelligent BSC. What are the advantages of this split?
2. Why does the VLR exist? What would happen if every operation required a query to the HLR?
3. In the GSM authentication process, why does the Ki never leave the SIM or AuC? What would be the consequence if it did?
4. Explain why circuit switching is fundamentally inefficient for bursty data traffic like web browsing.
5. What is the role of GTP tunneling between SGSN and GGSN? Why can’t packets just be routed normally?
6. Why does EDGE fall back to GMSK in poor radio conditions instead of always using 8PSK?
7. If EDGE can achieve ~236 kbps, why was 3G (UMTS) still needed?

References

- 3GPP TS 03.02: Network Architecture (GSM)
- 3GPP TS 23.060: GPRS Service Description (Stage 2)
- 3GPP TS 43.064: GERAN Overall Description (EDGE)
- 3GPP TS 45.001: Physical Layer on the Radio Path
- Halonen, Romero, Melero: “GSM, GPRS and EDGE Performance” (Wiley, 2003)
- Eberspächer, Vögel, Bettstetter: “GSM – Architecture, Protocols and Services” (Wiley, 2009)



In This Module

1	Circuit Switching Fundamentals	47
2	SS7 Signaling Concepts	48
3	Call Setup Process	49
4	Call Routing	53
5	Location Update Procedure	54
6	Authentication Procedure	57
7	Roaming	60
8	Call Release	61
9	Why Circuit Switching?	62
	Key Terms Summary	64

1 Circuit Switching Fundamentals

Concept: Dedicated Path

In circuit switching, a **dedicated communication path** is established between two endpoints for the entire duration of the call. This path is reserved exclusively — no other user can use those resources until the call is released.

How It Works in GSM

Step	Action
1	Caller initiates a call
2	Network allocates a dedicated timeslot on the air interface (TDMA)
3	A fixed path is reserved through BTS → BSC → MSC → trunk to destination
4	Resources remain allocated for the entire call duration
5	Upon hangup, all resources are released

Resource Reservation

- **Air Interface:** A dedicated TDMA timeslot (one of 8 per carrier frequency) is assigned
- **Abis Interface** (BTS↔BSC): A 16 kbps or 64 kbps channel on E1/T1 links
- **A Interface** (BSC↔MSC): A 64 kbps timeslot on E1/T1 links
- **Trunk** (MSC↔PSTN/other MSC): A 64 kbps circuit on PCM links

Key Characteristics

- **Connection-oriented:** Path established before data flows
- **Guaranteed bandwidth:** Fixed 64 kbps per direction once connected
- **Constant delay:** Predictable latency (~150ms end-to-end target)
- **Inefficient for bursty data:** Resources wasted during silence periods

2 SS7 Signaling Concepts

What is SS7?

Signaling System 7 (SS7) is the out-of-band signaling protocol stack used by telecom networks to:

- Set up and tear down calls
- Manage subscriber databases
- Enable roaming and handovers
- Provide supplementary services (call forwarding, SMS, etc.)

SS7 signaling is **separate** from the voice path — this is called **Common Channel Signaling (CCS)**.

SS7 Protocol Stack (Simplified)

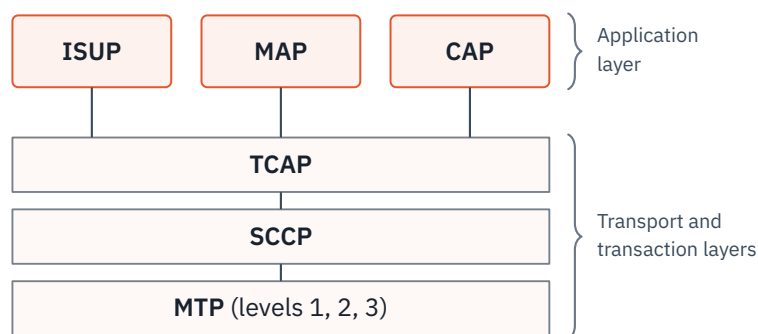


Figure 2.1 Simplified SS7 protocol stack

Key Protocols

ISUP (ISDN User Part)

- **Purpose:** Call setup, management, and release between exchanges (MSCs)
- **Messages:** IAM (Initial Address Message), ACM (Address Complete), ANM (Answer), REL (Release), RLC (Release Complete)
- **Used for:** Establishing the voice circuit between switches

MAP (Mobile Application Part)

- **Purpose:** Communication between GSM network elements (MSC, HLR, VLR, AuC)
- **Operations:** Location updates, authentication data retrieval, subscriber info queries, SMS routing
- **Used for:** Mobility management, roaming, subscriber data exchange

TCAP (Transaction Capabilities Application Part)

- **Purpose:** Provides a framework for MAP and other applications to exchange structured data
- **Model:** Request-response transactions (invoke, result, error)

- **Used for:** Carrying MAP operations over the SS7 network

■ Signaling Points

Element	Role
SSP (Service Switching Point)	MSC/GMSC — originates/terminates signaling
STP (Signal Transfer Point)	Routes signaling messages between SSPs
SCP (Service Control Point)	Database nodes (HLR, AuC)

3 Call Setup Process

3.1 Mobile-Originated Call (MOC)

A Mobile-Originated Call is initiated by the mobile subscriber.

■ Sequence of Events

1. **UE dials number** → sends CHANNEL REQUEST on RACH
2. **BTS** forwards to BSC
3. **BSC** assigns a dedicated channel (SDCCH) → IMMEDIATE ASSIGNMENT
4. **UE** sends CM SERVICE REQUEST (service type = MOC) to MSC via BSC
5. **MSC** authenticates the subscriber (see Section 6)
6. **MSC** activates ciphering
7. **UE** sends SETUP message with called party number
8. **MSC** sends CALL PROCEEDING to UE
9. **MSC** analyzes digits, determines routing
10. **MSC** sends IAM (ISUP) toward GMSC/PSTN
11. **GMSC/PSTN** returns ACM (Address Complete) → ringback
12. **MSC** sends ALERTING to UE
13. **Called party answers** → ANM (Answer Message) returned
14. **MSC** sends CONNECT to UE
15. **BSC** assigns a TCH (Traffic Channel) for voice
16. **Voice path established** — conversation begins

■ MOC Sequence Diagram

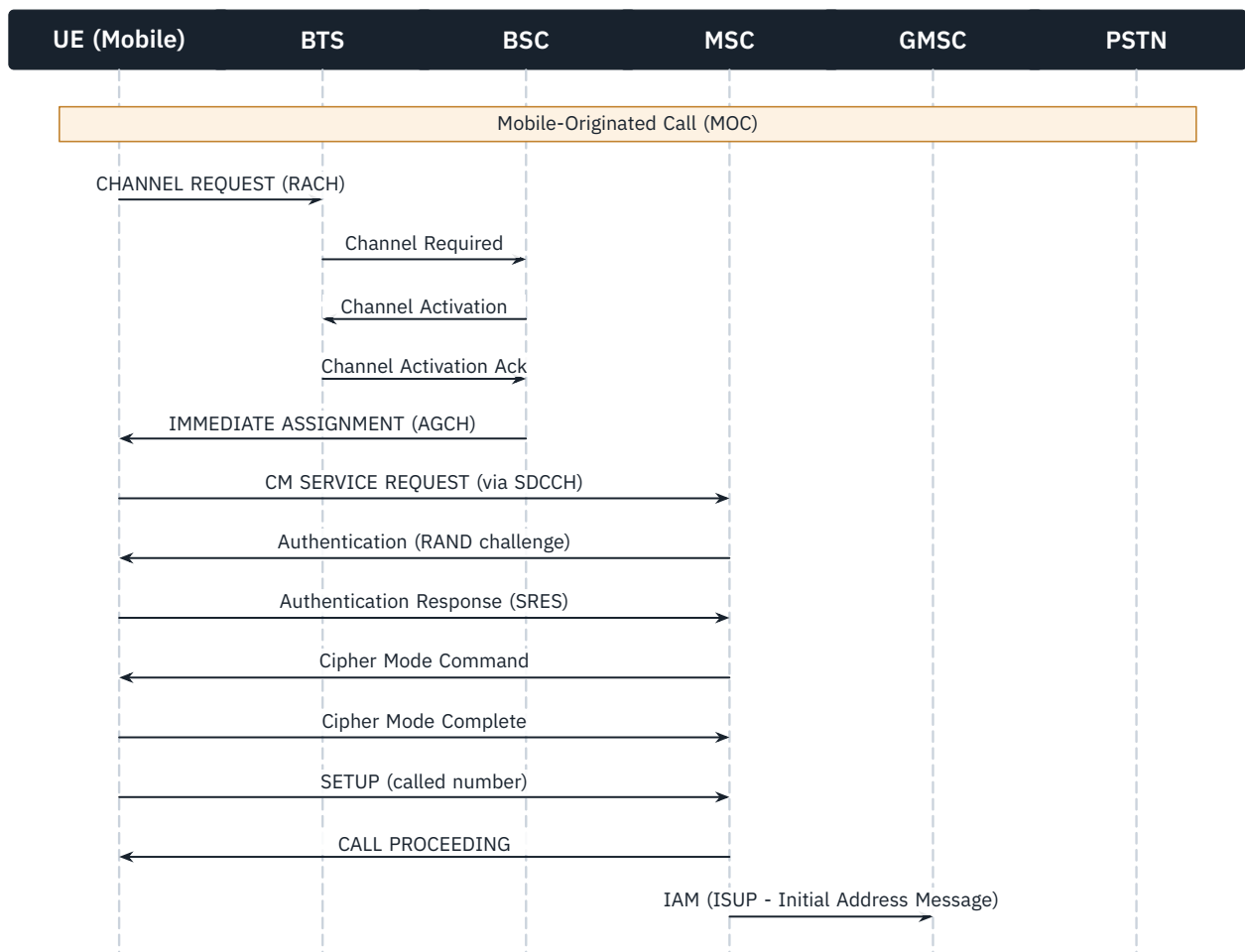


Figure 2.2 Mobile-originated call (MOC) set-up (part 1 of 2)

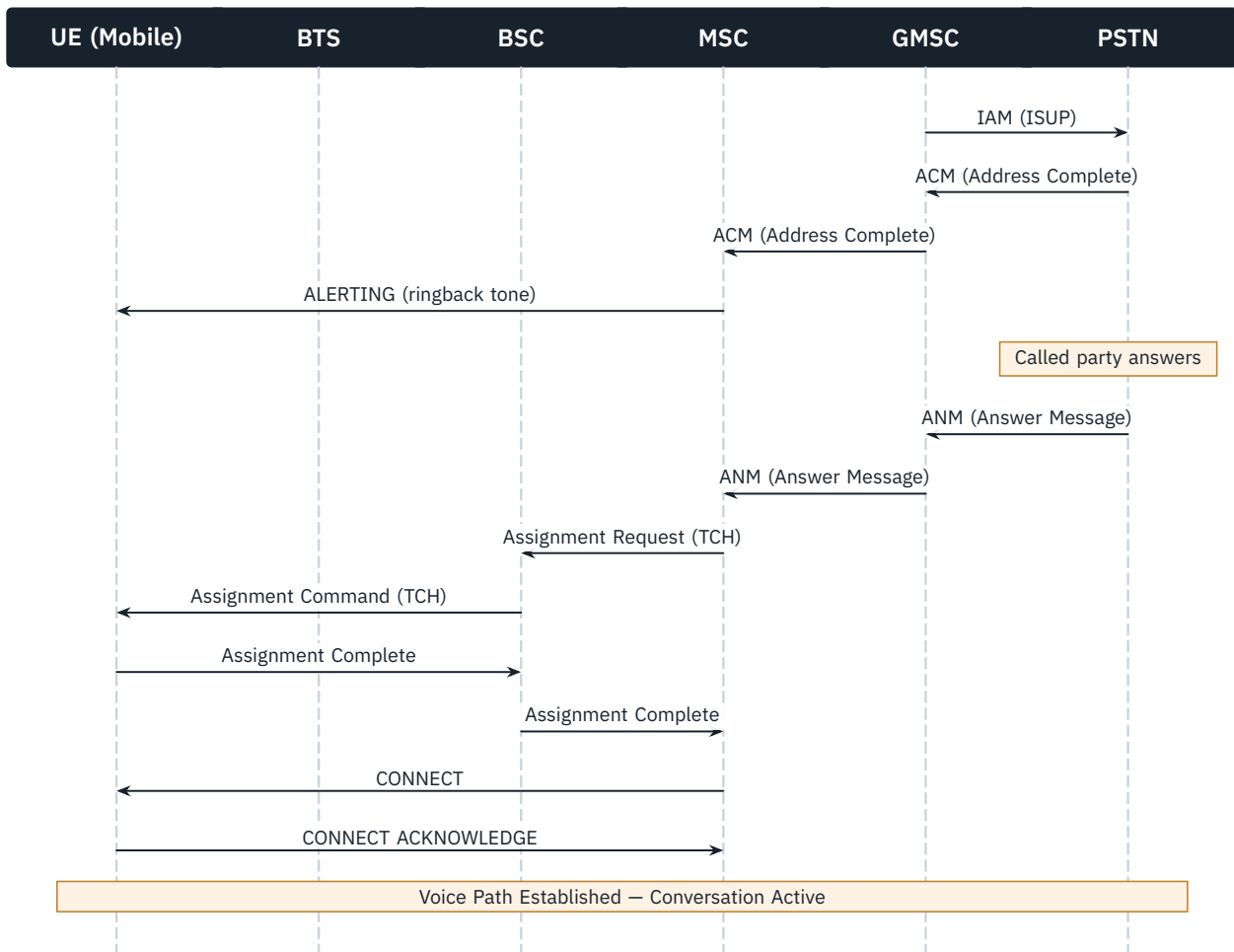


Figure 2.3 Mobile-originated call (MOC) set-up (part 2 of 2)

3.2 Mobile-Terminated Call (MTC)

A Mobile-Terminated Call is a call arriving for a mobile subscriber.

■ Sequence of Events

1. **PSTN/calling party** sends IAM to GMSC with called MSISDN
2. **GMSC** queries HLR: “Where is this subscriber?” (MAP: Send Routing Info)
3. **HLR** returns MSRN (Mobile Station Roaming Number) pointing to serving MSC
4. **GMSC** routes call to serving MSC using MSRN (IAM)
5. **MSC** queries VLR for subscriber state and location area
6. **MSC** pages the UE in the Location Area
7. **UE** responds to paging on RACH
8. **BSC** assigns SDCCH → authentication & ciphering
9. **MSC** sends SETUP to UE
10. **UE** sends CALL CONFIRMED
11. **UE** alerts user (ringtone)
12. **User answers** → UE sends CONNECT
13. **MSC** sends ANM back toward GMSC/PSTN
14. **Traffic channel assigned** — voice path active

■ MTC Sequence Diagram

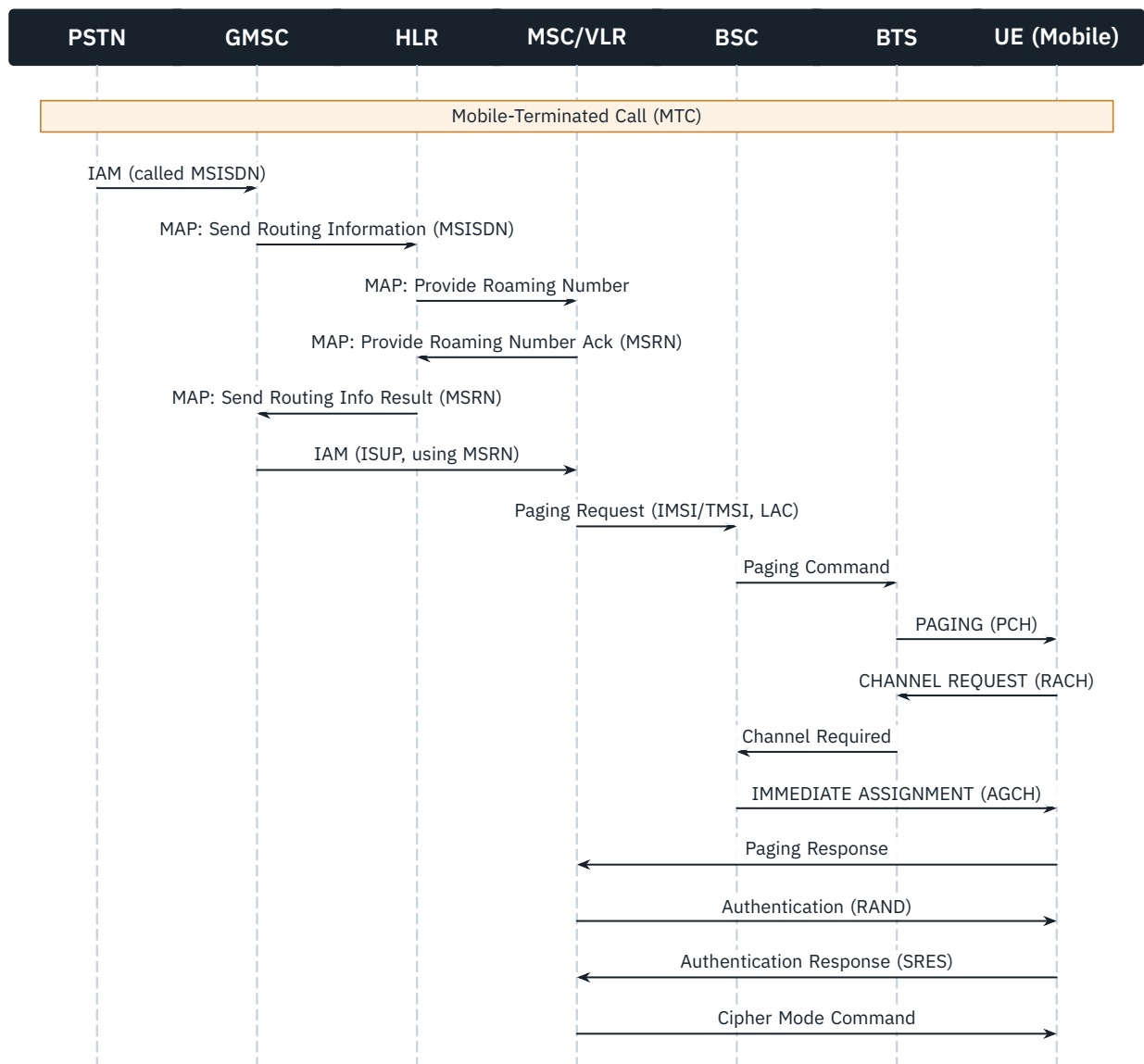


Figure 2.4 Mobile-terminated call (MTC) set-up (part 1 of 2)

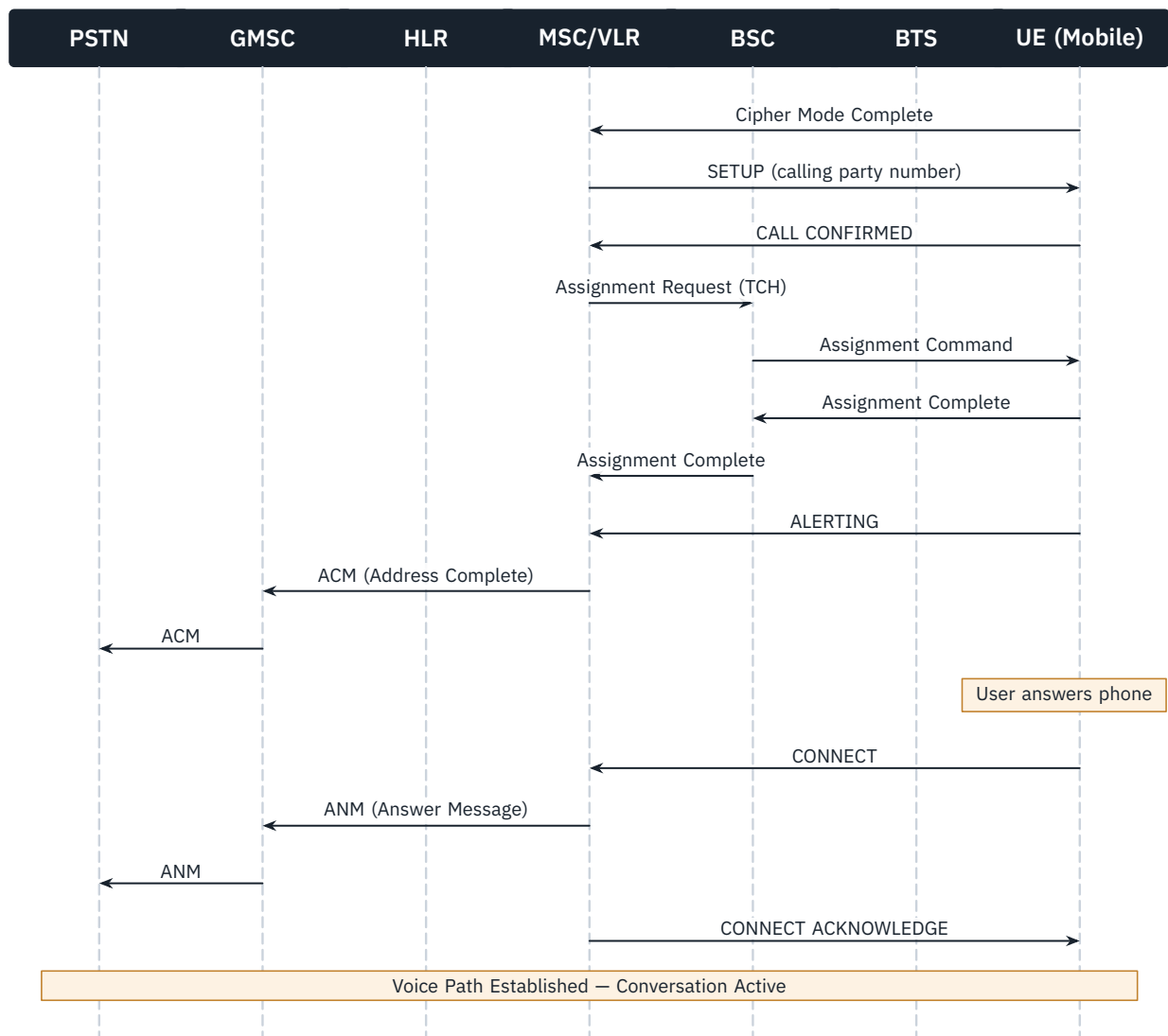


Figure 2.5 Mobile-terminated call (MTC) set-up (part 2 of 2)

4 Call Routing

How MSC/GMSC Route Calls Using HLR

The HLR is the **master database** of all subscribers. It knows which MSC/VLR currently serves each subscriber.

Routing a Mobile-Terminated Call

```

Step 1: PSTN delivers call to GMSC (based on MSISDN number plan)
Step 2: GMSC sends MAP "Send Routing Information" to HLR
Step 3: HLR identifies the serving MSC/VLR for that subscriber
Step 4: HLR asks serving VLR to allocate an MSRN (temporary routing number)
Step 5: VLR returns MSRN to HLR
Step 6: HLR returns MSRN to GMSC
Step 7: GMSC uses MSRN to route the ISUP IAM to the correct MSC
    
```

MSRN (Mobile Station Roaming Number)

- Temporary number in E.164 format
- Looks like a regular phone number in the serving network's numbering plan
- Allows GMSC to route the call using standard ISUP without knowing GSM internals
- Allocated per call, released after routing

Routing a Mobile-Originated Call

For MOC, routing is simpler:

1. MSC receives the dialed digits from the UE
2. MSC performs digit analysis (number plan, prefixes, supplementary services)
3. MSC routes the call via ISUP toward the appropriate trunk/GMSC/PSTN gateway
4. If calling another mobile on same network → MSC may query HLR for MSRN

5 Location Update Procedure

Why Location Updates Are Needed

The network must know **where a subscriber is** to deliver calls and messages. Without location tracking:

- The network would have to page every cell in the entire country
- This would overwhelm the paging channel capacity
- Call setup time would be unacceptable

Location updates allow the network to **narrow paging to a specific Location Area** (group of cells).

Types of Location Updates

Type	Trigger
IMSI Attach	Phone powers on — informs network subscriber is reachable
Normal LU	UE moves to a new Location Area (detects new LAI on BCCH)
Periodic LU	Timer-based update to confirm UE is still reachable
IMSI Detach	Phone powers off gracefully — informs network

Location Update Sequence Diagram

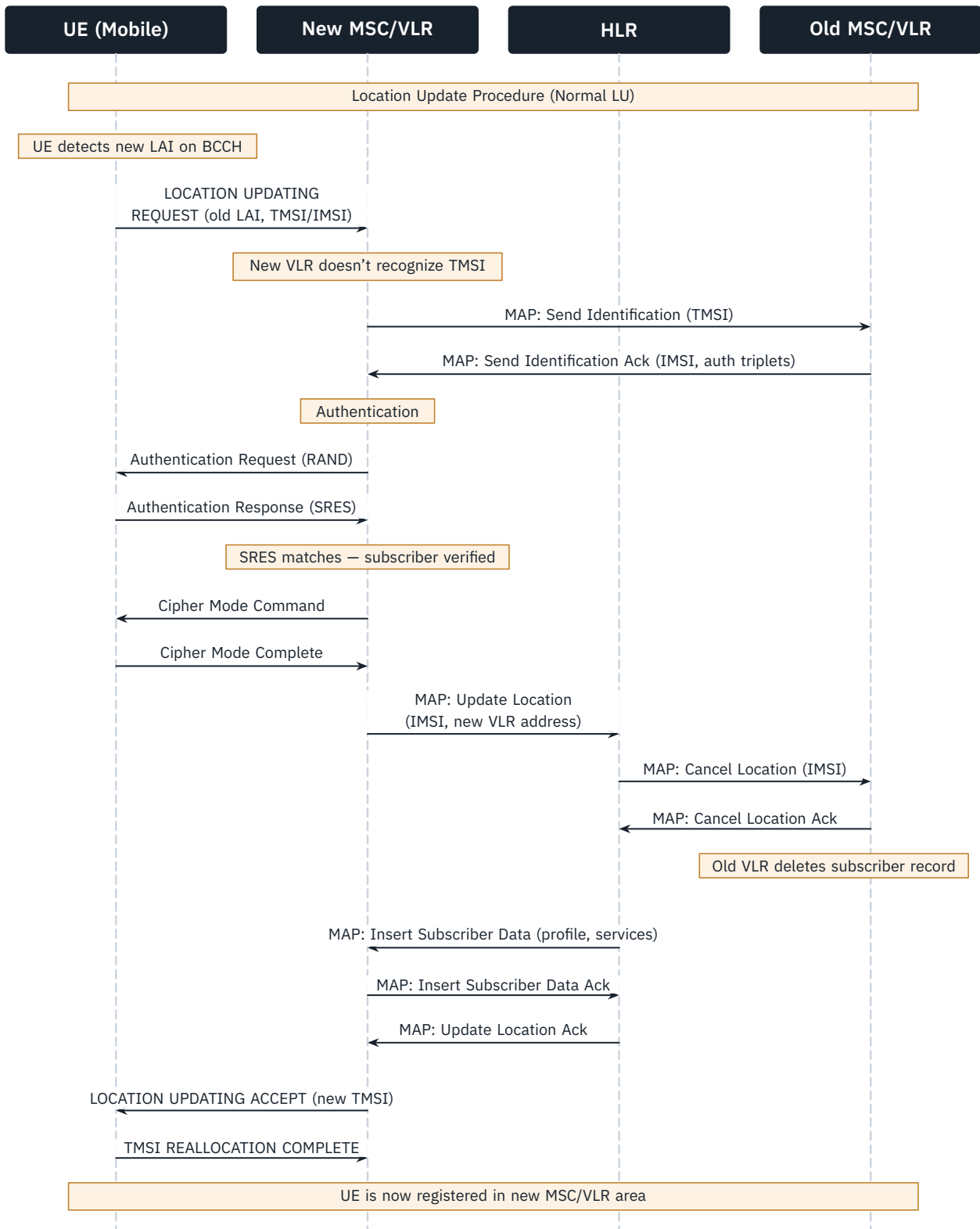


Figure 2.6 Location update with a change of MSC/VLR

IMSI Attach

When a phone powers on:

1. UE reads BCCH to get LAI and network information
2. UE sends LOCATION UPDATING REQUEST with attach flag
3. Same procedure as normal LU follows
4. VLR marks subscriber as “attached” (reachable)

Periodic Location Update

- Network broadcasts a timer value (T3212) on BCCH
- UE must send a LU before the timer expires, even if it hasn’t moved
- If the timer expires without an LU, VLR marks subscriber as “implicitly detached”
- Purpose: detect phones that powered off without IMSI Detach (battery died, etc.)

6 Authentication Procedure

Purpose

Authentication ensures that the subscriber is who they claim to be, preventing:

- Unauthorized network access
- Cloned SIM fraud
- Eavesdropping (when combined with ciphering)

The Authentication Triplet

The AuC (Authentication Center, co-located with HLR) generates **authentication triplets**:

Element	Size	Purpose
RAND	128 bits	Random challenge sent to UE
SRES	32 bits	Expected response (computed from RAND + Ki)
Kc	64 bits	Cipher key for encrypting the air interface

How Triplets Are Generated

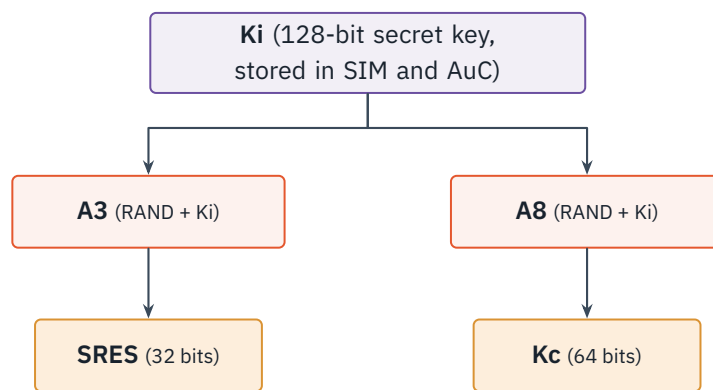


Figure 2.7 Generation of an authentication triplet (RAND, SRES, Kc)

- **Ki** never leaves the SIM card or the AuC
- The same computation happens independently on both sides
- If SRES from UE matches SRES from triplet → subscriber is authenticated

Authentication Sequence Diagram

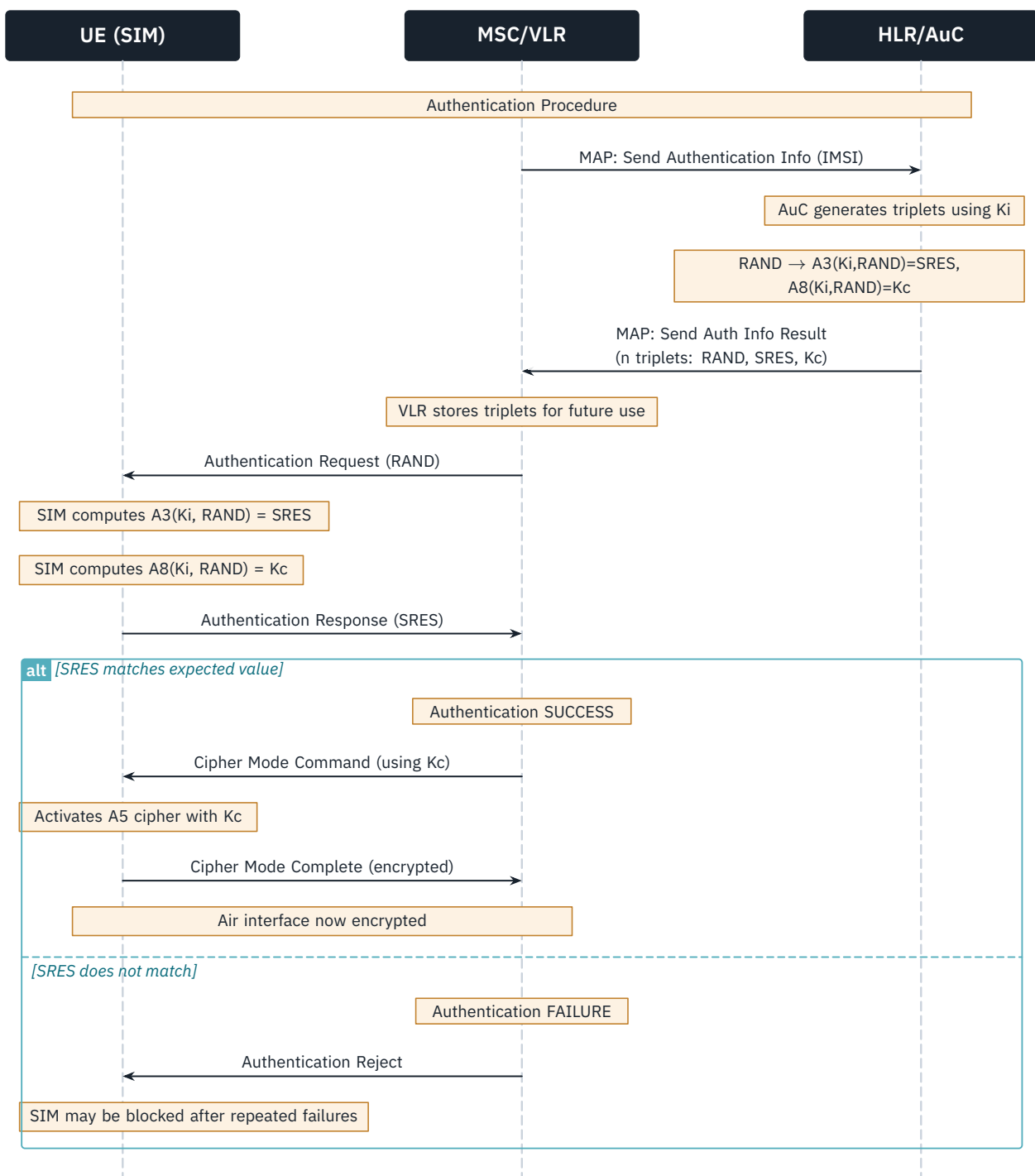


Figure 2.8 GSM authentication and ciphering

Key Points

- VLR stores **multiple triplets** (typically 5) to avoid querying HLR for every transaction
- Authentication is performed at: call setup, location update, SMS, supplementary service activation
- GSM authentication is **one-way** – network authenticates the UE, but UE does not authenticate the network (vulnerability fixed in 3G/UMTS)

7 Roaming

Concepts

Term	Definition
HPLMN	Home PLMN — the network where the subscriber has their subscription
VPLMN	Visited PLMN — a foreign network the subscriber is using while roaming
Roaming Agreement	Commercial + technical agreement between two operators
GRX/IPX	Interconnect network for signaling between PLMNs

How Roaming Works

1. **UE powers on in foreign country** → selects VPLMN (based on SIM's preferred PLMN list)
2. **UE performs Location Update** to VPLMN's MSC/VLR
3. **VPLMN's VLR** contacts **HPLMN's HLR** via SS7/MAP (through international signaling links)
4. **HLR** authenticates subscriber (sends triplets to VPLMN)
5. **HLR** sends subscriber profile to VPLMN's VLR
6. **HLR** records the VPLMN's VLR address as the subscriber's current location
7. **Subscriber can now make/receive calls** through VPLMN

Signaling Path for Roaming

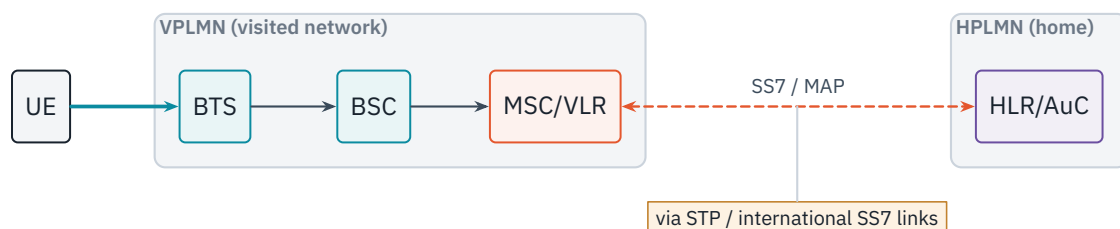


Figure 2.9 Signalling path for roaming: visited MSC/VLR to home HLR/AuC over SS7/MAP

Incoming Call to a Roaming Subscriber

1. Call arrives at HPLMN's GMSC
2. GMSC queries HLR → learns subscriber is in VPLMN
3. HLR requests MSRN from VPLMN's VLR
4. VPLMN's VLR returns MSRN
5. HLR gives MSRN to GMSC
6. GMSC routes call to VPLMN's MSC using MSRN (via international trunks)
7. VPLMN's MSC pages and connects the subscriber

Outgoing Call from Roaming Subscriber

1. UE initiates call in VPLMN
2. VPLMN's MSC handles the call locally
3. Call is routed from VPLMN to destination (may transit through HPLMN or go direct)
4. CAMEL/CAP protocol may be used for prepaid control from HPLMN

8 Call Release

Normal Call Release (Mobile-Initiated)

1. **UE** sends DISCONNECT to MSC
2. **MSC** sends REL (Release) via ISUP toward the far end
3. **Far end** responds with RLC (Release Complete)
4. **MSC** sends RELEASE to UE
5. **UE** sends RELEASE COMPLETE
6. **BSC** deallocates the TCH (traffic channel)
7. **All resources freed**: timeslots, trunks, circuit references

Network-Initiated Release

Can be triggered by:

- Far-end hangup
- Radio link failure (UE out of coverage)
- BSC/MSC timer expiry
- Administrative disconnect

ISUP Release Messages

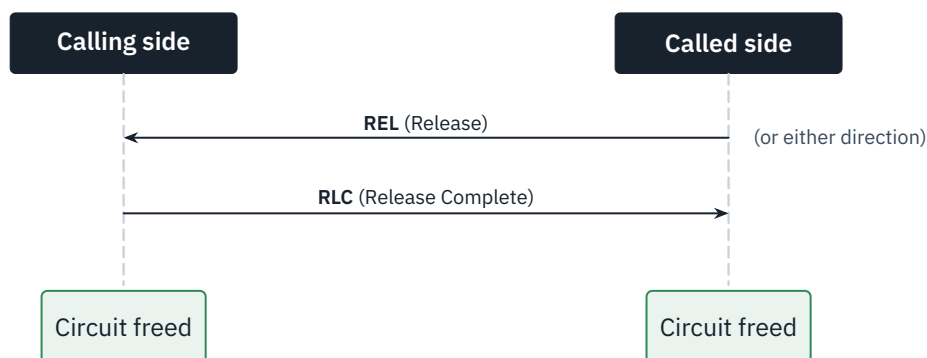


Figure 2.10 ISUP call release

9 Why Circuit Switching?

Why Circuit Switching Was Used for Voice

Voice Requirements

- **Constant bit rate:** 13 kbps (full rate) or 6.5 kbps (half rate) — continuous stream
- **Low and predictable delay:** < 150ms one-way for acceptable conversation quality
- **No jitter tolerance:** Human ear is very sensitive to variable delay
- **Symmetric traffic:** Both parties speak roughly equally

Why Circuit Switching Fits Voice

Requirement	How CS Addresses It
Low delay	Dedicated path = no queuing delay
No jitter	Fixed timeslot allocation = constant delay
Guaranteed bandwidth	Reserved resources = no congestion drops
Simple processing	Once path is set up, switching is trivial
Real-time	No store-and-forward delays

Historical Context (1980s-90s)

- Processing power was expensive — simple switching was preferred
- Memory was expensive — no buffering/packetization
- Existing PSTN was circuit-switched — natural integration
- Voice was the **only** service expected from mobile phones

Why Circuit Switching Became a Limitation

The Problem with Data

Issue	Explanation
Resource waste	A 64 kbps circuit is held even during silence (50-60% of a voice call)
No statistical multiplexing	Cannot share resources among multiple users dynamically
Fixed bandwidth	Cannot burst above 64 kbps even when network is idle
Call setup delay	3-7 seconds to establish a circuit — unacceptable for web browsing
Per-minute charging	Inefficient for checking email (30 seconds of data, charged 1 minute)
Scalability	Each data session consumes a full voice circuit

Data Traffic Characteristics vs. Voice

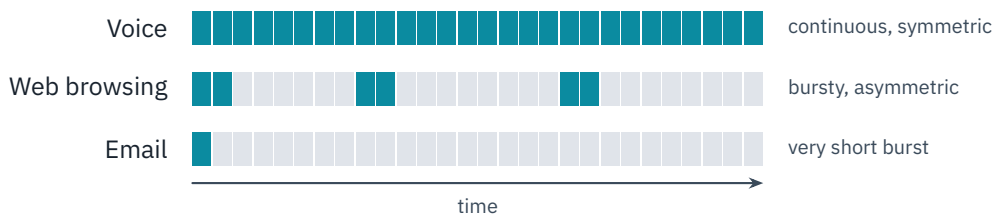


Figure 2.11 Occupancy of a reserved circuit: continuous voice versus bursty data

Circuit switching **cannot efficiently handle bursty traffic** because resources are reserved but unused most of the time.

The Evolution Path

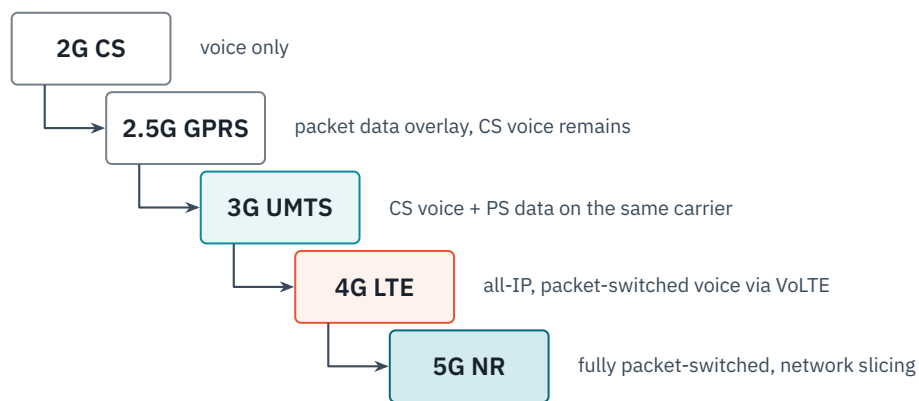


Figure 2.12 From circuit switching to fully packet-switched networks

Summary

Circuit switching was the **right choice** for voice in the 1990s — it provided the quality guarantees that voice communication demands with the technology available at that time. However, as mobile usage shifted from voice-only to data-centric services, the inefficiency of holding dedicated circuits for bursty, asymmetric data traffic made packet switching the inevitable successor.

Key Terms Summary

Term	Definition
MOC	Mobile-Originated Call
MTC	Mobile-Terminated Call
MSRN	Mobile Station Roaming Number — temporary number for call routing
IAM	Initial Address Message — ISUP message to initiate a call
ACM	Address Complete Message — called party is being alerted
ANM	Answer Message — called party has answered
REL/RLC	Release / Release Complete — call teardown
LAI	Location Area Identity — identifies a group of cells
TMSI	Temporary Mobile Subscriber Identity — privacy protection
Ki	Secret key shared between SIM and AuC
SRES	Signed Response — authentication proof
Kc	Cipher key derived during authentication
VPLMN	Visited PLMN — foreign network during roaming
HPLMN	Home PLMN — subscriber's home network

Next Module: Module 3 — GPRS and Packet Switching

2G/3G Core Networks

شبکه‌های هسته 2G/3G

CS DOMAIN

PS DOMAIN

In This Module

3.1 Evolution from 2G to 3G — The Driving Forces	66
3.2 UMTS Architecture Overview	66
3.3 UTRAN: NodeB and RNC	68
3.4 3G Core Network: CS Domain + PS Domain	69
3.5 The Iu Interface	70
3.6 Release 99 vs Release 5+ (HSPA Evolution)	71
3.7 IMS (IP Multimedia Subsystem) — Introduction	73
3.8 Major Comparison Table: Mobile Network Generations	74
3.9 The CS/PS Split — And Why LTE Killed It	75
3.10 Summary: Key Takeaways	77
Review Questions	77

3.1 Evolution from 2G to 3G – The Driving Forces

Why Did We Need 3G?

The transition from 2G (GSM) to 3G (UMTS) was driven by fundamental market and technology demands:

1. **Exploding Data Demand:** SMS and basic WAP browsing were insufficient. Users wanted email, web browsing, and file downloads on mobile devices.
2. **Multimedia Services:** Video calling, streaming audio/video, and rich content required bandwidth far beyond GSM's 9.6 kbps (or even GPRS's theoretical 171 kbps).
3. **Higher Data Rates:** The ITU's IMT-2000 vision targeted:
 - 144 kbps for high mobility (vehicular)
 - 384 kbps for pedestrian
 - 2 Mbps for stationary/indoor
4. **Global Roaming:** A unified worldwide standard (unlike fragmented 2G: GSM vs CDMA vs PDC).
5. **Spectrum Efficiency:** WCDMA's spreading codes allow more users per cell with graceful degradation.

The Evolution Path

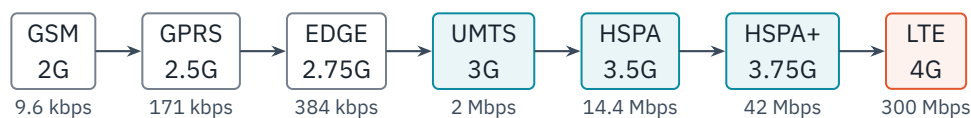


Figure 3.1 Generations and their peak data rates

3.2 UMTS Architecture Overview

UMTS (Universal Mobile Telecommunications System) is built on two major domains:

- **UTRAN** (UMTS Terrestrial Radio Access Network) — the radio/air interface side
- **Core Network (CN)** — the intelligence and connectivity backbone

Key Design Principles

- Backward compatibility with GSM/GPRS core
- Separation of radio access from core network (allowing independent evolution)
- Split core into Circuit-Switched (CS) and Packet-Switched (PS) domains
- Support for simultaneous voice and data

■ UMTS Architecture Diagram

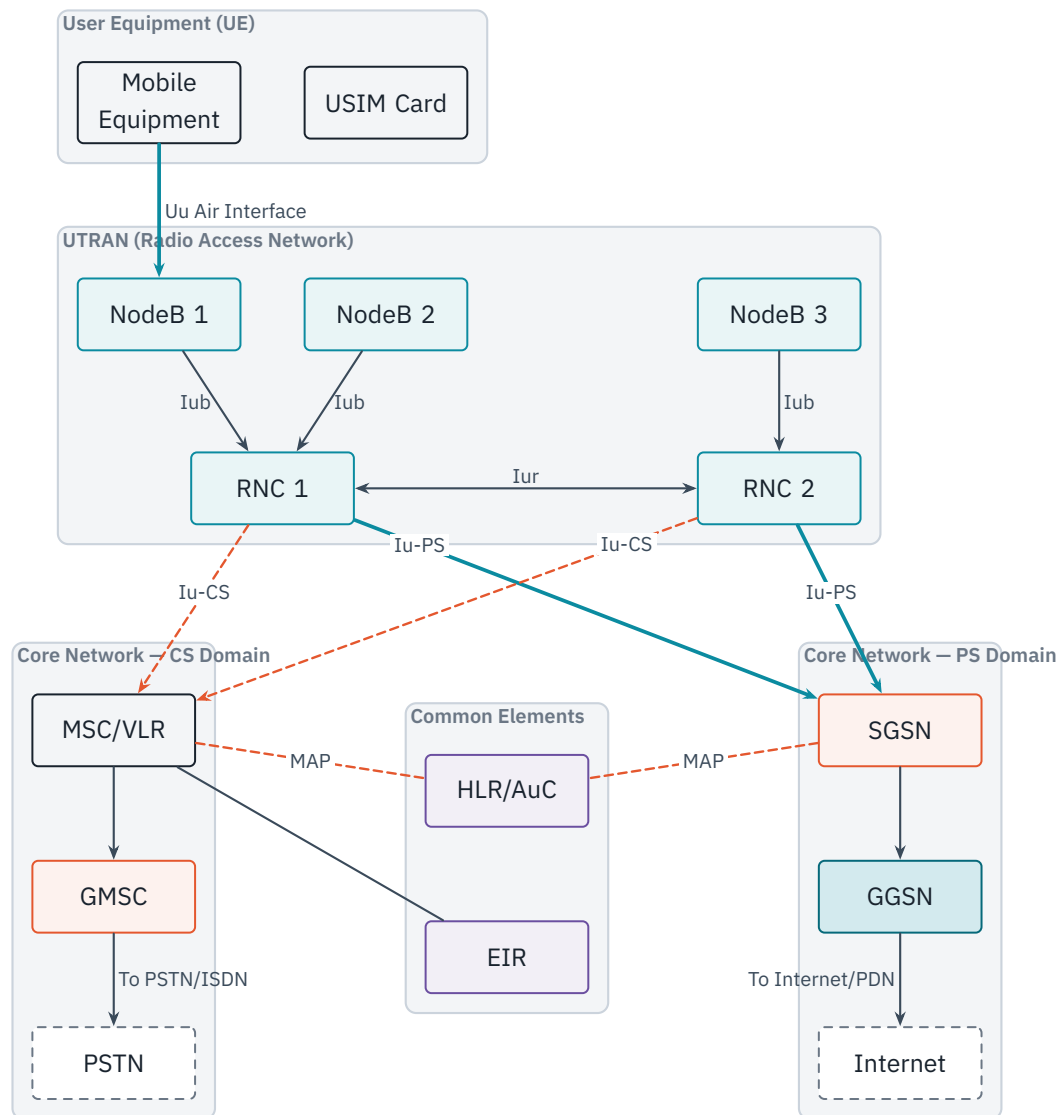


Figure 3.2 UMTS architecture: UTRAN with the CS and PS core domains

3.3 UTRAN: NodeB and RNC

NodeB (Base Station)

The NodeB is the 3G equivalent of the GSM BTS (Base Transceiver Station):

Aspect	Function
Primary Role	Radio transmission/reception over the Uu air interface
Spreading/Despreading	Applies WCDMA spreading codes
Power Control	Inner-loop power control (fast, 1500 Hz)
Soft Handover	Supports macro-diversity combining
Channel Coding	Encoding/decoding of transport channels
Interface	Connected to RNC via Iub interface (ATM or IP-based)

Key Difference from GSM BTS: NodeB has MORE intelligence — it handles fast power control and soft handover combining locally, reducing latency.

RNC (Radio Network Controller)

The RNC is the 3G equivalent of the GSM BSC, but with significantly more responsibilities:

Aspect	Function
Radio Resource Management	Admission control, code allocation, load control
Handover Decision	Soft/softer/hard handover management
Outer-Loop Power Control	SIR target adjustment
Ciphering/Integrity	Security functions for user data and signaling
Protocol Termination	RRC (Radio Resource Control) protocol endpoint
Macro-Diversity Combining	Combines signals from multiple NodeBs
Segmentation/Reassembly	RLC layer processing

RNC Roles (One RNC Can Play Multiple Roles)

- **Serving RNC (S-RNC):** Maintains the RRC connection with UE, performs outer-loop power control
- **Drift RNC (D-RNC):** Provides radio resources when UE moves but S-RNC doesn't change
- **Controlling RNC (C-RNC):** Controls the logical resources of its NodeBs

UTRAN Interfaces

Interface	Connects	Protocol Stack	Purpose
Uu	UE ↔ NodeB	WCDMA air interface	Radio transmission
Iub	NodeB ↔ RNC	NBAP over ATM/IP	NodeB control and user data
Iur	RNC ↔ RNC	RNSAP over ATM/IP	Inter-RNC soft handover, relocation
Iu-CS	RNC ↔ MSC	RANAP over ATM	Circuit-switched traffic
Iu-PS	RNC ↔ SGSN	RANAP over IP	Packet-switched traffic

3.4 3G Core Network: CS Domain + PS Domain

The 3G core network inherits the GSM/GPRS split architecture, maintaining two parallel domains:

Circuit-Switched (CS) Domain

Handles real-time voice calls and video telephony (same as GSM core, upgraded for 3G).

Element	Role
MSC (Mobile Switching Centre)	Call routing, handover between RNCs, call control signaling
VLR (Visitor Location Register)	Temporary storage of subscriber data for visitors in the MSC area
GMSC (Gateway MSC)	Interface to external networks (PSTN, ISDN), handles incoming calls
HLR (Home Location Register)	Permanent subscriber database (shared with PS domain)
AuC (Authentication Centre)	Stores authentication keys (Ki), generates security vectors
EIR (Equipment Identity Register)	IMEI blacklist/whitelist for stolen device blocking

CS Domain Call Flow:



Figure 3.3 CS domain call flow

■ Packet-Switched (PS) Domain

Handles all data services (web, email, streaming) — evolved from GPRS.

Element	Role
SGSN (Serving GPRS Support Node)	Packet routing, mobility management, session management, ciphering
GGSN (Gateway GPRS Support Node)	Interface to external packet networks (Internet), IP address allocation, billing anchor

PS Domain Data Flow:

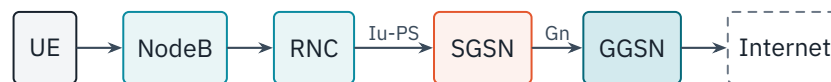


Figure 3.4 PS domain data flow

■ Why Two Domains?

The split CS/PS approach was a **pragmatic engineering decision**:

1. **Legacy Compatibility:** Voice infrastructure (SS7, PSTN interconnect) was mature and reliable — no reason to rebuild it
2. **QoS Guarantees:** Circuit-switching provides deterministic delay for voice (guaranteed 20ms frame delivery)
3. **Incremental Deployment:** Operators could add PS domain without disrupting existing voice revenue
4. **Risk Mitigation:** IP networks in 2000 couldn't guarantee voice quality — packet loss and jitter were real problems

3.5 The Iu Interface

The Iu interface is the critical boundary between UTRAN and the Core Network.

■ Iu-CS (Circuit-Switched)

- **Transport:** ATM (AAL2 for user plane, AAL5 for control plane)
- **Control Plane:** RANAP → SCCP → M3UA → SCTP (or MTP3b)
- **User Plane:** Iu UP protocol (framing for AMR voice codec frames)
- **Purpose:** Carries voice calls and video telephony

Iu-PS (Packet-Switched)

- **Transport:** IP-based (GTP-U for user plane)
- **Control Plane:** RANAP → SCCP → M3UA → SCTP
- **User Plane:** GTP-U (GPRS Tunneling Protocol - User plane)
- **Purpose:** Carries all packet data (web, email, streaming)

RANAP (Radio Access Network Application Part)

Key procedures over Iu:

- **RAB (Radio Access Bearer) Assignment:** Sets up the end-to-end bearer
- **Relocation:** Moves Serving RNC role to another RNC (hard handover)
- **Paging:** Core network pages UE through UTRAN
- **Security Mode:** Activates ciphering and integrity protection
- **Direct Transfer:** Passes NAS messages transparently (e.g., MM, CC, SM)

3.6 Release 99 vs Release 5+ (HSPA Evolution)

Release 99 (R99) – Original UMTS

- **Downlink:** Up to 384 kbps (dedicated channels)
- **Uplink:** Up to 384 kbps
- **Channel Type:** Dedicated Channel (DCH) — one code per user
- **Scheduling:** Done at RNC level (slow, ~100ms TTI)
- **Power Control:** Primary mechanism for link adaptation
- **Limitation:** Inefficient for bursty data (keeps resources allocated even during silence)

Release 5 – HSDPA (High Speed Downlink Packet Access)

Feature	Improvement
Peak Rate	14.4 Mbps (DL)
Shared Channel	HS-DSCH — time-multiplexed among users
Scheduling	Moved to NodeB (2ms TTI — fast, channel-aware)
AMC	Adaptive Modulation & Coding (QPSK → 16QAM → 64QAM)
HARQ	Hybrid ARQ with soft combining at NodeB
No Power Control	Transmits at full power, adapts rate instead

Release 6 – HSUPA (High Speed Uplink Packet Access)

Feature	Improvement
Peak Rate	5.76 Mbps (UL)
Channel	E-DCH (Enhanced Dedicated Channel)
Scheduling	NodeB-controlled with 2ms/10ms TTI
HARQ	Uplink soft combining
Benefit	Better latency, higher throughput for uploads

Release 7/8 – HSPA+

- 64QAM downlink, 16QAM uplink
- MIMO (2×2) for downlink
- Dual-carrier operation (2 × 5 MHz)
- Peak rates: **42 Mbps DL, 11.5 Mbps UL**
- **Flat architecture option:** Direct NodeB-to-GGSN (bypassing RNC)

The Shift in Philosophy

R99: RNC controls everything → Centralized, slow adaptation
R5+: NodeB makes fast decisions → Distributed, rapid adaptation (2ms vs 100ms)

This shift foreshadowed LTE's design where the eNodeB handles almost all radio decisions.

3.7 IMS (IP Multimedia Subsystem) – Introduction

What is IMS?

IMS is a **framework for delivering IP-based multimedia services** over mobile networks. Introduced in 3GPP Release 5, it represents the vision of converging ALL services (voice, video, messaging, presence) onto IP.

Why IMS?

Problem	IMS Solution
CS domain can't evolve (legacy SS7)	All-IP signaling using SIP
Adding new services requires network upgrades	Application servers plug in modularly
Voice and data are separate worlds	Unified session control for all media
No service interaction (voice OR data)	Combinational services (voice + video + sharing)

Key IMS Components

- **P-CSCF (Proxy-CSCF)**: First contact point, SIP proxy, policy enforcement
- **I-CSCF (Interrogating-CSCF)**: Entry point from other networks, routes to correct S-CSCF
- **S-CSCF (Serving-CSCF)**: Core session control, service trigger logic, SIP registrar
- **HSS (Home Subscriber Server)**: Evolution of HLR, stores user profiles and service triggers
- **Application Servers**: Provide actual services (VoIP, conferencing, messaging)

IMS Signaling Protocol: SIP

IMS uses **SIP (Session Initiation Protocol)** instead of SS7:

- Text-based, extensible
- Works natively over IP
- Supports any media type (voice, video, gaming, IoT)
- Enables service composition and mashups

IMS's Role in the 2G→4G Journey

2G: Voice = CS (SS7)	Data = None/CSD	
3G: Voice = CS (SS7)	Data = PS (GTP)	IMS = Optional overlay
4G: Voice = VoLTE via IMS	Data = PS (GTP)	IMS = Mandatory for voice
5G: Voice = VoNR via IMS	Data = PS (GTP/SBA)	IMS = Still essential

3.8 Major Comparison Table: Mobile Network Generations

Feature	2G (GSM)	2.5G (GPRS)	3G (UMTS R99)	3.5G (HSPA)	4G (LTE)
Radio Access	TDMA/FDMA (200 kHz channels, 8 timeslots)	Same as GSM (shared timeslots)	WCDMA (5 MHz carrier, spreading codes)	WCDMA + shared channels (HS-DSCH/E-DCH)	OFDMA DL / SC-FDMA UL (1.4–20 MHz)
Core Network	MSC/VLR/HLR (SS7-based)	MSC + SGSN/ GGSN added	MSC/VLR (CS) + SGSN/GGSN (PS)	Same as 3G (optional flat arch in HSPA+)	EPC: MME/ S-GW/P-GW (all-IP, flat)
Switching Type	Circuit-switched only	CS (voice) + PS (data)	CS (voice) + PS (data)	CS (voice) + PS (data)	Packet-switched ONLY
Peak Data Rate (DL)	9.6 kbps (CSD) / 14.4 kbps (HSCSD)	171.2 kbps (theoretical) / ~40 kbps (practical)	384 kbps (pedestrian) / 2 Mbps (indoor)	14.4 Mbps (HSDPA) / 42 Mbps (HSPA+)	300 Mbps (Cat 5) / 3 Gbps (Cat 20)
Peak Data Rate (UL)	9.6 kbps	171.2 kbps (theoretical) / ~20 kbps (practical)	384 kbps	5.76 Mbps (HSUPA) / 11.5 Mbps (HSPA+)	75 Mbps (Cat 5) / 1.5 Gbps (Cat 20)
Latency (RTT)	~500 ms (data)	~300–600 ms	~150–200 ms	~50–100 ms (HSPA) / ~30 ms (HSPA+)	~10–30 ms (user plane) / ~50 ms (control)
Mobility Support	Handover (hard) up to 250 km/h	Same as GSM	Soft handover + hard handover, up to 500 km/h	Same as UMTS	Hard handover only, up to 500 km/h
Signaling Protocol	SS7 (MAP, ISUP, BSSAP)	SS7 + GTP-C	SS7 (RANAP, MAP) + GTP-C	Same as 3G	Diameter + GTP-C v2 (all IP-based)
Main Innovation	Digital voice, SMS, roaming, SIM	Always-on data, IP connectivity, shared radio	Wideband data, video calls, multimedia, soft handover	Fast scheduling at NodeB, AMC, HARQ, shared channels	All-IP flat architecture, OFDMA, MIMO, VoLTE
Main Limitation	No real data capability	Low throughput, high latency, shared with voice timeslots	Dedicated channels waste resources for bursty data	Still CS voice, complex dual-stack	No native voice (needs VoLTE/IMS), complex handover to 2G/3G

3.9 The CS/PS Split – And Why LTE Killed It

■ The Problem with Dual Domains

Running CS and PS domains in parallel meant:

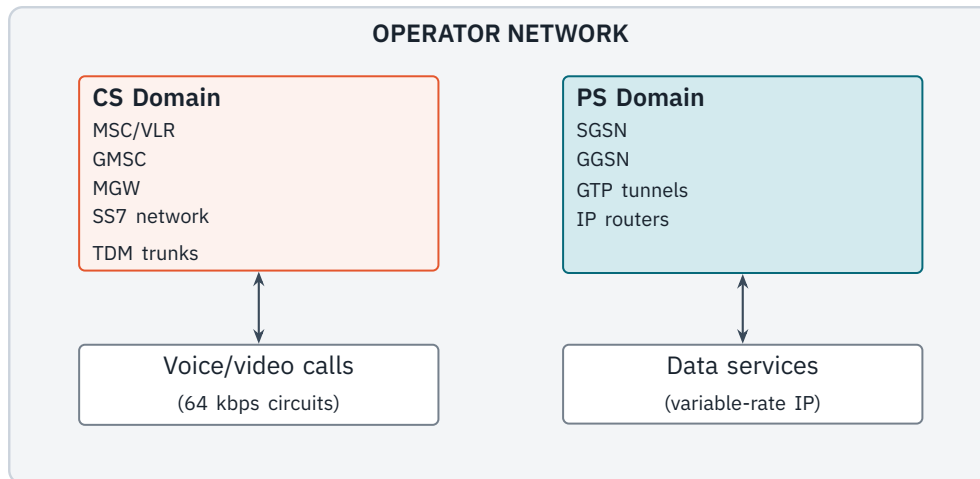


Figure 3.5 Two parallel core domains in the operator network

■ Costs of the Split Architecture

Problem	Impact
Duplicate infrastructure	Two complete core networks to maintain, upgrade, and staff
Inefficient spectrum use	CS reserves capacity even during silence (60% of a call is silence)
No service convergence	Voice and data can't easily interact (e.g., sharing a photo during a call)
SS7 is legacy	Hard to extend, proprietary equipment, no IP ecosystem benefits
Separate billing	Voice minutes + data MB = complex charging systems
Scaling limitations	CS scales with ports/trunks (expensive), PS scales with bandwidth (cheap)

■ Why LTE Unified to Packet-Only (EPC)

LTE's Evolved Packet Core (EPC) made a radical decision: **NO circuit-switching at all.**

■ The Technical Enablers (What Changed by 2008):

1. **IP Networks Matured:** QoS mechanisms (DiffServ, MPLS, dedicated bearers) could now guarantee voice quality

2. **Codec Efficiency:** AMR-WB and EVS codecs work well over IP with minimal bandwidth (~12 kbps)
3. **Low Latency Achieved:** LTE's ~10ms RTT is actually BETTER than CS voice (~50ms codec delay anyway)
4. **VoIP Proven:** Skype, Vonage proved IP voice was commercially viable
5. **IMS Was Ready:** After years of development, IMS could replace SS7 call control with SIP
6. **Economics:** IP equipment costs 1/10th of equivalent TDM/SS7 equipment

■ The EPC Architecture (All Packet):

Element	Replaces	Function
MME	MSC (control plane)	Mobility, authentication, bearer management — signaling only
S-GW	SGSN (user plane)	Local mobility anchor, data forwarding between eNodeBs
P-GW	GGSN	External network gateway, IP allocation, policy/charging
HSS	HLR + AuC	Subscriber data, authentication vectors
PCRF	— (new)	Policy and Charging Rules — enables QoS per-flow

■ How Voice Works Without CS (VoLTE):

Traditional 3G voice — dedicated 64 kbps circuit



LTE voice (VoLTE) — IP packets on a QCI 1 dedicated bearer, about 40 kbps AMR-WB

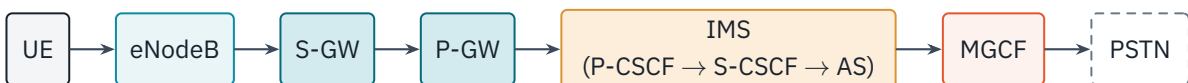


Figure 3.6 Voice path in 3G (circuit) versus LTE (VoLTE over IMS)

■ Benefits of Unified Packet Architecture:

1. **Single infrastructure** to maintain → 50%+ OPEX reduction
2. **Statistical multiplexing** → no wasted capacity during silence
3. **Service convergence** → voice + video + data on same bearer simultaneously
4. **Rapid service innovation** → new services via Application Servers, not network upgrades
5. **HD Voice** → AMR-WB (wideband) sounds dramatically better than GSM's narrowband codec
6. **Network simplification** → fewer interfaces, fewer protocol stacks, fewer points of failure

■ The Trade-off

Aspect	CS (2G/3G)	PS-only (LTE)
Voice reliability	Extremely high (30+ years proven)	Depends on coverage and VoLTE implementation
Call setup time	~3–5 seconds	~1–2 seconds (SIP is faster)
Emergency calls	Always works (CS fallback)	Requires VoLTE or CSFB to 2G/3G
Complexity	Simple for voice, complex overall	Simple overall, complex for voice QoS
Coverage gaps	Voice still works at cell edge	Voice drops if LTE coverage drops (needs CSFB)

CSFB (Circuit-Switched Fallback): Early LTE networks fell back to 3G/2G for voice calls. This was a transitional solution until VoLTE coverage was ubiquitous.

3.10 Summary: Key Takeaways

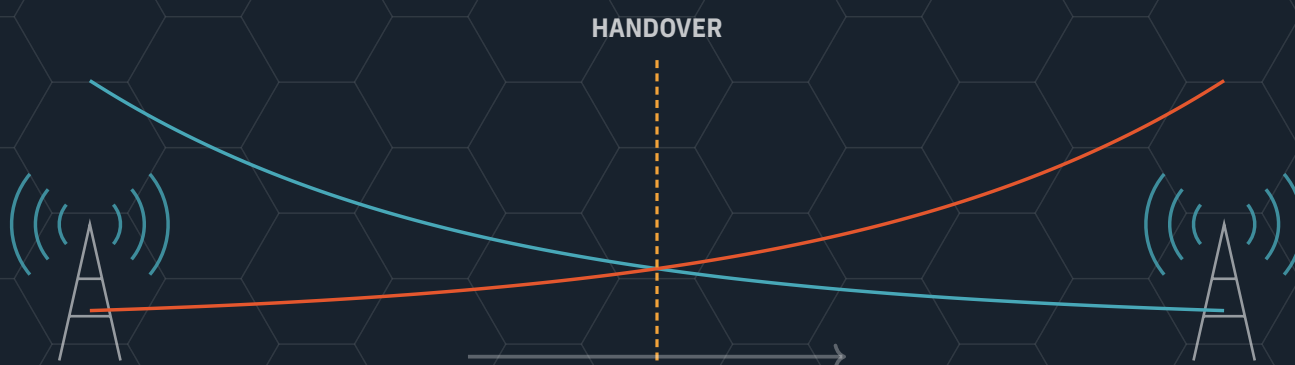
1. **3G (UMTS)** introduced WCDMA radio with dedicated channels and maintained the 2G CS/PS split core
2. **UTRAN** separates radio functions: NodeB (physical layer) and RNC (radio resource management)
3. **The Iu interface** cleanly separates radio access from core — enabling independent evolution
4. **HSPA (R5/R6)** moved intelligence to NodeB (fast scheduling, HARQ, AMC) — a preview of LTE’s philosophy
5. **IMS** provides the all-IP service layer that eventually replaces SS7 for voice
6. **LTE unified everything to packets** because IP matured enough to guarantee voice QoS, and the economics of maintaining dual infrastructure became unjustifiable
7. **The evolution was gradual:** each generation solved real limitations of the previous one

Review Questions

1. Explain why UMTS maintained separate CS and PS domains instead of going all-packet from the start.
2. What is the difference between a Serving RNC and a Drift RNC?
3. How does HSDPA’s NodeB-based scheduling improve over R99’s RNC-based scheduling?
4. Draw the protocol stack for the Iu-PS interface.
5. Why was IMS necessary for LTE to eliminate the CS domain?
6. Compare the role of SGSN in 3G with S-GW in LTE — what changed and why?

Module 3 — Advanced Communication Networks — 2G/3G Core Networks

Next: Module 4 — LTE/EPC Architecture



In This Module

1	Why Mobility Management Is Needed	79
2	Location Management: LA vs RA vs TA	80
3	Cell Reselection (Idle Mode Mobility)	81
4	Handover Types	83
5	Handover Decision Process	87
6	Handover Procedure — LTE X2 Handover	91
7	Handover Failures	96
8	Evolution of Handover: 2G → 3G → LTE → 5G	101
9	Engineering Examples with Typical Parameter Values	107
10	Summary Diagrams	112
	Key Formulas Summary	114
	Review Questions	114

Learning Objectives

By the end of this module, you will be able to:

- Explain WHY mobility management is essential in cellular networks
- Distinguish between Location Areas, Routing Areas, and Tracking Areas
- Describe cell reselection mechanisms in idle mode
- Classify and compare handover types across generations
- Analyze the handover decision process using measurement events
- Trace the LTE X2 handover procedure step by step
- Identify handover failures and their mitigation strategies
- Compare mobility evolution from 2G through 5G

1 Why Mobility Management Is Needed

WHY

Users are mobile — they walk, drive, and travel while expecting uninterrupted voice calls, video streams, and data sessions. Without mobility management, a user crossing a cell boundary would experience immediate service disconnection.

WHAT

Mobility management is the set of network procedures that:

- **Track** the user's location (so the network can reach them for incoming calls/data)
- **Maintain** service continuity when the user moves between cells (handover)
- **Optimize** radio resource usage by connecting users to the best-serving cell

HOW

The network achieves this through two complementary mechanisms:

State	Mechanism	Purpose
Idle mode	Cell reselection + location/tracking updates	Network knows user's approximate location
Connected mode	Handover	Seamless transfer of active connection

■ The Fundamental Trade-off

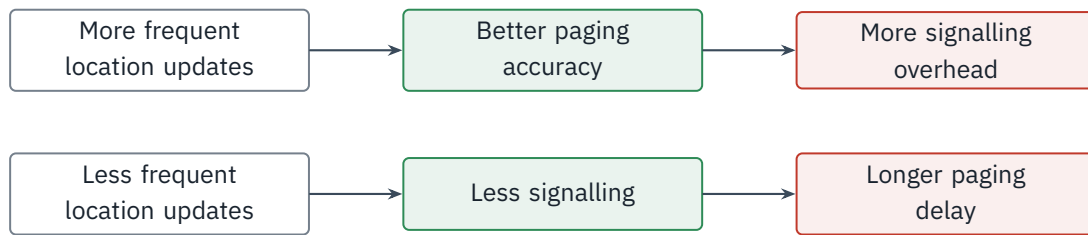


Figure 4.1 The location-update trade-off: paging accuracy versus signalling load

The network must balance:

- **Signaling load** (updates consume radio resources)
- **Paging load** (larger areas = more cells to page)
- **Service latency** (time to reach the user for incoming services)

2 Location Management: LA vs RA vs TA

■ WHY

The network needs to know WHERE a user is (approximately) to deliver incoming calls or data. Different generations use different grouping strategies to balance signaling vs. paging efficiency.

■ WHAT

Generation	Area Concept	Used For	Typical Size
2G (GSM)	Location Area (LA)	Circuit-switched services	20–100 cells
2.5G (GPRS)	Routing Area (RA)	Packet-switched services	Subset of LA, 10–50 cells
4G (LTE)	Tracking Area (TA)	All services (all-IP)	30–80 cells
5G (NR)	Tracking Area + RNA	All services + RAN-level	Configurable per UE

■ HOW – Location Area (2G)

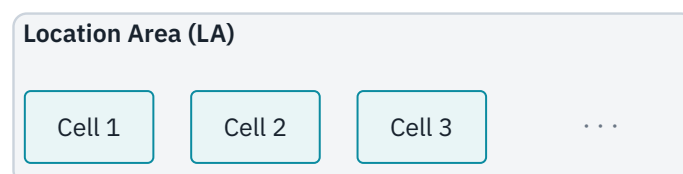


Figure 4.2 A Location Area groups many cells

- UE performs **Location Area Update (LAU)** when crossing LA boundary
- Network pages ALL cells in the LA for incoming call
- Managed by MSC/VLR

■ HOW — Routing Area (GPRS)

- RA is a **subset** of an LA (one LA contains multiple RAs)
- UE performs **Routing Area Update (RAU)** for packet services
- Managed by SGSN
- Allows finer-grained location for data users

■ HOW — Tracking Area (LTE/5G)

- **Tracking Area List (TAL)**: UE is assigned a LIST of TAs — no update needed when moving within the list
- Reduces signaling for high-mobility users (larger TAL)
- Low-mobility users get smaller TAL to reduce paging area

```
UE TAL = {TA1, TA2, TA3, TA4}
```

```
Moving TA1→TA2→TA3→TA4: NO update needed
```

```
Moving TA4→TA5 (outside TAL): Tracking Area Update triggered
```

■ 5G Enhancement: RAN-based Notification Area (RNA)

- In **RRC_INACTIVE** state (new in 5G), the gNB manages a smaller RNA
- Reduces core network signaling
- Two-level mobility: RNA (RAN-managed) + TA (core-managed)

3 Cell Reselection (Idle Mode Mobility)

■ WHY

Even when not in an active call/session, the UE must camp on the best cell to:

- Receive paging messages efficiently
- Ensure good signal quality when a connection is needed
- Avoid unnecessary location updates

WHAT

Cell reselection is the **UE-autonomous** process of selecting a new cell to camp on while in idle mode. The network provides parameters via System Information Broadcasts (SIBs), but the UE makes the decision independently.

HOW – Cell Reselection Criteria

Ranking Criterion (S-criterion)

A cell is suitable for camping if it meets the S-criterion:

$$S_{\text{rxlev}} = Q_{\text{rxlevmeas}} - (Q_{\text{rxlevmin}} + Q_{\text{rxlevminoffset}}) - P_{\text{compensation}} > 0$$

$$S_{\text{qual}} = Q_{\text{qualmeas}} - (Q_{\text{qualmin}} + Q_{\text{qualminoffset}}) > 0$$

Where:

- $Q_{\text{rxlevmeas}}$: Measured RSRP (dBm)
- Q_{rxlevmin} : Minimum required RSRP (-124 to -44 dBm, typical: -124 dBm)
- Q_{qualmeas} : Measured RSRQ (dB)
- Q_{qualmin} : Minimum required RSRQ (-20 to -3 dB, typical: -20 dB)

Cell Ranking (R-criterion)

For intra-frequency reselection, cells are ranked:

$$R_s = Q_{\text{meas},s} + Q_{\text{Hyst}} \quad (\text{serving cell})$$

$$R_n = Q_{\text{meas},n} - Q_{\text{offset}} \quad (\text{neighbour cell})$$

Reselect to the neighbour if $R_n > R_s$ holds for the duration $T_{\text{reselection}}$.

Priority-Based Reselection (Inter-frequency/Inter-RAT)

Priority	Action
Higher priority layer	Reselect if neighbor meets ThreshX,High
Equal priority layer	Use ranking (R-criterion)
Lower priority layer	Reselect only if serving < ThreshServing,Low AND neighbor > ThreshX,Low

HOW – Hysteresis and Timers

Hysteresis (QHyst) prevents ping-pong reselection:

- Typical values: 2–6 dB
- Adds “stickiness” to the serving cell

Treselection timer (typically 1–4 seconds):

- Neighbor must remain better for this duration before reselection occurs

- Prevents reselection due to momentary fading

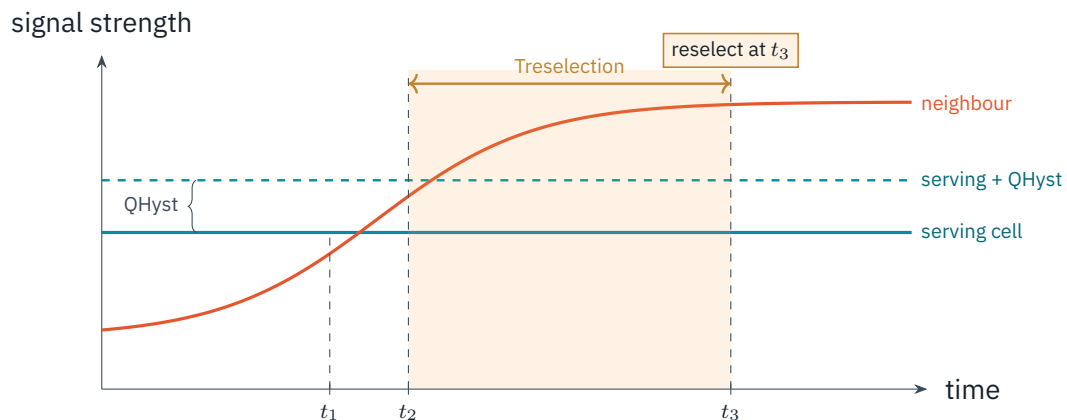


Figure 4.3 Cell reselection: the neighbour must stay above serving + QHyst for Tresselection

Typical Cell Reselection Parameters

Parameter	Typical Value	Purpose
QHyst	4 dB	Serving cell hysteresis
Tresselection	2 s	Stability timer
Qrxlevmin	-124 dBm	Minimum RSRP for cell selection
Qqualmin	-20 dB	Minimum RSRQ for cell selection
ThreshServing,Low	4 (= 4 dB)	Threshold to leave serving (to lower priority)
ThreshX,High	8 (= 8 dB)	Threshold to enter higher priority layer

4 Handover Types

WHY

Different mobility scenarios require different handover strategies. A user walking between co-channel cells needs a different mechanism than a user on a high-speed train crossing between LTE and 3G coverage areas.

■ WHAT – Classification Overview

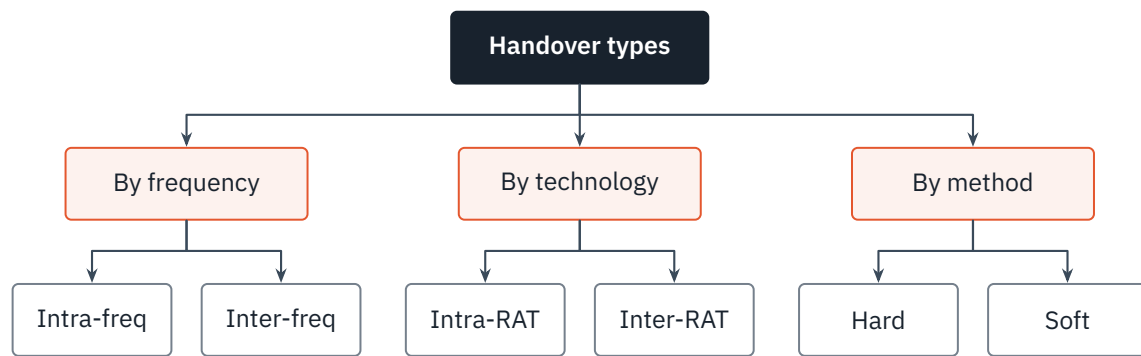


Figure 4.4 Classification of handovers

4.1 Intra-Frequency Handover

■ WHY

Most common handover — occurs when user moves between cells operating on the **same carrier frequency**. This is the typical scenario in dense urban deployments.

■ WHAT

- Source and target cells use the same frequency (e.g., both on 1800 MHz Band 3)
- UE can measure neighbors **WITHOUT** measurement gaps
- Fastest and simplest handover type

■ HOW

- UE continuously measures RSRP/RSRQ of serving and intra-frequency neighbors
- No interruption to measurements needed (same frequency = can measure while receiving)
- Triggered by Event A3: Neighbor becomes offset better than serving

Typical scenario: Walking between adjacent cells in a shopping mall (all cells on same frequency)

4.2 Inter-Frequency Handover

■ WHY

Networks deploy multiple frequency layers for capacity (e.g., macro on 800 MHz for coverage, small cells on 2600 MHz for capacity). Users must move between these layers.

■ WHAT

- Source and target cells operate on **different carrier frequencies**
- UE needs **measurement gaps** to tune its receiver to other frequencies
- Slightly longer interruption than intra-frequency

■ HOW

- Network configures measurement gaps (6 ms every 40 or 80 ms)
- UE measures inter-frequency neighbors during gaps
- Triggered by Event A4 (neighbor on other frequency becomes better than threshold) or A5 (serving degrades AND neighbor is good)

Typical scenario: User moves from indoor small cell (2600 MHz) to outdoor macro (800 MHz)

4.3 Inter-RAT Handover

■ WHY

Not all areas have LTE/5G coverage. When LTE signal degrades, the UE may need to fall back to 3G (UMTS) or even 2G (GSM) to maintain service. Also used for CS Fallback (voice calls).

■ WHAT

- Handover between different radio access technologies
- Examples: LTE → UMTS, LTE → GSM, NR → LTE
- Most complex type — different protocols, different core networks

■ HOW

- UE measures inter-RAT neighbors during measurement gaps
- Triggered by Event B1 (inter-RAT neighbor above threshold) or B2 (serving below threshold AND inter-RAT above threshold)
- Requires inter-RAT coordination between eNB and RNC/BSC

Typical scenario:

- VoLTE not available → CS Fallback to 3G for voice call
- User drives into rural area with only 3G coverage

4.4 Hard Handover (Break-Before-Make)

■ WHY

When simultaneous connection to two cells is not possible (different frequencies, different technologies, or system design choice), the old connection must be released before the new one is established.

■ WHAT

- Connection to source cell is **released** before connecting to target
- Brief interruption in service (typically 20–50 ms in LTE)
- Used in: GSM, LTE, 5G NR

■ HOW

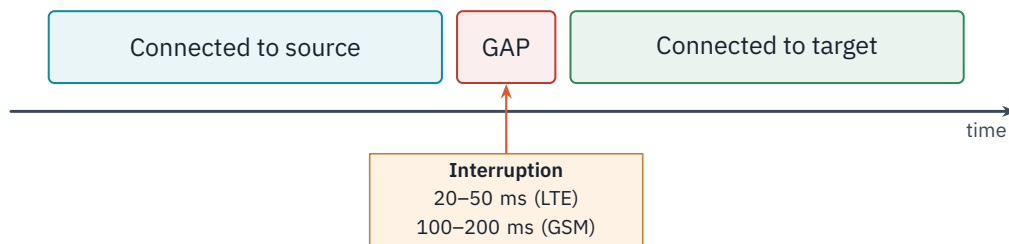


Figure 4.5 Hard handover: break before make

- **Advantage:** Simpler, no need for combining/splitting
- **Disadvantage:** Brief interruption (acceptable for data, noticeable for voice in 2G)

4.5 Soft Handover (Make-Before-Break) – 3G UMTS Specific

■ WHY

CDMA-based 3G systems can receive the same signal from multiple cells simultaneously (same frequency, code division). This enables seamless handover with NO interruption.

■ WHAT

- UE is connected to **multiple cells simultaneously** (Active Set)
- Signals from multiple cells are **combined** for diversity gain
- Unique to CDMA/WCDMA (3G) — NOT used in LTE or 5G

■ HOW – Active Set Management

Active Set Events (3GPP UMTS):

- Event 1A: Add cell to Active Set (neighbor strong enough)
- Event 1B: Remove cell from Active Set (cell too weak)
- Event 1C: Replace cell in Active Set (better candidate found)

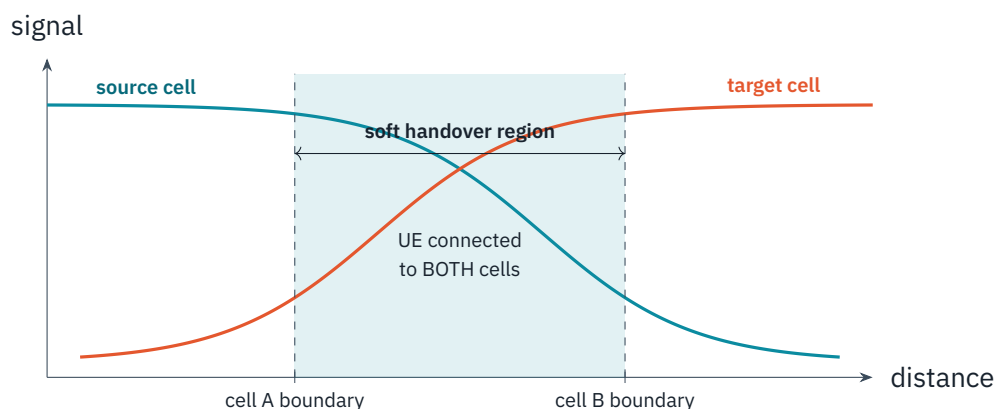


Figure 4.6 Soft handover region: the UE is connected to both cells

- **Active Set:** Set of cells currently serving the UE (max 3–6 cells)
- **Soft handover:** Between cells of same Node B or different Node Bs, same frequency
- **Softer handover:** Between sectors of the SAME Node B (MRC combining at Node B)

Advantages:

- No interruption
- Macro-diversity gain (2–3 dB)
- Reduced ping-pong

Disadvantages:

- Uses resources in multiple cells simultaneously
- Increased backhaul and processing requirements
- Adds complexity to power control

5 Handover Decision Process

WHY

The network must decide WHEN to handover, to WHICH cell, and ensure the decision is robust against fading and noise. A poorly timed handover causes drops; a well-tuned one is imperceptible to the user.

WHAT

The handover decision in LTE is based on:

1. **UE Measurement Reports** — periodic or event-triggered
2. **Measurement Events** — conditions that trigger the report
3. **Hysteresis and Time-to-Trigger** — filtering to avoid false triggers
4. **Network decision** — eNB evaluates report and decides

5.1 Measurement Reports (RSRP and RSRQ)

■ WHAT

Metric	Full Name	Range	Purpose
RSRP	Reference Signal Received Power	-140 to -44 dBm	Signal strength (coverage)
RSRQ	Reference Signal Received Quality	-20 to -3 dB	Signal quality (accounts for interference + noise)
SINR	Signal-to-Interference-plus-Noise Ratio	-23 to 40 dB	Overall link quality

■ HOW

RSRP = power of one resource element carrying the reference signal

$$\text{RSRQ} = \frac{N \times \text{RSRP}}{\text{RSSI}}$$

where N is the number of RBs and RSSI is the total received wideband power.

- RSRP answers “how strong is **my** cell’s signal?”
- RSRQ answers “how good is **my** cell’s signal relative to everything else I hear?”

When to use which:

- RSRP-based handover: Good for coverage-limited scenarios (rural)
- RSRQ-based handover: Good for interference-limited scenarios (dense urban)

5.2 LTE Measurement Events (A1–A6)

■ WHAT

Event	Condition	Use Case
A1	Serving > Threshold	Stop inter-freq measurements (serving is good enough)
A2	Serving < Threshold	Start inter-freq/inter-RAT measurements
A3	Neighbor > Serving + Offset	Primary intra-freq handover trigger
A4	Neighbor > Threshold	Inter-frequency handover
A5	Serving < Thresh1 AND Neighbor > Thresh2	Inter-freq/inter-RAT handover
A6	Neighbor > SCell + Offset	Secondary cell (CA) change

■ Event B (Inter-RAT):

Event	Condition	Use Case
B1	Inter-RAT neighbor > Threshold	Redirect to other RAT
B2	Serving < Thresh1 AND Inter-RAT > Thresh2	Fallback (e.g., LTE→3G)

■ HOW – Event A3 in Detail (Most Important)

Event A3 entering condition:

$$M_n + O_{fn} + O_{cn} - H_{ys} > M_s + O_{fs} + O_{cs} + O_{ff}$$

Symbol	Meaning
M_n	Measurement of the neighbour cell (RSRP or RSRQ)
M_s	Measurement of the serving cell
O_{fn}	Frequency-specific offset for the neighbour
O_{fs}	Frequency-specific offset for the serving cell
O_{cn}	Cell-specific offset for the neighbour
O_{cs}	Cell-specific offset for the serving cell
H_{ys}	Hysteresis (positive value)
O_{ff}	A3 offset (the main tunable parameter)

Simplified (same frequency, no cell-specific offsets):

$$RSRP_{neighbour} - Hysteresis > RSRP_{serving} + A3_Offset$$

5.3 Hysteresis and Time-to-Trigger

■ WHY

Radio signals fluctuate due to fading. Without filtering, the network would trigger handovers on every temporary fade, causing excessive signaling and ping-pong.

■ WHAT

Two mechanisms prevent false triggering:

Parameter	Function	Typical Value
Hysteresis (Hys)	Signal must exceed threshold by this margin	1–3 dB
Time-to-Trigger (TTT)	Condition must be sustained for this duration	40–640 ms

■ HOW – Visualization

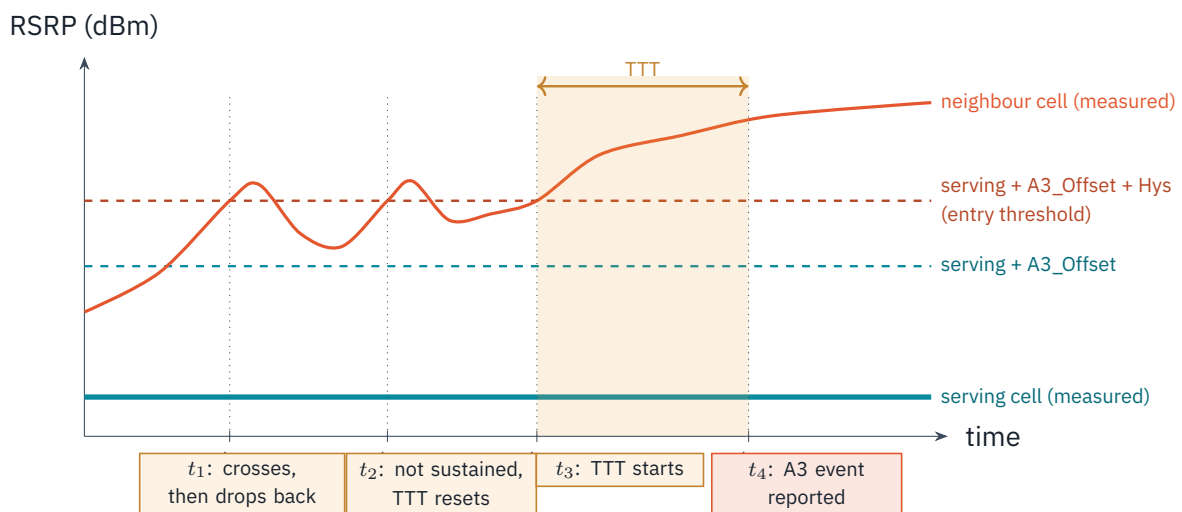


Figure 4.7 Event A3: entry threshold, hysteresis and time-to-trigger

Combined effect:

- Hysteresis provides amplitude filtering (dB domain)
- TTT provides time filtering (time domain)
- Together they provide robust handover triggering

■ Engineering Trade-off

	Low Hys + Short TTT	High Hys + Long TTT
Handover speed	Fast handover	Slow handover
Main risk	Ping-pong	Too-late HO
Good for	Slow users	Fast users
Signalling	More signalling	Less signalling
Typical setting	Urban: Hys = 2 dB, TTT = 160 ms	Highway: Hys = 3 dB, TTT = 80 ms

5.4 Handover Margin

■ WHAT

The **total handover margin** is the combined effect of A3 offset, hysteresis, and TTT that determines how much better a neighbor must be before handover occurs.

$$\text{Effective HO margin} \approx \text{A3_Offset} + \text{Hysteresis} + \underbrace{(\text{UE speed} \times \text{TTT})}_{\text{signal change during TTT}}$$

■ Example Calculation

```

Given:
  A3 Offset = 3 dB
  Hysteresis = 2 dB
  TTT = 320 ms
  UE speed = 50 km/h → path loss changes ~0.5 dB during TTT

Effective margin = 3 + 2 + 0.5 = 5.5 dB

Meaning: Neighbor must be 5.5 dB better than serving
         before handover is triggered
    
```

6 Handover Procedure – LTE X2 Handover

■ WHY

The X2-based handover is the **preferred** handover mechanism in LTE because it enables direct eNB-to-eNB communication without routing through the core network (MME), reducing latency and signaling load.

WHAT

- Flat architecture: eNBs communicate directly via X2 interface
- UE-assisted, network-controlled: UE measures, eNB decides
- Preparation phase ensures target has resources BEFORE handover command
- Total interruption time: **20–50 ms** (imperceptible for most applications)

■ HOW – X2 Handover Sequence

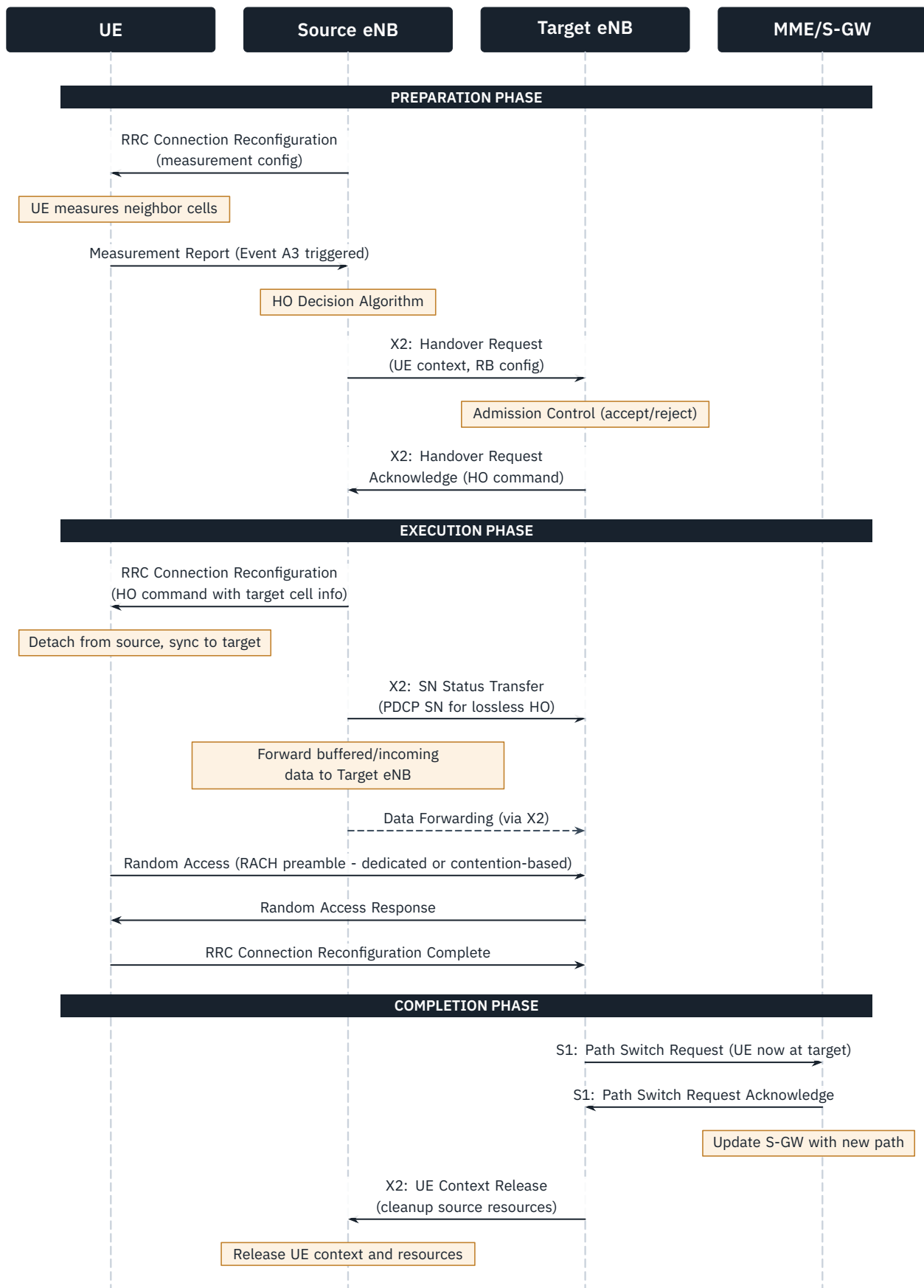


Figure 4.8 LTE X2 handover: preparation, execution and completion

■ Phase-by-Phase Breakdown

■ Phase 1: Preparation (~50–200 ms)

Step	Message	Purpose
1	Measurement Report	UE informs source about neighbor quality
2	HO Decision	Source eNB runs algorithm based on report
3	Handover Request (X2)	Source asks target to prepare resources
4	Admission Control	Target checks if it can accept the UE
5	HO Request Ack (X2)	Target confirms, provides HO command

■ Phase 2: Execution (~20–50 ms interruption)

Step	Message	Purpose
6	RRC Reconfiguration	Source commands UE to handover
7	SN Status Transfer	Ensures no data loss (PDCP SN alignment)
8	Data Forwarding	Buffered data forwarded to target
9	Random Access	UE synchronizes to target cell
10	Reconfig Complete	UE confirms successful handover

■ Phase 3: Completion (~20–100 ms)

Step	Message	Purpose
11	Path Switch Request	Informs MME/S-GW of new location
12	Path Switch Ack	Core confirms path update
13	UE Context Release	Source releases resources

■ Key Timing Values

Parameter	Typical Value
Total preparation time	50–200 ms
User-plane interruption	20–50 ms
Path switch completion	20–100 ms
Total end-to-end	100–350 ms
RACH preamble	Dedicated (no contention) or contention-based

■ S1-based Handover (Fallback)

When X2 interface is NOT available (no direct connection between eNBs):

- All messages route through MME
- Longer preparation time (~100–500 ms)
- Used for inter-eNB handover without X2, or inter-MME handover
- Same UE experience but higher network latency

7 Handover Failures

■ WHY

Handover failures directly impact user experience — dropped calls, interrupted video, lost data. Understanding failure modes is essential for network optimization.

■ WHAT

Four primary failure modes exist, each with distinct causes and solutions:

7.1 Too-Late Handover

■ WHY it happens

The UE moves too fast or the handover margin is too large, so the serving cell signal drops below usable level BEFORE the handover completes.

■ WHAT happens

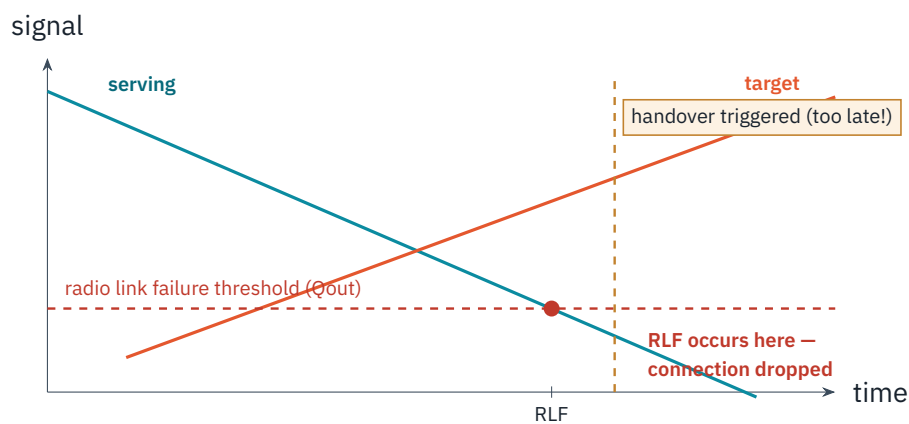


Figure 4.9 Too-late handover: the serving link fails before the handover

■ Indicators

- Radio Link Failure (RLF) in source cell BEFORE HO command received
- High RLF counter on source cell
- RLF report shows strong neighbor at time of failure

■ Mitigation

- **Decrease** A3 offset (trigger HO earlier)
- **Decrease** TTT (react faster)
- **Decrease** hysteresis
- Add early measurement reporting (A2 event with lower threshold)

7.2 Too-Early Handover

■ WHY it happens

Handover triggers too quickly (parameters too aggressive). UE is handed to a cell that is not yet strong enough to sustain the connection.

■ WHAT happens

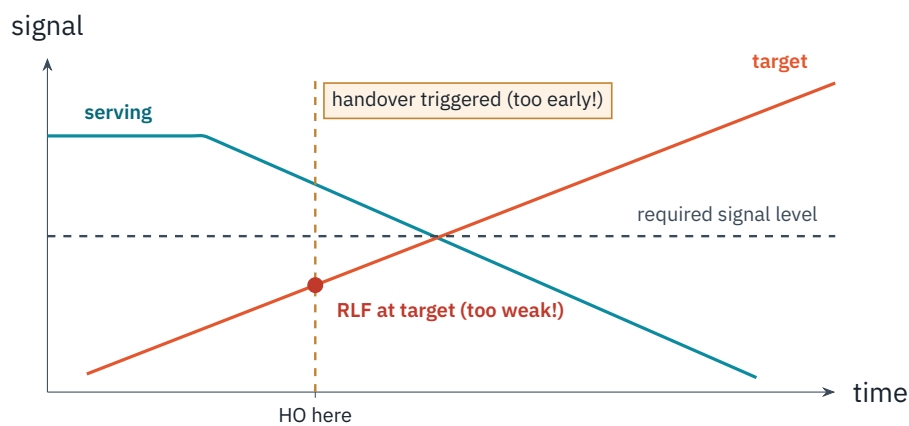


Figure 4.10 Too-early handover: the target cell cannot yet sustain the link

■ Indicators

- RLF occurs at TARGET cell shortly after handover completion
- UE attempts re-establishment back at source cell
- Short time between HO completion and RLF

■ Mitigation

- **Increase** A3 offset (require bigger difference)
- **Increase** TTT (wait longer)
- **Increase** hysteresis
- Verify neighbor cell coverage footprint

7.3 Ping-Pong Effect

■ WHY it happens

Two cells have similar signal strength in an overlap area. With aggressive handover parameters, the UE bounces repeatedly between them.

■ WHAT happens

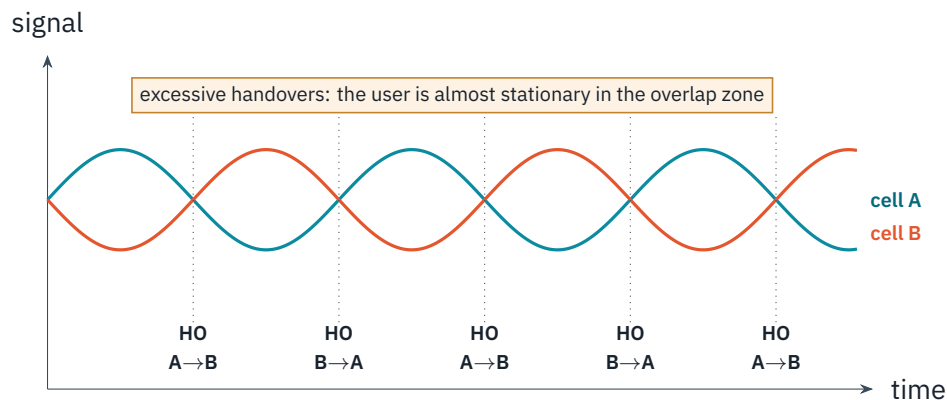


Figure 4.11 Ping-pong: repeated handovers between two similar cells

■ Impact

- Excessive signaling load
- Interrupted user data (each HO = 20-50 ms break)
- Wasted radio resources for HO preparation
- Poor QoE for the user

■ Indicators

- Same UE performing multiple handovers between same cell pair within short time
- High handover rate with low mobility
- A→B followed by B→A within seconds

■ Mitigation

Strategy	How it helps
Increase hysteresis	Larger margin needed to trigger → fewer triggers
Increase TTT	Must sustain longer → transient fluctuations filtered
Cell-Individual Offset (CIO)	Bias one cell to be preferred in overlap
Increase A3 offset	Neighbor must be significantly better
Antenna tilt optimization	Reduce overlap zone size

Rule of thumb: If >30% of handovers between a cell pair are ping-pong (HO back within 5s), increase TTT or hysteresis for that pair.

7.4 Handover to Wrong Cell

■ WHY it happens

The best target cell is not the one selected. This can occur due to:

- Missing neighbor relation (target not in neighbor list)
- Measurement report doesn't include the actual best cell
- Propagation anomaly (signal from far cell momentarily strong)

■ WHAT happens

- UE is handed to Cell X, but Cell Y would have been the correct target
- UE quickly experiences poor signal at Cell X
- May trigger another handover (to Cell Y) or RLF

■ Mitigation

- **Automatic Neighbor Relation (ANR):** Automatically discover and add neighbors
- **Comprehensive neighbor lists:** Ensure all potential targets are configured
- **Cell-specific offsets:** Deprioritize cells known to cause issues
- **Layer-3 filtering:** Average measurements to reduce anomalies

7.5 Handover Failure State Diagram

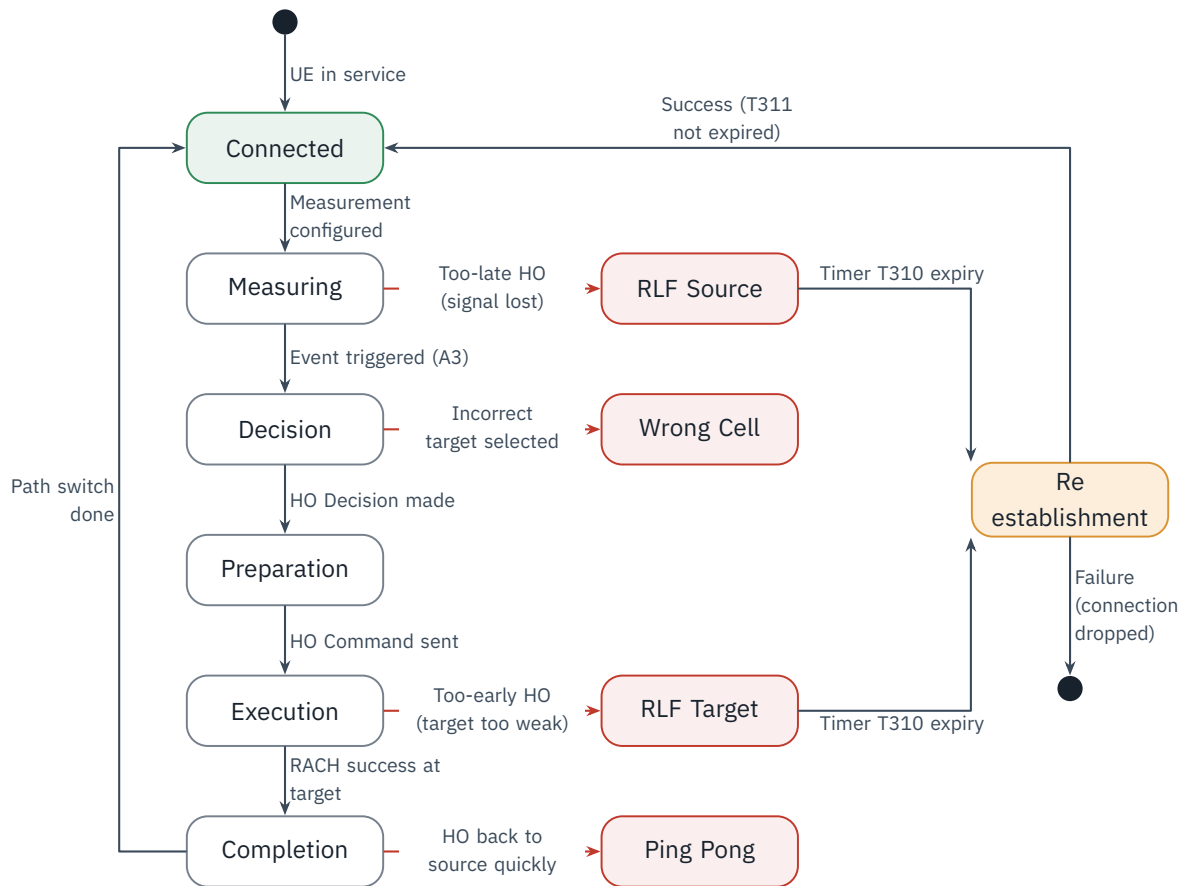


Figure 4.12 Handover states and failure modes

8 Evolution of Handover: 2G → 3G → LTE → 5G

WHY

Each generation improved handover mechanisms to support higher speeds, lower latency, and more complex network architectures. Understanding evolution reveals the design rationale.

8.1 GSM (2G): Hard Handover, BSC-Controlled

Architecture

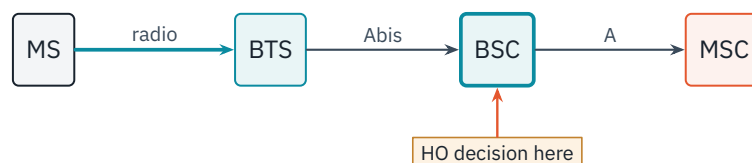


Figure 4.13 GSM: hard handover decided by the BSC

■ Characteristics

Aspect	2G GSM
Handover type	Hard only (break-before-make)
Decision maker	BSC (Base Station Controller)
Measurements	MS measures up to 6 neighbors (BCCH carriers)
Measurement rate	Every SACCH period (480 ms)
Interruption	100–200 ms
Criteria	RXLEV (signal strength), RXQUAL (BER-based quality)
Power control	Combined with handover decision

■ How it worked

1. MS continuously measures serving + 6 strongest neighbors (from BA list)
2. MS reports measurements every 480 ms via SACCH
3. BSC applies averaging (HREQAVE, HREQT parameters)
4. BSC decides based on multiple criteria:
 - Power budget (PBGT) → coverage-based
 - RXQUAL → quality-based (interference)
 - RXLEV → level-based (signal too weak/strong)
 - Distance → timing advance too large
 - Traffic → congestion relief

■ Limitations

- Hard handover: audible click on voice calls
- BSC-controlled: all intelligence in network, slow adaptation
- Limited measurements: only 6 neighbors, slow reporting
- No soft handover: no diversity gain at cell edge

8.2 UMTS (3G): Soft Handover, RNC-Controlled

■ Architecture

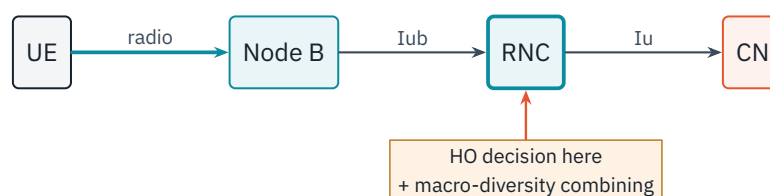


Figure 4.14 UMTS: handover decided by the RNC

■ Characteristics

Aspect	3G UMTS
Handover type	Soft + Softer + Hard
Decision maker	RNC (Radio Network Controller)
Measurements	CPICH Ec/No, RSCP
Active Set	Up to 3–6 cells simultaneously
Interruption	0 ms (soft HO), 20–50 ms (hard HO)
Criteria	Events 1A–1C (active set), 2A–2F (inter-freq/RAT)

■ Soft Handover Mechanism

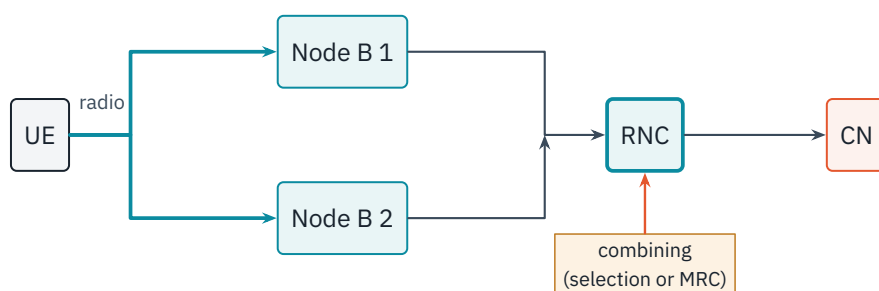


Figure 4.15 Soft handover: two radio legs combined in the RNC

○ Active Set Management:

- Event 1A: $\text{Pilot} > \text{Best_Pilot} - \text{Reporting_Range} + \text{Hysteresis} \rightarrow \text{ADD cell}$
- Event 1B: $\text{Pilot} < \text{Best_Pilot} - \text{Reporting_Range} - \text{Hysteresis} \rightarrow \text{REMOVE cell}$
- Event 1C: $\text{New pilot} > \text{Worst_in_ActiveSet} + \text{Hysteresis} \rightarrow \text{REPLACE cell}$

○ Macro-diversity gain: 2–3 dB at cell edge

○ 20–40% of connections in soft handover in typical network

■ Key Innovation

Soft handover enabled seamless mobility but at the cost of:

- Multiple Channelization Codes consumed (one per active set member)
- Increased Iub backhaul bandwidth
- Complex power control (must satisfy all links)

8.3 LTE (4G): Hard Handover, UE-Assisted Network-Controlled, X2-Based

■ Architecture

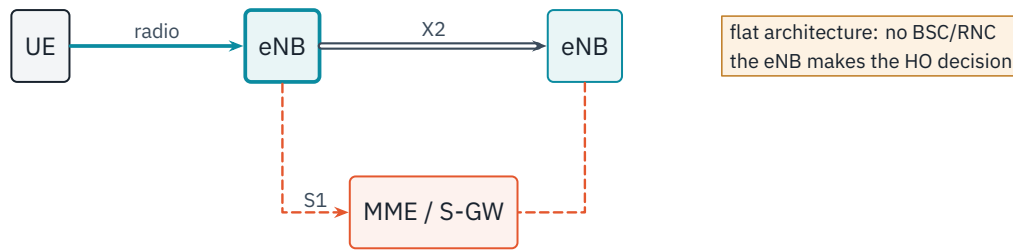


Figure 4.16 LTE: flat architecture, the eNB decides and uses X2

■ Characteristics

Aspect	4G LTE
Handover type	Hard only (break-before-make)
Decision maker	Source eNB (distributed decision)
Measurements	RSRP, RSRQ (events A1–A6, B1–B2)
Interruption	20–50 ms
Interface	X2 (direct eNB-to-eNB)
Key feature	UE-assisted, network-controlled

■ Why LTE Dropped Soft Handover

1. **OFDMA** (not CDMA): Different cells use different subcarriers → cannot combine same signal from multiple cells
2. **Flat architecture**: No central RNC to perform combining
3. **Hard HO is fast enough**: 20–50 ms interruption is acceptable for data
4. **Simplicity**: Significantly simpler than managing Active Sets
5. **Efficiency**: No wasted resources for multiple simultaneous links

■ Key Innovations in LTE

- **X2 interface**: Direct eNB-eNB → fast preparation, data forwarding
- **Event-based reporting**: Only report when conditions met (less signaling)
- **Layer-3 filtering**: Configurable averaging to smooth measurements
- **Automatic Neighbor Relations (ANR)**: Self-configuring neighbor lists
- **Mobility Robustness Optimization (MRO)**: Self-healing handover parameters

8.4 5G NR: Conditional Handover, DAPS, Dual Connectivity

■ Architecture

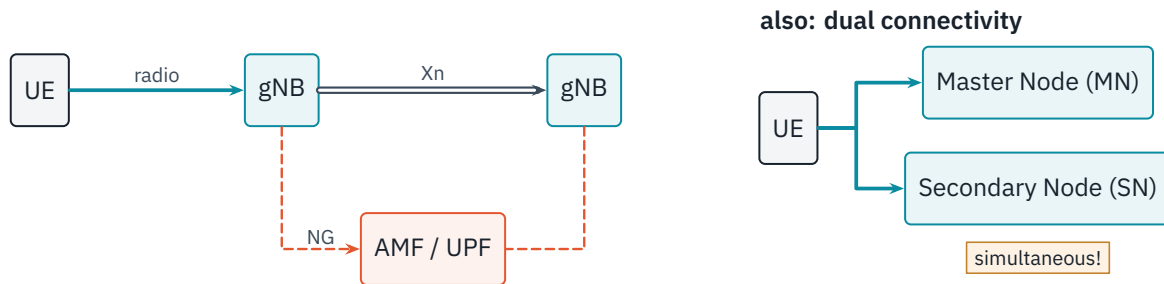


Figure 4.17 5G NR: Xn handover and dual connectivity

■ Characteristics

Aspect	5G NR
Handover type	Hard, Conditional , DAPS
Decision maker	Source gNB (+ CPC for conditional)
Measurements	SS-RSRP, SS-RSRQ, SS-SINR, CSI-RS based
Interruption	0 ms (DAPS), ~0 ms (DC), 20 ms (hard)
Key features	CHO, DAPS, Dual Connectivity, MR-DC

■ Conditional Handover (CHO) – 5G Innovation

WHY: In legacy LTE, the eNB must wait for measurement report, make decision, prepare target, then send command. If radio conditions degrade quickly (high speed, mmWave), this is too slow.

WHAT: Network **pre-prepares** multiple target cells and gives the UE conditions to trigger the handover **autonomously**.

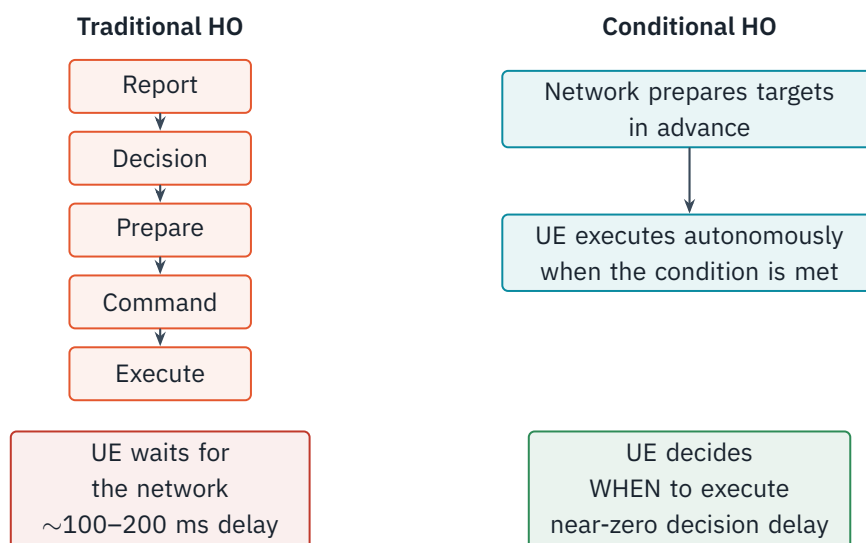


Figure 4.18 Traditional versus conditional handover

HOW:

1. Network configures CHO: “If Event A3 for Cell X OR Cell Y, execute HO to that cell”
2. Target cells are pre-prepared (resources reserved)
3. UE monitors conditions
4. When condition met → UE immediately executes (no round-trip to network)
5. **Benefit:** Eliminates handover preparation delay → fewer failures at high speed

■ Dual Active Protocol Stack (DAPS) – 0 ms Interruption

WHY: Even 20 ms interruption matters for URLLC (Ultra-Reliable Low-Latency Communication).

WHAT: UE maintains connection to BOTH source and target simultaneously during handover — similar concept to 3G soft handover but implemented differently.

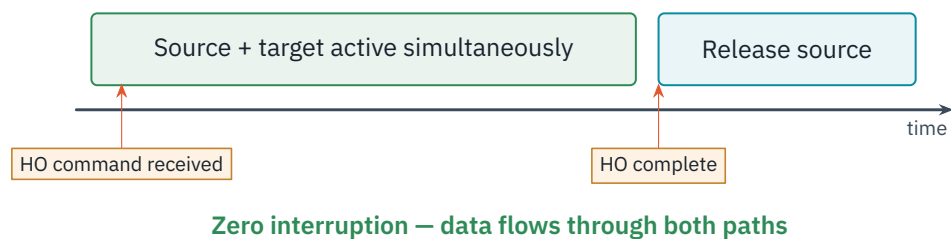


Figure 4.19 DAPS handover: make before break

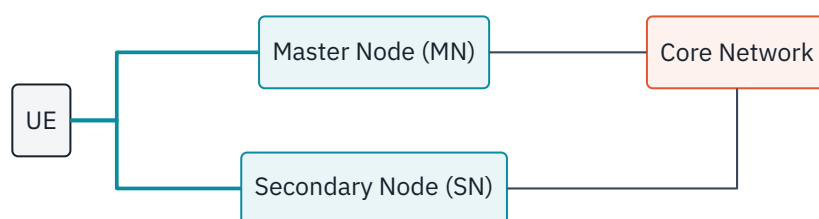
HOW:

- UE has two active protocol stacks (hence the name)
- Data is sent/received through both paths during transition
- Once target confirms successful connection, source is released
- **Trade-off:** Increased UE complexity and power consumption

■ Dual Connectivity (MR-DC) – Multi-RAT

WHAT: UE simultaneously connected to two different nodes:

- **EN-DC** (E-UTRA-NR Dual Connectivity): LTE master + NR secondary
- **NR-DC**: NR master + NR secondary
- **NE-DC**: NR master + LTE secondary



Both links active simultaneously → seamless mobility
SN change ≠ handover (no interruption to MN)

Figure 4.20 Dual connectivity: two radio links at the same time

8.5 Generation Comparison Summary

Feature	2G GSM	3G UMTS	4G LTE	5G NR
HO Type	Hard	Soft + Hard	Hard	Hard + CHO + DAPS
Controller	BSC	RNC	eNB	gNB
Interruption	100-200 ms	0 ms (soft)	20-50 ms	0 ms (DAPS/DC)
Measurement	RXLEV, RXQUAL	CPICH RSCP, Ec/No	RSRP, RSRQ	SS-RSRP, CSI-RS
Intelligence	Network-only	Network-only	UE-assisted	UE-autonomous (CHO)
Architecture	Hierarchical	Hierarchical	Flat	Flat + DC
Self-optimization	Manual	Manual	SON (MRO)	AI/ML-based

9 Engineering Examples with Typical Parameter Values

9.1 Example: Urban Macro Network – LTE Handover Optimization

■ Scenario

- Dense urban area, 500m inter-site distance
- Users: mix of pedestrians (3 km/h) and vehicles (50 km/h)
- Problem: 5% handover failure rate, excessive ping-pong

■ Initial Parameters (before optimization)

Parameter	Value	Issue
A3 Offset	2 dB	Too aggressive → too-early HO
Hysteresis	1 dB	Not enough filtering
TTT	100 ms	Too short → ping-pong
Layer-3 filter (k)	1	Not enough averaging

■ Optimized Parameters

Parameter	New Value	Rationale
A3 Offset	3 dB	More margin before triggering
Hysteresis	2 dB	Better fading resistance
TTT	256 ms	Eliminates most ping-pong for pedestrians
Layer-3 filter (k)	4	Smooths fast fading (~200 ms averaging)
CIO (cell pair with ping-pong)	+3 dB	Biases HO for known problem pair

■ Result

- Handover failure rate: 5% → 1.5%
- Ping-pong rate: 12% → 3%
- Trade-off: Mean HO delay increased by ~150 ms (acceptable)

9.2 Example: High-Speed Rail – 350 km/h

■ Challenge

At 350 km/h, the UE crosses a cell (1 km coverage) in ~10 seconds. Standard parameters are too slow.

■ Parameter Tuning

Parameter	Standard	High-Speed
A3 Offset	3 dB	2 dB
Hysteresis	2 dB	1 dB
TTT	256 ms	40 ms
Layer-3 filter (k)	4	1
A2 threshold	-120 dBm	-110 dBm

■ Additional Measures

- Dedicated RF planning along rail corridor
- Directional antennas pointing along track
- Cell overlap zone: 200–300 m minimum
- Consider **Conditional Handover (5G)** for this scenario

■ Speed Sensitivity Calculation

```

At 350 km/h = 97 m/s
Path loss change rate ≈ 0.3 dB per 10 ms (at 2 GHz, Hata model)

With TTT = 256 ms: Signal changes by ~7.5 dB during TTT!
→ This is unacceptable – must reduce TTT

With TTT = 40 ms: Signal changes by ~1.2 dB during TTT
→ Acceptable margin
    
```

9.3 Example: Inter-RAT Handover – LTE to 3G CSFB

■ Scenario

VoLTE not deployed; voice calls require CS Fallback to 3G.

■ Configuration

Parameter	Value	Purpose
B2 Threshold1 (serving LTE)	-110 dBm RSRP	Trigger when LTE weak
B2 Threshold2 (UMTS target)	-95 dBm RSCP	Only if 3G is adequate
TTT (B2)	640 ms	Avoid unnecessary fallback
A2 Threshold (start measuring 3G)	-105 dBm RSRP	Begin inter-RAT measurements
IRAT measurement gap	6 ms / 40 ms	Gap pattern for measuring UMTS

■ Procedure

1. UE on LTE, RSRP drops below -105 dBm → A2 event → start measuring UMTS
2. LTE RSRP drops below -110 dBm AND UMTS RSCP > -95 dBm for 640 ms → B2 event
3. eNB triggers inter-RAT handover to UMTS
4. UE performs hard handover to 3G, establishes CS domain connection
5. Voice call proceeds on 3G circuit-switched
6. After call ends: UE returns to LTE (fast return / redirection)

9.4 Example: Measurement Event Configuration (Complete)

■ Full A3 Event Configuration for Intra-Frequency HO

```
RRC Connection Reconfiguration (measurement config):
{
  measObjectEUTRA: {
    carrierFreq: 1850 (EARFCN for Band 3, 1800 MHz)
    allowedMeasBandwidth: 50 RBs
    neighCellConfig: 01 (partial measurement)
    offsetFreq: 0 dB
  }

  reportConfigEUTRA: {
    triggerType: event
    eventId: A3
    a3-Offset: 3 dB (= 6 in IE, 0.5 dB steps)
    hysteresis: 2 dB (= 4 in IE, 0.5 dB steps)
    timeToTrigger: 256 ms
    triggerQuantity: RSRP
    reportQuantity: both (RSRP + RSRQ)
    maxReportCells: 4
    reportInterval: 480 ms
    reportAmount: infinity
  }

  measIdToAddList: {
    measId: 1
    measObjectId: 1
    reportConfigId: 1
  }

  quantityConfig: {
    filterCoefficientRSRP: 4 (k=4, ~200ms averaging)
    filterCoefficientRSRQ: 4
  }
}
```

■ Layer-3 Filtering Formula

$$F_n = (1 - a) F_{n-1} + a M_n, \quad a = \left(\frac{1}{2}\right)^{k/4}$$

where k is the filterCoefficient (0, 1, 2, ..., 8, ...):

- $k = 0$: $a = 1.0 \rightarrow$ no filtering (raw measurement used)
- $k = 4$: $a = 0.5 \rightarrow$ 50% new + 50% old $\rightarrow$ about 200 ms memory
- $k = 8$: $a = 0.25 \rightarrow$ 25% new + 75% old $\rightarrow$ about 400 ms memory

9.5 Typical Handover KPIs (Operator Benchmarks)

KPI	Target	Good	Poor
Handover Success Rate	>98%	>99%	<95%
Ping-pong Rate	<5%	<2%	>10%
RLF Rate (HO-related)	<1%	<0.5%	>3%
Mean Interruption Time	<50 ms	<30 ms	>100 ms
Too-early HO ratio	<1%	<0.5%	>3%
Too-late HO ratio	<2%	<1%	>5%
Unnecessary HO (ping-pong + wrong cell)	<5%	<3%	>10%

10 Summary Diagrams

■ Measurement Event A3 Triggering – With Hysteresis and TTT

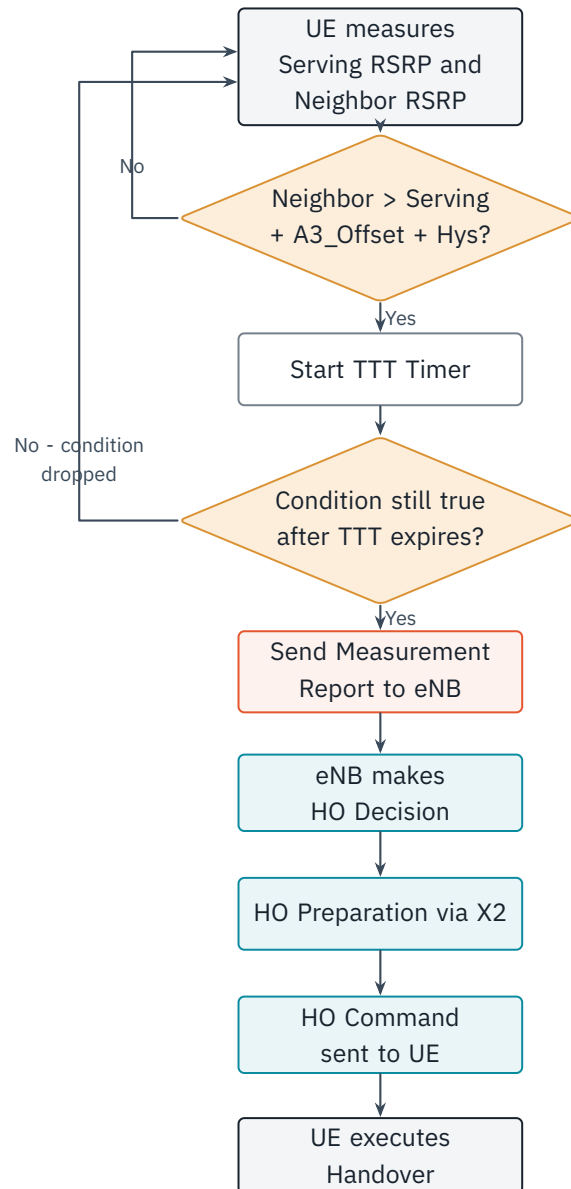


Figure 4.21 Event A3 triggering with hysteresis and time-to-trigger

Handover States – Complete Flow

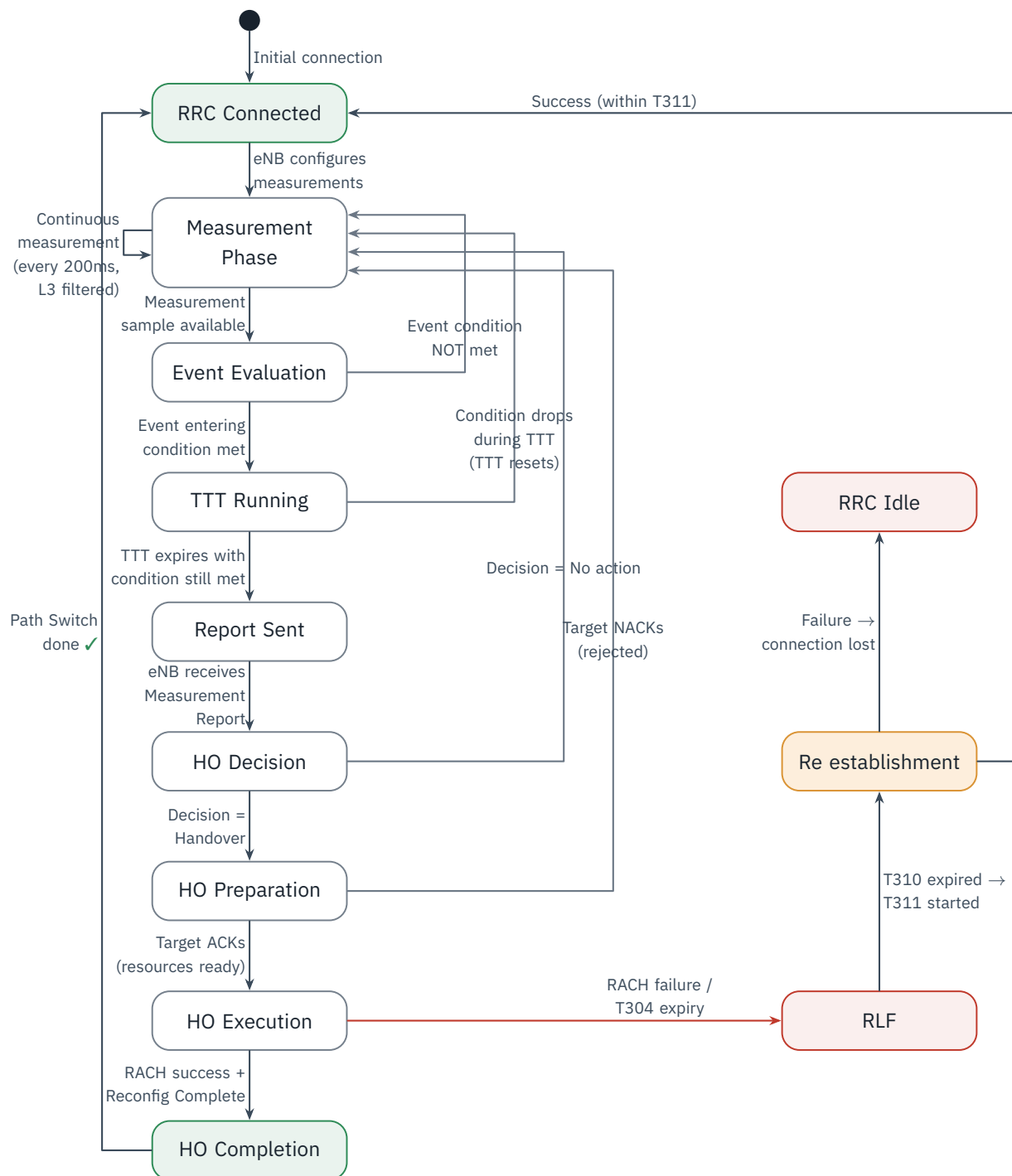


Figure 4.22 Complete handover state flow

LTE Handover Parameters – Decision Tree

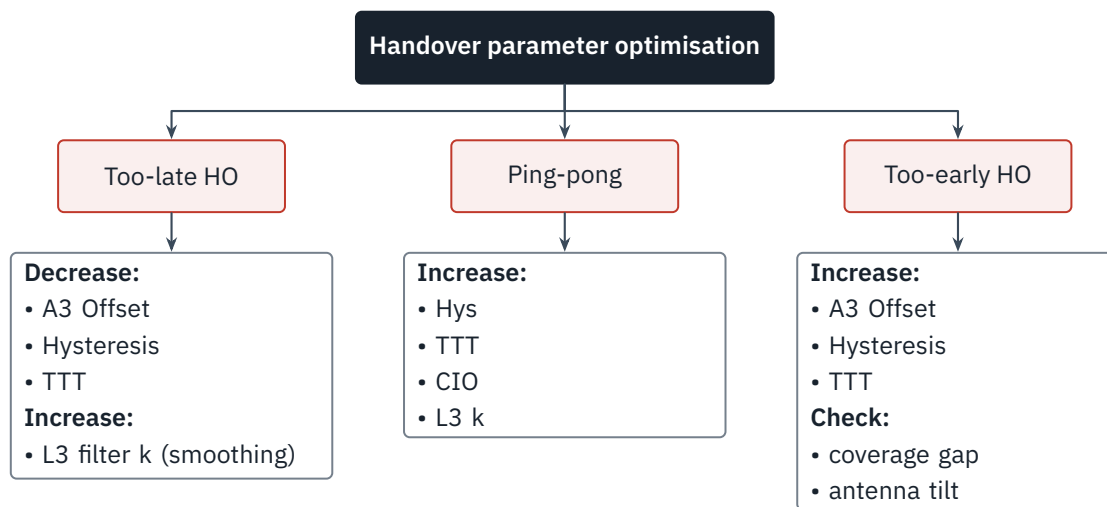


Figure 4.23 Handover parameter optimisation decision tree

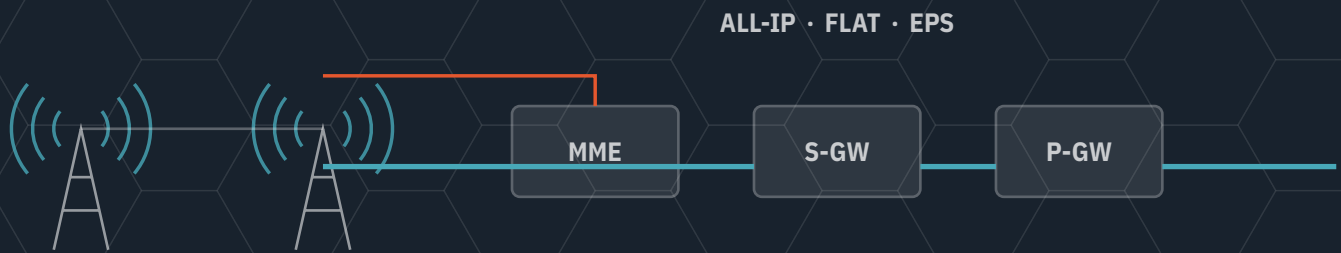
Key Formulas Summary

Formula	Expression	Use
A3 Entry	$M_n - Hys > M_s + A3_Off$	Intra-freq HO trigger
A3 Leave	$M_n + Hys < M_s + A3_Off$	Cancel pending report
Cell Ranking (Rs)	$Q_{meas,s} + Q_{Hyst}$	Serving cell reselection rank
Cell Ranking (Rn)	$Q_{meas,n} - Q_{offset}$	Neighbor reselection rank
S-criterion	$S_{rxlev} = Q_{rxlevmeas} - Q_{rxlevmin} > 0$	Cell suitability
L3 Filter	$F_n = (1 - a) F_{n-1} + a M_n$	Measurement smoothing
RSRQ	$N \times RSRP / RSSI$	Signal quality metric
HO Margin	$A3_Off + Hys + speed \times TTT$	Effective trigger margin

Review Questions

1. **Explain** why LTE uses hard handover instead of soft handover like 3G.
2. **Calculate** the effective handover margin for: A3 offset = 4 dB, Hys = 2 dB, TTT = 320 ms, UE speed = 120 km/h.
3. **Draw** the timeline of a too-late handover and identify which parameter to adjust.
4. **Compare** Tracking Areas (LTE) with Location Areas (2G). What is the advantage of Tracking Area Lists?
5. **Design** handover parameters for a highway scenario (130 km/h). Justify each choice.
6. **Explain** how Conditional Handover in 5G solves the high-mobility problem that LTE struggles with.
7. **Analyze** a scenario where ping-pong rate is 15% between two cells. List three mitigation strategies in order of preference.
8. **Describe** the X2 handover procedure and identify where data loss could occur without SN Status Transfer.

Module 4 complete. Next: Module 5 — Radio Resource Management and Scheduling



In This Module

5.1	Why LTE Was Developed	117
5.2	LTE Design Principles	118
5.3	LTE Architecture Overview	119
5.4	Network Elements — Detailed	120
5.5	LTE Interfaces — Reference Points	126
5.6	IMS Overview (IP Multimedia Subsystem)	132
5.7	Control Plane vs User Plane Separation (CUPS)	134
5.8	Key Differences from 3G	135
5.9	Control Plane vs User Plane Paths	137
5.10	Interface Summary Table	140
	Key Takeaways	141
	References	141

5.1 Why LTE Was Developed

■ Limitations of 3G (UMTS/HSPA)

The 3rd Generation Partnership Project (3GPP) initiated the Long Term Evolution (LTE) project in 2004 to address fundamental limitations in 3G networks:

Architectural Complexity:

- 3G UTRAN required a hierarchical structure: UE → NodeB → RNC → CN
- The Radio Network Controller (RNC) was a single point of failure and bottleneck
- Complex protocol stacks between NodeB and RNC (Iub interface) added processing overhead
- Soft handover required coordination between multiple NodeBs via the RNC

Latency Issues:

- 3G round-trip times (RTT): 50–100 ms typical, 150+ ms under load
- The RNC added processing hops in both control and user planes
- Circuit-switched voice added further complexity and delay
- Real-time applications (gaming, video calls) suffered noticeably

Spectrum Efficiency:

- WCDMA/HSPA approaching theoretical limits
- Demand for mobile data growing exponentially (smartphones era beginning)
- Need for flexible bandwidth: 1.4 MHz to 20 MHz channel widths

All-IP Target:

- 3G still maintained circuit-switched domain for voice (MSC, MGW)
- Dual-stack (CS + PS) increased OPEX and complexity
- Industry consensus: everything must converge to IP

Performance Targets for LTE (3GPP Release 8):

Parameter	Target
Peak DL rate	100 Mbps (20 MHz)
Peak UL rate	50 Mbps (20 MHz)
User plane latency	< 5 ms
Control plane latency	< 100 ms (idle→active)
Spectrum efficiency	3–4x improvement over HSPA
Mobility	Up to 350 km/h

5.2 LTE Design Principles

LTE was designed from scratch with these guiding principles:

■ Flat Architecture

- **Eliminate the RNC** — distribute its functions to the eNodeB and core network
- Fewer network nodes = fewer hops = lower latency
- Each eNodeB connects directly to the core network (EPC)

■ All-Packet (No Circuit Switching)

- **Pure packet-switched network** — no circuit-switched domain
- Voice handled via VoLTE (Voice over LTE) using IMS
- Single protocol stack simplifies operations

■ No Soft Handover

- 3G soft handover required simultaneous connections to multiple NodeBs
- LTE uses **hard handover** only — simpler, more spectrum-efficient
- Handover is fast enough (~20–50 ms interruption) to be imperceptible

■ Simplified Protocol Stack

- Fewer layers, cleaner separation of concerns
- User plane: PDCP → RLC → MAC → PHY
- Control plane: NAS (between UE and MME) + RRC (between UE and eNB)

■ Shared Channel Access

- No dedicated channels — all resources shared dynamically
- Scheduler in eNodeB makes 1 ms TTI decisions
- Enables statistical multiplexing gains

■ Self-Organizing Network (SON) Features

- Automatic neighbor relations (ANR)
- Self-configuration of new eNodeBs
- Self-optimization and self-healing capabilities

5.3 LTE Architecture Overview

LTE architecture consists of two main domains:

1. **E-UTRAN** (Evolved UTRAN) — the radio access network
2. **EPC** (Evolved Packet Core) — the core network

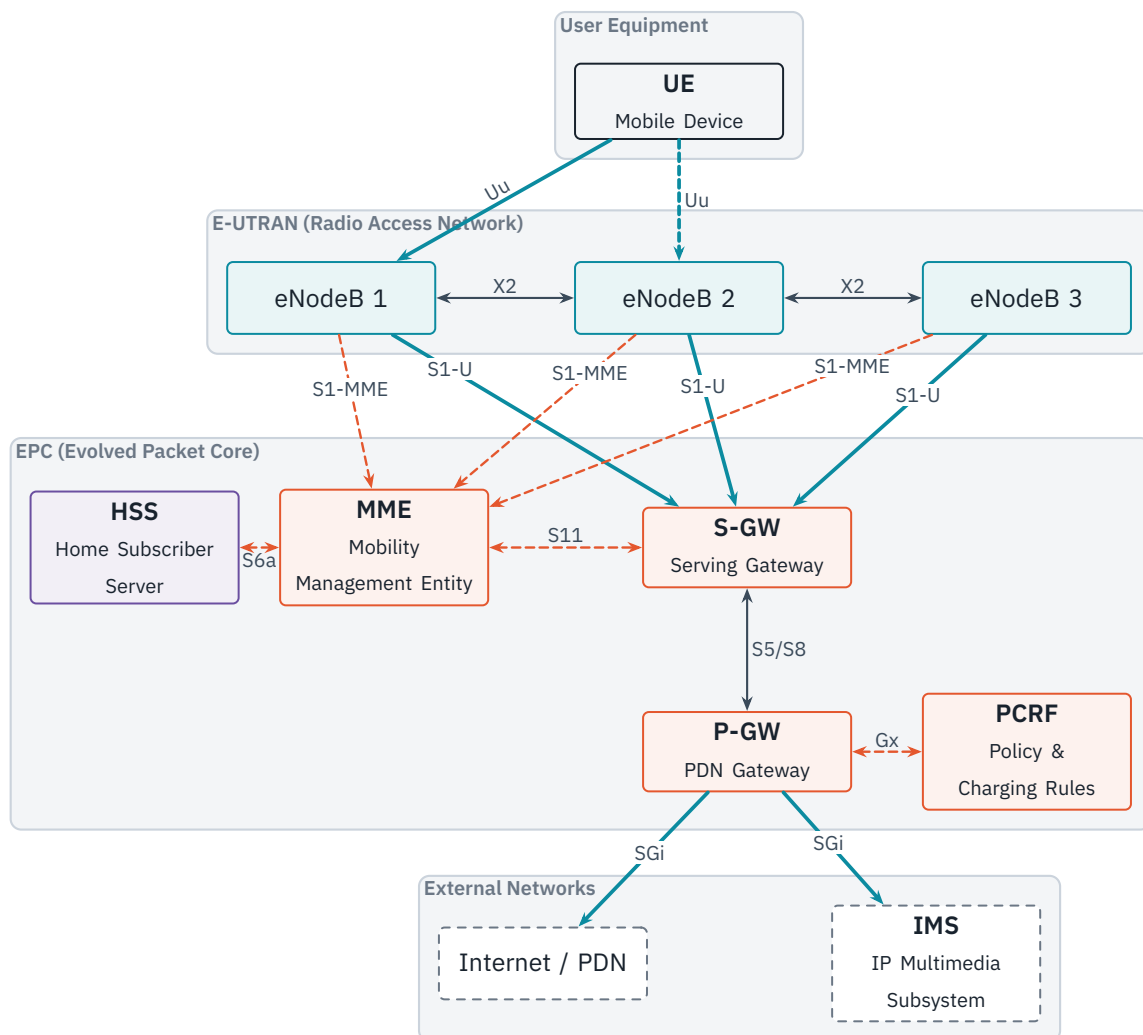


Figure 5.1 LTE architecture: UE, E-UTRAN, EPC and external networks

E-UTRAN: The Radio Access Network

Key insight: E-UTRAN consists of eNodeBs only — there is NO RNC!

- The eNodeB handles ALL radio-related functions previously split between NodeB and RNC
- eNodeBs connect directly to the EPC via the S1 interface
- eNodeBs connect to each other via the X2 interface for handover and load balancing
- This is the “flat architecture” — one node type in the RAN

■ EPC: The Evolved Packet Core

The EPC is a pure packet core with separated control and user planes:

Element	Primary Role
MME	Control plane — signaling, mobility, security
S-GW	User plane anchor within E-UTRAN
P-GW	User plane anchor to external networks
HSS	Subscriber database and authentication
PCRF	Policy and charging rules

5.4 Network Elements — Detailed

5.4.1 UE (User Equipment)

Purpose: The mobile device that communicates over the LTE air interface.

Components:

- **ME (Mobile Equipment):** The physical device (phone, tablet, modem)
- **USIM (Universal Subscriber Identity Module):** Stores subscriber identity (IMSI), security keys (K), and authentication algorithms

UE Categories (3GPP Release 8–12):

Category	Max DL (Mbps)	Max UL (Mbps)	DL MIMO	Notes
Cat 1	10	5	None	IoT, basic
Cat 2	51	25	2×2	Entry smartphones
Cat 3	100	50	2×2	Mainstream
Cat 4	150	50	2×2	High-end
Cat 5	300	75	4×4	Flagship
Cat 6	300	50	2×2	Carrier Aggregation
Cat 9	450	50	2×2	3CC CA
Cat 12	600	100	4×4	Advanced
Cat 16	980	150	4×4	LTE-Advanced Pro

Key UE Capabilities:

- Carrier Aggregation (CA): combine multiple component carriers
- MIMO support (2×2 or 4×4 antenna configurations)
- VoLTE support (IMS registration, SIP signaling)
- Dual connectivity (connect to two eNodeBs simultaneously — Release 12+)

5.4.2 eNodeB (Evolved NodeB)

Purpose: The single RAN node that handles ALL radio access functions — replaces both NodeB AND RNC from 3G.

Problem it solves: Eliminates the RNC bottleneck, reduces latency by one hop, enables faster scheduling decisions.

Functions absorbed from 3G RNC:

- Radio Resource Management (RRM) — scheduling, admission control
- Radio Bearer Control — setup, maintenance, release of bearers
- Connection Mobility Control — handover decisions
- Dynamic Resource Allocation (Scheduler) — 1 ms TTI granularity
- Inter-cell Interference Coordination (ICIC)
- Header compression (ROHC via PDCP)
- Encryption and integrity protection (PDCP layer)
- Segmentation and reassembly (RLC layer)

Functions unique to eNodeB:

- S1 interface termination (to MME and S-GW)
- X2 interface termination (to neighboring eNodeBs)
- Paging message distribution
- Broadcast information (SIBs)
- Measurement configuration and reporting

Scheduler — The Brain of eNodeB:

- Makes resource allocation decisions every 1 ms (TTI)
- Assigns Resource Blocks (RBs) to UEs based on:
 - Channel quality (CQI reports from UEs)
 - QoS requirements (GBR vs non-GBR bearers)
 - Fairness algorithms (Proportional Fair, Round Robin, Max Throughput)
 - Buffer status and priority

5.4.3 MME (Mobility Management Entity)

Purpose: The control plane anchor of the EPC — handles all signaling and control functions, touches NO user data.

Problem it solves: Separates control from user plane, enabling independent scaling. Centralizes mobility and security management.

Key Functions:

Function	Description
NAS Signaling	Manages Non-Access Stratum protocols (EMM, ESM) between UE and MME
Authentication	Generates authentication vectors with HSS (EPS-AKA)
Security	NAS encryption/integrity; derives keys for AS security
Mobility Management	Tracks UE location (Tracking Areas), handles TAU
Bearer Management	Establishes, modifies, releases EPS bearers
Paging	Initiates paging when downlink data arrives for idle UE
S-GW Selection	Selects appropriate Serving Gateway for UE
P-GW Selection	Selects PDN Gateway (during initial attach or new PDN connection)
Handover Control	Inter-eNB and inter-MME handover signaling
Idle Mode	Manages UE state transitions (connected ↔ idle)

NAS Procedures:

- **Attach:** UE registers with the network (identity, authentication, bearer setup)
- **Detach:** UE deregisters (voluntary or network-initiated)
- **TAU (Tracking Area Update):** UE reports new location when crossing TA boundary
- **Service Request:** Idle UE requests to become active (for data transfer)

5.4.4 S-GW (Serving Gateway)

Purpose: The user plane anchor within the E-UTRAN — all user data passes through S-GW.

Problem it solves: Provides a stable user plane anchor point during intra-LTE mobility (handover between eNodeBs). Only one tunnel switch needed at S-GW during handover instead of rerouting through external networks.

Key Functions:

- **Local mobility anchor:** During inter-eNB handover, only the S1-U tunnel is switched at S-GW
- **Packet routing and forwarding:** Routes packets between eNB and P-GW
- **Data buffering during handover:** Buffers DL packets while handover completes
- **Downlink data notification:** Triggers paging via MME when DL data arrives for idle UE
- **Lawful intercept:** Interface for legal interception of user traffic
- **Charging data collection:** Usage data for offline charging (CDRs)
- **Transport level marking:** DSCP marking for QoS in transport network
- **Idle mode buffering:** Buffers first DL packet and notifies MME for paging

Deployment Note: In many networks, S-GW and P-GW are combined into a single node (SAE-GW) for simplicity.

5.4.5 P-GW (PDN Gateway)

Purpose: The point of connection to external packet data networks (Internet, IMS, enterprise VPNs). Allocates IP addresses and enforces policy.

Problem it solves: Provides a single, stable IP anchor point for the UE regardless of mobility. The UE keeps the same IP address even when moving between S-GWs.

Key Functions:

- **IP address allocation:** Assigns IPv4 and/or IPv6 addresses to UEs (via DHCPv4, SLAAC, or static)
- **Policy enforcement:** Applies QoS rules received from PCRF (via Gx interface)
- **Charging:** Online charging (Gy to OCS) and offline charging (Gz to OFCS)
- **Packet filtering:** Deep packet inspection, TFT (Traffic Flow Template) enforcement
- **Per-SDF (Service Data Flow) treatment:** Apply different QoS/charging per traffic flow
- **Inter-operator mobility anchor:** When roaming, P-GW remains in home network (home-routed) or visited network (local breakout)
- **NAT (if needed):** Network Address Translation for IPv4
- **GTP/PMIP anchor:** Terminates S5/S8 tunnel from S-GW

APN (Access Point Name):

- Each PDN connection is identified by an APN
- P-GW selection based on APN (e.g., “internet”, “ims”, “enterprise.vpn”)
- Multiple simultaneous PDN connections possible (e.g., Internet + IMS for VoLTE)

5.4.6 HSS (Home Subscriber Server)

Purpose: The central database containing all subscriber information and security credentials. Evolution of 3G HLR/AuC.

Problem it solves: Centralizes subscriber management, provides authentication vectors, and stores subscription data (allowed APNs, QoS profiles, roaming permissions).

Key Functions:

- **Subscriber data storage:**
 - IMSI (permanent identity)
 - MSISDN (phone number)
 - Subscribed APNs and associated QoS profiles
 - Roaming restrictions
 - Subscribed Tracking Areas
- **Authentication vector generation:**
 - Stores permanent key K (shared with USIM)
 - Generates authentication vectors (RAND, AUTN, XRES, K_{ASME}) using EPS-AKA
 - Provides vectors to MME for UE authentication
- **Location management:**
 - Stores current serving MME for each subscriber
 - Used for terminating calls/sessions (where to page the UE)
- **Authorization:**
 - Determines which services a subscriber can access
 - Enforces subscription limits (max bitrate, allowed APNs)

Interfaces:

- S6a to MME (Diameter protocol)
- Cx/Dx to IMS (for VoLTE subscriber data)
- Sh to Application Servers

5.4.7 PCRF (Policy and Charging Rules Function)

Purpose: The policy brain of the network — makes real-time decisions about QoS and charging rules applied to each data flow.

Problem it solves: Enables dynamic, per-flow QoS (e.g., guarantee bandwidth for a VoLTE call while best-effort for web browsing). Enables flexible charging models (zero-rating, tiered plans, sponsored data).

Key Functions:

- **Policy decision:** Determines QoS parameters (QCI, ARP, MBR, GBR) for each service data flow
- **Charging rules:** Specifies how each flow is charged (rating group, metering method)

- **PCC (Policy and Charging Control) rules:** Combined policy + charging instructions sent to P-GW
- **Dynamic policy updates:** Can modify rules in real-time (e.g., when user starts a video call)
- **Subscription-based policy:** Considers user's subscription tier
- **Network-condition-based policy:** Can throttle based on congestion

Example PCC Rule:

```
Rule: "VoLTE_Voice"
- Service Data Flow: SIP/RTP traffic to IMS
- QCI: 1 (Conversational Voice)
- GBR: 40 kbps UL / 40 kbps DL
- ARP: Priority 1, pre-emption capable
- Charging: Included in voice plan (no data deduction)
```

Interfaces:

- Gx to P-GW (installs/modifies PCC rules)
- Rx from IMS/AF (receives session info, triggers policy)
- Sp to SPR (Subscription Profile Repository)

5.5 LTE Interfaces – Reference Points

Interface Map

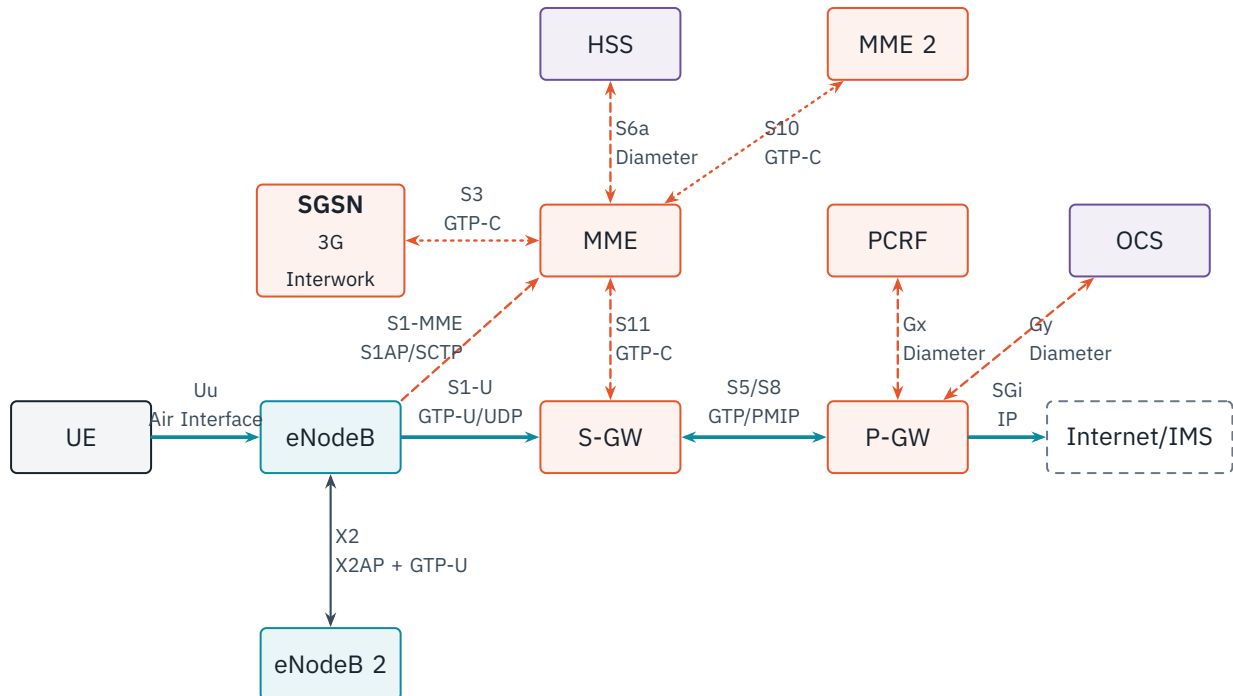


Figure 5.2 LTE interface map with the protocol of each reference point

5.5.1 S1-MME Interface

Property	Value
Endpoints	eNodeB ↔ MME
Plane	Control Plane
Protocol Stack	S1AP / SCTP / IP
Purpose	All control signaling between RAN and core

What flows over S1-MME:

- Initial UE messages (Attach, TAU, Service Request)
- UE context setup/release
- Handover preparation and execution (inter-eNB via MME)
- Paging requests from MME to eNB
- Bearer setup/modification/release commands
- NAS message transport (MME ↔ UE via eNB as relay)
- Reset and error indication procedures

Why SCTP (not TCP):

- Multi-homing: survives link failures without session drop
- Multi-streaming: avoids head-of-line blocking
- Message-oriented (not byte-stream): natural fit for signaling
- Built-in heartbeat for connection monitoring

5.5.2 S1-U Interface

Property	Value
Endpoints	eNodeB ↔ S-GW
Plane	User Plane
Protocol Stack	GTP-U / UDP / IP
Purpose	Carries user IP packets in GTP tunnels

What flows over S1-U:

- All user data (encapsulated in GTP-U tunnels)
- One GTP tunnel per EPS bearer per UE
- End marker packets during handover (signals last packet from old path)

GTP-U Tunnel Identification:

- Each tunnel identified by TEID (Tunnel Endpoint Identifier) + IP address
- TEIDs allocated independently at each end
- Enables multiplexing many UE bearers over single transport connection

Why GTP-U (not plain IP):

- Tunneling allows mobility transparency (IP address doesn't change)
- Per-bearer QoS enforcement
- Simple encapsulation: GTP-U header (8+ bytes) + inner IP packet

5.5.3 X2 Interface

Property	Value
Endpoints	eNodeB ↔ eNodeB
Plane	Control + User Plane
Protocol Stack	X2AP/SCTP (control) + GTP-U/UDP (user)
Purpose	Direct inter-eNB handover and load management

Control Plane (X2AP) Functions:

- Handover preparation (request, acknowledge, cancel)
- SN (Sequence Number) status transfer
- Load indication (exchange load information between eNBs)

- Resource status reporting
- Inter-cell interference coordination (ICIC) information exchange

User Plane (GTP-U) Functions:

- Forward DL data from source eNB to target eNB during handover
- Prevents data loss during handover gap
- Temporary tunnel — torn down after handover completes

X2 Handover Advantage:

- Direct eNB-to-eNB signaling — no need to involve MME for preparation
- Only MME notified after handover completes (path switch)
- Faster than S1-based handover (used when X2 is available)

5.5.4 S5/S8 Interface

Property	Value
Endpoints	S-GW ↔ P-GW
Plane	User + Control Plane
Protocol Stack	GTP-C + GTP-U (or PMIP alternative)
Purpose	Connects serving gateway to PDN gateway

S5 vs S8:

- **S5**: When S-GW and P-GW are in the **same** PLMN (non-roaming or home-routed)
- **S8**: When S-GW and P-GW are in **different** PLMNs (roaming scenario)
- Technically identical protocol, different name indicates inter-PLMN boundary

Functions:

- Tunnel management (create, update, delete sessions)
- Per-bearer GTP tunnels for user plane
- Triggered by MME (via S11) during attach or bearer modification
- GTP-C handles session/bearer management signaling
- GTP-U carries encapsulated user data

Protocol Options:

- **GTP-based S5/S8**: Full GTP-C signaling + GTP-U tunnels (most common)
- **PMIP-based S5/S8**: Proxy Mobile IPv6 signaling (rare in practice)

5.5.5 S6a Interface

Property	Value
Endpoints	MME ↔ HSS
Plane	Control Plane
Protocol Stack	Diameter / SCTP (or TCP) / IP
Purpose	Subscriber authentication and data retrieval

Diameter Commands on S6a:

Command	Direction	Purpose
Authentication-Information-Request (AIR)	MME→HSS	Request auth vectors
Authentication-Information-Answer (AIA)	HSS→MME	Return auth vectors (RAND, AUTN, XRES, K_{ASME})
Update-Location-Request (ULR)	MME→HSS	Register MME as serving node
Update-Location-Answer (ULA)	HSS→MME	Return subscription data
Cancel-Location-Request (CLR)	HSS→MME	Deregister UE from old MME
Insert-Subscriber-Data-Request (IDR)	HSS→MME	Push updated subscription data
Delete-Subscriber-Data-Request (DSR)	HSS→MME	Remove subscription data
Purge-UE-Request (PUR)	MME→HSS	Inform HSS that UE is unreachable

Why Diameter (not SS7/MAP):

- IP-native (aligns with all-IP architecture)
- Extensible via AVPs (Attribute-Value Pairs)
- Better security (TLS/IPsec support)
- Larger message sizes (supports complex subscription data)

5.5.6 S11 Interface

Property	Value
Endpoints	MME ↔ S-GW
Plane	Control Plane
Protocol Stack	GTP-C v2 / UDP / IP
Purpose	EPS bearer/session management between MME and S-GW

Key Messages:

- **Create Session Request/Response:** During attach — establishes default bearer
- **Modify Bearer Request/Response:** During handover — updates S1-U tunnel info
- **Delete Session Request/Response:** During detach — tears down bearers
- **Create Bearer Request/Response:** MME-initiated dedicated bearer setup
- **Release Access Bearers:** When UE goes idle (release S1-U but keep S5/S8)
- **Downlink Data Notification:** S-GW tells MME that data arrived for idle UE → triggers paging

Why GTP-C v2:

- Already proven in GPRS/3G (GTP-C v1)
- Efficient binary encoding
- Built-in TEID-based message routing
- Supports piggybacking (multiple IEs in one message)

5.5.7 SGi Interface

Property	Value
Endpoints	P-GW ↔ External Networks (Internet, IMS, Enterprise)
Plane	User Plane
Protocol Stack	IP (standard Internet protocols)
Purpose	Connects LTE network to the outside world

Characteristics:

- Equivalent to the Gi interface in 2G/3G GPRS
- Standard IP interface — no GTP or mobile-specific protocols
- P-GW performs NAT here (if needed) or routes native IPv6
- This is where the UE's IP address is “visible” to the internet
- Firewall, DPI, and value-added services typically deployed here

Connected Networks:

- Public Internet
- IMS core (for VoLTE, VoWiFi)

- Enterprise VPNs (via APN-based routing)
- Content Delivery Networks (CDNs)
- Walled garden services

5.5.8 Gx Interface

Property	Value
Endpoints	PCRF ↔ P-GW
Plane	Control Plane
Protocol Stack	Diameter / SCTP or TCP / IP
Purpose	Install, modify, remove PCC (Policy and Charging Control) rules

How it works:

1. P-GW contacts PCRF when a new IP-CAN session is established (UE attaches)
2. PCRF evaluates subscription data + network policy + application input
3. PCRF sends PCC rules to P-GW specifying per-flow QoS and charging
4. PCRF can push updated rules at any time (e.g., when VoLTE call starts)

Diameter Commands:

- **CC-Request/Answer (CCR/CCA):** P-GW reports events, PCRF responds with rules
- **Re-Auth-Request/Answer (RAR/RAA):** PCRF pushes new rules to P-GW

PCC Rule Contents:

- Service Data Flow (SDF) filter (5-tuple: src/dst IP, ports, protocol)
- QoS parameters (QCI, ARP, MBR, GBR)
- Charging key (rating group)
- Metering method (volume, time, event)
- Gate status (open/close a flow)

5.5.9 Gy Interface

Property	Value
Endpoints	P-GW ↔ OCS (Online Charging System)
Plane	Control Plane (charging signaling)
Protocol Stack	Diameter / SCTP or TCP / IP
Purpose	Real-time credit control for prepaid/online charging

How it works:

1. When a prepaid user starts a data session, P-GW requests quota from OCS
2. OCS grants units (bytes, seconds, or events)
3. P-GW meters usage against granted quota
4. When quota runs low, P-GW requests more (or session is terminated)
5. Enables real-time balance deduction and spend limits

Diameter Commands:

- **Credit-Control-Request (CCR):** P-GW requests/reports usage
 - INITIAL: Session start, request first quota
 - UPDATE: Quota running low, request more
 - TERMINATION: Session end, report final usage
- **Credit-Control-Answer (CCA):** OCS grants quota or denies

Use Cases:

- Prepaid data plans with real-time balance
- Fair usage policies (throttle after threshold)
- Sponsored data (third party pays)
- Zero-rating specific applications

5.6 IMS Overview (IP Multimedia Subsystem)

Why VoLTE Needs IMS

LTE is a **pure packet network** — there is no circuit-switched domain for voice. Without IMS:

- Voice calls would fall back to 2G/3G (CSFB — Circuit-Switched Fallback)
- CSFB adds 2–4 seconds call setup delay and drops LTE data temporarily
- No way to provide rich communication services (video calls, messaging)

IMS provides the session control layer that makes VoLTE possible.

IMS Basic Architecture

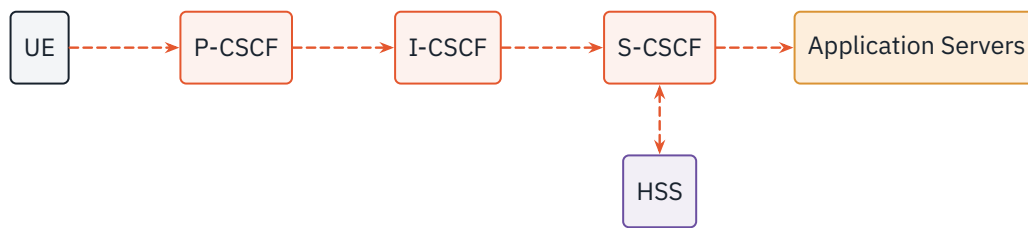


Figure 5.3 IMS signalling path from the UE to the application servers

Key IMS Elements:

Element	Role
P-CSCF	First contact point; SIP proxy in visited network; provides security (IPsec)
I-CSCF	Entry point in home network; queries HSS for S-CSCF assignment
S-CSCF	Core SIP registrar and session controller; applies service logic
Application Servers	VoLTE TAS (Telephony AS), conference, messaging services
MGCF/MGW	Media Gateway — interworks with PSTN for legacy calls

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VoLTE Call Flow (Simplified)

1. UE registers with IMS via SIP REGISTER (through P-CSCF → S-CSCF)
2. UE initiates call via SIP INVITE
3. IMS signals PCRF (via Rx interface) to request dedicated bearer
4. PCRF instructs P-GW (via Gx) to create GBR bearer (QCI=1 for voice)
5. MME/eNB establish dedicated bearer with guaranteed bitrate
6. RTP media flows over the dedicated bearer
7. Result: HD voice with guaranteed QoS, ~100 ms end-to-end

IMS Integration with LTE

- P-CSCF discovered by UE during PDN connection to IMS APN
- UE maintains two PDN connections: Internet APN + IMS APN
- Rx interface: IMS → PCRF (triggers dedicated bearer for voice/video)
- VoLTE requires: UE support + IMS core + PCRF + dedicated bearers

5.7 Control Plane vs User Plane Separation (CUPS)

The Concept

CUPS (3GPP Release 14, TS 23.214) separates the control and user plane functions of S-GW and P-GW into independent entities that can be scaled and deployed independently.

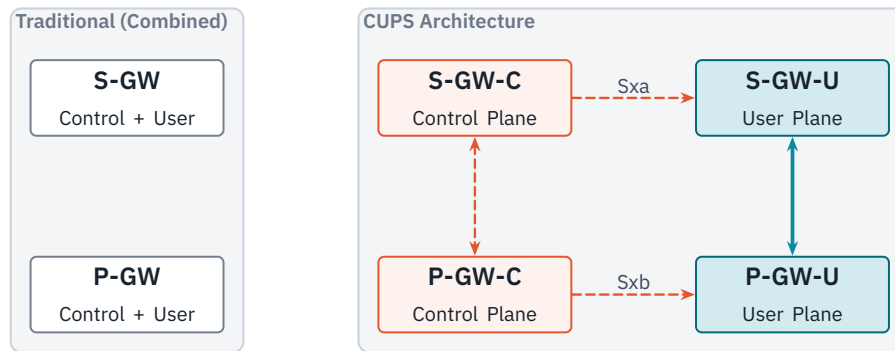


Figure 5.4 Combined gateways versus CUPS (control/user plane separation)

Why CUPS Matters

Aspect	Without CUPS	With CUPS
Scaling	Scale entire gateway (wasteful)	Scale UP and CP independently
Deployment	GW in central DC only	UP at edge, CP centralized
Latency	All traffic through central GW	UP at edge = lower latency
Cost	Expensive combined HW	Cheaper, commodity UP hardware
Evolution	Monolithic	Direct path to 5G UPF architecture

CUPS Interface: Sx (PFCP)

- Protocol: PFCP (Packet Forwarding Control Protocol)
- Sxa: SGW-C ↔ SGW-U
- Sxb: PGW-C ↔ PGW-U
- Sxc: TDF-C ↔ TDF-U
- CP installs forwarding rules (PDRs, FARs, QERs, URRs) into UP
- UP executes packet forwarding based on installed rules

CUPS as Bridge to 5G

CUPS is the architectural precursor to 5G's fully separated architecture:

- SGW-C + PGW-C → 5G SMF (Session Management Function)
- SGW-U + PGW-U → 5G UPF (User Plane Function)

- PFCP protocol reused in 5G (N4 interface = Sx evolution)

5.8 Key Differences from 3G

Aspect	3G UMTS/HSPA	4G LTE
RAN Architecture	Hierarchical: NodeB → RNC	Flat: eNodeB only
RAN Controller	RNC (centralized)	None — functions in eNodeB
Core Network	CS domain + PS domain	EPC (PS only, all-IP)
Voice	Circuit-switched (MSC)	VoLTE via IMS (packet)
Handover	Soft handover (macro diversity)	Hard handover only
Air Interface	WCDMA / HSPA	OFDMA (DL) / SC-FDMA (UL)
Channel Type	Dedicated + shared	All shared (no dedicated channels)
Scheduling	2 ms TTI (HSPA)	1 ms TTI
Bearer Concept	RAB + PDP Context	EPS Bearer (unified)
Max Bandwidth	5 MHz fixed	1.4–20 MHz flexible
Peak Rate	42 Mbps (HSPA+)	300+ Mbps (Cat 5+)
Latency (RTT)	50–100 ms	10–20 ms
Duplexing	FDD only (most)	FDD + TDD
Mobility Anchor	SGSN (control+user)	MME (control) + S-GW (user) — separated
Subscriber DB	HLR (SS7/MAP)	HSS (Diameter)
Policy Control	Limited (PCRF later addition)	Native PCRF (PCC from day one)

Architecture Comparison Diagram

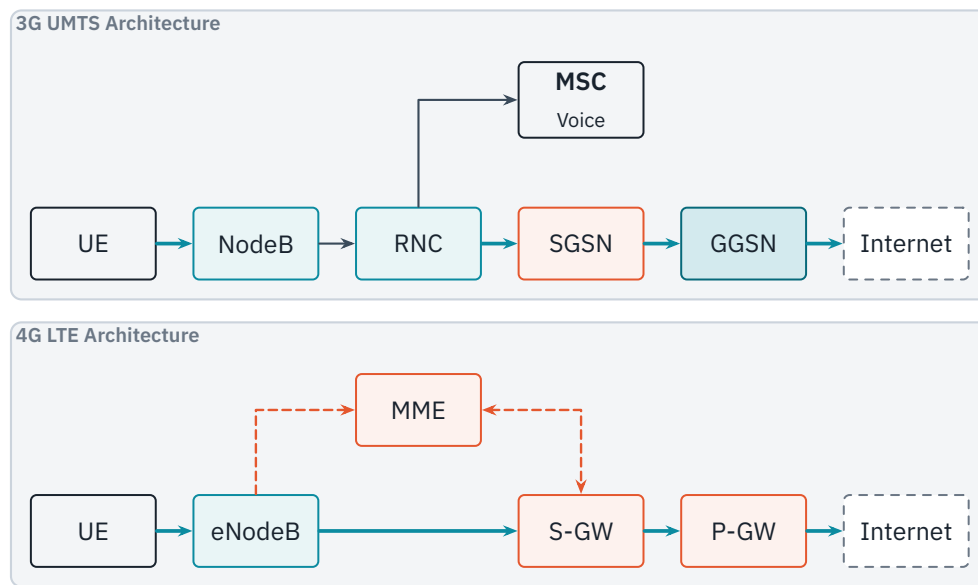


Figure 5.5 3G UMTS versus 4G LTE architecture

Key Architectural Wins

1. Elimination of RNC:

- Removed single point of failure
- Reduced latency by one hop
- Enabled distributed scheduling (faster response)
- Simplified RAN planning and scaling

2. All-IP / No Circuit Switching:

- Single protocol stack to manage
- Reduced OPEX (no MSC, MGW, VLR infrastructure)
- Statistical multiplexing for voice (VoLTE more efficient)
- Unified charging and policy for all services

3. EPS Bearer Model:

- Unified bearer from UE to P-GW (end-to-end QoS)
- Replaces fragmented 3G model (RAB + PDP Context + separate QoS at each interface)
- Default bearer always on (always connected experience)
- Dedicated bearers for specific QoS needs (VoLTE, video)

4. Separation of Control and User Planes:

- MME handles only signaling (scales with number of UEs)
- S-GW handles only data (scales with traffic volume)
- Independent evolution and scaling

5.9 Control Plane vs User Plane Paths

Control Plane Path

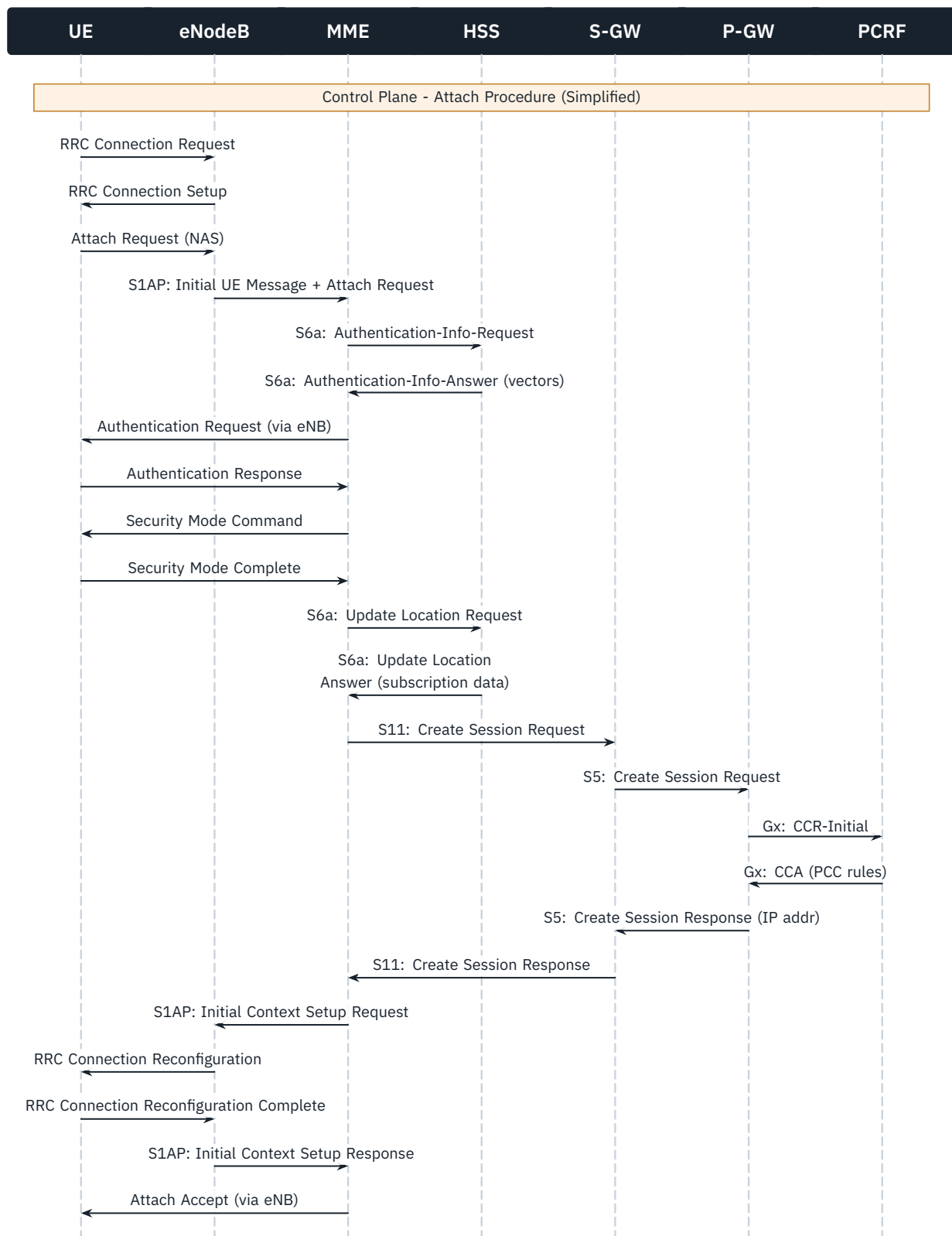


Figure 5.6 Control-plane path: the simplified attach procedure

User Plane Path



Figure 5.7 User-plane path from the UE to the Internet

User Plane Protocol Stack:

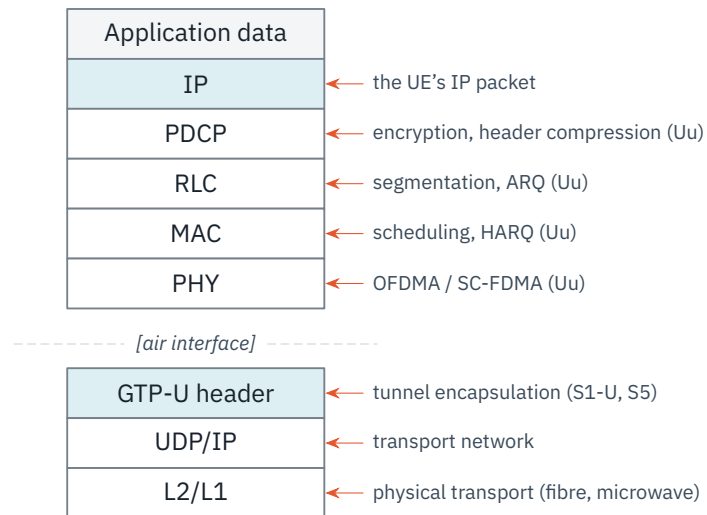


Figure 5.8 User-plane protocol stack of an LTE packet

5.10 Interface Summary Table

Interface	Endpoints	Plane	Protocol	Purpose
Uu	UE ↔ eNodeB	Control + User	LTE-Uu (OFDMA/SC-FDMA)	Air interface — radio communication
S1-MME	eNodeB ↔ MME	Control	S1AP / SCTP	RAN-Core control signaling (NAS relay, handover, bearer mgmt)
S1-U	eNodeB ↔ S-GW	User	GTP-U / UDP / IP	User data tunneling between RAN and core
X2	eNodeB ↔ eNodeB	Control + User	X2AP/SCTP + GTP-U/UDP	Inter-eNB handover signaling and data forwarding
S5/S8	S-GW ↔ P-GW	Control + User	GTP-C + GTP-U (or PMIP)	Session/bearer management and user data between gateways
S6a	MME ↔ HSS	Control	Diameter / SCTP	Authentication vectors and subscriber data retrieval
S11	MME ↔ S-GW	Control	GTP-C v2 / UDP	Bearer/session lifecycle management
SGi	P-GW ↔ Internet/IMS	User	IP (native)	External PDN connectivity — UE traffic exits here
Gx	PCRF ↔ P-GW	Control	Diameter	Policy and charging rule installation/modification
Gy	P-GW ↔ OCS	Control	Diameter	Online/real-time credit control (prepaid charging)
Rx	IMS/AF ↔ PCRF	Control	Diameter	Application-triggered QoS/policy requests
S10	MME ↔ MME	Control	GTP-C v2	Inter-MME handover (UE context transfer)
S3	MME ↔ SGSN	Control	GTP-C v2	3G/4G inter-RAT mobility

Interface	Endpoints	Plane	Protocol	Purpose
SBc	MME ↔ CBC	Control	SBc-AP / SCTP	Cell broadcast (emergency alerts, CMAS/ETWS)

Key Takeaways

1. **LTE = Flat RAN + All-IP Core** — simplicity drives performance
2. **eNodeB is the only RAN node** — absorbed all RNC functions
3. **EPC separates control (MME) from user plane (S-GW/P-GW)** — independent scaling
4. **Every interface has a clear, single purpose** — modular, replaceable
5. **GTP tunnels provide mobility transparency** — IP address never changes regardless of movement
6. **PCRF enables dynamic QoS** — different treatment per application flow
7. **IMS replaces circuit-switched voice** — VoLTE = SIP + dedicated bearer
8. **CUPS is the bridge to 5G** — same separation philosophy, evolved into 5G SBA

References

- 3GPP TS 23.401: “GPRS Enhancements for E-UTRAN Access” (EPC architecture)
- 3GPP TS 36.300: “E-UTRA and E-UTRAN; Overall Description” (RAN architecture)
- 3GPP TS 23.214: “Architecture Enhancements for CUPS”
- 3GPP TS 29.274: “GTPv2-C” (S11, S5/S8 control)
- 3GPP TS 29.281: “GTP-U” (S1-U, S5/S8 user)
- 3GPP TS 36.413: “S1AP” (S1-MME interface)
- 3GPP TS 36.423: “X2AP” (X2 interface)
- 3GPP TS 29.272: “S6a/S6d” (MME-HSS Diameter interface)
- 3GPP TS 29.212: “Gx” (PCRF-PCEF Diameter interface)

Module 5 Complete — Next: Module 6: 5G NR Architecture

E-UTRAN — LTE Radio Access Network

E-UTRAN — شبکه دسترسی رادیویی LTE



In This Module

6.1 E-UTRAN Architecture	143
6.2 eNodeB Functions	144
6.3 LTE Protocol Stack	145
6.4 Protocol Layers in Detail	146
6.5 LTE Channel Structure	151
6.6 S1 Interface	154
6.7 X2 Interface	156
6.8 RRC States and Transitions	157
Summary	160

6.1 E-UTRAN Architecture

Why No RNC? The Flat Architecture Decision

In UMTS (3G), the Radio Network Controller (RNC) sat between the NodeBs and the Core Network, centralizing radio resource management, handover decisions, and protocol termination. LTE **eliminated the RNC entirely**, creating a flat, single-node radio access network.

Reasons for removing the RNC:

Concern	3G (with RNC)	LTE (flat, no RNC)
Latency	Extra hop through RNC adds ~5–10 ms	Direct eNB↔EPC path, sub-5 ms user-plane latency
Single point of failure	RNC failure affects all connected NodeBs	Each eNB is independent
Scalability	RNC capacity limits number of NodeBs	No bottleneck; eNBs scale independently
Complexity	Three-node RAN (NodeB→RNC→CN)	Two-node RAN (eNB→EPC)
Cost	Expensive, proprietary RNC hardware	Distributed intelligence reduces CAPEX
Handover speed	RNC-mediated = slower	Direct eNB-to-eNB via X2 = faster

Key architectural principle: All radio-related intelligence is pushed into the **eNodeB (eNB)**, which communicates directly with the Evolved Packet Core (EPC) via the S1 interface and with neighboring eNBs via the X2 interface.

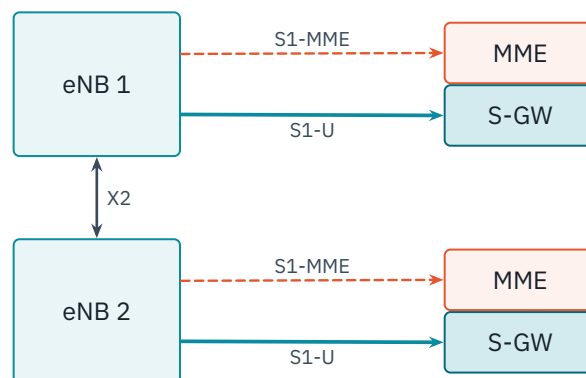


Figure 6.1 Each eNB connects to the MME over S1-MME and to the S-GW over S1-U

6.2 eNodeB Functions

The eNodeB is the **sole node** in E-UTRAN. It performs all functions previously split between NodeB and RNC:

■ Radio Resource Management (RRM)

- **Radio Bearer Control:** Establishment, maintenance, and release of radio bearers
- **Radio Admission Control (RAC):** Accept/reject new radio bearer requests based on interference and load
- **Connection Mobility Control:** Handover decisions based on UE measurements
- **Dynamic Resource Allocation (Scheduling):** Assigns resource blocks to UEs every TTI (1 ms)

■ Scheduling

- Operates every **1 ms TTI** (Transmission Time Interval)
- Frequency-domain and time-domain scheduling
- Algorithms: Round Robin, Proportional Fair, Max Throughput
- Uses **CQI** (Channel Quality Indicator) reports from UEs
- Controls both DL and UL resource allocation

■ HARQ (Hybrid Automatic Repeat Request)

- Stop-and-wait HARQ with **8 parallel processes**
- Synchronous in UL, asynchronous in DL
- Soft combining: Chase Combining or Incremental Redundancy
- Round-trip time: 8 ms (4 ms each way)

■ Channel Encoding

- **Turbo coding** (rate 1/3) for data channels
- **Convolutional coding** for control channels
- Rate matching and HARQ buffer management
- CRC attachment for error detection

■ Other Functions

- IP header compression (ROHC)
- Ciphering and integrity protection
- Paging message distribution
- Broadcast information (SIBs, MIBs)
- Measurement configuration and reporting

6.3 LTE Protocol Stack

Protocol Stack Diagram

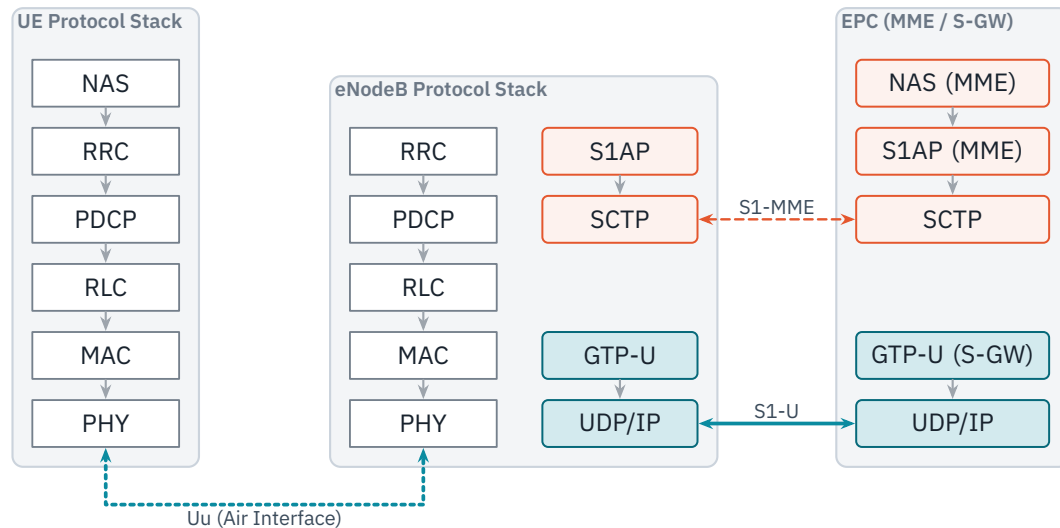


Figure 6.2 LTE protocol stacks of the UE, the eNodeB and the EPC

Control Plane Stack

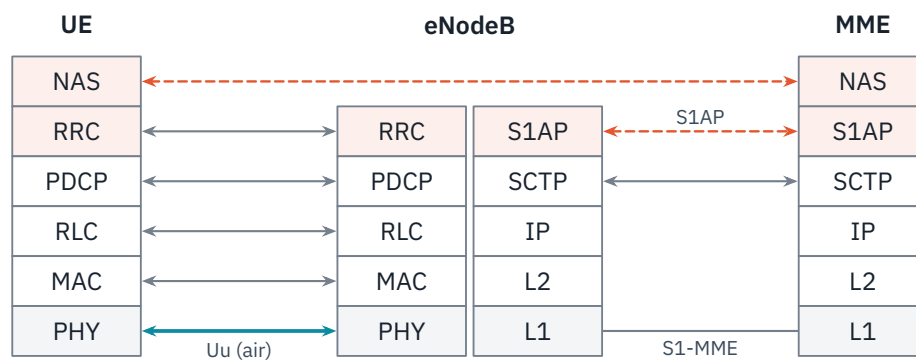


Figure 6.3 LTE control-plane protocol stack

User Plane Stack

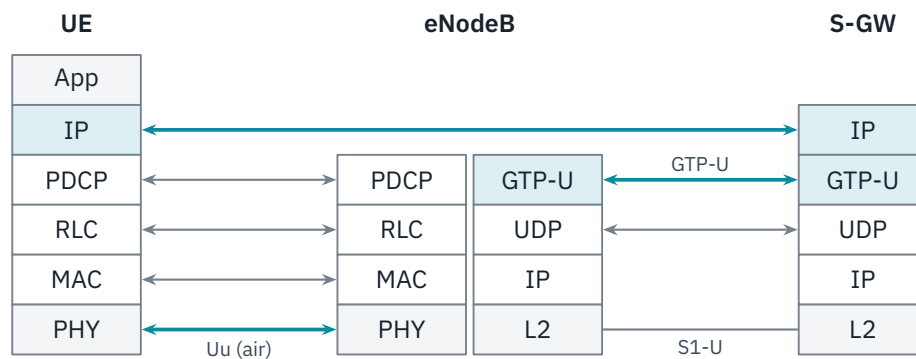


Figure 6.4 LTE user-plane protocol stack

6.4 Protocol Layers in Detail

6.4.1 Physical Layer (PHY)

Downlink: OFDMA (Orthogonal Frequency Division Multiple Access)

- Divides bandwidth into many narrow **subcarriers** (15 kHz spacing)
- Each subcarrier is orthogonal → no inter-carrier interference
- Robust against multipath fading (cyclic prefix absorbs delay spread)
- Enables flexible frequency-domain scheduling

Resource Grid:

- Resource Element (RE):** 1 subcarrier × 1 OFDM symbol = smallest allocatable unit
- Resource Block (RB):** 12 subcarriers × 7 OFDM symbols (1 slot = 0.5 ms) = **84 REs**
- Resource Block Pair:** 2 RBs = 1 subframe (1 ms TTI)

Bandwidth	Number of RBs	Subcarriers
1.4 MHz	6	72
3 MHz	15	180
5 MHz	25	300
10 MHz	50	600
15 MHz	75	900
20 MHz	100	1200

Frame Structure (FDD):

- 1 radio frame = 10 ms = 10 subframes
- 1 subframe = 1 ms = 2 slots

- 1 slot = 0.5 ms = 7 OFDM symbols (normal CP)

■ Uplink: SC-FDMA (Single Carrier FDMA)

- Also called DFT-spread OFDM (DFT-s-OFDM)
- Lower **PAPR** (Peak-to-Average Power Ratio) than OFDMA → better for battery-powered UEs
- Single-carrier property achieved by DFT precoding before IFFT
- Same resource grid structure as DL (12 subcarriers per RB)
- **Constraint:** UL allocation must be contiguous in frequency

■ Key PHY Procedures

- Cell search (PSS/SSS synchronization)
- Channel estimation (reference signals: CRS, DM-RS, SRS)
- MIMO: up to 4×4 in DL, 1×2 or 2×2 in UL (Rel-8/10)
- Link adaptation via CQI/PMI/RI feedback
- Power control (open-loop and closed-loop)

6.4.2 MAC Layer (Medium Access Control)

The MAC layer sits between RLC and PHY, responsible for **real-time scheduling** and data multiplexing.

■ Key Functions

1. Scheduling (DL and UL)

- Operates every 1 ms TTI
- Assigns RBs based on CQI, buffer status, QoS priority, fairness
- DL: eNB decides; UL: eNB grants resources to UEs

2. Multiplexing/Demultiplexing

- Multiplexes data from multiple logical channels into one transport block
- Uses Logical Channel Prioritization (LCP) in UL

3. HARQ Entity

- 8 parallel stop-and-wait processes
- DL: Asynchronous adaptive HARQ
- UL: Synchronous HARQ (fixed timing)
- Generates ACK/NACK; soft combining at receiver

4. Random Access Procedure

- Contention-based (4-step) or contention-free (2-step)
- Used for initial access, handover, UL synchronization recovery

5. BSR (Buffer Status Report)

- UE informs eNB about pending UL data
- Triggers UL scheduling grants

6. PHR (Power Headroom Report)

- UE reports remaining power margin
- Helps eNB with UL scheduling decisions

■ Channel Mapping at MAC

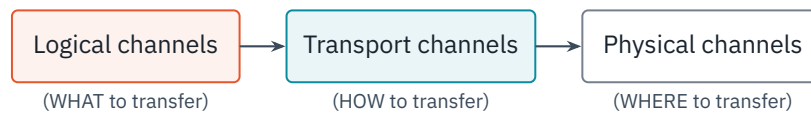


Figure 6.5 Three levels of LTE channels

6.4.3 RLC Layer (Radio Link Control)

RLC provides segmentation, reassembly, and (optionally) ARQ.

■ Three Modes

Mode	Full Name	ARQ	Segmentation	Use Case
TM	Transparent Mode	No	No	BCCH, PCCH, CCCH (small, fixed-size messages)
UM	Unacknowledged Mode	No	Yes	VoIP, real-time video (delay-sensitive)
AM	Acknowledged Mode	Yes	Yes	TCP traffic, signaling (reliability needed)

■ Key Functions

- **Segmentation and Reassembly:** Splits PDCP PDUs into RLC-sized segments fitting transport block size
- **Concatenation:** Packs multiple small SDUs into one PDU
- **ARQ (AM only):** Retransmits lost RLC PDUs based on STATUS reports
 - Polling mechanism: RLC sender requests STATUS from receiver
 - Operates independently from HARQ (RLC ARQ catches what HARQ misses)
- **Reordering (UM/AM):** Delivers SDUs in order to upper layers
- **Duplicate Detection (AM):** Discards duplicate PDUs

■ RLC vs. HARQ

- HARQ: Fast (8 ms RTT), limited retransmissions (typically 4)
- RLC ARQ: Slower, but catches residual errors after HARQ exhaustion
- Combined: achieves target BLER of 10^{-6} for AM bearers

6.4.4 PDCP Layer (Packet Data Convergence Protocol)

PDCP sits above RLC, providing security and efficiency functions.

■ Key Functions

1. Header Compression (User Plane only)

- Uses **ROHC** (Robust Header Compression, RFC 3095/4995)
- Compresses IP/UDP/RTP headers from ~40–60 bytes → 1–4 bytes
- Critical for VoIP: 40-byte header on 32-byte payload → ~56% overhead without ROHC
- Multiple ROHC profiles for different protocol stacks

2. Ciphering (User Plane + Control Plane)

- Encrypts all user data and signaling
- Algorithms: EEA0 (null), 128-EEA1 (SNOW 3G), 128-EEA2 (AES-CTR), 128-EEA3 (ZUC)
- Uses COUNT (sequence number + HFN) + BEARER + DIRECTION as input

3. Integrity Protection (Control Plane only)

- Protects RRC signaling from tampering
- Algorithms: EIA1 (SNOW 3G), EIA2 (AES-CMAC), EIA3 (ZUC)
- Generates 32-bit MAC-I appended to each RRC message
- **Not applied to user plane in LTE** (added in 5G NR for user plane)

4. Sequence Numbering and Reordering

- Assigns PDCP SN to each SDU
- Reorders out-of-sequence PDUs (important during handover)
- Detects duplicates

5. Handover Support

- In-sequence delivery and duplicate elimination during handover
- Status reporting: tells source eNB which PDUs were successfully delivered
- Enables lossless handover for AM bearers

■ PDCP PDU Format

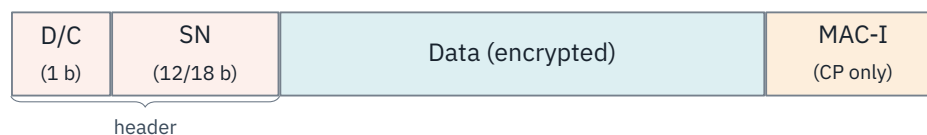


Figure 6.6 PDCP PDU format

6.4.5 RRC Layer (Radio Resource Control)

RRC is the **control plane protocol** between UE and eNodeB, managing the radio connection.

■ Key Functions

1. Connection Management

- RRC Connection Setup / Reconfiguration / Release
- Security activation (ciphering + integrity)
- SRB (Signaling Radio Bearer) establishment

2. Measurement Configuration and Reporting

- Configures UE to measure serving and neighbor cells
- Events: A1–A6 (intra/inter-frequency), B1–B2 (inter-RAT)
- Reporting: periodic or event-triggered
- Quantities: RSRP, RSRQ, SINR

3. Handover Control

- eNB makes handover decision based on measurement reports
- Sends `RRCConnectionReconfiguration` with mobility control info
- UE performs handover, sends `RRCConnectionReconfigurationComplete`

4. System Information Broadcasting

- **MIB**: Transmitted on PBCH every 40 ms (bandwidth, SFN, PHICH config)
- **SIB1**: Cell access info, scheduling of other SIBs (80 ms period)
- **SIB2**: Common channel config (RACH, paging, UL power control)
- **SIB3–SIB8**: Cell reselection, inter-frequency/inter-RAT info
- **SIB9–SIB19**: Additional info (home eNB, MBMS, etc.)

5. Paging

- Notifies UEs in IDLE of incoming calls/data
- DRX-based: UE wakes at specific paging occasions

6. NAS Message Transfer

- Transparently carries NAS messages between UE and MME

■ Signaling Radio Bearers (SRBs)

SRB	Purpose	RLC Mode
SRB0	RRC messages on CCCH (before security)	TM
SRB1	RRC messages + piggybacked NAS (after setup)	AM
SRB2	NAS messages (lower priority than SRB1)	AM

6.5 LTE Channel Structure

Channel Types Overview

LTE defines three categories of channels, each at a different layer of abstraction:

Category	Layer	Defined By	Question Answered
Logical Channels	MAC (upper)	What type of data	What is being transferred?
Transport Channels	MAC (lower)	How data is transferred	How is it characterized?
Physical Channels	PHY	Where in the resource grid	Where are the bits placed?

Logical Channels

Control Channels

Channel	Name	Direction	Purpose
BCCH	Broadcast Control Channel	DL	System information (MIB, SIBs)
PCCH	Paging Control Channel	DL	Paging messages for IDLE UEs
CCCH	Common Control Channel	DL/UL	Messages before RRC connection (Msg3/Msg4)
DCCH	Dedicated Control Channel	DL/UL	Dedicated signaling (RRC after connection)
MCCH	Multicast Control Channel	DL	MBMS control information

Traffic Channels

Channel	Name	Direction	Purpose
DTCH	Dedicated Traffic Channel	DL/UL	User data (IP packets)
MTCH	Multicast Traffic Channel	DL	MBMS user data

■ Transport Channels

Channel	Name	Direction	Purpose
BCH	Broadcast Channel	DL	Carries MIB; fixed format, 40 ms period
PCH	Paging Channel	DL	Carries paging; supports DRX
DL-SCH	Downlink Shared Channel	DL	Main DL data; supports HARQ, dynamic scheduling
UL-SCH	Uplink Shared Channel	UL	Main UL data; supports HARQ, dynamic scheduling
RACH	Random Access Channel	UL	Random access preambles
MCH	Multicast Channel	DL	MBMS data; fixed scheduling

■ Physical Channels

■ Downlink Physical Channels

Channel	Name	Purpose
PDSCH	Physical DL Shared Channel	User data + system info (SIBs, paging)
PDCCH	Physical DL Control Channel	DCI (scheduling grants, power control)
PBCH	Physical Broadcast Channel	MIB transmission
PMCH	Physical Multicast Channel	MBMS data
PCFICH	Physical CFI Channel	Indicates # of OFDM symbols for PDCCH
PHICH	Physical HARQ Indicator Channel	UL HARQ ACK/NACK

■ Uplink Physical Channels

Channel	Name	Purpose
PUSCH	Physical UL Shared Channel	UL user data + control info
PUCCH	Physical UL Control Channel	CQI, HARQ ACK/NACK, SR
PRACH	Physical Random Access Channel	Random access preambles

Channel Mapping

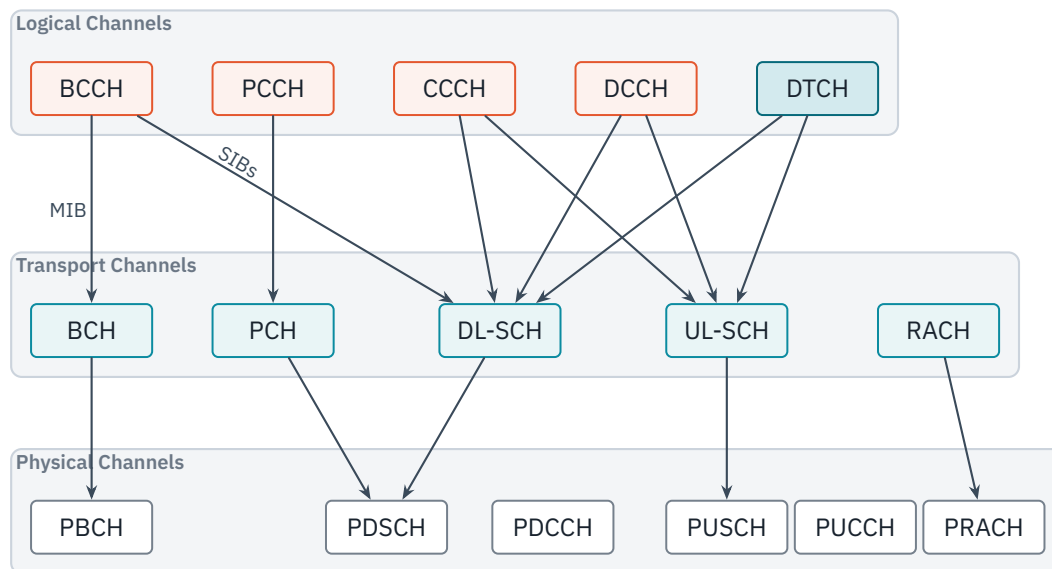


Figure 6.7 Mapping of logical, transport and physical channels

Complete Channel Mapping Table

Logical Channel	Transport Channel	Physical Channel	Direction
BCCH (MIB)	BCH	PBCH	DL
BCCH (SIBs)	DL-SCH	PDSCH	DL
PCCH	PCH	PDSCH	DL
CCCH	DL-SCH	PDSCH	DL
CCCH	UL-SCH	PUSCH	UL
DCCH	DL-SCH	PDSCH	DL
DCCH	UL-SCH	PUSCH	UL
DTCH	DL-SCH	PDSCH	DL
DTCH	UL-SCH	PUSCH	UL
— (HARQ feedback)	—	PHICH	DL
— (DCI/grants)	—	PDCCH	DL
— (CFI)	—	PCFICH	DL
— (CQI/SR/ACK)	—	PUCCH	UL
— (RA preamble)	RACH	PRACH	UL

6.6 S1 Interface

The S1 interface connects the eNodeB to the EPC. It is split into:

- **S1-MME (S1-C):** Control plane → eNB ↔ MME
- **S1-U:** User plane → eNB ↔ S-GW

■ S1-MME Protocol Stack

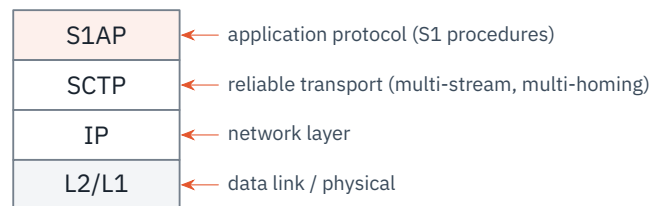


Figure 6.8 S1-MME protocol stack

Why SCTP (not TCP)?

- Multi-streaming: avoids head-of-line blocking
- Multi-homing: failover between IP addresses
- Message-oriented (preserves boundaries)

S1AP Procedures

UE-Associated Procedures (per-UE context)

Procedure	Direction	Purpose
Initial UE Message	eNB → MME	Carries first NAS message (Attach, Service Request)
Downlink NAS Transport	MME → eNB	Delivers NAS message to UE
Uplink NAS Transport	eNB → MME	Forwards NAS message from UE
Initial Context Setup	MME → eNB	Establishes security context + default bearer
UE Context Release	MME ↔ eNB	Releases UE context (Request/Command/Complete)
Handover Preparation	Source eNB → MME	S1-based handover when X2 unavailable
Handover Resource Allocation	MME → Target eNB	Allocates resources at target
Path Switch Request	Target eNB → MME	Updates S-GW path after X2 handover
E-RAB Setup/Modify/Release	MME → eNB	Bearer management

Non-UE-Associated Procedures (global)

Procedure	Purpose
S1 Setup	eNB registers with MME (TAC, PLMN, capacity)
Reset	Error recovery (partial or full)
Paging	MME sends paging to eNBs in tracking area
eNB/MME Configuration Update	Capacity/configuration changes
Overload Start/Stop	MME signals congestion
Error Indication	Report protocol errors

S1-U Protocol Stack

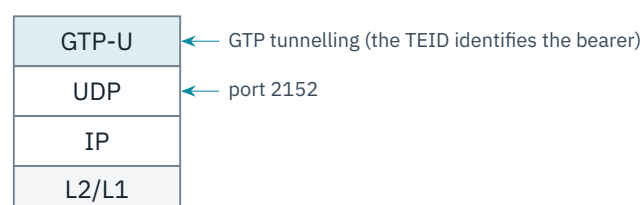


Figure 6.9 S1-U protocol stack

- Each E-RAB has a **GTP tunnel** identified by TEID (Tunnel Endpoint ID)
- One-to-one mapping: EPS bearer ↔ S1 GTP tunnel ↔ radio bearer

6.7 X2 Interface

The X2 interface connects **neighboring eNodeBs** directly for:

- Fast handover (no MME involvement for intra-LTE mobility)
- Inter-cell load management
- Inter-cell interference coordination (ICIC)

X2 Protocol Stack

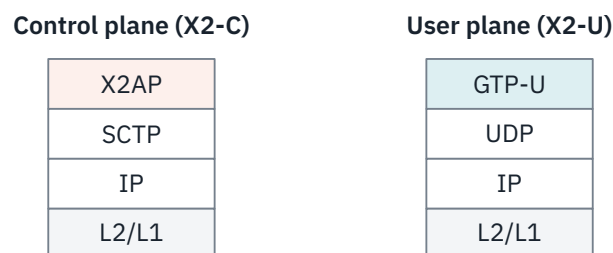


Figure 6.10 X2 protocol stacks

X2AP Procedures

■ Mobility Management (Handover)

X2 Handover Sequence:

1. **Source eNB** sends **Handover Request** → Target eNB (includes UE context, bearer info)
2. **Target eNB** performs admission control → sends **Handover Request Acknowledge**
3. **Source eNB** sends **RRConnectionReconfiguration** to UE (with target cell info)
4. **UE** synchronizes to target cell, sends **RRConnectionReconfigurationComplete**
5. **Target eNB** sends **Path Switch Request** to MME (updates core network path)
6. **Source eNB** receives **UE Context Release** → releases resources

During handover: Data forwarded source→target via X2-U GTP tunnel (avoids data loss).

■ Load Management

Procedure	Purpose
Resource Status Reporting	Request/report load information (PRB usage, composite load)
Load Indication	Inform neighbor about UL interference (HII, OI)
Mobility Change Request	Negotiate handover parameters (CIO adjustment)

■ Inter-Cell Interference Coordination (ICIC)

- **Load Information message** carries:
 - **HII (High Interference Indication)**: UL RBs where cell plans to schedule cell-edge UEs
 - **OI (Overload Indicator)**: UL RBs experiencing high interference
 - **RNTP (Relative Narrowband TX Power)**: DL RBs where cell will exceed power threshold

■ X2 Setup and Configuration

- X2 link established via **X2 Setup procedure** (or via MME using S1 ENB Configuration Transfer)
- Automatic Neighbor Relation (ANR): eNB discovers neighbors and establishes X2
- Multiple eNBs can connect forming a mesh topology

6.8 RRC States and Transitions

■ RRC State Machine

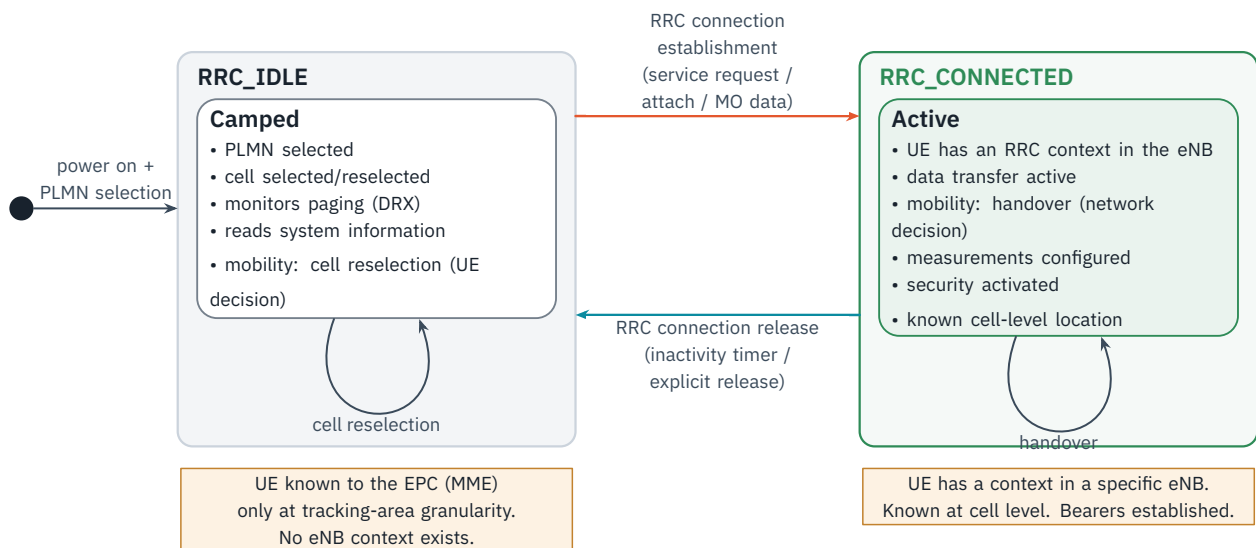


Figure 6.11 LTE RRC state machine

■ RRC_IDLE State

Characteristic	Description
Context	No eNB context; registered with MME only
Location	Known at Tracking Area level
Mobility	Cell reselection (UE-controlled, based on SIBs)
Data	No data transfer possible
Paging	UE monitors paging channel with DRX cycle
System Info	UE acquires MIB and relevant SIBs
Power	Low power consumption (long DRX sleep)

■ RRC_CONNECTED State

Characteristic	Description
Context	Full UE context in serving eNB
Location	Known at cell level
Mobility	Handover (network-controlled, measurement-based)
Data	Active data transfer (DL + UL)
Measurements	Configured by eNB (events A1–A6, B1–B2)
DRX	Connected-mode DRX possible (shorter cycles)
Security	AS ciphering + integrity active

■ State Transitions

■ IDLE → CONNECTED (Connection Establishment)

1. UE sends **RRC Connection Request** (on CCCH via SRB0)
2. eNB responds with **RRC Connection Setup** (configures SRB1)
3. UE sends **RRC Connection Setup Complete** (carries initial NAS message)
4. eNB forwards NAS to MME via S1AP **Initial UE Message**
5. MME initiates **Initial Context Setup** (security + bearer establishment)

■ CONNECTED → IDLE (Connection Release)

- MME triggers via S1AP **UE Context Release Command**
- eNB sends **RRC Connection Release** to UE
- UE deletes AS context, returns to idle
- Triggers: inactivity timer, explicit NAS detach, radio link failure recovery failure

5G Extension: RRC_INACTIVE State

In 5G NR (and late LTE Rel-15+), a third state is introduced:

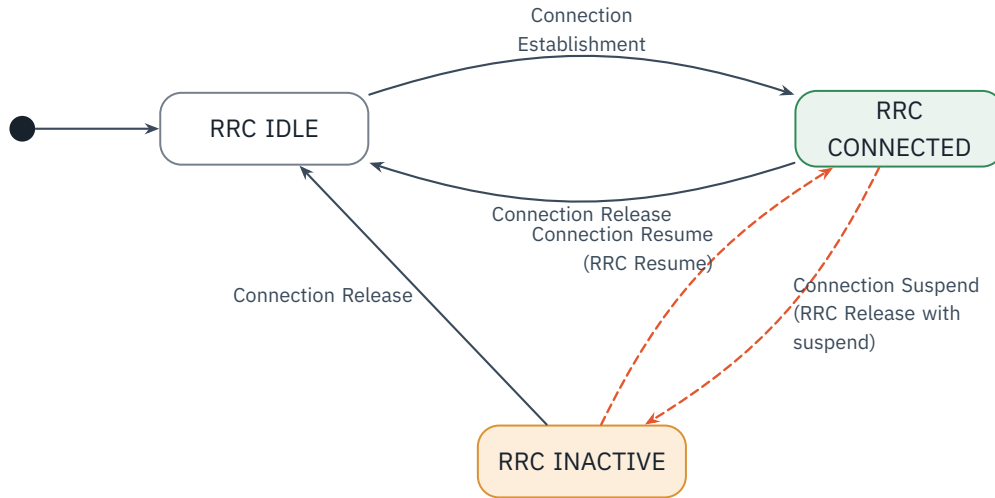


Figure 6.12 5G NR RRC states with RRC_INACTIVE

State	UE Context	Location Granularity	Mobility	CN Connection
IDLE	None in RAN	Tracking Area	Cell reselection	None
INACTIVE	Stored in gNB (suspended)	RAN Notification Area	Cell reselection + RAN paging	Maintained with CN
CONNECTED	Active in gNB	Cell level	Handover	Active

Benefits of INACTIVE state:

- Faster resumption than IDLE→CONNECTED (skip NAS procedures)
- Power saving (no connected-mode signaling)
- Reduced signaling load on core network
- AS context preserved → faster resume with single RRC Resume message
- Ideal for IoT devices with intermittent small data transmissions

Summary

Component	Key Takeaway
Architecture	Flat (no RNC); eNB handles all RAN functions
eNodeB	Scheduling, RRM, HARQ, handover decisions, encoding
PHY	OFDMA (DL) / SC-FDMA (UL); resource blocks as basic unit
MAC	1 ms scheduling, HARQ, multiplexing, random access
RLC	TM/UM/AM; segmentation + ARQ for reliability
PDCP	ROHC compression, ciphering, integrity (CP), reordering
RRC	Connection/mobility management, measurement, system info
Channels	Logical→Transport→Physical mapping; shared channels dominate
S1	eNB↔EPC; S1AP (SCTP) for control, GTP-U for user plane
X2	eNB↔eNB; fast handover, load mgmt, interference coordination
RRC States	IDLE (low power) ↔ CONNECTED (active); INACTIVE in 5G

Module 6 — E-UTRAN (LTE Radio Access Network) — Advanced Communication Networks



1	Why Scheduling Matters	162
2	The LTE Resource Grid	162
3	Scheduling Granularity	164
4	Downlink Scheduling	164
5	Uplink Scheduling	166
6	Channel Quality Indicator (CQI)	167
7	Link Adaptation	168
8	HARQ (Hybrid Automatic Repeat Request)	170
9	Scheduling Algorithms	172
10	Engineering Trade-offs	173
11	Numerical Example	174
	Key Takeaways	176
	References	177

1 Why Scheduling Matters

In LTE, the radio channel is a **shared resource**. Multiple users compete for limited time-frequency resources on the same carrier. The scheduler is the central intelligence at the eNodeB that decides:

- **Who** gets to transmit/receive
- **When** (which subframe)
- **Where** (which resource blocks)
- **How** (which modulation and coding scheme)

Without scheduling, users would collide, resources would be wasted, and QoS guarantees would be impossible. The scheduler must balance:

- **Throughput maximization** — use resources efficiently
- **Fairness** — don't starve cell-edge users
- **Latency** — meet delay budgets for real-time services
- **QoS** — honor GBR (Guaranteed Bit Rate) bearers

2 The LTE Resource Grid

The resource grid is the fundamental 2D structure that organizes all LTE transmissions in time and frequency.

Time Domain Hierarchy

Structure	Duration	Composition
Radio Frame	10 ms	10 subframes
Subframe	1 ms	2 slots
Slot	0.5 ms	7 OFDM symbols (normal CP)
OFDM Symbol	~71.4 μ s	1 symbol period + cyclic prefix

Frequency Domain Hierarchy

Structure	Bandwidth	Composition
Subcarrier	15 kHz	Smallest frequency unit
Resource Block (RB)	180 kHz	12 subcarriers
System Bandwidth	1.4–20 MHz	6–100 RBs

Key Definitions

- **Resource Element (RE):** 1 subcarrier $\times$ 1 OFDM symbol — the smallest allocatable unit
- **Resource Block (RB):** 12 subcarriers $\times$ 7 symbols = **84 REs** (one slot)
- **Resource Block Pair:** 12 subcarriers $\times$ 14 symbols = **168 REs** (one subframe / TTI)

Resource Grid Visualization

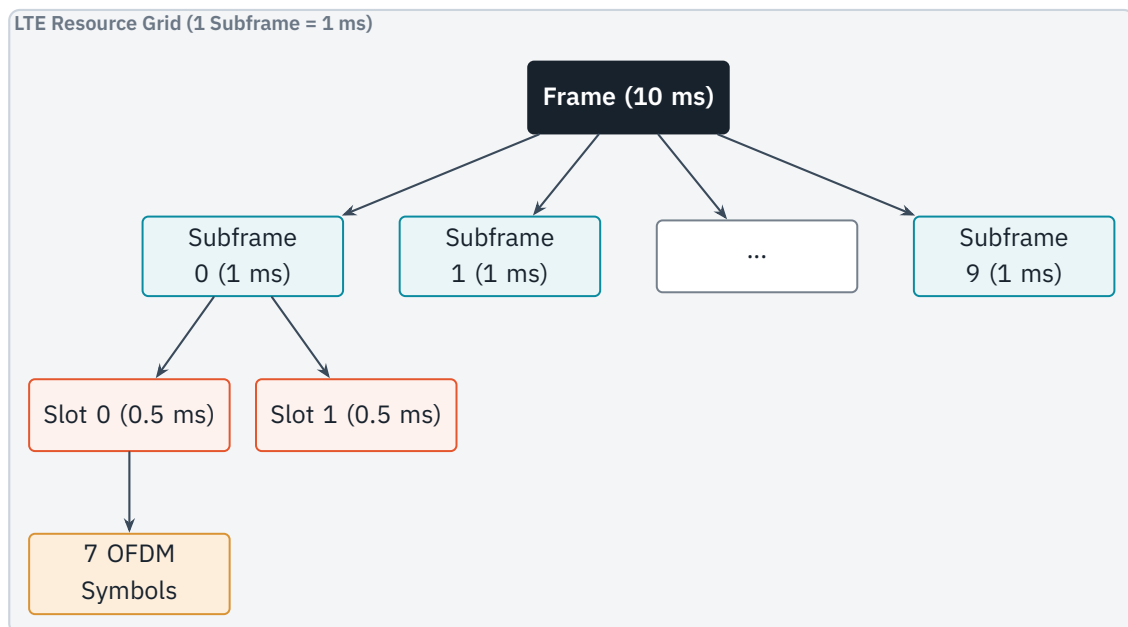


Figure 7.1 LTE time structure: frame, subframe, slot and OFDM symbols

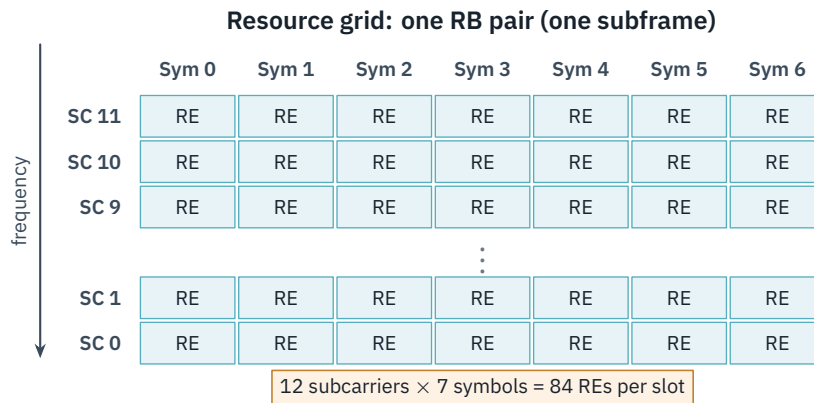


Figure 7.2 Resource elements of one resource block (12 subcarriers × 7 symbols)

3 Scheduling Granularity

Parameter	Value
Time granularity	1 subframe = 1 ms (Transmission Time Interval, TTI)
Frequency granularity	1 Resource Block = 180 kHz (12 subcarriers)
Minimum allocation	1 RB pair per TTI per user
Maximum allocation	All RBs in a TTI to one user
Scheduling decision rate	1000 decisions/second

The scheduler makes a fresh decision every TTI (1 ms), assigning RBs to UEs based on channel conditions, buffer status, QoS requirements, and fairness metrics.

4 Downlink Scheduling

Overview

The eNodeB scheduler controls all downlink resource assignments. It informs UEs of their allocations via **DCI (Downlink Control Information)** messages carried on the **PDCCH (Physical Downlink Control Channel)**.

DCI Contents

A DCI message tells the UE:

- Which RBs are allocated (resource assignment)
- Which MCS to use for decoding
- HARQ process number and redundancy version

- Power control commands
- MIMO-related parameters (precoding, layers)

Resource Allocation Types

Type	Method	Flexibility	Overhead
Type 0	Bitmap of RB Groups (RBGs)	Distributed, non-contiguous	Medium
Type 1	Subset + shift + bitmap	Distributed within subset	Higher
Type 2	Start RB + length (RIV)	Contiguous only	Lowest

RBG size depends on system bandwidth:

System BW (RBs)	RBG Size (RBs)
6	1
15	2
25	2
50	3
75	4
100	4

MCS Selection

The scheduler selects MCS based on:

1. UE reports **CQI** (Channel Quality Indicator)
2. Scheduler maps CQI → appropriate MCS index
3. Applies outer-loop adjustments based on ACK/NACK history
4. Encodes DCI with selected MCS for the UE to decode PDSCH

5 Uplink Scheduling

■ Key Differences from Downlink

Unlike downlink, uplink scheduling must account for:

- **SC-FDMA constraint:** UEs must be allocated **contiguous** RBs (single-carrier property)
- **UE power limitations:** Distance from eNodeB affects achievable rate
- **Buffer awareness:** eNodeB doesn't directly know UE buffer status

■ Scheduling Grant

The eNodeB sends an **uplink grant** via DCI (Format 0/4) on PDCCH, containing:

- RB allocation (contiguous)
- MCS index
- Power control command
- HARQ information

■ Buffer Status Reports (BSR)

UEs inform the eNodeB about queued data via BSR:

BSR Type	Trigger
Regular BSR	New data arrives in a logical channel group with higher priority than currently reported
Periodic BSR	Timer-based periodic reporting
Padding BSR	Sent when UL grant has spare capacity

BSR reports data volume per **Logical Channel Group (LCG)**, allowing priority-aware scheduling.

■ Power Headroom Report (PHR)

PHR tells the eNodeB how much **additional transmit power** the UE has available:

$$PH = P_{MAX} - P_{PUSCH}$$

- If $PH > 0$: UE can support higher MCS or more RBs
- If $PH \leq 0$: UE is power-limited, scheduler should reduce allocation

SC-FDMA Contiguous Allocation

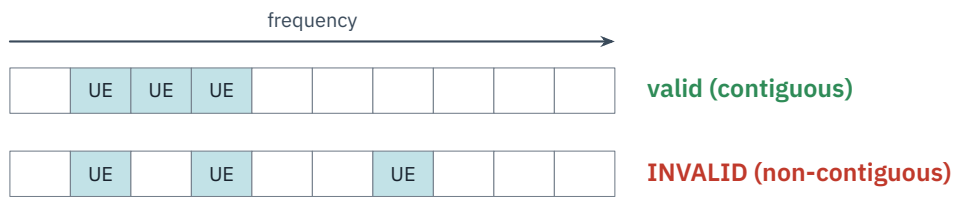


Figure 7.3 SC-FDMA uplink allocations must be contiguous in frequency

6 Channel Quality Indicator (CQI)

What CQI Represents

CQI is a **4-bit index** (1–15) reported by the UE indicating the highest MCS that can be received with a transport block error rate (BLER) $\leq 10\%$.

CQI = 0 means “out of range” (channel too poor for any transmission).

CQI Table (3GPP TS 36.213, Table 7.2.3-1)

CQI Index	Modulation	Code Rate $\times 1024$	Spectral Efficiency (bps/Hz)
1	QPSK	78	0.1523
2	QPSK	120	0.2344
3	QPSK	193	0.3770
4	QPSK	308	0.6016
5	QPSK	449	0.8770
6	QPSK	602	1.1758
7	16QAM	378	1.4766
8	16QAM	490	1.9141
9	16QAM	616	2.4063
10	64QAM	466	2.7305
11	64QAM	567	3.3223
12	64QAM	666	3.9023
13	64QAM	772	4.5234
14	64QAM	873	5.1152
15	64QAM	948	5.5547

Wideband vs Subband CQI

Reporting Mode	Description	Use Case
Wideband CQI	Single CQI for entire bandwidth	Low overhead, coarse adaptation
Subband CQI	CQI per group of RBs	Frequency-selective scheduling, higher gain
UE-selected subband	UE reports best-M subbands	Compromise between overhead and gain

Subband CQI enables **frequency-selective scheduling**, where the scheduler assigns each UE the RBs where their channel is best.

7 Link Adaptation

Adaptive Modulation and Coding (AMC)

AMC dynamically adjusts the modulation order and code rate to match instantaneous channel conditions:

- **Good channel** → Higher-order modulation + higher code rate → More bits/RE
- **Poor channel** → Lower-order modulation + lower code rate → Fewer bits/RE but robust

Modulation Schemes

Modulation	Bits per Symbol	Required SNR (approx.)	Usage
QPSK	2	< 10 dB	Cell-edge users
16QAM	4	10–17 dB	Mid-range users
64QAM	6	17–25 dB	Near-cell users
256QAM	8	> 25 dB	Cat-11+ (Rel-12), best conditions

■ MCS Index Table (3GPP TS 36.213, Table 7.1.7.1-1)

MCS Index	Modulation Order	TBS Index	Notes
0	2 (QPSK)	0	Lowest rate
1	2 (QPSK)	1	
2	2 (QPSK)	2	
3	2 (QPSK)	3	
4	2 (QPSK)	4	
5	2 (QPSK)	5	
6	2 (QPSK)	6	
7	2 (QPSK)	7	
8	2 (QPSK)	8	
9	2 (QPSK)	9	
10	4 (16QAM)	9	Modulation transition
11	4 (16QAM)	10	
12	4 (16QAM)	11	
13	4 (16QAM)	12	
14	4 (16QAM)	13	
15	4 (16QAM)	14	
16	4 (16QAM)	15	
17	6 (64QAM)	15	Modulation transition
18	6 (64QAM)	16	
19	6 (64QAM)	17	
20	6 (64QAM)	18	
21	6 (64QAM)	19	
22	6 (64QAM)	20	
23	6 (64QAM)	21	
24	6 (64QAM)	22	
25	6 (64QAM)	23	
26	6 (64QAM)	24	
27	6 (64QAM)	25	
28	6 (64QAM)	26	Highest rate
29	2 (QPSK)	—	Reserved (retx)
30	4 (16QAM)	—	Reserved (retx)
31	6 (64QAM)	—	Reserved (retx)

Outer Loop Link Adaptation (OLLA)

The inner loop (CQI → MCS) can be inaccurate due to:

- CQI reporting delay
- UE measurement errors
- Interference variability

OLLA adjusts the effective CQI with an offset:

$$MCS_{effective} = f(CQI + \Delta_{OLLA})$$

The offset is updated based on HARQ feedback:

- **ACK received:** $\Delta_{OLLA} += \text{step_up}$ (e.g., +0.1 dB) → try higher MCS
- **NACK received:** $\Delta_{OLLA} -= \text{step_down}$ (e.g., -1.0 dB) → fall back to lower MCS

Target: maintain **10% BLER** at first transmission (ratio step_down/step_up ≈ 9:1).

8 HARQ (Hybrid Automatic Repeat Request)

Overview

HARQ combines **FEC (Forward Error Correction)** with **ARQ (retransmission)** for reliable delivery with low latency.

LTE HARQ Parameters

Parameter	Downlink	Uplink
Type	Asynchronous	Synchronous
# Processes	8	8
Timing	Flexible retx subframe	Fixed: retx at n+8
Feedback	ACK/NACK on PUCCH/PUSCH	PHICH (ACK/NACK)
RTT	8 ms	8 ms

Stop-and-Wait with 8 Processes

Each process handles one transport block at a time (stop-and-wait). With **8 parallel processes**, the pipeline stays full:

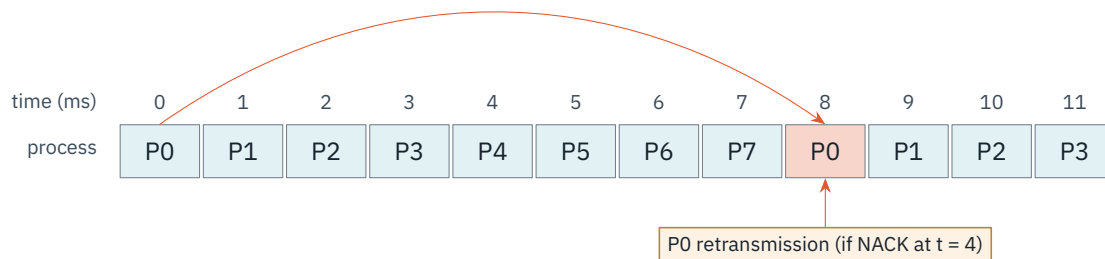


Figure 7.4 Eight parallel stop-and-wait HARQ processes

RTT = 8 ms: UE receives at time t → decodes → sends ACK/NACK at $t + 4$ → eNB receives → retransmits at $t + 8$.

Combining Methods

Method	Description	Advantage
Chase Combining (CC)	Retransmit identical bits, soft-combine at receiver	Simple, ~3 dB gain
Incremental Redundancy (IR)	Retransmit different redundancy version (new parity bits)	Higher coding gain, better performance

LTE uses **incremental redundancy** by default with 4 redundancy versions (RV 0, 2, 3, 1).

9 Scheduling Algorithms

Round Robin (RR)

Principle: Allocate equal resources to all users in turn.

$$\text{User}_{\text{selected}}(t) = (t \bmod N_{\text{users}})$$

Pros	Cons
Simple to implement	Ignores channel conditions
Perfectly fair in resources	Unfair in throughput (edge users get less)
No feedback required	Poor spectral efficiency
Low computational cost	No QoS differentiation

Proportional Fair (PF)

Principle: Maximize the ratio of instantaneous rate to average throughput.

$$\text{User}_{\text{selected}} = \arg \max_k \frac{R_k^{\text{inst}}(t)}{\bar{R}_k(t)}$$

Where:

- $R_k^{\text{inst}}(t)$ = instantaneous achievable rate for user k (from CQI)
- $\bar{R}_k(t)$ = exponential moving average of user k's throughput

Average update:

$$\bar{R}_k(t+1) = (1 - \frac{1}{t_c})\bar{R}_k(t) + \frac{1}{t_c} \cdot r_k(t)$$

where t_c is the averaging window (typically 100–1000 TTIs).

Pros	Cons
Exploits multi-user diversity	More complex
Good throughput-fairness balance	Requires CQI feedback
Schedules users on their peaks	Tuning t_c affects behavior
Maximizes sum log(throughput)	Not optimal for strict delay

Maximum C/I (Max Throughput)

Principle: Always schedule the user with the best channel.

$$\text{User}_{\text{selected}} = \arg \max_k R_k^{\text{inst}}(t)$$

Pros	Cons
Maximizes cell throughput	Starves cell-edge users
Best spectral efficiency	Extremely unfair
Simple decision	Violates QoS for weak users

QoS-Aware Scheduling

Principle: Guarantee minimum rates for GBR bearers, prioritize by QCI.

Steps:

1. **Satisfy GBR bearers first** — allocate minimum required RBs
2. **Priority ordering** — rank by QCI priority level
3. **Remaining resources** — distribute via PF among non-GBR bearers
4. **Delay budget check** — boost priority for packets approaching deadline

10 Engineering Trade-offs

The Scheduling Trilemma

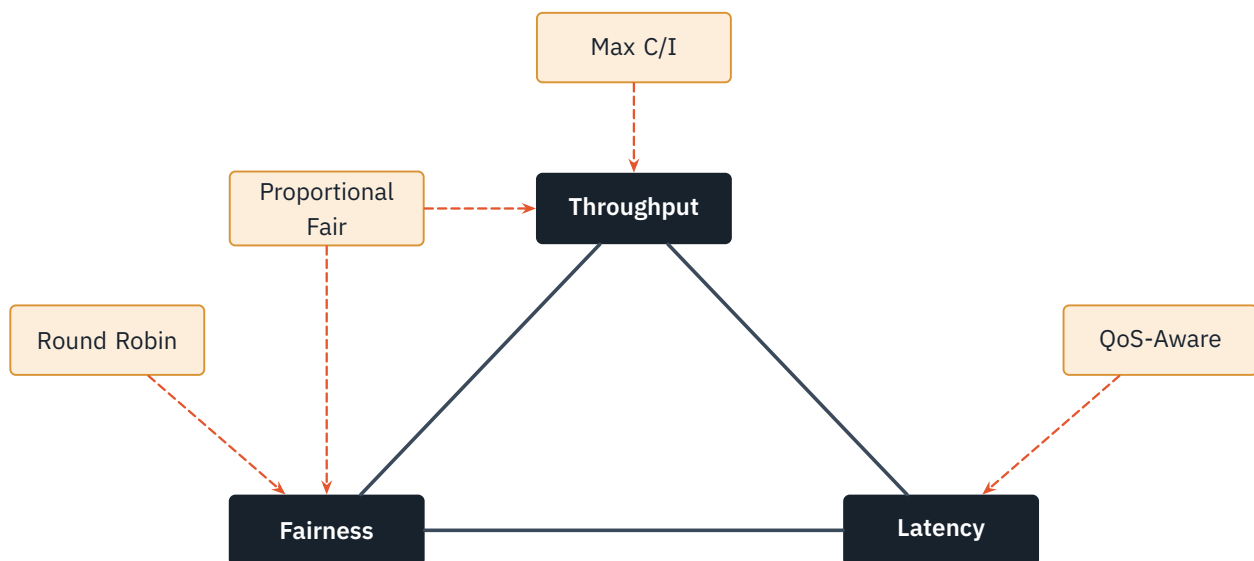


Figure 7.5 The scheduling trilemma and where each algorithm sits

Algorithm	Throughput	Fairness	Latency	Complexity
Round Robin	Low	High (resource)	Medium	Very Low
Proportional Fair	High	Medium	Medium	Medium
Max C/I	Highest	Very Low	Low (for best users)	Low
QoS-Aware	Medium-High	Configurable	Guaranteed	High

Design Considerations

Factor	Impact on Scheduling
User density	More users → more multi-user diversity → PF gains increase
Traffic type	VoLTE needs strict latency; FTP tolerates delay
Cell size	Large cells → big CQI spread → fairness more critical
Frequency band	Higher bands → more path loss variation → more scheduling gain
MIMO mode	MU-MIMO allows co-scheduling users on same RBs

11 Numerical Example

System Parameters

- **Bandwidth:** 10 MHz → **50 RBs** available per TTI
- **Users:** 5 UEs with different channel conditions
- **Overhead:** Assume 10 data symbols per subframe (after control/reference signals)

User CQI and Spectral Efficiency

User	CQI	Modulation	Spectral Efficiency (bps/Hz)	Bits per RB per TTI
UE1	4	QPSK	0.6016	$0.6016 \times 180\text{kHz} \times 1\text{ms} = 108$
UE2	7	16QAM	1.4766	$1.4766 \times 180\text{kHz} \times 1\text{ms} = 266$
UE3	10	64QAM	2.7305	$2.7305 \times 180\text{kHz} \times 1\text{ms} = 491$
UE4	13	64QAM	4.5234	$4.5234 \times 180\text{kHz} \times 1\text{ms} = 814$
UE5	15	64QAM	5.5547	$5.5547 \times 180\text{kHz} \times 1\text{ms} = 1000$

Note: Bits per RB per TTI $\approx$ Spectral Efficiency $\times$ 12 subcarriers $\times$ 15 kHz spacing $\times$ 1 ms $\approx$ SE $\times$ 180 (simplified; actual TBS lookup gives discrete values).

Round Robin Allocation

Each user gets equal RBs: $50 \text{ RBs} \div 5 \text{ users} = 10 \text{ RBs/user}$

User	RBs	Throughput per TTI	Throughput (Mbps)
UE1	10	$10 \times 108 = 1,080 \text{ bits}$	1.08
UE2	10	$10 \times 266 = 2,660 \text{ bits}$	2.66
UE3	10	$10 \times 491 = 4,910 \text{ bits}$	4.91
UE4	10	$10 \times 814 = 8,140 \text{ bits}$	8.14
UE5	10	$10 \times 1,000 = 10,000 \text{ bits}$	10.00
Total	50	26,790 bits	26.79

Jain's Fairness Index (resource): 1.0 (perfectly fair in RBs)

Jain's Fairness Index (throughput): ~ 0.69

Maximum C/I Allocation

All 50 RBs go to UE5 (best channel):

User	RBs	Throughput (Mbps)
UE1	0	0
UE2	0	0
UE3	0	0
UE4	0	0
UE5	50	50.00
Total	50	50.00

Cell throughput: 50 Mbps (maximum possible)

Jain's Fairness Index: 0.20 (worst case for 5 users)

Proportional Fair Allocation

Assuming all users start with equal average throughput, PF initially behaves like Max C/I. After convergence, it balances by scheduling users on their relative peaks.

Steady-state approximation (PF tends toward equal throughput in log domain):

The PF scheduler approximately equalizes $\log(\bar{R}_k)$, leading to a throughput distribution between RR and Max C/I:

User	Approx. RBs	Throughput (Mbps)	Metric ($R_{\text{inst}}/R_{\text{avg}}$)
UE1	15	1.62	Boosted (low R_{avg})
UE2	12	3.19	Moderate
UE3	10	4.91	Baseline
UE4	8	6.51	Slightly reduced
UE5	5	5.00	Reduced (high R_{avg})
Total	50	21.23	—

Note: Exact PF allocation depends on fading dynamics and tc window. In flat-fading (no frequency selectivity), PF gives more resources to weaker users over time. With frequency-selective fading, all users benefit from being scheduled on their best subbands.

Comparison Summary

Metric	Round Robin	Max C/I	Proportional Fair
Cell Throughput (Mbps)	26.79	50.00	~21–35
UE1 Throughput (Mbps)	1.08	0	~1.6
UE5 Throughput (Mbps)	10.00	50.00	~5.0
Fairness (Jain's, resource)	1.0	0.20	~0.75
Spectral Efficiency	Low	Maximum	High
Starvation?	No	Yes (4 users)	No

Key Takeaways

1. **The resource grid** is a time-frequency matrix — scheduling operates at RB granularity every 1 ms TTI
2. **CQI feedback** is the scheduler's eyes — it drives MCS selection and resource allocation
3. **HARQ** provides reliability with 8 parallel processes and 8 ms RTT
4. **Link adaptation (AMC + OLLA)** continuously matches rate to channel conditions
5. **No single algorithm wins** — the choice depends on deployment scenario, traffic mix, and operator priorities
6. **Proportional Fair** is the de-facto standard in commercial networks as it provides a good throughput-fairness compromise
7. **SC-FDMA** constrains uplink to contiguous allocation, limiting scheduling flexibility compared to downlink OFDMA

References

- 3GPP TS 36.213: Physical layer procedures (CQI/MCS tables)
- 3GPP TS 36.321: MAC protocol (BSR, PHR, HARQ)
- 3GPP TS 36.212: Multiplexing and channel coding (TBS tables)
- Sesia, Toufik, Baker: “LTE – The UMTS Long Term Evolution” (Wiley)

Random Access Channel (RACH)

کانال دسترسی تصادفی (RACH)

PRACH PREAMBLES

COLLISION

In This Module

8.1	Why Random Access is Needed	179
8.2	PRACH (Physical Random Access Channel)	179
8.3	Preamble Sequences	180
8.4	Contention-Based Random Access (4-Step CBRA)	182
8.5	Contention-Free Random Access (CFRA)	184
8.6	Timing Advance (TA)	185
8.7	Collisions and Backoff	186
8.8	RACH Overload	189
8.9	5G NR Random Access	191
8.10	Summary and Key Equations	193
	Review Questions	193
	References	194

8.1 Why Random Access is Needed

When a UE (User Equipment) powers on or transitions from idle mode, it faces fundamental challenges:

Challenge	Description
No uplink timing alignment	UE doesn't know its propagation delay to the base station
No allocated uplink resources	UE has no scheduled grants for transmission
No unique identity	Network hasn't assigned a C-RNTI yet

Scenarios Requiring Random Access

1. **Initial Access** — UE powering on, first connection to network
2. **Handover** — Moving to a new cell, need synchronization with target eNodeB (eNB)
3. **DL Data Arrival (Idle Mode)** — UE paged for incoming data, must re-establish UL sync
4. **UL Data Arrival** — UE in connected mode lost UL synchronization
5. **Positioning** — Timing measurements for location services

Key Insight: Random access is the ONLY mechanism for a UE to initiate uplink communication when it has no prior scheduling from the network.

8.2 PRACH (Physical Random Access Channel)

Time-Frequency Location in Resource Grid

The PRACH occupies a specific position in the LTE resource grid:

- **Frequency domain:** 6 consecutive PRBs (1.08 MHz bandwidth)
- **Time domain:** Configurable subframes based on PRACH configuration index
- **Location:** Specified by `prach-FreqOffset` (PRB offset from lowest UL PRB)

PRACH Configuration Index

The configuration index (0–63) determines:

- **Which subframes** carry PRACH opportunities
- **Periodicity** of PRACH occasions (every 1, 2, 5, or 10 ms)
- **Preamble format** used

Config Index	Preamble Format	Subframe(s)	Density
0	0	1	1 per frame
3	0	1, 4, 7	3 per frame
15	0	0–9	10 per frame
48	4 (short)	1	1 per frame

■ Preamble Formats

Format	Sequence Length (Ts)	CP Length (Ts)	Total Duration	Max Cell Radius
0	24576 (0.8 ms)	3168 (0.1 ms)	~1 ms	~14.5 km
1	24576 (0.8 ms)	21024 (0.68 ms)	~2 ms	~77.3 km
2	2×24576 (1.6 ms)	6240 (0.2 ms)	~2 ms	~29.5 km
3	2×24576 (1.6 ms)	21024 (0.68 ms)	~3 ms	~100.2 km
4 (short)	4096 (0.13 ms)	448 (0.015 ms)	~0.15 ms	~1.4 km

Design Trade-off: Longer CP → larger cell radius but fewer PRACH opportunities per frame.

8.3 Preamble Sequences

■ Zadoff-Chu (ZC) Sequences

The PRACH preamble uses Zadoff-Chu sequences defined as:

$$x_u(n) = e^{-j\pi un(n+1)/N_{ZC}}$$

where:

- u = root sequence index ($1 \leq u \leq N_{ZC} - 1$)
- N_{ZC} = sequence length (839 for long, 139 for short)

Why Zadoff-Chu?

Property	Benefit
Constant amplitude (zero PAPR)	Maximizes PA efficiency at cell edge
Ideal periodic autocorrelation	Enables precise timing estimation
Zero cross-correlation between roots	Different root → zero interference
Flat frequency spectrum	Robust in frequency-selective channels

64 Preambles Per Cell

Each cell has exactly **64 available preambles**, derived from:

1. **Root Sequence Index** (`rootSequenceIndex`, broadcast in SIB2)
2. **Cyclic Shifts** applied to root sequences

$$\text{Preamble}(v) = \text{CyclicShift}(x_u, v \times N_{CS})$$

Where:

- N_{CS} = cyclic shift size (`zeroCorrelationZoneConfig`)
- Number of preambles per root = $\lfloor N_{ZC}/N_{CS} \rfloor$
- Multiple roots used if one root doesn't provide 64 preambles

Cell Coverage and Cyclic Shift

The cyclic shift N_{CS} determines the **zero-correlation zone**, which must exceed the maximum round-trip delay:

$$N_{CS} > \frac{2 d_{\max}}{c T_{s,\text{seq}}}$$

where $d_{\max}$ is the maximum cell radius, c the speed of light (3×10^8 m/s) and $T_{s,\text{seq}}$ the sequence sample duration.

Example: For $N_{CS} = 13$ (format 0):

- Max distinguishable delay = $13 \times (1/1.25 \text{ MHz}) = 10.4 \mu\text{s}$
- Max cell radius = $10.4 \mu\text{s} \times c/2 \approx 1.56 \text{ km}$
- Preambles per root = $\lfloor 839/13 \rfloor = 64 \rightarrow$ single root sufficient

8.4 Contention-Based Random Access (4-Step CBRA)

Procedure Overview

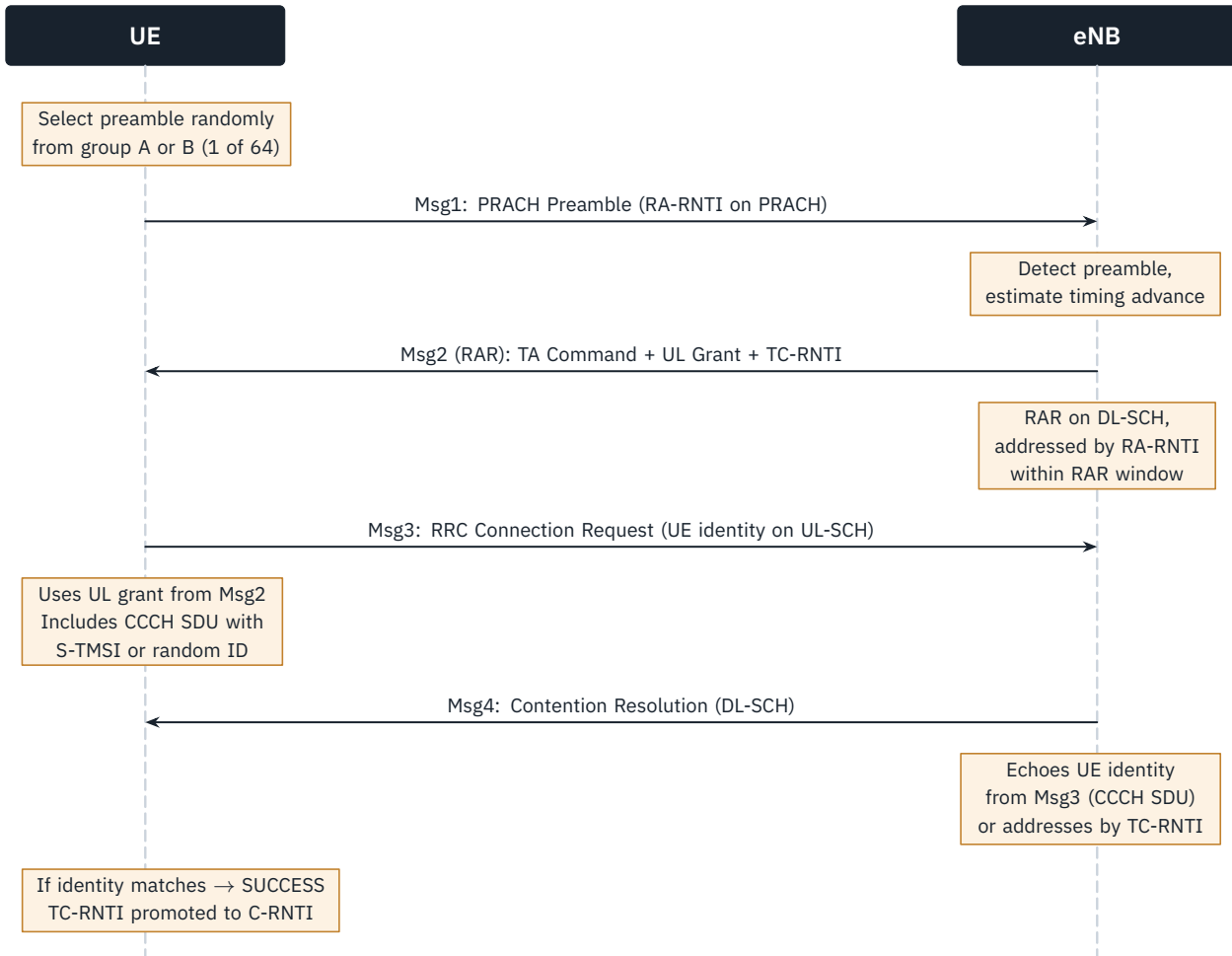


Figure 8.1 Four-step contention-based random access

Step-by-Step Details

■ Step 1 – Msg1: Preamble Transmission

- UE selects a **random preamble** from the available set (contention preambles)
- Transmits on PRACH at configured time-frequency resources
- **RA-RNTI** derived from PRACH occasion: $RA-RNTI = 1 + t_id + 10 \times f_id$
 - t_id = subframe index (0–9)
 - f_id = frequency index (0–5)

■ Step 2 – Msg2: Random Access Response (RAR)

RAR contains (per responding preamble):

Field	Bits	Purpose
Timing Advance	11	UL timing correction (0–1282, step = $16 \times T_s$)
UL Grant	20	Resources for Msg3
TC-RNTI	16	Temporary identity
Backoff Indicator	4	Backoff if overloaded

- **RAR Window:** 2–10 subframes (configurable by `ra-ResponseWindowSize`)
- If no RAR received → back off and retry

■ Step 3 – Msg3: Scheduled Transmission

- UE transmits on UL-SCH using the grant from Msg2
- Contains **RRC Connection Request** with:
 - UE identity (S-TMSI or random value)
 - Establishment cause
- HARQ enabled (max 5 retransmissions for Msg3)

■ Step 4 – Msg4: Contention Resolution

- eNB echoes the UE identity received in Msg3
- UE checks if the echoed identity matches its own:
 - **Match** → RACH successful, TC-RNTI becomes C-RNTI
 - **No match** → Collision detected, restart from Step 1 with backoff

8.5 Contention-Free Random Access (CFRA)

When Used

Scenario	Reason
Handover	Fast sync needed, can't afford collision delay
DL data arrival (connected)	eNB knows UE, can pre-assign preamble
Positioning (OTDOA)	Precise timing required
Beam failure recovery	Quick recovery needed

Procedure

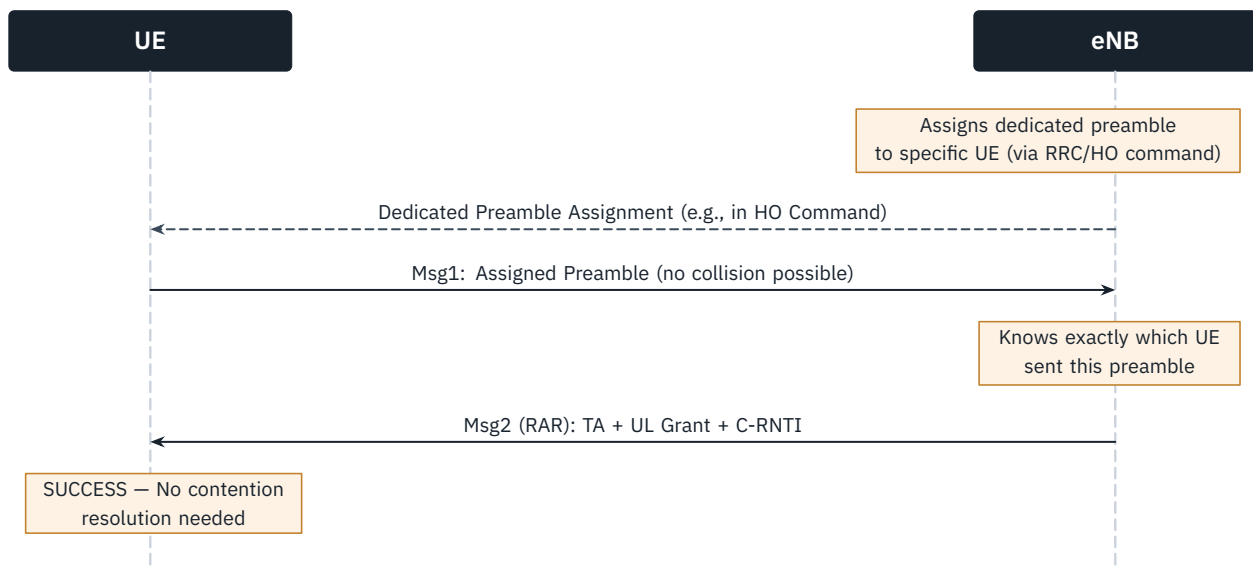


Figure 8.2 Contention-free random access

Key Differences from CBRA

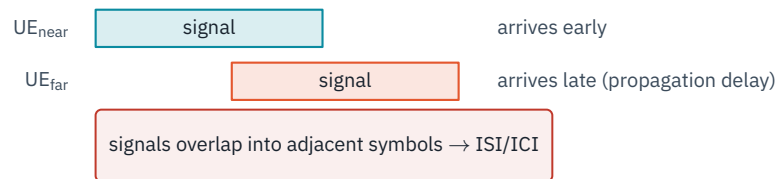
Aspect	Contention-Based	Contention-Free
Steps	4	2
Preamble selection	Random	Assigned by eNB
Collision possible	Yes	No
Latency	~15 ms (success)	~5 ms
Use case	Initial access	Handover, DL data
Preamble pool	Shared (e.g., 54)	Dedicated (e.g., 10)

8.6 Timing Advance (TA)

Why Timing Advance is Needed

In OFDMA uplink (SC-FDMA), all UE signals must arrive at the eNB **aligned within the cyclic prefix** to maintain orthogonality:

Without TA



With TA

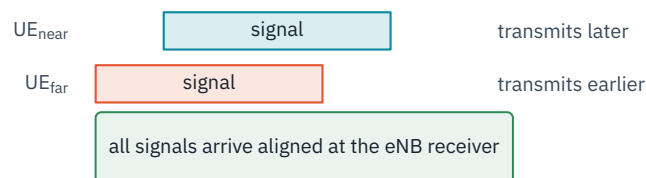


Figure 8.3 Timing advance aligns the uplink signals of near and far UEs

TA Mechanism

Initial TA (from RAR - Msg2):

- 11-bit value: 0–1282
- Step size: $16 \times T_s = 16 \times (1/30.72 \text{ MHz}) \approx 0.52 \mu\text{s}$
- Represents one-way advance applied to UL transmission

TA Maintenance (MAC CE - Timing Advance Command):

- 6-bit value: 0–63
- Applied as: $\text{TA}_{\text{new}} = \text{TA}_{\text{old}} + (\text{TA}_{\text{command}} - 31) \times 16 \times T_s$

- Sent periodically via MAC Control Element
- If no TA update received within `timeAlignmentTimer` → UL sync lost

Cell Radius from Timing Advance

The maximum TA corrects for the **round-trip propagation delay**:

$$\text{Round-trip delay} = \text{TA} \times 16 \times T_s$$

$$\text{One-way delay} = \text{TA} \times 16 \times T_s / 2$$

$$\text{Max distance} = c \times \text{one-way delay}$$

Numerical Example: Maximum Cell Radius

For initial TA (11-bit, max = 1282):

$$\begin{aligned} \text{Max round-trip delay} &= 1282 \times 16 \times T_s = 1282 \times 16 \times \frac{1}{30.72 \times 10^6} \\ &= 1282 \times 0.5208 \mu\text{s} = 667.7 \mu\text{s} \end{aligned}$$

$$\text{Max one-way delay} = 667.7 / 2 = 333.8 \mu\text{s}$$

$$\text{Max cell radius} = 3 \times 10^8 \times 333.8 \times 10^{-6} = 100.1 \text{ km}$$

Verification with preamble format:

Format	Guard Time (CP)	Max RTD	Max Cell Radius
0	99.5 μs ($3168 \times T_s$)	99.5 μs	14.9 km
1	684.4 μs ($21024 \times T_s$)	684.4 μs	102.6 km
2	203.1 μs ($6240 \times T_s$)	203.1 μs	30.5 km
3	684.4 μs ($21024 \times T_s$)	684.4 μs	102.6 km

Note: The actual cell radius is limited by the **smaller** of: TA range OR preamble CP duration.

8.7 Collisions and Backoff

Collision Probability

When N UEs simultaneously attempt random access with 64 available preambles:

Probability that a specific UE collides with at least one other UE:

$$P_{\text{collision}} = 1 - \left(1 - \frac{1}{64}\right)^{N-1}$$

■ Numerical Examples

N (simultaneous UEs)	P(collision) per UE	Expected colliding UEs
2	1.56%	0.03
5	6.10%	0.31
10	13.0%	1.30
20	25.5%	5.10
30	36.8%	11.0
50	54.0%	27.0
64	63.2%	40.4

Example Calculation (N=10):

$$\begin{aligned}
 P(\text{collision}) &= 1 - \left(1 - \frac{1}{64}\right)^{10-1} = 1 - \left(\frac{63}{64}\right)^9 \\
 &= 1 - (0.98438)^9 = 1 - 0.8681 = 0.1319 \approx 13.2\%
 \end{aligned}$$

Collision Scenario with Resolution

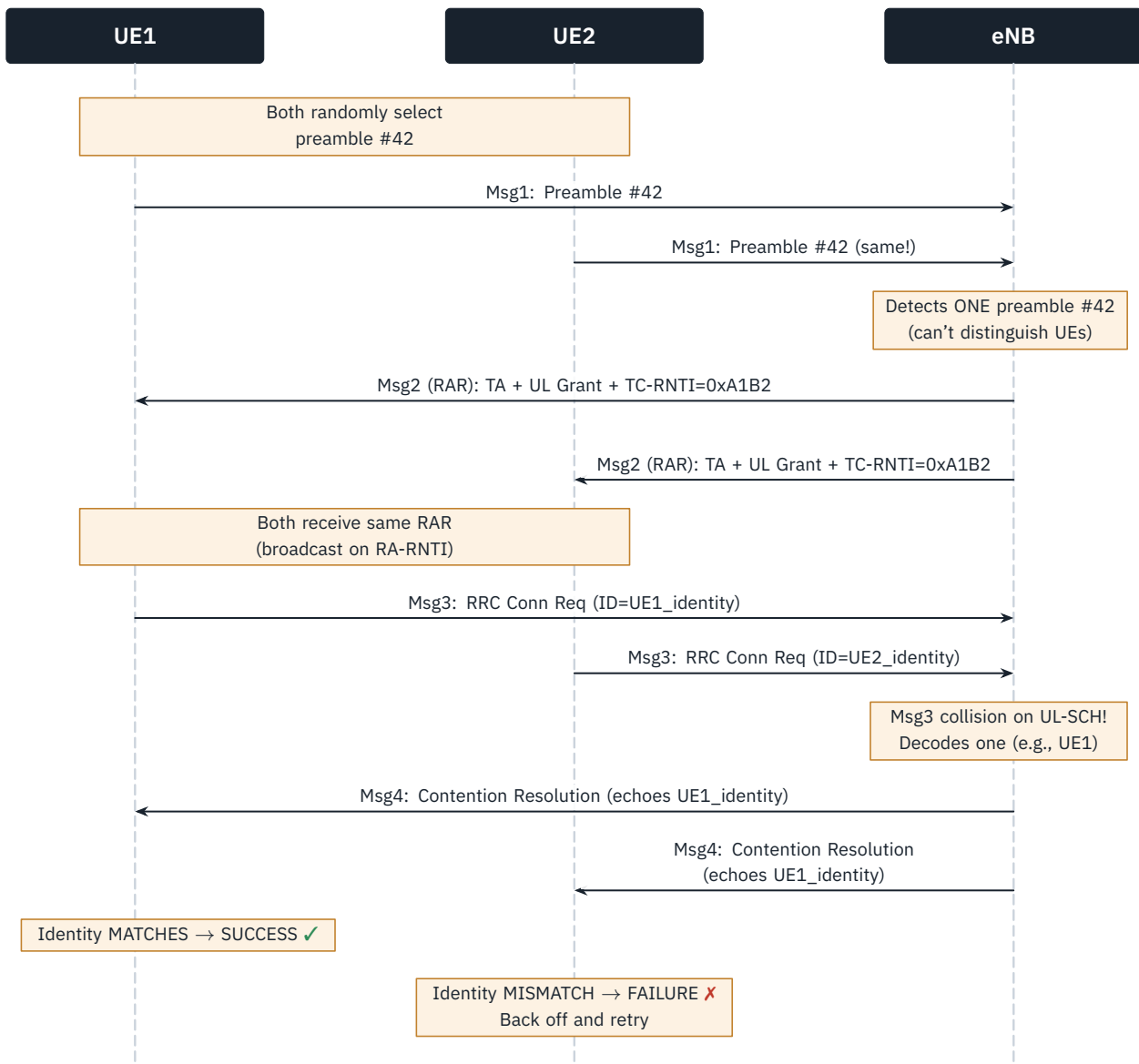


Figure 8.4 Preamble collision and its resolution in Msg4

Backoff Mechanism

When a RACH attempt fails, the UE must wait before retrying:

1. Backoff Indicator (BI): Broadcast in RAR subheader

- Values: 0, 10, 20, 30, 40, 60, 80, 120, 160, 240, 320, 480, 960 ms
- UE selects random backoff: uniform [0, BI]

2. Power Ramping: Each failed attempt increases transmit power

- Step size: `powerRampingStep` (2 dB typical)
- $P_{\text{PRACH}}(\text{attempt}) = P_{\text{initial}} + (\text{attempt} - 1) \times \text{powerRampingStep}$
- Maximum: `preambleInitialReceivedTargetPower` + ramping
- Max attempts: `preambleTransMax` (3, 4, 5, 6, 7, 8, 10, 20, 50, 100, 200)

3. Preamble re-selection: New random preamble chosen for each attempt

RACH Attempt Timeline

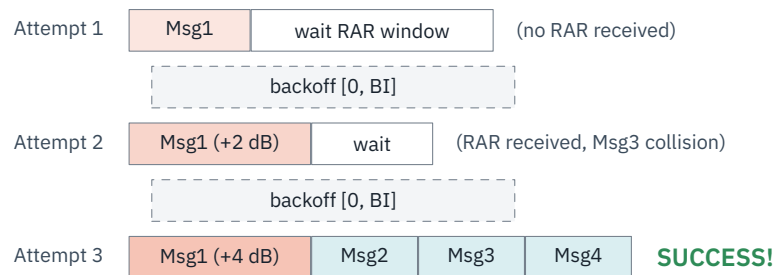


Figure 8.5 RACH attempts with power ramping and backoff

8.8 RACH Overload

Causes of RACH Overload

Cause	Description	Scale
M2M/IoT	Massive simultaneous device activation (e.g., smart meters reporting hourly)	10,000–30,000 devices
Paging storms	Major event triggers mass DL data (emergency alert, popular content)	Thousands of UEs
Disaster/emergency	Everyone calls simultaneously after earthquake/event	Extreme burst
Cell reselection	Many UEs hand over simultaneously (train, stadium)	Hundreds
Power outage recovery	All devices reconnect simultaneously	Thousands

Overload Symptoms

- High collision rate (>50%)
- Excessive preamble retransmissions → increased interference
- RAR capacity exhausted (limited DL resources for RAR)
- Msg3/Msg4 resource starvation
- Access delays from seconds to minutes

■ Solutions

■ Access Class Barring (ACB)

- UEs assigned to Access Classes 0–9 (plus special classes 11–15)
- eNB broadcasts barring factor (0–1) and barring time
- UE draws random number; if > barring factor → barred for barring time
- Spreads access attempts over time

■ Extended Access Barring (EAB)

- For delay-tolerant M2M/IoT devices (AC 0–9)
- eNB broadcasts EAB bitmap (one bit per AC)
- Barred UEs wait for barring to be lifted
- More aggressive than ACB for low-priority traffic

■ Back-off Tuning

- Dynamically increase BI (up to 960 ms) during overload
- Effectively rate-limits RACH attempts
- Trade-off: individual access delay vs. system throughput

■ Additional 3GPP Solutions

Release	Solution	Mechanism
R10	ACB	Probabilistic barring
R11	EAB	IoT/M2M specific barring
R12	ACDC	App-level congestion control
R13	Group paging	Reduce paging-triggered RACH
R14	EDT (Early Data Transmission)	Data in Msg3 → fewer full connections

8.9 5G NR Random Access

Key Enhancements over LTE

2-Step RACH (Rel-16): MsgA + MsgB

Combines the 4-step procedure into 2 steps for **ultra-low latency**:

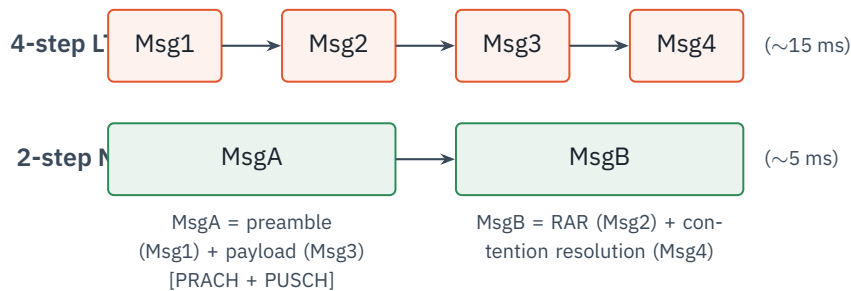


Figure 8.6 Four-step LTE RACH versus two-step NR RACH

When to use 2-step vs 4-step:

- 2-step: Small cells, low load, URLLC requirements
- 4-step: Large cells (need TA before UL data), high load scenarios
- Fallback: If MsgB not received → fall back to 4-step

Beam-Based RACH (SSB-to-RACH Mapping)

In 5G NR with beamforming, the RACH procedure is **beam-aware**:

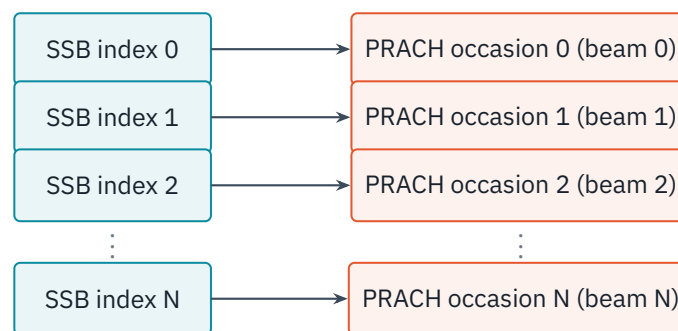


Figure 8.7 SSB-to-RACH mapping: one PRACH occasion per beam

- UE selects **best SSB beam** (highest RSRP)
- Transmits preamble on **associated RACH occasion**
- gNB knows which beam direction the UE is located → **beam correspondence**
- Enables directional RAR transmission

Configuration:

- `ssb-perRACH-OccasionAndCB-PreamblesPerSSB`: maps SSBs to RACH occasions

- Multiple SSBs can share one RACH occasion, or one SSB can span multiple

■ Supplementary Uplink (SUL) RACH

- Lower frequency band (e.g., 700 MHz) for extended coverage
- UE selects SUL PRACH when $rsrp-ThresholdSSB-SUL$ condition met
- Separate preamble format and configuration for SUL
- Enables coverage extension without increasing NR UL power

■ NR Preamble Formats

Format	Sequence Length	Subcarrier Spacing	Use Case
0	839 (long)	1.25 kHz	Large cells (FR1)
1	839 (long)	1.25 kHz	Very large cells
2	839 (long)	1.25 kHz	Extreme coverage
3	839 (long)	5 kHz	Medium cells
A1	139 (short)	15/30 kHz	Small cells (FR1)
A2	139 (short)	15/30 kHz	Normal cells (FR1)
A3	139 (short)	15/30 kHz	Normal cells (FR1)
B4	139 (short)	60/120 kHz	FR2 (mmWave)
C0	139 (short)	15/30 kHz	Msg3 early TX
C2	139 (short)	15/30 kHz	Msg3 early TX

■ Beam Failure Recovery (BFR)

A special RACH-like procedure in NR:

- UE detects beam failure (RSRP below threshold for all configured beams)
- UE selects a **new candidate beam** (from measured SSBs/CSI-RS)
- UE transmits **dedicated CFRA preamble** on PRACH occasion of new beam
- gNB responds with BFR response → reconfigure beam pair

8.10 Summary and Key Equations

Critical Formulas

Parameter	Formula
Collision probability	$P_c = 1 - (63/64)^{N-1}$
RA-RNTI	$1 + t_{id} + 10 \times f_{id}$
Timing Advance (initial)	$TA = N_{TA} \times 16 \times T_s$
Max cell radius	$d_{max} = c \times N_{TA,max} \times 16 \times T_s / 2$
Power ramping	$P_n = P_{init} + (n - 1) \times \Delta_{ramp}$
Preambles per root	$\lfloor N_{ZC} / N_{CS} \rfloor$

Design Trade-offs

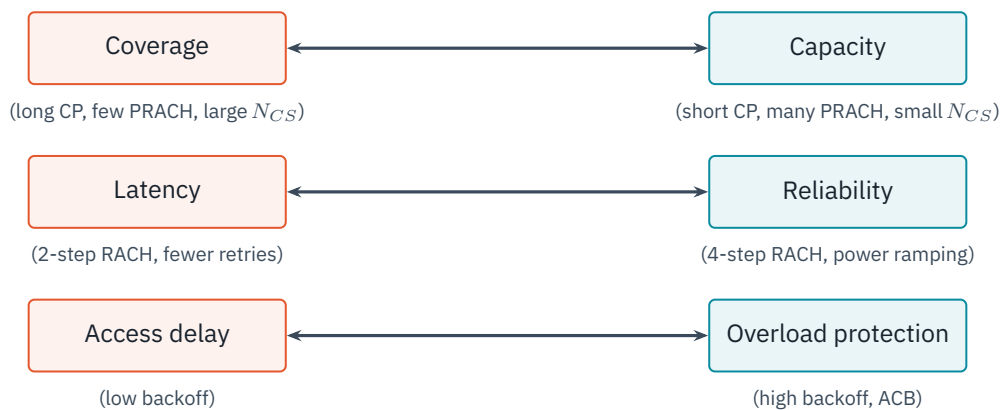


Figure 8.8 RACH design trade-offs

Review Questions

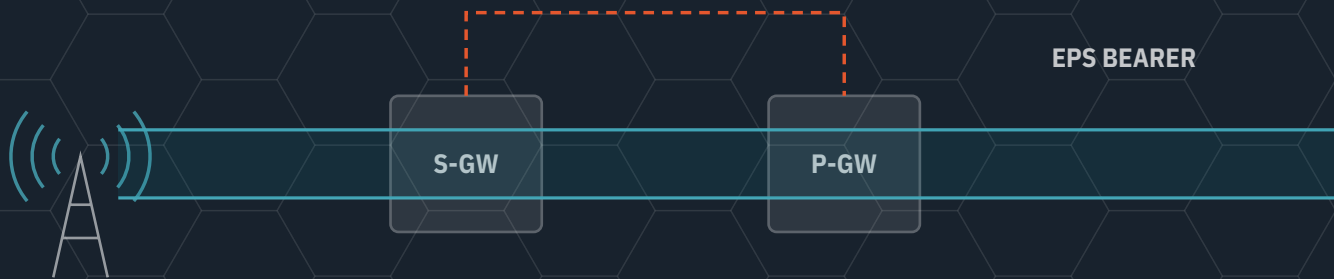
1. Why can't the UE simply transmit on PUSCH without performing RACH first?
2. Calculate the collision probability when 25 UEs attempt simultaneous access.
3. What is the maximum cell radius for PRACH format 3? Show your calculation.
4. Explain why contention-free RACH is used during handover instead of contention-based.
5. A cell has `zeroCorrelationZoneConfig` = 11 ($N_{CS} = 15$). How many root sequences are needed to generate 64 preambles?
6. Describe the beam failure recovery procedure in 5G NR and its relation to RACH.
7. An IoT network has 10,000 devices that simultaneously attempt RACH. Describe two mechanisms to handle this overload.
8. Compare the latency of 2-step NR RACH vs. 4-step LTE RACH. When would you choose each?

References

- 3GPP TS 36.211 §5.7: Physical Random Access Channel (LTE)
- 3GPP TS 36.213 §6: Random Access Procedure (LTE)
- 3GPP TS 36.321 §5.1: Random Access Procedure (MAC)
- 3GPP TS 38.211 §6.3.3: PRACH (NR)
- 3GPP TS 38.213 §8: Random Access Procedure (NR)
- 3GPP TS 38.321 §5.1: Random Access Procedure (NR MAC)

Evolved Packet Core (EPC)

هسته بسته تکامل یافته (EPC)



In This Module

9.1 Why EPC Was Designed	196
9.2 EPC Architecture Overview	197
9.3 EPC Network Functions	198
9.4 EPC Bearers	205
9.5 Authentication in EPC	208
9.6 Session Management	211
9.7 Mobility Management	212
9.8 Failure Scenarios	215
9.9 Summary	217

9.1 Why EPC Was Designed

The Evolved Packet Core (EPC) was introduced in 3GPP Release 8 as the core network for LTE/4G. Its design was driven by fundamental limitations in legacy core networks (2G/3G circuit-switched + packet-switched domains).

Design Principles

Principle	Legacy Problem	EPC Solution
All-IP Architecture	Separate CS and PS domains, TDM transport	Single flat IP-based core for all services (voice via VoLTE/IMS)
Flat Architecture	Deep hierarchy (RNC, SGSN, GGSN) adding latency	Reduced node count; eNB connects directly to EPC
Control/User Plane Separation	Combined CP/UP in SGSN/GGSN	MME (CP only), S-GW/P-GW (UP with minimal CP)
Multiple RAT Support	Separate cores per RAT	Single EPC serves LTE, WCDMA, CDMA2000, Wi-Fi
Simplified Mobility	Complex handover with many nodes	GTP-based anchoring with fewer hops
Policy & Charging Convergence	Fragmented policy/charging	Integrated PCC architecture (PCRF/PCEF)

Key Benefits

- **Lower latency:** User plane path is shorter (eNB → S-GW → P-GW → Internet)
- **Higher throughput:** No RNC bottleneck; distributed scheduling at eNB
- **Scalability:** Stateless forwarding nodes; CP scales independently
- **Service agility:** Dynamic QoS via dedicated bearers and PCC rules
- **Cost efficiency:** IP transport throughout; commodity hardware

9.2 EPC Architecture Overview

Complete Architecture Diagram

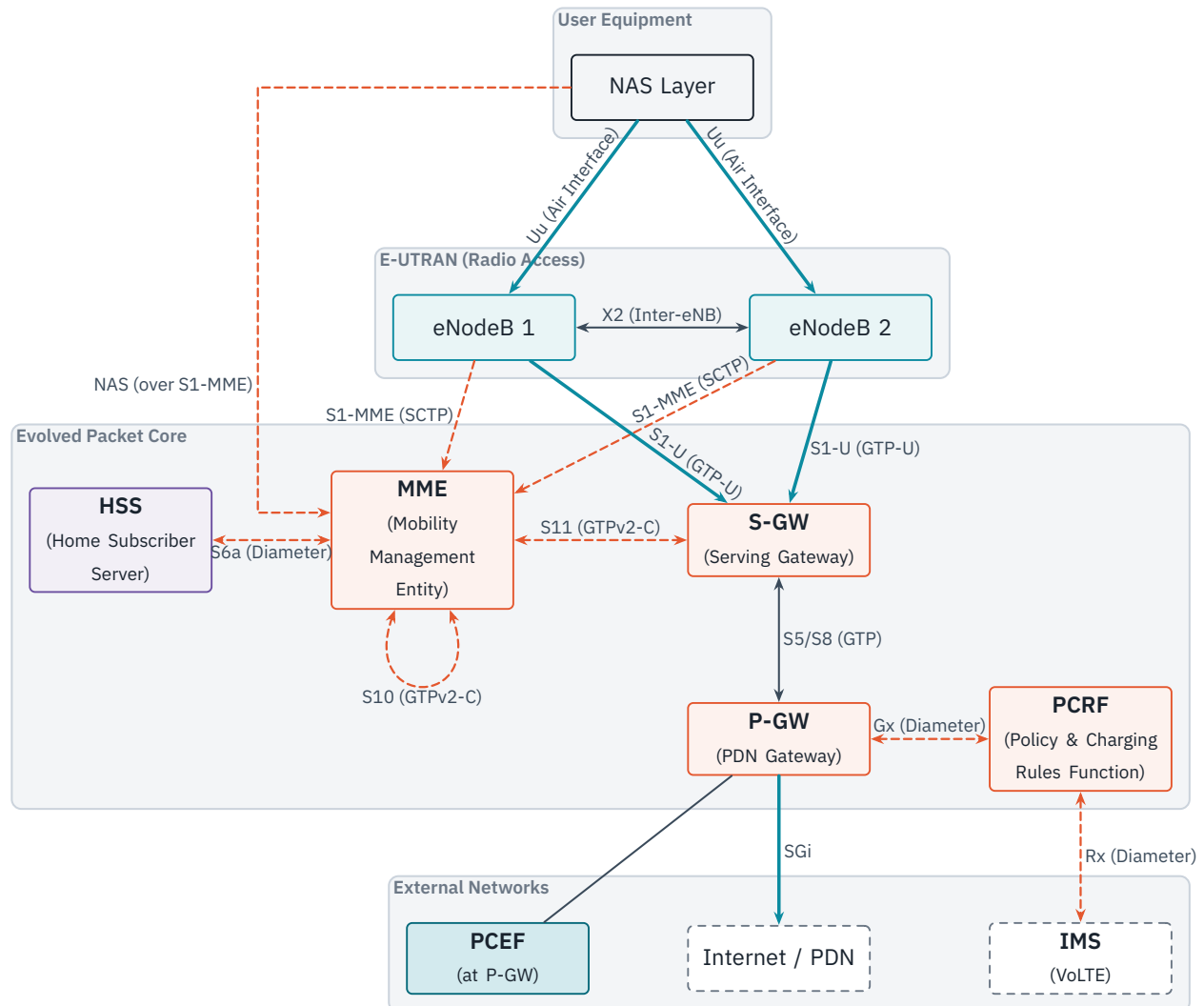


Figure 9.1 Complete EPC architecture with its interfaces and protocols

Interface Summary

Interface	Endpoints	Protocol	Purpose
S1-MME	eNB ↔ MME	S1AP over SCTP	Control plane signaling
S1-U	eNB ↔ S-GW	GTP-U over UDP	User plane data
X2	eNB ↔ eNB	X2AP + GTP-U	Handover, load mgmt
S11	MME ↔ S-GW	GTPv2-C	Session/bearer mgmt
S5/S8	S-GW ↔ P-GW	GTP (S5) or PMIP (S8)	User plane + CP
S6a	MME ↔ HSS	Diameter	Authentication, subscription
S10	MME ↔ MME	GTPv2-C	MME relocation
S3	MME ↔ SGSN	GTPv2-C	Inter-RAT mobility
Gx	PCRF ↔ P-GW	Diameter	Policy & charging rules
Gy	P-GW ↔ OCS	Diameter	Online charging
Rx	PCRF ↔ AF/IMS	Diameter	Service-level QoS request
SGi	P-GW ↔ PDN	IP	External data network

9.3 EPC Network Functions

9.3.1 MME – Mobility Management Entity

■ Purpose

The MME is the **primary control plane node** in EPC. It handles all signaling between the UE and the core network, managing mobility, security, and session establishment. The MME never touches user plane data.

■ Key Functions

1. **NAS Signaling:** Terminates Non-Access Stratum protocols (EMM and ESM) with the UE
2. **Authentication & Security:** Initiates EPS-AKA, derives keys ($K_{ASME} \rightarrow K_{eNB}$), selects NAS security algorithms
3. **Mobility Management:** Tracks UE location (TAI), manages ECM states, handles handover signaling
4. **Bearer Management:** Creates/modifies/deletes EPS bearers via S11 signaling to S-GW
5. **Paging:** Pages UEs in ECM-IDLE state when downlink data arrives
6. **MME Selection & Pooling:** Multiple MMEs serve a pool area; load balanced via S1AP weighting
7. **Inter-RAT Mobility:** Handles handover to/from 2G/3G via S3 interface to SGSN
8. **Lawful Intercept:** Provides signaling for LI triggers

■ Interfaces

Interface	Peer	Protocol	Purpose
S1-MME	eNB	S1AP/SCTP	RAN control plane
S11	S-GW	GTPv2-C	Bearer/session management
S6a	HSS	Diameter	Auth vectors, subscription data
S10	MME	GTPv2-C	MME relocation (handover)
S3	SGSN	GTPv2-C	Inter-RAT mobility
S13	EIR	Diameter	Equipment identity check
NAS	UE	EMM/ESM (over S1AP)	Direct UE signaling

■ Protocols

- **S1AP**: Application protocol for S1 interface (ASN.1 encoded, SCTP transport)
- **GTPv2-C**: GPRS Tunnelling Protocol v2 Control (UDP port 2123)
- **Diameter**: For S6a (authentication), S13 (EIR check)
- **NAS**: EMM (EPS Mobility Management) + ESM (EPS Session Management)

■ Failure Impact

- **Immediate**: All UEs served by the failed MME lose signaling connectivity
- **S1 connections released**: eNBs detect failure via SCTP heartbeat timeout
- **Recovery**: UEs in ECM-CONNECTED move to ECM-IDLE, then re-attach to a different MME in the pool
- **Mitigation**: MME pooling ensures surviving MMEs absorb load; UE context can be retrieved from HSS

9.3.2 HSS — Home Subscriber Server

■ Purpose

The HSS is the **central subscriber database** for EPC. It is the evolution of the HLR/AuC from 2G/3G, storing all permanent subscriber information, authentication credentials, and service profiles.

■ Key Functions

1. **Subscriber Database**: Stores IMSI, MSISDN, service subscriptions, APN configurations
2. **Authentication Vector Generation**: Generates EPS authentication vectors (RAND, AUTN, XRES, K_{ASME}) using the permanent key K and sequence number SQN
3. **Location Management**: Maintains current MME address per subscriber; handles registration/deregistration
4. **Subscription Profiles**: Stores allowed APNs, QoS profiles (subscribed AMBR, default QCI), roaming restrictions
5. **Charging Characteristics**: Per-subscriber charging config
6. **Service Authorization**: Determines which services a UE is permitted

■ Interfaces

Interface	Peer	Protocol	Purpose
S6a	MME	Diameter	Auth vectors, location update, subscription data
S6d	SGSN	Diameter	2G/3G equivalent of S6a
Cx/Dx	IMS (CSCF)	Diameter	IMS registration, routing
Sh	AS	Diameter	Service data for application servers

■ Protocols

- **Diameter:** All HSS interfaces use Diameter (RFC 6733) with specific applications:
 - S6a: 3GPP Diameter application (vendor-specific AVPs)
 - Authentication-Information-Request/Answer (AIR/AIA)
 - Update-Location-Request/Answer (ULR/ULA)
 - Cancel-Location-Request (CLR) for purging old MME

■ Failure Impact

- **New attaches fail:** No authentication vectors available for new UEs
- **Existing sessions survive:** MME caches auth vectors (typically 5 per UE)
- **Handovers impacted:** If new MME cannot fetch subscriber profile
- **Mitigation:** HSS is typically deployed as a geo-redundant mated pair with database replication

9.3.3 S-GW – Serving Gateway

■ Purpose

The S-GW is the **user plane anchor** between the radio network and the core. It acts as a local mobility anchor for inter-eNB handovers, ensuring user plane continuity while the UE moves between base stations.

■ Key Functions

1. **User Plane Forwarding:** Routes user data between eNB and P-GW via GTP-U tunnels
2. **Local Mobility Anchor:** During X2 handover, S-GW remains unchanged; only the downlink GTP-U endpoint is switched to the target eNB
3. **Downlink Data Buffering:** When UE is in ECM-IDLE, S-GW buffers incoming packets and triggers MME to page the UE
4. **Per-UE Charging:** Collects charging data (CDRs) per bearer, reports to charging system
5. **Lawful Intercept:** User plane copy point for legal interception
6. **Inter-RAT Anchor:** Anchors user plane during handover between LTE and 2G/3G (with SGSN)
7. **Transport Level Marking:** Applies DSCP marking based on bearer QCI

■ Interfaces

Interface	Peer	Protocol	Purpose
S1-U	eNB	GTP-U	User plane from RAN
S11	MME	GTPv2-C	Bearer/tunnel management
S5/S8	P-GW	GTP-U + GTPv2-C	User plane + signaling to P-GW
S4	SGSN	GTP	Inter-RAT user plane (2G/3G)
S12	UTRAN	GTP-U	Direct tunnel (3G)

■ Protocols

- **GTP-U** (UDP port 2152): Encapsulates user IP packets in GTP headers with TEID
- **GTPv2-C** (UDP port 2123): Create/Modify/Delete Session messages from MME
- **DSCP**: Outer IP header marking for transport QoS

■ Failure Impact

- **Active sessions disrupted**: Ongoing data flows interrupted
- **Path switch required**: MME detects failure, selects a new S-GW, re-establishes GTP tunnels
- **Buffered data lost**: Any DL packets buffered for IDLE UEs are lost
- **Mitigation**: S-GW typically deployed in N+1 redundancy; ICGW (Idle-mode Signaling Reduction) minimizes state

9.3.4 P-GW – PDN Gateway

■ Purpose

The P-GW is the **point of interconnection** between the EPC and external packet data networks (PDNs). It is the IP anchor point for UE sessions — the UE's IP address remains constant as long as the PDN connection exists, regardless of mobility.

■ Key Functions

1. IP Address Allocation:

- IPv4: DHCPv4 or static assignment from configured pools
- IPv6: Stateless Address Autoconfiguration (SLAAC) via Router Advertisement; assigns /64 prefix
- Dual-stack: Both simultaneously

2. Policy Enforcement (PCEF):

- Applies PCC rules received from PCRF
- Traffic Flow Templates (TFT): Packet filters mapping IP flows to bearers
- Service Data Flow (SDF) filters: 5-tuple matching for QoS/charging

3. Charging:

- Offline (Gz interface → CDR generation)

- Online (Gy interface → real-time credit control with OCS)
- 4. **SGi Interface:** Connects to external networks (Internet, IMS, enterprise VPN)
- 5. **NAT/NAPT:** Optional Network Address Translation for IPv4
- 6. **GTP Anchoring:** Terminates S5/S8 GTP tunnels from S-GW

■ Interfaces

Interface	Peer	Protocol	Purpose
S5/S8	S-GW	GTP	User plane + control plane
Gx	PCRF	Diameter	PCC rule download
Gy	OCS	Diameter (Ro)	Online charging
Gz	OFCS	GTP' (CDRs)	Offline charging
SGi	PDN/Internet	IP	External connectivity
S2a/S2b	ePDG/TWAG	GTP/PMIP	Wi-Fi offload

■ Protocols

- **GTP-U/GTPv2-C:** Tunnel and session management toward S-GW
- **Diameter Gx:** Policy and Charging Control (Install/Remove PCC rules)
- **Diameter Gy (Ro):** Credit-Control-Request/Answer for online charging
- **DHCPv4/ICMPv6:** IP address allocation to UE
- **RADIUS/Diameter:** Possible AAA for enterprise APNs

■ Failure Impact

- **Session loss:** All PDN connections through the failed P-GW are lost (IP addresses released)
- **UE must re-establish:** New PDN connectivity request → new IP address (breaks ongoing TCP sessions)
- **Critical impact for VoLTE:** Active voice calls dropped
- **Mitigation:** P-GW clustering, session replication, ICSR (Inter-Chassis Session Recovery)

9.3.5 PCRF – Policy and Charging Rules Function

■ Purpose

The PCRF is the **policy decision point** in the PCC architecture. It makes real-time decisions about QoS authorization, gating, and charging rules based on subscriber profile, network conditions, and application requirements.

■ Key Functions

1. **PCC Rule Generation:** Creates Policy and Charging Control rules containing:
 - SDF filters (which traffic)
 - QoS parameters (QCI, GBR, MBR)

- Charging keys and methods
- Gate status (open/close)
- 2. **QoS Authorization:** Validates requested QoS against subscription and network policy
- 3. **Dynamic Policy:** Real-time policy changes (e.g., when VoLTE call starts → install GBR bearer rule)
- 4. **Rx Interface Processing:** Receives session information from AF/IMS (media description, bandwidth) and translates to PCC rules
- 5. **Spending Limits:** Interfaces with OCS for policy decisions based on remaining credit
- 6. **Subscription-Based Policy:** Applies rules based on subscriber tier (e.g., gold/silver/bronze plans)

■ Interfaces

Interface	Peer	Protocol	Purpose
Gx	P-GW (PCEF)	Diameter	Install/remove PCC rules
Rx	AF / IMS (P-CSCF)	Diameter	Service-level info (SDP)
Sp	SPR	Diameter/proprietary	Subscriber profile retrieval
Sy	OCS	Diameter	Spending limit reporting
Sd	TDF	Diameter	Traffic Detection Function rules

■ Protocols

- **Diameter Gx Application:**
 - CC-Request/Answer (CCR/CCA): Initial, Update, Terminate
 - Re-Auth-Request (RAR): Push rule changes to PCEF
- **Diameter Rx Application:**
 - AA-Request/Answer (AAR/AAA): AF session establishment
 - Session-Termination-Request (STR): AF session teardown
 - Abort-Session-Request (ASR): Network-initiated removal

■ Failure Impact

- **Static rules continue:** Previously installed PCC rules at P-GW remain active
- **No dynamic policy:** New service requests (e.g., VoLTE) cannot get dedicated bearer authorization
- **Default QoS only:** New sessions get only default bearer with static policy
- **Mitigation:** PCRF deployed in active/standby pairs; Diameter routing agent (DRA) for failover

9.3.6 PCEF – Policy and Charging Enforcement Function

■ Purpose

The PCEF is the **policy enforcement point**, co-located with the P-GW. It enforces the PCC rules received from the PCRF by applying packet filtering, QoS enforcement, and charging actions on every user plane packet.

■ Key Functions

1. **SDF Filter Enforcement:** Matches incoming/outgoing IP packets against Service Data Flow filters (5-tuple: src IP, dst IP, src port, dst port, protocol)
2. **Gate Control:** Opens or closes the gate for specific SDFs (blocking/allowing traffic)
3. **QoS Enforcement:** Applies rate limiting per-flow:
 - GBR (Guaranteed Bit Rate): Minimum rate ensured
 - MBR (Maximum Bit Rate): Hard cap per bearer
 - AMBR (Aggregate Maximum Bit Rate): Cap across all non-GBR bearers
4. **Bearer Binding:** Maps SDF filters to appropriate EPS bearers
5. **Charging Enforcement:**
 - Applies charging keys to traffic
 - Volume/time measurement per rating group
 - Online charging: Reports usage, enforces credit limits
 - Offline charging: Generates CDRs
6. **Usage Monitoring:** Reports volume thresholds to PCRF for policy decisions

■ Interfaces

The PCEF is integrated within the P-GW, so its external interface is primarily:

Interface	Peer	Protocol	Purpose
Gx	PCRF	Diameter	Receives PCC rules
Gy	OCS	Diameter (Ro)	Online charging credit control
Gz	OFCS	GTP'	Offline CDR delivery

■ Protocols

- **Diameter Gx:** Receives Charging-Rule-Install/Remove AVPs containing PCC rules
- **Deep Packet Inspection (DPI):** Optional; identifies applications for application-aware charging
- **Token Bucket / Leaky Bucket:** Rate enforcement algorithms for GBR/MBR policing

■ Failure Impact

- Since PCEF is co-located with P-GW, PCEF failure = P-GW failure (see Section 9.3.4)
- **Degraded mode:** If only policy engine fails, traffic may pass unpoliced (fail-open) or be blocked (fail-close) depending on implementation

- **Mitigation:** Hardware redundancy within P-GW chassis; hitless software upgrades

9.4 EPC Bearers

■ Bearer Concept

An EPS bearer is an **end-to-end logical channel** between the UE and the P-GW with specific QoS characteristics. Every bearer is defined by three components:

$$\text{Bearer} = \{ \text{TFT} + \text{QoS parameters} + \text{GTP tunnel(s)} \}$$

- **TFT (Traffic Flow Template):** Packet filters (5-tuple) that classify which IP packets belong to this bearer
- **QoS Parameters:** QCI, ARP, GBR/MBR (for GBR bearers), AMBR (aggregate)
- **GTP Tunnel:** Identified by TEID (Tunnel Endpoint Identifier) on each hop:
 - S1-U: eNB ↔ S-GW
 - S5/S8: S-GW ↔ P-GW

■ Bearer Structure Diagram

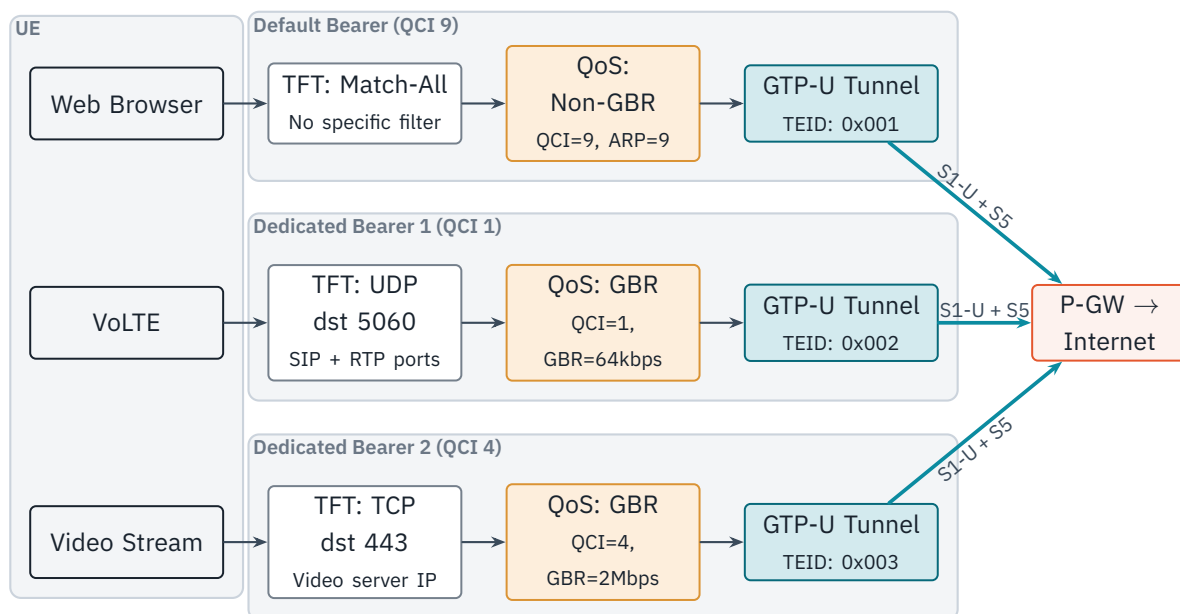


Figure 9.2 Default and dedicated bearers: TFT, QoS and GTP-U tunnel per bearer

9.4.1 Default Bearer

- **Created automatically** during PDN connection (Attach or additional PDN connectivity request)
- **Always-on:** Exists for the entire duration of the PDN connection
- **Non-GBR:** No guaranteed bit rate; best-effort with AMBR cap

- **Match-all TFT:** Carries all traffic that doesn't match any dedicated bearer's TFT
- **QoS from subscription:** QCI and AMBR values come from HSS subscription profile
- **One per PDN connection:** Each APN gets exactly one default bearer (EBI = EPS Bearer ID assigned)

9.4.2 Dedicated Bearer

- **Created on-demand:** Triggered by service request (PCRF rule) or UE/network request
- **Specific QoS:** Can be GBR (guaranteed) or Non-GBR (prioritized best-effort)
- **Specific TFT:** Packet filters direct matching traffic to this bearer
- **Linked to default bearer:** Shares the same PDN connection (same IP address)
- **Lifecycle:** Created via Bearer Resource Allocation → PCRF decision → P-GW initiates dedicated bearer activation
- **Example triggers:** VoLTE call start (QCI 1), video streaming (QCI 4), gaming (QCI 3)

9.4.3 QCI Table (Standardized Values)

QCI	Resource Type	Priority	Packet Delay Budget	Packet Error Loss Rate	Example Services
1	GBR	2	100 ms	10^{-2}	Conversational Voice (VoLTE)
2	GBR	4	150 ms	10^{-3}	Conversational Video (live)
3	GBR	3	50 ms	10^{-3}	Real-time Gaming
4	GBR	5	300 ms	10^{-6}	Non-conversational Video (buffered streaming)
5	Non-GBR	1	100 ms	10^{-6}	IMS Signaling (SIP)
6	Non-GBR	6	300 ms	10^{-6}	Video (buffered), TCP-based services (web, email)
7	Non-GBR	7	100 ms	10^{-3}	Voice, Video (live), Interactive Gaming
8	Non-GBR	8	300 ms	10^{-6}	Video (buffered), TCP-based (premium)
9	Non-GBR	9	300 ms	10^{-6}	Video (buffered), TCP-based (default/best-effort)

Key QoS Parameters:

- **QCI (QoS Class Identifier):** Scalar value (1-9) mapping to standardized packet treatment

- **ARP (Allocation and Retention Priority):** Priority level (1-15), pre-emption capability/vulnerability
- **GBR:** Guaranteed minimum bit rate (only for QCI 1-4)
- **MBR:** Maximum bit rate (caps GBR bearers)
- **UE-AMBR:** Aggregate MBR across all non-GBR bearers of a UE
- **APN-AMBR:** Aggregate MBR across all non-GBR bearers of one APN

9.5 Authentication in EPC

9.5.1 EPS-AKA (EPS Authentication and Key Agreement)

EPS-AKA provides **mutual authentication** between the UE and the network. Both parties prove knowledge of the permanent key K (stored in USIM and HSS) without transmitting it.

■ EPS-AKA Flow

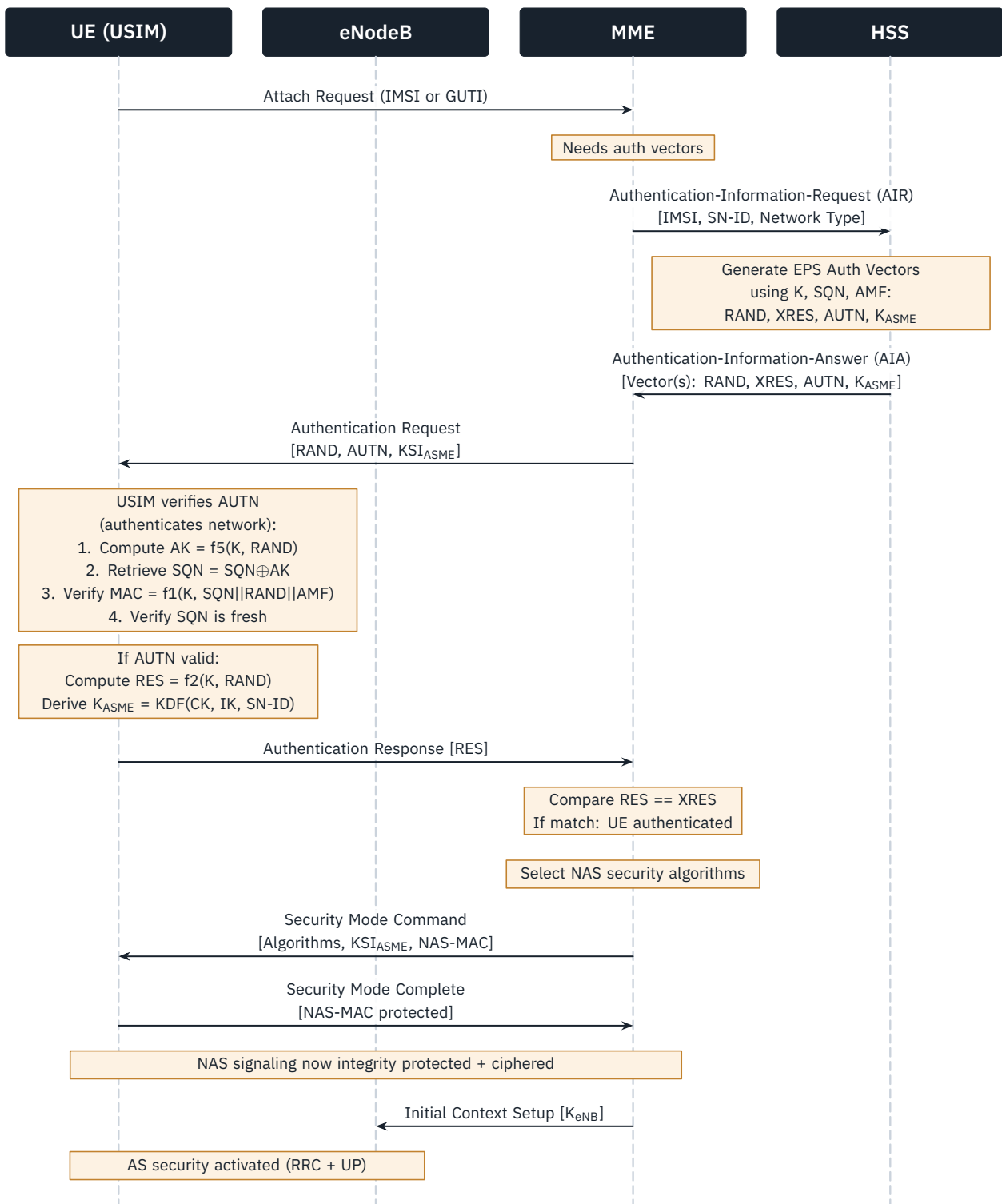


Figure 9.3 EPS-AKA authentication and NAS/AS security activation

■ Authentication Failure Cases

- **MAC failure** (UE rejects AUTN): Network is not authentic → UE sends Authentication Failure (cause: MAC failure)
- **SQN out of range**: Synchronization failure → UE sends AUTS for resync
- **XRES mismatch**: UE is not authentic → MME sends Authentication Reject

9.5.2 K_{ASME} Derivation

The K_{ASME} (Key for Access Security Management Entity) is the master key for all EPC security:

$$K_{ASME} = \text{KDF}(\text{CK} \parallel \text{IK}, \text{SN-ID}, \text{SQN} \oplus \text{AK})$$

where:

- KDF = HMAC-SHA-256 based key derivation function
- CK = cipher key = $f_3(K, \text{RAND})$
- IK = integrity key = $f_4(K, \text{RAND})$
- SN-ID = serving network identity (MCC || MNC)
- SQN = sequence number
- AK = anonymity key = $f_5(K, \text{RAND})$

■ Key Hierarchy

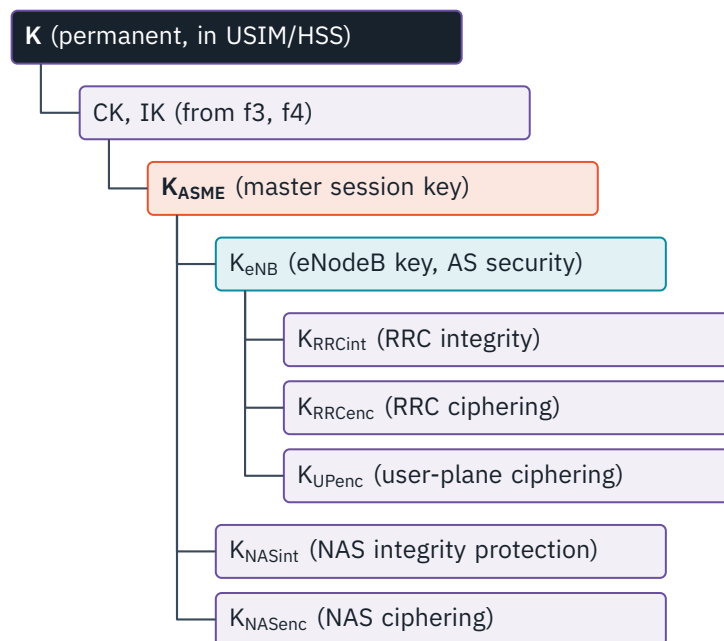


Figure 9.4 EPS key hierarchy

9.5.3 NAS Security

NAS security operates between UE and MME (transparent to eNB):

Protection	Algorithm Options	Purpose
Integrity (mandatory)	EIA1 (SNOW 3G), EIA2 (AES), EIA3 (ZUC)	Prevents message tampering
Ciphering (optional)	EEA0 (null), EEA1 (SNOW 3G), EEA2 (AES), EEA3 (ZUC)	Prevents eavesdropping

- **NAS COUNT:** 32-bit counter prevents replay attacks (separate UL/DL counts)
- **Security Mode Command:** Establishes algorithms; itself integrity-protected
- **Protected messages:** All NAS messages after SMC are integrity-protected; most are ciphered

9.5.4 AS Security (Access Stratum)

AS security operates between UE and eNB:

- K_{eNB} : Derived from K_{ASME} at MME, sent to eNB in Initial Context Setup
- **RRC protection:** Both integrity and ciphering (SRB1, SRB2)
- **User plane:** Ciphering only (no integrity for performance reasons in LTE)
- **Key refresh:** New K_{eNB} derived at handover ($K_{eNB}^* = \text{KDF}(K_{eNB}, \text{PCI}, \text{freq})$)
- **Activated via:** SecurityModeCommand (RRC level, distinct from NAS SMC)

9.6 Session Management

9.6.1 PDN Connectivity (Default Bearer Establishment)

PDN connectivity creates the **default EPS bearer** and assigns an IP address to the UE.

■ Procedure Flow

1. **UE → MME:** ESM PDN Connectivity Request (APN, PDN type: IPv4/IPv6/IPv4v6)
2. **MME → S-GW:** Create Session Request (IMSI, bearer QoS from HSS profile, APN)
3. **S-GW → P-GW:** Create Session Request (forwarded with S-GW F-TEID)
4. **P-GW:** Allocates IP address, contacts PCRF (CCR-Initial on Gx), installs default PCC rules
5. **P-GW → S-GW:** Create Session Response (IP address, bearer context, TFT)
6. **S-GW → MME:** Create Session Response (S-GW F-TEID, bearer context)
7. **MME → eNB:** Initial Context Setup Request (bearer context, K_{eNB} , NAS: Attach Accept + Activate Default Bearer)
8. **eNB → UE:** RRC Connection Reconfiguration (DRB setup) + NAS messages
9. **UE → eNB → MME:** Attach Complete / Activate Default EPS Bearer Context Accept
10. **MME → S-GW:** Modify Bearer Request (eNB F-TEID for downlink)

Result: End-to-end bearer established: UE ↔ eNB ↔ S-GW ↔ P-GW ↔ Internet

9.6.2 Dedicated Bearer Activation

Dedicated bearers are **network-initiated** (triggered by PCRF rule or operator policy):

1. **PCRF** → **P-GW**: Re-Auth-Request (RAR) with PCC rule (QCI, GBR, TFT filters)
2. **P-GW** → **S-GW**: Create Bearer Request (EBI, TFT, QoS, S5 F-TEID)
3. **S-GW** → **MME**: Create Bearer Request (forwarded with S1-U F-TEID)
4. **MME** → **eNB** → **UE**: Bearer Setup Request (NAS: Activate Dedicated Bearer Context Request)
5. **UE**: Accepts, stores TFT for uplink classification
6. **UE** → **eNB** → **MME**: Activate Dedicated EPS Bearer Context Accept
7. **MME** → **S-GW** → **P-GW**: Create Bearer Response (all TEIDs now bound)

9.6.3 Bearer Modification

Bearer modification changes QoS or TFT of an existing bearer:

- **Network-initiated**: PCRF updates PCC rule → P-GW sends Update Bearer Request
- **UE-initiated**: UE sends Bearer Resource Modification Request to MME
- **Examples**: Upgrade video quality (increase GBR), add new SDF filter, change gate status

9.6.4 Bearer Deactivation

- **Network-initiated**: PCRF removes PCC rule → P-GW sends Delete Bearer Request
- **UE-initiated**: UE sends Deactivate EPS Bearer Context Request
- **Default bearer deactivation**: Deletes ALL associated dedicated bearers + releases IP address
- **PDN disconnection**: Triggered when last default bearer is deactivated

9.7 Mobility Management

9.7.1 ECM States

State	Description	UE Location Known?	Resources
ECM-IDLE	No NAS signaling connection; UE monitors paging	TA-level only (TAI list)	No S1 connection, no DRB; S-GW buffers DL data
ECM-CONNECTED	Active NAS signaling via S1-MME	Cell-level (ECGI)	S1-U tunnel active; DRBs established

State Transitions:

- **IDLE** → **CONNECTED**: Service Request (UE-triggered) or Paging Response (network-triggered)
- **CONNECTED** → **IDLE**: S1 Release (inactivity timer expiry or explicit release)

9.7.2 Tracking Area Update (TAU)

TAU keeps the network informed of the UE's approximate location in ECM-IDLE state:

When TAU is triggered:

- UE enters a new Tracking Area (TA) not in its TAI list
- Periodic TAU timer expires (configurable, typically 54 minutes)
- After ISR (Idle-mode Signaling Reduction) activation

TAU Procedure:

1. UE → MME: Tracking Area Update Request (old GUTI, TAI visited)
2. MME: Verifies identity, optionally authenticates
3. MME → HSS: Update Location (if new MME)
4. MME → old MME: Context transfer (if MME changed)
5. MME → UE: TAU Accept (new GUTI, new TAI list)
6. MME → S-GW: Modify Bearer (if S-GW changed)

TAI List: MME assigns a list of TAs to reduce TAU frequency (load balancing vs. paging cost tradeoff)

9.7.3 X2 Handover (Intra-LTE, Without MME Change)

The **fastest handover** in LTE — direct eNB-to-eNB signaling without MME involvement for data forwarding:

1. **Source eNB:** Measurement reports indicate target cell is better
2. **Source → Target eNB (X2):** Handover Request (UE context, bearers, QoS)
3. **Target eNB:** Admits UE, allocates resources
4. **Target → Source eNB (X2):** Handover Request Acknowledge (DL forwarding address)
5. **Source eNB → UE:** RRC Connection Reconfiguration (handover command)
6. **Source eNB:** Starts forwarding buffered + new DL data to target eNB via X2
7. **UE:** Detaches from source, synchronizes to target cell
8. **UE → Target eNB:** RRC Connection Reconfiguration Complete
9. **Target eNB → MME:** Path Switch Request (new ECGI, new TEIDs)
10. **MME → S-GW:** Modify Bearer Request (target eNB address)
11. **S-GW:** Switches DL path to target eNB (end marker to source)
12. **MME → Target eNB:** Path Switch Request Acknowledge
13. **Target → Source eNB:** UE Context Release (cleanup)

Key point: S-GW is unchanged (local mobility anchor). Only the S1-U endpoint moves.

9.7.4 S1 Handover (With MME/S-GW Change)

Used when X2 is unavailable or when MME/S-GW relocation is needed:

1. **Source eNB → Source MME:** Handover Required (target ID, cause)
2. **Source MME → Target MME:** Forward Relocation Request (UE context) [if MME changes]
3. **Target MME → Target eNB:** Handover Request (bearer setup)

4. **Target eNB** → **Target MME**: Handover Request Acknowledge
5. **Target MME** → **Source MME**: Forward Relocation Response
6. **Source MME** → **Source eNB**: Handover Command
7. **Source eNB** → **UE**: RRC Connection Reconfiguration
8. **UE** → **Target eNB**: Handover Confirm
9. **Target eNB** → **Target MME**: Handover Notify
10. **Target MME** → **S-GW**: Modify Bearer / Create Session (if S-GW changes)

Indirect Data Forwarding: DL data forwarded Source eNB → Source S-GW → Target S-GW → Target eNB during handover gap.

9.7.5 Inter-RAT Mobility (LTE ↔ 3G)

■ LTE → 3G (PS Handover)

1. MME triggers handover to UTRAN (measurement reports or coverage)
2. MME → SGSN (S3): Forward Relocation Request (PDP contexts mapped from EPS bearers)
3. SGSN → RNC: Relocation Request
4. After handover: S-GW reroutes user plane via S4 to SGSN (or direct tunnel via S12 to RNC)
5. Bearer QoS mapped: QCI → UMTS Traffic Class (Conversational/Streaming/Interactive/Background)

■ 3G → LTE

1. SGSN detects LTE capability; initiates handover
2. SGSN → MME (S3): Forward Relocation Request
3. MME activates bearers, sets up S1 path
4. PDP contexts mapped to EPS bearers

9.8 Failure Scenarios

9.8.1 MME Failure

Aspect	Impact
Detection	eNB detects via SCTP association failure (heartbeat timeout ~30s)
Immediate effect	All S1-MME connections to failed MME are lost
ECM-CONNECTED UEs	Move to ECM-IDLE; eNB releases radio resources
Recovery action	UE sends Service Request / Attach to new MME (selected via S1AP weighting from MME pool)
Context recovery	New MME fetches subscription from HSS; active bearers re-established via new Create Session to S-GW
Data loss	Momentary — ongoing TCP sessions may survive (IP address preserved at P-GW); UDP/real-time sessions interrupted

Mitigation Strategies:

- MME pooling (multiple MMEs per tracking area)
- Active/Standby with session state replication
- S10-based context transfer from backup node
- DNS-based MME selection for load distribution

9.8.2 S-GW Failure

Aspect	Impact
Detection	MME detects via GTPv2-C Echo timeout on S11; eNB detects via GTP-U path failure on S1-U
Immediate effect	User plane interrupted for all UEs served by failed S-GW
Recovery action	MME selects new S-GW, sends Modify Bearer to P-GW with new S-GW address
Path switch	P-GW redirects S5 tunnel to new S-GW; MME updates eNB with new S1-U TEID
Data loss	Buffered DL data (for IDLE UEs) at failed S-GW is lost

Mitigation Strategies:

- S-GW pool with DNS-based selection (weighted records)
- GTP-U path failure detection (Echo Request/Response)
- N+1 redundancy; ICSR between chassis pairs

- Pre-established backup GTP contexts

9.8.3 P-GW Failure

Aspect	Impact
Detection	S-GW detects via GTP path failure on S5; PCRF detects via Diameter transport failure on Gx
Immediate effect	All PDN connections through failed P-GW are lost
IP address loss	UE IP addresses are released — cannot be recovered
Session impact	All TCP/UDP sessions terminated; VoLTE calls dropped
Recovery action	UE must establish new PDN connectivity (new default bearer, new IP address)
Application impact	DNS-based services recover; long-lived sessions (VPN, SSH) broken

Mitigation Strategies:

- Inter-Chassis Session Recovery (ICSR): Active/Standby P-GW pair with session state sync
- GTP-C restart counter: S-GW detects P-GW restart, reports to MME
- IP address pool replication between paired P-GWs
- P-GW selection via DNS with health monitoring

9.8.4 HSS Unreachable

Aspect	Impact
Detection	MME detects via Diameter transport failure or watchdog timeout on S6a
Existing sessions	Continue normally — bearer state is at MME/S-GW/P-GW
Authentication	MME uses cached authentication vectors (typically 5 vectors per UE fetched at last AIR)
New attaches	Fail once cached vectors exhausted — no new UEs can authenticate
Handover to new MME	Impacted — new MME cannot fetch subscription profile
Duration tolerance	Depends on cache depth; typically sustains operations for minutes to hours

Mitigation Strategies:

- Geo-redundant HSS pair (active/active or active/standby)
- Diameter Routing Agent (DRA) for automatic failover
- Increased auth vector batch size (e.g., 10 instead of 5)
- MME-level subscription caching with TTL

9.9 Summary

Component	Plane	Primary Role	Key Interface
MME	Control	Signaling, Auth, Mobility	S1-MME, S11, S6a
HSS	Control	Subscriber DB, Auth vectors	S6a
S-GW	User + Control	Local UP anchor, buffering	S1-U, S5/S8
P-GW	User + Control	IP anchor, policy enforcement	S5/S8, SGi, Gx
PCRF	Control	Policy decision	Gx, Rx
PCEF	User	Policy enforcement	Gx (at P-GW)

Key Takeaways

1. **EPC eliminates circuit-switching:** All services (including voice) over IP bearers
2. **Control/User plane split:** Enables independent scaling and evolution toward 5G
3. **Bearer-based QoS:** Deterministic quality for real-time services (QCI 1-4 = GBR)
4. **Security by design:** Mutual authentication (EPS-AKA), layered key derivation, mandatory integrity
5. **Mobility without disruption:** GTP anchoring at S-GW (local) and P-GW (global) preserves IP sessions
6. **Policy architecture:** PCRF/PCEF enable real-time, subscriber-aware traffic treatment
7. **Resilience through pooling:** MME pools, S-GW pools, and HSS redundancy minimize single points of failure

Module 9 – End

Diameter Protocol

Diameter پروتکل

10



CLIENT

AGENT

SERVER

HEADER · AVP · AVP · AVP

In This Module

1	Why Diameter Was Introduced	219
2	Diameter vs RADIUS Comparison	220
3	Diameter Architecture	220
4	Diameter Message Structure	221
5	Diameter Commands	224
6	Important Diameter Applications in LTE	227
7	Diameter in LTE Authentication (S6a Deep Dive)	230
	Gx Policy Flow (CCR-I/CCA-I at Session Start)	234
8	Diameter vs HTTP/2 in 5G	236
9	Diameter = Control Plane Only	237
	Summary	238

1 Why Diameter Was Introduced

RADIUS Limitations

RADIUS (Remote Authentication Dial-In User Service) was designed in the 1990s for dial-up access. As networks evolved toward 3G/4G, RADIUS could not meet modern requirements:

Limitation	Impact
UDP transport (unreliable)	Messages can be lost silently; no built-in retransmission
No server-initiated messages	Server cannot push policy changes or disconnect users proactively
Limited AVP space	8-bit attribute type field = max 255 attribute types
No application-layer ACK	Cannot confirm message delivery at application level
Weak security model	Shared secret + MD5 only; no hop-by-hop TLS built-in
No capability negotiation	Peers cannot discover supported features
Client-server only	No peer-to-peer; no agent/relay architecture

Diameter's Answer

Diameter (the name implies “twice the RADIUS”) was defined in **RFC 6733** to address all these limitations:

- Reliable transport (TCP/SCTP)
- Server-initiated messages (push model)
- 32-bit AVP code space (4+ billion attribute types)
- Built-in capability exchange and error handling
- Agent architecture for scalable routing
- Native TLS/DTLS support

⚠ **Key Concept:** Diameter is a **CONTROL PLANE** protocol only. It carries signaling (authentication, authorization, accounting, policy) — never user data traffic.

2 Diameter vs RADIUS Comparison

Feature	RADIUS	Diameter
RFC	RFC 2865/2866	RFC 6733
Transport	UDP (unreliable)	TCP/SCTP (reliable)
Port	1812/1813	3868
Connection	Per-request	Persistent peer connections
Direction	Client→Server only	Peer-to-peer (both can initiate)
AVP code space	8-bit (255 max)	32-bit (4 billion+)
Security	Shared secret + MD5	TLS/DTLS, IPsec
Max message size	4096 bytes	16 MB (24-bit length)
Failover	Application-dependent	Built-in (transport-level detection)
Capability negotiation	None	CER/CEA exchange
Server-initiated	No	Yes (e.g., RAR, ASR)
Agent support	Limited (proxy)	Relay, Proxy, Redirect, Translation
Accounting	Interim updates only	Real-time with guaranteed delivery
Vendor AVPs	Limited	Native vendor-ID support
Used in	Wi-Fi, VPN, ISP	LTE/4G core, IMS, VoLTE

3 Diameter Architecture

3.1 Diameter Nodes

Node Type	Role
Diameter Client	Generates requests (e.g., MME, P-GW)
Diameter Server	Processes requests, returns answers (e.g., HSS, PCRF, OCS)
Diameter Agent	Intermediary that routes/processes messages

■ Agent Types

Agent	Function
Relay	Routes messages based on realm; does NOT modify AVPs
Proxy	Routes messages AND may modify AVPs (policy enforcement)
Redirect	Returns routing information to sender (does not forward)
Translation	Translates between Diameter and other protocols (e.g., RADIUS↔Diameter)

3.2 Peer Connections

- **Transport:** TCP (default) or SCTP (preferred for multi-homing)
- **Port:** 3868 (plaintext) or 5868 (TLS)
- **Persistent:** Connection stays up; monitored by watchdog (DWR/DWA)
- **Peer-to-peer:** Both endpoints can initiate messages
- **Capability Exchange:** CER/CEA must succeed before application messages

3.3 Realm-Based Routing

Diameter routes messages based on **realms** (domain-like identifiers):

- Each node is configured with a realm (e.g., `epc.mnc001.mcc208.3gppnetwork.org`)
- Routing table maps realms → peer connections
- Agents use **Destination-Realm** AVP to determine next hop
- Enables multi-operator and roaming scenarios

4 Diameter Message Structure

4.1 Message Header (20 bytes fixed)

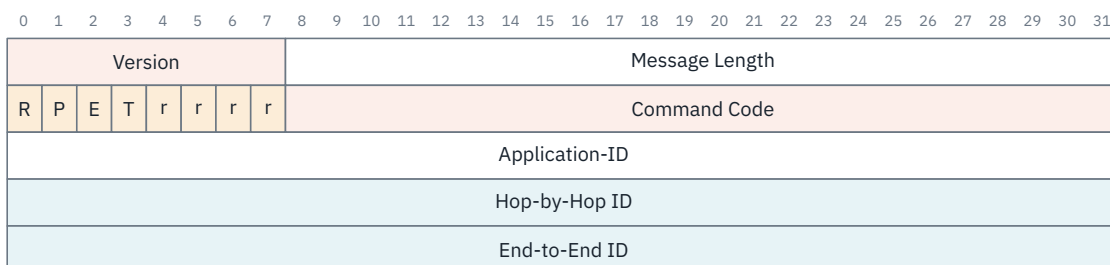


Figure 10.1 Diameter message header (20 bytes)

Field	Size	Description
Version	1 byte	Always 1
Message Length	3 bytes	Total message length including header
Flags	1 byte	R=Request, P=Proxiable, E=Error, T=Retransmit
Command Code	3 bytes	Identifies the command (e.g., 318 = AIR)
Application-ID	4 bytes	Identifies the application (e.g., 16777251 = S6a)
Hop-by-Hop ID	4 bytes	Matches request/answer on a single hop (unique per connection)
End-to-End ID	4 bytes	Matches request/answer end-to-end (unique globally)

■ Header Flags

Flag	Bit	Meaning
R	0	Request (1) or Answer (0)
P	1	Proxiable — can be relayed/proxied
E	2	Error — answer contains error
T	3	Retransmitted message

4.2 AVP (Attribute-Value Pair) Structure

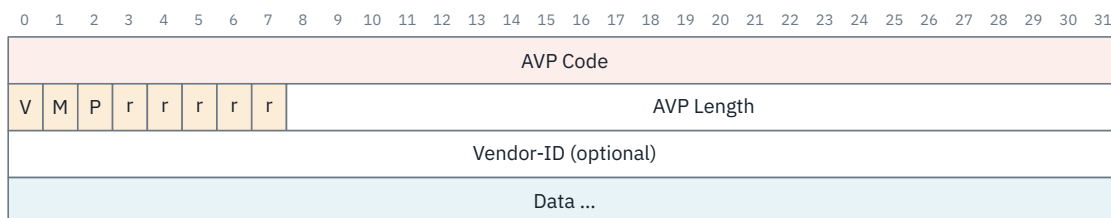


Figure 10.2 Diameter AVP structure

Field	Description
AVP Code	32-bit identifier for the attribute
V flag	Vendor-specific — Vendor-ID field is present
M flag	Mandatory — receiver MUST understand this AVP
P flag	Protected — end-to-end encryption needed
AVP Length	Total AVP length (header + data)
Vendor-ID	32-bit vendor identifier (present only if V=1)
Data	The actual value (various types)

■ AVP Data Types

Type	Description	Example
OctetString	Raw bytes	RAND, AUTN
UTF8String	Text string	User-Name
Unsigned32	32-bit unsigned integer	Result-Code
Unsigned64	64-bit unsigned integer	Accounting counters
Address	IP address (IPv4/IPv6)	Framed-IP-Address
Time	Seconds since Jan 1, 1900	Event-Timestamp
Enumerated	Named integer values	Auth-Session-State
Grouped	Contains other AVPs	Subscription-Data

4.3 Grouped AVPs

Grouped AVPs contain other AVPs as their data — enabling hierarchical structures:

```
Subscription-Data (Grouped AVP)
├── MSISDN: +33612345678
├── Access-Restriction-Data: 0x00000000
├── Subscriber-Status: SERVICE_GRANTED
├── APN-Configuration-Profile (Grouped)
│   ├── Context-Identifier: 1
│   ├── APN-Configuration (Grouped)
│   │   ├── Service-Selection: "internet"
│   │   ├── PDN-Type: IPv4v6
│   │   └── EPS-Subscribed-QoS-Profile (Grouped)
│   │       ├── QoS-Class-Identifier: 9
│   │       └── Allocation-Retention-Priority (Grouped)
│   │           ├── Priority-Level: 15
│   │           └── Pre-emption-Capability: NOT_PRE_EMPT
```

Key AVPs Table

AVP Name	Code	Type	Used In
Session-Id	263	UTF8String	All messages
Origin-Host	264	DiameterIdentity	All messages
Origin-Realm	296	DiameterIdentity	All messages
Destination-Host	293	DiameterIdentity	Requests
Destination-Realm	283	DiameterIdentity	Requests
Result-Code	268	Unsigned32	All answers
Auth-Session-State	277	Enumerated	Auth messages
User-Name	1	UTF8String	User identification (IMSI)
Visited-PLMN-Id	1407	OctetString	S6a messages
Subscription-Data	1400	Grouped	S6a (ULA)
Charging-Rule-Install	1001	Grouped	Gx (CCA)
CC-Request-Type	416	Enumerated	Gy/Gx (CCR)

5 Diameter Commands

5.1 Naming Convention

All Diameter commands follow the pattern:

- **Request:** X-Request (abbreviated XR) — R flag = 1
- **Answer:** X-Answer (abbreviated XA) — R flag = 0

Every request MUST receive exactly one answer.

5.2 Base Protocol Commands (Application-ID = 0)

Command	Code	Abbreviation	Purpose
Capabilities-Exchange-Request	257	CER	Initiate peer connection, exchange capabilities
Capabilities-Exchange-Answer	257	CEA	Respond with own capabilities
Device-Watchdog-Request	280	DWR	Heartbeat — verify peer is alive
Device-Watchdog-Answer	280	DWA	Confirm alive status
Disconnect-Peer-Request	282	DPR	Gracefully close connection
Disconnect-Peer-Answer	282	DPA	Acknowledge disconnection
Accounting-Request	271	ACR	Send accounting record
Accounting-Answer	271	ACA	Acknowledge accounting record
Abort-Session-Request	274	ASR	Server requests session termination
Abort-Session-Answer	274	ASA	Client acknowledges termination
Re-Auth-Request	258	RAR	Server requests re-authentication
Re-Auth-Answer	258	RAA	Client acknowledges re-auth

5.3 Diameter Message Exchange Flow

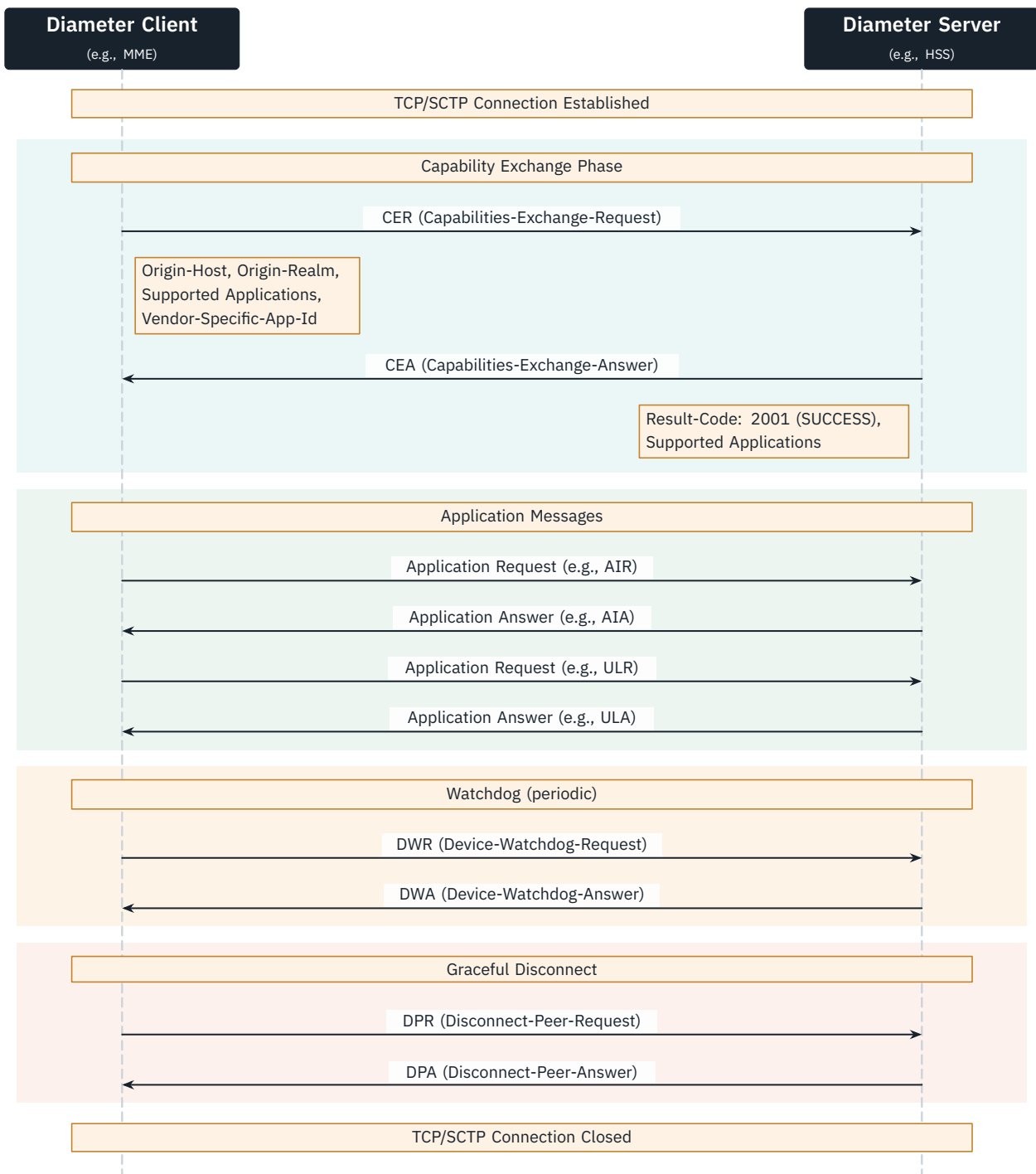


Figure 10.3 Diameter peer life cycle: capability exchange, application messages, watchdog and disconnect

5.4 Result Codes

Code	Name	Meaning
2001	DIAMETER_SUCCESS	Request processed successfully
3xxx	Protocol Errors	Redirect, unable to deliver
4xxx	Transient Failures	Retry may succeed (e.g., 4012 = DIAMETER_UNABLE_TO_COMPLY)
5xxx	Permanent Failures	Do not retry (e.g., 5001 = DIAMETER_AVP_UNSUPPORTED)
5004	DIAMETER_UNKNOWN_SESSION_ID	Session not found
5012	DIAMETER_UNABLE_TO_COMPLY	General permanent failure

6 Important Diameter Applications in LTE

6.1 Overview of LTE Diameter Interfaces

Interface	Endpoints	Application-ID	Purpose
S6a	MME ↔ HSS	16777251	Authentication, subscriber data
Gx	PCRF ↔ P-GW	16777238	Policy and Charging Control (PCC)
Gy	OCS ↔ P-GW	4 (Credit-Control)	Online charging
Rx	AF ↔ PCRF	16777236	Application-level QoS/media authorization
S13	MME ↔ EIR	16777252	Equipment identity check (IMEI)
S6d	SGSN ↔ HSS	16777251	2G/3G authentication via Diameter
SWx	3GPP AAA ↔ HSS	16777265	Non-3GPP access authentication
Sh	AS ↔ HSS	16777217	IMS service data

6.2 S6a Interface (MME ↔ HSS)

The most critical Diameter interface in LTE — handles all subscriber authentication and data management.

Command	Code	Direction	Purpose
Authentication-Information-Request	318	MME→HSS	Request authentication vectors
Authentication-Information-Answer	318	HSS→MME	Return auth vectors (RAND, AUTN, XRES, K_{ASME})
Update-Location-Request	316	MME→HSS	Register UE location at MME
Update-Location-Answer	316	HSS→MME	Return subscription data
Purge-UE-Request	321	MME→HSS	Inform HSS that UE data purged
Purge-UE-Answer	321	HSS→MME	Acknowledge purge
Cancel-Location-Request	317	HSS→MME	HSS tells MME to detach UE
Cancel-Location-Answer	317	MME→HSS	Acknowledge cancel
Notify-Request	323	MME→HSS	Notify HSS of terminal info
Notify-Answer	323	HSS→MME	Acknowledge notification

6.3 Gx Interface (PCRF ↔ P-GW)

Controls **Policy and Charging Control (PCC)** — decides what QoS and charging rules apply to each session.

Command	Code	Direction	Purpose
CC-Request (Initial)	272	P-GW→PCRF	New IP-CAN session — request PCC rules
CC-Answer (Initial)	272	PCRF→P-GW	Install PCC rules (QoS, gates, charging)
CC-Request (Update)	272	P-GW→PCRF	Session modification (e.g., new bearer)
CC-Answer (Update)	272	PCRF→P-GW	Updated PCC rules
CC-Request (Termination)	272	P-GW→PCRF	Session ended
CC-Answer (Termination)	272	PCRF→P-GW	Acknowledge termination
Re-Auth-Request	258	PCRF→P-GW	Push new PCC rules mid-session
Re-Auth-Answer	258	P-GW→PCRF	Acknowledge rule update

CC-Request-Type values: INITIAL(1), UPDATE(2), TERMINATION(3), EVENT(4)

6.4 Gy Interface (OCS ↔ P-GW)

Handles **online charging** — credit reservation and debit in real-time.

Command	Code	Direction	Purpose
CC-Request (Initial)	272	P-GW→OCS	Reserve initial credit quota
CC-Answer (Initial)	272	OCS→P-GW	Grant credit (units/time/volume)
CC-Request (Update)	272	P-GW→OCS	Report usage, request more credit
CC-Answer (Update)	272	OCS→P-GW	Grant additional credit or deny
CC-Request (Termination)	272	P-GW→OCS	Final report, release reservation
CC-Answer (Termination)	272	OCS→P-GW	Acknowledge final

Gx vs Gy: **Gx** = policy rules (what QoS to apply), **Gy** = charging (how much credit remains)

6.5 Rx Interface (AF ↔ PCRF)

Allows application functions (e.g., IMS P-CSCF for VoLTE) to request specific QoS from the network.

Command	Code	Direction	Purpose
AA-Request	265	AF→PCRF	Request media authorization (SDP info)
AA-Answer	265	PCRF→AF	Confirm media session authorized
Session-Termination-Request	275	AF→PCRF	Media session ended
Session-Termination-Answer	275	PCRF→AF	Acknowledge
Re-Auth-Request	258	PCRF→AF	Notify AF of bearer changes
Abort-Session-Request	274	PCRF→AF	PCRF aborts media session

6.6 S13 Interface (MME ↔ EIR)

Equipment identity check — validates IMEI against blacklist/greylist.

Command	Code	Direction	Purpose
ME-Identity-Check-Request	324	MME→EIR	Send IMEI for validation
ME-Identity-Check-Answer	324	EIR→MME	Return equipment status (white/grey/black)

7 Diameter in LTE Authentication (S6a Deep Dive)

7.1 Authentication Flow

When a UE attaches to the LTE network, the MME uses S6a to:

1. **Obtain authentication vectors** from the HSS (AIR/AIA)
2. **Register UE location** at the MME (ULR/ULA)

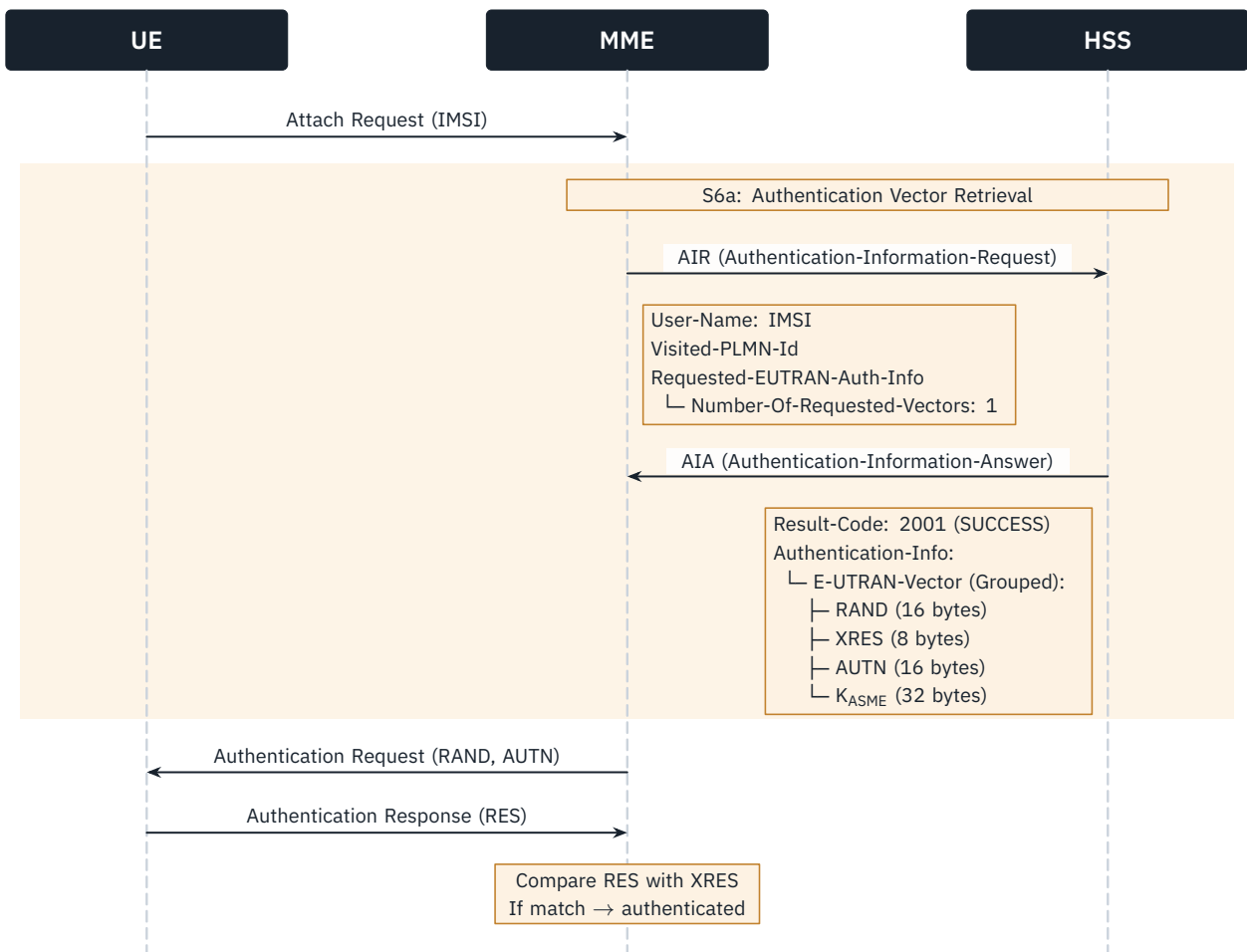


Figure 10.4 S6a messages and their AVPs during attach (part 1 of 2)

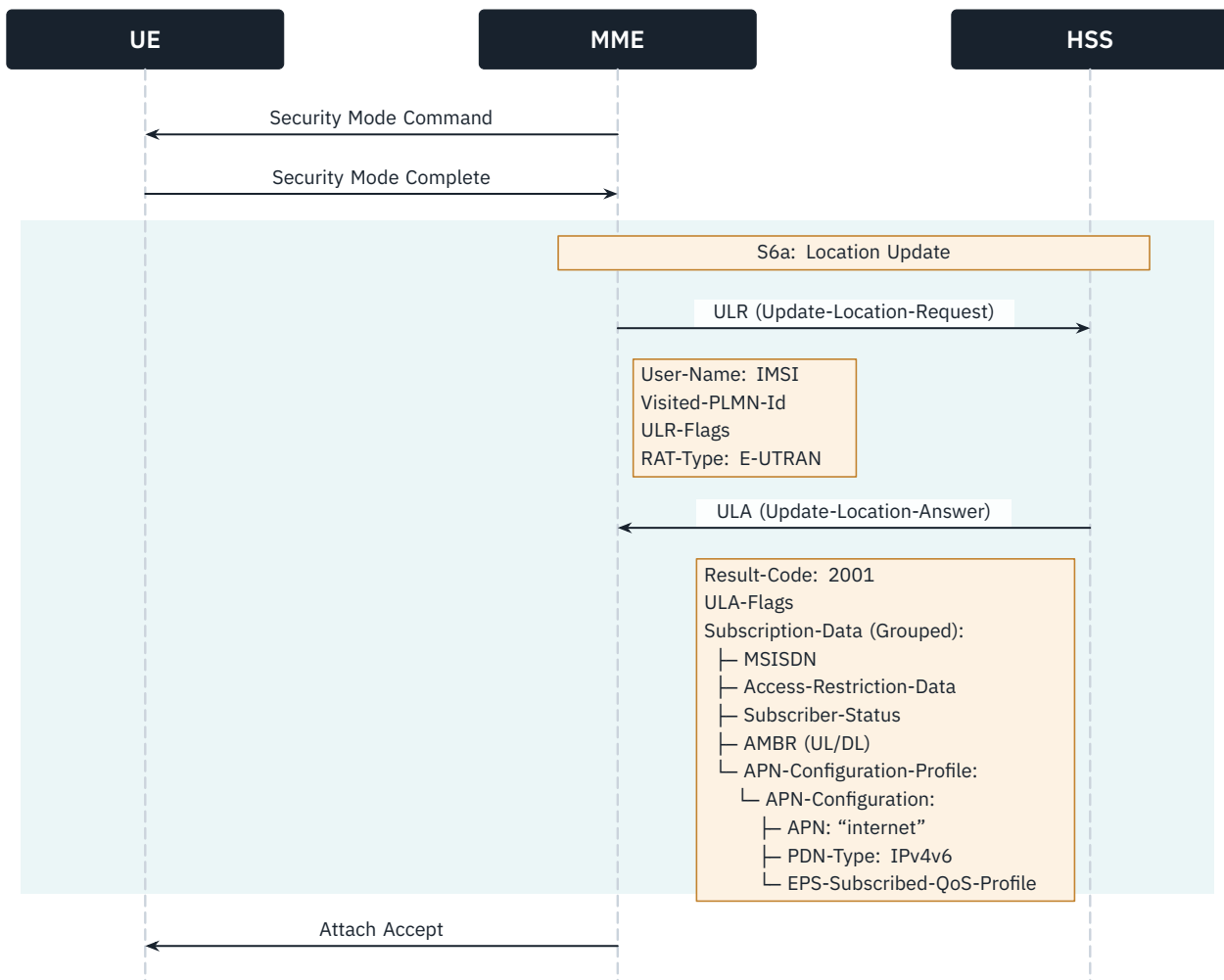


Figure 10.5 S6a messages and their AVPs during attach (part 2 of 2)

7.2 AIR (Authentication-Information-Request) Details

Purpose: MME requests fresh authentication vectors from HSS

Key AVPs in AIR:

| AVP | Purpose |

|--|----|

| User-Name | IMSI of the subscriber |

| Visited-PLMN-Id | 3-byte MCC+MNC of visited network |

| Requested-EUTRAN-Authentication-Info | Specifies how many vectors needed |

| Number-Of-Requested-Vectors | Typically 1-5 |

| Re-Synchronization-Info | Sent if previous auth failed (SQN sync) |

7.3 AIA (Authentication-Information-Answer) Details

Purpose: HSS returns E-UTRAN authentication vectors

Key AVPs in AIA:

| AVP | Purpose |

|--|----|

| Result-Code | 2001=success, 5001=user unknown |

- | Authentication-Info (Grouped) | Contains the vectors |
- | E-UTRAN-Vector (Grouped) | One complete auth vector |
- | └─ RAND | 128-bit random challenge |
- | └─ XRES | Expected Response (for verification) |
- | └─ AUTN | Authentication Token (for mutual auth) |
- | └─ K_{ASME} | Key for NAS/AS security derivation |

7.4 ULR (Update-Location-Request) Details

Purpose: MME registers itself as serving MME for this subscriber

Key AVPs in ULR:

- | AVP | Purpose |
- |---|----|
- | User-Name | IMSI |
- | Visited-PLMN-Id | Serving network identity |
- | RAT-Type | E-UTRAN (LTE) |
- | ULR-Flags | Indicates initial attach vs TAU |

7.5 ULA (Update-Location-Answer) Details

Purpose: HSS acknowledges and returns complete subscription profile

Key AVPs in ULA:

- | AVP | Purpose |
- |---|----|
- | Result-Code | Success or failure |
- | Subscription-Data (Grouped) | Complete subscriber profile |
- | └─ MSISDN | Phone number |
- | └─ AMBR | Aggregate Max Bit Rate (UL + DL) |
- | └─ APN-Configuration-Profile | All configured APNs |
- | └─ Subscriber-Status | SERVICE_GRANTED or BARRED |

Gx Policy Flow (CCR-I/CCA-I at Session Start)

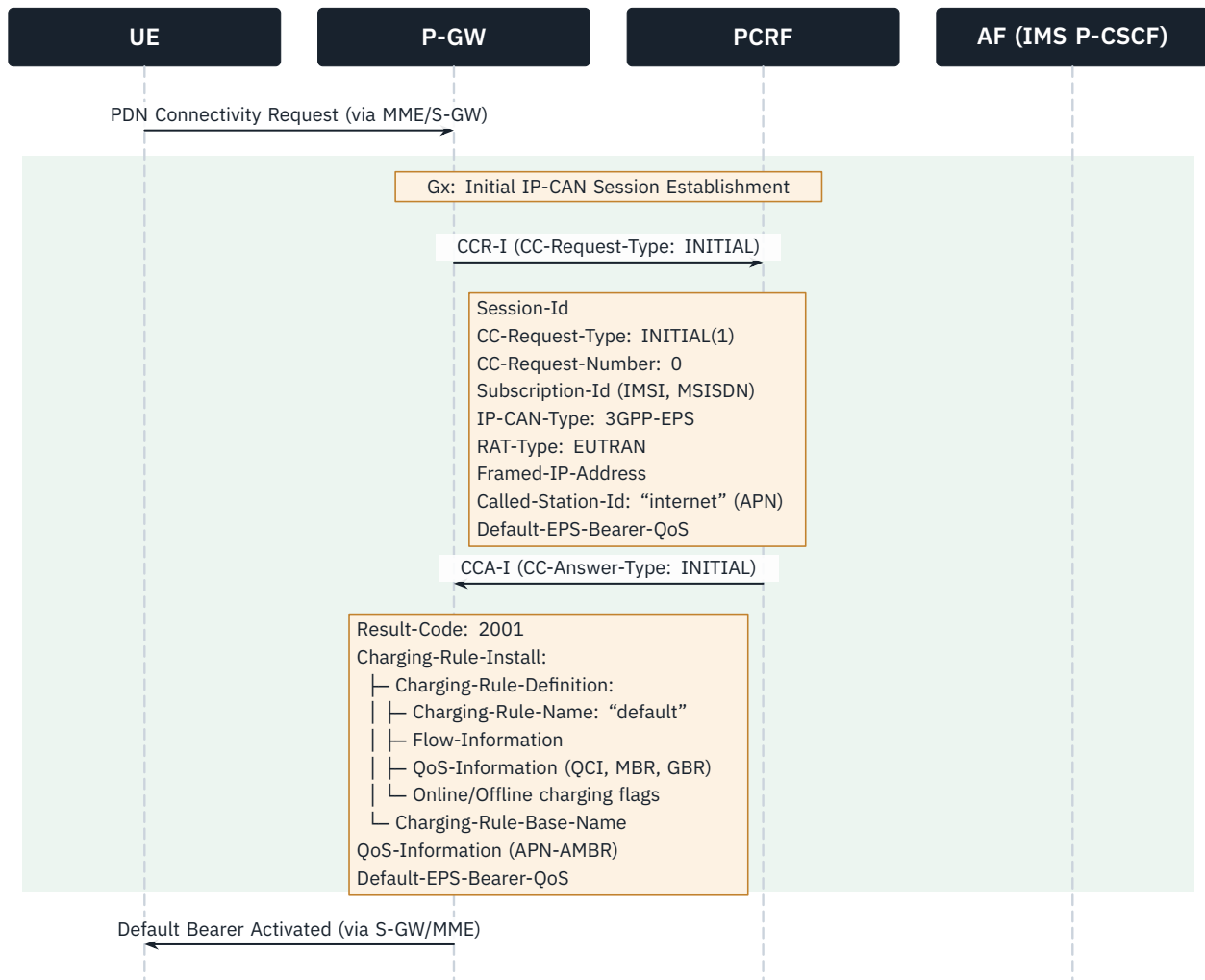


Figure 10.6 Gx and Rx policy flow for a default bearer and a VoLTE call (part 1 of 2)

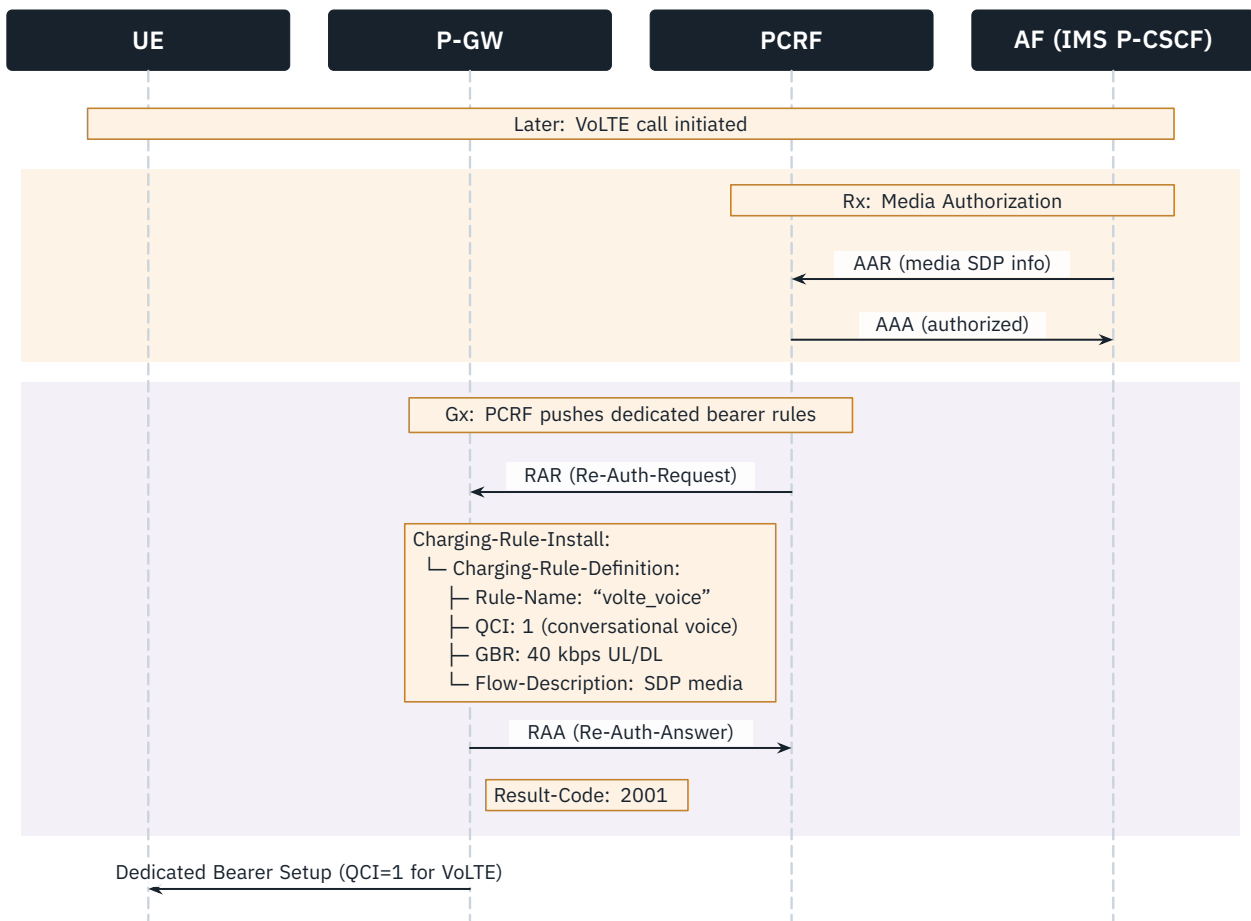


Figure 10.7 Gx and Rx policy flow for a default bearer and a VoLTE call (part 2 of 2)

8 Diameter vs HTTP/2 in 5G

8.1 The Transition

Aspect	4G (Diameter)	5G (HTTP/2 + JSON)
Protocol	Diameter (binary, TCP/SCTP)	HTTP/2 (text-based headers, binary framing)
Data format	AVPs (binary TLV)	JSON (human-readable)
Architecture	Point-to-point peers	Service-Based Interface (SBI)
Discovery	Static peer config / DRA	NRF (Network Repository Function)
Interface naming	S6a, Gx, Gy, Rx	Nausf, Npcf, Nchf, Naf
Session state	Stateful (session-id maintained)	Stateless (REST-like)
Connection model	Persistent peer connections	HTTP request/response
Scaling	Vertical (bigger peers)	Horizontal (cloud-native, k8s)
Message routing	Diameter Routing Agents (DRA)	Service Mesh / API Gateway
Standard	IETF RFC 6733 + 3GPP	3GPP + IETF HTTP/2 (RFC 7540)

8.2 Why 5G Moved Away from Diameter

1. **Simpler tooling:** HTTP/2 + JSON leverages existing web infrastructure (proxies, load balancers, API gateways)
2. **Web-native:** Standard REST APIs — easier for developers, standard debugging tools
3. **Stateless design:** Enables cloud-native microservices, horizontal scaling, Kubernetes orchestration
4. **Cloud-friendly:** No persistent peer connections; standard load balancing works
5. **Flexibility:** JSON schema evolution is easier than binary AVP definitions
6. **Ecosystem:** Massive HTTP/2 ecosystem (libraries, tools, monitoring)

8.3 Diameter Equivalent Functions in 5G

4G Diameter Interface	5G SBI Equivalent	NF Involved
S6a (MME↔HSS)	Nudm (UDM services)	AMF ↔ UDM/AUSF
Gx (PCRF↔P-GW)	Npcf (Policy services)	SMF ↔ PCF
Gy (OCS↔P-GW)	Nchf (Charging services)	SMF ↔ CHF
Rx (AF↔PCRF)	Npcf + Naf	AF ↔ PCF (via NEF)

8.4 Migration: 4G/5G Interworking

Diameter is NOT gone — it persists at interworking boundaries:

- **Diameter Routing Agent (DRA)** still used in 4G
- **Interworking Function (IWF)** translates Diameter ↔ HTTP/2 at 4G/5G border
- NSA (Non-Standalone) 5G still uses EPC with Diameter
- Full migration to SA (Standalone) 5G eliminates Diameter within the core

9 Diameter = Control Plane Only

■ Critical Distinction

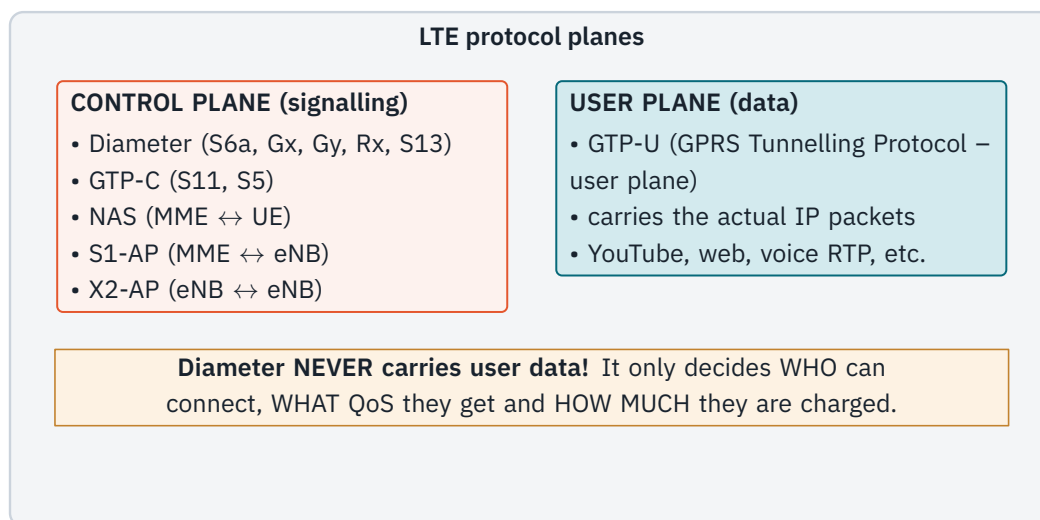


Figure 10.8 Diameter belongs to the control plane only

What Diameter Controls vs What Carries Data

Function	Protocol	Plane
Authenticate subscriber	Diameter S6a	Control
Authorize QoS policy	Diameter Gx	Control
Check credit balance	Diameter Gy	Control
Authorize media session	Diameter Rx	Control
Validate IMEI	Diameter S13	Control
Carry YouTube video	GTP-U	User
Carry VoLTE voice (RTP)	GTP-U	User
Carry web browsing	GTP-U	User

★ **Remember:** Diameter tells the network *what to do*. GTP-U *does it*.

Summary

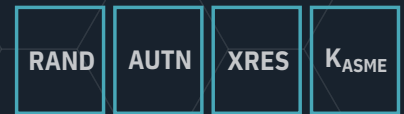
Topic	Key Takeaway
Why Diameter	RADIUS too limited for mobile core (no reliable transport, no push, limited AVPs)
Architecture	Client/Server/Agent model with persistent TCP/SCTP peers and realm routing
Messages	20-byte header + AVPs; every Request gets exactly one Answer
S6a	Authentication (AIR/AIA) + Location Update (ULR/ULA) — the backbone of LTE mobility
Gx	Policy control — installs QoS rules on bearers (CCR/CCA + RAR/RAA)
Gy	Online charging — real-time credit control
Rx	Application QoS — IMS/VoLTE media authorization
5G transition	HTTP/2 + JSON replaces Diameter; IWF for interworking
Scope	Control plane ONLY — never carries user data

Module 10 — Diameter Protocol | Advanced Communication Networks

Home Subscriber Server (HSS)

سرور مشترک خانگی (HSS)

11



In This Module

1	Why a Centralized Subscriber Database?	240
2	HSS Role in LTE/EPC	240
3	HSS Data Model	242
4	Authentication Vector Generation	244
5	HSS Interaction with MME (S6a / Diameter)	245
6	HSS Interaction with Other Network Elements	247
7	Authentication Scenario (Step-by-Step)	249
8	Evolution to 5G: HSS Decomposition	252
	Summary	254
	References	255

1 Why a Centralized Subscriber Database?

In any mobile network, thousands to millions of subscribers connect, move, and consume services simultaneously. A centralized subscriber database is essential for:

Function	Purpose
Authentication	Verify subscriber identity before granting network access
Authorization	Determine what services/resources a subscriber is allowed to use
Location Tracking	Know which network element currently serves each subscriber
Subscription Management	Store and enforce service plans, QoS profiles, and restrictions

Without a centralized database:

- Any device could impersonate a subscriber
- The network cannot route incoming calls/data to the correct location
- Service differentiation (premium vs. basic plans) would be impossible
- Roaming between networks would not function

The **HSS (Home Subscriber Server)** is the evolution of the 2G/3G HLR (Home Location Register), re-designed for all-IP LTE/EPC networks using the Diameter protocol instead of SS7/MAP.

2 HSS Role in LTE/EPC

The HSS is the **master database** for all subscriber and subscription data in the Evolved Packet Core (EPC):

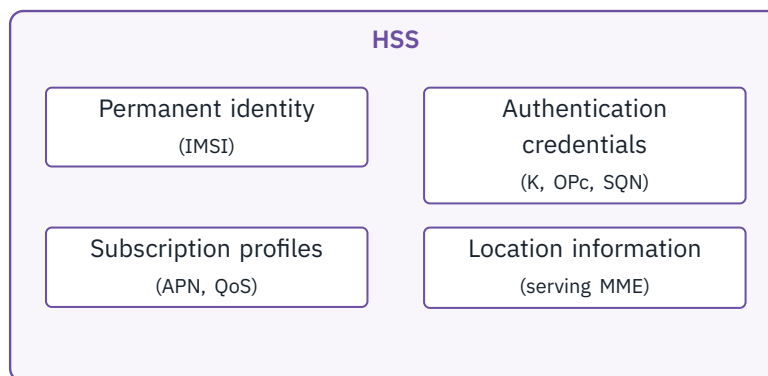


Figure 11.1 What the HSS stores

■ Key Responsibilities:

1. Stores Permanent Subscriber Identity (IMSI)

- International Mobile Subscriber Identity — globally unique 15-digit identifier
- Format: MCC (3 digits) + MNC (2-3 digits) + MSIN (9-10 digits)

2. Stores Authentication Credentials (K, OPc)

- **K**: 128-bit permanent secret key (shared only between USIM and HSS)
- **OPc**: Operator-specific constant derived from OP (operator key) and K
- Never transmitted over the air — used to generate temporary credentials

3. Stores Subscription Profiles (APN, QoS, Allowed Services)

- Which APNs the subscriber can connect to
- Maximum bitrates (AMBR) for uplink/downlink
- Quality of Service class identifiers (QCI)
- Allowed services and restrictions

4. Tracks UE Location (Serving MME)

- Records which MME currently serves the UE
- Enables routing of incoming services (MT-SMS, paging)
- Updated every time UE performs a Tracking Area Update (TAU) or attach

3 HSS Data Model

■ HSS Data Model Diagram

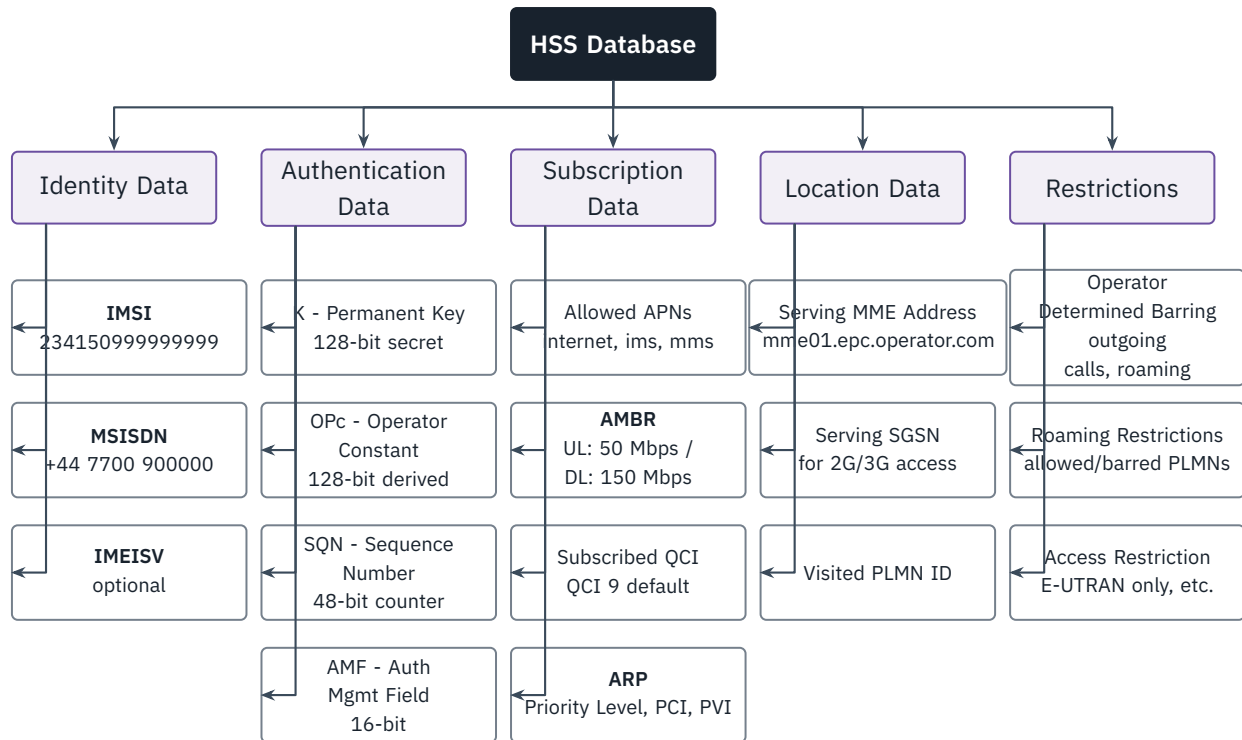


Figure 11.2 HSS data model: identity, authentication, subscription, location and restrictions

Detailed Data Fields

Category	Field	Description	Example
Identity	IMSI	Unique subscriber ID	234150999999999
Identity	MSISDN	Phone number (E.164)	+447700900000
Auth	K	Permanent 128-bit secret key	0x465B5CE8...(hex)
Auth	OPc	Operator variant constant	0xE8ED289D...(hex)
Auth	SQN	Sequence number (anti-replay)	000000000021
Auth	AMF	Authentication Management Field	0x8000
Subscription	Default APN	Primary data connection	"internet"
Subscription	APN list	All allowed APNs	internet, ims, mms
Subscription	UE-AMBR	Max aggregate bitrate	DL:150M, UL:50M
Subscription	APN-AMBR	Per-APN max bitrate	DL:100M, UL:40M
Subscription	QCI	QoS Class Identifier	9 (best effort)
Subscription	ARP	Allocation/Retention Priority	Level 8, PCI=yes, PVI=no
Location	MME Identity	Serving MME hostname/address	mme01.epc.mnc015.mcc234
Location	MME Realm	Diameter realm of MME	epc.mnc015.mcc234.3gppnetwork.org
Location	Purged flag	UE detached/unreachable	true/false
Restrictions	ODB	Operator Determined Barring	Barring of outgoing intl calls
Restrictions	Roaming	Allowed/barred PLMNs	Home only / EU roaming
Restrictions	RAT restriction	Allowed access technologies	E-UTRAN, UTRAN

4 Authentication Vector Generation

■ EPS-AKA (Evolved Packet System - Authentication and Key Agreement)

The HSS generates **Authentication Vectors (AVs)** that enable mutual authentication between UE and network without ever transmitting the permanent key K .

Each Authentication Vector contains:

Component	Size	Purpose
RAND	128 bits	Random challenge (generated fresh by HSS)
AUTN	128 bits	Authentication Token (proves network identity to UE)
XRES	32-128 bits	Expected Response (what UE should reply)
K_{ASME}	256 bits	Base key for deriving all session keys

■ Milenage Algorithm Set

The **Milenage** algorithm set is the standard cryptographic toolkit (3GPP TS 35.205-208):

Inputs: K (128-bit), $RAND$ (128-bit), SQN (48-bit), AMF (16-bit), OPc (128-bit)

Milenage function	Output	Purpose
$f1(K, RAND, SQN, AMF)$	MAC-A	Network authentication
$f1^*(K, RAND, SQN, AMF)$	MAC-S	Resynchronisation
$f2(K, RAND)$	RES / XRES	Response
$f3(K, RAND)$	CK	Cipher key
$f4(K, RAND)$	IK	Integrity key
$f5(K, RAND)$	AK	Anonymity key
$f5^*(K, RAND)$	AK	Anonymity key for resynchronisation

Vector derivation process:

1. HSS generates random 128-bit **RAND**
2. HSS increments **SQN** (sequence number)
3. Compute: $MAC-A = f1(K, RAND, SQN, AMF)$
4. Compute: $XRES = f2(K, RAND)$
5. Compute: $CK = f3(K, RAND)$
6. Compute: $IK = f4(K, RAND)$
7. Compute: $AK = f5(K, RAND)$
8. Construct: $AUTN = (SQN \oplus AK) || AMF || MAC-A$
9. Derive: $KASME = KDF(CK, IK, SN-ID, SQN \oplus AK)$

■ SQN Management (Replay Protection)

- **SQN** is a 48-bit counter maintained independently by HSS and USIM
- Prevents replay of old authentication vectors
- UE verifies: received SQN > last accepted SQN (within a window)
- If SQN is out of range → UE sends **AUTS** (resynchronization token)
- HSS uses f1* and f5* to resynchronize its SQN with the USIM

■ Vector Delivery to MME

- HSS typically generates **multiple vectors** (batch) per request
- MME stores unused vectors for subsequent re-authentications
- Reduces signaling load (MME doesn't need to contact HSS every time)
- Typical batch size: 3-5 vectors per AIR request

5 HSS Interaction with MME (S6a / Diameter)

The **S6a interface** uses the **Diameter protocol** (RFC 6733) with the 3GPP Diameter application for EPS (application ID: 16777251).

■ Message Flows

Procedure	Request → Answer	Direction	Purpose
Authentication	AIR → AIA	MME → HSS	Get auth vectors
Location Update	ULR → ULA	MME → HSS	Register MME, get subscription
Purge	PUR → PUA	MME → HSS	UE explicitly detached
Cancel Location	CLR → CLA	HSS → MME	Force detach (e.g., UE moved)
Insert Sub Data	IDR → IDA	HSS → MME	Push subscription changes
Delete Sub Data	DSR → DSA	HSS → MME	Remove subscription data
Notification	NOR → NOA	MME → HSS	Notify terminal info

■ Attach Procedure (AIR/AIA + ULR/ULA)

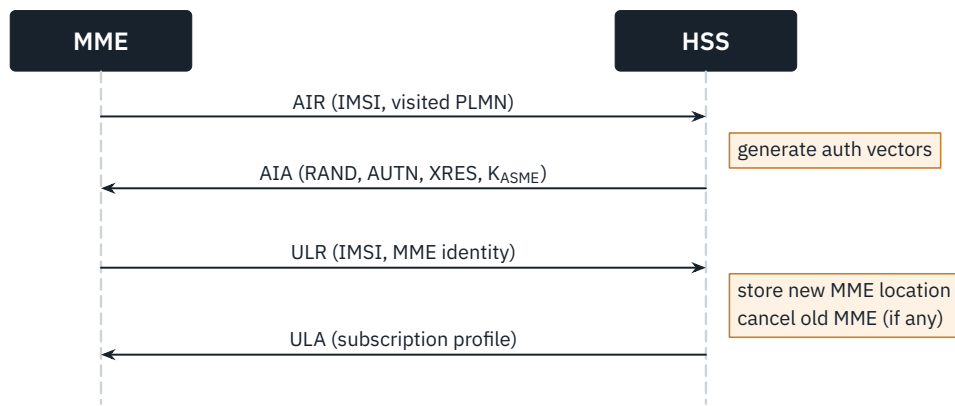


Figure 11.3 S6a exchange between the MME and the HSS during attach

AIR (Authentication-Information-Request):

- IMSI of the subscriber
- Visited PLMN ID
- Number of requested vectors
- Resynchronization info (AUTS, if SQN mismatch)

AIA (Authentication-Information-Answer):

- One or more EPS Authentication Vectors (RAND, AUTN, XRES, K_{ASME})
- Result code (success/failure)

ULR (Update-Location-Request):

- IMSI
- MME identity (hostname + realm)
- RAT type
- ULR flags (initial attach, etc.)

ULA (Update-Location-Answer):

- Subscription data (all APNs, AMBR, QoS profiles)
- Result code
- Separation of default APN from additional APNs

■ Detach / Purge (PUR/PUA or CLR/CLA)

Explicit detach (UE-initiated):

- MME sends **PUR** → HSS marks subscriber as “not attached”
- HSS responds with **PUA**

Network-initiated cancel (HSS pushes):

- HSS sends **CLR** to old MME (e.g., UE registered at new MME)
- Old MME removes UE context, responds with **CLA**
- CLR reasons: UE moved, subscription withdrawn, operator policy

■ Subscription Modification (IDR/IDA)

When operator modifies a subscription (e.g., customer upgrades plan):

- HSS sends **IDR** (Insert-Subscriber-Data-Request) to serving MME
- MME updates local copy of subscription data
- MME responds with **IDA**
- Changes take effect immediately (no re-attach needed)

6 HSS Interaction with Other Network Elements

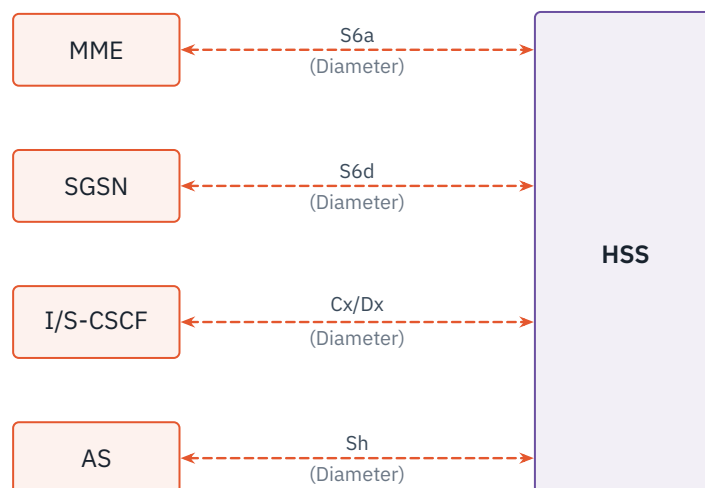


Figure 11.4 Diameter interfaces of the HSS

Interface Details

Interface	Peer Entity	Protocol	Purpose
S6a	MME	Diameter	EPS authentication, location, subscription
S6d	SGSN	Diameter	2G/3G interworking (same functions as S6a for GERAN/UTRAN)
Cx	I-CSCF / S-CSCF	Diameter	IMS registration, user authorization
Dx	I-CSCF	Diameter	IMS location query (find S-CSCF for user)
Sh	Application Server	Diameter	Read/write user data for services
S13	EIR	Diameter	Equipment identity check (IMEI verification)

S6d (SGSN – 2G/3G interworking)

- Same message set as S6a but adapted for GERAN/UTRAN access
- Used when UE is connected via 2G/3G radio
- Enables seamless handover between 4G and 2G/3G (IRAT)

Cx/Dx (IMS – VoLTE)

- Cx:** Registration/de-registration of IMS users
 - UAR/UAA: User-Authorization (during REGISTER)
 - SAR/SAA: Server-Assignment (assign S-CSCF)
 - MAR/MAA: Multimedia-Auth (get IMS auth vectors)
 - LIR/LIA: Location-Info (find assigned S-CSCF)
- Used for VoLTE call setup and IMS registration

Sh (Application Servers)

- Allows AS to read/update user-specific data
- Used for supplementary services, presence, messaging
- Supports subscriptions for notifications on data changes

7 Authentication Scenario (Step-by-Step)

Full EPS-AKA Authentication Flow

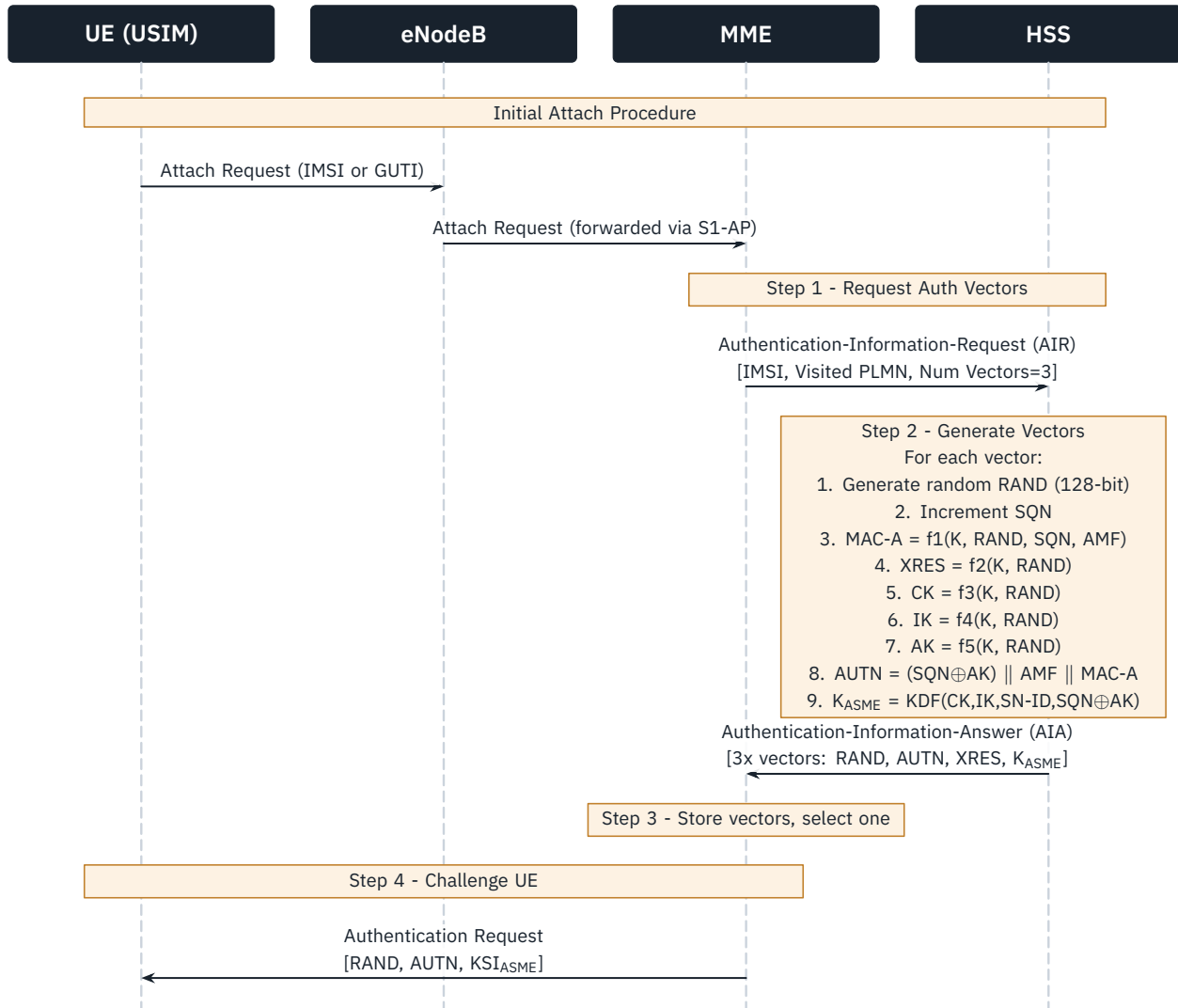


Figure 11.5 Full EPS-AKA authentication flow with the key computations (part 1 of 2)

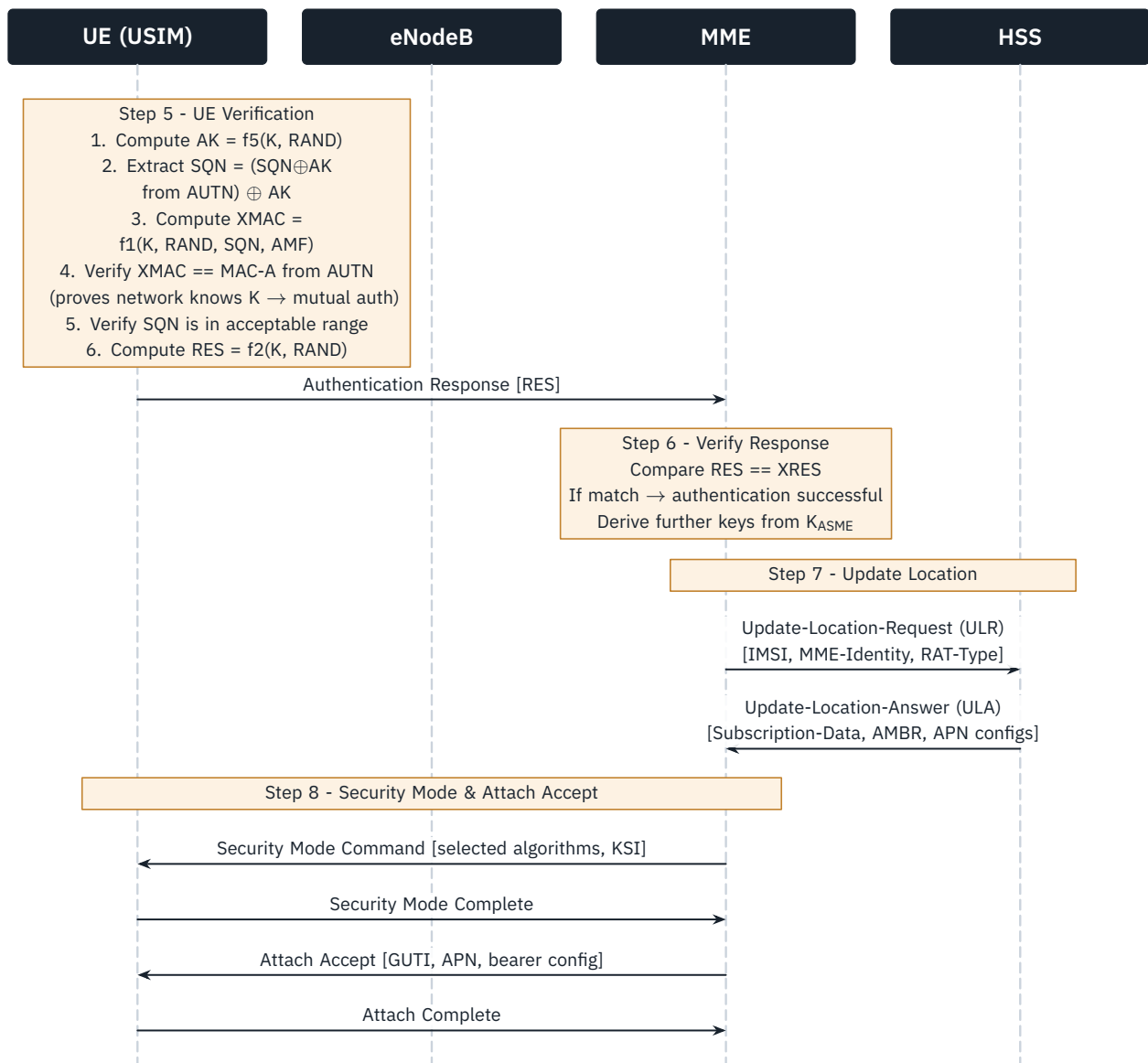


Figure 11.6 Full EPS-AKA authentication flow with the key computations (part 2 of 2)

■ Step-by-Step Explanation

Step	Action	Details
1	UE sends Attach Request	Contains IMSI (first time) or old GUTI (subsequent)
2	MME requests vectors	AIR sent to HSS with subscriber IMSI
3	HSS generates vectors	Uses $K + OPc + RAND + SQN$ through Milenage
4	HSS returns vectors	AIA contains $RAND, AUTN, XRES, K_{ASME}$ (batch of 3-5)
5	MME challenges UE	Sends $RAND + AUTN$ to UE in NAS Authentication Request
6	UE verifies network	Checks MAC in AUTN using its own $K \rightarrow$ mutual authentication
7	UE computes RES	Sends response back to MME
8	MME verifies UE	Compares RES against XRES from HSS
9	Location update	MME registers itself at HSS as serving MME
10	Session setup	Keys derived, bearers established

■ Authentication Failure Scenarios

Scenario	Cause	Recovery
MAC failure at UE	Network doesn't know correct K	UE sends Auth Failure (cause: MAC failure)
SQN out of range	SQN desynchronized	UE sends Auth Failure + AUTS for resync
XRES mismatch at MME	UE doesn't know correct K	MME rejects attach, may send Auth Reject
Timeout	Network/UE unreachable	Retry with backoff

8 Evolution to 5G: HSS Decomposition

Architecture Comparison

In 5G (3GPP Release 15+), the monolithic HSS is decomposed into specialized Network Functions:

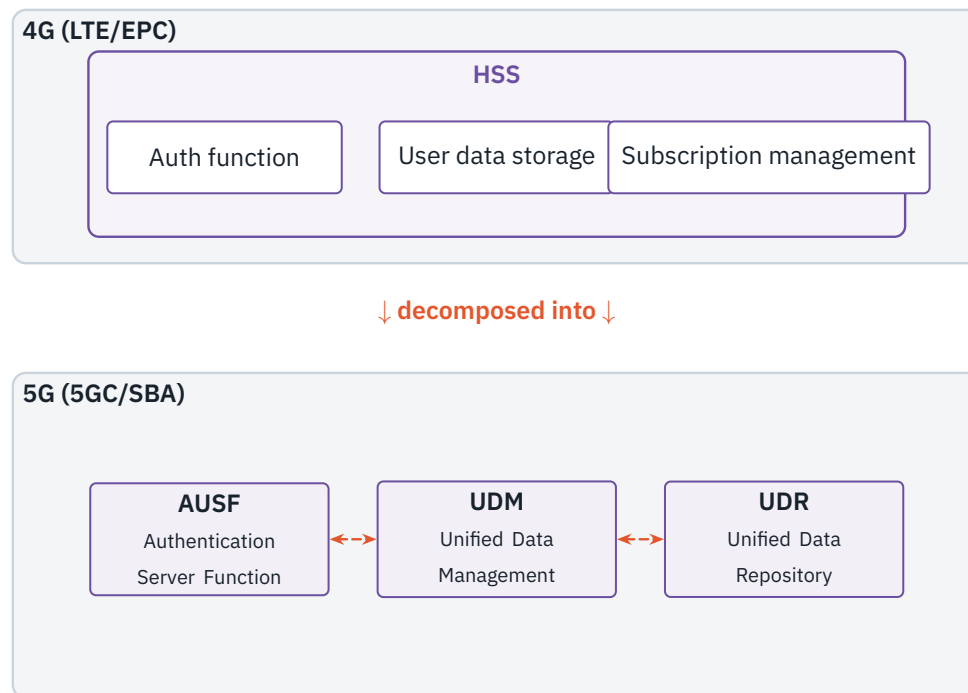


Figure 11.7 The HSS is decomposed into AUSF, UDM and UDR in 5G

Detailed Comparison: HSS (4G) vs UDM + AUSF + UDR (5G)

Aspect	HSS (4G)	UDM + AUSF + UDR (5G)
Architecture	Monolithic	Decomposed, microservices
Protocol	Diameter (S6a)	HTTP/2 + JSON (SBI)
Auth Function	Built into HSS	Separate AUSF (Authentication Server Function)
Data Management	Built into HSS	Separate UDM (Unified Data Management)
Data Storage	Internal DB	Separate UDR (Unified Data Repository)
Identity	IMSI (sent in clear)	SUPI/SUCI (SUCI = encrypted SUPI)
Privacy	IMSI exposed on air	SUCI conceals identity (ECIES encryption)
Auth Protocol	EPS-AKA only	5G-AKA or EAP-AKA' (extensible)
Key Hierarchy	$K_{ASME} \rightarrow K_{eNB} \rightarrow \dots$	$K_{AUSF} \rightarrow K_{SEAF} \rightarrow K_{AMF} \rightarrow K_{gNB} \rightarrow \dots$
Scalability	Scale entire HSS	Scale each NF independently
Interfaces	Point-to-point Diameter	Service-Based Interface (REST APIs)
Discovery	Static config / DNS	NRF (Network Repository Function)
Redundancy	Active/Standby	Cloud-native, stateless NFs + shared UDR

5G Identity: SUPI and SUCI

SUPI (Subscription Permanent Identifier)
 = IMSI (same format as 4G: MCC+MNC+MSIN)
 (never sent over the air in 5G!)

SUCI (Subscription Concealed Identifier)
 = Home Network ID + Routing Indicator + Protection Scheme + Encrypted MSIN

Encryption: ECIES (Elliptic Curve Integrated Encryption Scheme)

- Home network publishes public key
- UE encrypts MSIN portion with home network public key
- Only home network (UDM) can decrypt → reveals SUPI

5G Authentication Methods

5G-AKA:

- Enhanced version of EPS-AKA
- Adds home network confirmation (HRES/HXRES)
- Serving network proves to home network that auth completed
- Prevents certain MITM attacks possible in 4G

EAP-AKA':

- Extensible Authentication Protocol variant

- Binds authentication to serving network name
- Better suited for non-3GPP access (Wi-Fi, fixed)
- Uses AT_KDF_INPUT for network binding

5G Decomposed Functions

5G NF	Role	4G Equivalent
AUSF	Handles authentication procedures, generates keys	HSS auth function
UDM	Manages subscription data, generates auth vectors, handles registration	HSS data management
UDR	Stores all data (subscriber, policy, application)	HSS internal database
SIDF	Decrypts SUCI → SUPI (part of UDM)	N/A (no equivalent in 4G)

Summary

The HSS is the security and identity anchor of the LTE/EPC network. It:

1. **Guards the keys** — K never leaves the HSS (and USIM)
2. **Proves identity** — Mutual authentication via EPS-AKA
3. **Controls access** — Subscription profiles determine what users can do
4. **Tracks location** — Knows which MME serves each subscriber
5. **Evolves gracefully** — Decomposed into UDM/AUSF/UDR in 5G for cloud-native deployment

Understanding the HSS is critical for:

- Network security analysis
- Debugging authentication failures
- Planning subscriber provisioning
- Understanding the transition from 4G to 5G core

References

- 3GPP TS 29.272: S6a/S6d Diameter-based interface (MME/SGSN–HSS)
- 3GPP TS 33.401: EPS security architecture
- 3GPP TS 35.205-208: Milenage algorithm specification
- 3GPP TS 23.501: 5G System Architecture
- 3GPP TS 33.501: 5G Security Architecture and Procedures
- 3GPP TS 29.509: AUSF Services (5G)
- 3GPP TS 29.503: UDM Services (5G)

Mobility Management Entity (MME)

موجودیت مدیریت تحرک (MME)

12



In This Module

1	Why MME Exists	257
2	MME Functions	257
3	NAS Protocol Stack	258
4	Detailed Procedures	260
5	Bearer Management	268
6	MME Selection	269
7	S1-Flex and MME Pooling	269
8	Evolution: MME → AMF (5G)	270

1 Why MME Exists

The MME is the **central control-plane entity** in the LTE/EPC architecture. It exists to:

- **Offload the RAN from core signaling:** In 2G/3G, the RNC/BSC handled significant core-network signaling. LTE's flat architecture moved all NAS (Non-Access Stratum) signaling termination to the MME, simplifying eNodeB (eNB) design.
- **Centralize control-plane intelligence:** Authentication, mobility management, bearer setup, and paging are all orchestrated by the MME.
- **Decouple control and user planes:** The MME handles no user-plane traffic — it purely manages signaling, enabling independent scaling of CP and UP.

Key Interfaces

Interface	Peer Node	Protocol	Purpose
S1-MME	eNB	S1-AP (SCTP)	NAS transport, handover signaling
S6a	HSS	Diameter	Authentication vectors, subscription data
S11	S-GW	GTPv2-C	Bearer/session management
S10	Other MME	GTPv2-C	Inter-MME handover, context transfer
S3	SGSN	GTPv2-C	2G/3G ↔ LTE mobility

2 MME Functions

2.1 NAS Signaling Termination

- The MME is the NAS endpoint — it terminates all NAS messages from the UE (transported transparently through eNB via S1-AP).
- NAS messages are integrity-protected and ciphered between UE and MME.

2.2 Mobility Management

- Tracking Area (TA) management and TA List assignment
- Handover signaling (S1-based and X2 with MME involvement)
- Inter-RAT mobility (to/from 2G/3G via S3/S10)

2.3 Authentication and Security

- Retrieves authentication vectors from HSS (via S6a)
- Runs EPS-AKA (Authentication and Key Agreement)
- Derives NAS keys (K_{NASint} , K_{NASenc}) and AS keys (K_{eNB})
- Initiates NAS Security Mode Command

2.4 Bearer Management

- Establishes default EPS bearers during attach
- Coordinates dedicated bearer activation/modification/deactivation
- Interacts with S-GW (S11) and indirectly P-GW

2.5 Paging

- Pages UEs in ECM-IDLE state when downlink data arrives
- Sends S1-AP Paging messages to all eNBs in the UE's registered TA list

2.6 Idle-Mode Reachability

- Maintains UE context even when UE is IDLE (ECM-IDLE)
- Knows which Tracking Areas the UE may be in
- Triggers paging when data or signaling arrives

2.7 MME Selection, Pooling, and Rebalancing

- Multiple MMEs serve the same area (MME Pool)
- Load balancing via relative capacity (weight factor) advertised to eNBs
- Rebalancing: graceful transfer of UE contexts when adding/removing MMEs

3 NAS Protocol Stack

NAS (Non-Access Stratum) operates between UE and MME, transparent to eNB.

3.1 EMM — EPS Mobility Management

Manages UE registration and mobility:

- Attach / Detach
- Tracking Area Update (TAU)
- Authentication (EPS-AKA)
- Security Mode Control
- GUTI Reallocation

- Service Request

EMM States:

- EMM-DEREGISTERED: UE not registered
- EMM-REGISTERED: UE attached to network

3.2 ESM – EPS Session Management

Manages PDN connections and EPS bearers:

- PDN Connectivity Request/Accept
- Bearer Resource Allocation
- Bearer Modify / Deactivate
- PDN Disconnect

ESM operates piggybacked on EMM during attach (ESM message embedded in Attach Request).

4 Detailed Procedures

4.1 LTE Initial Attach (Full Flow)

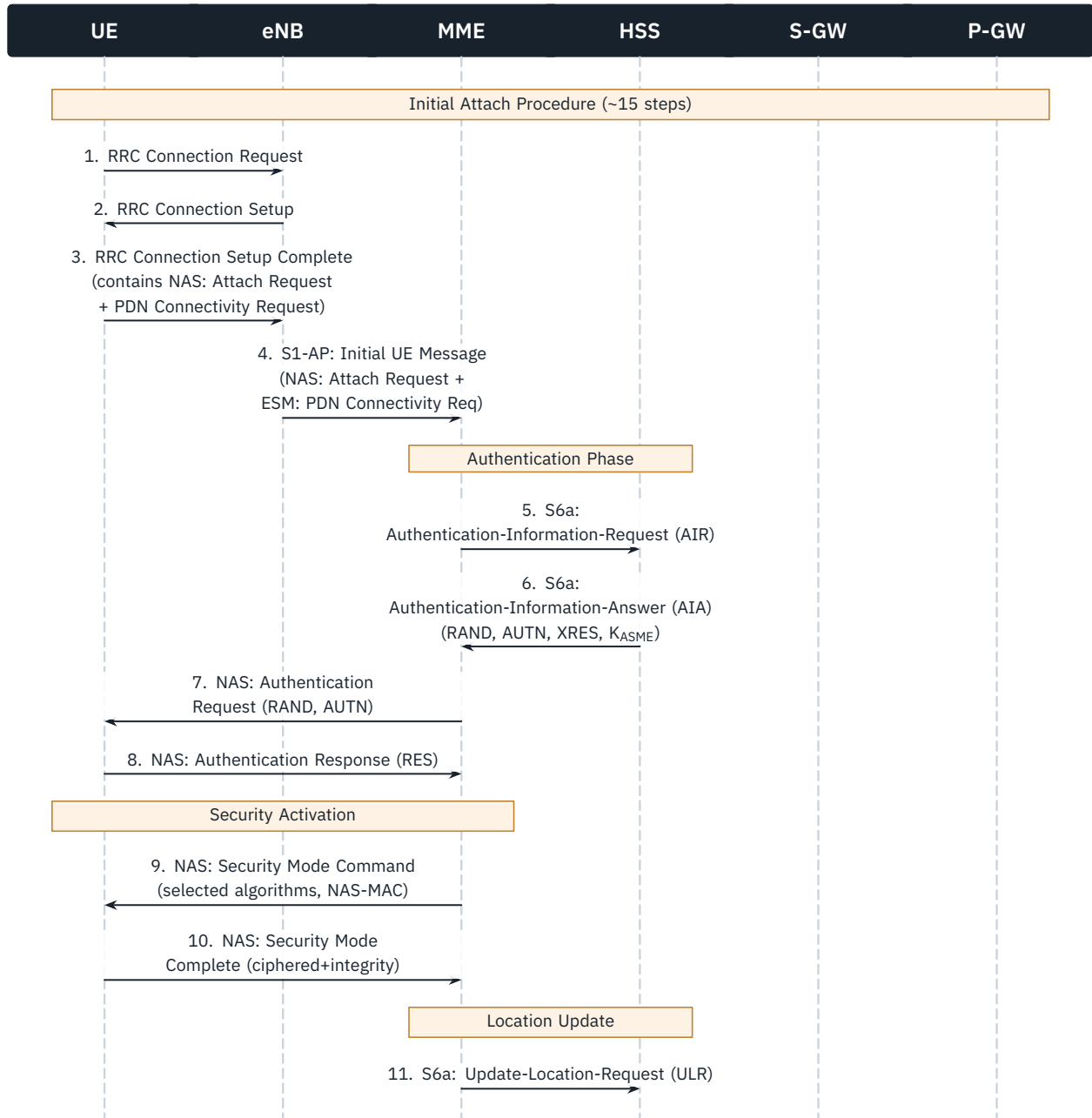


Figure 12.1 LTE initial attach (full flow) (part 1 of 2)

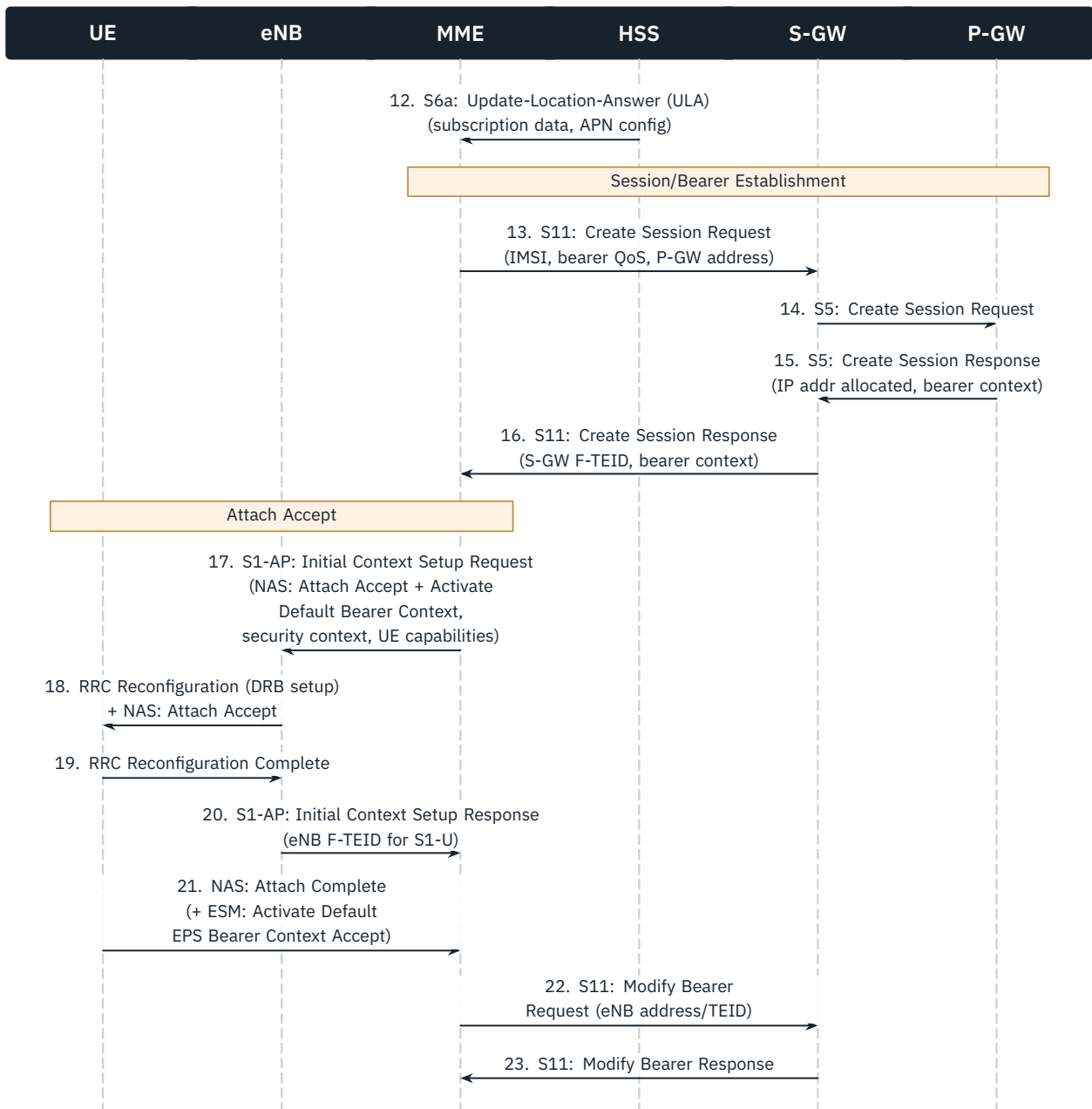


Figure 12.2 LTE initial attach (full flow) (part 2 of 2)

Key Points:

- Steps 7-10: EPS-AKA + NAS security activation
- Steps 13-16: GTP-C session establishment across S11/S5
- The Attach Request carries a piggybacked PDN Connectivity Request (ESM)
- The UE gets its IP address from P-GW (step 15)

4.2 Authentication and Security Mode

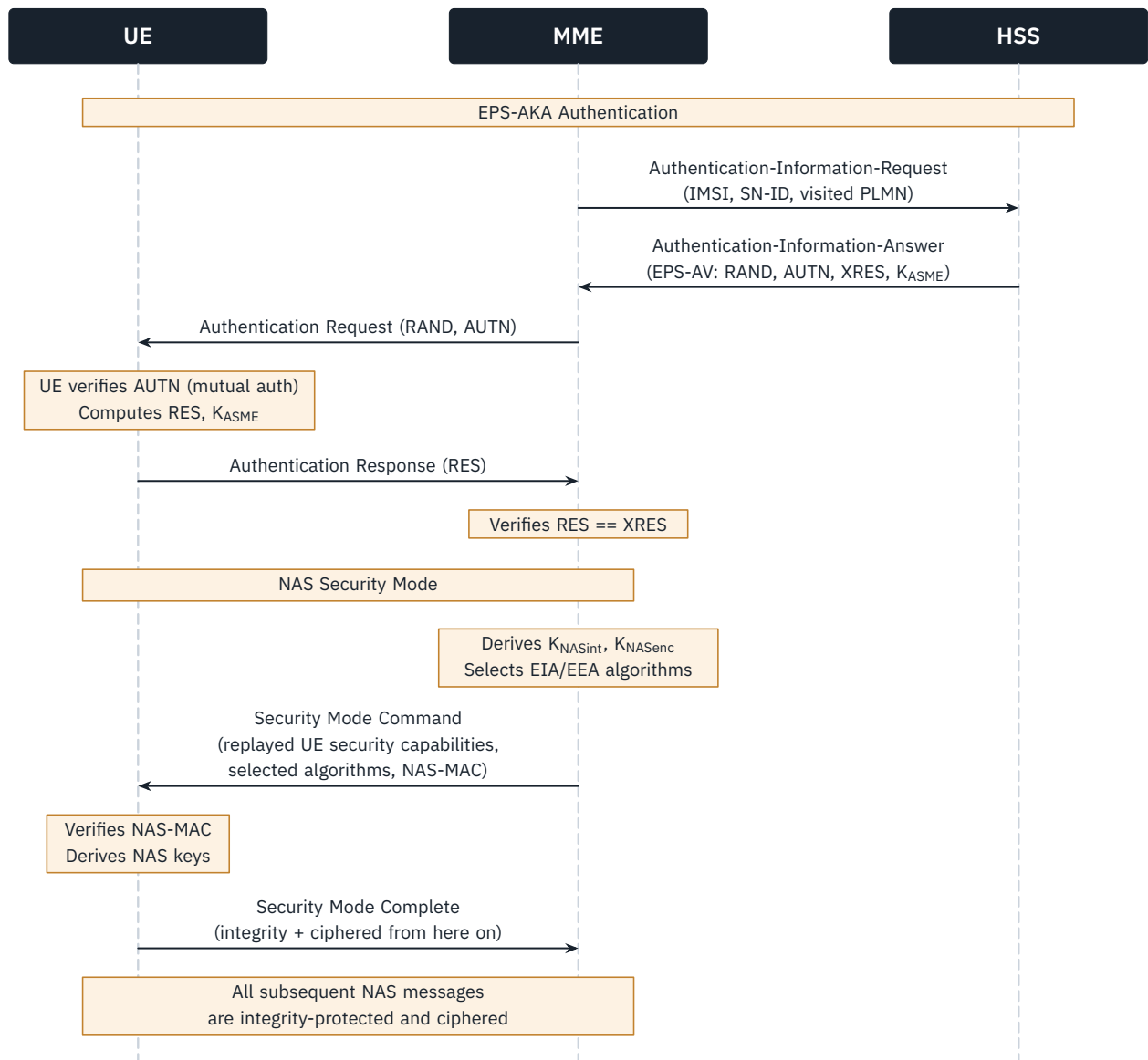


Figure 12.3 EPS-AKA authentication and NAS security mode

Key Derivation Hierarchy:

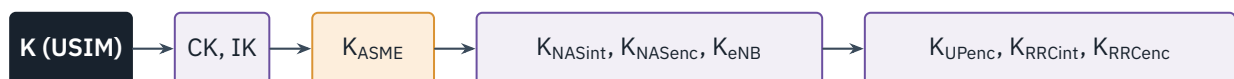


Figure 12.4 Key derivation hierarchy

4.3 Tracking Area Update (TAU)

Triggers:

- **Periodic TAU:** Timer T3412 expires (default ~54 min) — confirms UE is still reachable
- **Mobility-triggered TAU:** UE moves to a cell whose TAC is not in its TA List

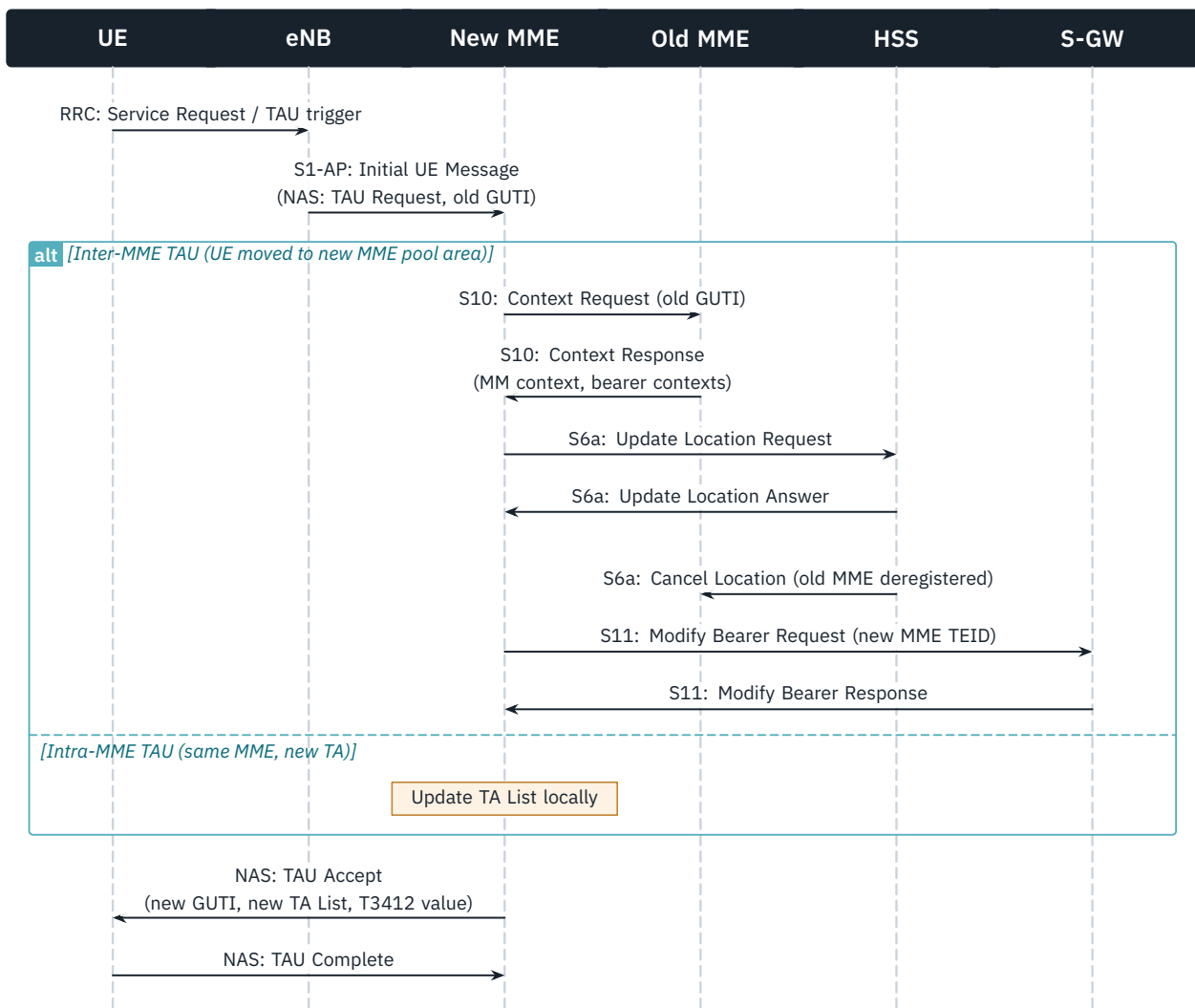


Figure 12.5 Tracking area update, inter-MME and intra-MME

4.4 Detach Procedure

■ UE-Initiated Detach

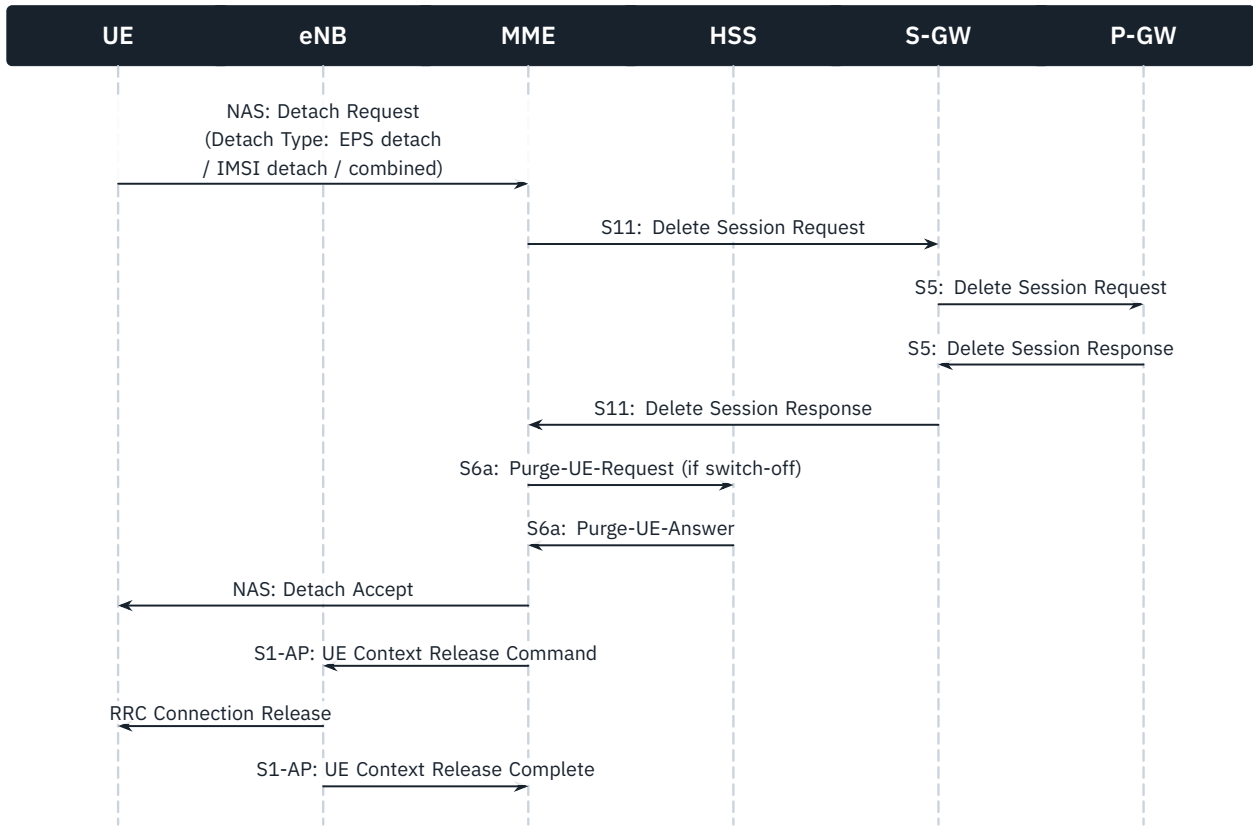


Figure 12.6 UE-initiated detach

■ Network-Initiated Detach

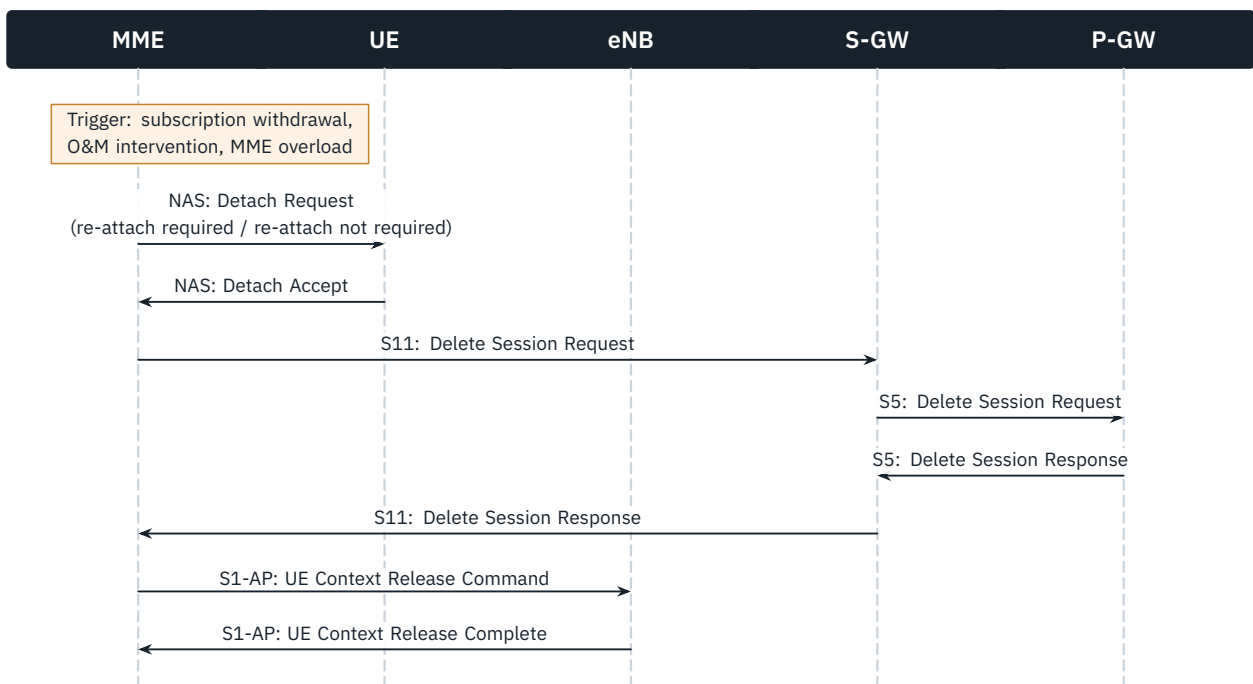


Figure 12.7 Network-initiated detach

4.5 Paging (S1 Paging Procedure)

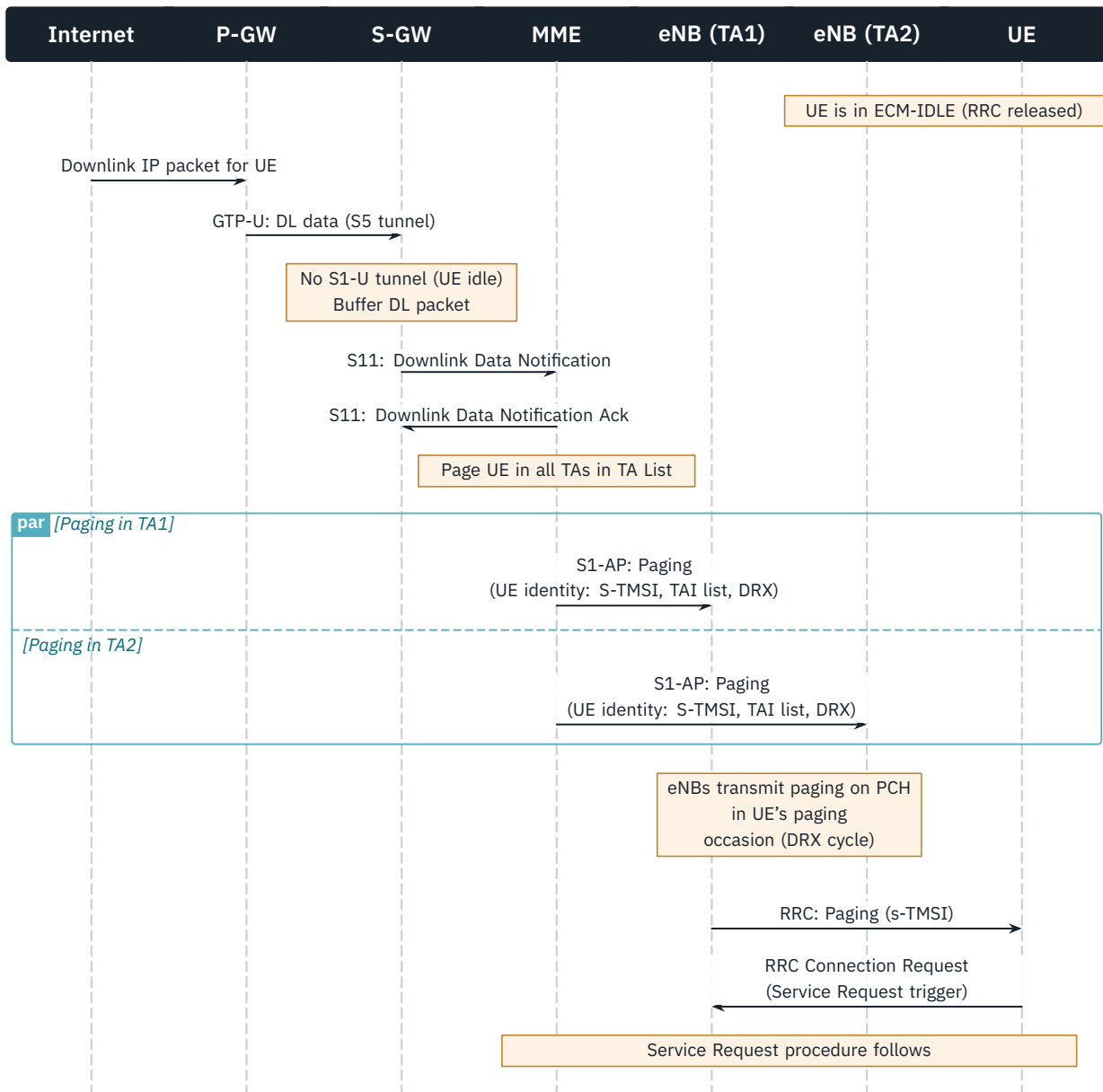


Figure 12.8 S1 paging of an idle UE

4.6 Service Request (UE in IDLE → CONNECTED)

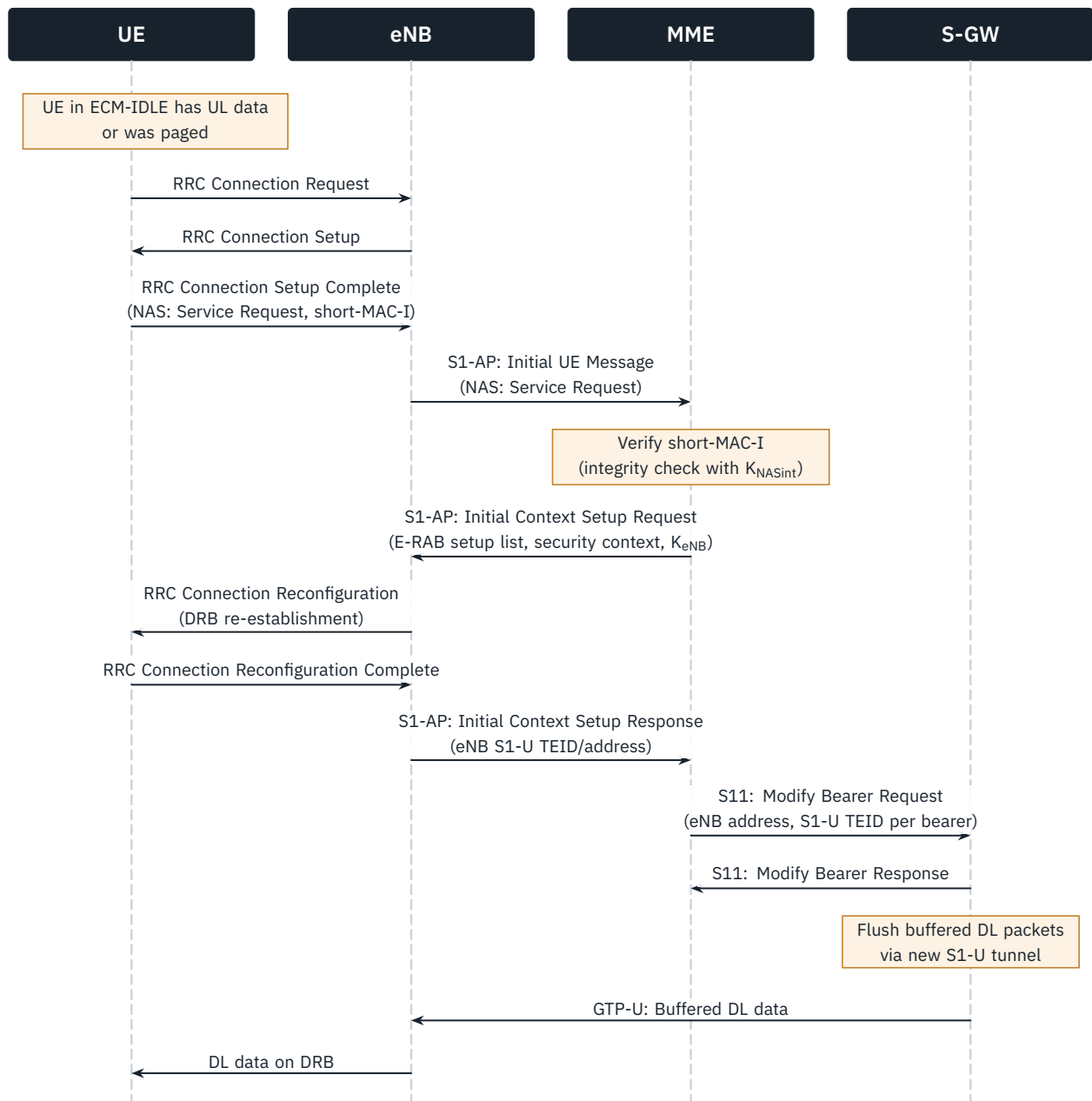


Figure 12.9 Service request (ECM-IDLE → ECM-CONNECTED)

5 Bearer Management

5.1 Default Bearer Setup

- Created automatically during Initial Attach (part of Create Session)
- One default bearer per PDN connection (always-on, non-GBR)
- QCI typically 9 (best-effort Internet) or 5 (IMS signaling)
- Cannot be deactivated without disconnecting the PDN

5.2 Dedicated Bearer Activation

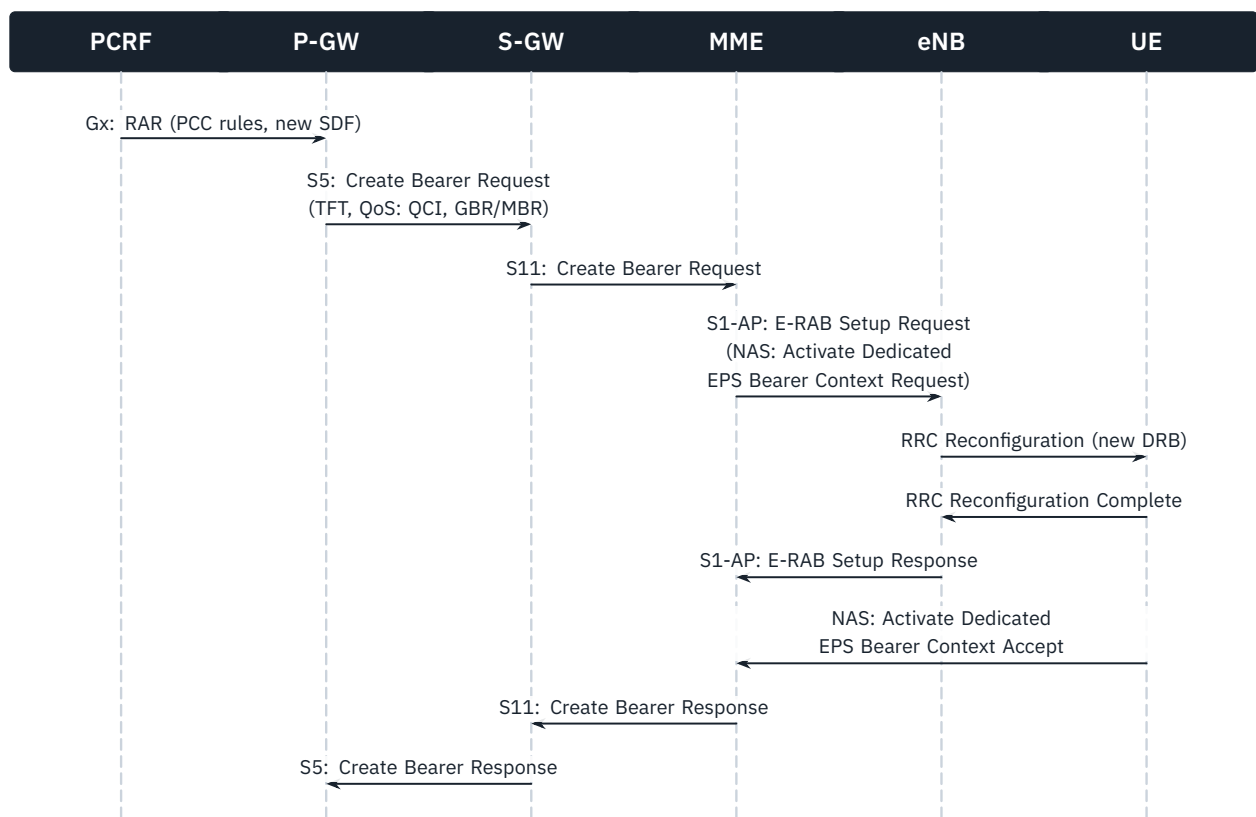


Figure 12.10 Network-initiated dedicated bearer activation

5.3 Dedicated Bearer Modification

- Triggered by PCRF policy change (e.g., QoS upgrade during VoLTE call)
- Uses Update Bearer Request/Response across S5 → S11 → S1-AP

5.4 Dedicated Bearer Deactivation

- Triggered by PCRF (policy removal), P-GW, or MME
- Uses Delete Bearer Request/Response
- Default bearer deactivation = PDN disconnect (all bearers for that PDN removed)

6 MME Selection

6.1 DNS-Based Selection

- eNB resolves MME FQDN from TAI → DNS returns list of MME IPs
- Format: `tac-XXXX.tac-YYYY.mme.epc.mnc<MNC>.mcc<MCC>.3gppnetwork.org`

6.2 Load Balancing

- Each MME advertises a **Relative MME Capacity** (weight, 0-255) via S1 Setup
- eNB distributes new UEs proportionally across pool members
- Higher capacity value = more UEs directed to that MME

6.3 GUMMEI – Globally Unique MME Identifier

$GUMMEI = MCC + MNC + MME \text{ Group ID (MMEGI)} + MME \text{ Code (MMEC)}$

- MMEGI identifies the MME pool
- MMEC identifies a specific MME within the pool
- GUTI = GUMMEI + M-TMSI (temporary UE identity)

7 S1-Flex and MME Pooling

I Concept

- **S1-Flex**: Each eNB connects to **multiple MMEs** in a pool (not just one)
- **MME Pool Area**: Geographic region served by a set of MMEs
- Benefits:
 - **Redundancy**: If one MME fails, others serve UEs
 - **Load sharing**: New UEs distributed by weight
 - **Geo-redundancy**: Pool members can be in different data centers
 - **Graceful scaling**: Add/remove MMEs without service interruption

MME Overload and Rebalancing

- MME sends **Overload Start** to eNBs → eNB redirects new connections to other pool members
- **Rebalancing**: MME requests UEs to re-attach to redistribute load (via implicit detach with re-attach indication)

8 Evolution: MME → AMF (5G)

Aspect	LTE MME	5G AMF
Architecture	Monolithic NE	Microservice (SBI-based)
Protocol	GTPv2-C, Diameter	HTTP/2, JSON (SBI), NGAP
NAS	EMM + ESM (combined)	5G-NAS (MM only; SM separated to SMF)
Session Mgmt	MME handles bearer signaling	Delegated entirely to SMF
Slicing	N/A	Slice-aware, routes to correct SMF per slice
Registration	Attach/TAU	Registration (unified)
State	Stateful	Designed for stateless (UDSF)
Interface to RAN	S1-AP	NGAP (N2)
Interface to UDM	Diameter S6a	Nudm (SBI)

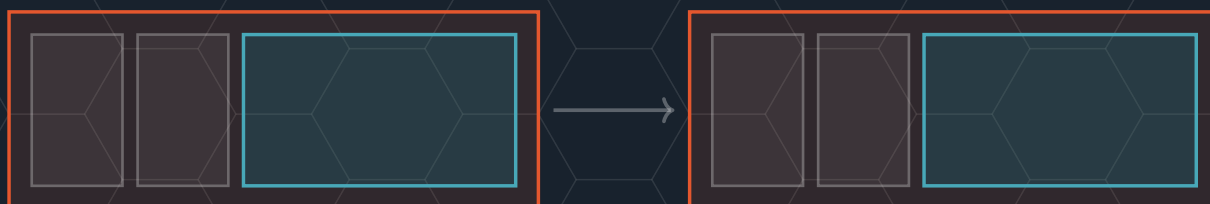
Key Changes:

1. **Session management decoupled**: MME handled both mobility + bearers; in 5G, AMF handles only mobility/security, SMF handles sessions
2. **Service-based interface**: AMF exposes Namf services, consumed by other NFs via HTTP/2
3. **Network slicing**: AMF performs slice selection (NSSF interaction) and routes to slice-specific SMF
4. **Stateless design**: UE context can be stored externally (UDSF) for resilience
5. **Unified registration**: No separate Attach vs TAU — single Registration procedure handles both

S-GW (Serving Gateway) and P-GW (PDN Gateway)

S-GW (دروازه خدمات‌دهنده) و P-GW (دروازه PDN)

OUTER IP | UDP | GTP-U | USER IP PACKET



In This Module

1	Why Separate S-GW and P-GW?	272
2	S-GW (Serving Gateway)	272
3	P-GW (PDN Gateway)	274
4	GTP Protocol	277
5	Packet Path Trace (End-to-End with Encapsulation)	279
6	CUPS: Control and User Plane Separation	282
7	Evolution: S-GW + P-GW → UPF (5G)	283
	Summary	285

1 Why Separate S-GW and P-GW?

LTE deliberately splits the user-plane gateway into two distinct nodes:

Role	S-GW	P-GW
Anchoring	Local mobility anchor (intra-LTE)	PDN/IP anchor (inter-RAT, global)
Scope	Per-MME pool area	Per-PDN connection (could span MME areas)
Mobility	Changes during inter-MME HO	Stays the same for entire PDN session
Placement	Close to RAN (low latency)	Centralized (policy, charging)

Design Rationale:

1. **Different lifetimes:** S-GW can relocate during mobility; P-GW remains fixed for IP continuity
2. **Independent scaling:** S-GW scales with local traffic; P-GW scales with subscribers/policies
3. **CUPS preparation:** The split enables Control/User Plane Separation — you can distribute S-GW-U at the edge while keeping S-GW-C centralized
4. **Roaming:** In home-routed roaming, S-GW is in visited PLMN; P-GW is in home PLMN (S8 interface)

2 S-GW (Serving Gateway)

2.1 User-Plane Forwarding

The S-GW is essentially a **GTP-U tunnel switch**:

- Receives DL packets from P-GW on S5/S8 GTP-U tunnel
- Forwards them to eNodeB (eNB) on S1-U GTP-U tunnel
- Receives UL packets from eNB on S1-U and forwards to P-GW on S5/S8

Each bearer has **separate GTP-U tunnels** on each interface (identified by TEID).

2.2 Local Mobility Anchor

During **X2 handover** (direct eNB-to-eNB):

- S-GW does NOT change
- Source eNB forwards buffered/in-transit packets to target eNB via X2
- After handover, MME sends **Modify Bearer Request** to S-GW with new eNB address/TEID
- S-GW switches S1-U tunnel endpoint from source eNB to target eNB (**path switch**)

This avoids tearing down the S5 tunnel to P-GW — only the S1-U leg changes.

2.3 Downlink Data Buffering

When UE is in **ECM-IDLE** (no S1-U tunnel exists):

1. DL packets arrive from P-GW on S5 tunnel
2. S-GW has no S1-U endpoint → **buffers packets**
3. S-GW sends **Downlink Data Notification** to MME (via S11)
4. MME initiates paging
5. After UE responds (Service Request), MME sends Modify Bearer Request with new eNB TEID
6. S-GW flushes buffered packets to eNB

Buffer limits: Configurable per operator; typically 1-10 packets per bearer. Excess dropped.

2.4 Per-UE/Per-Bearer Charging

S-GW generates **CDRs (Charging Data Records)** with:

- UL/DL volume per bearer
- Duration
- QoS information (QCI, ARP, GBR/MBR)
- Cause of record closure (HO, time limit, volume limit)

Sent to **offline charging system (CDF)** via Rf/Gz interface.

2.5 Lawful Interception

S-GW can mirror user-plane traffic for lawful interception (LI):

- X2 interface (ETSI standard) to Law Enforcement Monitoring Facility (LEMF)
- Triggered by warrant, activated via X1 interface from LI administration
- Duplicates packets without affecting UE service

2.6 S-GW Interfaces

Interface	Peer	Protocol	Function
S1-U	eNB	GTP-U (UDP)	User-plane to/from RAN
S5	P-GW (same PLMN)	GTP-U + GTPv2-C	UP forwarding + CP session mgmt
S8	P-GW (roaming)	GTP-U + GTPv2-C	Same as S5 but inter-PLMN
S11	MME	GTPv2-C	CP: bearer/session management
S12	UTRAN (3G)	GTP-U	Direct tunnel for 3G interworking

3 P-GW (PDN Gateway)

3.1 PDN Connectivity and IP Address Allocation

The P-GW is the **UE's gateway to the Internet/PDN** — it assigns the UE its IP address:

IPv4 Allocation Methods:

- **Static:** Pre-configured in HSS subscription data
- **Dynamic (DHCPv4):** P-GW acts as DHCP server or relays to external DHCP
- **Dynamic (internal pool):** P-GW allocates from local address pool (most common)

IPv6 Allocation:

- P-GW assigns a /64 prefix via **Router Advertisement (SLAAC)**
- UE generates Interface ID (EUI-64 or privacy extensions)
- Prefix delivered in PCO (Protocol Configuration Options) during bearer activation

Dual-Stack (IPv4v6):

- Both IPv4 address and IPv6 prefix assigned simultaneously
- Requires PDN Type = IPv4v6 in subscription

3.2 Policy Enforcement

P-GW enforces policies received from **PCRF** via the **Gx interface**:

- **PCC Rules (Policy and Charging Control):**
 - SDF (Service Data Flow) filters — 5-tuple matching
 - QoS parameters (QCI, GBR, MBR, ARP)
 - Charging keys and rating groups
 - Gate status (open/close specific flows)

- **TFT (Traffic Flow Template):**

- **DL TFT:** At P-GW — maps incoming DL packets to correct bearer
- **UL TFT:** Sent to UE — UE maps UL packets to correct bearer
- Packet filters: source/dest IP, port range, protocol, ToS/DSCP

3.3 Charging

Type	Interface	Node	Use Case
Online (OCS)	Gy (Diameter)	OCS	Prepaid; real-time credit control
Offline (OFCS)	Gz (GTP')	CDF/CGF	Postpaid; CDR generation

Charging triggers:

- Volume threshold (e.g., every 1 MB)
- Time threshold (e.g., every 60 seconds)
- QoS change, bearer modification
- Re-authorization from OCS

3.4 Packet Filtering and Routing

- **DL:** Internet → P-GW: Match destination IP to UE session, apply SDF filter to select bearer, encapsulate in GTP-U toward S-GW
- **UL:** UE → P-GW: Decapsulate GTP-U, apply NAT (if private addressing), route to Internet
- **NAT/NAPT:** P-GW often performs NAT for IPv4 (private UE addresses → public)
- **Header compression:** Not at P-GW (done at PDCP layer in RAN)

3.5 SGi Interface

- SGi connects P-GW to **external PDNs** (Internet, IMS, enterprise VPNs)
- IP-based interface (no GTP)
- Equivalent to Gi in 3G (GGSN → Internet)
- Multiple PDNs supported (each identified by APN)
- May connect to:
 - Internet (public APN)
 - IMS core (ims APN)
 - Enterprise VPN (private APN)
 - Content/CDN platforms

3.6 UL/DL TFT – Mapping Packets to Bearers

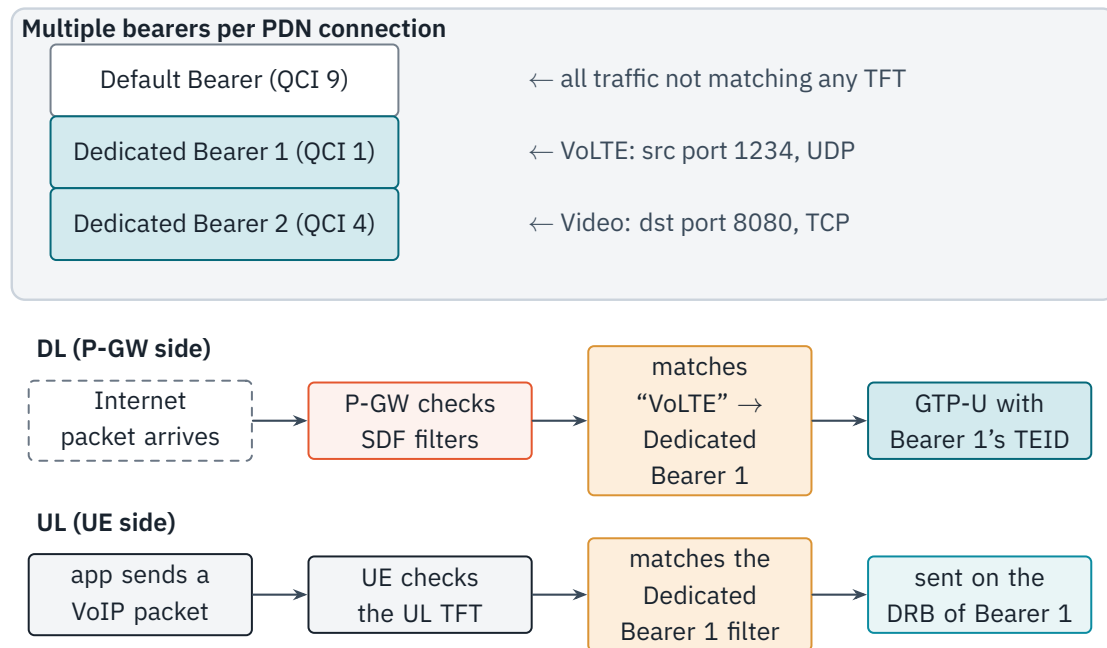


Figure 13.1 Several bearers per PDN connection and how packets are mapped to them

4 GTP Protocol

4.1 GTP-C (GTP Control Plane) – GTPv2-C

GTP-C manages tunnel/session lifecycle on S11, S5/S8, S10, S3 interfaces.

Key Messages:

Message	Direction	Purpose
Create Session Request/Response	MME→S-GW→P-GW	Establish PDN connection + default bearer
Modify Bearer Request/Response	MME→S-GW	Update tunnel endpoints (HO, service request)
Delete Session Request/Response	MME→S-GW→P-GW	Tear down PDN connection
Create Bearer Request/Response	P-GW→S-GW→MME	Dedicated bearer activation
Update Bearer Request/Response	P-GW→S-GW→MME	QoS modification
Delete Bearer Request/Response	P-GW→S-GW→MME	Dedicated bearer deactivation
Downlink Data Notification	S-GW→MME	Trigger paging for idle UE
Release Access Bearers	MME→S-GW	UE goes idle, release S1-U

GTPv2-C Header:

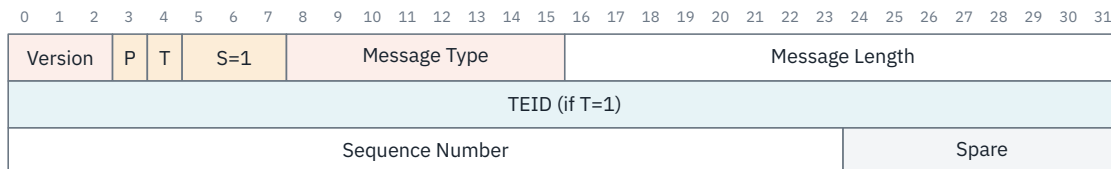


Figure 13.2 GTPv2-C header

TEID (Tunnel Endpoint Identifier):

- 32-bit identifier, locally unique per node
- Each side assigns its own TEID and communicates it to the peer
- Allows multiplexing multiple tunnels on same IP:port

4.2 GTP-U (GTP User Plane)

GTP-U encapsulates user IP packets for transport between EPC nodes.

GTP-U Header:

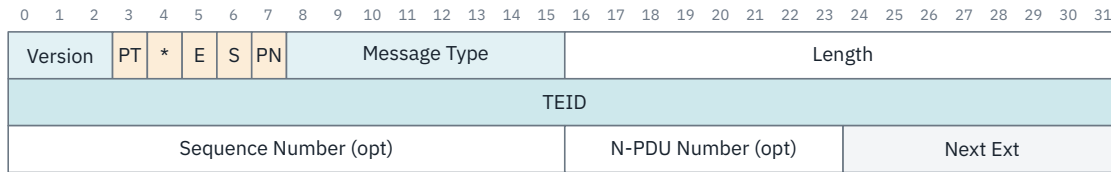


Figure 13.3 GTP-U header

- **Message Type 0xFF (255):** T-PDU — contains user data (most common)
- **Transport:** UDP port 2152
- **T-PDU:** The encapsulated user IP packet
- **Sequence Number:** Optional, used for reordering during path switch/handover

4.3 GTP-C vs GTP-U Comparison

Aspect	GTP-C (GTPv2-C)	GTP-U (GTPv1-U)
Purpose	Session/tunnel management	User data transport
Version	v2 (in LTE)	v1 (unchanged from 3G)
Transport	UDP port 2123	UDP port 2152
Reliability	Request/Response + retransmission	No ACK (relies on upper layers)
TEID	Identifies control session	Identifies user data tunnel (per-bearer)
Messages	Create/Modify/Delete Session, etc.	T-PDU (0xFF), Echo, Error Indication
Interfaces	S11, S5/S8, S10, S3	S1-U, S5/S8 (user plane)
Payload	IE (Information Elements) structured	Raw IP packet (T-PDU)
Header size	12 bytes (min)	8 bytes (min), 12 with options
Statefulness	Stateful (sessions)	Stateless (per-packet forwarding)

5 Packet Path Trace (End-to-End with Encapsulation)

5.1 Downlink: Internet → UE

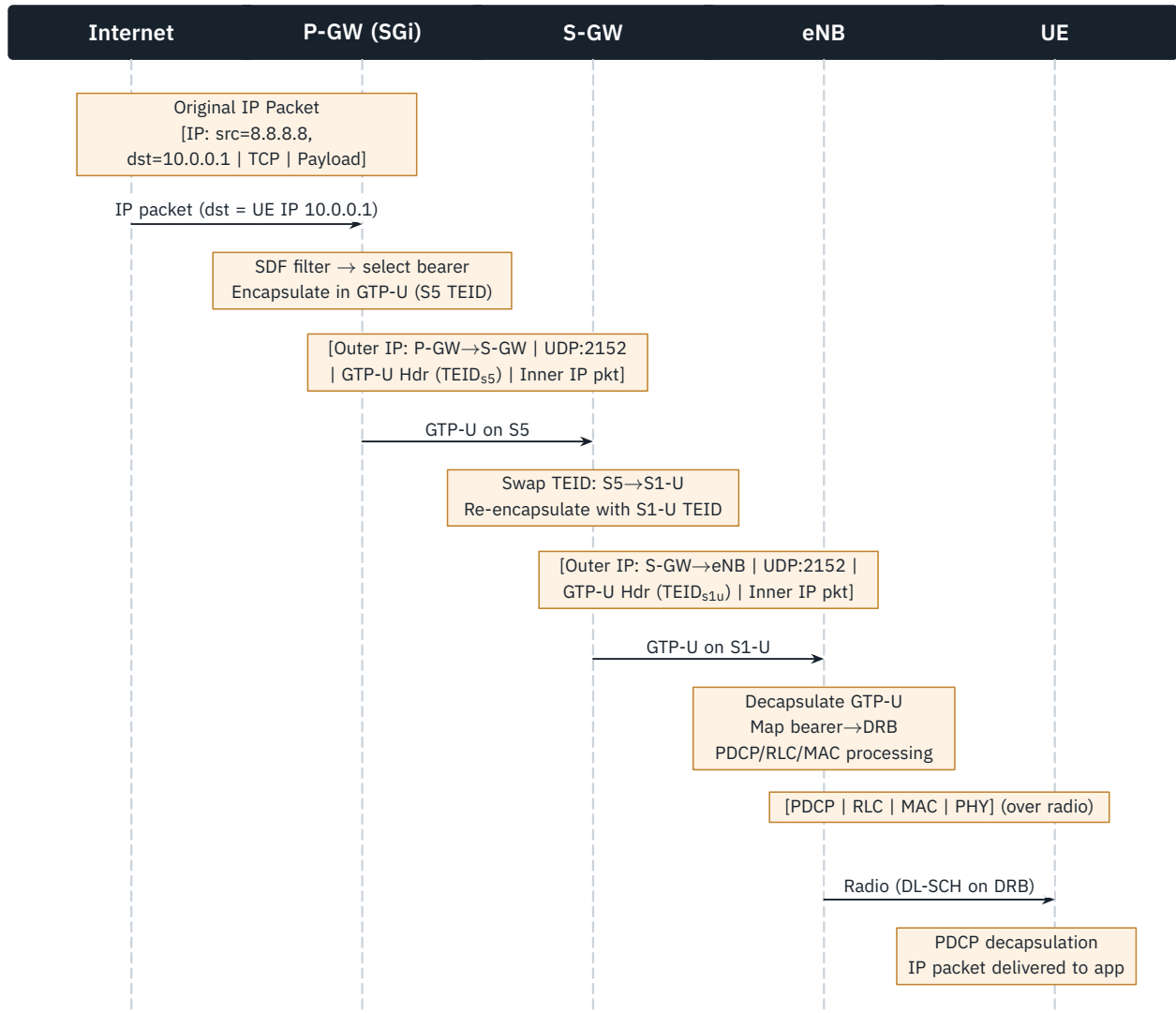


Figure 13.4 Downlink packet trace from the Internet to the UE with each encapsulation

5.2 Headers at Each Point

Segment 1: Internet → P-GW (SGi)

IP header src=8.8.8.8 dst=10.0.0.1	TCP	Application data
---	------------	------------------

Segment 2: P-GW → S-GW (S5/S8 GTP-U tunnel)

Outer IP src=P-GW IP dst=S-GW IP	UDP dst=2152	GTP-U header TEID=0xABCD Type=0xFF (TPDU)	Inner IP src=8.8.8.8 dst=10.0.0.1	TCP	Data
---	------------------------	--	--	------------	------

Segment 3: S-GW → eNB (S1-U GTP-U tunnel)

Outer IP src=S-GW IP dst=eNB IP	UDP dst=2152	GTP-U header TEID=0x1234 Type=0xFF (TPDU)	Inner IP src=8.8.8.8 dst=10.0.0.1	TCP	Data
--	------------------------	--	--	------------	------

Segment 4: eNB → UE (radio / air interface)

PHY	MAC	RLC	PDCP (ciphered)	Inner IP packet (original IP packet delivered to the UE)
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Figure 13.5 Headers of a downlink packet on each segment

5.3 GTP Tunnel Structure Diagram

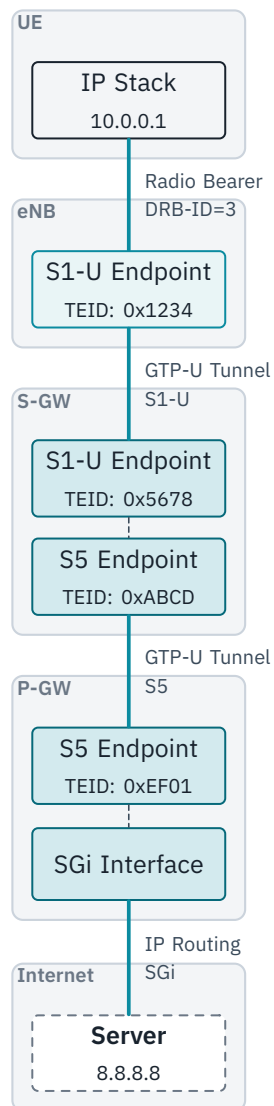


Figure 13.6 GTP tunnel endpoints and TEIDs along the user-plane path

Note: Each GTP-U tunnel segment has **two TEIDs** — one for each direction:

- S1-U UL: eNB→S-GW uses S-GW's TEID (0x5678)
- S1-U DL: S-GW→eNB uses eNB's TEID (0x1234)

6 CUPS: Control and User Plane Separation

6.1 Concept (3GPP Release 14+)

CUPS splits S-GW and P-GW each into control-plane and user-plane components:

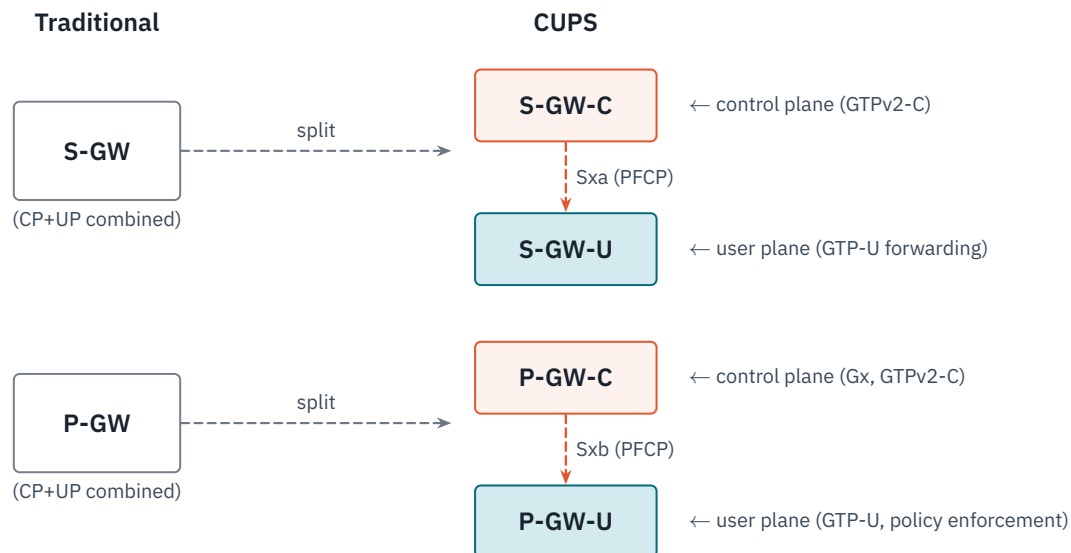


Figure 13.7 CUPS splits S-GW and P-GW into control-plane and user-plane parts

6.2 Sx Interface and PFCP Protocol

PFCP (Packet Forwarding Control Protocol) — 3GPP TS 29.244:

- Runs between CP and UP components (Sxa for S-GW, Sxb for P-GW, Sxc for TDF)
- **Session Establishment:** CP tells UP to create forwarding rules
- **Session Modification:** CP updates rules (e.g., new TEID after handover)
- **Session Deletion:** CP instructs UP to remove forwarding state

PFCP Key Concepts:

| Element | Purpose |

| --- | --- |

| PDR (Packet Detection Rule) | Match criteria (TEID, IP, port) |

| FAR (Forwarding Action Rule) | What to do (forward, buffer, drop) |

| QER (QoS Enforcement Rule) | Rate limiting (GBR/MBR) |

| URR (Usage Reporting Rule) | Volume/time measurement for charging |

| BAR (Buffering Action Rule) | Buffering behavior for idle UE |

6.3 Benefits of CUPS

1. **Distributed user plane:** Deploy S-GW-U / P-GW-U at edge (low latency for MEC)
2. **Centralized control:** Keep S-GW-C / P-GW-C in central DC (simpler management)
3. **Independent scaling:** Scale UP with traffic volume, CP with signaling load

4. **Flexible deployment:** Multiple UP instances per CP, or vice versa
5. **5G readiness:** CUPS architecture directly maps to 5G SMF (CP) + UPF (UP) split

6.4 CUPS Architecture Diagram

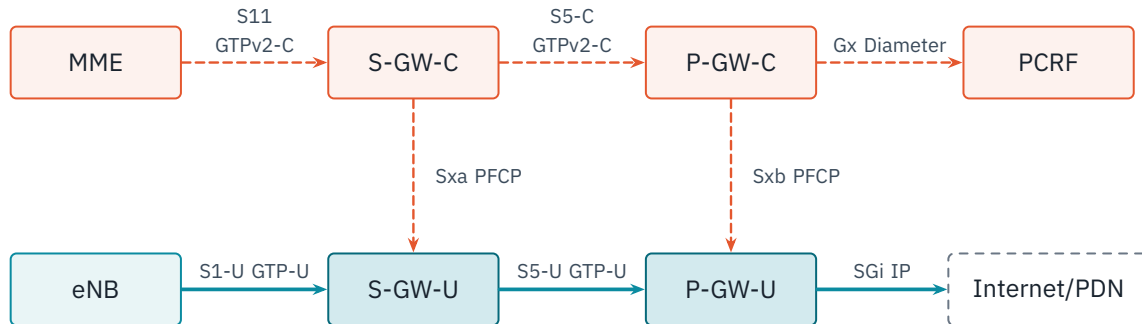


Figure 13.8 CUPS: S-GW and P-GW split into control and user planes

Blue = Control Plane | Green = User Plane

7 Evolution: S-GW + P-GW → UPF (5G)

7.1 Why Merged into UPF?

In 5G, S-GW and P-GW user-plane functions are **merged into a single UPF (User Plane Function)**:

Aspect	LTE (S-GW + P-GW)	5G (UPF)
Nodes	2 separate (S-GW, P-GW)	1 unified (UPF)
GTP hops	2 (S1-U + S5)	1 (N3 only, or N3+N9 for chaining)
Control	S-GW-C + P-GW-C	SMF (single CP for UP)
CP-UP protocol	Sxa/Sxb PFCP	N4 PFCP (same protocol, evolved)
Deployment	Typically centralized	Distributed (edge UPF, anchor UPF)
Mobility anchor	Fixed P-GW	Flexible (SSC modes 1/2/3)
Slicing	N/A	UPF per slice

7.2 Why the Merge Makes Sense

1. **Eliminates one GTP-U hop:** S1-U + S5 → just N3 (+ optional N9 between UPFs)
 - Lower latency, fewer encap/decap operations
2. **Simplified forwarding:** One node handles both local mobility and PDN anchoring

3. **Edge deployment:** UPF placed close to user for MEC/URLLC — no need for separate S-GW-U at edge
4. **Flexible anchoring (SSC modes):**
 - SSC 1: Fixed UPF (like LTE P-GW) — IP continuity
 - SSC 2: UPF can change — new IP address (for non-persistent sessions)
 - SSC 3: Multi-homing — old + new UPF active simultaneously during transition
5. **UPF chaining:** Multiple UPFs in series (I-UPF as intermediate, PSA-UPF as anchor) replaces S-GW→P-GW chain with more flexible topology

7.3 Interface Mapping

LTE	5G
S1-U (eNB ↔ S-GW)	N3 (gNB ↔ UPF)
S5/S8 (S-GW ↔ P-GW)	N9 (UPF ↔ UPF, if chained)
SGi (P-GW ↔ Internet)	N6 (UPF ↔ DN)
S11 (MME ↔ S-GW)	N4 (SMF ↔ UPF) [PFCP]
Gx (PCRF ↔ P-GW)	N7 (PCF ↔ SMF) [SBI]

Summary

Function	S-GW	P-GW
User-plane forwarding	✓	✓
GTP-U termination	S1-U + S5	S5 + SGi
IP address allocation	✗	✓
Policy enforcement (PCC)	✗	✓
Charging (online)	✗	✓ (Gy)
Charging (offline)	✓ (CDRs)	✓ (Gz)
DL buffering + paging trigger	✓	✗
Local mobility anchor	✓	✗
PDN/IP anchor	✗	✓
Lawful interception	✓	✓
NAT	✗	✓
TFT/SDF enforcement	✗	✓

5G Networks Introduction

مقدمه‌ای بر شبکه‌های 5G



In This Module

1	Why 5G? — Limitations of LTE/4G	287
2	5G Requirements — ITU IMT-2020 Targets	288
3	Three Pillars — 5G Use Cases	289
4	5G NR (New Radio) — Physical Layer	291
5	5G NSA vs SA — Deployment Strategies	293
6	5G Architecture Overview	295
7	gNB Architecture — CU/DU Split	297
8	Timeline — 3GPP Releases	299
	Summary — Key Takeaways	300
	References	301

1 Why 5G? – Limitations of LTE/4G

Despite the success of LTE (4G), several fundamental limitations drove the need for a new generation:

1.1 Latency Floor (~10 ms)

- LTE's Transmission Time Interval (TTI) is fixed at **1 ms** subframe
- With scheduling, HARQ, and processing delays, the practical **round-trip latency floor is ~10 ms**
- This is insufficient for:
 - Autonomous vehicle control (requires < 5 ms)
 - Remote surgery / haptic feedback (requires ~1 ms)
 - Industrial real-time control loops

1.2 Capacity Ceiling

- LTE peak: ~1 Gbps (Cat-16, 3CA, 4×4 MIMO, 256-QAM) — theoretical
- Practical throughput: 100–300 Mbps in ideal conditions
- Spectrum below 6 GHz is congested; limited bandwidth per carrier (20 MHz max)
- 4×4 MIMO is the practical antenna limit in LTE form factors

1.3 IoT Scale

- LTE was designed for **human-centric** broadband communication
- LTE-M / NB-IoT added IoT support, but:
 - Connection density limited to ~10⁴ devices/km²
 - Not natively designed for **10⁶ devices/km²** required for smart city/industrial IoT
 - Signaling overhead per device is too high for massive deployments

1.4 Summary of LTE Limitations

Parameter	LTE Capability	5G Target
Peak data rate (DL)	~1 Gbps	20 Gbps
User-plane latency	~10 ms	1 ms
Connection density	~10 ⁴ /km ²	10 ⁶ /km ²
Spectral efficiency	30 bps/Hz	90 bps/Hz
Mobility	350 km/h	500 km/h

2 5G Requirements – ITU IMT-2020 Targets

The International Telecommunication Union (ITU) defined **IMT-2020** as the framework for 5G with the following key performance indicators (KPIs):

2.1 Peak Data Rate

- **Downlink:** 20 Gbps
- **Uplink:** 10 Gbps
- Achieved through wider bandwidth (up to 400 MHz per carrier) + higher-order MIMO + 256-QAM

2.2 Latency

- **User-plane latency:** 1 ms (URLLC scenario)
- **Control-plane latency:** 20 ms (idle to connected)
- Enables real-time control applications previously impossible on cellular

2.3 Connection Density

- **10⁶ devices per km²** (mMTC scenario)
- 100× improvement over LTE
- Support for sensors, actuators, meters with minimal overhead

2.4 User-Experienced Data Rate

- **100 Mbps** available everywhere (cell edge, dense urban)
- **50 Mbps UL** everywhere
- Represents the “guaranteed minimum” user experience

2.5 Reliability

- 99.999% (five nines) for URLLC
- Packet delivery within latency budget with 10^{-5} error probability
- Critical for safety-of-life applications

2.6 Additional IMT-2020 Targets

KPI	Target
Area traffic capacity	10 Mbps/m ²
Network energy efficiency	100× improvement
Spectrum efficiency	3× over IMT-Advanced
Mobility	500 km/h

3 Three Pillars – 5G Use Cases

5G is defined by three fundamental service categories (use-case pillars):

■ 5G Use Case Triangle

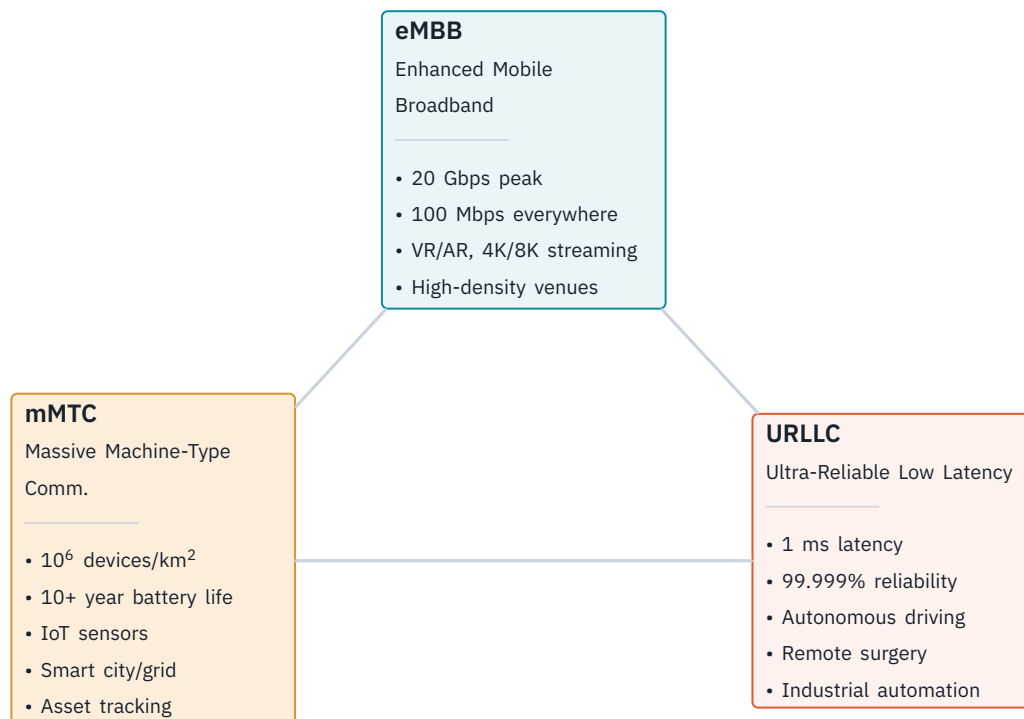


Figure 14.1 The 5G usage scenarios of ITU IMT-2020

3.1 eMBB – Enhanced Mobile Broadband

Focus: High data rates, capacity, coverage

Characteristic	Details
Peak rate	20 Gbps DL / 10 Gbps UL
User experience	100 Mbps everywhere
Key tech	Massive MIMO, mmWave, CA

Applications:

- Virtual Reality (VR) / Augmented Reality (AR) — requires >100 Mbps, <20 ms
- 4K/8K video streaming
- High-density event venues (stadiums: 100k+ users)
- Fixed Wireless Access (FWA) as fiber replacement

3.2 URLLC – Ultra-Reliable Low Latency Communications

Focus: Deterministic latency, extreme reliability

Characteristic	Details
Latency	1 ms (user plane, one way)
Reliability	99.999% (10^{-5} BLER)
Key tech	Mini-slots, pre-emption, redundancy

Applications:

- **Autonomous driving:** V2X communication, cooperative perception
- **Remote surgery:** Haptic feedback over cellular
- **Industry 4.0:** Real-time robotic control, motion control
- **Smart grid:** Protection and fault isolation

3.3 mMTC – Massive Machine-Type Communications

Focus: Scalability, energy efficiency, low cost

Characteristic	Details
Density	10^6 devices/km ²
Battery life	10–15 years on coin cell
Key tech	Grant-free access, NOMA, deep coverage

Applications:

- IoT sensor networks (environmental, agricultural)
- Smart city infrastructure (parking, waste, lighting)
- Smart metering (electricity, water, gas)
- Asset tracking and logistics
- Wearables

4 5G NR (New Radio) – Physical Layer

5G NR is the radio access technology defined in 3GPP Release 15+, designed from the ground up for flexibility.

4.1 Flexible Numerology

Unlike LTE (fixed 15 kHz SCS, 1 ms TTI), NR supports **multiple subcarrier spacings (SCS)**:

Numerology (μ)	SCS (kHz)	Slot Duration	CP Type	Typical Use
0	15	1 ms	Normal	FR1, eMBB (low freq)
1	30	0.5 ms	Normal	FR1, typical deployment
2	60	0.25 ms	Normal/Extended	FR1/FR2, URLLC
3	120	0.125 ms	Normal	FR2 (mmWave)
4	240	62.5 μ s	Normal	FR2 (SSB only)

Key relationships:

- Slot duration = $1 \text{ ms} / 2^\mu$
- Symbols per slot = 14 (normal CP) or 12 (extended CP)
- Higher SCS → shorter slot → lower latency (but wider subcarrier → more phase noise tolerance needed for mmWave)

4.2 Frequency Ranges

Range	Band	Frequencies	Max BW/carrier	Typical SCS
FR1	Sub-7 GHz	410 MHz – 7.125 GHz	100 MHz	15/30/60 kHz
FR2	mmWave	24.25 – 52.6 GHz	400 MHz	60/120/240 kHz

- FR1 provides coverage and capacity (n77, n78 = C-band 3.3–4.2 GHz)
- FR2 provides extreme capacity in short range (n257 = 28 GHz, n258 = 26 GHz, n260 = 39 GHz, n261 = 28 GHz)
- 3GPP Rel-17 introduced **FR2-2**: 52.6–71 GHz

4.3 Bandwidth Parts (BWP)

- A **Bandwidth Part** is a contiguous subset of the carrier bandwidth
- UE can be configured with up to **4 BWPs per serving cell** (1 active at a time)
- Allows UE power saving: narrow BWP for monitoring, wide BWP for data
- Each BWP can have its own numerology (SCS, CP)

4.4 Mini-Slots

- Standard NR slot = 14 symbols
- Mini-slot** = 2, 4, or 7 OFDM symbols
- Enables sub-slot scheduling for **URLLC** without waiting for slot boundary
- Can **pre-empt** ongoing eMBB transmissions for urgent URLLC traffic

4.5 Massive MIMO and Beamforming

Feature	LTE	5G NR
Max antenna ports	16 (Rel-13 FD-MIMO)	256+
Beamforming	Fixed / semi-static	Dynamic, per-slot beam management
Beam management	N/A	SSB beams, CSI-RS, beam tracking
Panel architecture	Single panel	Multi-panel (FR2)

5G NR beam management procedures:

- P1**: Initial beam acquisition (SSB sweep)
- P2**: gNB Tx beam refinement (CSI-RS)
- P3**: UE Rx beam refinement
- Beam failure recovery**: Detect failure → select new candidate → RACH-based recovery

4.6 Key Differences from LTE PHY

Aspect	LTE	5G NR
Subcarrier spacing	Fixed 15 kHz	15/30/60/120/240 kHz
Max bandwidth	20 MHz	100 MHz (FR1) / 400 MHz (FR2)
Waveform DL	OFDMA	CP-OFDMA
Waveform UL	SC-FDMA	CP-OFDMA (+ DFT-s-OFDMA)
Reference signals	CRS always-on	No CRS; on-demand DMRS/CSI-RS
HARQ timing	Fixed (8 ms RTT)	Flexible, configurable
Scheduling	Slot-based only	Slot + mini-slot
Duplex	FDD or TDD (fixed)	Dynamic TDD, self-contained slot
Control channel	PDCCH in first 1-3 symbols	CORESET (flexible in time/freq)
DC subcarrier	Yes (LTE center)	No DC subcarrier

5 5G NSA vs SA – Deployment Strategies

5.1 Non-Standalone (NSA) – Option 3/3a/3x

Architecture: NR radio (gNB) + LTE core (EPC)

- The LTE eNB acts as the **master node** (anchor)
- NR gNB is the **secondary node** (adds capacity)
- Control plane goes through LTE (eNB → EPC)
- User plane can be split or switched

Sub-options:

- **Option 3:** User plane through eNB (split at eNB)
- **Option 3a:** User plane directly from gNB to EPC
- **Option 3x:** User plane split at gNB (gNB handles NR data + some LTE data)

5.2 Standalone (SA) – Option 2

Architecture: NR radio (gNB) + 5G Core (5GC)

- gNB connects directly to 5G Core via **NG interface**
- Full 5G functionality available
- No dependency on LTE infrastructure

5.3 Why NSA Was Deployed First

1. **Faster time-to-market:** Reuse existing EPC infrastructure
2. **Lower CAPEX:** No need to deploy 5G Core immediately
3. **Coverage anchor:** LTE provides ubiquitous coverage; NR adds throughput
4. **Proven core:** EPC is mature, well-understood
5. **Spectrum strategy:** Early NR deployments in limited bands

5.4 Why SA Is the Target

1. **Network Slicing:** Requires 5G Core SBA
2. **URLLC:** End-to-end low latency needs 5GC + MEC
3. **mMTC:** Native 5GC support for IoT optimizations
4. **Edge Computing:** UPF placement flexibility
5. **Independence:** No LTE dependency, cleaner architecture
6. **Service-Based Architecture:** Enables rapid service creation

5.5 NSA vs SA Architecture Comparison

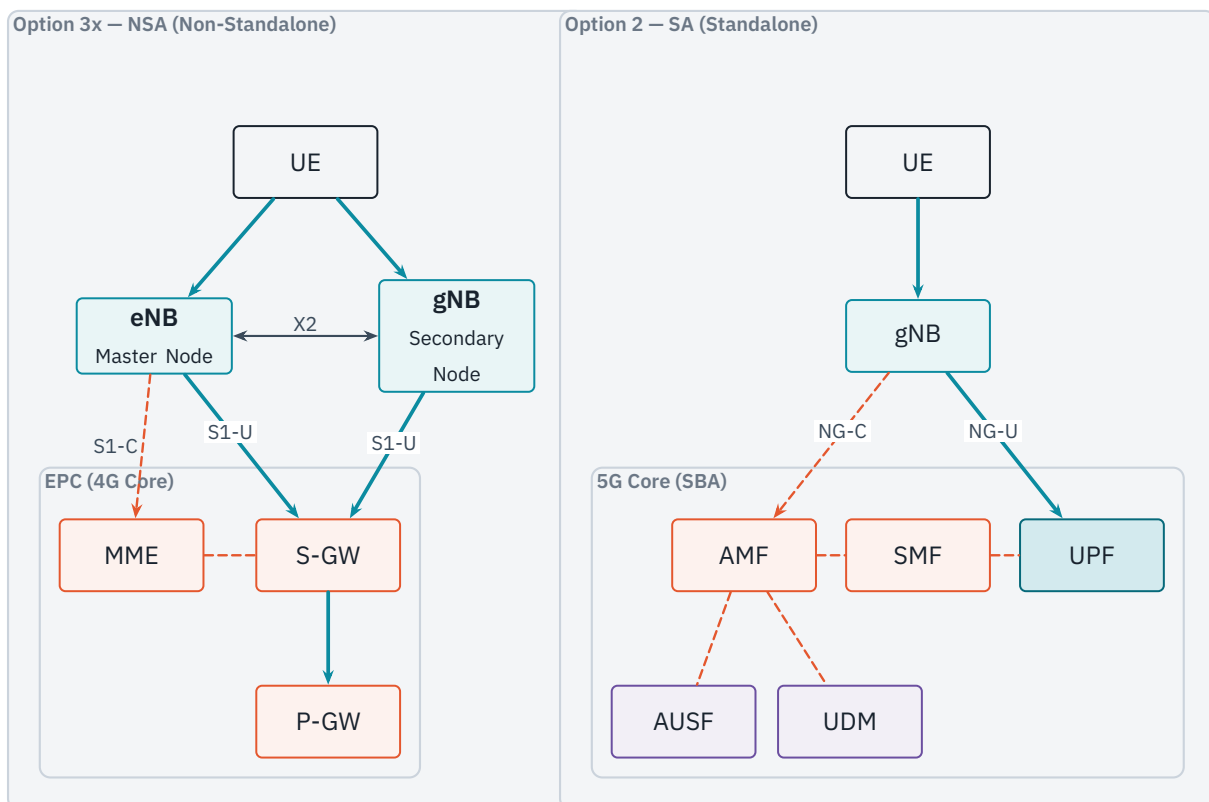


Figure 14.2 NSA Option 3x versus SA Option 2

5.6 Deployment Options Overview

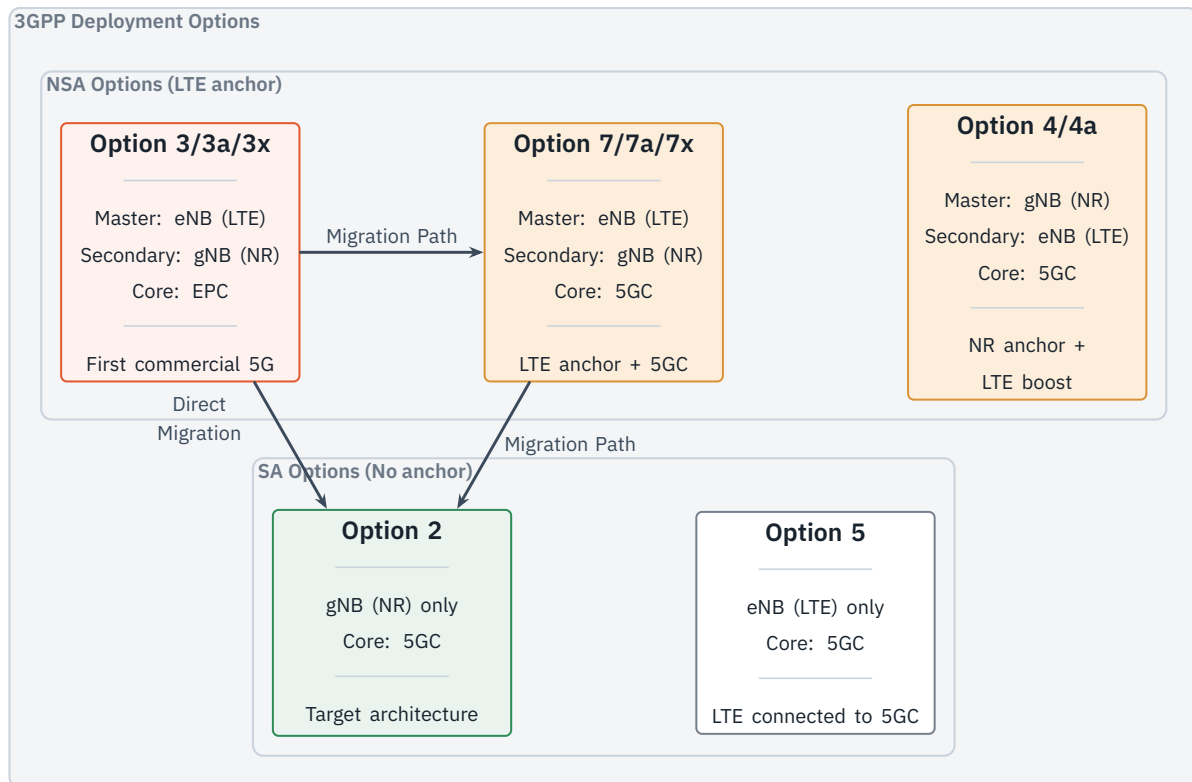


Figure 14.3 3GPP deployment options and migration paths

Migration path: Most operators follow **Option 3** → **Option 7** → **Option 2** or directly **Option 3** → **Option 2**.

6 5G Architecture Overview

6.1 NG-RAN (Next Generation Radio Access Network)

Node	Full Name	Description
gNB	Next Generation Node B	5G NR base station (SA)
en-gNB	E-UTRA-NR gNB	NR node in NSA (secondary to eNB)
ng-eNB	Next Generation eNB	LTE node connected to 5GC

Interfaces:

- **Xn:** Between gNBs (and ng-eNBs) — handover, dual connectivity
- **NG:** Between gNB and 5G Core (NG-C to AMF, NG-U to UPF)
- **F1:** Internal gNB split (CU ↔ DU)
- **E1:** CU-CP ↔ CU-UP

6.2 5G Core – Service-Based Architecture (SBA)

The 5G Core replaces the monolithic EPC with **microservices**:

Network Function	Role
AMF (Access & Mobility Mgmt)	Registration, connection, mobility, NAS security
SMF (Session Management)	PDU session establishment, QoS, UPF selection
UPF (User Plane Function)	Packet routing, inspection, QoS enforcement
AUSF (Authentication Server)	Authentication procedures
UDM (Unified Data Mgmt)	Subscription data, similar to HSS
PCF (Policy Control)	Policy rules, replaces PCRF
NSSF (Network Slice Selection)	Slice selection and management
NEF (Network Exposure)	API exposure to external applications
NRF (NF Repository)	Service discovery and registration

Key SBA principles:

- NFs communicate via **HTTP/2 REST APIs** (service-based interfaces)
- **Stateless** design: compute and storage separated
- **Cloud-native**: containerized, horizontally scalable
- **Service discovery**: NRF enables dynamic NF registration

6.3 Separation of Radio Technology from Network Architecture

A critical 5G design principle:

Radio Access Technology (RAT) $\neq$ Core Network Architecture

- LTE radio can connect to 5G Core (ng-eNB → Option 5/7)
- NR radio can connect to EPC (en-gNB → Option 3)
- **Decoupling** enables flexible migration and coexistence

7 gNB Architecture – CU/DU Split

7.1 Functional Split

The gNB can be decomposed into:

Component	Function	Location
CU (Central Unit)	RRC, SDAP, PDCP	Centralized (edge DC)
DU (Distributed Unit)	RLC, MAC, High-PHY	Near antenna site
RU (Radio Unit)	Low-PHY, RF	At antenna

7.2 CU-CP / CU-UP Split

The CU is further split:

- **CU-CP** (Control Plane): RRC + control part of PDCP
- **CU-UP** (User Plane): SDAP + user part of PDCP

This enables:

- Independent scaling of control and user plane
- Multiple CU-UPs per CU-CP
- Flexible placement (CU-UP closer to edge for low latency)

7.3 gNB CU/DU Split Architecture

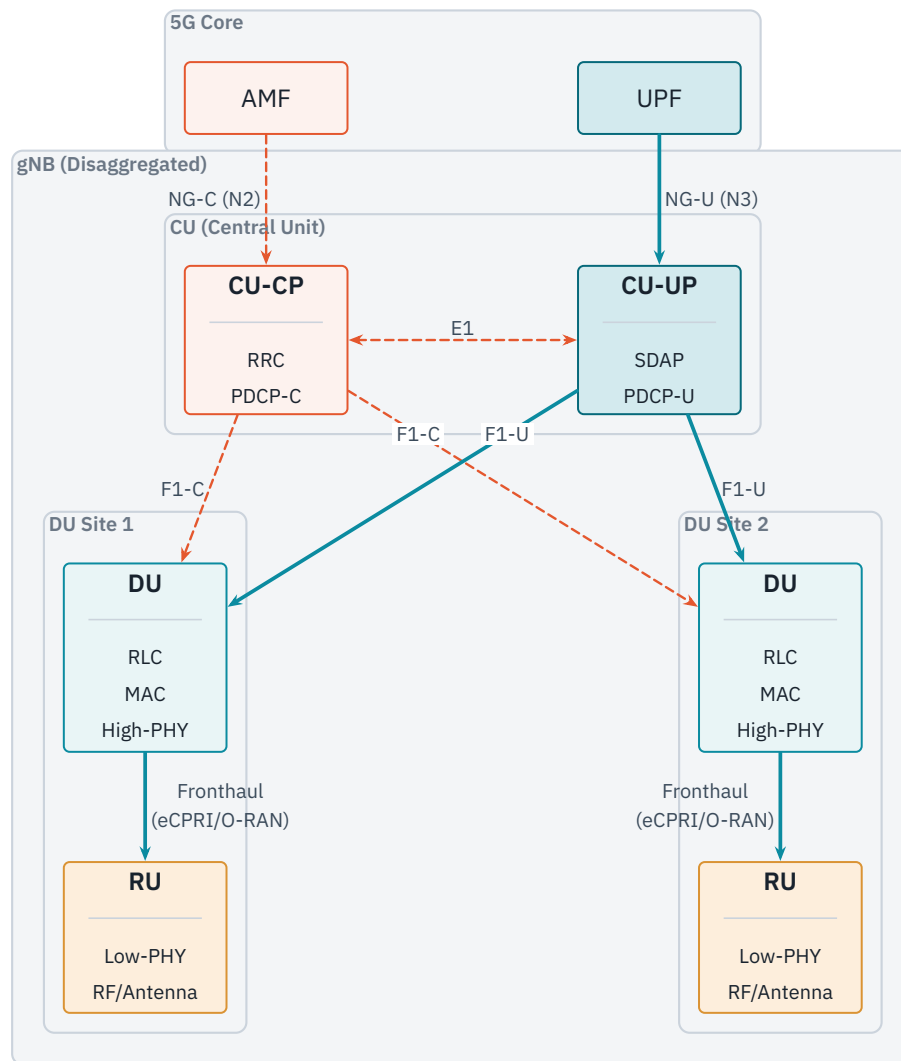


Figure 14.4 Disaggregated gNB: CU-CP, CU-UP, DU and RU

7.4 Interface Summary

Interface	Endpoints	Protocol	Purpose
NG-C (N2)	gNB ↔ AMF	NGAP/SCTP	Control plane signaling
NG-U (N3)	gNB ↔ UPF	GTP-U/UDP	User data transport
Xn-C	gNB ↔ gNB	XnAP/SCTP	Handover, dual connectivity
Xn-U	gNB ↔ gNB	GTP-U/UDP	Data forwarding during HO
F1-C	CU-CP ↔ DU	F1AP/SCTP	RRC message transport, UE context
F1-U	CU-UP ↔ DU	GTP-U/UDP	User data between CU-UP and DU
E1	CU-CP ↔ CU-UP	E1AP/SCTP	Bearer context management

7.5 Deployment Scenarios

Scenario	CU Location	DU Location	Fronthaul	Use Case
Co-located	Cell site	Cell site	Internal	Rural, simple
Centralized CU	Edge DC	Cell site	Midhaul (F1)	Urban, pooling
Full disaggregation	Regional DC	Aggregation	Mid + Fronthaul	Cloud RAN, O-RAN

8 Timeline – 3GPP Releases

8.1 Release Overview

Release	Freeze Date	Key Features
Rel-15	2018 (Late Drop: 2019)	First 5G NR specification; NSA & SA; eMBB focus
Rel-16	2020 (Q3)	URLLC enhancements; V2X; NR-U (unlicensed); IIoT; NR positioning
Rel-17	2022 (Q1)	FR2-2 (52-71 GHz); RedCap (reduced capability); NTN (satellite); NR sidelink; MBS
Rel-18	2024 (Q1)	5G-Advanced; AI/ML for air interface; XR optimization; duplex evolution; network energy saving

8.2 Evolution Path

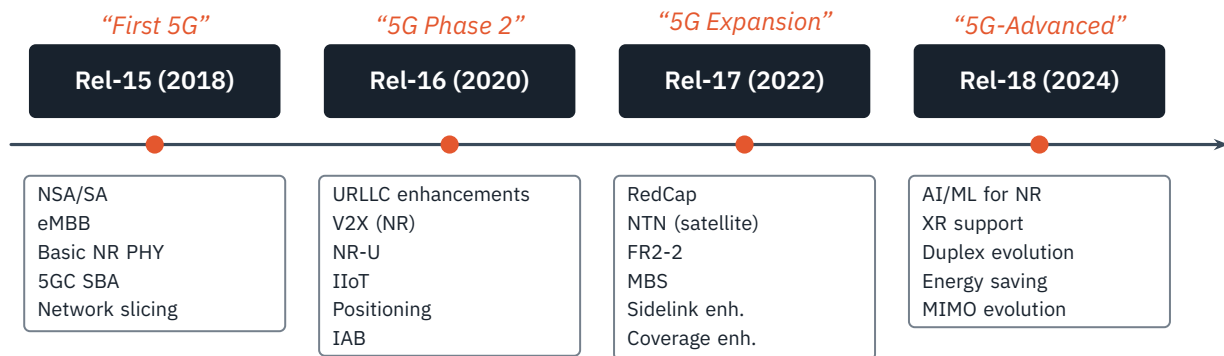


Figure 14.5 Evolution path from Release 15 to Release 18

8.3 Key Milestones

- **Dec 2017:** Rel-15 NSA (Option 3) specification completed (early drop)
- **Jun 2018:** Rel-15 SA (Option 2) specification completed
- **Apr 2019:** First commercial 5G NSA networks launched (South Korea, US)
- **2020:** First 5G SA commercial deployments (T-Mobile US, China operators)
- **2022:** Rel-17 freeze; RedCap enables mid-tier IoT on NR
- **2024:** Rel-18 (5G-Advanced) freeze; bridge to 6G research

Summary – Key Takeaways

1. **5G addresses three fundamental LTE limitations:** latency, capacity, and IoT scale
2. **IMT-2020 targets** define quantitative goals: 20 Gbps, 1 ms, 10^6 devices/km²
3. **Three pillars** (eMBB/URLLC/mMTC) serve different verticals with different KPIs
4. **5G NR** introduces flexible numerology, mmWave support, and dynamic beamforming
5. **NSA enables quick deployment** by reusing LTE/EPC; **SA unlocks full 5G** potential
6. **SBA-based 5G Core** is cloud-native, microservice-based, enabling slicing and edge
7. **gNB disaggregation** (CU/DU/RU) enables flexible, scalable deployments
8. **3GPP releases** progressively add features: 15→basic, 16→URLLC, 17→expansion, 18→5G-Advanced

References

- 3GPP TS 38.300: NR and NG-RAN Overall Description
- 3GPP TS 38.401: NG-RAN Architecture Description
- 3GPP TS 23.501: System Architecture for 5G System
- ITU-R M.2083: IMT Vision — Framework and Overall Objectives of the Future Development of IMT for 2020 and Beyond
- 3GPP TS 38.211: NR Physical Channels and Modulation
- 3GPP TS 38.104: NR Base Station Radio Transmission and Reception

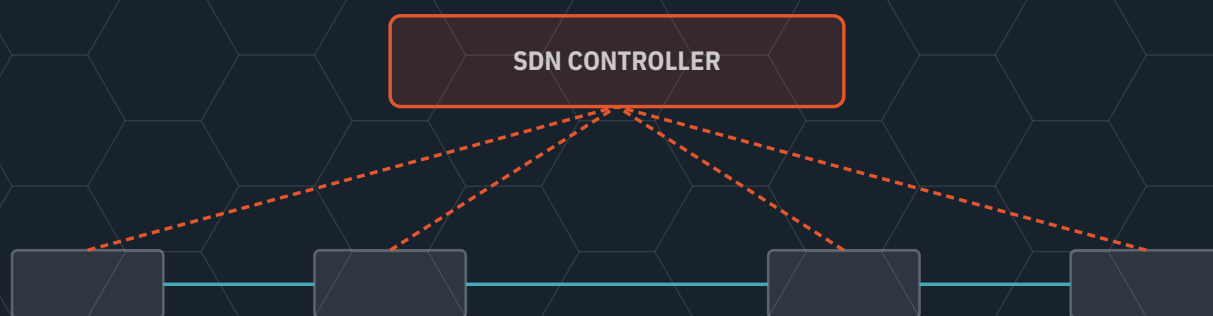
Module 14 — Advanced Communication Networks Course

Last updated: August 2026

Software Defined Networking (SDN)

شبکه سازی نرم افزار محور (SDN)

15



In This Module

1	Why SDN?	303
2	Traditional vs SDN Architecture	303
3	SDN Architecture Layers	304
4	SDN Controller Functions	305
5	Southbound Interfaces	305
6	Northbound Interfaces	306
7	OpenFlow Basics	308
8	SDN in Telecom	309
9	Benefits of SDN	311
10	Challenges of SDN	311
	Key Takeaways	312

1 Why SDN?

Traditional networks suffer from fundamental limitations that hinder innovation and agility:

Problem	Description
Distributed Control	Every device runs its own control plane independently. No single entity has a global view of the network state.
Vendor Lock-in	Proprietary hardware + software bundles. Migrating between vendors is costly and complex.
Slow Provisioning	Manual, box-by-box configuration via CLI. Weeks to deploy a new service. Human errors cause outages.
Rigid Architecture	Adding new protocols requires vendor firmware updates. Innovation cycles measured in years.
Inconsistent Policies	Policies must be manually replicated across hundreds of devices. Drift is inevitable.

The Core Insight: If we separate the “brain” (control plane) from the “muscle” (data plane), we can centralize intelligence, automate operations, and program the network like software.

2 Traditional vs SDN Architecture

Traditional Architecture

- Control plane embedded in every device
- Each device independently computes forwarding decisions
- Distributed protocols (OSPF, BGP, STP) synchronize state
- Convergence takes time; no global optimization

SDN Architecture

- Control plane extracted and centralized
- Switches become simple forwarding devices
- Controller has a **global view** of the entire network
- Programmable via APIs (northbound and southbound)

Aspect	Traditional	SDN
Control Plane	Distributed (per device)	Centralized (controller)
Data Plane	Coupled with control	Separated, simple forwarding
Configuration	CLI, box-by-box	API-driven, programmable
Vendor	Locked-in	Open, multi-vendor
Innovation Speed	Years (firmware cycle)	Days (software development)
Visibility	Per-device, fragmented	Global, real-time
Automation	Scripts/templates	Native API + intent

3 SDN Architecture Layers

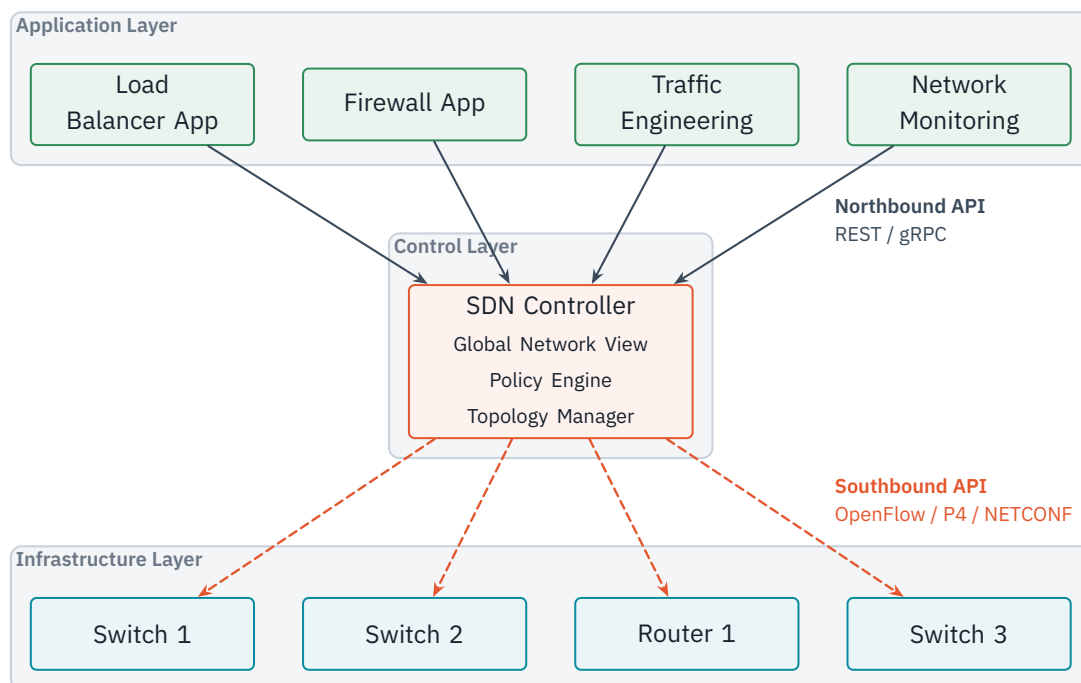


Figure 15.1 SDN architecture: application, control and infrastructure layers

3.1 Infrastructure Layer (Data Plane)

- Physical or virtual **switches and routers**
- Performs packet forwarding based on **flow tables** programmed by the controller
- No independent decision-making — follows instructions from the control layer
- Hardware: white-box switches, programmable ASICs, TCAM for fast lookup

3.2 Control Layer (SDN Controller)

- The “**brain**” of the network
- Maintains a **global view** of topology, links, hosts, and traffic
- Translates high-level policies into low-level flow rules
- Examples: ONOS, OpenDaylight (ODL), Ryu, Floodlight, Nokia NSP, Cisco ACI

3.3 Application Layer

- **Business logic and policies** expressed as software applications
- Consumes the controller’s northbound API
- Examples: traffic engineering, security (micro-segmentation), load balancing, monitoring, intent-based networking

4 SDN Controller Functions

Function	Description
Topology Discovery	Discovers all network devices, links, and hosts. Builds a graph representation. Uses LLDP, OFDP.
Path Computation / Routing	Computes optimal paths using global view. Can apply constraints (latency, bandwidth, cost).
Policy Enforcement	Translates high-level policies into device-level flow rules.
API Exposure	Provides northbound REST/gRPC APIs for applications to consume network services.
State Management	Maintains consistent network state (flow tables, counters, port status).
Multi-tenancy	Network slicing and isolation between tenants.
Fault Management	Detects link/device failures and reroutes traffic in milliseconds.

5 Southbound Interfaces

The southbound interface connects the **controller to infrastructure devices** (data plane).

5.1 OpenFlow

- Most well-known SDN southbound protocol
- Defined by **Open Networking Foundation (ONF)**
- Provides a standard way to program **flow tables** in switches
- Controller installs/modifies/deletes flow entries

- Switch reports events (topology changes, packet-in) to controller
- Versions: 1.0 → 1.3 (most deployed) → 1.5

5.2 P4 (Programming Protocol-independent Packet Processors)

- **Domain-specific language** for programming the data plane itself
- Goes beyond OpenFlow: defines **what** the switch can match on
- Programmable packet parsing and processing pipelines
- Target-independent: compiles to different hardware (Tofino, FPGA, software switches)
- Use case: custom headers, in-network computing, INT (In-band Network Telemetry)

5.3 ForCES (Forwarding and Control Element Separation)

- IETF standard (RFC 5810)
- Separates Control Element (CE) from Forwarding Element (FE)
- Defines a protocol for CE-FE communication
- Less adopted than OpenFlow in practice

5.4 Other Southbound Protocols

Protocol	Use Case
NETCONF/YANG	Device configuration (model-driven)
gNMI/gNOI	Streaming telemetry and operations
BGP-LS	Topology export from traditional networks to SDN controller
PCEP	Path computation for MPLS/SR networks

6 Northbound Interfaces

The northbound interface connects **applications to the controller**.

6.1 REST APIs

- Most common northbound interface
- HTTP-based, JSON/XML payloads
- CRUD operations on network resources (flows, topologies, policies)
- Example:

```
JSON
POST /api/v1/flows
{
  "switch": "s1",
  "match": {"src_ip": "10.0.0.1", "dst_ip": "10.0.0.2"},
  "action": {"output": "port3"},
  "priority": 100
}
```

6.2 Intent-Based Networking (IBN)

- Higher abstraction: express **what** you want, not **how** to achieve it
- Controller translates intent into device-level configuration
- Example intent: *“Ensure video traffic between Site A and Site B has latency < 10ms”*
- The controller figures out the path, QoS policies, and failover
- Implementations: ONOS intent framework, Cisco DNA Center, Nokia NSP

6.3 gRPC / Protobuf

- High-performance, streaming-capable API
- Used in modern controllers for real-time telemetry and configuration
- Strongly typed (protobuf schemas)

7 OpenFlow Basics

7.1 Flow Table Structure

Each switch maintains one or more **flow tables**. Each entry contains:

Field	Description
Match Fields	Criteria to match packets (src/dst MAC, IP, port, VLAN, MPLS label)
Priority	Higher priority entries matched first
Counters	Packet/byte count for statistics
Instructions/Actions	What to do with matched packets
Timeouts	Idle timeout, hard timeout (auto-expire)
Cookie	Opaque identifier set by controller

7.2 Actions

Action	Description
OUTPUT(port)	Forward packet to specified port
DROP	Discard the packet
SET_FIELD	Modify header fields (rewrite MAC, IP, VLAN)
PUSH/POP_VLAN	Add or remove VLAN tags
GROUP	Send to a group table (multipath, failover)
METER	Rate-limit the flow

7.3 Group Tables

Enable advanced forwarding:

Group Type	Use Case
All	Multicast/broadcast (copy to all buckets)
Select	Load balancing (round-robin or weighted)
Indirect	Single next-hop (efficient pointer)
Fast Failover	First live bucket used (link protection)

7.4 Meter Tables

- Rate limiting per flow
- Defined in **bands**: rate threshold + action (drop or DSCP remark)
- Enables QoS enforcement at the switch level

7.5 Packet Processing Pipeline

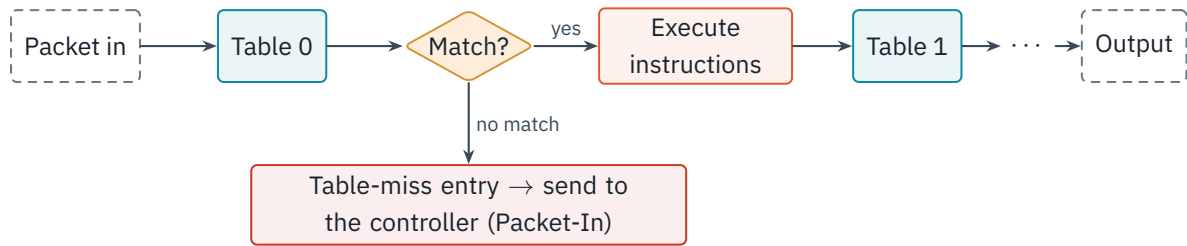


Figure 15.2 OpenFlow packet-processing pipeline

8 SDN in Telecom

8.1 Transport SDN (WAN Optimization)

- Centralized control of optical/MPLS transport networks
- Dynamic bandwidth allocation based on traffic demand
- Multi-layer optimization (IP over optical)
- Examples: Nokia NSP, Ciena Blue Planet, Cisco WAE

8.2 SD-WAN (Software-Defined Wide Area Network)

- SDN principles applied to enterprise WAN
- Centralized controller manages branch CPE devices
- Overlays on top of multiple underlay transports (MPLS, Internet, LTE)
- Application-aware routing (route voice over MPLS, web over Internet)
- Zero-touch provisioning for branch sites
- Vendors: Cisco Viptela, VMware VeloCloud, Fortinet, Versa

8.3 5G Integration: SMF as SDN Controller for UPF

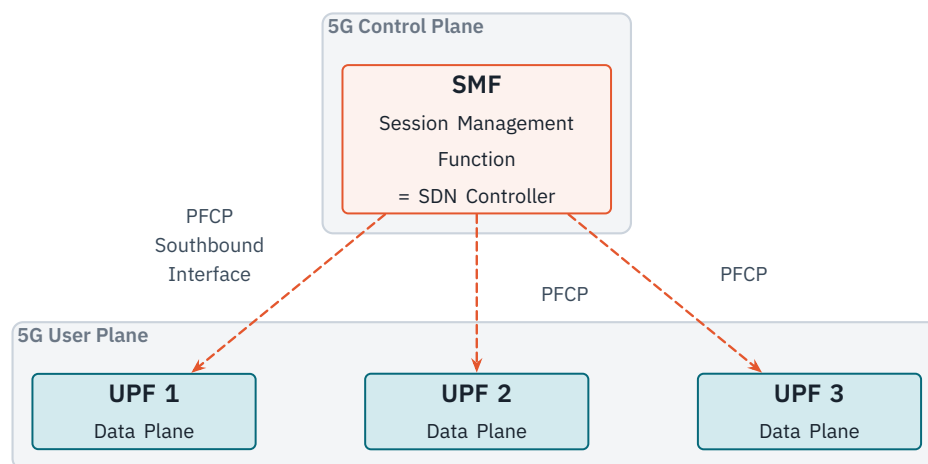


Figure 15.3 The SMF acting as an SDN controller for the UPFs

The 5G User Plane is inherently SDN-like:

SDN Concept	5G Equivalent
SDN Controller	SMF (Session Management Function)
Data Plane Switch	UPF (User Plane Function)
Southbound Protocol	PFCP (Packet Forwarding Control Protocol)
Flow Rules	PDRs (Packet Detection Rules) + FARs (Forwarding Action Rules)
QoS Enforcement	QERs (QoS Enforcement Rules)

8.4 UPF Programmability via PFCP

PFCP (3GPP TS 29.244) is the protocol between SMF and UPF:

- **PDR (Packet Detection Rule)**: Match criteria (source interface, IP, GTP TEID)
- **FAR (Forwarding Action Rule)**: Action (forward, drop, buffer, encapsulate)
- **QER (QoS Enforcement Rule)**: Rate limiting (MBR, GBR)
- **URR (Usage Reporting Rule)**: Traffic measurement and reporting

This is functionally equivalent to OpenFlow's match-action model!

9 Benefits of SDN

Benefit	Description
Programmability	Network is software-controlled; automate everything via APIs
Automation	Zero-touch provisioning, auto-remediation, CI/CD for network
Multi-vendor	Open interfaces allow mixing hardware vendors (white-box revolution)
Fast Innovation	New services deployed in hours/days vs. months
Global Optimization	Centralized view enables optimal path computation
Network Slicing	Dynamic resource allocation per tenant/service
Cost Reduction	Commodity hardware + open-source software vs. proprietary boxes

10 Challenges of SDN

10.1 Controller Scalability

- Single controller cannot handle thousands of switches + millions of flows
- **Solution:** Controller clustering (ONOS, ODL support distributed controllers)
- Trade-off: consistency vs. availability (CAP theorem applies)

10.2 Single Point of Failure

- If the controller fails, no new flows can be installed
- Data plane continues forwarding existing flows (proactive mode)
- **Mitigation:** Controller redundancy (active-standby, active-active)

10.3 East-West Interface

- Communication between controller instances in a cluster
- No standard protocol (each controller has proprietary clustering)
- Challenges: state synchronization, split-brain scenarios, consistency

10.4 Security

Threat	Impact
Controller compromise	Attacker controls entire network
Man-in-the-middle (controller↔switch)	Flow rule injection/modification
DoS on controller	Overwhelm with Packet-In messages
Malicious application	Rogue app installs harmful flows via NBI

Mitigations: TLS on OpenFlow channel, RBAC for northbound API, rate-limiting Packet-In, application sandboxing.

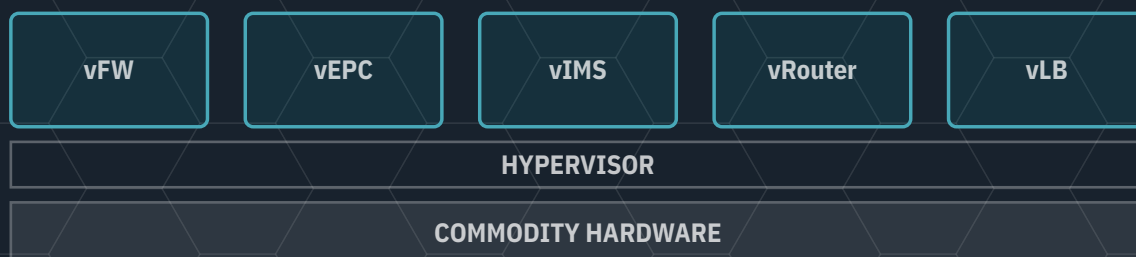
Key Takeaways

1. SDN **separates** the control plane from the data plane
2. A **centralized controller** provides a global view and programmability
3. **OpenFlow** is the foundational southbound protocol (match-action paradigm)
4. **P4** extends programmability to the data plane pipeline itself
5. 5G architecture is **inherently SDN-like** (SMF controls UPF via PFCP)
6. SDN enables **network automation, slicing, and rapid service deployment**
7. Challenges remain around **scalability, security, and standardization**

Network Function Virtualization (NFV)

مجازی سازی توابع شبکه (NFV)

16



In This Module

1	Why NFV?	314
2	Traditional vs NFV Comparison	314
3	ETSI NFV Architecture	316
4	NFV Infrastructure (NFVI)	317
5	VNF vs CNF	318
6	MANO (Management and Orchestration)	319
7	NFV in Telecom (Real Examples)	320
8	Evolution: Physical → Virtual → Cloud-Native	321
9	Benefits of NFV	322
10	Challenges of NFV	322
	Key Takeaways	323

1 Why NFV?

Traditional telecom networks rely on **proprietary hardware appliances** for every network function:

Problem	Description
High CAPEX	Each function requires dedicated, expensive hardware (firewall appliance, load balancer, DPI box, etc.)
High OPEX	Power, cooling, rack space, vendor maintenance contracts for each appliance
Slow Scaling	Need more capacity? Order hardware, wait weeks for delivery, rack-and-stack, configure
Underutilization	Hardware sized for peak load; average utilization often 20-30%
Vendor Dependency	Locked into vendor hardware lifecycle, upgrade timelines, and pricing
Rigid Deployment	Cannot easily move functions, resize, or replicate on demand

The Core Insight: If network functions (firewall, router, load balancer) are just software, why not run them on standard servers as VMs or containers?

NFV was initiated by a group of telecom operators (AT&T, BT, Deutsche Telekom, etc.) who published the ETSI NFV white paper in 2012.

2 Traditional vs NFV Comparison

Traditional Approach

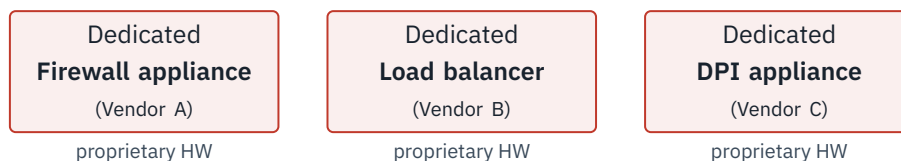


Figure 16.1 Traditional approach: one proprietary appliance per function

NFV Approach

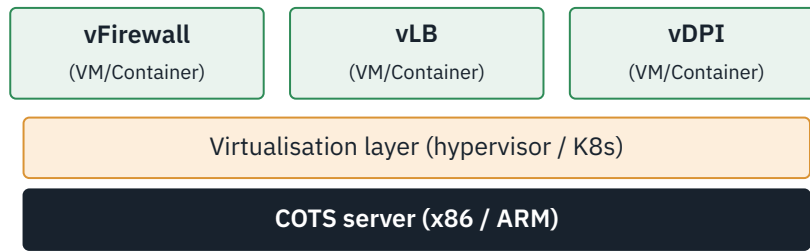


Figure 16.2 NFV approach: virtual functions on shared commodity hardware

Aspect	Traditional	NFV
Hardware	Proprietary, purpose-built	COTS (Commercial Off-The-Shelf) servers
Software	Embedded in appliance	Decoupled, runs as VM/container
Scaling	Buy more hardware	Spin up more instances (minutes)
Cost	High CAPEX + OPEX	Lower (commodity HW + SW licenses)
Flexibility	Fixed function	Any function on any server
Lifecycle	5-7 year hardware refresh	Continuous software updates

3 ETSI NFV Architecture

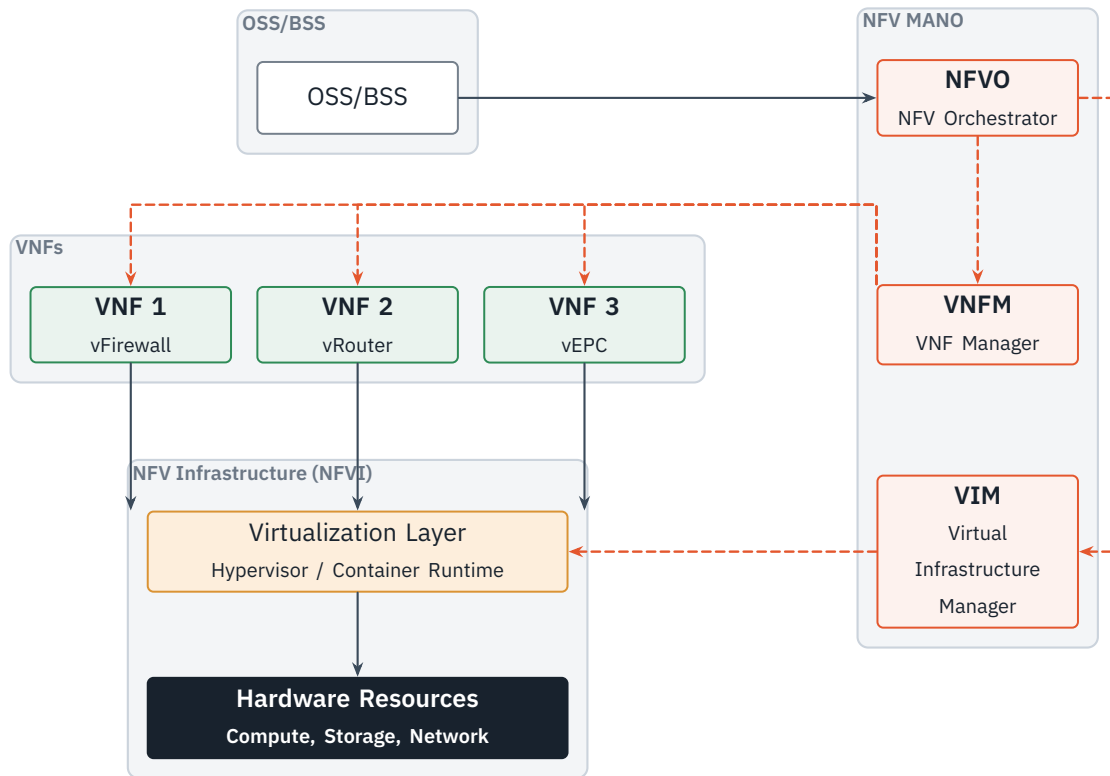


Figure 16.3 ETSI NFV reference architecture

■ Three Main Components:

Component	Role
NFVI	The infrastructure: hardware + virtualization layer that hosts VNFs
VNF	Software implementation of a network function running on NFVI
MANO	Management and Orchestration: lifecycle, resources, service chaining

4 NFV Infrastructure (NFVI)

NFVI provides the environment where VNFs execute.

4.1 Hardware Resources

Resource	Description
Compute	x86/ARM CPUs, GPU accelerators, NUMA topology
Storage	Local SSD/NVMe, SAN, distributed storage (Ceph)
Network	NICs (10/25/100G), SmartNICs, switches, SR-IOV capable

4.2 Virtualization Layer

Technology	Description
Hypervisor	KVM, VMware ESXi, Xen — creates VMs
Container Runtime	Docker, containerd, CRI-O — creates containers
Kubernetes	Container orchestration platform (for CNFs)

4.3 COTS Servers

- **Commercial Off-The-Shelf** servers (Dell, HP, Supermicro)
- Standard x86 architecture — no proprietary hardware
- High-volume, competitive pricing
- Performance enhanced with: SR-IOV, DPDK, huge pages, CPU pinning, NUMA awareness

4.4 Acceleration Technologies

Technology	Purpose
SR-IOV	Bypass hypervisor for direct NIC access (low latency)
DPDK	User-space packet processing (bypass kernel)
SmartNIC/DPU	Offload networking to NIC hardware
FPGA	Programmable hardware acceleration
Huge Pages	Reduce TLB misses for high-throughput workloads

5 VNF vs CNF

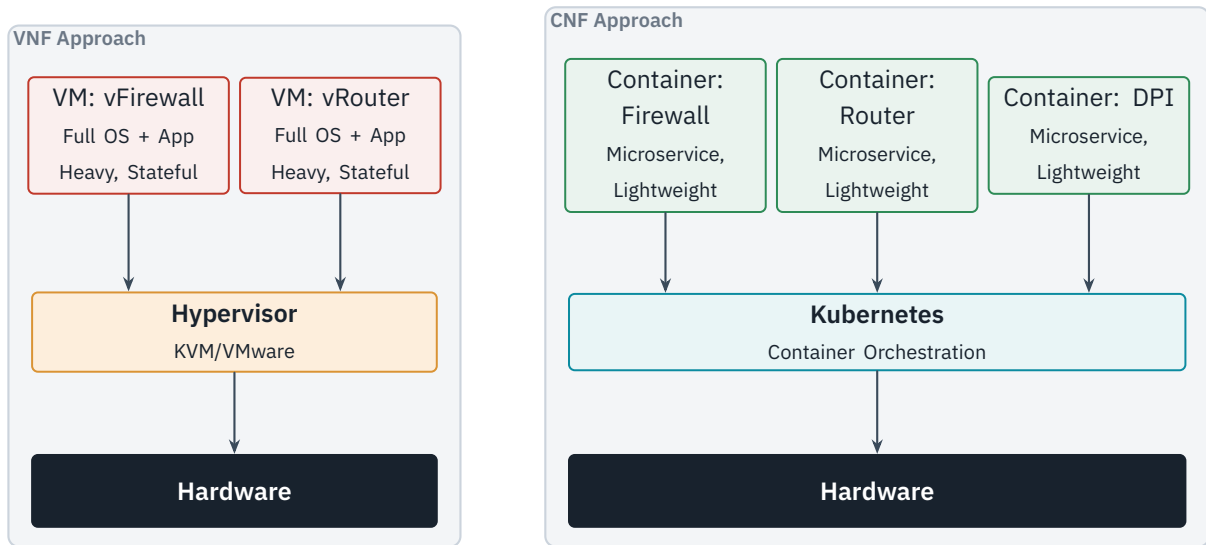


Figure 16.4 VNF (virtual machines) versus CNF (containers)

Comparison

Aspect	VNF (VM-based)	CNF (Container-based)
Runtime	Hypervisor (KVM, VMware)	Container runtime (K8s)
Size	GBs (full OS image)	MBs (shared kernel)
Boot Time	Minutes	Seconds
Architecture	Monolithic	Microservices
State	Stateful (local state in VM)	Stateless (external state store)
Scaling	Vertical (bigger VM) or slow horizontal	Fast horizontal (replicas)
Updates	Disruptive (VM restart)	Rolling updates, canary
Lifecycle	VNFM manages	Kubernetes manages (Helm, Operators)
Density	Low (few VMs per server)	High (hundreds of containers)
CI/CD	Difficult	Native (GitOps, pipelines)

5G Core = CNFs

The 3GPP 5G Core (5GC) is designed as a **Service-Based Architecture (SBA)** with cloud-native principles:

- Each Network Function (AMF, SMF, UPF, NRF, etc.) is a **CNF**
- Deployed on **Kubernetes**
- Communicates via **HTTP/2 + REST APIs**

- Stateless design with external data stores (e.g., UDSF)
- Horizontal scaling, rolling upgrades, self-healing

6 MANO (Management and Orchestration)

6.1 NFVO (NFV Orchestrator)

Function	Description
Service Orchestration	Composes end-to-end network services from multiple VNFs
Lifecycle Management	Onboard, instantiate, scale, update, terminate services
Resource Coordination	Requests resources from VIM across multiple data centers
Catalog Management	Stores NSD (Network Service Descriptors) and VNF packages
Policy Management	Placement, affinity, anti-affinity rules

Examples: ONAP, OSM (Open Source MANO), Cloudify, Nokia CloudBand

6.2 VNFM (VNF Manager)

Function	Description
Instantiate	Deploy VNF from package (image + descriptor)
Scale	Scale out (add instances) or scale in (remove)
Heal	Detect failure and restart/replace VNF instances
Terminate	Graceful shutdown and resource release
Update/Upgrade	Apply new software version
Configure	Day-0 (initial), Day-1 (service), Day-2 (operational) config

6.3 VIM (Virtualized Infrastructure Manager)

Function	Description
Compute Management	Create/delete VMs or containers, CPU/memory allocation
Network Management	Virtual networks, subnets, ports, security groups
Storage Management	Volumes, snapshots, object storage
Resource Monitoring	Utilization, capacity, alarms

Examples:

- **OpenStack** — most common VIM for VNFs (Nova, Neutron, Cinder)
- **Kubernetes** — VIM for CNFs (native container orchestration)
- **VMware vCenter** — enterprise virtualization

7 NFV in Telecom (Real Examples)

7.1 vEPC (Virtualized Evolved Packet Core)

Component	Traditional	Virtualized
MME	Dedicated chassis	VM running MME software
S-GW	Proprietary appliance	VM/container with S-GW logic
P-GW	Proprietary appliance	VM/container with P-GW logic
HSS	Dedicated server	VM with database

Benefits: scale MME independently during mass events, deploy regional instances

7.2 vIMS (Virtualized IP Multimedia Subsystem)

- P-CSCF, S-CSCF, I-CSCF as VNFs
- Scales voice/video capacity on demand
- Deploy in new markets without shipping hardware

7.3 5G Core as CNFs on Kubernetes

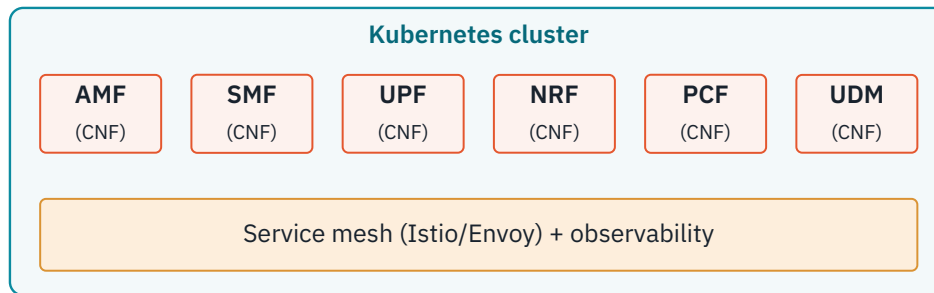


Figure 16.5 5G core functions as CNFs on Kubernetes

- Each 5G NF is a Kubernetes deployment with multiple replicas
- Helm charts or Operators manage lifecycle
- GitOps pipelines for CI/CD
- Horizontal Pod Autoscaler for dynamic scaling

8 Evolution: Physical → Virtual → Cloud-Native

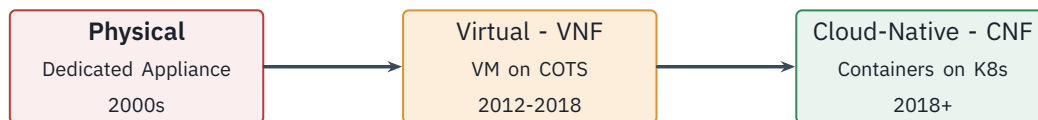


Figure 16.6 From physical appliances to cloud-native functions

Generation	Form Factor	Orchestration	Scaling	Updates
Physical	Proprietary appliance	Manual / NMS	Buy hardware	Firmware upgrade (downtime)
Virtual (VNF)	VM on hypervisor	MANO (NFVO/VNFM)	New VMs (minutes)	VM image replacement
Cloud-Native (CNF)	Container on K8s	Kubernetes + GitOps	Pod replicas (seconds)	Rolling / Canary / Blue-Green

Why the Industry Moved to CNF:

1. **Faster scaling:** Seconds vs. minutes
2. **Higher density:** More functions per server
3. **DevOps native:** CI/CD, GitOps, automated testing
4. **Resilience:** Self-healing, stateless design
5. **Portability:** Run anywhere Kubernetes runs (on-prem, public cloud, edge)

9 Benefits of NFV

Benefit	Description
Flexibility	Deploy any function on any server; relocate workloads
Elastic Scaling	Scale out/in based on demand (auto-scaling)
Multi-tenancy	Multiple tenants share infrastructure securely
CI/CD	Continuous integration and deployment of network functions
Cost Reduction	40-60% CAPEX reduction, 30-50% OPEX reduction (industry estimates)
Speed to Market	New services in days/weeks vs. months/years
Vendor Diversity	Mix VNFs from different vendors on same infrastructure
Geographic Flexibility	Deploy at edge, regional DC, or central cloud
Resource Efficiency	Better utilization through sharing and dynamic allocation

10 Challenges of NFV

10.1 Performance (Packet Processing)

- VMs/containers add overhead vs. bare-metal ASIC processing
- **Mitigation:** DPDK, SR-IOV, SmartNICs, CPU pinning, huge pages
- UPF and other data-plane VNFs are most sensitive

10.2 Orchestration Complexity

- MANO stack is complex to deploy and operate
- Multi-vendor VNF interoperability issues
- Descriptor standardization still evolving (TOSCA, SOL001/006)
- Day-2 operations (monitoring, healing, scaling policies) are hard

10.3 State Management

- VNFs often designed as stateful monoliths (ported from appliances)
- Stateful VMs are hard to scale horizontally or migrate
- CNF approach: externalize state (Redis, etcd, databases)
- Challenge: latency of external state access for high-throughput functions

10.4 Reliability and Availability

- Software failures more frequent than hardware failures
- Need robust health monitoring, auto-healing, redundancy
- Carrier-grade availability (99.999%) requires careful design

10.5 Security

- Larger attack surface (hypervisor, orchestration APIs, shared infrastructure)
- Multi-tenant isolation must be enforced at multiple layers
- Container escape vulnerabilities

10.6 Skills Gap

- Operators must learn cloud technologies (OpenStack, K8s, CI/CD)
- Telecom + IT convergence requires new organizational models

Key Takeaways

1. NFV **decouples network functions from proprietary hardware** → software on COTS
2. ETSI NFV architecture: **NFVI + VNF + MANO**
3. **VNFs** (VM-based) are being replaced by **CNFs** (container-based, cloud-native)
4. **5G Core is designed as CNFs** on Kubernetes from the start
5. **MANO** orchestrates the full lifecycle: instantiate, scale, heal, terminate
6. Evolution: Physical → Virtual (VNF) → **Cloud-Native (CNF)**
7. Key challenges: **performance, orchestration complexity, and state management**
8. NFV + SDN together enable **fully programmable, automated telecom networks**

3GPP Standardization for Advanced Communication Networks

استانداردسازی 3GPP برای شبکه‌های ارتباطی پیشرفته

17



In This Module

1	What is 3GPP?	325
2	3GPP Organizational Structure	326
3	Specification Types	327
4	Release System	329
5	How to Navigate 3GPP Specifications	331
6	Study Items vs Work Items	333
7	Practical Guide: Referencing 3GPP Specs in Academic Work	335
8	Important Specification Series Table	336
	Summary	337

1 What is 3GPP?

The **3rd Generation Partnership Project (3GPP)** is a global collaboration of telecommunications standards organizations that develops technical specifications and reports for mobile communication systems. Founded in December 1998, 3GPP unites seven regional Organizational Partners (OPs) to produce globally applicable standards covering radio access, core network architecture, and service capabilities.

Organizational Partners

Partner	Region	Full Name
ETSI	Europe	European Telecommunications Standards Institute
ARIB	Japan	Association of Radio Industries and Businesses
ATIS	North America	Alliance for Telecommunications Industry Solutions
CCSA	China	China Communications Standards Association
TTA	South Korea	Telecommunications Technology Association
TTC	Japan	Telecommunication Technology Committee
TSDSI	India	Telecommunications Standards Development Society, India

Key Characteristics

- **Not a legal entity** — 3GPP is a partnership project governed by its OPs
- **Market Representation Partners (MRPs)** provide industry input (e.g., GSMA, 5GAA, NGMN)
- **Individual Members** are companies participating through their regional OP
- Specifications are **transposed** into regional standards by each OP (e.g., ETSI publishes them as EN/ES/TS)
- Covers **2G (GSM/GPRS/EDGE), 3G (UMTS), 4G (LTE/LTE-A), and 5G (NR)** systems

2 3GPP Organizational Structure

■ Organizational Structure Diagram

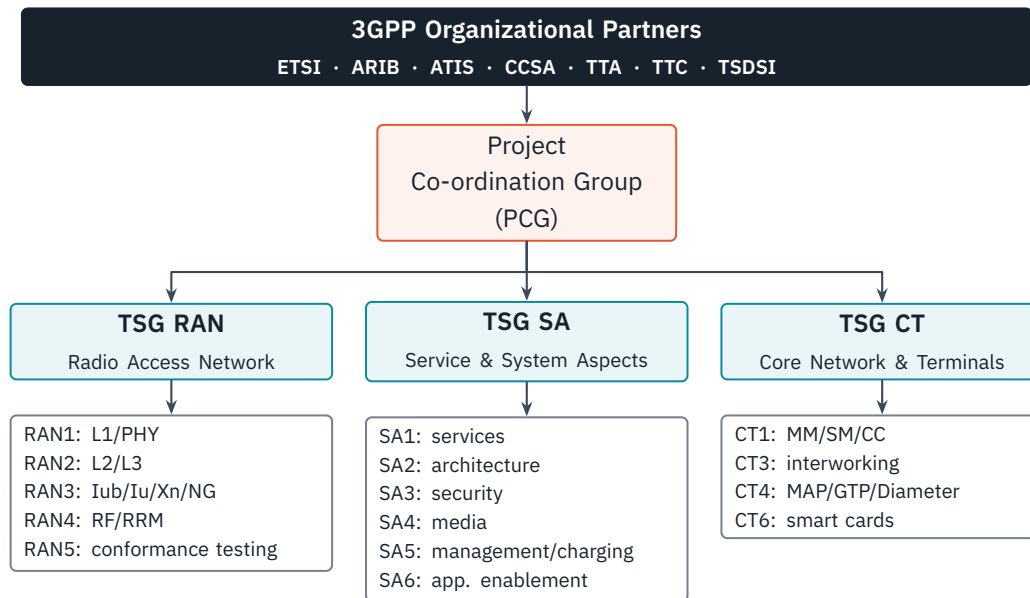


Figure 17.1 3GPP organisational structure

■ Technical Specification Groups (TSGs)

■ TSG RAN – Radio Access Network

Working Group	Responsibility
RAN1	Physical layer (PHY) specifications
RAN2	Layer 2 (MAC, RLC, PDCP, SDAP) and Layer 3 (RRC)
RAN3	Network architecture interfaces (Xn, NG, F1, E1)
RAN4	Radio performance, RF parameters, RRM requirements
RAN5	UE conformance testing

■ TSG SA – Service and System Aspects

Working Group	Responsibility
SA1	Service requirements and features
SA2	System architecture and network functions
SA3	Security and privacy
SA4	Media codecs, streaming, MBMS
SA5	Management, orchestration, and charging
SA6	Application layer enablement (SEAL, CAPIF, EDGEAPP)

■ TSG CT – Core Network and Terminals

Working Group	Responsibility
CT1	NAS protocols (MM, SM, call control)
CT3	Interworking with external networks
CT4	Control plane protocols (GTP-C, Diameter, HTTP/2-based SBI)
CT6	Smart card application aspects (USIM, ISIM)

3 Specification Types

3.1 Types of Documents

Type	Name	Nature	Purpose
TS	Technical Specification	Normative	Defines mandatory requirements, procedures, protocols
TR	Technical Report	Informational	Documents studies, feasibility analyses, best practices

- **TS documents** contain “shall” requirements — they define what implementations **MUST** do
- **TR documents** contain “should/may” language — they inform decisions but are not binding
- TRs often precede TSs (a study item produces a TR, which leads to a work item producing/updating TSs)

3.2 Numbering System

The specification number format is: **xx.yyy** (or xx.yyy for series ≤ 13) and **xx.yyy** for all series.

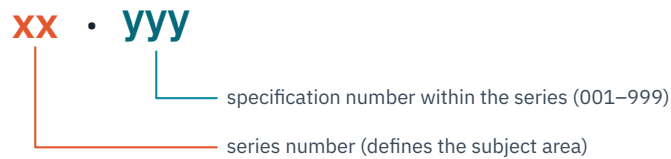


Figure 17.2 Specification numbering: series and number

Example: 38.331

- **38** = 5G NR radio technology series
- **331** = RRC protocol specification

■ Series 00–13: Legacy GSM/GPRS Series (originally numbered without leading digits)

These map to the older GSM specification numbering and are maintained for backward compatibility.

■ Series 21–38: UMTS and Beyond

Series	Subject Area
21	Requirements
22	Service aspects (“stage 1”)
23	Technical realization (“stage 2” — architecture)
24	Signalling protocols (“stage 3” — UE to network)
25	UTRAN radio technology (3G)
26	Codecs
27	Data (AT commands, etc.)
28	Signalling protocols (“stage 3” — SS, OAM)
29	Core network protocols (stage 3 — network-to-network)
30	Programme management (Rel-4 old numbering)
31	UICC / USIM / ISIM
32	OAM&P (management and charging)
33	Security
34	UE test specifications
35	Security algorithms
36	LTE (Evolved UTRA) radio technology
37	Radio measurement and protocol conformance (multi-RAT)
38	5G NR radio technology

3.3 Stage Methodology (ITU-T approach)

Stage	Description	Typical Series
Stage 1	Service description from user's perspective	22.xxx
Stage 2	Logical analysis / architecture	23.xxx
Stage 3	Protocol implementation / detailed procedures	24.xxx, 29.xxx

4 Release System

4.1 Release Lifecycle

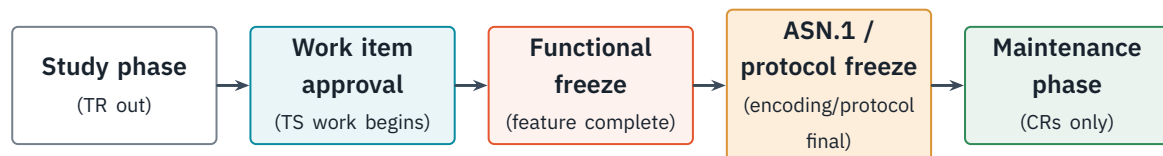


Figure 17.3 Life cycle of a 3GPP release

Phase Details:

1. **Study Phase** — Feasibility and technical investigations; output is a TR
2. **Work Item Approval** — PCG/TSG approves; specification drafting begins
3. **Functional Freeze** — All features for the release are complete; no new functionality added
4. **ASN.1 / Protocol Freeze** — Protocol encoding (ASN.1, JSON, etc.) is frozen; only essential corrections
5. **Maintenance** — Only critical bug-fix Change Requests (CRs) accepted

4.2 Key Releases

Release	Year	Generation	Key Features
R99	2000	3G (UMTS)	WCDMA air interface, CS/PS core, 384 kbps DL
Rel-4	2001	3G	TD-SCDMA, MSC server/MGW split
Rel-5	2002	3G/3.5G	HSDPA (14.4 Mbps DL), IMS introduction
Rel-6	2005	3.5G	HSUPA (5.76 Mbps UL), MBMS, WLAN interworking
Rel-7	2007	3.5G+	HSPA+ (MIMO, 64QAM), VoIP over HSPA
Rel-8	2009	4G (LTE)	OFDMA/SC-FDMA, flat architecture (eNB/EPC), 100 Mbps DL
Rel-9	2010	4G	eMBMS, dual-layer beamforming, HeNB
Rel-10	2011	4G (LTE-A)	Carrier aggregation, 8×8 MIMO, relay nodes, 1 Gbps DL
Rel-11	2013	4G+	CoMP, enhanced ICIC, carrier aggregation enhancements
Rel-12	2015	4G+	D2D (ProSe), MTC/M2M, dual connectivity, 256QAM DL
Rel-13	2016	4G+	LAA, LTE-U, NB-IoT, eMTC (Cat-M1)
Rel-14	2017	4G+/pre-5G	V2X (LTE), eLAA, further IoT enhancements
Rel-15	2019	5G Phase 1	NR NSA/SA, eMBB, mmWave, 5GC (SBA), network slicing
Rel-16	2020	5G Phase 2	URLLC enhancements, NR V2X, IAB, NR-U, IIoT, NPN
Rel-17	2022	5G	NTN (satellite), RedCap, NR sidelink enh., XR, multicast
Rel-18	2024	5G-Advanced	AI/ML for NR, duplex evolution, NR-light enh., XR enh.
Rel-19	2025-26	5G-Advanced+	AI-native air interface studies, ambient IoT, further NTN

5 How to Navigate 3GPP Specifications

5.1 Spec-Numbering Convention

Understanding the number tells you immediately what a spec covers:

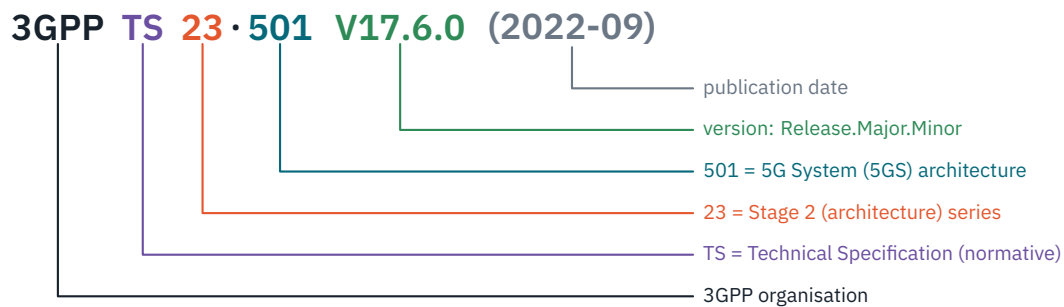


Figure 17.4 Reading a specification reference

Version numbering: The first digit of the version maps to the Release:

- Version 3.x.x → Release 99
- Version 4.x.x → Release 4
- ...
- Version 15.x.x → Release 15
- Version 17.x.x → Release 17

5.2 How to Find Specifications

Resource	URL	Use
Spec Finder	https://www.3gpp.org/specifications/specification-numbering	Search by number
Specification series	https://www.3gpp.org/specifications	Browse by series
Portal (latest + archive)	https://portal.3gpp.org/	Advanced search, CR tracking
FTP server	https://www.3gpp.org/ftp/Specs/	Direct download of .zip files
Work Plan	https://www.3gpp.org/specifications/work-plan	Track ongoing items

Practical Search Workflow:

- Go to **portal.3gpp.org** → “Specifications” tab
- Enter spec number (e.g., “23.501”) or keyword

3. Filter by Release and version
4. Download the latest approved version (.docx or .pdf)

5.3 Reading a 3GPP Specification

Every 3GPP spec follows a standard structure:

Section	Content
1. Scope	What the spec covers and does NOT cover
2. References	Normative and informative references to other specs
3. Definitions and abbreviations	Precise term definitions; abbreviations list
4. General / Overview	High-level system description
5+. Technical content	Procedures, architecture, message flows
Annexes	Additional information, ASN.1, examples

Tips for Reading:

- Start with **Scope** to confirm you have the right document
- Use **References** to trace dependencies (specs reference each other heavily)
- The **Definitions** section resolves ambiguity — always consult it
- Look for “**shall**” (mandatory), “**should**” (recommended), “**may**” (optional)
- Use Ctrl+F for specific procedure names or information elements

5.4 Key Specifications Every Telecom Engineer Should Know

Spec	Title	Why It Matters
TS 23.501	System architecture for the 5G System (5GS)	Defines 5GC architecture, network functions, reference points
TS 23.502	Procedures for the 5G System	Registration, PDU session, handover procedures
TS 24.501	NAS protocol for 5GS (UE ↔ AMF)	5G NAS message definitions, state machines
TS 38.300	NR and NG-RAN overall description	NR system overview, protocol stack, channels
TS 38.331	NR RRC protocol specification	RRC procedures, measurements, reconfiguration
TS 38.211	NR physical channels and modulation	OFDM parameters, resource grid, reference signals
TS 38.213	NR physical layer control procedures	DCI formats, HARQ, power control
TS 38.214	NR physical layer data procedures	Transport block size, CQI/MCS tables, scheduling
TS 33.501	Security architecture for 5GS	5G security framework, key hierarchy, algorithms
TS 28.541	5G NRM (Network Resource Model)	Network management and orchestration model

6 Study Items vs Work Items

6.1 Overview

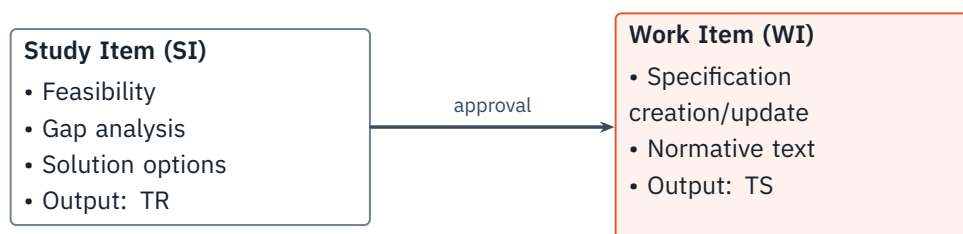


Figure 17.5 Study item versus work item

6.2 Study Item (SI)

- **Purpose:** Investigate feasibility of a new feature or technology
- **Output:** Technical Report (TR) documenting findings, options, conclusions
- **Duration:** Typically 1–4 TSG meetings (9–18 months)
- **Process:**

1. Company submits SI Description (SID) to TSG
2. TSG approves the study
3. Working group conducts the study (contributions from member companies)
4. TR is produced with conclusions and recommendations
5. If feasible → WI proposal submitted

6.3 Work Item (WI)

- **Purpose:** Create or modify normative specifications
- **Output:** New or updated Technical Specifications (TS)
- **Duration:** Tied to the target Release timeline
- **Process:**
 1. Company submits Work Item Description (WID) based on SI conclusions
 2. TSG approves the WI (assigns to working group, sets milestones)
 3. Working group drafts Change Requests (CRs) to existing specs or new specs
 4. CRs are reviewed, revised, and approved at WG/TSG level
 5. Approved CRs are incorporated into the spec version for that Release

6.4 SI → WI Flow Example

Example: NR Sidelink Relay (Release 17)

SI Phase (2019-2020):

- └─ TR 38.836: "Study on NR sidelink relay"
 - └─ Conclusions: Relay architectures A and B feasible
 - └─ Recommended for WI

WI Phase (2020-2022):

- └─ WID approved at TSG RAN#90
 - └─ Updates to: TS 38.300, TS 38.331, TS 38.321, TS 24.587
 - └─ Feature complete at Rel-17 functional freeze

7 Practical Guide: Referencing 3GPP Specs in Academic Work

7.1 Standard Citation Format

3GPP, "Title of Specification," 3GPP TS/TR xx.yyy, Version Vx.y.z, Month Year.

Example (IEEE style):

[1] 3GPP, "System architecture for the 5G System (5GS); Stage 2,"
3GPP TS 23.501, Version 17.6.0, Sep. 2022.

Example (APA style):

3rd Generation Partnership Project. (2022). System architecture for the 5G System (5GS); Stage 2 (3GPP TS 23.501 V17.6.0). <https://www.3gpp.org/>

7.2 Best Practices

Do	Don't
✓ Cite specific version (Vx.y.z)	✗ Cite without version (specs change every quarter)
✓ Include the Release number	✗ Assume "latest" is stable
✓ Reference the exact section/clause	✗ Cite entire 500-page spec without clause
✓ Use official 3GPP document titles	✗ Paraphrase titles
✓ Include access URL/date for online refs	✗ Link to unofficial mirrors

7.3 Where to Download for Citation

- **Official archive:** <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=xxxx>
- **Direct spec URL pattern:** [https://www.3gpp.org/ftp/Specs/archive/\[series\]/\[spec_number\]/](https://www.3gpp.org/ftp/Specs/archive/[series]/[spec_number]/)
- Always cite the **approved** version (not drafts or working documents)

7.4 Citing Change Requests (CRs) and Contributions

For referencing specific proposals or changes:

3GPP TSG RAN WG1, "Title," Tdoc R1-21xxxxx, Meeting #10x, Month Year.

8 Important Specification Series Table

Series	Area	Description	Example Specs
21	Requirements	Overall system requirements	21.915 (Rel-15 summary)
22	Services (Stage 1)	Service descriptions, use cases	22.261 (5G service requirements)
23	Architecture (Stage 2)	System architecture, functional descriptions	23.501 (5GS arch), 23.502 (5GS procedures)
24	UE-Network Signalling (Stage 3)	NAS protocols, SIP, AT commands	24.501 (5G NAS), 24.229 (IMS SIP)
25	UTRAN (3G Radio)	WCDMA physical layer, RLC, RRC	25.331 (UTRAN RRC)
26	Codecs	Speech, audio, video codecs	26.171 (AMR-WB), 26.346 (MBMS)
27	Data	Terminal adaptation, AT commands	27.007 (AT commands)
28	Signalling (Stage 3 - SS)	Supplementary services, OAM	28.541 (5G NRM)
29	CN Protocols (Stage 3)	Core network protocols (N-N)	29.500 (5GC HTTP APIs), 29.244 (PFCP)
30	Programme management	(Legacy - Rel-4 era)	—
31	UICC/USIM/ISIM	SIM card specifications	31.102 (USIM)
32	OAM & Charging	Management, performance, charging	32.255 (5G charging)
33	Security	Security architecture, algorithms	33.501 (5G security)
34	UE Testing	Test specifications, conformance	34.229 (SIP conformance)
35	Algorithms	Cryptographic algorithm specs	35.231 (TUAK algorithm)
36	LTE (E-UTRA)	LTE radio technology	36.331 (LTE RRC), 36.211 (LTE PHY)
37	Multi-RAT	Radio measurements across RATs	37.340 (MR-DC)
38	NR (5G Radio)	5G New Radio technology	38.331 (NR RRC), 38.211 (NR PHY)

Summary

3GPP is the cornerstone of global mobile telecommunications standardization. Understanding its structure, specification numbering, and release system is essential for:

- **Engineers** implementing standards-compliant equipment
- **Researchers** building on or extending current technology
- **Product managers** planning feature roadmaps aligned with release timelines
- **Regulators** referencing normative requirements

Quick Reference Card

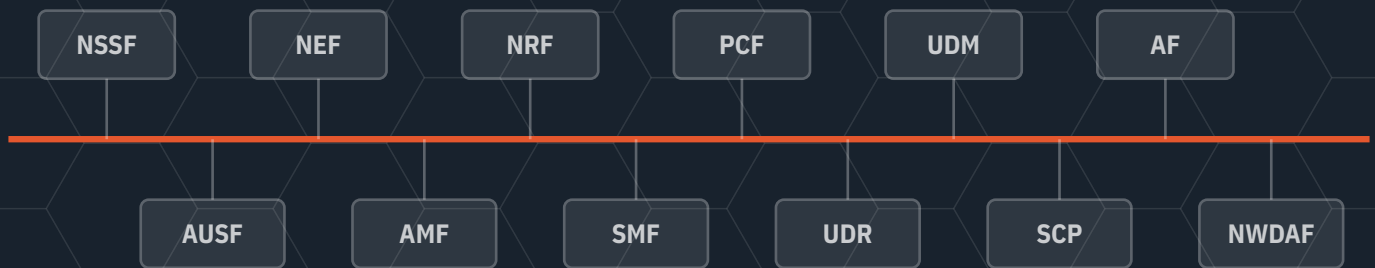
Question	Answer
What does 3GPP stand for?	3rd Generation Partnership Project
How many OPs?	7 (ETSI, ARIB, ATIS, CCSA, TTA, TTC, TSDSI)
How many TSGs?	3 (RAN, SA, CT)
TS vs TR?	TS = normative (shall); TR = informational (study)
Current 5G release?	Rel-18 (5G-Advanced), Rel-19 in progress
Key 5G arch spec?	TS 23.501
Key 5G NR spec?	TS 38.300 (overview), TS 38.331 (RRC)
Spec format?	xx.yyy where xx = series, yyy = spec number
Where to find?	portal.3gpp.org or 3gpp.org/specifications

End of Module 17

5G Service-Based Architecture (SBA)

معماری خدمات محور 5G (SBA)

18



In This Module

1	Why 5G Moved Away from EPC Point-to-Point Interfaces	339
2	SBA Design Principles	340
3	5G Core Network Functions	342
4	NRF — The Key Enabler of SBA	346
5	Service Discovery and Communication Flow	349
6	SBI Protocol Stack	351
7	Comparison: EPC Interfaces vs 5G SBA	352
	Summary	354
	Review Questions	354
	References	354

Learning Objectives

By the end of this module, you will be able to:

- Explain why 5G abandoned EPC's point-to-point interface model
- Describe SBA design principles and the role of HTTP/2, JSON, and REST
- Identify all 5G Core Network Functions and their services
- Explain the NRF's critical role in service discovery
- Compare direct vs indirect (SCP) communication models
- Contrast EPC architecture with 5G SBA

1 Why 5G Moved Away from EPC Point-to-Point Interfaces

1.1 The EPC Problem

In 4G/LTE, the Evolved Packet Core (EPC) uses **point-to-point reference points** between network elements:

Element Pair	Interface
MME ↔ S-GW	S11
MME ↔ HSS	S6a
S-GW ↔ P-GW	S5/S8
MME ↔ eNB	S1-MME
P-GW ↔ PCRF	Gx

The scaling problem:

- With N network elements, you need up to $N \times (N-1)/2$ unique reference points
- Each interface has its own protocol specification (Diameter, GTP-C, etc.)
- Adding a new network element requires defining new interfaces to every existing element
- Rigid coupling makes evolution extremely difficult
- Vendor interoperability requires testing every interface combination

Example: EPC with 6 core elements → up to 15 unique interfaces, each with its own protocol, message formats, and procedures.

1.2 The SBA Solution

5G introduces **Service-Based Architecture** where:

- Network Functions (NFs) expose **services** via a common interface framework
- Any NF can **discover and consume** any other NF's services
- Adding a new NF only requires registering its services — no new interfaces needed
- Common protocol stack (HTTP/2 + JSON) for all control plane communication

1.3 Cloud-Native Principles

SBA embraces cloud-native design:

Principle	Implementation in 5G SBA
Microservices	Each NF is decomposed into independent services
Stateless	NF instances don't store session state locally (externalized to data stores)
Horizontal scaling	Add more NF instances behind load balancers
Containerization	NFs deployed as containers in Kubernetes
CI/CD	Independent NF lifecycle management
Service mesh	SCP provides routing, load balancing, observability

2 SBA Design Principles

2.1 Service-Based Interfaces (SBI) vs Reference Points

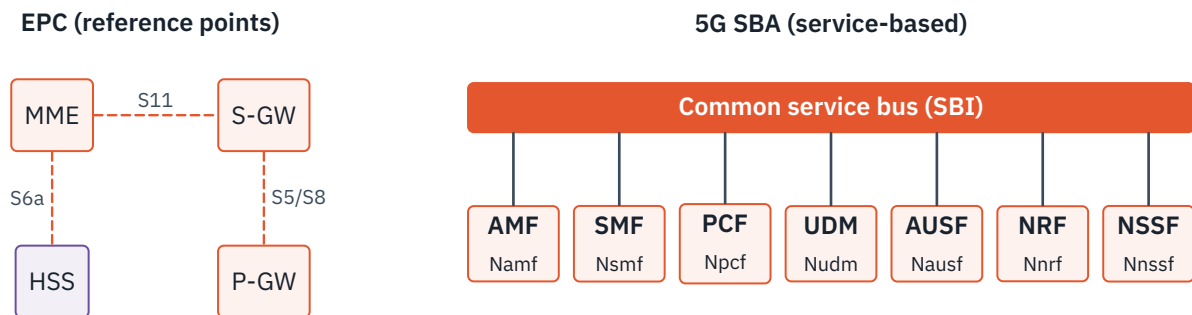


Figure 18.1 EPC reference points versus the 5G service-based bus

Reference Points (4G): Each line is a unique protocol/interface specification

Service-Based Interfaces (5G): All NFs connect to a common bus using the same protocol framework

2.2 NF Service Interaction Models

NFs communicate using two patterns:

1. Request-Response (Synchronous)

- NF Consumer sends a request to NF Producer
- Producer processes and returns a response
- Example: AMF queries UDM for subscriber data

2. Subscribe-Notify (Asynchronous)

- NF Consumer subscribes to events from NF Producer
- Producer sends notifications when events occur

- Example: AMF subscribes to UDM for subscription data changes

2.3 HTTP/2 as Transport Protocol

5G SBA chose HTTP/2 over HTTP/1.1 for critical reasons:

Feature	Benefit for 5G
Multiplexing	Multiple requests/responses over single TCP connection — reduces latency
Header compression (HPACK)	Reduces overhead for repetitive signaling headers
Server push	Enables efficient notification delivery
Binary framing	More efficient parsing than text-based HTTP/1.1
Stream prioritization	Critical signaling (e.g., emergency calls) gets priority

2.4 JSON as Data Format

Reason	Explanation
Human-readable	Easy debugging, logging, troubleshooting
Web-native	Leverages massive web ecosystem tooling
Tooling	Validators, editors, parsers available in every language
Schema support	OpenAPI/JSON Schema for API definition
Lightweight	Less verbose than XML (used in older 3GPP specs)

2.5 REST API Design Principles

5G SBA APIs follow RESTful conventions:

- **Resources** identified by URIs: <https://{nfInstanceId}.5gc.mnc{MNC}.mcc{MCC}/nsmf-pdusession/v1/sm-contexts>
- **HTTP methods** map to operations: GET (read), POST (create), PUT (update), DELETE (remove), PATCH (partial update)
- **Stateless** requests — each request contains all necessary context
- **HATEOAS** — responses include links to related resources
- **API versioning** — version in URI path (v1, v2)

3 5G Core Network Functions

■ SBA Architecture Diagram

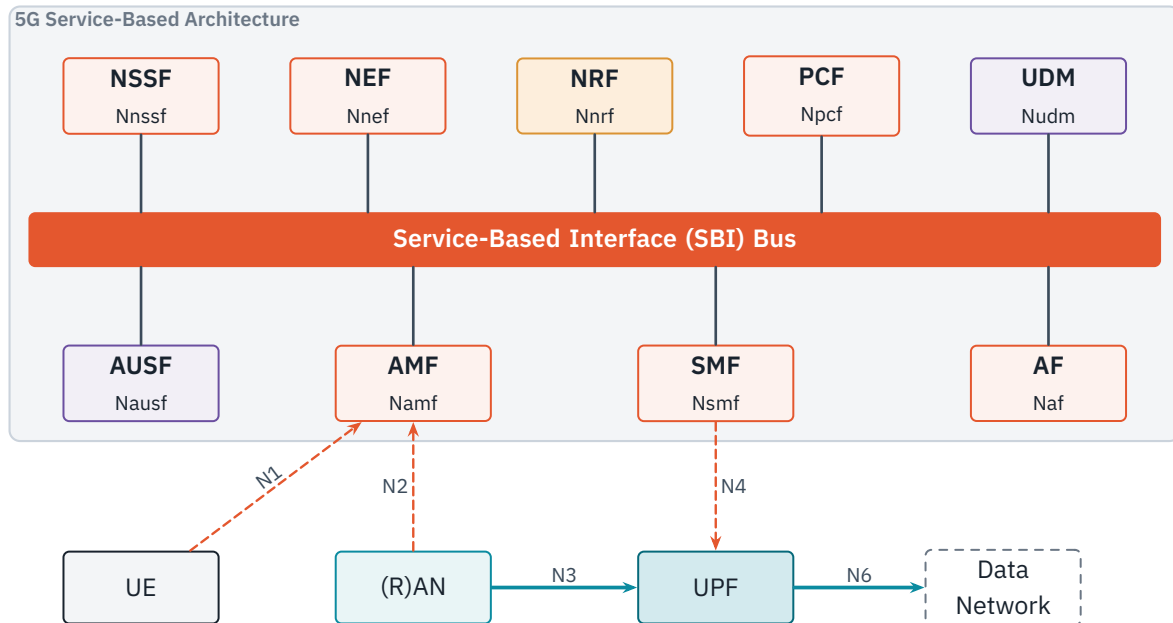


Figure 18.2 5G service-based architecture around the SBI

3.1 AMF – Access and Mobility Management Function

Purpose: Handles all access and mobility-related signaling for the UE.

Service Interface: [Namf](#)

Key Services:

Service	Description
Namf_Communication	N1/N2 message transport, UE-specific signaling
Namf_EventExposure	Expose mobility events (location, reachability)
Namf_MT	Mobile-terminated message delivery
Namf_Location	UE location services

Key Responsibilities:

- UE registration and deregistration
- Connection management (idle ↔ connected)
- Mobility management (handover, tracking area updates)
- Security context management (NAS encryption/integrity)
- Relay NAS-SM messages between UE and SMF

3.2 SMF – Session Management Function

Purpose: Manages PDU (data) sessions between UE and Data Network.

Service Interface: `Nsmf`

Key Services:

Service	Description
<code>Nsmf_PDUSession</code>	Create, modify, release PDU sessions
<code>Nsmf_EventExposure</code>	PDU session events (e.g., IP address change)

Key Responsibilities:

- PDU session establishment, modification, release
- UE IP address allocation and management
- UPF selection and control (via N4/PFCP)
- QoS flow management and enforcement
- Downlink data notification
- Traffic steering and routing configuration

3.3 UPF – User Plane Function

Purpose: Forwards user data packets between RAN and Data Networks.

⚠ **Note:** UPF does NOT use SBI. It uses reference points:

- **N3** – between RAN and UPF (GTP-U tunnel)
- **N4** – between SMF and UPF (PFCP protocol)
- **N6** – between UPF and Data Network (IP traffic)
- **N9** – between UPFs (inter-UPF GTP-U tunnel)

Key Responsibilities:

- Packet routing and forwarding
- QoS handling (marking, policing, shaping)
- Traffic usage reporting
- Uplink traffic verification
- Lawful intercept (user plane)
- Buffering of downlink packets (UE in idle mode)

3.4 UDM – Unified Data Management

Purpose: Manages subscriber data and computes authentication credentials.

Service Interface: `Nudm`

Key Services:

Service	Description
<code>Nudm_SubscriberDataManagement</code>	Retrieve/update subscription data

[Nudm_UEContextManagement](#)	Track which AMF/SMF serves a UE
[Nudm_UEAuthentication](#)	Generate authentication vectors
[Nudm_EventExposure](#)	Subscription data change notifications

Key Responsibilities:

- Stores subscription data (authorized slices, QoS profiles, DNN permissions)
- Computes authentication credentials (using SUPI, K, OPc)
- Manages UE context registration (serving AMF, serving SMF)
- De-concealment of SUCI → SUPI

3.5 AUSF – Authentication Server Function

Purpose: Executes the authentication procedure for UEs.

Service Interface: [Nausf](#)

Key Services:

Service	Description
—	—
[Nausf_UEAuthentication](#)	Perform 5G-AKA or EAP-AKA' authentication
[Nausf_SoRProtection](#)	Steering of Roaming security

Key Responsibilities:

- Authenticates UEs during registration
- Supports 5G-AKA and EAP-AKA' methods
- Stores intermediate key material (K_{AUSF})
- Interfaces with UDM for authentication vectors

3.6 PCF – Policy Control Function

Purpose: Provides policy rules governing network behavior and QoS.

Service Interface: [Npcf](#)

Key Services:

Service	Description
—	—
[Npcf_SMPolicyControl](#)	Session-level policies (QoS, charging, gating)
[Npcf_AMPolicyControl](#)	Access and mobility policies (RFSP, service area)
[Npcf_PolicyAuthorization](#)	AF-triggered policy (e.g., IMS requests QoS)
[Npcf_BDTPolicyControl](#)	Background data transfer policies
[Npcf_EventExposure](#)	Policy-related event notifications

Key Responsibilities:

- QoS policy decisions
- Charging policy rules
- Access control and gating decisions
- Network slicing policies

- Usage monitoring control

3.7 NRF — Network Repository Function

Purpose: The **central registry** of all NF instances and their services — the backbone of SBA.

Service Interface: `Nnrf`

Key Services:

| Service | Description |

|—|—|

| `Nnrf_NFManagement` | NF registration, update, deregistration |

| `Nnrf_NFDiscovery` | Query available NFs by type, slice, location |

| `Nnrf_AccessToken` | Issue OAuth2.0 tokens for NF-to-NF authorization |

★ **NRF is the critical enabler of SBA** — without it, NFs cannot discover each other and the service-based model collapses back to static configuration.

3.8 NSSF — Network Slice Selection Function

Purpose: Selects the appropriate network slice instance for a UE.

Service Interface: `Nnssf`

Key Services:

| Service | Description |

|—|—|

| `Nnssf_NSSelection` | Determine allowed slices and target AMF set |

| `Nnssf_NSSAIAvailability` | Manage NSSAI availability per tracking area |

Key Responsibilities:

- Determines the Allowed NSSAI for UE registration
- Selects target AMF Set/AMF based on requested slices
- Manages per-TA NSSAI availability information

3.9 NEF — Network Exposure Function

Purpose: Securely exposes 5G network capabilities to external applications.

Service Interface: `Nnef`

Key Services:

| Service | Description |

|—|—|

| `Nnef_EventExposure` | Expose network events to AFs |

| `Nnef_PFDManagement` | Packet Flow Description management |

| `Nnef_TrafficInfluence` | Allow AF to influence traffic routing |

| `Nnef_ParameterProvision` | Provision expected UE behavior |

Key Responsibilities:

- API gateway between 5G core and external world
- Translates internal 5GC information to external representations
- Enforces security policies on external access
- Rate limiting and API management

3.10 AF — Application Function

Purpose: Represents third-party applications that interact with the 5G core.

Service Interface: `Naf`

Key Responsibilities:

- Requests QoS for application flows (e.g., video streaming)
- Influences traffic routing (e.g., edge computing)
- Subscribes to network events (e.g., UE location)
- Can access 5GC directly (trusted) or via NEF (untrusted)

Examples: IMS (VoNR), streaming platforms, enterprise applications, MEC orchestrators.

4 NRF — The Key Enabler of SBA

The NRF is the **most critical component** that makes SBA work. Without NRF, 5G would revert to static, pre-configured connections like EPC.

4.1 NF Registration

When an NF instance starts, it **registers** with the NRF by providing its **NF Profile**:

```

JSON
{
  "nfInstanceId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "nfType": "SMF",
  "nfStatus": "REGISTERED",
  "heartBeatTimer": 30,
  "plmnList": [{ "mcc": "310", "mnc": "014" }],
  "sNssai": [{ "sst": 1, "sd": "000001" }],
  "fqdn": "smf1.5gc.mnc014.mcc310.3gppnetwork.org",
  "ipv4Addresses": ["10.10.10.5"],
  "nfServices": [
    {
      "serviceInstanceId": "nsmf-pdusession-01",
      "serviceName": "nsmf-pdusession",
      "versions": [{ "apiVersionInUri": "v1", "apiFullVersion": "1.2.0" }],
      "scheme": "https",
      "nfServiceStatus": "REGISTERED",
      "capacity": 100,
      "load": 25,
      "priority": 1
    }
  ],
  "smfInfo": {
    "sNssaiSmfInfoList": [{ "sNssai": { "sst": 1 }, "dnnSmfInfoList": [{ "dnn":
      ↪ "internet" }] } ]
  },
  "capacity": 100,
  "load": 25,
  "locality": "DC-East-1"
}

```

NF Profile contains:

- **NF Type** — what kind of NF (AMF, SMF, PCF, etc.)
- **Services offered** — list of service names and versions
- **Capacity** — relative capacity for load balancing (0-65535)
- **Load** — current load percentage (0-100)
- **Locality** — geographic/data center location for proximity-based selection
- **PLMN** — which networks this NF serves
- **S-NSSAI** — which slices this NF supports
- **IP/FQDN** — how to reach this NF

4.2 NF Discovery

When an NF needs a service, it queries the NRF:

```

GET /nrf-disc/v1/nf-instances?target-nf-type=SMF&requester-nf-type=AMF&snssai={ "sst":1, "sd": "000001" }&dnn=internet

```

Discovery query parameters:

| Parameter | Purpose |

|—|—|

| **target-nf-type** | Type of NF being searched (mandatory) |

| **requester-nf-type** | Type of the requesting NF (mandatory) |

| **service-names** | Specific services needed |

| **snssai** | Network slice filter |

| **dnn** | Data Network Name filter |

| **tai** | Tracking area for locality |

| **preferred-locality** | Prefer NFs in same data center |

NRF returns: Ranked list of matching NF instances with endpoints and load information.

4.3 NF Status Notification

NFs can **subscribe** to NRF for status events:

- NF registered / deregistered
- NF profile changed (load, capacity)
- NF service status changed

This enables:

- **AMF** to be notified when new SMFs become available
- **Load balancers** to update routing when NF load changes
- **Fault detection** when NF heartbeat expires

NF Discovery Flow Diagram

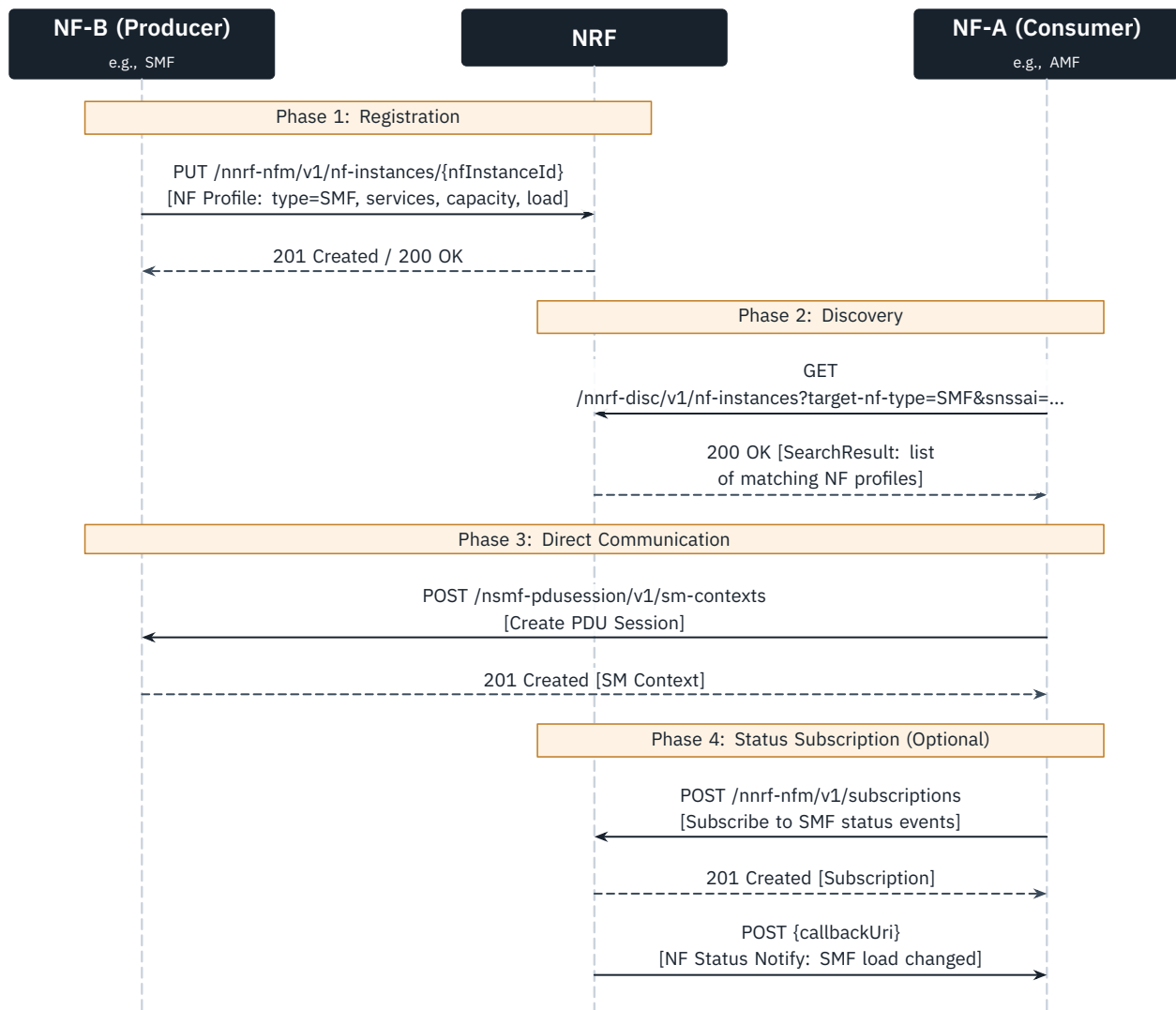


Figure 18.3 NF registration, discovery and subscription through the NRF

5 Service Discovery and Communication Flow

5.1 Direct Communication

In the **direct model**, NF-A communicates directly with NF-B after discovery:

1. NF-A → NRF: "Find me an SMF for slice 1, DNN internet"
2. NRF → NF-A: "Here are 3 SMFs: SMF-1 (load=20%), SMF-2 (load=45%), SMF-3 (load=80%)"
3. NF-A selects SMF-1 (lowest load)
4. NF-A → SMF-1: Direct service request

Advantages: Low latency (no intermediate hop), simple

Disadvantages: NF-A must handle load balancing, retries, failover logic

5.2 Indirect Communication via SCP

The **Service Communication Proxy (SCP)** acts as an intermediary:

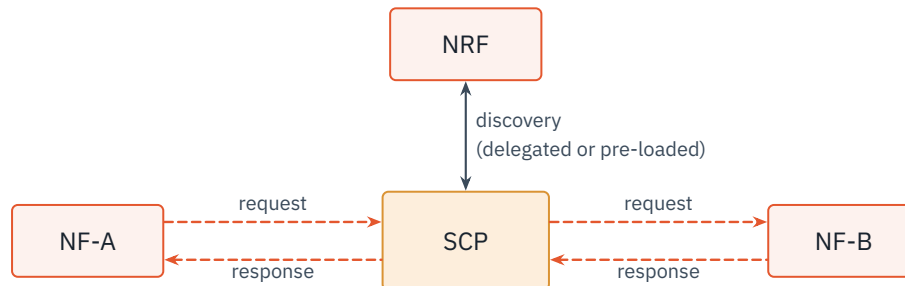


Figure 18.4 Indirect communication through the SCP

SCP provides:

- Delegated discovery (SCP queries NRF on behalf of NF-A)
- Load balancing across NF-B instances
- Message routing and topology hiding
- Retry and failover logic
- Monitoring and observability
- Connection pooling

Communication Models (3GPP TS 23.501):

| Model | Discovery | Routing |

|—|—|—|

| Model A | NF-A via NRF | Direct NF-A → NF-B |

| Model B | NF-A via NRF | Via SCP (NF-A → SCP → NF-B) |

| Model C | Delegated to SCP | Via SCP (NF-A → SCP → NF-B) |

| Model D | Delegated to SCP | Direct NF-A → NF-B |

6 SBI Protocol Stack

Complete Protocol Stack

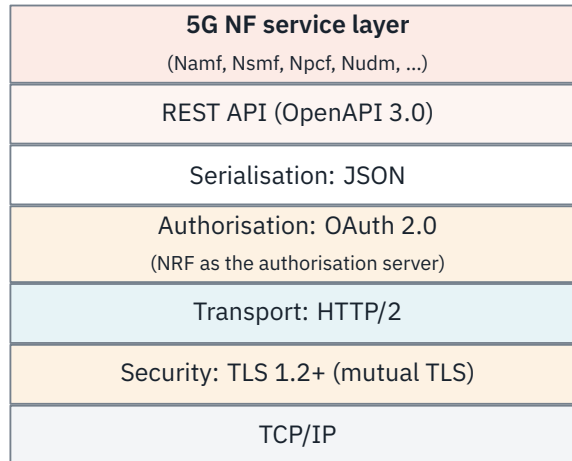


Figure 18.5 SBI protocol stack

OAuth 2.0 for NF Authorization

Before NF-A can call NF-B's service, it must obtain an **access token** from the NRF:

1. NF-A → NRF: POST /oauth2/token

```
{
  "grant_type": "client_credentials",
  "nfInstanceId": "{NF-A ID}",
  "nfType": "AMF",
  "targetNfType": "SMF",
  "scope": "nsmf-pdusession"
}
```
2. NRF → NF-A: Access Token (JWT)

```
{
  "access_token": "eyJ...",
  "token_type": "Bearer",
  "expires_in": 3600,
  "scope": "nsmf-pdusession"
}
```
3. NF-A → NF-B: Service Request + Authorization: Bearer eyJ...
4. NF-B validates the token (signature, expiry, scope)

7 Comparison: EPC Interfaces vs 5G SBA

■ EPC vs SBA Comparison Diagram

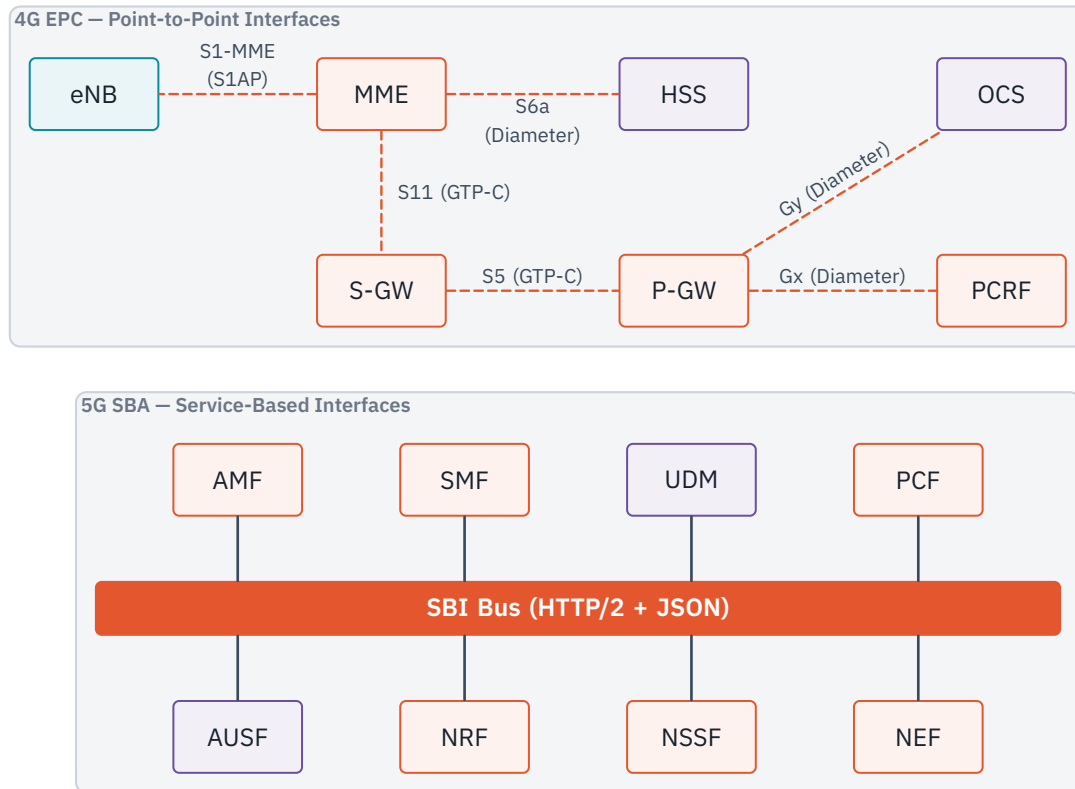


Figure 18.6 Point-to-point EPC interfaces versus the SBA bus

Detailed Comparison Table

Aspect	4G EPC	5G SBA
Architecture	Point-to-point reference points	Service-based interfaces on common bus
Protocols	Diameter, GTP-C, S1AP (multiple)	HTTP/2 + JSON (unified)
Coupling	Tight — each pair has unique interface	Loose — common framework for all
Adding new NF	Define new interfaces to all peers	Register services with NRF
Discovery	Static configuration	Dynamic via NRF
Scaling	Scale entire monolithic element	Scale individual NF services independently
Deployment	Hardware appliances / VMs	Cloud-native containers (Kubernetes)
State	Stateful network elements	Stateless NFs (externalized state)
Redundancy	Active-standby pairs	Horizontal scaling + NRF re-selection
Load balancing	DNS-based or static	NRF-based (capacity, load, locality)
Authorization	Hop-by-hop TLS	End-to-end OAuth 2.0 tokens
API definition	3GPP-specific ASN.1/XML	OpenAPI 3.0 (industry standard)
Extensibility	New release = new interfaces	New services registered dynamically
Vendor ecosystem	Telecom-specific vendors	Cloud/IT vendors can participate

Functional Mapping: EPC → 5G

EPC Element	5G Equivalent	Key Difference
MME	AMF + SMF	Separated access/mobility from session mgmt
S-GW + P-GW	UPF	Simplified to single user plane element
HSS	UDM + AUSF	Separated data storage from authentication
PCRF	PCF	Same role, now on SBI
— (new)	NRF	Service registry (no EPC equivalent!)
— (new)	NSSF	Slice selection (no slicing in 4G)
— (new)	NEF	Replaces SCEF with richer API exposure

Summary

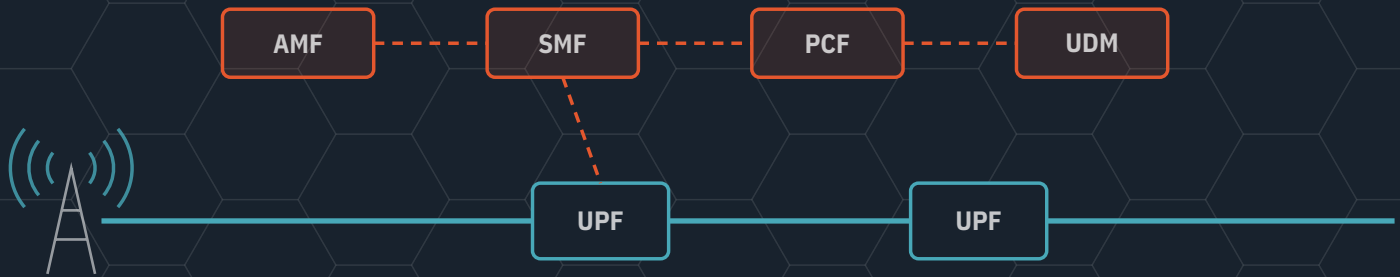
Concept	Key Takeaway
SBA motivation	EPC's $N \times (N-1)/2$ interfaces don't scale; SBA uses a common bus
Design principles	HTTP/2, JSON, REST, microservices, stateless
NRF	Heart of SBA — registration, discovery, authorization
Communication	Direct (NF→NF) or Indirect (via SCP)
Security	mTLS + OAuth 2.0 tokens from NRF
Key difference	EPC = static, rigid, monolithic; SBA = dynamic, flexible, cloud-native

Review Questions

1. Calculate how many unique interfaces are needed for 8 EPC elements. How does SBA solve this?
2. Why was HTTP/2 chosen over HTTP/1.1 for SBI?
3. What information is included in an NF Profile during NRF registration?
4. Why is the UPF not connected to the SBI bus?
5. Compare SCP communication Models B and C — when would you use each?
6. How does OAuth 2.0 work in the 5G SBA context? What role does NRF play?
7. An operator wants to add a new Analytics NF (NWDAF). Compare the effort in EPC vs SBA architecture.
8. Why is NRF considered the “single most critical” component of SBA?

References

- 3GPP TS 23.501 — System Architecture for the 5G System (Stage 2)
- 3GPP TS 23.502 — Procedures for the 5G System (Stage 2)
- 3GPP TS 29.510 — NRF Services (Stage 3)
- 3GPP TS 29.500 — Technical Realization of Service-Based Architecture
- 3GPP TS 33.501 — Security Architecture and Procedures for 5G System



In This Module

1	5G Core Overview	356
2	Reference Architecture	357
3	Network Functions in Detail	358
4	Control Plane vs User Plane in 5G Core	368
5	Major Comparison: 4G EPC vs 5G Core	370
6	Reference Points	372
7	Deployment Options	373
	Summary	376

1 5G Core Overview

The 5G Core (5GC) network is defined in **3GPP TS 23.501** and represents a fundamental redesign of the mobile core network. Unlike the 4G EPC which used point-to-point interfaces between monolithic network elements, the 5GC adopts a **Service-Based Architecture (SBA)** where Network Functions (NFs) expose services via RESTful APIs over HTTP/2.

■ Key Design Principles

- **Service-Based Architecture (SBA):** NFs communicate via service-based interfaces (SBI) using HTTP/2 and JSON
- **Control and User Plane Separation (CUPS):** Complete decoupling of CP and UP
- **Stateless NF Design:** State stored externally (UDSF) enabling horizontal scaling
- **Network Slicing Native:** Built-in support for multiple logical networks on shared infrastructure
- **Cloud-Native:** Designed for containerized, microservice-based deployment
- **Registration-Based NF Discovery:** NFs register with NRF and discover peers dynamically

2 Reference Architecture

The 5GC architecture has two equivalent representations defined in TS 23.501:

2.1 Service-Based Interface (SBI) Representation

In this view, all control-plane NFs connect to a common service bus and expose/consume services.

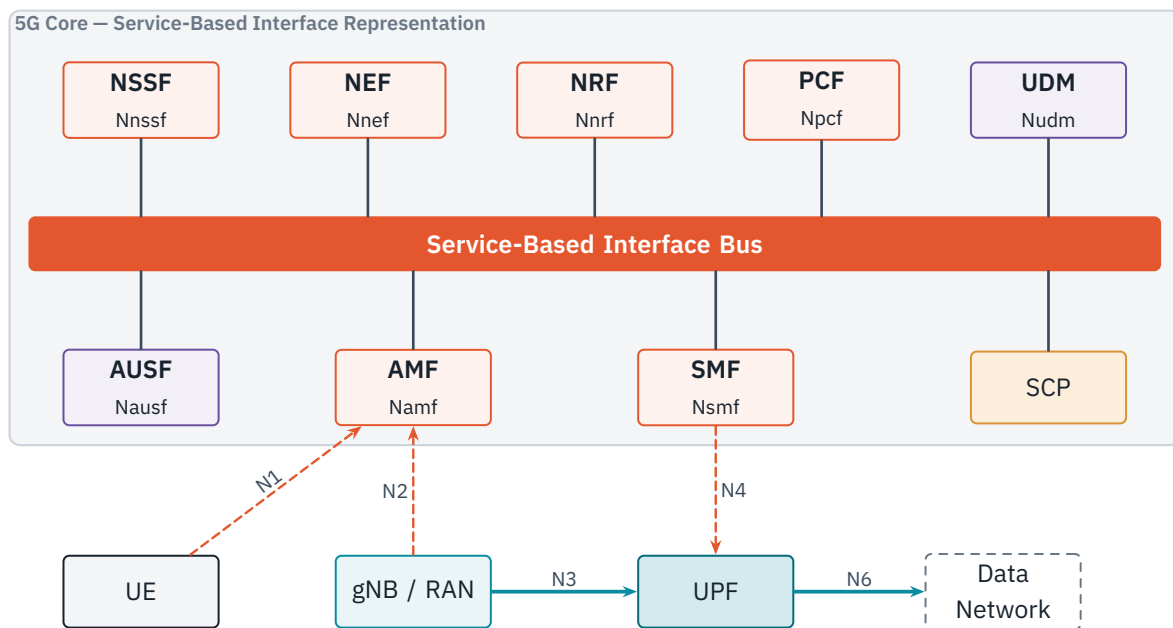


Figure 19.1 5G core in the service-based representation

2.2 Reference Point Representation

In this view, explicit point-to-point reference points are shown between NFs.

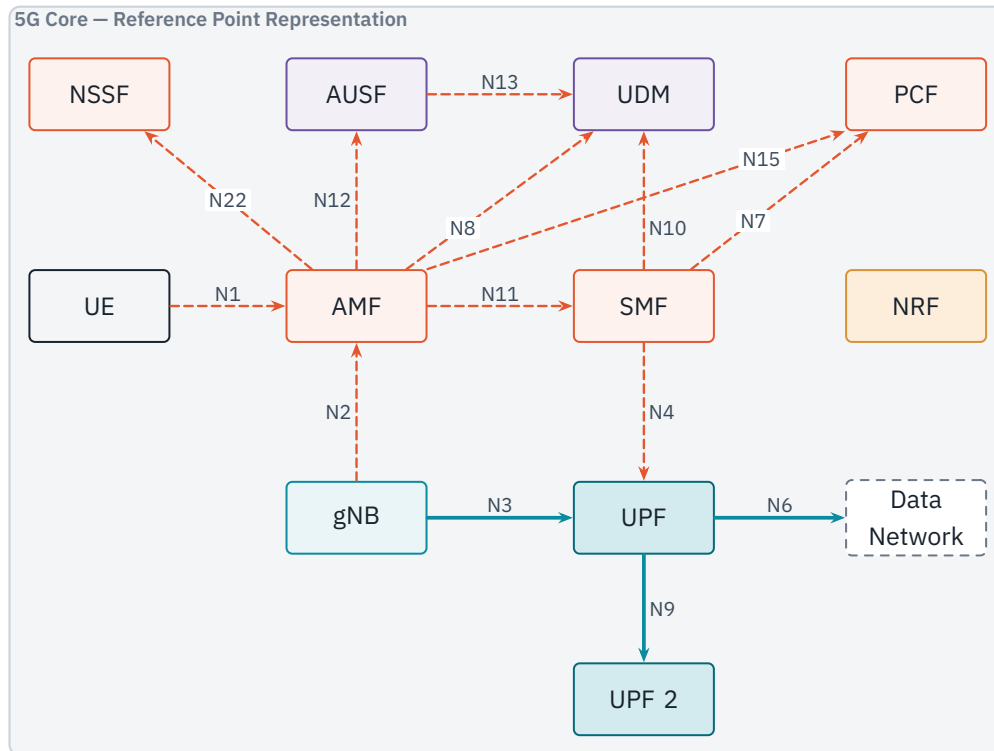


Figure 19.2 5G core in the reference-point representation

3 Network Functions in Detail

3.1 Access and Mobility Management Function (AMF)

The AMF is the single point of contact for all UE-related signaling from the RAN. It terminates the N1 (NAS) and N2 (NGAP) interfaces.

Key Responsibilities:

- **NAS Termination:** All NAS messages (Registration, Service Request, PDU Session Establishment) terminate at AMF
- **Registration Management:** Handles UE registration, deregistration, and periodic registration updates
- **Connection Management:** Manages RRC signaling context via N2 toward gNB
- **Mobility Management:** Handover signaling, tracking area management, paging coordination
- **Security:** Initiates authentication (delegates to AUSF), manages NAS security context (ciphering + integrity)
- **Slice Selection:** Interacts with NSSF to determine the appropriate network slice

Key Identifiers:

Identifier	Description
GUAMI	Globally Unique AMF Identifier = MCC + MNC + AMF Region ID + AMF Set ID + AMF Pointer
5G-GUTI	5G Globally Unique Temporary Identifier = GUAMI + 5G-TMSI (assigned to UE by AMF)
AMF Region ID	8 bits — identifies a region within a PLMN
AMF Set ID	10 bits — identifies an AMF Set within a region
AMF Pointer	6 bits — identifies a specific AMF within a set

Interfaces:

- **N1:** UE ↔ AMF (NAS signaling, carried transparently over RAN)
- **N2:** RAN ↔ AMF (NGAP protocol over SCTP)
- **N8:** AMF ↔ UDM (subscription data retrieval)
- **N11:** AMF ↔ SMF (session management messages)
- **N12:** AMF ↔ AUSF (authentication procedures)
- **N14:** AMF ↔ AMF (UE mobility between AMFs)
- **N15:** AMF ↔ PCF (access and mobility policy)
- **N22:** AMF ↔ NSSF (slice selection)

Registration Area:

- Composed of one or more Tracking Areas (TAs)
- UE only performs Registration Update when moving outside its Registration Area
- Larger registration areas → less signaling, more paging load
- AMF allocates the Registration Area based on UE mobility pattern

3.2 Session Management Function (SMF)

The SMF manages PDU sessions end-to-end. It selects and controls the UPF(s) that form the user plane path.

Key Responsibilities:

- **PDU Session Management:** Establishment, modification, and release of PDU sessions
- **UPF Selection and Control:** Selects appropriate UPF(s), programs forwarding rules via N4/PCF
- **IP Address Allocation:** Assigns IPv4/IPv6 addresses to UEs (or delegates to external DHCP)
- **QoS Enforcement:** Installs QoS rules in UPF based on PCF policies
- **Roaming:** Handles home-routed and local breakout scenarios
- **Downlink Data Notification:** Triggers paging via AMF when downlink data arrives for idle UE

PDU Session Types:

Type	Description	Use Case
IPv4	Traditional IPv4 connectivity	Legacy internet access
IPv6	IPv6-only connectivity	IoT, modern networks
IPv4v6	Dual-stack IPv4 + IPv6	Default smartphone access
Ethernet	Layer-2 Ethernet frames	Industrial LAN extension, TSN
Unstructured	Raw IP or non-IP data	IoT sensors, point-to-point tunnels

N4 Interface and PFCP:

- **N4** connects SMF to UPF using **PFCP (Packet Forwarding Control Protocol)** — 3GPP TS 29.244
- PFCP is a UDP-based protocol (port 8805) with:
 - **Session Establishment:** Create forwarding context in UPF
 - **Session Modification:** Update rules (e.g., handover, QoS change)
 - **Session Deletion:** Remove forwarding context
 - **Heartbeat:** Detect UPF failure

Session Anchor:

- The **PDU Session Anchor (PSA)** is the UPF that maintains a stable point of attachment to the Data Network
- Provides IP address continuity during UE mobility
- Does NOT change during handover (unlike intermediate UPFs)
- SMF may insert Intermediate UPFs (I-UPF) for local routing while keeping PSA stable

3.3 User Plane Function (UPF)

The UPF is the **only user-plane NF** in the 5G Core. It handles all packet processing, forwarding, and policy enforcement for user data.

Key Responsibilities:

- **Packet Routing and Forwarding:** Routes packets between RAN and Data Network
- **Traffic Detection and Reporting:** Performs DPI for application identification
- **QoS Handling:** Enforces per-flow QoS (marking, policing, shaping)
- **Usage Reporting:** Measures traffic volume/duration for charging
- **Uplink Classifier (UL-CL):** Steers traffic to different data networks based on traffic matching
- **Branching Point:** Supports multi-homed PDU sessions
- **Buffering:** Buffers downlink packets for idle UEs, triggers DDN to SMF

Interfaces:

Interface	Connection	Protocol	Purpose
N3	gNB ↔ UPF	GTP-U over UDP	User plane tunnel from RAN
N6	UPF ↔ DN	IP / Ethernet	Exit point to Data Network
N9	UPF ↔ UPF	GTP-U over UDP	Inter-UPF forwarding (chaining)
N4	SMF ↔ UPF	PCFP over UDP	Control of UPF from SMF

PCFP Rules (programmed by SMF via N4):

Rule	Full Name	Purpose
PDR	Packet Detection Rule	Matches incoming packets (source interface, IP filters, GTP TEID, QFI)
FAR	Forwarding Action Rule	Defines what to do (forward, drop, buffer, duplicate); specifies destination
QER	QoS Enforcement Rule	Rate limiting (MBR/GBR), gating (open/close), reflective QoS
URR	Usage Reporting Rule	Volume/time/event-based measurement triggers for charging
BAR	Buffering Action Rule	Controls buffering behavior, notification parameters

Traffic Detection Methods:

- **SDF Filter:** 5-tuple IP filter (src/dst IP, src/dst port, protocol)
- **Application ID:** Pre-configured or dynamically learned via DPI
- **GTP-U TEID:** Identifies specific GTP tunnels
- **QFI (QoS Flow Identifier):** Maps to specific QoS flows

3.4 Unified Data Management (UDM)

The UDM provides subscriber data management, authentication credential handling, and subscription authorization. It is the 5G equivalent of the HSS.

Key Responsibilities:

- **SUPI Storage:** Stores and manages the Subscription Permanent Identifier (SUPI = IMSI or NAI format)
- **Authentication Credential Computation:** Generates authentication vectors (AV) using subscriber's long-term key (K) and operator parameters (OP/OPc)
- **Subscription Profile Management:** Maintains subscribed services, DNN authorizations, QoS profiles
- **Registration Management:** Tracks serving AMF/SMF per subscriber (NF registration)

- **DNN Authorization:** Validates whether a UE is authorized to access a specific DNN/slice combination
- **SUCI De-concealment:** Resolves SUCI back to SUPI (using SIDF — Subscription Identifier De-concealing Function)

Key Data Stored:

- SUPI, SUCI resolution keys
- Subscriber authentication credentials (K, OP/OPc, SQN)
- Subscribed S-NSSAI and DNN list
- Default session QoS parameters
- Serving NF registrations (AMF, SMF)
- Access and mobility subscription data

Interfaces:

- **Nudm:** Service-based interface exposed by UDM
- **N8 (via Nudm):** AMF retrieves subscription/mobility data
- **N10 (via Nudm):** SMF retrieves session management subscription data
- **N13 (via Nudm):** AUSF retrieves authentication credentials

Relationship to UDR:

- UDM contains application logic (auth vector computation, SUCI de-concealment)
- UDR is the pure data store backend
- UDM accesses UDR via **Nudr** interface

3.5 Authentication Server Function (AUSF)

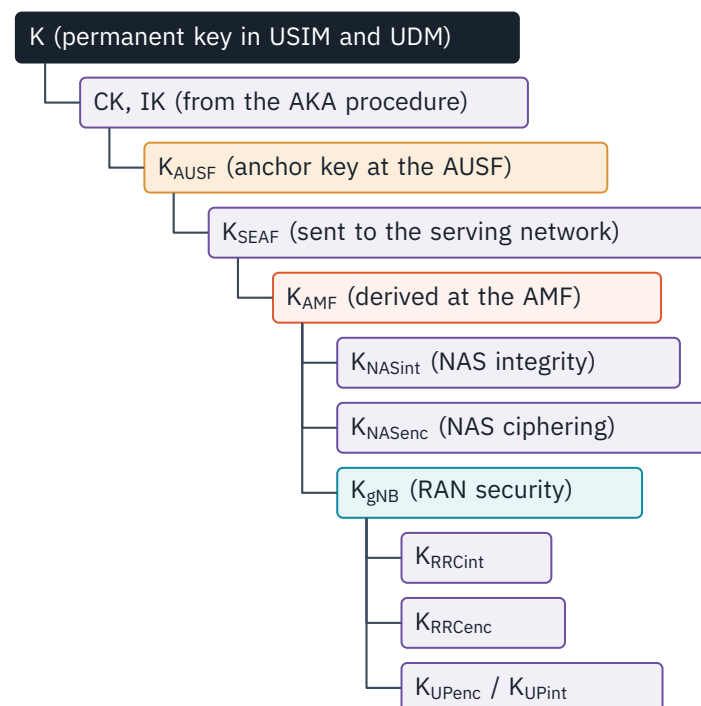
The AUSF acts as the authentication server for the 5G network. It executes authentication procedures between the UE and the home network.

Key Responsibilities:

- **Authentication Execution:** Runs 5G-AKA or EAP-AKA' procedures
- **K_{AUSF} Derivation:** Derives the anchor key (K_{AUSF}) from which all subsequent keys are derived
- **Authentication Result Storage:** Stores authentication status for future reference
- **Home Network Authority:** Ensures authentication decision is made by the home network (even in roaming)

Authentication Methods:

Method	Description	Use Case
5G-AKA	5G Authentication and Key Agreement — enhanced AKA with home network confirmation	Primary method for USIM-based UEs
EAP-AKA'	Extensible Authentication Protocol with AKA prime — EAP framework-based	Roaming, non-3GPP access, IoT

Key Hierarchy from AUSF:**Figure 19.3** 5G key hierarchy rooted at the AUSF**Interfaces:**

- **Nausf:** Service-based interface exposed by AUSF
- **N12 (via Nausf):** AMF initiates authentication
- **N13 (via Nudm):** AUSF retrieves authentication vectors from UDM

3.6 Policy Control Function (PCF)

The PCF is the centralized policy engine of the 5GC. It provides policy rules to control network behavior for access, sessions, and UE-specific policies.

Key Responsibilities:

- **PCC Rule Generation:** Creates Policy and Charging Control rules for QoS and charging
- **AM Policy:** Access and Mobility policies (RFSP index, service area restrictions, RAT/frequency selection)

- **SM Policy:** Session Management policies (QoS, gating, charging, traffic steering)
- **UE Policy:** Policies delivered to the UE (URSP rules — UE Route Selection Policy, ANDSP)
- **Policy Decision:** Makes real-time policy decisions based on subscription, network conditions, AF requests

Policy Types:

Policy	Scope	Delivered To	Purpose
AM Policy	Access & Mobility	AMF (N15)	Service area, RFSP, RAT restriction
SM Policy	Session	SMF (N7)	QoS flows, PCC rules, charging
UE Policy	UE behavior	UE (via AMF)	URSP, ANDSP, WLANSF

PCC Rule Contents:

- Rule identifier and precedence
- Traffic flow template (SDF filter or application ID)
- QoS parameters (5QI, ARP, MBR, GBR)
- Gating status (open/close for UL/DL)
- Charging method (online/offline) and rating group
- Traffic steering policy

Interfaces:

- **Npcf:** Service-based interface exposed by PCF
- **N7 (via Npcf):** SMF retrieves SM policy decisions
- **N15 (via Npcf):** AMF retrieves AM policy decisions
- **N5 (via Npcf):** AF influences policy via NEF or directly

3.7 Network Slice Selection Function (NSSF)

The NSSF assists in selecting the appropriate network slice instance and AMF set for a UE.

Key Responsibilities:

- **Allowed NSSAI Determination:** Determines which S-NSSAIs the UE is permitted to use in the serving PLMN
- **S-NSSAI Mapping:** Maps Requested NSSAI to Configured NSSAI (home↔serving PLMN mapping in roaming)
- **AMF Redirection:** When the current AMF cannot serve the selected slice, NSSF identifies a target AMF set that can
- **NRF Selection:** Helps select appropriate NRF instance for slice-specific NF discovery

Key Concepts:

Term	Description
S-NSSAI	Single NSSAI = SST (8-bit Slice/Service Type) + optional SD (24-bit Slice Differentiator)
Requested NSSAI	S-NSSAIs sent by UE in Registration Request
Allowed NSSAI	S-NSSAIs permitted by network (returned in Registration Accept)
Configured NSSAI	S-NSSAIs provisioned in the UE by the HPLMN/VPLMN
Subscribed S-NSSAI	Slice subscription in UDM

Standard SST Values:

- SST=1: eMBB (enhanced Mobile Broadband)
- SST=2: URLLC (Ultra-Reliable Low-Latency Communications)
- SST=3: MIoT (Massive IoT)
- SST=4: V2X (Vehicle-to-Everything)

3.8 Network Repository Function (NRF)

The NRF is the central registry for all NF instances in the 5GC. It enables service discovery and authorization.

Key Responsibilities:

- **NF Registration:** NFs register their profiles (type, capacity, supported services, PLMN, slice, location)
- **NF Discovery:** Consumer NFs query NRF to find producer NFs matching criteria
- **NF Status Notification:** Notifies subscribed NFs about NF status changes (registered, deregistered, suspended)
- **OAuth2 Token Issuance:** Issues access tokens for service-based interface authorization

NF Profile Contents:

- NF Instance ID (UUID)
- NF Type (AMF, SMF, UPF, etc.)
- NF Status (REGISTERED, SUSPENDED, UNDISCOVERABLE)
- PLMN ID(s) served
- S-NSSAI(s) supported
- FQDN or IP addresses
- Supported services and versions
- Capacity, priority, load information
- Geographic locality

Discovery Query Parameters:

- Target NF Type
- Requester NF Type
- Service names
- Target PLMN, S-NSSAI, DNN
- Preferred locality

OAuth2 in 5GC:

- NRF acts as OAuth2 Authorization Server
- Consumer NF requests token with target NF type and services
- Token included in SBI requests (Authorization header)
- Producer NF validates token before serving the request

3.9 Network Exposure Function (NEF)

The NEF provides a secure framework for exposing 5GC capabilities and events to external Application Functions (AFs).

Key Responsibilities:

- **Exposure Framework:** Exposes network capabilities to 3rd-party AFs via standardized APIs
- **Traffic Influence:** AF can request specific traffic routing (e.g., route traffic to local edge server)
- **Monitoring Events:** AF subscribes to UE events (reachability, location, connectivity status)
- **Parameter Provisioning:** AF provisions expected UE behavior, background data transfer policies
- **Translation:** Maps between external identifiers (GPSI, external group ID) and internal identifiers (SUPI)
- **Security Gateway:** Prevents direct AF access to internal NFs

Key Exposure Capabilities:

Capability	Description
Monitoring Events	UE reachability, location reporting, loss of connectivity, roaming status
Traffic Influence	Request traffic routing to local Data Network (edge computing)
Analytics Exposure	Expose NWDAF analytics to external AFs
Parameter Provisioning	Expected UE behavior, background data transfer
Device Triggering	Trigger specific action on IoT device
NIDD	Non-IP Data Delivery for IoT

3.10 Unified Data Repository (UDR)

The UDR is the backend data store for subscriber, policy, and exposure data. It is accessed by UDM, PCF, and NEF.

Key Responsibilities:

- **Subscription Data Storage:** Stores subscriber profiles accessed by UDM
- **Policy Data Storage:** Stores policy data (PCC rules, AM/SM policies) accessed by PCF
- **Exposure Data Storage:** Stores exposure subscriptions and data accessed by NEF
- **Application Data Storage:** Stores AF-provided data (e.g., PFDs — Packet Flow Descriptions)

Data Categories:

Data Type	Consumer NF	Examples
Subscription Data	UDM	Authentication credentials, subscribed S-NSSAI, DNN configs
Policy Data	PCF	PCC rule templates, sponsored data policies
Exposure Data	NEF	Monitoring subscriptions, AF influence requests
Application Data	NEF/PCF	PFDs, background data transfer policies

Interface:

- **Nudr:** Service-based interface (the only interface of UDR)
- Accessed by UDM (Nudr_DataManagement), PCF (Nudr_DataManagement), NEF (Nudr_DataManagement)

3.11 Unstructured Data Storage Function (UDSF)

The UDSF enables stateless design of NFs by providing external storage for NF context data.

Key Responsibilities:

- **Unstructured Data Storage:** Stores arbitrary NF context/state as opaque data blocks
- **Stateless NF Support:** NFs can recover state after restart or load-balance across instances
- **CRUD Operations:** Create, Read, Update, Delete operations on stored records

Why UDSF Matters:

- Without UDSF, each NF instance holds session state in memory
- With UDSF, any NF instance can serve any request by retrieving state from UDSF
- Enables true horizontal scaling, failover, and cloud-native deployment
- Each NF type may have its own UDSF instance

Interface:

- **Nudsf:** Service-based interface for storing/retrieving unstructured data

3.12 Service Communication Proxy (SCP)

The SCP is a routing and load-balancing element placed in the SBI communication path between NFs. Defined in TS 23.501 Release 16+.

Key Responsibilities:

- **Indirect Communication:** NFs send requests to SCP instead of directly to the target NF
- **Load Balancing:** Distributes requests across multiple NF instances
- **Message Routing:** Routes messages based on discovery parameters without consumer NF needing full discovery
- **Topology Hiding:** Hides internal NF topology from external networks (roaming)
- **Overload Control:** Implements congestion handling and request throttling

Communication Models:

Model	Description
Model A (Direct)	NF consumer discovers target via NRF, communicates directly
Model B (Indirect w/o delegated discovery)	NF discovers target via NRF, routes via SCP
Model C (Indirect w/ delegated discovery)	NF sends to SCP, SCP performs discovery and routing
Model D (Indirect w/ delegated discovery + selection)	SCP handles everything: discovery, selection, routing

Benefits:

- Simplifies NF implementation (NFs don't need complex discovery/retry logic)
- Centralized observability and telemetry
- Easier deployment of cross-cutting concerns (mTLS termination, tracing)
- Enables seamless NF scaling without reconfiguration

4 Control Plane vs User Plane in 5G Core

The 5GC enforces a strict separation between Control Plane (CP) and User Plane (UP), known as CUPS (Control and User Plane Separation).

Control Plane (CP)

- **All NFs except UPF** operate in the control plane
- Communication via **Service-Based Interfaces (SBI)** using HTTP/2 + JSON
- Protocol stack: HTTP/2 → TCP → IP (or via SCP)
- Functions: signaling, policy, authentication, registration, session management, discovery
- Deployed centrally or regionally depending on latency requirements
- AMF and SMF are the primary CP anchors for UE signaling and session management

■ User Plane (UP)

- Only **UPF** operates in the user plane
- Programmed by SMF via **N4 / PFCP** (control interface)
- User data protocol stack: App Data → PDU Layer → GTP-U → UDP → IP
- Functions: packet forwarding, QoS enforcement, traffic detection, usage measurement, buffering
- Can be deployed at edge, regional, or central locations depending on latency and data locality needs
- Multiple UPFs can be chained (via N9) for service function chaining or local breakout

■ CP/UP Interaction Summary

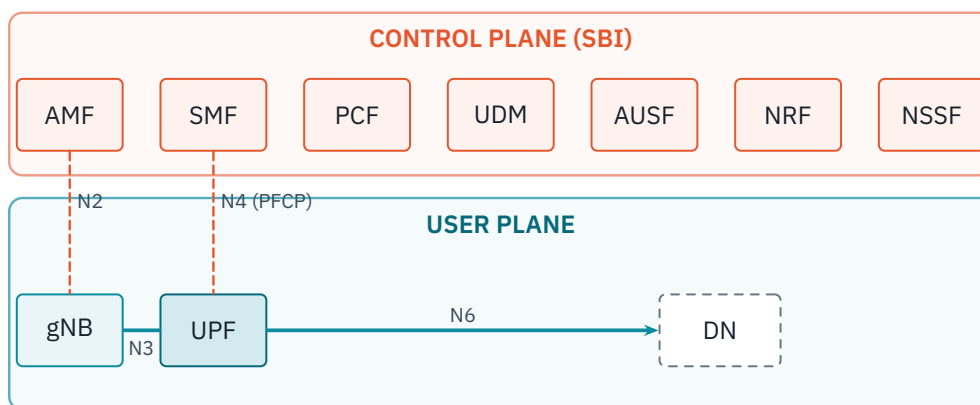


Figure 19.4 Control plane (SBI) and user plane of the 5G core

5 Major Comparison: 4G EPC vs 5G Core

Aspect	4G EPC	5G Core (5GC)
Architecture Style	Point-to-point interfaces between monolithic NEs	Service-Based Architecture (SBA), microservices
Signaling Protocol	GTP-C (v2), Diameter	HTTP/2 + JSON (RESTful APIs)
User Plane	S-GW + P-GW (combined or separated)	UPF (single user-plane NF)
Subscriber Database	HSS (Diameter-based)	UDM + UDR (SBI-based, separated logic/storage)
Authentication	EPS-AKA via HSS	5G-AKA / EAP-AKA' via AUSF + UDM
Policy	PCRF (Gx, Rx, Sd interfaces)	PCF (Npcf SBI, AM/SM/UE policies)
Session Management	MME + S-GW + P-GW combined	SMF (dedicated NF, decoupled from mobility)
Mobility Management	MME (combined with session signaling)	AMF (pure mobility/access, no session logic)
Network Slicing	Not natively supported	Native support (S-NSSAI, NSSF, per-slice NF instances)
Interface Style	Fixed reference points (S1, S5, S6a, S11...)	SBI services + reference points for UP
State Management	Stateful NEs	Stateless NFs + UDSF for external state
Deployment	Appliance-based, monolithic	Cloud-native, containerized, microservices
Discovery	Preconfigured or DNS-based	Dynamic via NRF (registration + discovery + OAuth2)
Edge Computing	Difficult (LIPA/SIPTO limited)	Native (distributed UPF, LADN, MEC integration)

4G to 5G Network Function Mapping

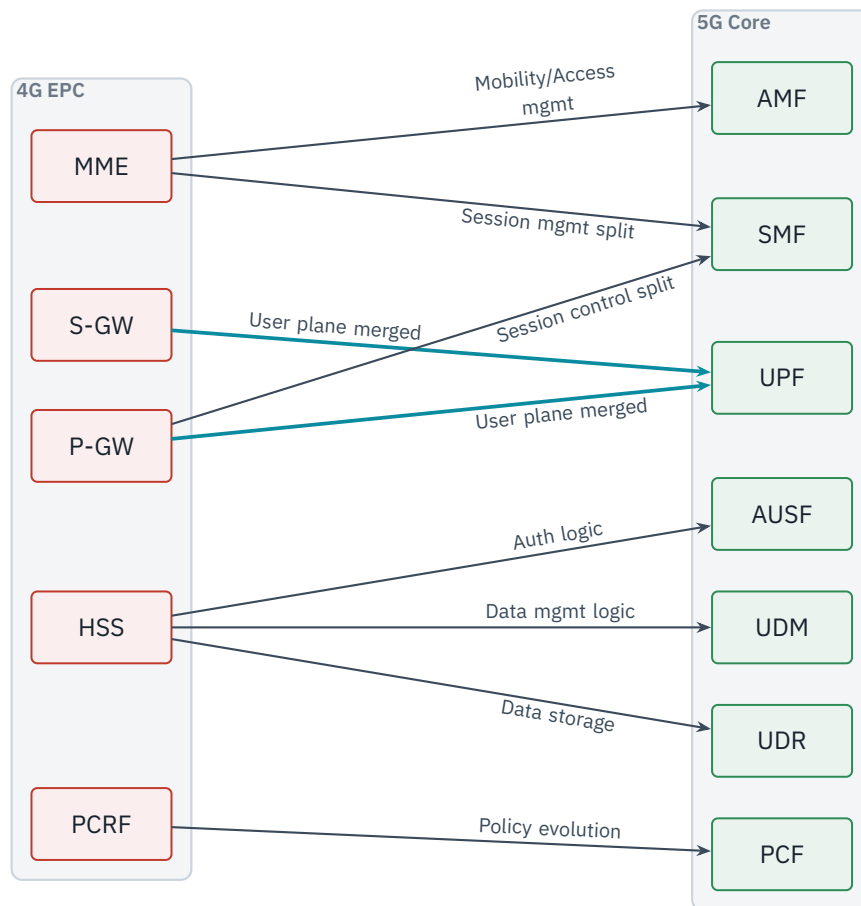


Figure 19.5 How the EPC functions map to 5G core network functions

6 Reference Points

Reference points define the interfaces between specific NF pairs. While SBI is used for CP-to-CP communication, reference points remain important for understanding protocol specifics.

Complete Reference Point Table

Reference Point	Connection	Protocol/Transport	Purpose
N1	UE ↔ AMF	NAS over RRC (via gNB)	NAS signaling (registration, session mgmt requests)
N2	RAN ↔ AMF	NGAP over SCTP	RAN-CN signaling (context setup, handover, paging)
N3	RAN ↔ UPF	GTP-U over UDP/IP	User plane tunnel carrying PDU session data
N4	SMF ↔ UPF	PCF over UDP (port 8805)	UPF programming (PDR, FAR, QER, URR rules)
N6	UPF ↔ DN	IP / Ethernet	User traffic exit to Data Network
N7	SMF ↔ PCF	SBI (HTTP/2)	SM policy (PCC rules, QoS decisions)
N8	AMF ↔ UDM	SBI (HTTP/2)	Access/mobility subscription data
N9	UPF ↔ UPF	GTP-U over UDP/IP	Inter-UPF user plane forwarding
N10	SMF ↔ UDM	SBI (HTTP/2)	Session management subscription data
N11	AMF ↔ SMF	SBI (HTTP/2)	PDU session lifecycle management
N12	AMF ↔ AUSF	SBI (HTTP/2)	Authentication initiation
N13	AUSF ↔ UDM	SBI (HTTP/2)	Authentication vector retrieval
N14	AMF ↔ AMF	SBI (HTTP/2)	Inter-AMF mobility (UE context transfer)
N15	AMF ↔ PCF	SBI (HTTP/2)	AM policy (access/mobility restrictions)
N22	AMF ↔ NSSF	SBI (HTTP/2)	Slice selection assistance
N27	NRF ↔ NRF	SBI (HTTP/2)	Inter-PLMN NF discovery (via SEPP)
N32	SEPP ↔ SEPP	PRINS / TLS	Inter-PLMN security (roaming)

■ Key Reference Point Details

N1 (UE ↔ AMF):

- Carries NAS PDUs (Registration Request/Accept, Authentication, Security Mode, PDU Session Establishment)
- NAS messages are transparently relayed by gNB (encapsulated in NGAP on N2)
- Supports NAS security (integrity + ciphering) after security activation

N2 (RAN ↔ AMF):

- NGAP protocol over SCTP
- Procedures: Initial UE Message, PDU Session Resource Setup, Handover Required/Request, Paging
- Carries UE-associated and non-UE-associated signaling

N3 (RAN ↔ UPF):

- GTP-U tunnel identified by TEID (Tunnel Endpoint Identifier)
- One GTP-U tunnel per PDU session per gNB-UPF pair
- QoS Flow Identifier (QFI) carried in GTP-U extension header to map QoS flows

N4 (SMF ↔ UPF):

- PFCP (Packet Forwarding Control Protocol) — 3GPP TS 29.244
- Session-level management (Establishment, Modification, Deletion)
- Node-level management (Association Setup, Heartbeat)
- Reports: Usage Reports, Downlink Data Notification, End Marker

N9 (UPF ↔ UPF):

- GTP-U tunnel for multi-hop user plane paths
- Used in UL-CL (Uplink Classifier) and branching point scenarios
- Enables service function chaining and distributed UPF deployment

7 Deployment Options

The 5GC's modular architecture allows flexible deployment topologies based on service requirements.

7.1 Centralized Deployment

All NFs (including UPF) deployed in a central data center.

Characteristics:

- Simple operations, single site
- Higher latency for user plane (all traffic traverses central site)

- Suitable for wide-area coverage with relaxed latency requirements
- Lower infrastructure cost

7.2 Distributed Deployment

Control plane centralized, user plane distributed closer to users.

Characteristics:

- CP NFs (AMF, SMF, PCF, etc.) in regional/central DC
- UPF deployed at regional aggregation points
- Reduced user-plane latency
- SMF manages multiple UPF instances across locations

7.3 Edge UPF Deployment (MEC Integration)

UPF instances deployed at the network edge (cell site or local aggregation point) for ultra-low latency.

Characteristics:

- Local breakout of traffic at the edge
- Enables Multi-access Edge Computing (MEC) — ETSI MEC framework
- SMF can insert/remove edge UPFs dynamically based on UE location
- **UL-CL (Uplink Classifier):** Steers specific traffic flows to edge UPF while rest goes to central UPF
- **Local Area Data Network (LADN):** Data Network available only in specific geographic area
- Typical use cases: AR/VR, autonomous driving, industrial automation, cloud gaming

Deployment Architecture Comparison

Aspect	Centralized	Distributed	Edge UPF
CP Location	Central DC	Central/Regional DC	Central/Regional DC
UP Location	Central DC	Regional PoP	Cell site / Edge DC
User Plane Latency	High (10-50ms)	Medium (5-20ms)	Ultra-low (1-5ms)
Complexity	Low	Medium	High
Use Cases	General broadband	Regional services	URLLC, MEC, V2X
UPF Count	Few	Moderate	Many

■ PDU Session User Plane Path

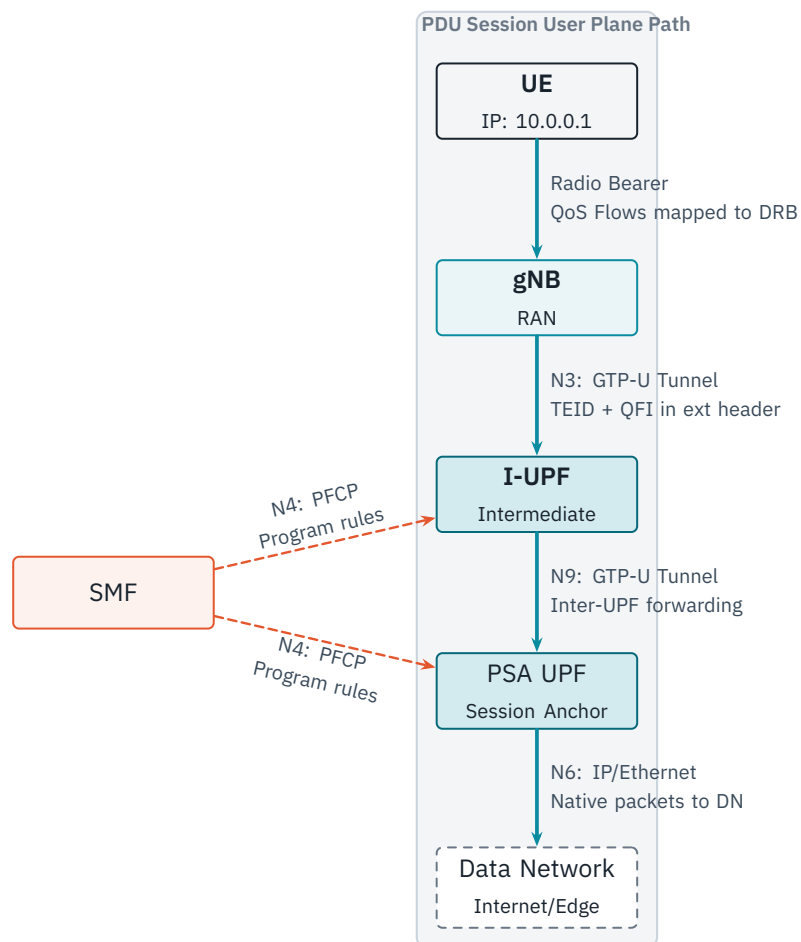


Figure 19.6 User-plane path of a PDU session with an intermediate UPF

PDU Session Establishment Flow (simplified):

1. UE → AMF: NAS PDU Session Establishment Request (via N1)
2. AMF → SMF: Nsmf_PDUSession_CreateSMContext (via N11)
3. SMF selects UPF, allocates IP address
4. SMF → UPF: PCFP Session Establishment (via N4) — installs PDR/FAR/QER
5. SMF → AMF: N1N2MessageTransfer (contains NAS accept + N2 info)
6. AMF → gNB: PDU Session Resource Setup Request (via N2)
7. gNB → UPF: GTP-U tunnel established (N3)
8. UE receives PDU Session Establishment Accept with QoS rules

Summary

The 5G Core represents a paradigm shift from the 4G EPC:

- **SBA replaces point-to-point interfaces** — enabling flexibility, reusability, and independent NF evolution
- **Complete CUPS** — UPF is the only user-plane entity, programmable via PFCP
- **Cloud-native by design** — stateless NFs, UDSF, NRF-based discovery, containerization
- **Native slicing** — S-NSSAI, NSSF, per-slice NF selection from the ground up
- **Security enhanced** — SUCI concealment, home-network-controlled authentication, SBI OAuth2
- **Edge-ready** — distributed UPF, UL-CL, LADN for ultra-low-latency services

References: 3GPP TS 23.501 (System Architecture), TS 23.502 (Procedures), TS 29.244 (PFCP), TS 33.501 (Security Architecture)



In This Module

1	Registration (Initial Registration)	378
2	Authentication (5G-AKA)	384
3	Security Establishment	390
4	PDU Session Establishment	394
5	PDU Session Release	400
6	Service Request	404
7	Paging	407
8	Xn Handover	410
9	N2 Handover (Inter-AMF)	416
	Summary: Procedure Comparison Table	421
	Key References	422

1 Registration (Initial Registration)

Purpose

The Initial Registration procedure allows a UE to register with the 5G Core Network (5GC) for the first time or after being deregistered. It establishes the UE's presence in the network, authenticates the subscriber, allocates a 5G-GUTI (Globally Unique Temporary Identifier), and optionally establishes a PDU session for data connectivity.

Preconditions

- UE is powered on and has a valid USIM with 5G subscription credentials
- UE has selected a PLMN and camped on a cell (gNB)
- UE is in RRC_IDLE or RRC_INACTIVE state
- UE has no valid 5G-GUTI (first-time registration) or previous GUTI has expired
- RRC connection has been established (Random Access procedure completed)

NFs Involved

Network Function	Role
UE	Initiates registration, provides identity (SUCI/5G-GUTI)
gNB	RAN node, forwards NAS messages between UE and AMF
AMF	Anchor for registration, selects AUSF, manages context
AUSF	Authentication server, validates credentials
UDM	Stores subscription data, generates auth vectors
ARPF	Authentication credential Repository (within UDM)
PCF	Provides AM policies
SMF	Session management (if PDU session requested)
UPF	User plane (if PDU session requested)
NRF	NF discovery and selection
NSSF	Network Slice selection

Messages

Step	Interface	Message
1	Uu (UE→gNB)	RRCSetupRequest
2	Uu (gNB→UE)	RRCSetup
3	Uu (UE→gNB)	RRCSetupComplete + NAS Registration Request
4	N2 (gNB→AMF)	NGAP Initial UE Message (NAS: Registration Request)
5	N12 (AMF→AUSF)	Nausf_UEAuthentication_Authenticate Request
6	N13 (AUSF→UDM)	Nudm_UEAuthentication_Get Request
7	N13 (UDM→AUSF)	Nudm_UEAuthentication_Get Response (AV)
8	N12 (AUSF→AMF)	Nausf_UEAuthentication_Authenticate Response
9	N2/NAS (AMF→UE)	Authentication Request (RAND, AUTN)
10	N2/NAS (UE→AMF)	Authentication Response (RES*)
11	N12 (AMF→AUSF)	Nausf_UEAuthentication_Authenticate (RES*)
12	N12 (AUSF→AMF)	Nausf_UEAuthentication_Authenticate (Success + K_{SEAF})
13	N2/NAS (AMF→UE)	Security Mode Command
14	N2/NAS (UE→AMF)	Security Mode Complete
15	N8 (AMF→UDM)	Nudm_UECM_Registration
16	N8 (AMF→UDM)	Nudm_SDM_Get (Subscription Data)
17	N15 (AMF→PCF)	Npcf_AMPolicyControl_Create
18	N2/NAS (AMF→UE)	Registration Accept (5G-GUTI, TAI List, Allowed NSSAI)
19	N2/NAS (UE→AMF)	Registration Complete
20	N2 (AMF→gNB)	NGAP Initial Context Setup Request

Step-by-Step Flow

- RRC Connection Establishment:** UE performs Random Access and sends RRCSetupRequest to gNB
- RRC Setup:** gNB responds with RRCSetup message
- Registration Request:** UE sends RRCSetupComplete containing NAS Registration Request (includes SUCI or 5G-GUTI, Registration Type, UE Security Capabilities, Requested NSSAI)

4. **AMF Selection:** gNB selects appropriate AMF (via NSSF if needed) and forwards via NGAP Initial UE Message
5. **Identity Request (conditional):** If AMF cannot resolve identity, sends Identity Request to UE
6. **SUCI Resolution:** AMF initiates authentication; sends SUCI to AUSF
7. **Auth Vector Request:** AUSF requests authentication vector from UDM/ARPF
8. **AV Generation:** UDM/ARPF generates 5G HE AV (RAND, AUTN, XRES*, K_{AUSF}), resolves SUCI→SUPI
9. **Auth Vector Response:** AUSF receives AV, computes HXRES, *stores* XRES
10. **Authentication Request:** AMF sends NAS Authentication Request to UE (RAND, AUTN)
11. **UE Verification:** UE verifies AUTN, computes RES*, CK, IK, derives K_{AUSF}
12. **Authentication Response:** UE sends Authentication Response (RES*) to AMF
13. **RES* Verification:** AMF sends RES* to AUSF; AUSF verifies HRES* matches HXRES*
14. **Authentication Confirmation:** AUSF confirms to AMF, provides K_{SEAF} ; AMF derives K_{AMF}
15. **NAS Security:** AMF sends Security Mode Command (selected algorithms); UE responds with Security Mode Complete
16. **UDM Registration:** AMF registers with UDM (Nudm_UECM_Registration)
17. **Subscription Retrieval:** AMF retrieves subscription data from UDM (Nudm_SDM_Get)
18. **Policy Association:** AMF establishes AM Policy association with PCF
19. **Registration Accept:** AMF sends Registration Accept (5G-GUTI, TAI List, Allowed NSSAI, Registration Area)
20. **Registration Complete:** UE acknowledges with Registration Complete; stores new 5G-GUTI

Detailed Sequence Diagram

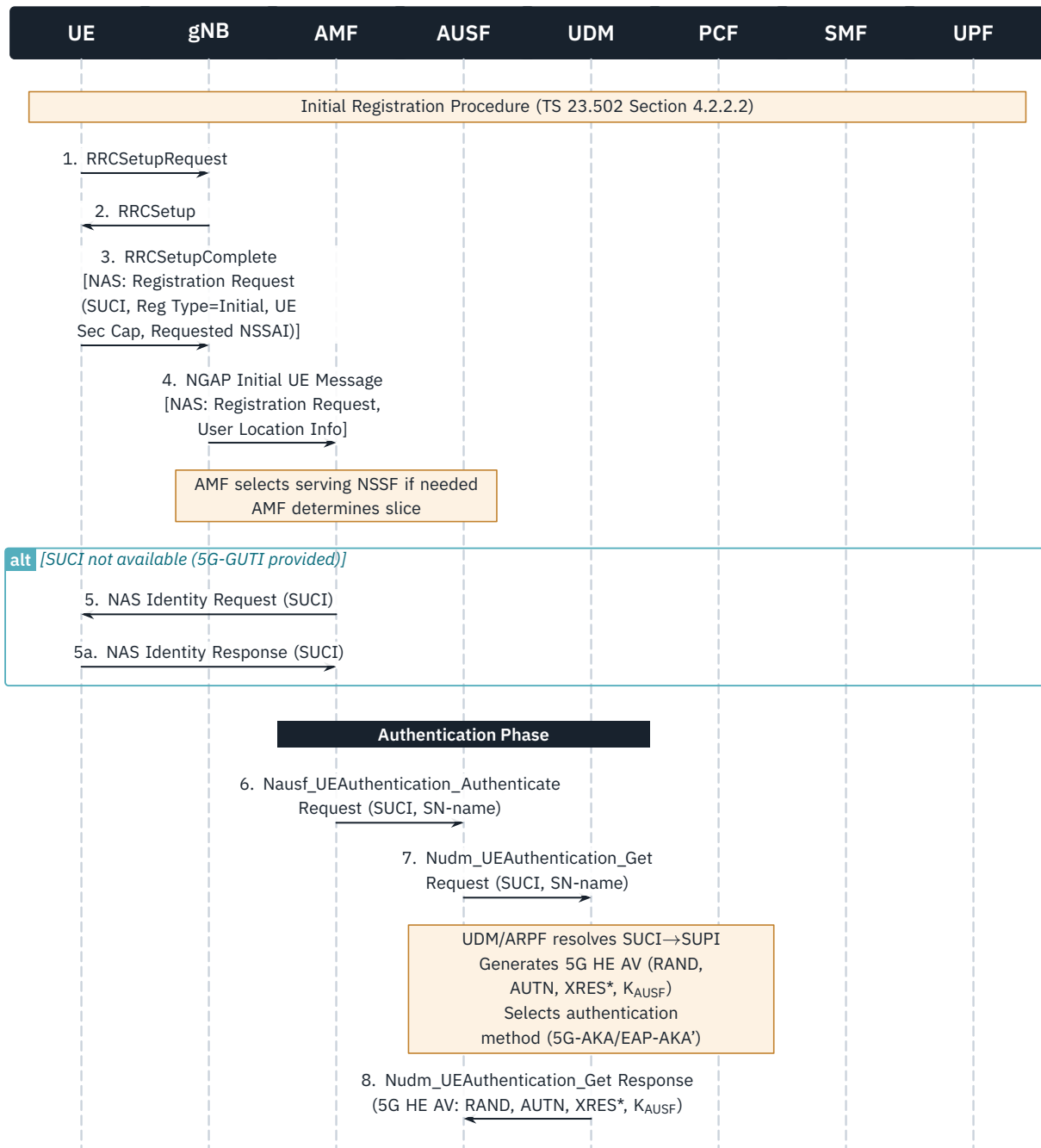


Figure 20.1 Initial registration (TS 23.502 §4.2.2.2) (part 1 of 3)

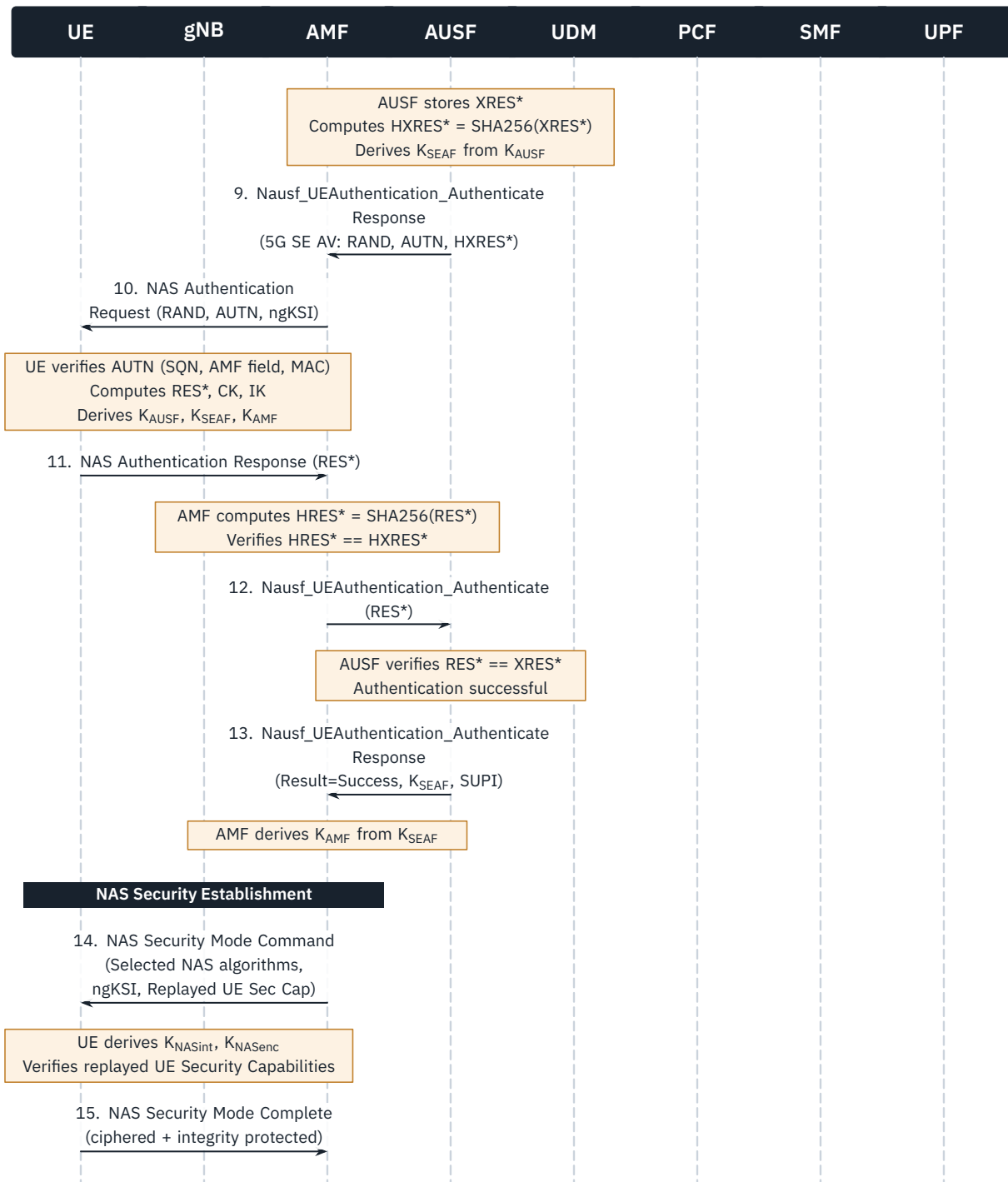


Figure 20.2 Initial registration (TS 23.502 §4.2.2.2) (part 2 of 3)

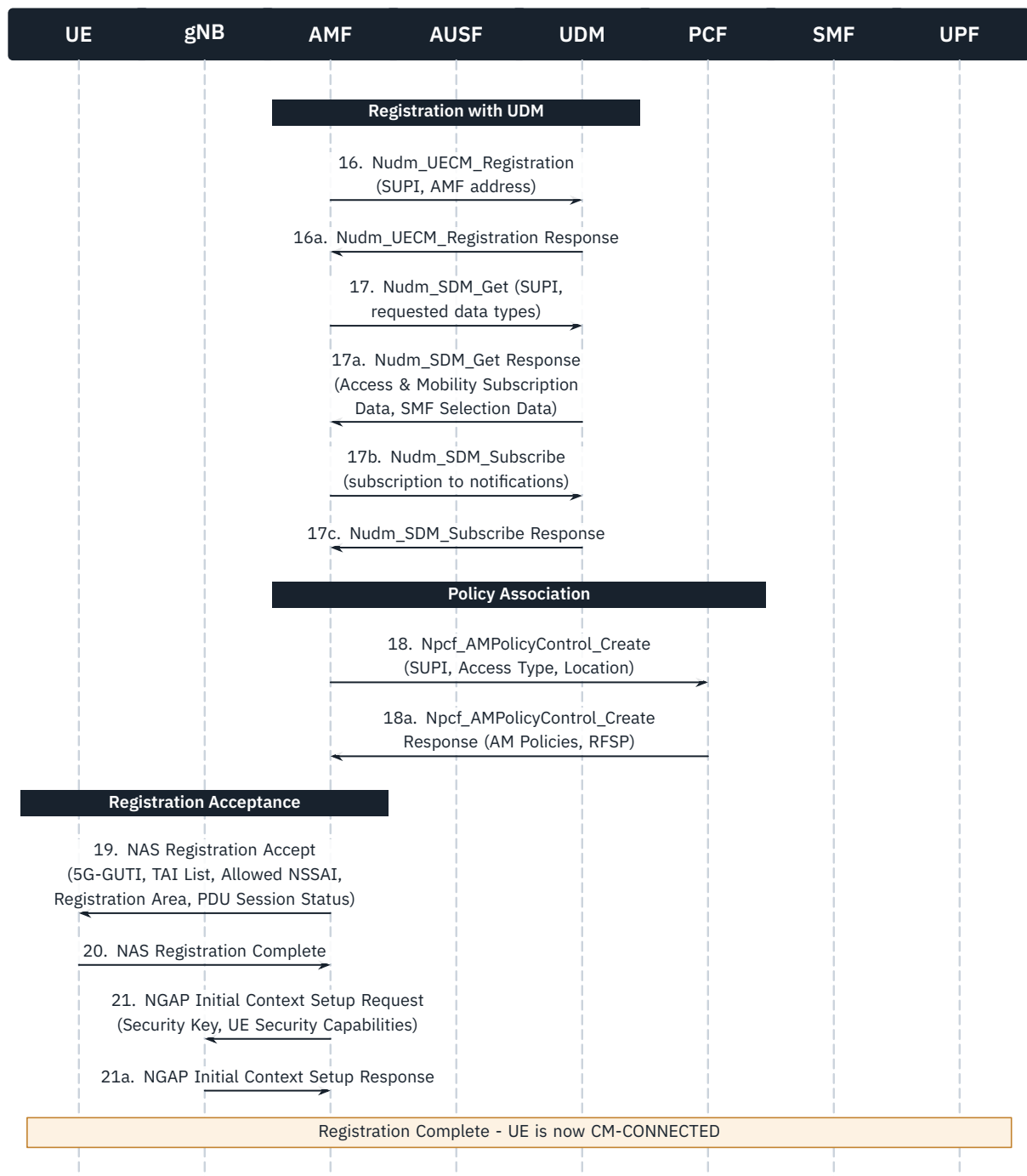


Figure 20.3 Initial registration (TS 23.502 §4.2.2.2) (part 3 of 3)

Failure Cases

Failure	Cause	Network Response
Authentication Failure	Invalid credentials, SQN out of sync	Authentication Reject or Sync Failure
Registration Reject	No subscription, roaming not allowed	Registration Reject (Cause: #11, #15)
PLMN Not Allowed	UE not subscribed for visited PLMN	Registration Reject (Cause: #11)
Congestion	Network overload	Registration Reject (Cause: #22) + back-off timer
Integrity Check Failure	NAS MAC failure	Reject silently or re-authenticate
UDM Unreachable	SBI communication failure	Registration Reject or AMF retry
Slice Not Available	Requested S-NSSAI not supported	Registration Accept with reduced Allowed NSSAI

2 Authentication (5G-AKA)

Purpose

The 5G-AKA (Authentication and Key Agreement) procedure provides mutual authentication between the UE and network, and establishes shared cryptographic keys. It ensures the network verifies the subscriber's identity and the UE verifies it is communicating with a legitimate network. The procedure generates the anchor key (K_{AUSF}) from which all subsequent security keys are derived.

Preconditions

- UE has initiated Registration or Re-authentication is triggered
- UE has valid USIM with long-term key K and OP/OPc
- AUSF is reachable from the serving AMF
- UDM/ARPF has valid subscription credentials for the subscriber
- SUCI has been provided (either from UE or resolved from 5G-GUTI)

■ NFs Involved

Network Function	Role
UE (USIM)	Holds permanent key K ; verifies AUTN; computes RES^*
AMF (SEAF)	Security Anchor; relays auth messages; derives K_{AMF}
AUSF	Verifies RES^* against $XRES^*$; derives K_{SEAF}
UDM/ARPF	Generates authentication vectors; holds permanent key K
SIDF (in UDM)	Decrypts SUCI to obtain SUPI

■ Messages

Step	Interface	Message	Key Parameters
1	N12	Nausf_UEAuthentication_Authenticate Request	SUCI/SUPI, Serving Network Name
2	N13	Nudm_UEAuthentication_Get Request	SUCI/SUPI, Serving Network Name
3	N13	Nudm_UEAuthentication_Get Response	5G HE AV (RAND, AUTN, $XRES^*$, K_{AUSF})
4	N12	Nausf_UEAuthentication_Authenticate Response	5G SE AV (RAND, AUTN, $HXRES^*$)
5	NAS	Authentication Request	RAND, AUTN, ngKSI
6	NAS	Authentication Response	RES^*
7	N12	Nausf_UEAuthentication_Authenticate	RES^*
8	N12	Nausf_UEAuthentication_Authenticate Response	Result, K_{SEAF} , SUPI

■ Step-by-Step Flow

1. **Trigger:** AMF determines authentication is needed (initial registration, periodic re-auth, or inter-AMF handover)
2. **AUSF Selection:** AMF selects AUSF based on SUPI home network (via NRF)
3. **Auth Initiation:** AMF sends SUCI and Serving Network Name (SN-name) to AUSF
4. **UDM Request:** AUSF forwards request to UDM with SUCI and SN-name
5. **SUCI Decryption:** SIDF (within UDM) decrypts SUCI using home network private key → obtains SUPI
6. **AV Generation:** ARPF generates 5G HE AV using:
 - RAND (random challenge)
 - $AUTN = SQN \oplus AK \parallel AMF \parallel MAC$ (network authentication token)
 - $XRES^* = KDF(CK, IK, SN-name, RAND, RES)$ (expected response)
 - $K_{AUSF} = KDF(CK, IK, SN-name, SQN \oplus AK)$ (anchor key)

7. **AV to AUSF:** UDM returns 5G HE AV to AUSF
8. **AUSF Processing:** AUSF stores XRES, computes $HXRES = \text{SHA-256}(XRES[128..255] || XRES)$, derives K_{SEAF} from K_{AUSF}
9. **SE AV to AMF:** AUSF sends 5G SE AV (RAND, AUTN, HXRES*) to AMF
10. **Challenge to UE:** AMF sends Authentication Request (RAND, AUTN, ngKSI) to UE
11. **UE Verification:** USIM verifies AUTN:
 - Extracts AK from RAND using f5, recovers SQN
 - Computes XMAC and verifies $XMAC == MAC$ (network authentication)
 - Verifies SQN is in acceptable range (replay protection)
12. **UE Key Derivation:** UE computes:
 - RES, CK, IK from RAND using f2, f3, f4
 - $RES^* = KDF(CK, IK, SN\text{-}name, RAND, RES)$
 - $K_{AUSF} = KDF(CK, IK, SN\text{-}name, SQN \oplus AK)$
 - $K_{SEAF} = KDF(K_{AUSF}, SN\text{-}name)$
 - $K_{AMF} = KDF(K_{SEAF}, SUPI, ABBA)$
13. **Response:** UE sends Authentication Response with RES* to AMF
14. **AMF Verification:** AMF computes HRES* and verifies $HRES^* == HXRES^*$
15. **AUSF Verification:** AMF forwards RES* to AUSF; AUSF verifies $RES^* == XRES^*$
16. **Success:** AUSF returns success, K_{SEAF} , and confirmed SUPI to AMF
17. **K_{AMF} Derivation:** AMF derives $K_{AMF} = KDF(K_{SEAF}, SUPI, ABBA)$

■ Key Hierarchy Derived

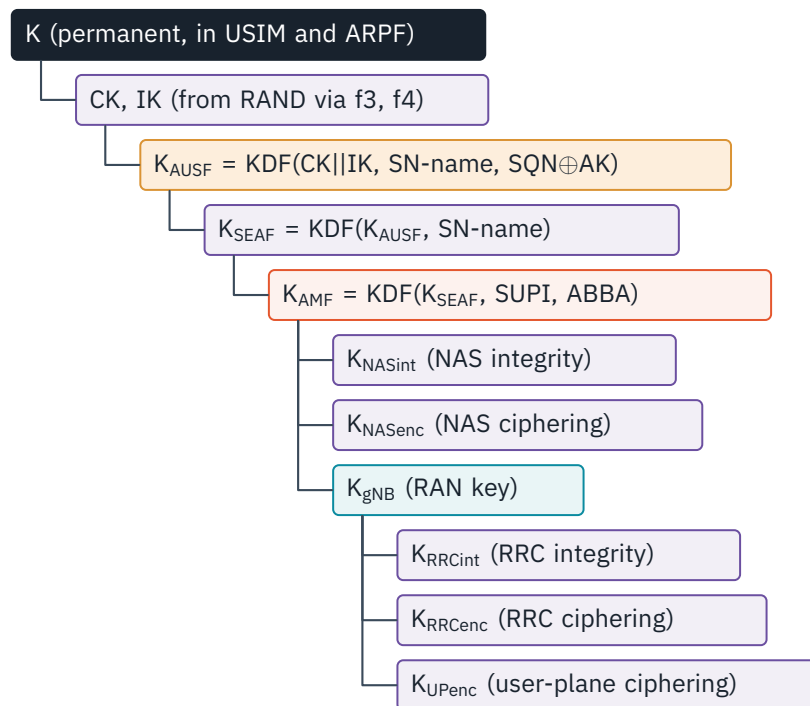


Figure 20.4 Key hierarchy derived during 5G-AKA

Detailed Sequence Diagram

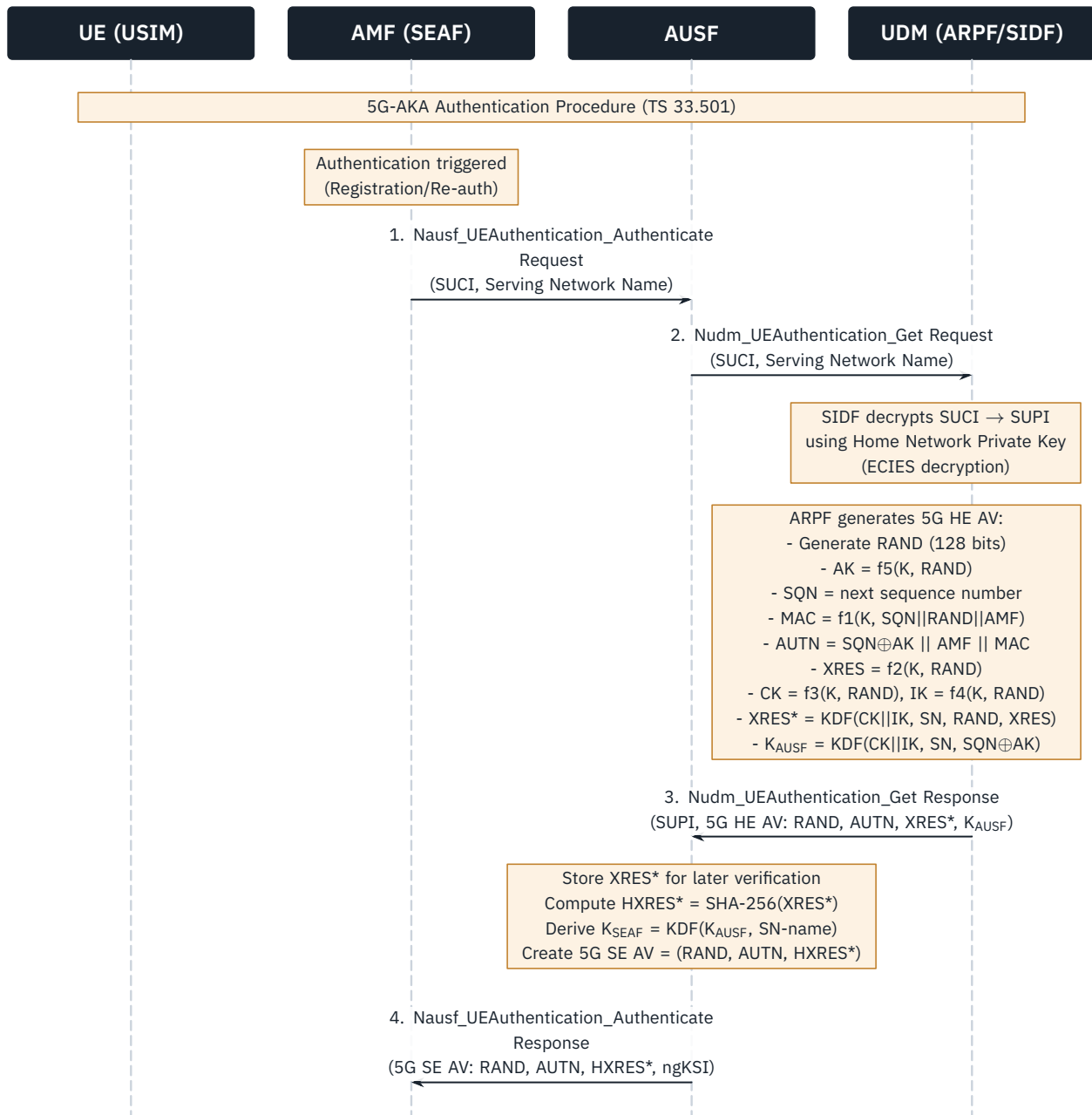


Figure 20.5 5G-AKA authentication (TS 33.501) (part 1 of 2)

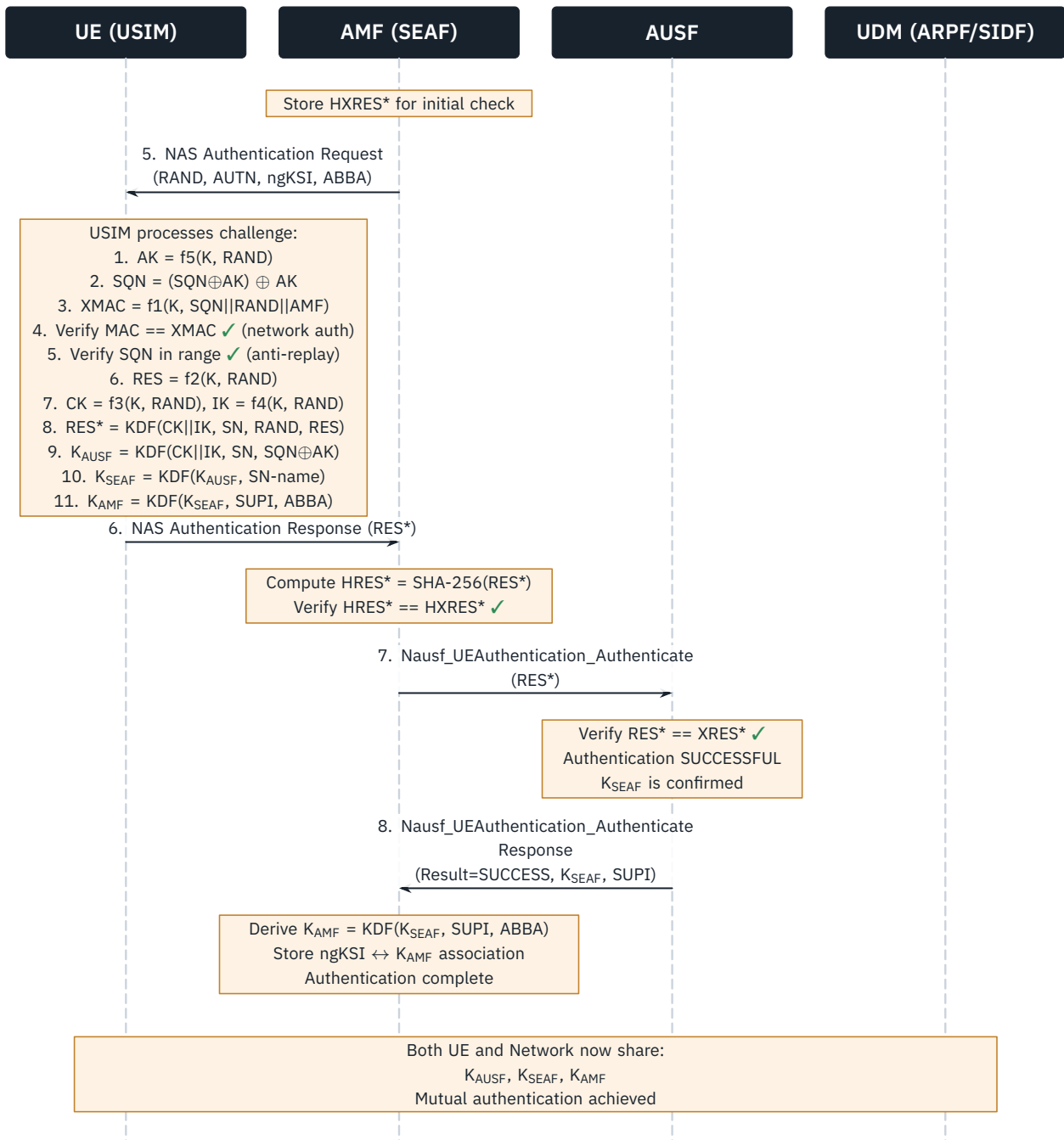


Figure 20.6 5G-AKA authentication (TS 33.501) (part 2 of 2)

Failure Cases

Failure	Cause	UE Action	Network Action
MAC Failure	AUTN verification fails (wrong K or tampered)	Send Auth Failure (MAC failure)	AMF informs AUSF; may retry
SQN Sync Failure	SQN out of acceptable range	Send Auth Failure (Synch failure) + AUTS	UDM resynchronizes SQN, generates new AV
Network Auth Failure	UE cannot verify network identity	Abort authentication	—
RES* Mismatch	UE response doesn't match expected	—	AUSF reports failure; AMF rejects registration
HXRES* Mismatch	Initial check at AMF fails	—	AMF still forwards to AUSF for final decision
AUSF Timeout	AUSF unreachable	—	AMF may select alternate AUSF or reject
SUCI Decryption Failure	Invalid SUCI format or wrong HN key	—	UDM reports error; registration fails

3 Security Establishment

Purpose

The Security Establishment procedure activates NAS and AS security between the UE and the network. It selects and activates ciphering and integrity protection algorithms for both NAS signaling (between UE and AMF) and AS signaling/user plane (between UE and gNB). This procedure ensures all subsequent communications are protected against eavesdropping and tampering.

Preconditions

- Authentication has been successfully completed
- K_{AMF} has been derived at both UE and AMF
- UE Security Capabilities have been provided to the AMF
- ngKSI is established (identifies the active security context)

NFs Involved

Network Function	Role
UE	Derives NAS/AS keys, verifies algorithm selection
AMF	Selects NAS algorithms, sends Security Mode Command, derives NAS keys
gNB	Selects AS algorithms, activates RRC/UP security

Messages

Step	Interface	Message	Key Parameters
1	NAS	Security Mode Command	NAS algorithms, ngKSI, Replayed UE Sec Cap, IMEISV request
2	NAS	Security Mode Complete	IMEISV (if requested), NAS message container
3	N2	Initial Context Setup Request	K_{gNB} , UE Security Capabilities
4	RRC	SecurityModeCommand	AS algorithms (ciphering + integrity)
5	RRC	SecurityModeComplete	—
6	N2	Initial Context Setup Response	—

Step-by-Step Flow

- NAS Algorithm Selection:** AMF selects NAS security algorithms based on UE Security Capabilities and AMF policy (highest priority algorithm supported by both)
- NAS Key Derivation (AMF):** AMF derives:
 - $K_{NASint} = \text{KDF}(K_{AMF}, \text{NAS-int-alg-ID}, \text{alg-type})$
 - $K_{NASenc} = \text{KDF}(K_{AMF}, \text{NAS-enc-alg-ID}, \text{alg-type})$
- Security Mode Command:** AMF sends NAS Security Mode Command (integrity protected with K_{NASint} but NOT ciphered):
 - Selected NAS ciphering algorithm (e.g., NEA1, NEA2)
 - Selected NAS integrity algorithm (e.g., NIA1, NIA2)
 - ngKSI, Replayed UE Security Capabilities, IMEISV request
- UE Verification:** UE verifies:
 - Replayed UE Security Capabilities match what was sent (anti-MITM)
 - Integrity protection of the message using derived K_{NASint}
- NAS Key Derivation (UE):** UE derives K_{NASint} and K_{NASenc} using same KDF

6. **Security Mode Complete:** UE sends NAS Security Mode Complete (integrity protected AND ciphered)
7. **NAS Security Active:** NAS security context is now active in both directions
8. **K_{gNB} Derivation:** AMF derives $K_{gNB} = \text{KDF}(K_{AMF}, \text{NAS UL COUNT})$ and sends to gNB via Initial Context Setup Request
9. **AS Algorithm Selection:** gNB selects AS algorithms based on UE Security Capabilities and gNB policy
10. **AS Key Derivation:** gNB derives:
 - $K_{RRCint} = \text{KDF}(K_{gNB}, \text{RRC-int-alg-ID})$
 - $K_{RRCenc} = \text{KDF}(K_{gNB}, \text{RRC-enc-alg-ID})$
 - $K_{UPenc} = \text{KDF}(K_{gNB}, \text{UP-enc-alg-ID})$
 - $K_{UPint} = \text{KDF}(K_{gNB}, \text{UP-int-alg-ID})$ [optional for DRBs]
11. **RRC Security Mode Command:** gNB sends RRC SecurityModeCommand to UE (integrity protected)
12. **UE AS Key Derivation:** UE derives same AS keys from K_{gNB}
13. **RRC Security Mode Complete:** UE responds with RRC SecurityModeComplete (ciphered + integrity protected)
14. **AS Security Active:** Full AS security is now active

■ Key Hierarchy Diagram

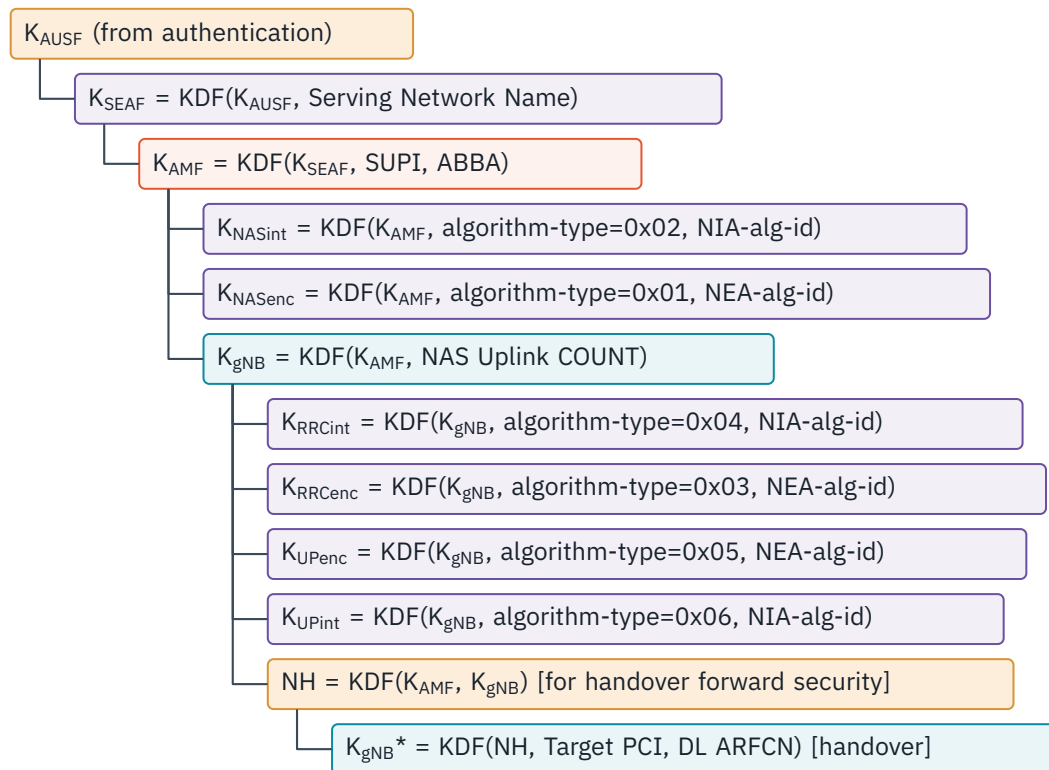


Figure 20.7 Derivation of the NAS and AS keys

Condensed Sequence Diagram

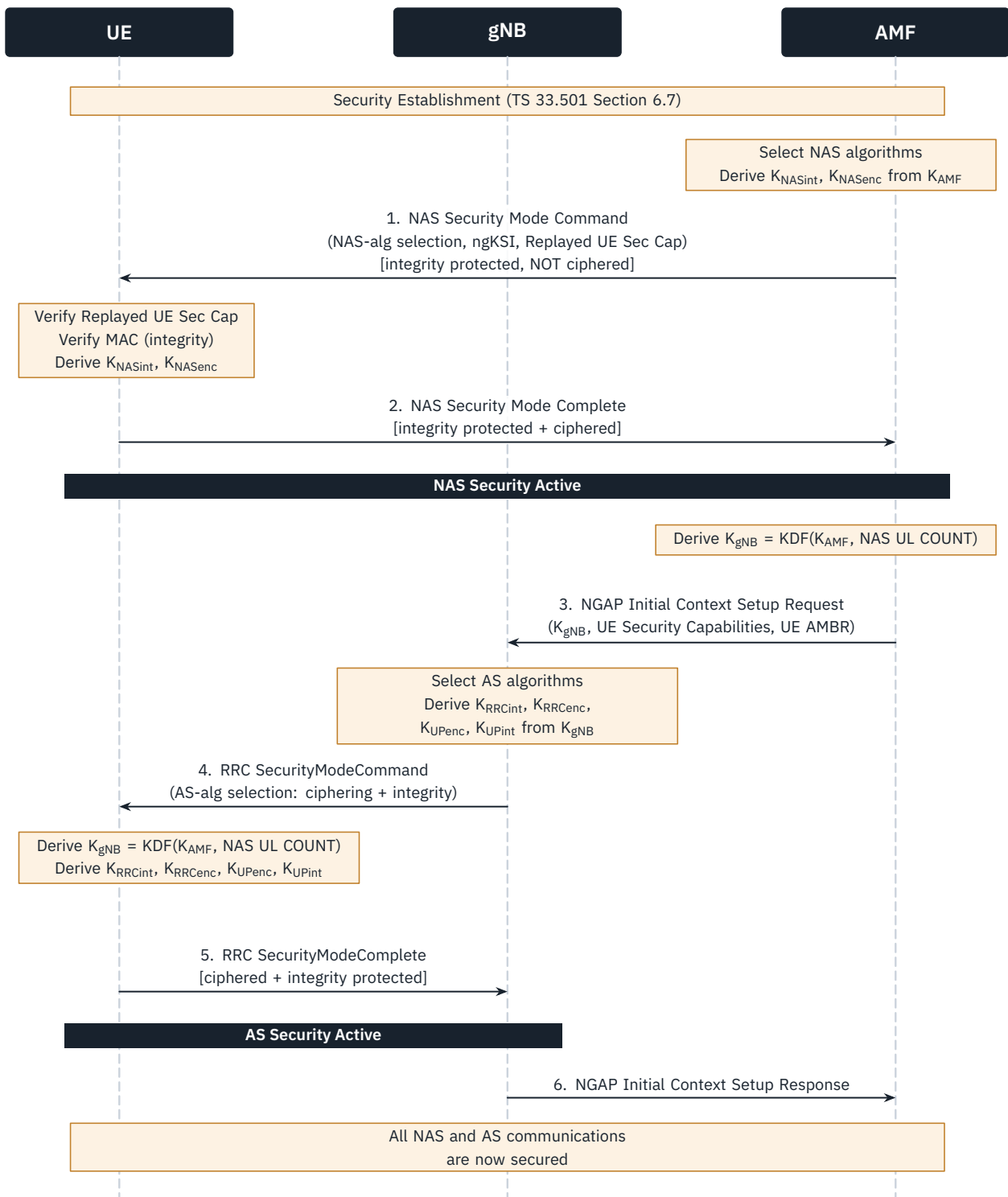


Figure 20.8 NAS and AS security establishment

Failure Cases

Failure	Cause	Response
UE Sec Cap Mismatch	Replayed capabilities don't match (bidding-down attack)	UE sends Security Mode Reject
Integrity Verification Failure	MAC check fails on Security Mode Command	UE sends Security Mode Reject
Algorithm Not Supported	Selected algorithm not in UE capabilities	UE sends Security Mode Reject
NAS Security Mode Reject	Any NAS SMC failure	AMF may re-attempt or reject registration
AS Security Mode Failure	RRC SMC integrity check fails	gNB reports failure; may release connection
Null Integrity Not Allowed	NIA0 selected when not permitted	UE rejects (NIA0 only for emergency)

4 PDU Session Establishment

Purpose

The PDU Session Establishment procedure creates a data connectivity session between the UE and a Data Network (DN). It allocates an IP address (or other PDU type), sets up QoS rules and flows, establishes GTP-U tunnels on N3 (gNB↔UPF) and N9 (UPF↔UPF) interfaces, and configures PFCP sessions on N4 (SMF↔UPF). This enables the UE to send and receive user plane data.

Preconditions

- UE is registered with the 5GC (Registration procedure completed)
- UE is in CM-CONNECTED state (RRC connection active)
- NAS security context is established
- UE has valid subscription for the requested DNN and S-NSSAI
- SMF and UPF resources are available for the requested slice

■ NFs Involved

Network Function	Role
UE	Initiates session request, provides DNN/S-NSSAI/PDU type
gNB	Forwards NAS, establishes N3 GTP-U tunnel
AMF	Routes SM NAS messages, selects SMF
SMF	Session management, IP allocation, QoS policy, UPF selection/control
UPF	User plane forwarding, packet detection/enforcement
PCF	Provides SM policies and PCC rules
UDM	Provides session management subscription data
DN-AAA	External authentication (if required by DN)
CHF	Charging function

■ Messages

Step	Interface	Message
1	NAS	PDU Session Establishment Request (in UL NAS Transport)
2	N11	Nsmf_PDUSession_CreateSMContext Request
3	N10	Nudm_SDM_Get (SM Subscription Data)
4	N7	Npcf_SMPolicyControl_Create
5	N4	PFCP Session Establishment Request
6	N4	PFCP Session Establishment Response
7	N11	Nsmf_PDUSession_CreateSMContext Response
8	N11	Namf_Communication_N1N2MessageTransfer
9	N2	NGAP PDU Session Resource Setup Request
10	RRC	RRCReconfiguration (DRB setup)
11	RRC	RRCReconfigurationComplete
12	N2	NGAP PDU Session Resource Setup Response
13	N11	Nsmf_PDUSession_UpdateSMContext (N2 Info)
14	N4	PFCP Session Modification Request (DL tunnel info)
15	NAS	PDU Session Establishment Accept

■ Step-by-Step Flow

- 1. UE Request:** UE sends PDU Session Establishment Request containing:
 - PDU Session ID, PDU Type (IPv4/IPv6/IPv4v6/Ethernet/Unstructured)
 - Requested DNN, S-NSSAI, SSC Mode
 - SM PDU DN Request Container (optional)

- Requested QoS Rules and QoS Flow Descriptions
- 2. **NAS Transport:** AMF receives UL NAS Transport message with SM NAS container
- 3. **SMF Selection:** AMF selects SMF based on DNN, S-NSSAI, PLMN, and NRF discovery
- 4. **SM Context Creation:** AMF sends Nsmf_PDUSession_CreateSMContext to selected SMF
- 5. **Subscription Data:** SMF retrieves session management subscription data from UDM (Nudm_SDM_Get)
- 6. **SMF Registration:** SMF registers with UDM for the PDU session
- 7. **Policy Association:** SMF creates SM Policy association with PCF (Npcf_SMPolicyControl_Create)
- 8. **PCF Response:** PCF returns PCC rules, QoS policies, and charging rules
- 9. **UPF Selection:** SMF selects appropriate UPF(s) based on UE location, DNN, slice, and data path requirements
- 10. **IP Allocation:** SMF allocates IP address/prefix (from local pool, UPF pool, or external DHCP/AAA)
- 11. **N4 Session Establishment:** SMF sends PFCP Session Establishment Request to UPF:
 - PDRs (Packet Detection Rules), FARs (Forwarding Action Rules)
 - QERs (QoS Enforcement Rules), URRs (Usage Reporting Rules)
 - UL tunnel endpoint (F-TEID on UPF for N3)
- 12. **UPF Response:** UPF responds with allocated F-TEID (N3 UL endpoint)
- 13. **N1N2 Message Transfer:** SMF sends to AMF:
 - N1 SM container: PDU Session Establishment Accept (IP addr, QoS rules, S-NSSAI)
 - N2 SM container: PDU Session Resource Setup Request Transfer (UPF N3 TEID, QoS Profile)
- 14. **RAN Setup:** AMF forwards to gNB via NGAP PDU Session Resource Setup Request
- 15. **DRB Establishment:** gNB maps QoS flows to DRBs, sends RRCReconfiguration to UE
- 16. **UE Configuration:** UE applies QoS rules, configures DRBs, responds with RRCReconfigurationComplete
- 17. **gNB Response:** gNB sends NGAP PDU Session Resource Setup Response (gNB N3 F-TEID for DL)
- 18. **Path Update:** AMF forwards gNB's DL tunnel info to SMF (Nsmf_PDUSession_UpdateSMContext)
- 19. **N4 Modification:** SMF sends PFCP Session Modification to UPF with DL tunnel endpoint (gNB's F-TEID)
- 20. **Session Active:** End-to-end user plane path is established: UE↔gNB↔UPF↔DN

Detailed Sequence Diagram

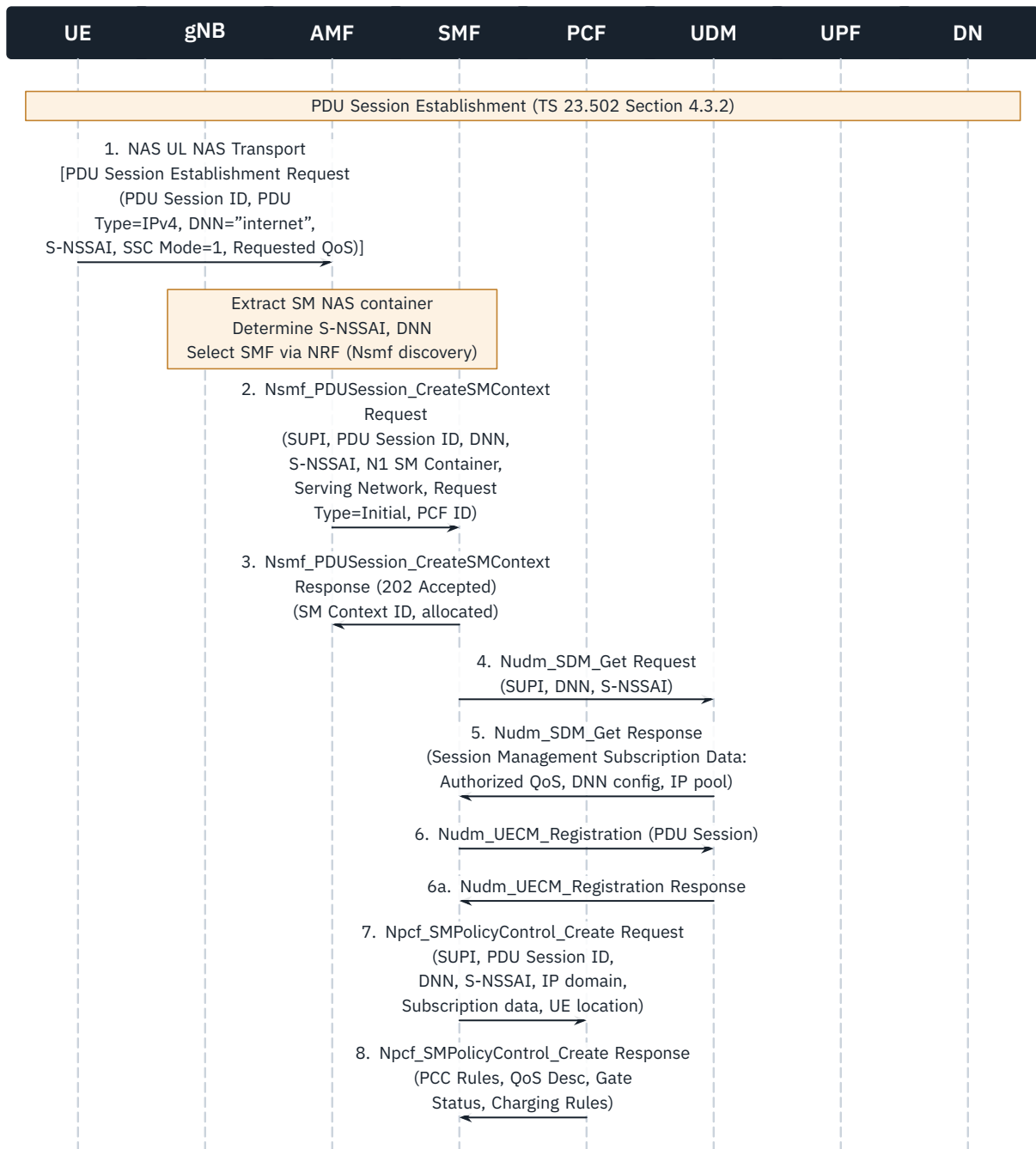


Figure 20.9 PDU session establishment (TS 23.502 §4.3.2) (part 1 of 3)

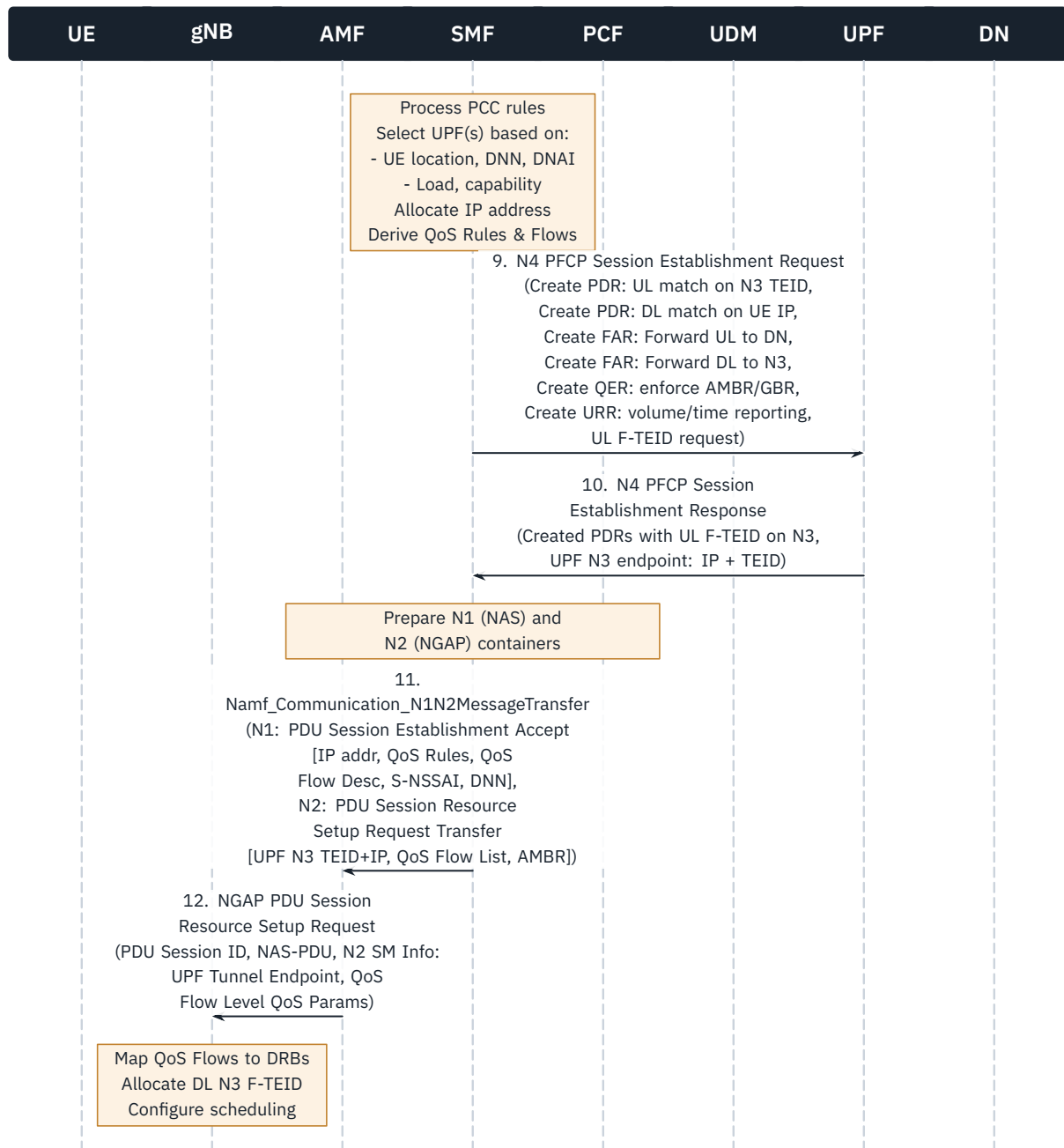


Figure 20.10 PDU session establishment (TS 23.502 §4.3.2) (part 2 of 3)

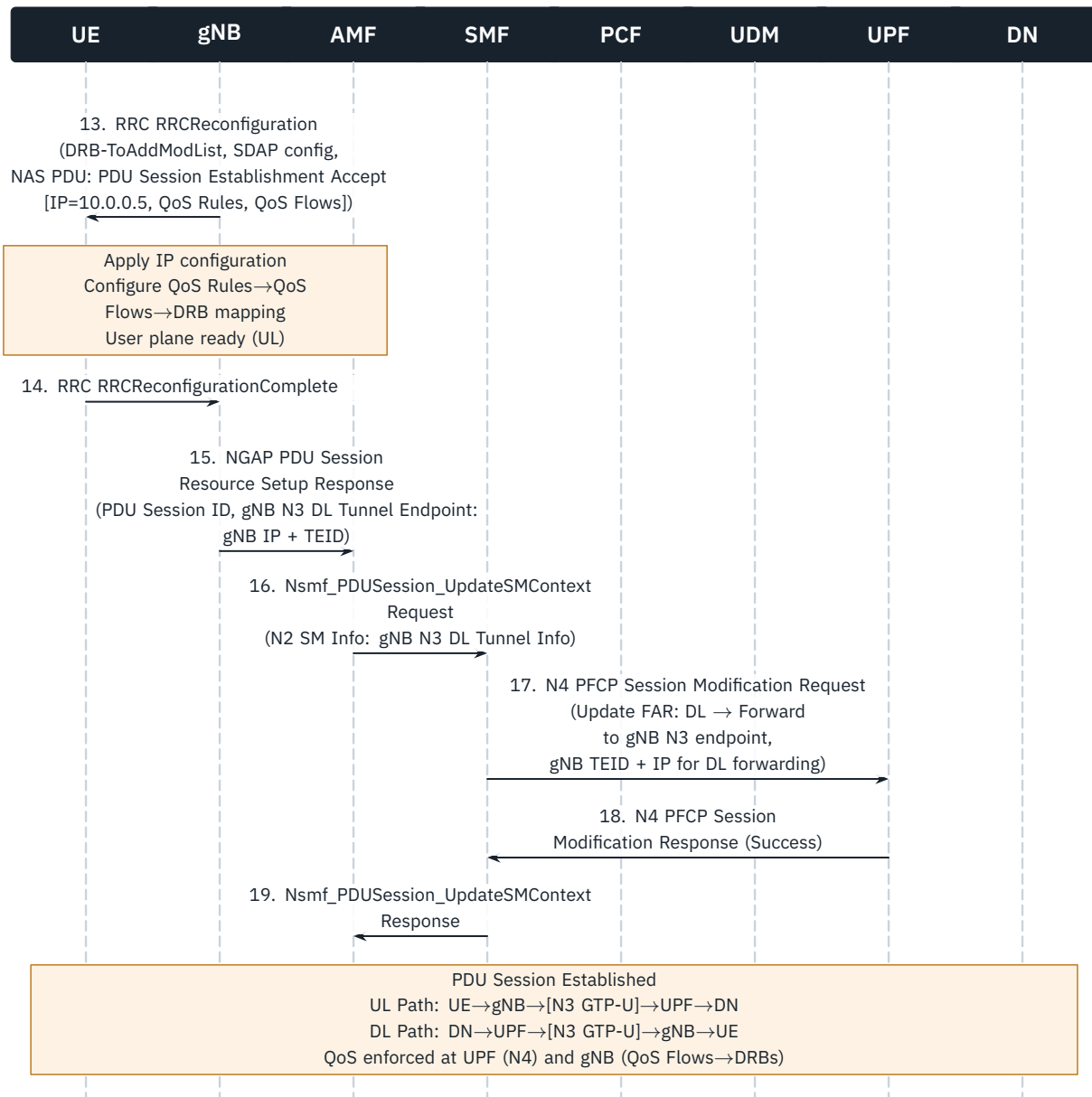


Figure 20.11 PDU session establishment (TS 23.502 §4.3.2) (part 3 of 3)

Failure Cases

Failure	Cause	Response
Insufficient Resources	No IP addresses available or UPF overloaded	PDU Session Establishment Reject (Cause: #26)
DNN Not Subscribed	UE not authorized for requested DNN	Reject (Cause: #27)
Invalid S-NSSAI	Slice not in Allowed NSSAI	Reject (Cause: #62)
SMF Unavailable	No SMF available for DNN/slice	AMF rejects request
UPF Selection Failure	No suitable UPF found	SMF rejects session
PCF Policy Denial	PCF rejects session based on policy	SMF rejects session
N4 Establishment Failure	PFCEP to UPF fails	SMF retries or rejects
DRB Setup Failure	gNB cannot allocate radio resources	Partial/failed resource setup reported to AMF
Max Sessions Reached	UE subscription limit exceeded	Reject (Cause: #65)
Network Failure	Transport network issue	Session establishment timeout

5 PDU Session Release

Purpose

The PDU Session Release procedure tears down an existing PDU session, releasing all associated resources including IP address, GTP-U tunnels (N3/N9), PFCEP sessions (N4), QoS flows, and DRBs. It can be initiated by the UE (user no longer needs connectivity), the SMF (inactivity timer, policy), or the AMF (deregistration).

Preconditions

- An active PDU Session exists (SM context in SMF, N4 session in UPF)
- UE is registered with the network
- NAS security context is active

■ NFs Involved

Network Function	Role
UE	Initiates release (UE-triggered) or acknowledges
gNB	Releases DRBs and N3 tunnel resources
AMF	Routes SM NAS messages
SMF	Orchestrates release, removes SM context
UPF	Removes PFCP session, releases tunnels
PCF	Terminates SM policy association
UDM	Deregisters PDU session
CHF	Final charging record

■ Messages

Step	Interface	Message
1	NAS	PDU Session Release Request (UE-initiated)
2	N11	Nsmf_PDUSession_UpdateSMContext (Release)
3	N4	PFCP Session Deletion Request
4	N4	PFCP Session Deletion Response
5	N7	Npcf_SMPolicyControl_Delete
6	N2	NGAP PDU Session Resource Release Command
7	NAS	PDU Session Release Command
8	RRC	RRCReconfiguration (DRB release)
9	N2	NGAP PDU Session Resource Release Response
10	NAS	PDU Session Release Complete

■ Step-by-Step Flow

■ UE-Initiated Release

1. UE sends PDU Session Release Request (PDU Session ID, Cause)
2. AMF forwards to SMF (Nsmf_PDUSession_UpdateSMContext)
3. SMF initiates resource cleanup
4. SMF sends PFCP Session Deletion Request to UPF
5. UPF releases tunnels, responds with PFCP Session Deletion Response
6. SMF terminates PCF policy association
7. SMF deregisters session from UDM
8. SMF sends PDU Session Release Command to UE (via AMF N1N2 transfer)
9. AMF sends NGAP PDU Session Resource Release Command to gNB
10. gNB releases DRBs, sends RRCReconfiguration to UE
11. gNB responds with PDU Session Resource Release Response

12. UE sends PDU Session Release Complete

■ Network-Initiated Release

1. SMF decides to release (inactivity, policy change, subscription withdrawal)
2. SMF sends PDU Session Release Command to UE via AMF
3. Remaining steps are similar to UE-initiated from step 9

Condensed Sequence Diagram

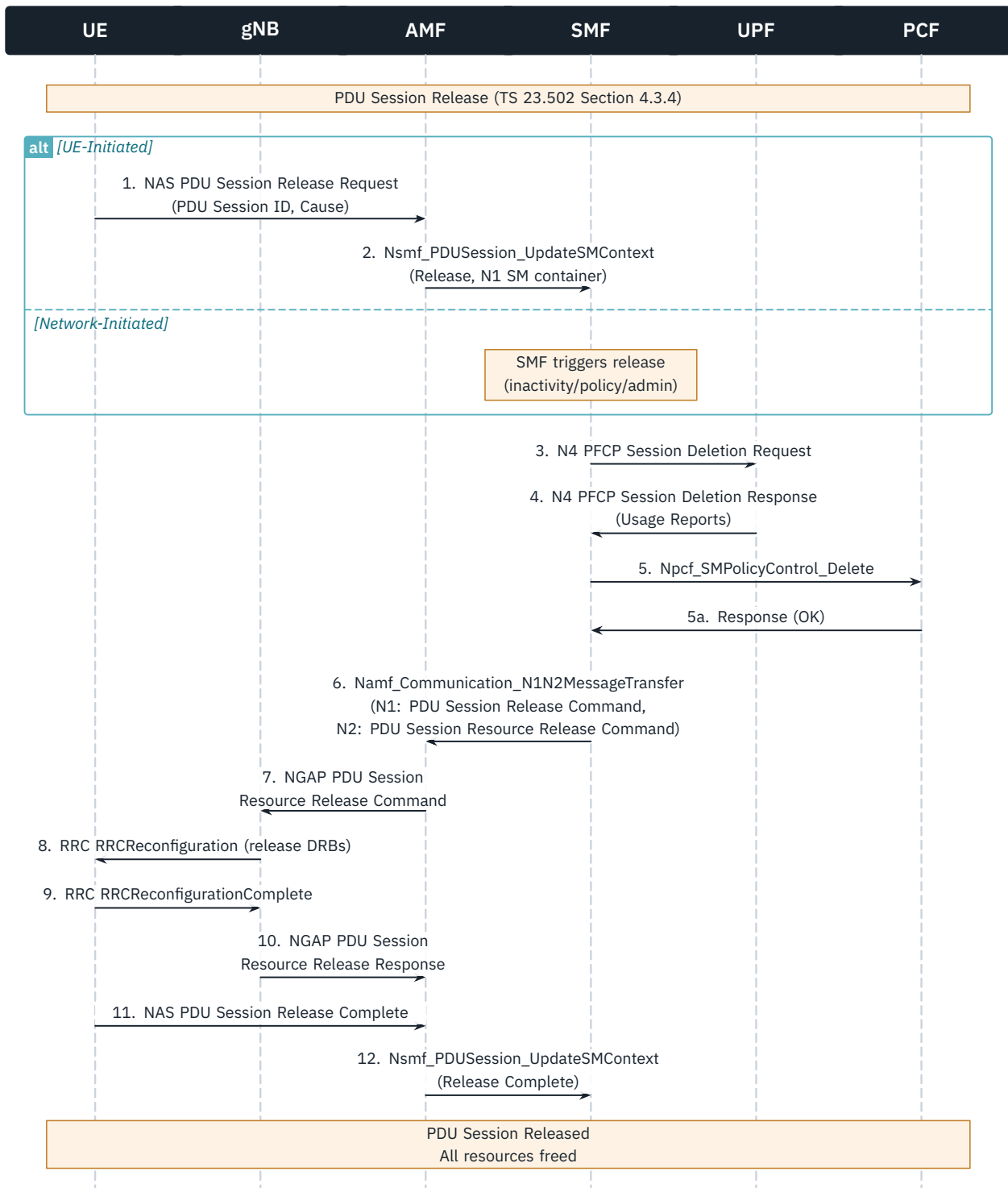


Figure 20.12 PDU session release

Failure Cases

Failure	Cause	Response
UPF Unreachable	N4 path failure	SMF retries; local cleanup after timeout
UE Unreachable (CM-IDLE)	UE in idle mode	SMF proceeds with network-side cleanup
Timer Expiry	No response from UE	Implicit release; local resource cleanup
Partial Release	Some resources fail to release	SMF performs cleanup of remaining resources

6 Service Request

Purpose

The Service Request procedure allows a UE in CM-IDLE state (or CM-CONNECTED with inactive user plane) to re-establish signaling and/or user plane connections when uplink data or signaling needs to be sent. It triggers transition from RRC_IDLE/RRC_INACTIVE to RRC_CONNECTED and reactivates the user plane path (N3 GTP-U tunnels) for existing PDU sessions.

Preconditions

- UE is registered with valid registration context
- UE is in CM-IDLE (RRC_IDLE or RRC_INACTIVE) state
- UE has at least one active PDU session (UP resources to reactivate)
- NAS security context exists (K_{AMF} , K_{NASint} , K_{NASenc} available)
- UE has uplink data or NAS signaling to send

■ NFs Involved

Network Function	Role
UE	Triggers service request when UL data arrives
gNB	Establishes RRC connection, allocates DRBs
AMF	Processes service request, coordinates UP reactivation
SMF	Updates N4 session with new DL tunnel info
UPF	Receives updated FAR for DL forwarding

■ Messages

Step	Interface	Message
1	Uu	RRC Resume/Setup Request (if RRC_INACTIVE/IDLE)
2	NAS	Service Request (ngKSI, PDU Session Status)
3	N2	NGAP Initial UE Message or UE Context Resume
4	N2	NGAP Initial Context Setup / PDU Session Resource Setup
5	N11	Nsmf_PDUSession_UpdateSMContext (UP activate)
6	N4	PCF Session Modification (new DL TEID)
7	RRC	RRCReconfiguration (DRB reactivation)
8	NAS	Service Accept

■ Step-by-Step Flow

1. **Trigger:** UE application generates uplink data; NAS layer detects CM-IDLE state
2. **RRC Connection:** UE initiates RRC connection (RRC Setup from IDLE or RRC Resume from INACTIVE)
3. **Service Request:** UE sends NAS Service Request (integrity protected with K_{NASint}):
 - ngKSI, Service Type (signaling/data/mobile-terminated)
 - PDU Session Status, Uplink Data Status
4. **AMF Processing:** AMF verifies integrity, identifies active PDU sessions
5. **Security Activation:** AMF sends new K_{gNB} to gNB (Initial Context Setup or UE Context Resume)
6. **UP Reactivation:** For each PDU session to reactivate:
 - AMF sends Nsmf_PDUSession_UpdateSMContext to SMF (new AN info needed)
 - gNB allocates DRBs, assigns DL N3 TEID
 - SMF updates UPF with new DL tunnel info (PCF Session Modification)
7. **DRB Setup:** gNB sends RRCReconfiguration to UE with DRB configuration
8. **Service Accept:** AMF sends Service Accept to UE
9. **Data Flow:** User plane path is re-established; buffered DL data is forwarded

Condensed Sequence Diagram

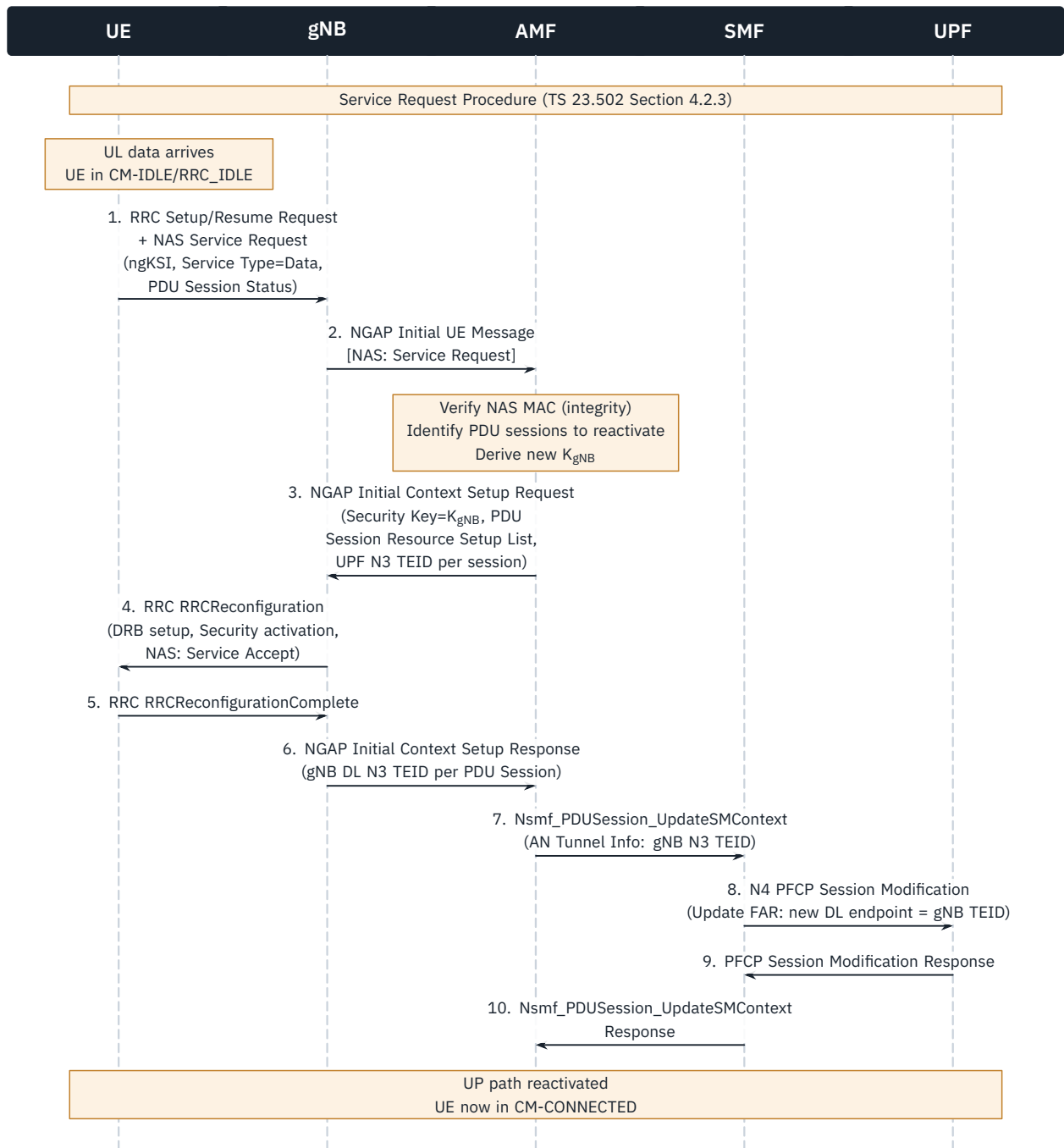


Figure 20.13 Service request

Failure Cases

Failure	Cause	Response
Integrity Failure	NAS MAC verification fails	AMF rejects; may trigger re-authentication
Context Not Found	AMF has no UE context (e.g., AMF restart)	Service Reject (Cause: #9) → UE re-registers
PDU Session Mismatch	Status inconsistency between UE and network	AMF triggers PDU session status sync
RAN Resources Unavailable	gNB cannot allocate DRBs	Partial acceptance or rejection
Security Context Invalid	ngKSI mismatch or expired context	Trigger re-authentication

7 Paging

Purpose

The Paging procedure allows the network to reach a UE that is in CM-IDLE state when downlink data or signaling arrives for that UE. The AMF distributes paging messages to all gNBs in the UE's Registration Area (set of Tracking Areas), and the UE responds with a Service Request to re-establish connectivity.

Preconditions

- UE is in CM-IDLE state (RRC_IDLE or RRC_INACTIVE)
- Downlink data arrives at UPF for the UE, or MT signaling is pending
- AMF has valid UE context with Registration Area information
- UE is monitoring paging occasions in its current cell

■ NFs Involved

Network Function	Role
UPF	Detects DL data for idle UE, sends notification
SMF	Receives data notification, triggers paging via AMF
AMF	Distributes paging to gNBs in Registration Area
gNB(s)	Broadcasts paging message on Uu interface
UE	Monitors paging, initiates Service Request

■ Messages

Step	Interface	Message
1	N4	Downlink Data Notification (UPF→SMF via PFCP)
2	N11	Nsmf_PDUSession_UpdateSMContext (DL data pending) or Namf_Communication_N1N2MessageTransfer
3	N2	NGAP Paging (to all gNBs in Registration Area)
4	Uu	RRC Paging message (5G-S-TMSI)
5	Uu	Service Request (UE response)

■ Step-by-Step Flow

- DL Data Arrival:** Downlink packet arrives at UPF for a UE in CM-IDLE state
- Data Notification:** UPF buffers the packet and sends Downlink Data Notification to SMF (via N4 PFCP)
- Paging Trigger:** SMF sends Namf_Communication_N1N2MessageTransfer or Nsmf notification to AMF indicating DL data pending
- Paging Distribution:** AMF sends NGAP Paging message to all gNBs in the UE's Registration Area:
 - 5G-S-TMSI (for UE identification)
 - TAI List (Tracking Area Identities)
 - Paging Priority (if applicable)
 - Paging DRX information
- RRC Paging:** Each gNB broadcasts RRC Paging message on PCCH at the UE's paging occasion
- UE Detection:** UE wakes up at its paging occasion (determined by 5G-S-TMSI and DRX cycle), detects its identity
- Service Request:** UE initiates Service Request procedure (see Section 6)
- Data Delivery:** After UP path is reactivated, UPF forwards buffered DL data
- Paging Retry:** If UE doesn't respond within T3513, AMF retries paging (up to configured retransmissions)

Condensed Sequence Diagram

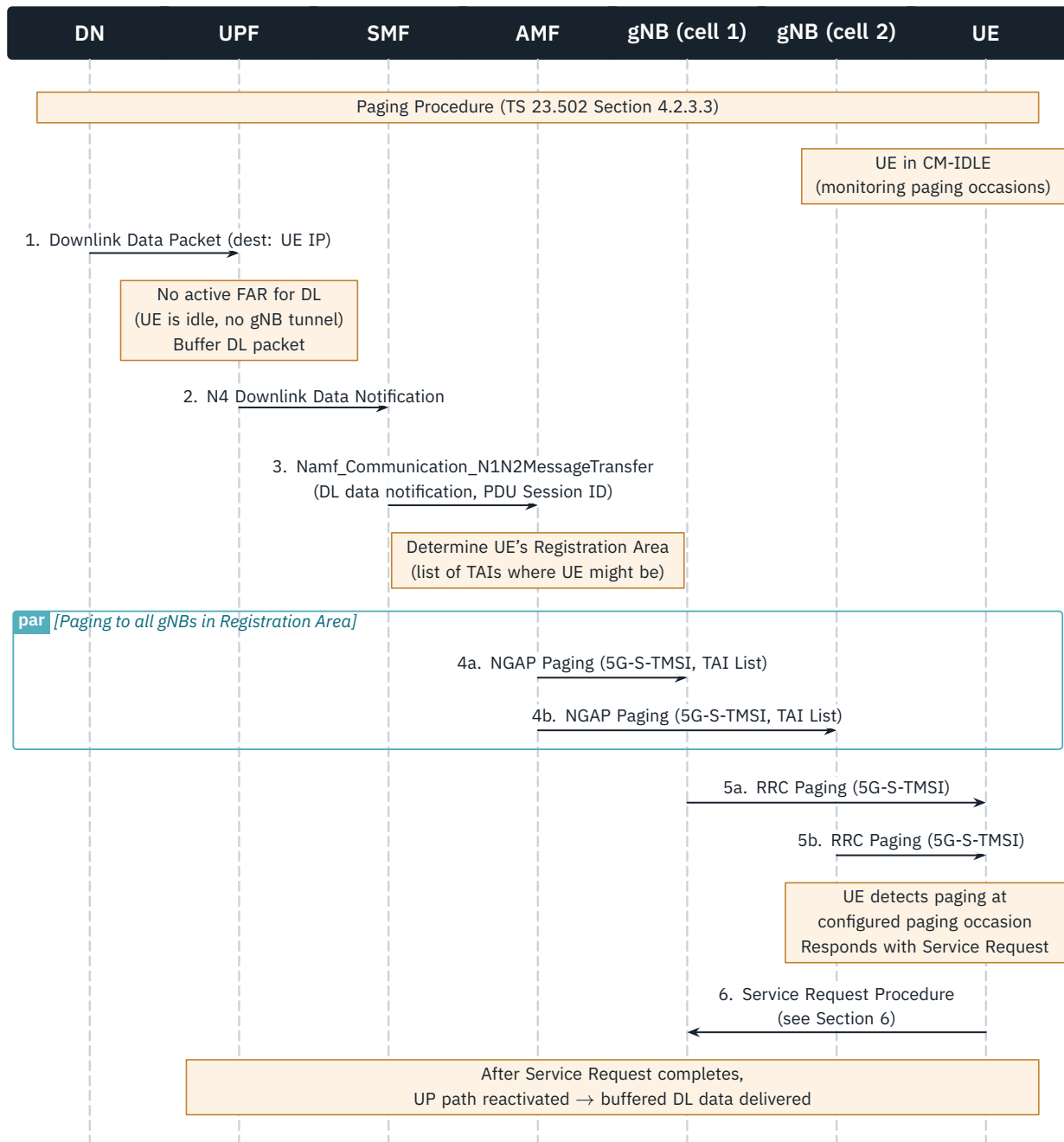


Figure 20.14 Paging of a CM-IDLE UE

Failure Cases

Failure	Cause	Response
UE Not Responding	UE powered off, out of coverage	AMF retries (T3513); eventually notifies SMF
Paging Timeout	Max retransmissions reached	AMF informs SMF; UPF may discard buffered data
Wrong Registration Area	UE moved without updating TA	Paging fails; UE will eventually re-register
UPF Buffer Overflow	Too much buffered DL data	UPF discards oldest packets
Paging Congestion	Too many paging requests	AMF applies paging restrictions/priorities

8 Xn Handover

Purpose

The Xn Handover procedure transfers a UE's connection from a source gNB to a target gNB over the Xn interface without involving the AMF in the handover decision. It provides seamless mobility with minimal service interruption. The source gNB makes the handover decision based on measurement reports, prepares the target gNB, and the user plane path is switched at the UPF. This is used when both gNBs are connected to the same AMF and an Xn interface exists between them.

Preconditions

- UE is in RRC_CONNECTED state with active PDU sessions
- Xn interface exists between source and target gNBs
- Both gNBs are served by the same AMF
- UE measurement reports indicate target cell is better than source
- Target cell belongs to same or compatible network slice
- Source gNB has the UE's security context and active AS keys

NFs Involved

Network Function	Role
UE	Sends measurement reports, performs handover to target
Source gNB	Makes HO decision, prepares target, forwards data
Target gNB	Admits UE, allocates resources, becomes new serving
AMF	Updates UE context with new serving gNB (path switch)
SMF	Updates N4 session for new downlink tunnel
UPF	Switches downlink path to target gNB

Messages

Step	Interface	Message
1	Uu	RRC MeasurementReport (UE→Source gNB)
2	Xn	Handover Request (Source→Target)
3	Xn	Handover Request Acknowledge (Target→Source)
4	Uu	RRC RRCReconfiguration (HO Command)
5	—	UE detaches from source, syncs to target
6	Uu	RRC RRCReconfigurationComplete (UE→Target)
7	Xn	SN Status Transfer (Source→Target)
8	N2	NGAP Path Switch Request (Target→AMF)
9	N11	Nsmf_PDUSession_UpdateSMContext (AMF→SMF)
10	N4	PCFP Session Modification (SMF→UPF)
11	N11	Nsmf_PDUSession_UpdateSMContext Response
12	N2	NGAP Path Switch Request Acknowledge (AMF→Target)
13	Xn	UE Context Release (Target→Source)

Step-by-Step Flow

- Measurement Configuration:** Source gNB configures UE with measurement objects and reporting criteria
- Measurement Report:** UE detects target cell signal meets threshold; sends MeasurementReport to source gNB
- HO Decision:** Source gNB evaluates measurement report, selects target cell, decides to handover
- Handover Preparation:**
 - Source gNB sends Handover Request to target gNB over Xn:
 - UE context (security, capabilities, PDU session info)
 - UE History Information
 - Source-to-Target transparent container (RRC config)

5. **Admission Control:** Target gNB performs admission control, allocates resources (DRBs, radio resources)
6. **Handover Request Ack:** Target gNB responds with:
 - Target-to-Source transparent container (RRC HO Command)
 - Allocated DL forwarding tunnels (for data forwarding during HO)
7. **HO Command:** Source gNB sends RRCReconfiguration (mobilityControlInfo) to UE containing:
 - Target cell ID, new C-RNTI
 - Target gNB security algorithm configuration
 - Radio resource configuration for target cell
8. **Data Forwarding:** Source gNB begins forwarding buffered/incoming DL data to target gNB (via Xn forwarding tunnel)
9. **SN Status Transfer:** Source gNB sends PDCP SN Status Transfer to target gNB (for lossless handover on AM DRBs)
10. **Random Access:** UE synchronizes to target cell (RACH procedure)
11. **HO Complete:** UE sends RRCReconfigurationComplete to target gNB
12. **Path Switch:** Target gNB sends NGAP Path Switch Request to AMF:
 - New security context info (NH/NCC if needed)
 - New N3 tunnel info for each PDU session (target gNB's F-TEID)
13. **UP Path Update:** AMF sends Nsmf_PDUSession_UpdateSMContext to SMF with new AN tunnel info
14. **UPF Update:** SMF sends PFCP Session Modification to UPF → update DL FAR to point to target gNB
15. **End Marker:** UPF sends end marker packet on old N3 tunnel (to source gNB) to indicate DL path switched
16. **Path Switch Ack:** AMF sends NGAP Path Switch Request Acknowledge to target gNB (new NH, NCC for future HO)
17. **UE Context Release:** Target gNB sends UE Context Release to source gNB over Xn
18. **Source Cleanup:** Source gNB releases UE context and resources

Detailed Sequence Diagram

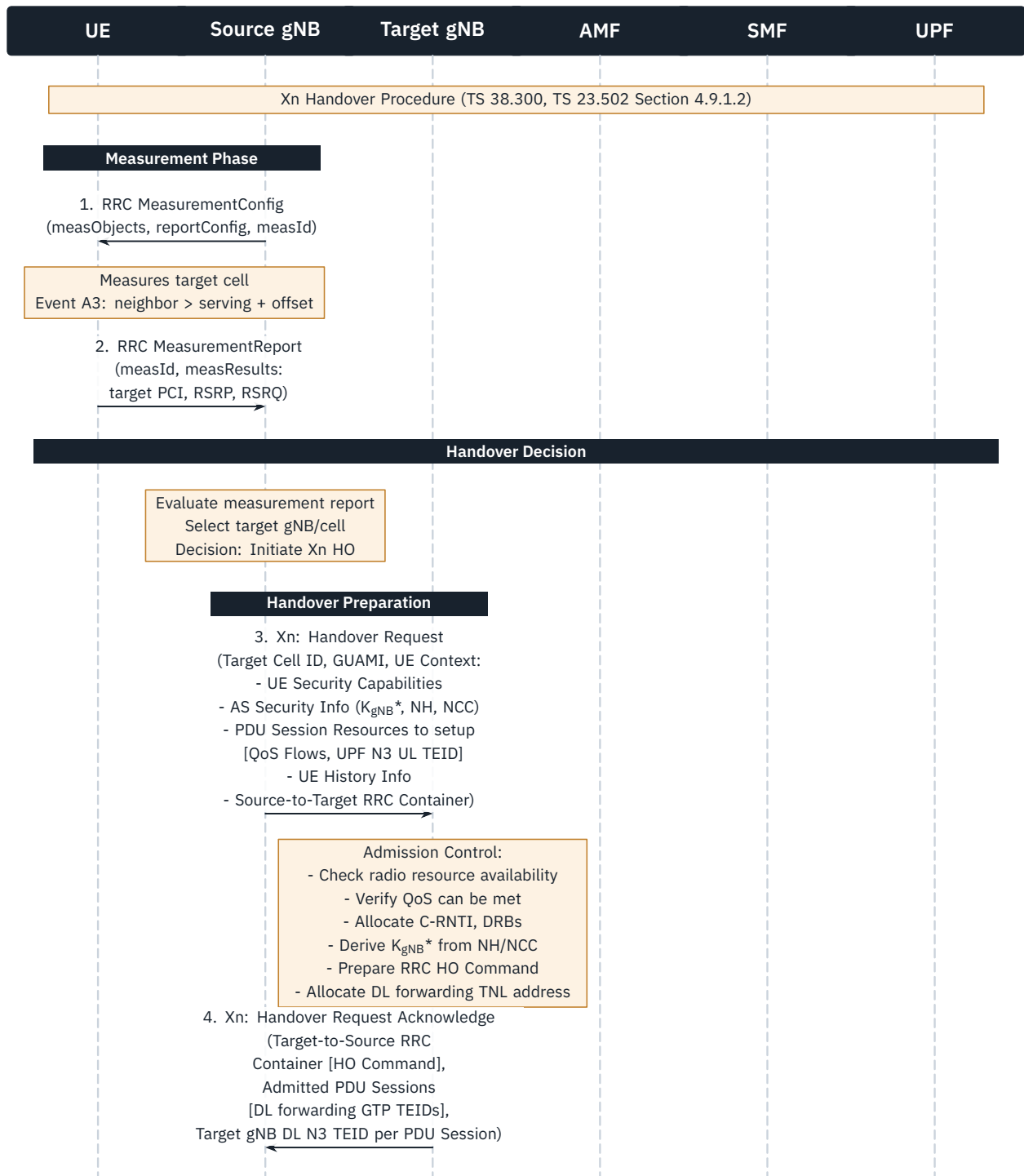


Figure 20.15 Xn handover (part 1 of 3)

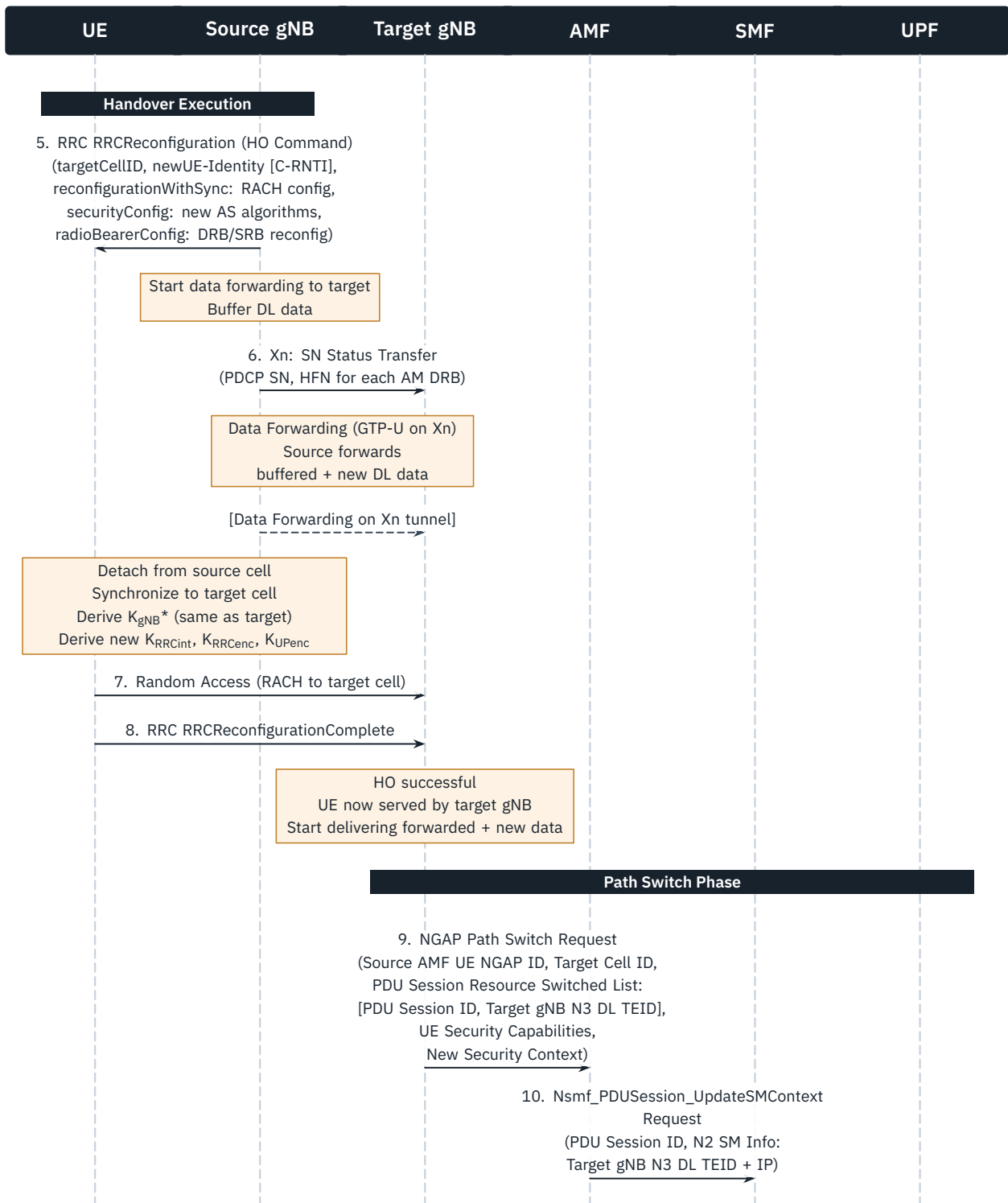


Figure 20.16 Xn handover (part 2 of 3)

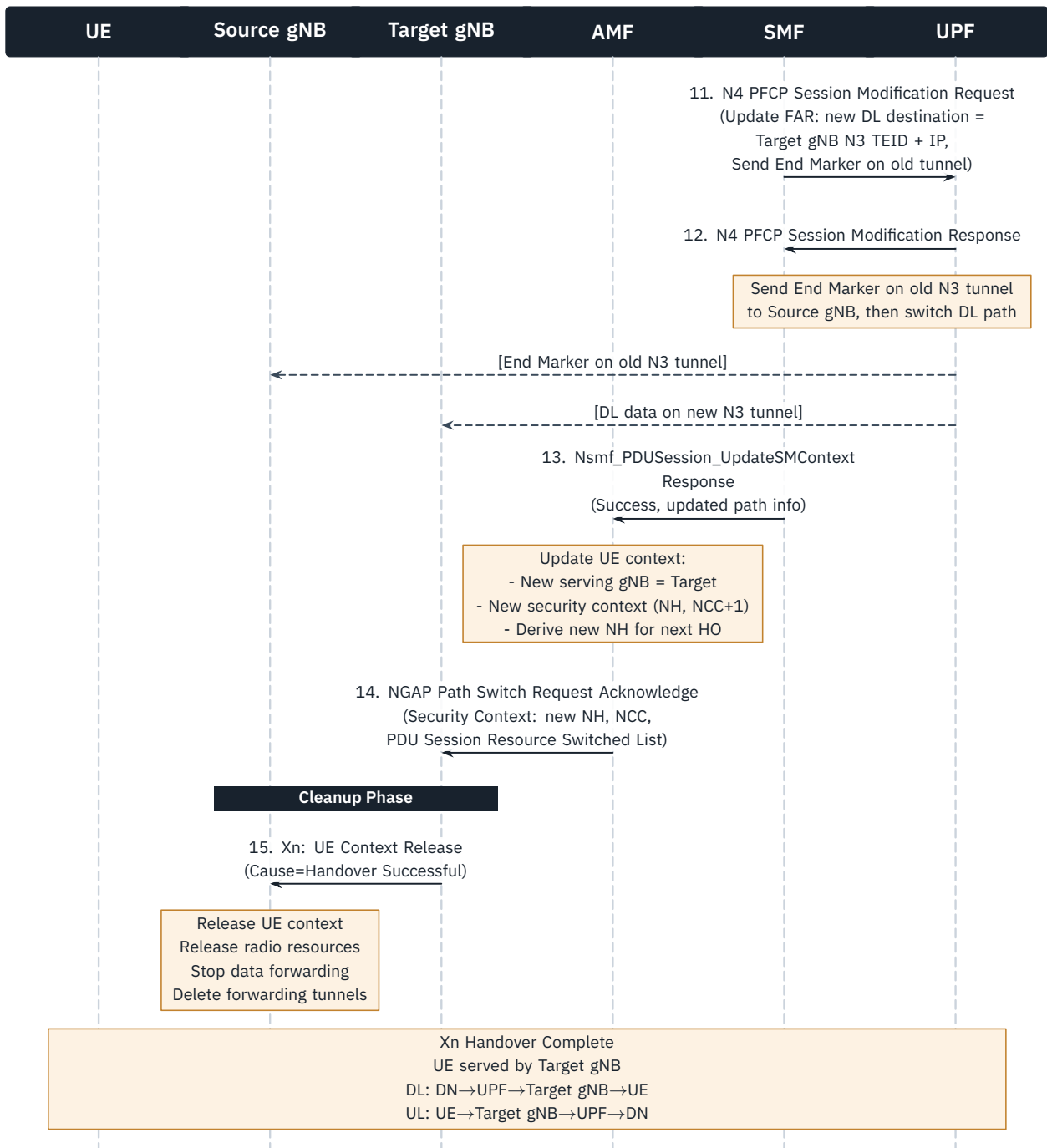


Figure 20.17 Xn handover (part 3 of 3)

Failure Cases

Failure	Cause	Response
HO Preparation Failure	Target rejects (no resources, admission control)	Source gNB tries another target or maintains connection
HO Failure (timer expiry)	UE fails RACH to target within T304	UE initiates RRC Re-establishment to source or other cell
RRC Re-establishment	Radio link failure during HO	UE attempts re-establishment; may trigger new HO
Path Switch Failure	AMF/SMF fails to switch path	AMF sends Path Switch Request Failure; target may release UE
Data Loss	Forwarding tunnel congestion	Some packets lost during HO gap (UM bearers)
Too-Late HO	UE loses source before HO command	RLF; RRC Re-establishment required
Too-Early HO	HO triggered before stable measurement	UE may fail at target; RRC Re-establishment back to source
Xn Interface Failure	Transport issue between gNBs	Fall back to N2 (inter-AMF) handover via AMF

9 N2 Handover (Inter-AMF)

Purpose

The N2 Handover (inter-AMF) procedure handles mobility when the target gNB is served by a different AMF than the source. This is more complex than Xn Handover because it requires UE context transfer between AMFs, re-establishment of NF associations (SMF, PCF), and potentially re-authentication. It is used when no Xn interface exists between source and target gNBs, or when the target falls under a different AMF's serving area.

Preconditions

- UE is in RRC_CONNECTED state with active PDU sessions
- Source gNB determines handover is needed (measurement reports)
- Target gNB is NOT served by the same AMF (different AMF region/set)
- OR no Xn interface exists between source and target gNBs
- N2 interface available between source gNB and source AMF

NFs Involved

Network Function	Role
UE	Performs handover to target cell
Source gNB	Initiates HO, sends HO Required to source AMF
Target gNB	Admits UE, allocates resources
Source AMF	Identifies target AMF, initiates context transfer
Target AMF	Receives UE context, becomes new serving AMF
SMF	Updates UP path for each PDU session
UPF	Switches DL tunnel to target gNB
UDM	Updates AMF registration (new AMF)

Messages

Step	Interface	Message
1	Uu	MeasurementReport (UE→Source gNB)
2	N2	NGAP Handover Required (Source gNB→Source AMF)
3	N14	Namf_Communication_CreateUEContext (Source AMF→Target AMF)
4	N2	NGAP Handover Request (Target AMF→Target gNB)
5	N2	NGAP Handover Request Acknowledge (Target gNB→Target AMF)
6	N14	Namf_Communication_CreateUEContext Response (Target AMF→Source AMF)
7	N2	NGAP Handover Command (Source AMF→Source gNB)
8	Uu	RRC RRCReconfiguration (HO Command to UE)
9	Uu	RRC RRCReconfigurationComplete (UE→Target gNB)
10	N2	NGAP Handover Notify (Target gNB→Target AMF)
11	N14/N2	UE Context Release (cleanup)

■ Step-by-Step Flow

1. **HO Decision:** Source gNB decides handover based on measurement reports
2. **HO Required:** Source gNB sends NGAP Handover Required to source AMF:
 - Target ID (Target gNB + Target TAI)
 - Source-to-Target RRC transparent container
 - PDU Session Resource List
3. **Target AMF Selection:** Source AMF determines target gNB is served by different AMF (based on TAI→AMF mapping)
4. **UE Context Transfer:** Source AMF sends Namf_Communication_CreateUEContext to target AMF:
 - UE context (SUPI, security context, PDU session list)
 - Source-to-Target container
 - MM context, SM contexts
5. **Target gNB Preparation:** Target AMF sends NGAP Handover Request to target gNB
6. **Admission Control:** Target gNB performs admission, allocates resources
7. **HO Request Ack:** Target gNB sends Handover Request Acknowledge (HO Command, DL tunnel info)
8. **Context Response:** Target AMF responds to source AMF with HO Command
9. **HO Command Forwarded:** Source AMF sends NGAP Handover Command to source gNB
10. **RRC HO Command:** Source gNB forwards HO Command to UE via RRCReconfiguration
11. **Data Forwarding:** Source gNB forwards data to target gNB (indirect via UPF or direct if possible)
12. **UE Handover:** UE detaches from source, attaches to target cell
13. **HO Complete:** UE sends RRCReconfigurationComplete to target gNB
14. **HO Notify:** Target gNB sends NGAP Handover Notify to target AMF
15. **Registration Update:** Target AMF registers with UDM as new serving AMF
16. **UP Path Switch:** Target AMF triggers Nsmf_PDUSession_UpdateSMContext for each PDU session
17. **N4 Update:** SMF updates UPF with new DL tunnel info (target gNB TEID)
18. **Source Cleanup:** Target AMF→Source AMF: UE Context Release; Source AMF→Source gNB: UE Context Release Command

Condensed Sequence Diagram

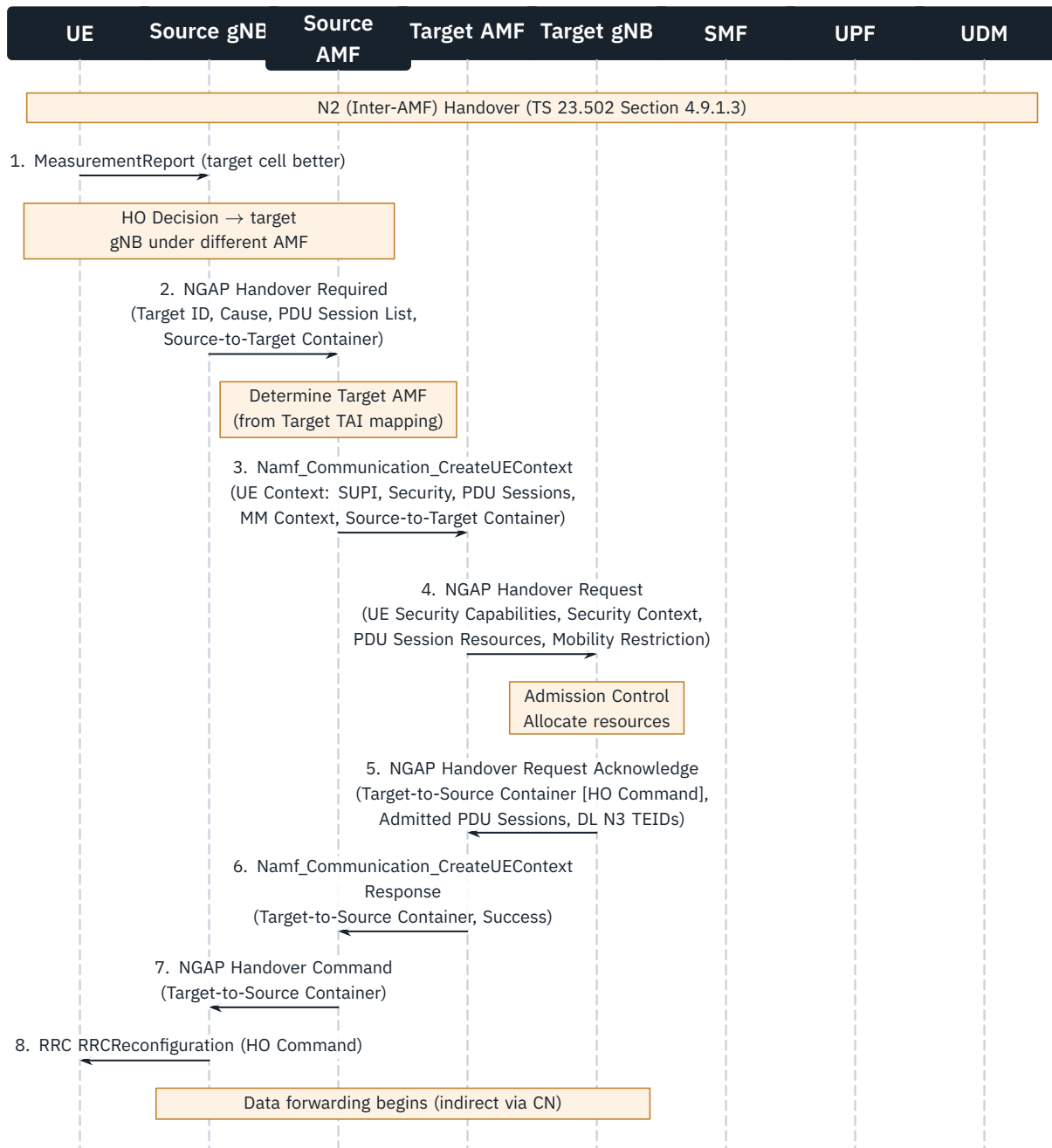


Figure 20.18 N2 (inter-AMF) handover (part 1 of 2)

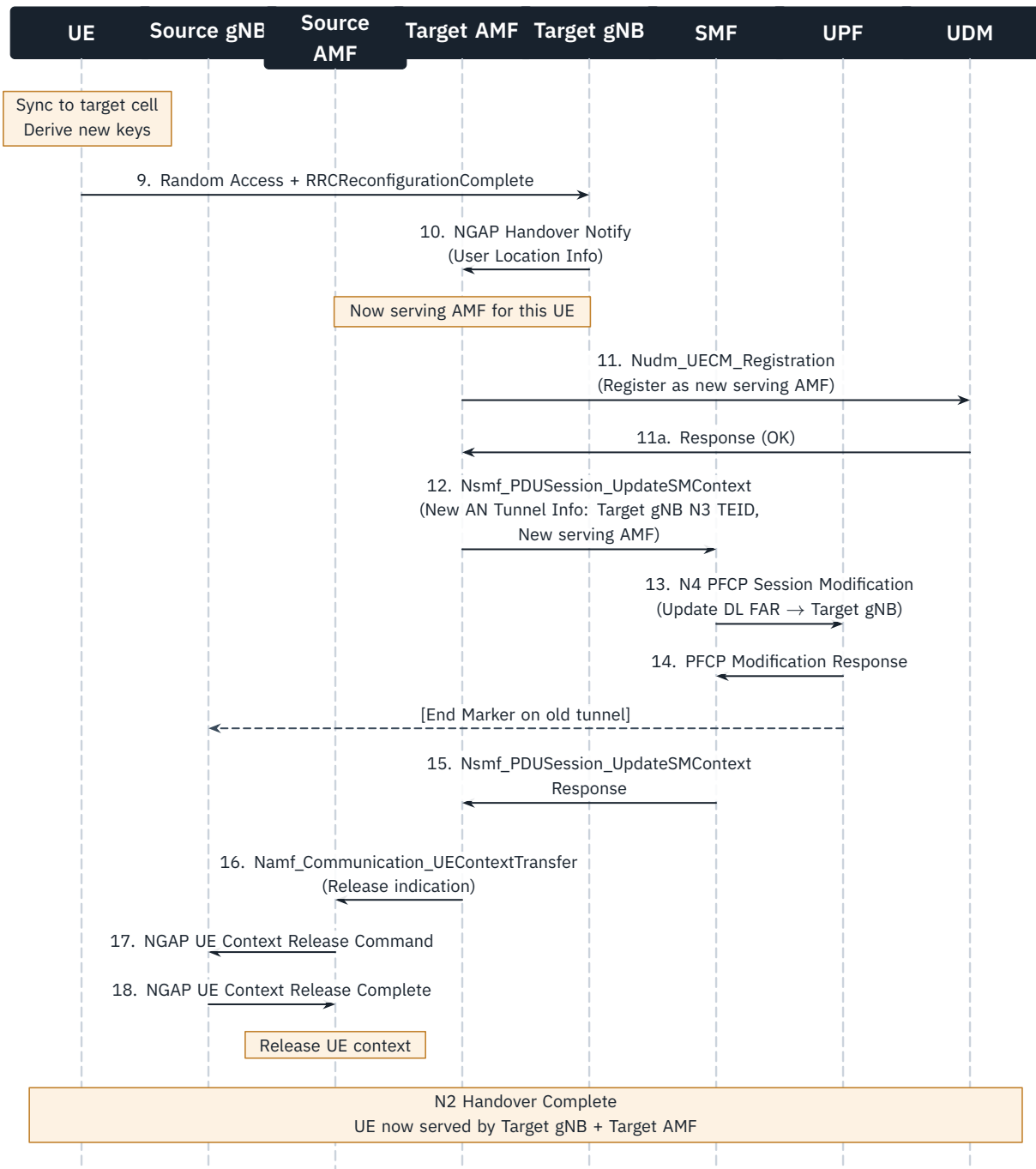


Figure 20.19 N2 (inter-AMF) handover (part 2 of 2)

Failure Cases

Failure	Cause	Response
Target AMF Unreachable	N14 communication failure	Source AMF sends HO Preparation Failure to gNB
Target gNB Rejects	Admission control failure at target	Target AMF informs Source AMF; HO Preparation Failure
HO Failure (UE side)	UE cannot reach target cell	UE attempts RRC Re-establishment
Context Transfer Failure	Incomplete/corrupted UE context	Target AMF rejects; source maintains UE
Timer Expiry (TRELOCprep)	Preparation takes too long	Source AMF cancels HO
Timer Expiry (TRELOCoverall)	Overall HO takes too long	Source AMF releases, target AMF cleans up
Path Switch Failure	SMF/UPF update fails	Handover may succeed but UP path broken; recovery needed
Security Context Mismatch	Keys don't match after transfer	Target AMF triggers re-authentication

Summary: Procedure Comparison Table

Procedure	Trigger	Key NFs	Latency Budget	3GPP Reference
Initial Registration	UE power-on / new PLMN	AMF, AUSF, UDM, PCF	Seconds	TS 23.502 §4.2.2.2
5G-AKA	Registration / Re-auth	AMF, AUSF, UDM	~100ms (NW side)	TS 33.501 §6.1
Security Establishment	Post-authentication	AMF, gNB	~50ms	TS 33.501 §6.7
PDU Session Establishment	UE data request	AMF, SMF, UPF, PCF	~200-500ms	TS 23.502 §4.3.2
PDU Session Release	UE/Network decision	SMF, UPF	~100ms	TS 23.502 §4.3.4
Service Request	UL data in IDLE	AMF, SMF, UPF	~50-100ms	TS 23.502 §4.2.3
Paging	DL data for IDLE UE	AMF, gNBs	~320ms-2s (DRX)	TS 23.502 §4.2.3.3
Xn Handover	Mobility (same AMF)	gNBs, AMF, SMF, UPF	~30-50ms interrupt	TS 38.300 §9.2.3
N2 Handover	Mobility (diff AMF)	gNBs, AMFs, SMF, UPF, UDM	~50-100ms interrupt	TS 23.502 §4.9.1.3

Key References

- **3GPP TS 23.501:** System Architecture for the 5G System (5GS)
- **3GPP TS 23.502:** Procedures for the 5G System (5GS)
- **3GPP TS 33.501:** Security architecture and procedures for 5G System
- **3GPP TS 38.300:** NR; NR and NG-RAN Overall Description
- **3GPP TS 38.331:** NR; Radio Resource Control (RRC) Protocol
- **3GPP TS 29.244:** Interface between the Control Plane and the User Plane (PCF)
- **3GPP TS 38.413:** NG-RAN; NG Application Protocol (NGAP)
- **3GPP TS 38.423:** NG-RAN; Xn Application Protocol (XnAP)



In This Module

1	Why Network Slicing?	424
2	Network Slicing Concept	424
3	Slice Identification	426
4	Slice Selection (NSSF Role)	428
5	Slice Architecture	429
6	Slice Isolation	431
7	Slice Lifecycle Management	432
8	Slice Use Cases	434
9	Slicing vs. QoS vs. DNN	435
10	RAN Slicing	438
	Summary	439
	Key Takeaways	439

1 Why Network Slicing?

Traditional mobile networks deploy a single, monolithic infrastructure designed for a “one-size-fits-all” service model. However, 5G must simultaneously serve radically different requirements:

Use Case	Requirement	Challenge
Enhanced Mobile Broadband (eMBB)	High throughput (Gbps), moderate latency	Bandwidth-hungry
Ultra-Reliable Low Latency (URLLC)	Sub-1ms latency, 99.999% reliability	Deterministic guarantees
Massive IoT (MIoT)	Millions of devices/km ² , ultra-low power	Scale and efficiency
Vehicle-to-Everything (V2X)	Low latency + high reliability + mobility	Safety-critical

Network slicing solves this by allowing **one physical network** to serve all these diverse requirements simultaneously through multiple logical networks, each optimized for its specific purpose.

Key drivers:

- **Economic efficiency:** Share expensive infrastructure (spectrum, towers, fiber) across services
- **Service differentiation:** Offer tailored SLAs to different verticals (automotive, healthcare, industry)
- **Operational agility:** Deploy new services without building new physical networks
- **Multi-tenancy:** Enable MVNOs and enterprises to operate isolated virtual networks

2 Network Slicing Concept

A **network slice** is a complete, logical end-to-end network built on top of shared physical infrastructure.

2.1 End-to-End Scope

Each slice spans the entire network:

- **RAN Slice:** Radio resource allocation, scheduling policies, beamforming configuration
- **Transport Slice:** Bandwidth guarantees, routing paths, latency bounds in backhaul/midhaul
- **Core Slice:** Dedicated or shared Network Functions (SMF, UPF, PCF, etc.)

2.2 Isolation Between Slices

Three dimensions of isolation:

1. **Resource Isolation:** Dedicated spectrum, compute, storage, and bandwidth per slice
2. **Security Isolation:** Separate security contexts, key hierarchies, access controls
3. **Failure Isolation:** A fault in one slice does not propagate to another

2.3 Customization Per Slice

Each slice can be independently customized:

- **Network Functions:** Choose which NFs are deployed (e.g., URLLC may skip certain NFs)
- **Topology:** Define the placement of NFs (e.g., edge UPF for URLLC)
- **QoS Parameters:** Configure slice-specific 5QI values, AMBR, priority levels
- **Policies:** Independent charging, access control, and mobility management rules

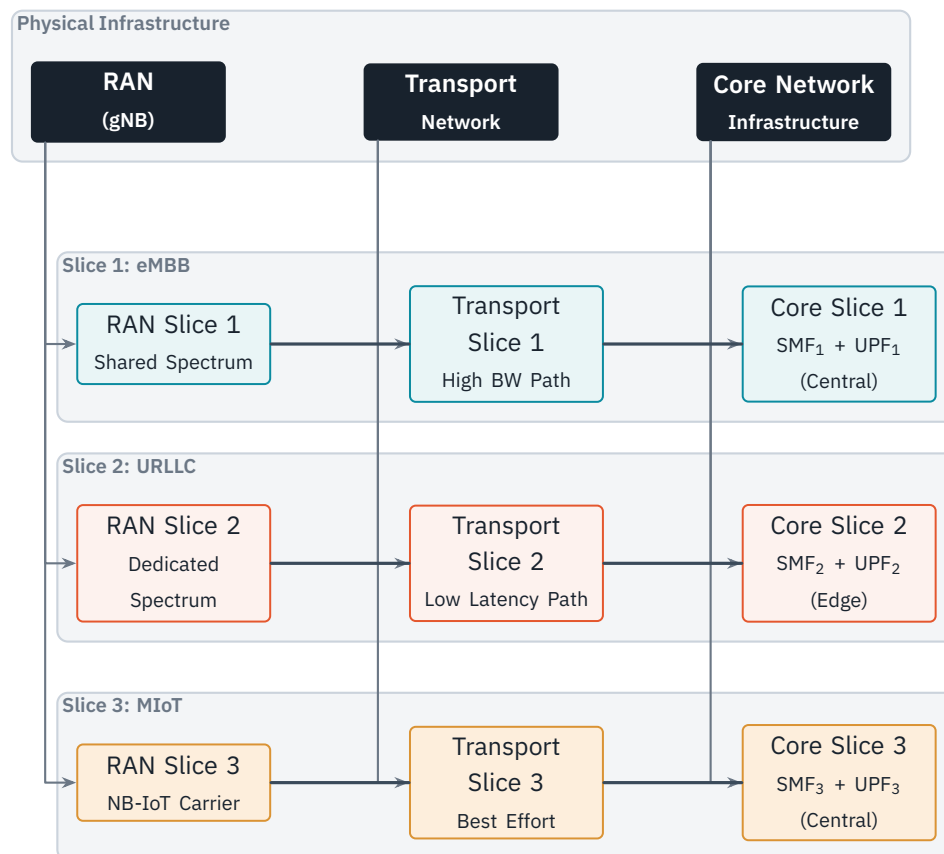


Figure 21.1 End-to-end slices across RAN, transport and core

3 Slice Identification

3.1 S-NSSAI (Single Network Slice Selection Assistance Information)

The fundamental identifier for a network slice consists of two components:

$$\text{S-NSSAI} = \text{SST} + \text{SD}$$

Field	Size	Description
SST (Slice/Service Type)	8 bits	Indicates the expected behavior of the slice (service characteristics)
SD (Slice Differentiator)	24 bits (optional)	Differentiates among slices of the same SST (e.g., different tenants)

3.2 Standardized SST Values

SST Value	Slice Type	Description
1	eMBB	Enhanced Mobile Broadband
2	URLLC	Ultra-Reliable Low Latency Communications
3	MIoT	Massive IoT
4	V2X	Vehicle-to-Everything
5–127	Reserved	For future standardized use
128–255	Operator-specific	Custom slice types

3.3 NSSAI (Network Slice Selection Assistance Information)

NSSAI = a collection (set) of S-NSSAIs. Multiple NSSAI types exist in the system:

NSSAI Type	Where Stored	Purpose
Configured NSSAI	UE (from HPLMN)	All S-NSSAIs the UE is subscribed to
Requested NSSAI	Sent by UE in Registration	S-NSSAIs the UE wants to use now
Allowed NSSAI	Returned by network	S-NSSAIs the UE is permitted to use in current PLMN
Rejected NSSAI	Returned by network	S-NSSAIs denied (with rejection cause)
Pending NSSAI	AMF internal	S-NSSAIs requiring additional authorization

3.4 Example

UE Subscription:

```
Configured NSSAI = { S-NSSAI(SST=1), S-NSSAI(SST=1,SD=0x000001),  
                    S-NSSAI(SST=2), S-NSSAI(SST=3) }
```

Registration Request:

```
Requested NSSAI = { S-NSSAI(SST=1), S-NSSAI(SST=2) }
```

Network Response:

```
Allowed NSSAI = { S-NSSAI(SST=1), S-NSSAI(SST=2) }
```

```
Rejected NSSAI = {}
```

4 Slice Selection (NSSF Role)

The **Network Slice Selection Function (NSSF)** is the dedicated NF responsible for determining the appropriate network slice instance(s) for a UE.

4.1 Slice Selection Procedure

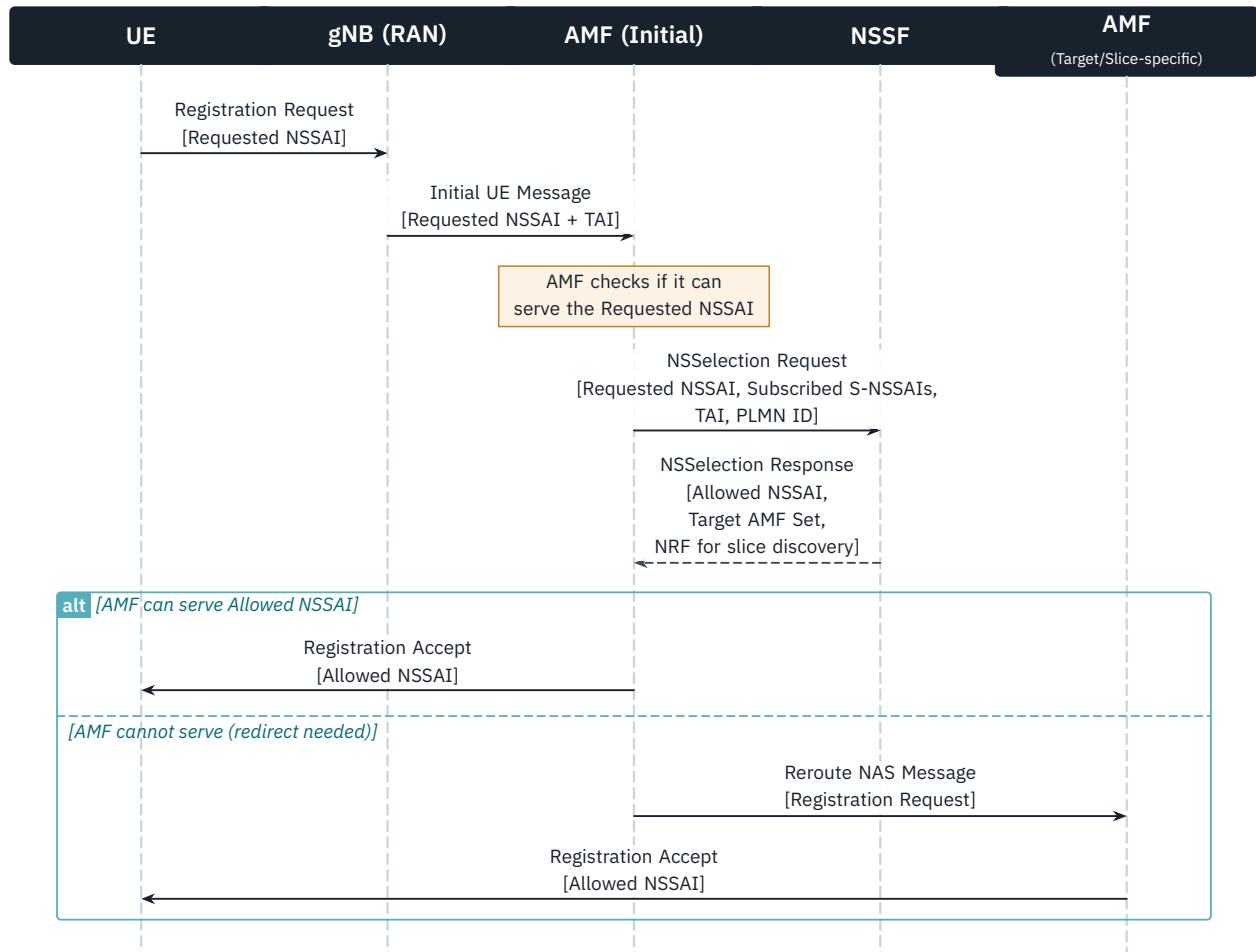


Figure 21.2 Slice selection with the NSSF and AMF re-allocation

4.2 NSSF Functions

Function	Description
Slice selection	Determine Allowed NSSAI based on subscription, PLMN policies, and availability
AMF set determination	Identify the set of AMFs capable of serving the selected slices
NRF redirection	Provide NRF endpoint for slice-specific NF discovery
Roaming support	Map between HPLMN and VPLMN S-NSSAIs

4.3 Slice Selection Inputs

The NSSF considers:

- **Requested NSSAI** from UE
- **Subscribed S-NSSAIs** from UDM (subscription data)
- **PLMN policies** (which slices are available in this area)
- **Tracking Area** (slice availability may vary by location)
- **Roaming agreements** (mapping between home and visited PLMN slices)

5 Slice Architecture

5.1 Dedicated vs. Shared NFs

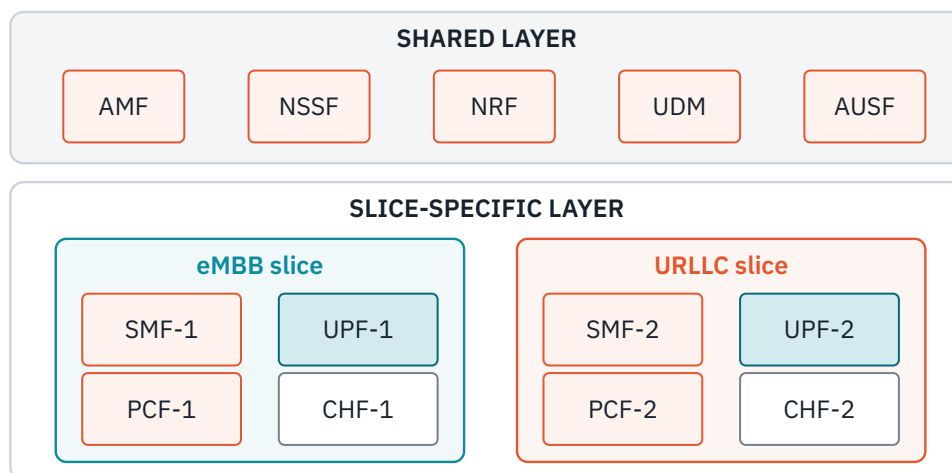


Figure 21.3 Shared network functions and slice-specific network functions

5.2 NF Deployment Model

NF	Typically Shared/Dedicated	Rationale
AMF	Shared (or per-slice set)	Handles initial access for all UEs; may have slice-specific AMF sets
NSSF	Shared	Central slice selection logic
NRF	Shared (with slice-aware discovery)	NF registry, supports slice-based queries
UDM/ AUSF	Shared	Subscription/authentication is per-UE, not per-slice
SMF	Dedicated per slice	Session management varies significantly between slice types
UPF	Dedicated per slice	Data plane customization (edge vs. central, throughput vs. latency)
PCF	Dedicated per slice	Policies differ drastically between eMBB and URLLC
CHF	Dedicated or shared	Charging models may differ per slice

5.3 Slice-Specific Customization

Aspect	eMBB Slice	URLLC Slice	MIoT Slice
UPF Placement	Central DC	Edge (MEC)	Central DC
QoS Priority	Medium (5QI=9)	Highest (5QI=80-82)	Low (5QI=79)
Redundancy	Standard	Dual connectivity, redundant paths	Minimal
Mobility	Full handover support	Conditional handover, DAPS	Rarely mobile
Charging	Volume-based	Latency SLA-based	Per-device flat rate

6 Slice Isolation

Slice isolation ensures that slices behave as independent networks despite sharing physical infrastructure.

6.1 Resource Isolation

Resource	Isolation Mechanism
Spectrum	Dedicated carriers, BWP (Bandwidth Parts), scheduling isolation
Compute	Dedicated VMs/containers, CPU pinning, NUMA allocation
Storage	Separate volumes, IOPS guarantees
Transport	VLAN/VxLAN segmentation, FlexE, dedicated wavelengths
Core NFs	Separate instances (pods/VMs) per slice

6.2 Security Isolation

- **Separate security domains:** Each slice can have independent security policies
- **Key hierarchy isolation:** Potential for separate key derivation per slice
- **Access control:** Slice-specific authentication/authorization (e.g., enterprise slice with additional EAP)
- **Traffic encryption:** Separate encryption contexts between slices
- **Audit trails:** Independent logging and compliance per slice

6.3 Failure Isolation

Failure Type	Isolation Approach
NF crash	Slice-dedicated NFs don't affect other slices
Overload	Resource quotas prevent one slice from starving others
Security breach	Compromised slice cannot access another slice's data/control
Configuration error	Slice-scoped config limits blast radius
Hardware failure	Slice-aware redundancy and recovery

6.4 Isolation Levels

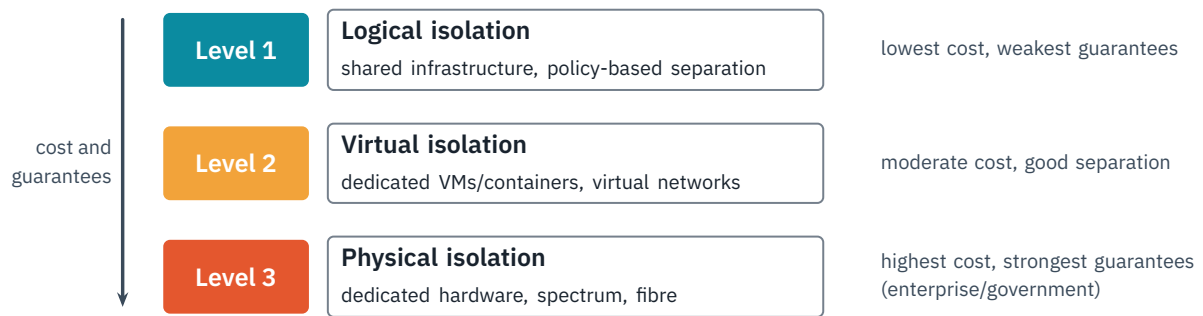


Figure 21.4 Three levels of slice isolation

7 Slice Lifecycle Management

7.1 Lifecycle Phases

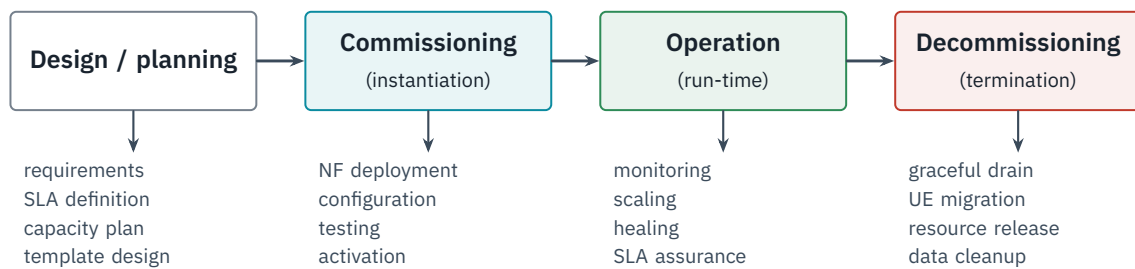
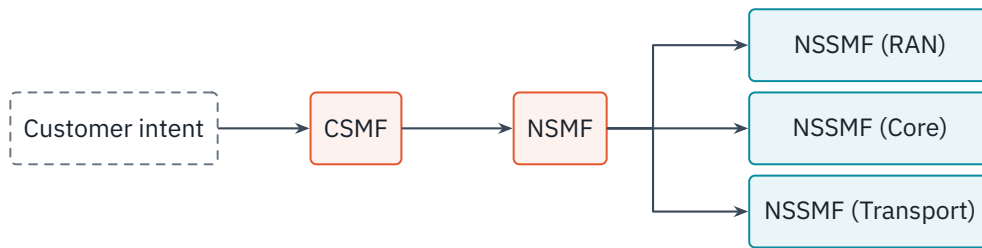


Figure 21.5 Network slice life cycle

7.2 Management Functions

Function	Role
CSMF (Communication Service Mgmt Function)	Translates customer requirements into slice requirements
NSMF (Network Slice Mgmt Function)	Manages the overall slice lifecycle
NSSMF (Network Slice Subnet Mgmt Function)	Manages individual slice subnets (RAN, Core, Transport)

Hierarchy:**Figure 21.6** Slice-management hierarchy**7.3 Intent-Based Slice Management**

Instead of specifying low-level parameters, operators express **intents**:

```

# Example Slice Intent
slice_intent:
  name: "Enterprise-Manufacturing"
  service_type: URLLC
  requirements:
    latency: "< 5ms"
    reliability: "99.999%"
    throughput: "> 100 Mbps"
    isolation: "dedicated"
    coverage: "Factory Floor Building A"
  constraints:
    max_devices: 500
    mobility: "low"
    security: "enhanced"
  
```

The management system automatically translates this intent into:

- NF deployment configurations
- Resource allocation decisions
- RAN scheduling parameters
- Transport path computation

8 Slice Use Cases

8.1 eMBB Slice

Attribute	Configuration
Goal	Maximum throughput for mobile broadband
SST	1
RAN	Shared spectrum, wide bandwidth (100 MHz+), MIMO
UPF	Central data center, high-capacity
Typical Services	Video streaming, AR/VR, web browsing
SLA	DL: 100 Mbps guaranteed, latency < 20ms

8.2 URLLC Slice

Attribute	Configuration
Goal	Deterministic ultra-low latency and high reliability
SST	2
RAN	Dedicated spectrum, mini-slots, configured grant
UPF	Edge deployment (MEC), co-located with gNB
Typical Services	Industrial automation, remote surgery, autonomous driving
SLA	Latency < 1ms, reliability 99.999%

8.3 MIoT Slice

Attribute	Configuration
Goal	Support massive number of low-power devices
SST	3
RAN	NB-IoT/LTE-M carrier, relaxed scheduling
UPF	Central, optimized for small packets
Typical Services	Smart meters, sensors, environmental monitoring
SLA	Supports 1M+ devices/km ² , battery life > 10 years

8.4 Enterprise Slice (Private Network)

Attribute	Configuration
Goal	Fully isolated private network for enterprise
SST	Operator-specific (e.g., 128)
RAN	Dedicated small cells on enterprise premises
UPF	On-premises (local breakout)
Typical Services	Campus connectivity, industrial IoT, private LAN
SLA	Full SLA customization, dedicated resources
Isolation	Level 3 (physical isolation where possible)

9 Slicing vs. QoS vs. DNN

9.1 Conceptual Comparison

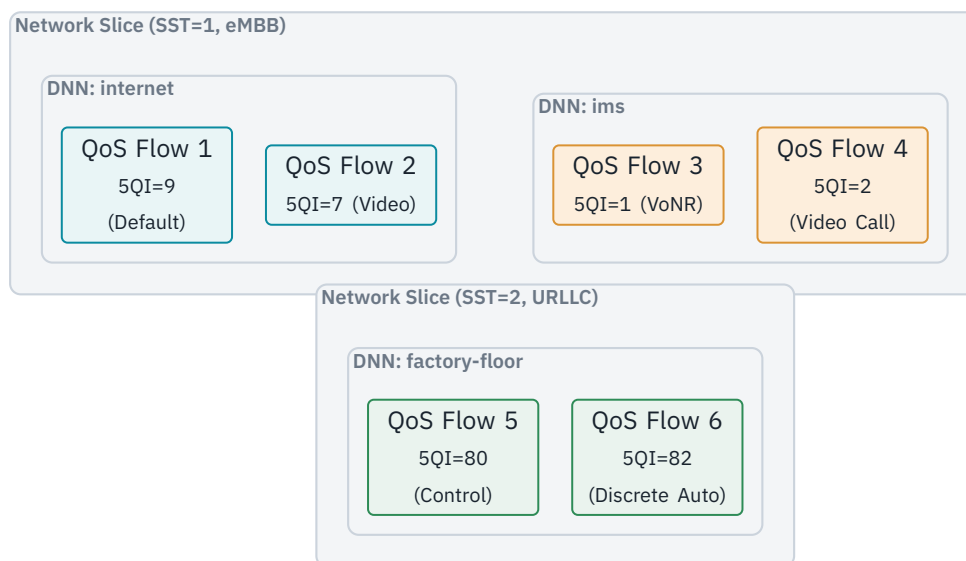


Figure 21.7 Slices contain DNNs, DNNs contain QoS flows

9.2 Detailed Comparison

Aspect	Network Slicing	QoS (5QI/QoS Flows)	DNN
Scope	End-to-end logical network	Per-flow packet treatment	Connectivity endpoint (PDU Session)
Granularity	Entire network (RAN + Core + Transport)	Individual data flows	Per PDU Session
Identifier	S-NSSAI (SST + SD)	5QI, QFI, ARP	DNN string (e.g., “internet”)
Isolation	Full (resource, security, failure)	No isolation (shared NFs)	No isolation
Customization	NF topology, placement, policies	Scheduling priority, rate, latency	Routing to specific data network
Analogy	Separate highway systems	Lane priorities on a highway	Highway exit (destination)
Who decides	Network (NSSF) at registration	PCF/SMF per session/flow	UE requests in PDU Session Establishment
Number	Few per PLMN (max 8 per UE)	Many per slice (up to 64 per session)	Multiple per slice
Relationship	Contains DNNs and QoS	Exists within a slice	Exists within a slice

9.3 Hierarchical Relationship

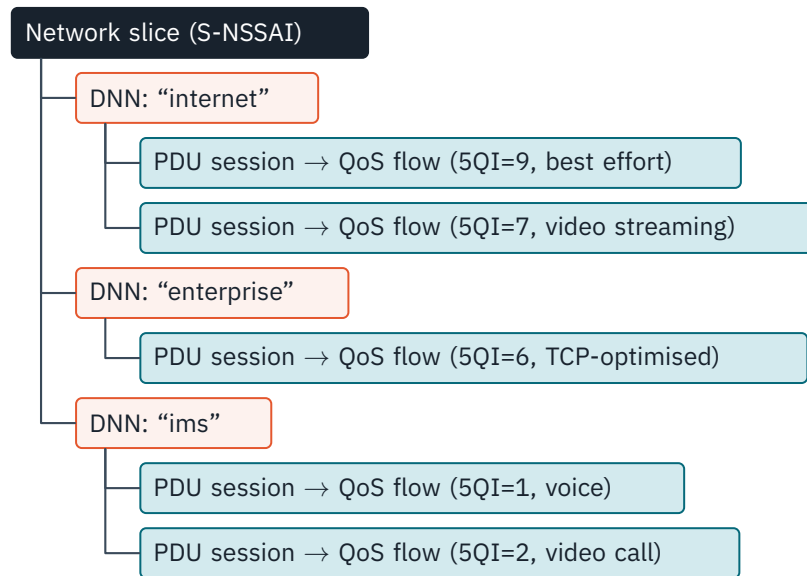


Figure 21.8 A slice contains DNNs, a DNN contains PDU sessions and QoS flows

9.4 When to Use What

Requirement	Solution
Different vertical industries on same network	Network Slicing
Prioritize voice over web traffic	QoS (different 5QI values)
Connect to different external data networks	DNN
Isolate enterprise traffic from consumer	Network Slicing
Differentiate video streaming from file download	QoS
Route to MEC application vs. central cloud	DNN (+ UPF selection)

10 RAN Slicing

10.1 Radio Resource Partitioning

RAN slicing divides radio resources among slices using different strategies:

Strategy	Description	Trade-off
Hard Partitioning	Dedicated PRBs/carriers per slice	Strong isolation, lower multiplexing gain
Soft Partitioning	Minimum guaranteed + shared pool	Good isolation + statistical multiplexing
Priority-based	Shared resources with priority scheduling	Maximum efficiency, weakest isolation
BWP-based	Assign Bandwidth Parts per slice	Clean separation at physical layer

10.2 Slice-Aware Scheduling

The gNB MAC scheduler must support multi-slice operation:

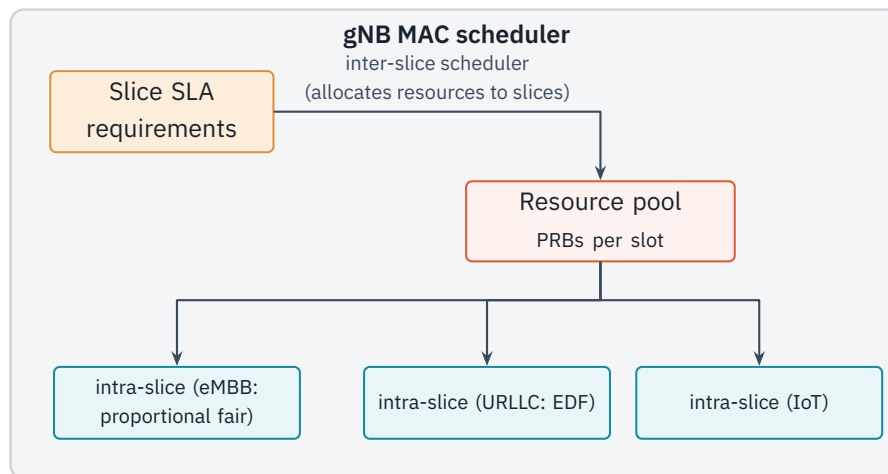


Figure 21.9 Two-level slice-aware scheduling in the gNB

Two-level scheduling:

1. **Inter-slice scheduler:** Allocates resources (PRBs, slots) to each slice based on SLA
2. **Intra-slice scheduler:** Schedules individual UEs within a slice using slice-appropriate algorithms

10.3 RAN Slice SLA Enforcement

SLA Parameter	Enforcement Mechanism
Minimum throughput	Reserved PRBs (guaranteed minimum)
Maximum latency	Priority scheduling, configured grants, mini-slots
Reliability target	HARQ configuration, repetition, diversity
Maximum UE count	Admission control per slice
Isolation level	BWP separation or hard partitioning

10.4 RAN Slicing Signaling

- **S-NSSAI in RRC:** UE provides S-NSSAI during RRC setup, gNB maps to appropriate slice
- **Slice-aware DRB mapping:** QoS flows from same slice map to common DRBs
- **RAN slice descriptors:** O-RAN defines slice-level radio resource management policies
- **NSSI (Network Slice Subnet Instance):** RAN maintains per-slice subnet instances

Summary

Concept	Key Point
Network Slice	Logical end-to-end network (RAN + Transport + Core) on shared infrastructure
S-NSSAI	SST (8-bit type) + SD (24-bit differentiator) identifies a slice
NSSF	Selects appropriate slice for UE during registration
Isolation	Resource + Security + Failure isolation between slices
Lifecycle	Design → Commission → Operate → Decommission (managed by NSMF/NSSMF)
RAN Slicing	Radio resource partitioning with two-level scheduling
Slice vs QoS vs DNN	Slice (whole network) $\supset$ DNN (connectivity) $\supset$ QoS (per-flow treatment)

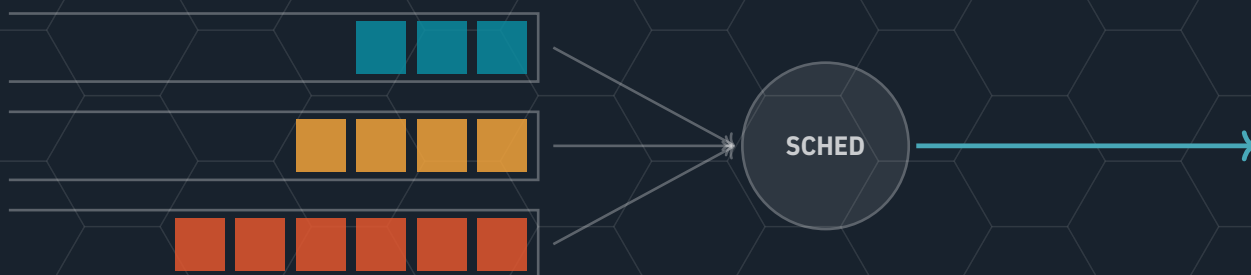
Key Takeaways

1. **Network slicing is the defining feature of 5G architecture** — it transforms a single physical network into multiple purpose-built logical networks
2. **S-NSSAI = SST + SD** is the fundamental slice identifier; UE provides Requested NSSAI, network returns Allowed NSSAI
3. **NSSF is the brain of slice selection** — it determines which slices a UE can access and which AMF should serve it
4. **Isolation is multi-dimensional** — resource, security, and failure isolation must all be addressed
5. **RAN slicing requires two-level scheduling** — inter-slice allocation followed by intra-slice per-UE scheduling
6. **Slicing $\neq$ QoS** — slicing provides end-to-end logical network separation; QoS provides per-flow treatment within a slice
7. **Lifecycle management** automates slice creation through decommissioning, driven by intent-based specifications

Quality of Service (QoS)

کیفیت خدمات (QoS)

22



In This Module

1	Why QoS?	442
2	QoS Evolution Across Generations	442
3	LTE Bearer-Based QoS	444
4	5G Flow-Based QoS	447
5	QoS Mapping and Enforcement	450
6	Bearer vs Flow Comparison	451
7	GBR vs Non-GBR vs Delay-Critical GBR	452
8	AMBR (Aggregate Maximum Bit Rate)	454
9	QoS Enforcement Architecture	455
	Summary: Key Takeaways	456
	QCI vs 5QI Quick Reference	456

1 Why QoS?

■ The Problem: Not All Traffic Is Equal

Different applications have fundamentally different network requirements:

Service	Delay Tolerance	Loss Tolerance	Bandwidth	Pattern
Voice (VoLTE)	Very Low (<150ms)	Low (<1%)	Low (64 kbps)	Constant
Video Call	Low (<300ms)	Low (<1%)	Medium (1-5 Mbps)	Variable
Video Streaming	Medium (<5s buffer)	Very Low	High (5-25 Mbps)	Bursty
IoT Sensor	High (seconds)	Very Low	Very Low (kbps)	Periodic
Web Browsing	Medium (<1s)	None (TCP retx)	Variable	Bursty
Email	High (seconds)	None (TCP retx)	Low	Bursty
Autonomous Driving	Ultra-Low (<10ms)	Ultra-Low	Medium	Constant

■ Without QoS:

- Voice calls get choppy during congestion
- Video freezes while background downloads continue
- Emergency services compete with social media
- Industrial control packets delayed by bulk transfers

■ QoS Goal:

Differentiate traffic and provide appropriate treatment to each service based on its requirements.

2 QoS Evolution Across Generations

2G	3G	LTE (4G)	5G NR
No QoS (all circuit-switched)	4 traffic classes	Bearer-based QoS (QCI + bearer)	Flow-based QoS (5QI + QoS flow)
Same quality for all	Conversational, streaming, interactive, background	9 standard QCIs; GBR / non-GBR; default + dedicated bearers	90 standard 5QIs; GBR / non-GBR / DC-GBR; reflective QoS; fine granularity

2G (GSM): No Real QoS

- All services are **circuit-switched**
- Each call gets a dedicated timeslot → same quality for everyone
- No differentiation needed (only voice + SMS)

3G (UMTS): Four Traffic Classes

Class	Delay	Example
Conversational	Strict (< 150ms)	Voice, video call
Streaming	Moderate (< 1s)	Video streaming
Interactive	Variable	Web browsing
Background	Best-effort	Email, downloads

LTE (4G): Bearer-Based QoS

- Introduced **EPS bearers** with specific QoS parameters
- **QCI** (QoS Class Identifier): standardized classes
- **ARP** (Allocation and Retention Priority)
- **GBR** (Guaranteed Bit Rate) vs **Non-GBR** bearers

5G NR: Flow-Based QoS

- **QoS Flows** identified by **QFI** (QoS Flow Identifier)
- **5QI** replaces QCI with more values
- **Reflective QoS** for simplified UL treatment
- Finer granularity: multiple flows per PDU session

3 LTE Bearer-Based QoS

■ Bearer Concept

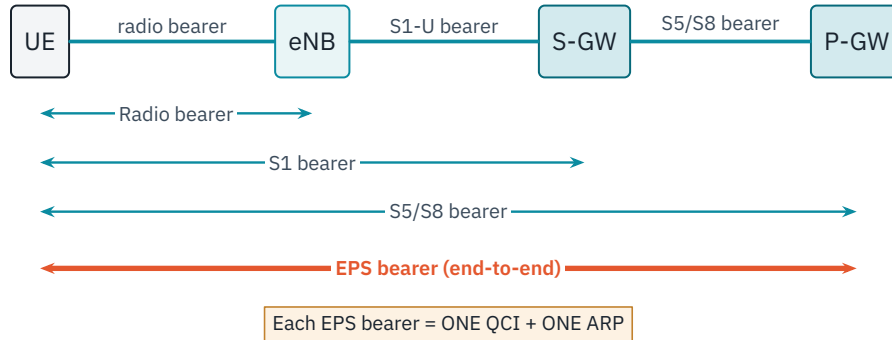


Figure 22.1 The EPS bearer is a chain of radio, S1 and S5/S8 bearers

Bearer = GTP-U tunnel with specific QoS parameters (QCI + ARP + bit rates)

■ Default vs Dedicated Bearers

Aspect	Default Bearer	Dedicated Bearer
Creation	At PDU connectivity (attach)	On-demand (triggered)
Lifetime	Always on (until detach)	Temporary (can be released)
QoS	Best-effort (QCI 9)	Specific QoS (any QCI)
Type	Always Non-GBR	GBR or Non-GBR
IP Address	Assigned with bearer	Shares IP of default
Purpose	Internet access	VoLTE, video, gaming

■ QCI Table (Standardized Values)

QCI	Resource Type	Priority	Packet Delay Budget	Packet Error Loss Rate	Example Services
1	GBR	2	100 ms	10^{-2}	Conversational Voice
2	GBR	4	150 ms	10^{-3}	Conversational Video
3	GBR	3	50 ms	10^{-3}	Real-Time Gaming
4	GBR	5	300 ms	10^{-6}	Non-Conv. Video (Buffered)
5	Non-GBR	1	100 ms	10^{-6}	IMS Signaling
6	Non-GBR	6	300 ms	10^{-6}	Video (Buffered), TCP web
7	Non-GBR	7	100 ms	10^{-3}	Voice, Live Video, Gaming
8	Non-GBR	8	300 ms	10^{-6}	Video (Buffered), TCP web
9	Non-GBR	9	300 ms	10^{-6}	Video, TCP, Email, Chat

■ Mission-Critical (MCX) QCI Values (65-82)

QCI	Resource Type	Priority	Delay	Example
65	GBR	0.7	75 ms	MC-PTT Voice
66	GBR	2	100 ms	MC-Video (non-real-time)
67	GBR	1.5	100 ms	MC-PTT Video
69	Non-GBR	0.5	60 ms	MC-PTT Signaling
70	Non-GBR	5.5	200 ms	MC Data
75	GBR	2.5	50 ms	V2X Messages
79	Non-GBR	6.5	50 ms	V2X Messages
82	GBR	1.9	10 ms	Discrete Automation
83	GBR	2.2	10 ms	Discrete Automation

■ ARP (Allocation and Retention Priority)

ARP determines whether a bearer can be established during congestion:

Parameter	Values	Purpose
Priority Level	1-15 (1=highest)	Relative importance
Pre-emption Capability	Yes/No	Can this bearer preempt others?
Pre-emption Vulnerability	Yes/No	Can this bearer be preempted?

Example Scenario:

Congestion situation - resources are full:

Bearer A: Priority=5, Pre-emption Capability=Yes

Bearer B: Priority=10, Pre-emption Vulnerability=Yes

→ Bearer A can preempt (remove) Bearer B to get resources

→ Emergency calls (Priority=1) can preempt regular calls

■ GBR vs Non-GBR Bearers

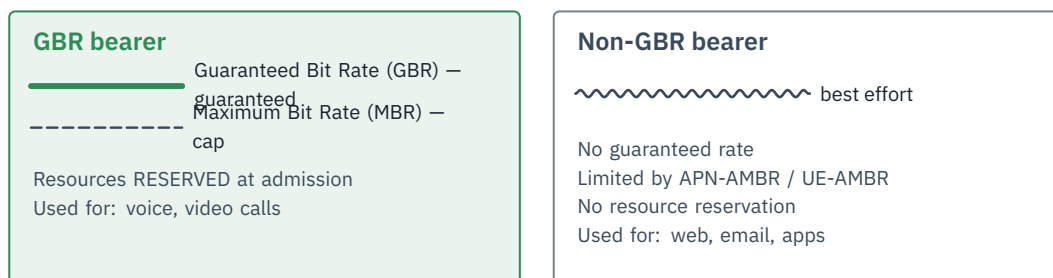


Figure 22.2 GBR and non-GBR bearers

■ LTE QoS Limitations

1. **Coarse granularity:** One QCI per bearer → one bearer per service type
2. **Bearer overhead:** Each new QCI requires a new GTP tunnel
3. **Signaling heavy:** Dedicated bearer setup requires full NAS/S1AP signaling
4. **Inflexible:** Cannot mix traffic with different QoS in one bearer
5. **Limited QCIs:** Only 9 standard values (extended to ~20 with MCX)

4 5G Flow-Based QoS

■ Key Concept: QoS Flow

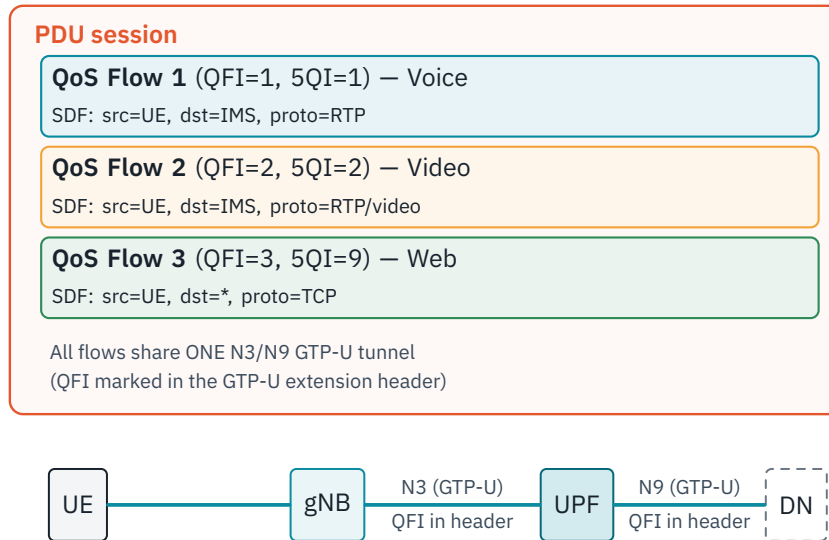


Figure 22.3 5G QoS flows inside one PDU session share one GTP-U tunnel

Key differences from LTE:

- Multiple QoS Flows share ONE PDU session (one GTP tunnel)
- QFI (6-bit) identifies the flow within the tunnel
- No need to create/delete bearers → less signaling
- Finer granularity with less overhead

5QI Table (Key Standardized Values)

5QI	Resource Type	Priority	Packet Delay	Packet Error Rate	Example Services
1	GBR	20	100 ms	10^{-2}	Conversational Voice
2	GBR	40	150 ms	10^{-3}	Conversational Video
3	GBR	30	50 ms	10^{-3}	Real-Time Gaming
4	GBR	50	300 ms	10^{-6}	Non-Conv. Video (Buffered)
5	Non-GBR	10	100 ms	10^{-6}	IMS Signaling
6	Non-GBR	60	300 ms	10^{-6}	Video, TCP (default)
7	Non-GBR	70	100 ms	10^{-3}	Voice, Video, Gaming
8	Non-GBR	80	300 ms	10^{-6}	Video, TCP web
9	Non-GBR	90	300 ms	10^{-6}	Video, TCP, Email (default)
65	GBR	7	75 ms	10^{-2}	MC-PTT Voice
66	GBR	20	100 ms	10^{-2}	Non-MC-PTT Voice
67	GBR	15	100 ms	10^{-3}	MC Video
69	Non-GBR	5	60 ms	10^{-6}	MC-PTT Signaling
70	Non-GBR	55	200 ms	10^{-6}	MC Data
79	Non-GBR	65	50 ms	10^{-2}	V2X Messages
80	Non-GBR	68	10 ms	10^{-6}	Low-Latency eMBB
82	DC-GBR	19	10 ms	10^{-4}	Discrete Automation (small)
83	DC-GBR	22	10 ms	10^{-4}	Discrete Automation (large)
84	DC-GBR	24	30 ms	10^{-5}	Intelligent Transport
85	DC-GBR	21	5 ms	10^{-5}	Electricity Distribution
86	GBR	18	5 ms	10^{-4}	V2X (platooning)
87	DC-GBR	25	5 ms	10^{-3}	Interactive Service (motion)
88	DC-GBR	23	10 ms	10^{-4}	Interactive Service (steering)
89	DC-GBR	26	15 ms	10^{-4}	Visual Content (cloud gaming)
90	DC-GBR	27	20 ms	10^{-4}	Visual Content (AR/ cloud XR)

DC-GBR = Delay-Critical GBR (new in 5G)

QoS Rules

QoS rules tell the UE how to classify uplink traffic:

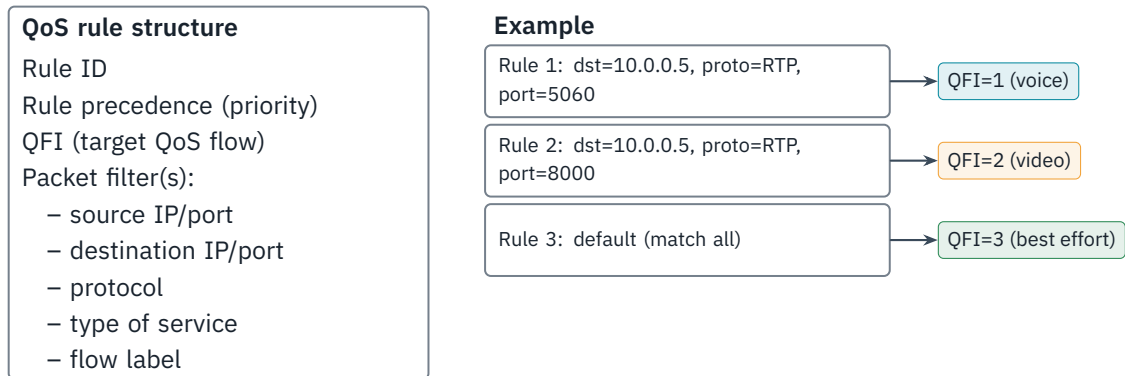


Figure 22.4 Structure of a QoS rule, with example rules

Reflective QoS

1. The network sends a DL packet with QFI = X and RQI = 1 (Reflective QoS Indication).
2. The UE observes the DL packet headers:
 - source IP/port → becomes the UL destination
 - destination IP/port → becomes the UL source
 - QFI = X → the UE uses the same QFI for its reply
3. The UE creates a “derived QoS rule” automatically:
 - no explicit QoS-rule signalling is needed
 - the rule is timer-based: valid for the RQ timer duration

Benefit: less signalling to establish QoS rules. **Use case:** the UE does not know the server IP in advance.

5 QoS Mapping and Enforcement

End-to-End QoS Mapping Diagram

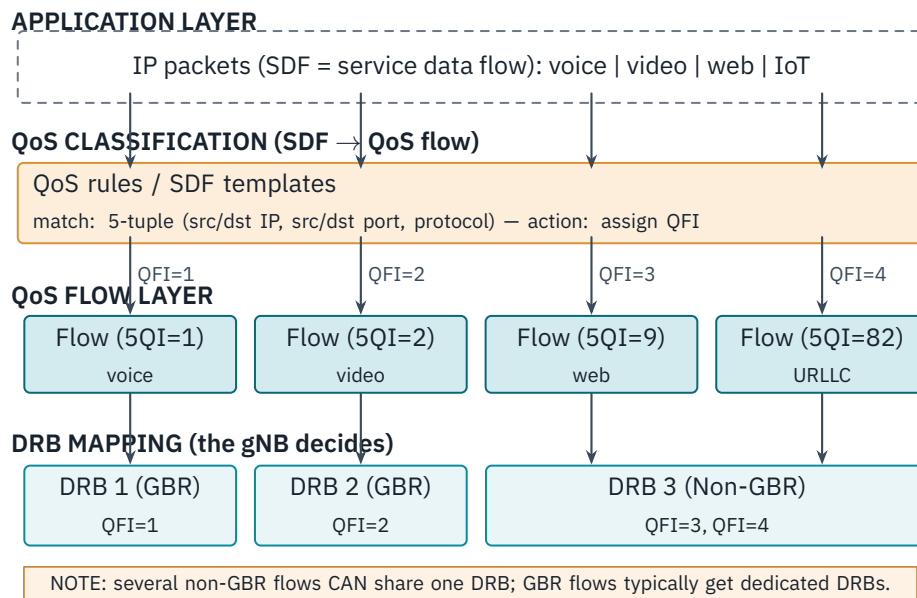


Figure 22.5 End-to-end 5G QoS mapping: SDF → QoS flow → DRB

Role of Each Network Element

Element	QoS Role	Actions
PCF	Policy Decision	Creates PCC rules, determines 5QI/ARP, sends to SMF
SMF	QoS Control	Derives QoS profiles, installs QoS rules, configures UPF
UPF	DL Enforcement	SDF detection, QFI marking, gating, policing, rate limiting
gNB	Radio Scheduling	Maps QoS flows→DRBs, schedules based on 5QI, enforces AMBR
UE	UL Classification	Applies QoS rules, maps UL packets to QoS flows

QoS Enforcement Architecture

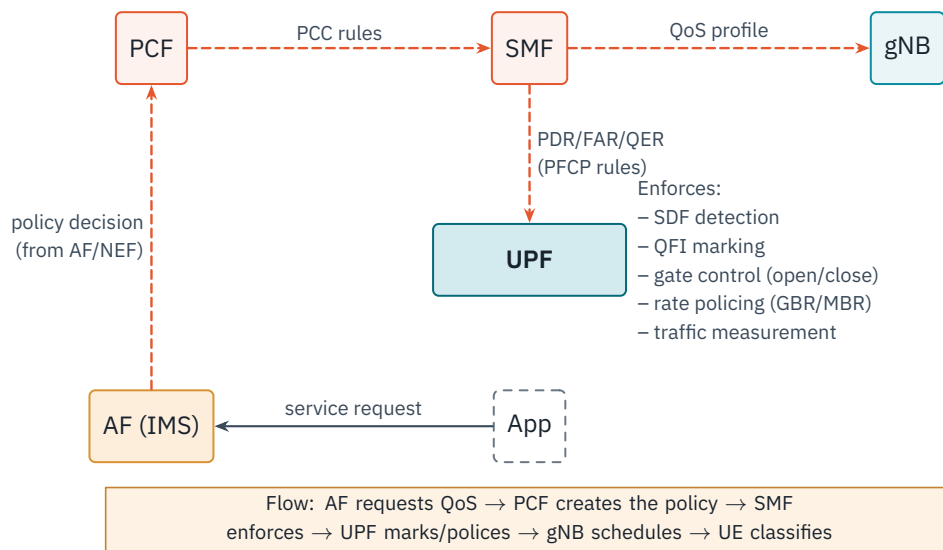


Figure 22.6 QoS policy and enforcement architecture

6 Bearer vs Flow Comparison

Side-by-Side Architecture

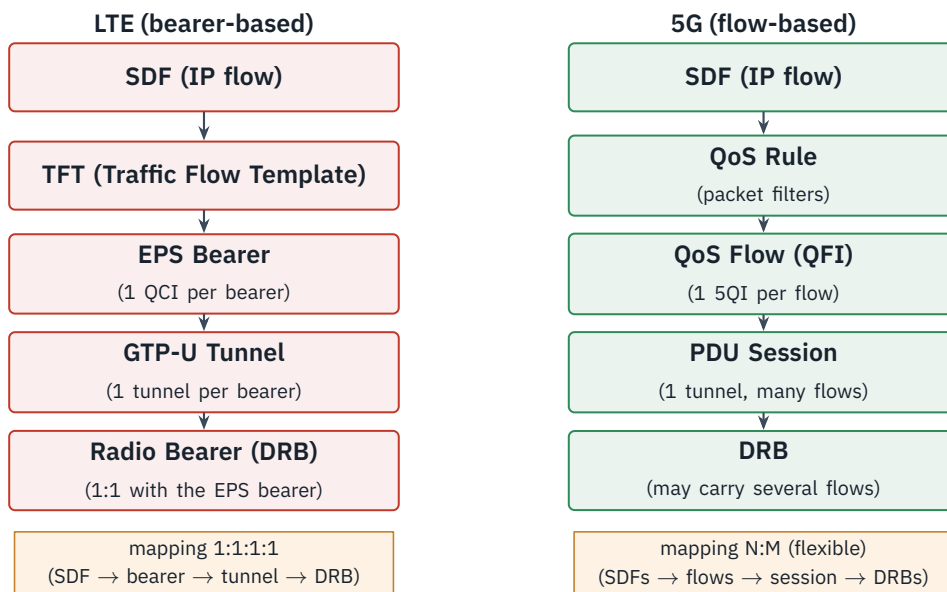


Figure 22.7 LTE bearer-based versus 5G flow-based QoS

Detailed Comparison Table

Aspect	LTE (Bearer-Based)	5G (Flow-Based)
QoS Unit	EPS Bearer	QoS Flow
Identifier	EBI (EPS Bearer ID)	QFI (QoS Flow ID, 6-bit)
QoS Class	QCI (1-9 + MCX)	5QI (1-90+)
Granularity	Coarse (1 QCI/bearer)	Fine (per-flow)
Tunnel	1 GTP-U per bearer	1 GTP-U per PDU session
Setup	NAS signaling per bearer	QoS rule install (lighter)
UL Classification	TFT (uplink packet filter)	QoS Rules
DL Classification	TFT at P-GW	SDF template at UPF
New QoS Class	New bearer (heavy signaling)	New flow (lightweight)
Sharing	No sharing between bearers	Flows share tunnel
Reflective QoS	Not supported	Supported (RQI)
Resource Types	GBR, Non-GBR	GBR, Non-GBR, DC-GBR

7 GBR vs Non-GBR vs Delay-Critical GBR

Three Resource Types in 5G

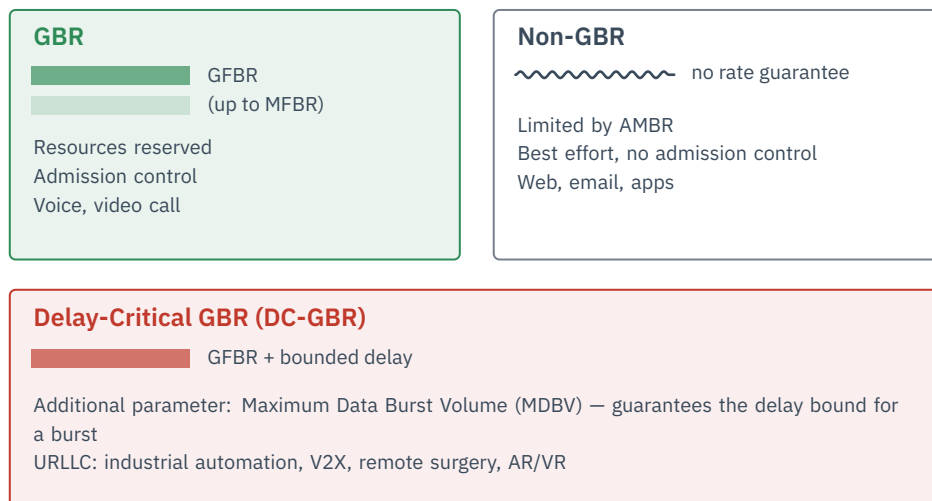


Figure 22.8 The three 5G QoS resource types

Parameters by Resource Type

Parameter	GBR	Non-GBR	DC-GBR
5QI	✓	✓	✓
ARP	✓	✓	✓
GFBP (Guaranteed Flow Bit Rate)	✓	✗	✓
MFBR (Maximum Flow Bit Rate)	✓	✗	✓
Session-AMBR	✗	✓	✗
Maximum Data Burst Volume	✗	✗	✓
Packet Delay Budget	✓	✓	✓ (strict)
Admission Control	Required	Not needed	Required
Resource Reservation	Yes	No	Yes

Rate Parameters Explained

- **GFBP = Guaranteed Flow Bit Rate**
 - the network **MUST** provide at least this rate
 - under congestion, GBR flows are protected up to GFBP
- **MFBR = Maximum Flow Bit Rate**
 - the network **MAY** provide up to this rate
 - traffic above MFBR is dropped or shaped

Example	GFBP	MFBR
VoLTE	40 kbps (AMR-NB codec rate)	80 kbps (with overhead margin)
Video call	1 Mbps (minimum acceptable quality)	4 Mbps (HD quality when resources allow)

8 AMBR (Aggregate Maximum Bit Rate)

■ AMBR Hierarchy

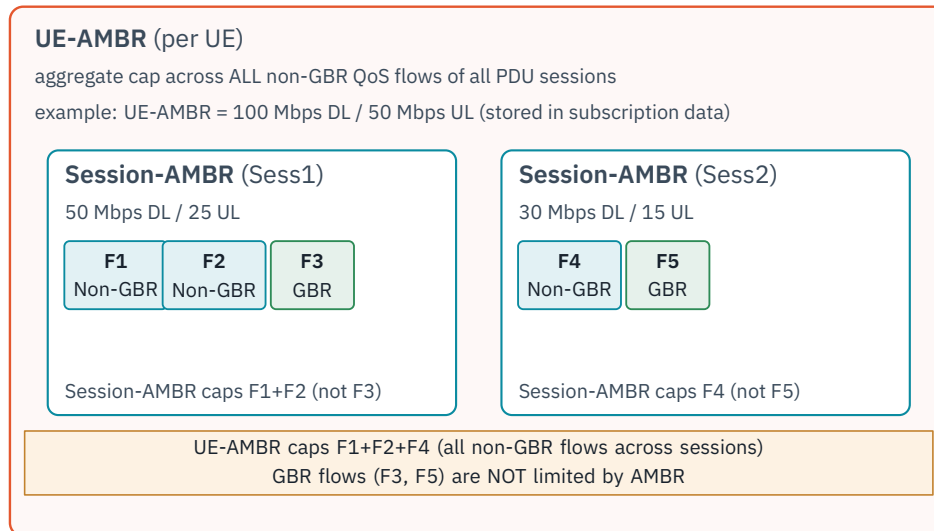


Figure 22.9 AMBR hierarchy: UE-AMBR above Session-AMBR; GBR flows are exempt

■ AMBR Enforcement Points

AMBR Type	Scope	Enforced At	Applies To
UE-AMBR	All sessions	gNB (RAN)	All Non-GBR flows, all sessions
Session-AMBR	Per PDU session	UPF	Non-GBR flows in that session

Key Rules:

- AMBR applies ONLY to Non-GBR flows
- GBR flows are limited by their own MFBR, not AMBR
- $\text{UE-AMBR} \geq \text{sum of Session-AMBRs (logically)}$
- Enforced separately for UL and DL

9 QoS Enforcement Architecture

Complete End-to-End QoS Signaling

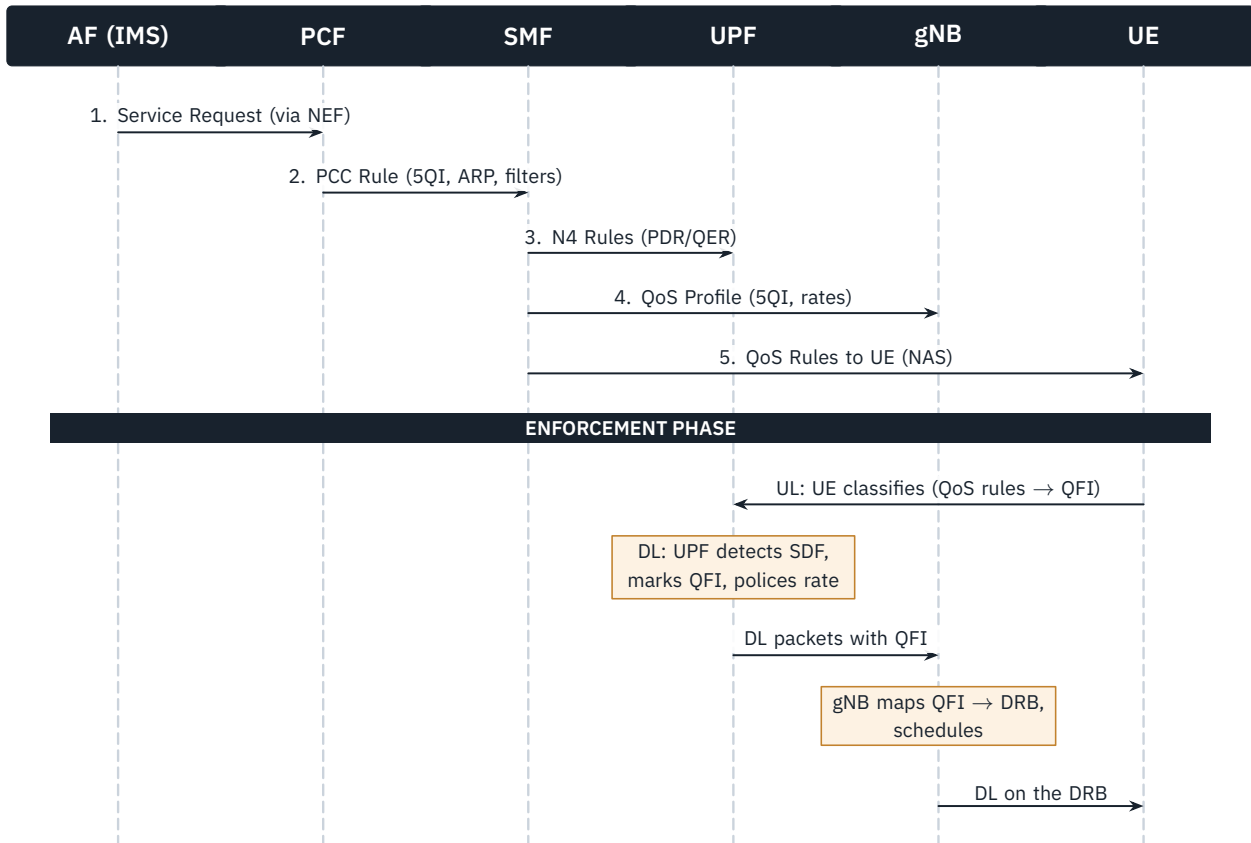


Figure 22.10 Complete end-to-end QoS signalling and enforcement

Summary of Enforcement Actions

Node	Uplink Actions	Downlink Actions
UE	Classify packets using QoS rules, mark QFI	Receive and process
gNB	Map QoS flows→DRBs, enforce UE-AMBR	Schedule based on 5QI priority, map DRB→flows
UPF	Verify QFI, enforce Session-AMBR	Detect SDF, mark QFI, gate/police/shape
SMF	Configure UPF (N4), signal gNB (N2), signal UE (N1)	Same
PCF	Generate PCC rules from policy	Same

Summary: Key Takeaways

Concept	Key Point
QoS Purpose	Different services need different network treatment
QCI (LTE)	9 standard classes, one per bearer
5QI (5G)	90+ classes, per QoS flow
Bearer (LTE)	1:1 mapping with QCI, separate GTP tunnel
QoS Flow (5G)	Many flows share one PDU session, lighter signaling
GBR	Guaranteed rate, admission control, resource reserved
Non-GBR	Best effort, limited by AMBR
DC-GBR	GBR + bounded delay (MDBV), for URLLC
AMBR	Aggregate cap on Non-GBR (UE-level and session-level)
Reflective QoS	UE derives UL QoS from DL (reduces signaling)
PCF	Policy brain, creates PCC rules
UPF	DL classifier and enforcer
gNB	Radio scheduler, DRB mapping, UE-AMBR

QCI vs 5QI Quick Reference

Feature	QCI (LTE)	5QI (5G)
Values	1-9 standard + 65-82 MCX	1-90+ standardized
Resource Types	GBR, Non-GBR	GBR, Non-GBR, DC-GBR
Priority Scale	1-9 (1=highest)	1-127 (1=highest)
Associated With	EPS Bearer	QoS Flow
Delay Range	50-300 ms	5-300 ms
New Category	—	Delay-Critical GBR
URLLC Support	Limited	Native (5ms delay, 10^{-5} loss)

Module 22 — Quality of Service (QoS) — Advanced Communication Networks

GTP-U · N3 · N9 · N6

In This Module

1	User Plane Concept	458
2	GTP-U Protocol (GPRS Tunneling Protocol — User Plane)	458
3	User Plane in LTE (4G)	460
4	User Plane in 5G	461
5	UPF Functions (3GPP TS 23.501)	463
6	Packet Trace Example: Downlink Data	464
7	User Plane Optimization	465
	Complete 5G User Plane Path Diagram	467
	Summary Table: User Plane Across Generations	468
	Key Takeaways	468

1 User Plane Concept

The **User Plane** (also called the **Data Plane** or **U-Plane**) is responsible for carrying actual user data through the mobile network. This includes:

- **IP packets** (web browsing, app data, file transfers)
- **Voice data** (VoLTE/VoNR encoded speech frames)
- **Video streams** (real-time video, streaming content)
- **Any application-layer payload**

The user plane is distinct from the control plane — while the control plane sets up, manages, and tears down connections, the user plane is the “pipe” through which data actually flows once those connections are established.

Key characteristics:

- High throughput requirement
- Low latency sensitivity
- Stateless forwarding (once tunnels are established)
- QoS-aware packet handling
- Transparent to signaling — carries data without interpreting it

2 GTP-U Protocol (GPRS Tunneling Protocol – User Plane)

GTP-U (3GPP TS 29.281) is the core tunneling protocol used to transport user data between network nodes across all generations (3G, 4G, 5G).

2.1 Header Structure

The GTP-U header has an **8-byte mandatory portion** plus optional extension headers:

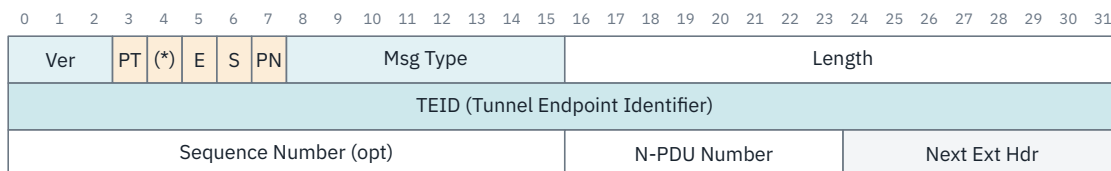


Figure 23.1 GTP-U header

Field	Bits	Description
Version	3	Always 001 (GTPv1)
PT (Protocol Type)	1	1 = GTP, 0 = GTP' (charging)
Reserved (*)	1	Must be 0
E (Extension Header)	1	Extension header follows
S (Sequence Number)	1	Sequence number field present
PN (N-PDU Number)	1	N-PDU number field present
Message Type	8	0xFF = T-PDU (user data), others = signaling
Length	16	Payload length (excludes mandatory 8 bytes)
TEID	32	Tunnel Endpoint Identifier

2.2 TEID (Tunnel Endpoint Identifier)

The **TEID** is a 32-bit value that uniquely identifies a GTP tunnel endpoint at a receiving node:

- Each direction of a tunnel has its own TEID
- The **receiving node** assigns the TEID (tells the sender “use this TEID when sending to me”)
- TEID = 0 is reserved for control plane initial messages
- A node may have thousands of TEIDs active simultaneously (one per bearer/session per direction)

Example:

```
gNB tells UPF: "For downlink data to UE-X, use TEID = 0x00A1B2C3 on N3"
UPF tells gNB: "For uplink data from UE-X, use TEID = 0x00D4E5F6 on N3"
```

2.3 Encapsulation / Decapsulation

At each GTP-U hop, the user's IP packet is **encapsulated** inside:

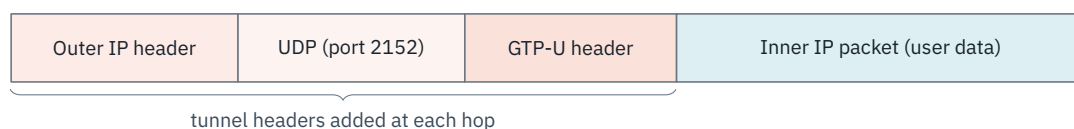


Figure 23.2 GTP-U encapsulation at each hop

- Encapsulation** (sending node): Prepends GTP-U header + UDP + outer IP
- Decapsulation** (receiving node): Strips outer IP + UDP + GTP-U header, forwards inner IP
- At intermediate nodes: may decapsulate and re-encapsulate with new TEID

2.4 Sequence Numbers and Extension Headers

Sequence Numbers (16-bit):

- Optional but used for reordering at tunnel endpoints
- Critical during handover (ensures no packet loss/reorder)
- Incremented per PDU per tunnel

Extension Headers:

- Chained via “Next Extension Header Type” field
- Each extension: [Length (1 byte)][Content (variable)][Next Ext Hdr Type (1 byte)]

Common extension headers:

Type	Name	Purpose
0x85	PDU Session Container	5G QoS Flow Identifier (QFI), reflective QoS
0xC0	PDCCP PDU Number	For lossless handover
0x40	UDP Port	Additional multiplexing

3 User Plane in LTE (4G)

3.1 End-to-End Path

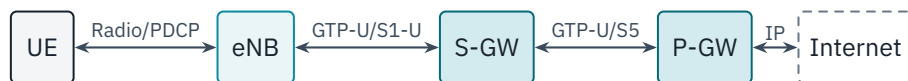


Figure 23.3 End-to-end LTE user-plane path

3.2 Packet Headers at Each Segment

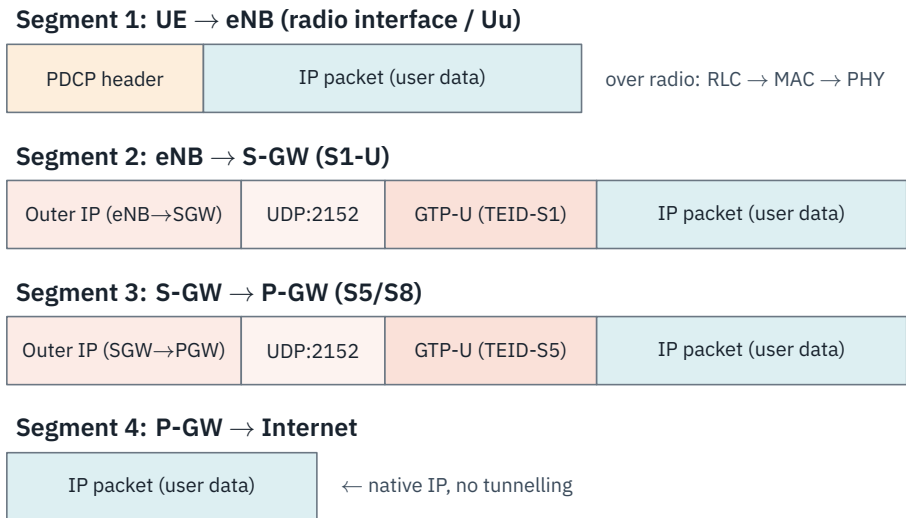


Figure 23.4 Packet headers on each LTE user-plane segment

3.3 LTE Bearer Model

- **Default Bearer:** Always-on, established at attach (one per PDN connection)
- **Dedicated Bearer:** Additional QoS, established on demand (e.g., VoLTE)
- Each bearer = unique set of TEIDs on S1-U and S5

4 User Plane in 5G

4.1 End-to-End Path

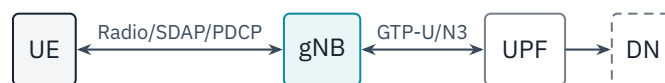


Figure 23.5 End-to-end 5G user-plane path

4.2 5G Reference Points for User Plane

Interface	Between	Transport	Purpose
N3	gNB ↔ UPF	GTP-U over UDP/IP	RAN to Core user data
N9	UPF ↔ UPF	GTP-U over UDP/IP	Inter-UPF forwarding
N6	UPF ↔ DN	Native IP	Exit to data network

4.3 SDAP Layer (Service Data Adaptation Protocol)

The **SDAP** layer is new in 5G NR (not present in LTE) and sits above PDCP:

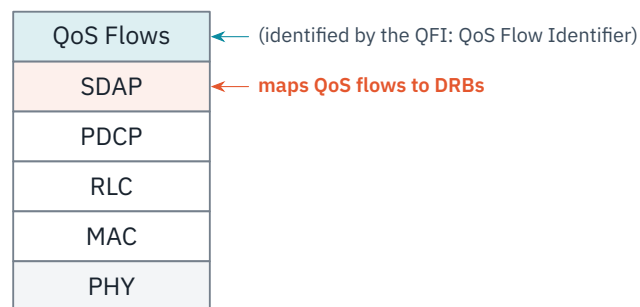


Figure 23.6 NR user-plane stack with the new SDAP layer

SDAP Functions:

- **Downlink:** QoS Flow → DRB mapping (based on QFI in GTP-U extension header)
- **Uplink:** DRB → QoS Flow marking (stamps QFI on uplink packets)
- **Reflective QoS:** UE mirrors the DL mapping for UL without explicit signaling

4.4 Multiple UPFs Architecture

5G supports **UPF chaining** for flexible traffic routing:

UPF Type	Role	Use Case
I-UPF (Intermediate)	Transit/branching point	Handover anchor, traffic steering
PSA-UPF (PDU Session Anchor)	Session anchor, IP allocation	Stable IP address point

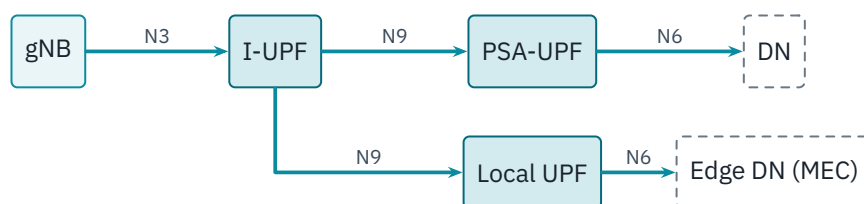


Figure 23.7 UPF chaining with a local breakout to the edge

5 UPF Functions (3GPP TS 23.501)

The UPF is controlled by the SMF via the **N4 interface** using the **PCFP protocol** (Packet Forwarding Control Protocol, TS 29.244).

5.1 PCFP Rules

Rule	Acronym	Function
Packet Detection Rule	PDR	Match incoming packets (by TEID, IP, port, QFI, etc.)
Forwarding Action Rule	FAR	What to do (forward, drop, buffer, duplicate)
QoS Enforcement Rule	QER	Rate limiting (MBR/GBR), gating, marking
Usage Reporting Rule	URR	Volume/time measurement, triggers for reporting
Buffering Action Rule	BAR	Buffer DL packets when UE is idle, notify SMF

5.2 How Rules Work Together

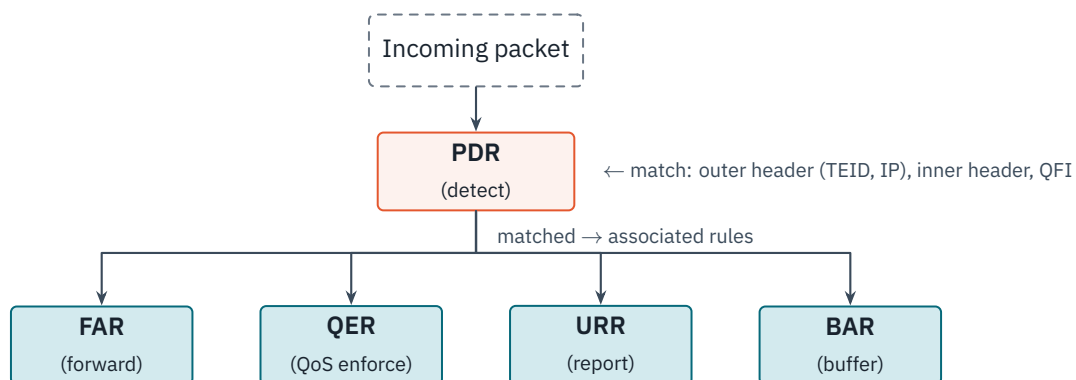


Figure 23.8 How the PCFP rules work together in the UPF

5.3 N4/PCFP Interface

- **SMF → UPF:** Session Establishment/Modification/Deletion Request
- **UPF → SMF:** Session Report (usage reports, DL data notification)
- PCFP uses **SEID** (Session Endpoint Identifier) — analogous to TEID but for control
- Runs over UDP port 8805

5.4 Complete UPF Function List

1. Packet routing & forwarding
2. Packet inspection (DPI for application detection)
3. UL/DL rate enforcement (per QoS flow)
4. Traffic usage reporting (for charging)
5. DL packet buffering & notification
6. QoS handling (DSCP marking, policing)
7. Lawful intercept (UP collection)
8. Transport level packet marking (outer header DSCP)

6 Packet Trace Example: Downlink Data

Scenario: Server on Internet sends data to UE

6.1 Step-by-Step

Step 1: Internet → UPF (N6)

IP header
Src: 93.184.216.34 (server)
Dst: 10.0.0.1 (UE IP)
TCP/UDP header
Application data (payload)

Step 2: UPF → gNB (N3, GTP-U)

Outer IP header
Src: 192.168.100.1 (UPF N3 IP)
Dst: 192.168.200.1 (gNB N3 IP)
UDP header (Src: ephemeral, Dst: 2152)
GTP-U header
TEID: 0x00A1B2C3 (assigned by the gNB)
Ext Hdr: PDU Session Container
QFI: 9 (QoS Flow ID)
Inner IP header
Src: 93.184.216.34 (server)
Dst: 10.0.0.1 (UE IP)
TCP/UDP + application data

Step 3: gNB → UE (radio)

SDAP header (QFI: 9)
PDCP header (SN, integrity info)
IP header
Src: 93.184.216.34
Dst: 10.0.0.1
TCP/UDP + application data

RLC segmentation → MAC mux
→ PHY (radio transmission)

Step 4: at the UE

IP header
Src: 93.184.216.34
Dst: 10.0.0.1
TCP/UDP + application data

delivered to the application layer

Figure 23.9 A downlink packet at each step from the Internet to the UE

6.2 Encapsulation Layer Diagram

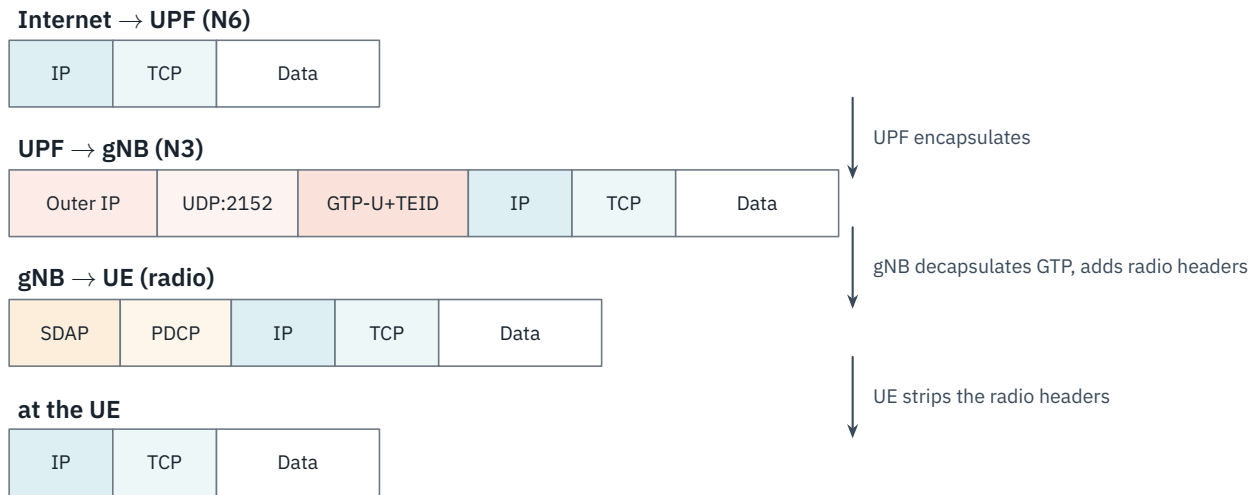


Figure 23.10 Headers of a downlink packet at each hop

7 User Plane Optimization

7.1 Dual Connectivity (DC) – Split Bearer

In **EN-DC** (E-UTRA–NR Dual Connectivity) or **NR-DC**:

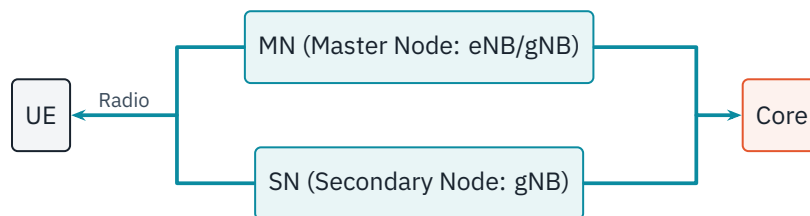


Figure 23.11 Dual connectivity with a split bearer

Split Bearer types:

Type	Description	GTP-U path
MCG Bearer	Data only via Master Node	UPF ↔ MN
SCG Bearer	Data only via Secondary Node	UPF ↔ SN
Split Bearer	Data split across both nodes	UPF ↔ MN ↔ SN (MN routes)

- Increases throughput by aggregating radio resources from two nodes
- PDCP layer at MN handles reordering for split bearers

7.2 Multi-Path UPF

Multiple UPFs in the data path for:

- **Load balancing:** Distribute traffic across UPFs
- **Redundancy:** N9 links between UPFs for resiliency
- **Geographic optimization:** Route traffic through nearest UPF

7.3 Edge Computing (Local Breakout)

5G enables **traffic steering** for MEC (Multi-access Edge Computing):

UL Classifier (UL CL):

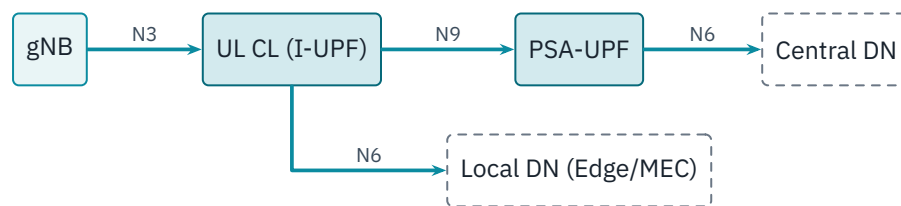


Figure 23.12 Uplink classifier for edge breakout

- UL CL inspects UL packets and routes matching traffic to local DN
- Non-matching traffic continues to central DN
- **Same UE IP address** for both paths (transparent to UE)

Branching Point:

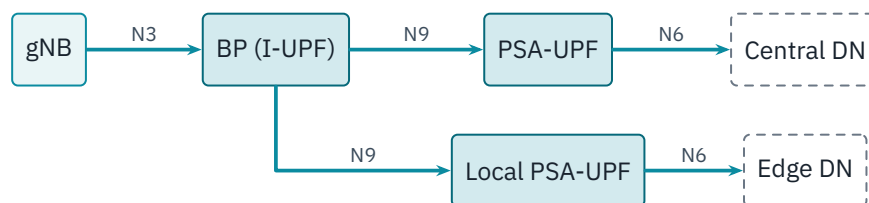


Figure 23.13 Branching point for IPv6 multi-homing

- Different PDU session anchors for local vs. remote
- May use different UE IP addresses

Benefits of Edge Computing:

- Ultra-low latency (< 10 ms RTT)
- Reduced backhaul bandwidth
- Local data processing (privacy, compliance)
- Use cases: AR/VR, autonomous driving, industrial IoT

Complete 5G User Plane Path Diagram

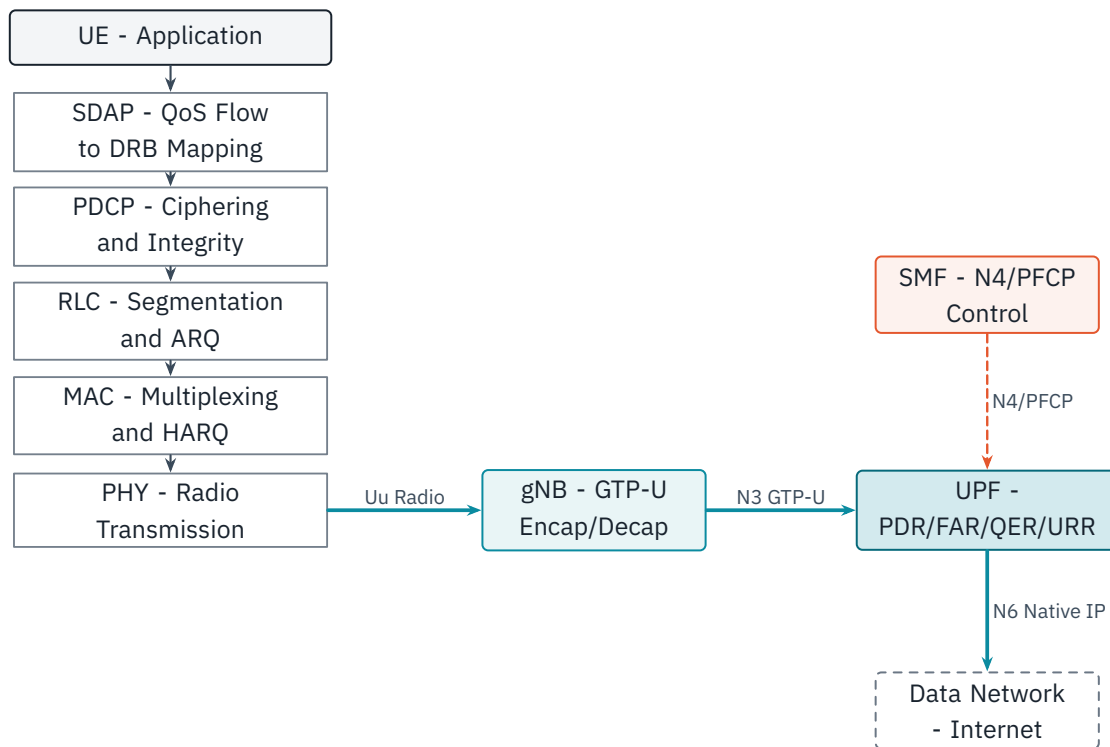


Figure 23.14 Complete 5G user-plane path

Summary Table: User Plane Across Generations

Aspect	3G (UMTS)	4G (LTE)	5G (NR)
Tunnel Protocol	GTP-U	GTP-U	GTP-U
RAN→Core Interface	Iu-PS	S1-U	N3
Core Internal	Gn (SGSN↔GGSN)	S5/S8 (SGW↔PGW)	N9 (UPF↔UPF)
Core→DN	Gi	SGi	N6
QoS Mechanism	Traffic classes	Bearers (QCI)	QoS Flows (QFI)
Radio QoS Layer	—	—	SDAP
Gateway Function	GGSN	P-GW	UPF
Control of UP	—	—	PCF (N4)
Edge Support	No	Limited	Native (UL CL, BP)

Key Takeaways

1. **GTP-U** is the universal user plane tunnel — same protocol from 3G to 5G
2. **TEID** identifies each tunnel endpoint — assigned by the receiver
3. **5G separates control and user plane** (CUPS) — SMF controls UPF via PCF/N4
4. **SDAP** is the new 5G layer that maps QoS flows to radio bearers
5. **UPF rules (PDR/FAR/QER/URR/BAR)** provide programmable packet processing
6. **Edge computing** is natively supported via UL Classifier and Branching Point
7. User plane optimization (DC, multi-UPF, edge) enables diverse 5G use cases



In This Module

1	Control Plane Concept	470
2	Control Plane Protocols — MAJOR Reference Table	470
3	NAS (Non-Access Stratum)	472
4	RRC (Radio Resource Control)	474
5	NGAP (NG Application Protocol) — TS 38.413	476
6	GTP-C (GTP Control Plane)	477
7	HTTP/2 SBI (Service-Based Interface)	478
8	Signaling Protocol Evolution: SS7 → Diameter → HTTP/2	480
	Complete Control Plane Architecture Diagram (5G)	482
	Summary: Control Plane vs. User Plane	483
	Key Takeaways	483

1 Control Plane Concept

The **Control Plane** (C-Plane) handles all **signaling** in the mobile network — the messages that set up, manage, modify, and tear down connections. It does NOT carry user data.

Control Plane responsibilities:

- **Authentication & Security:** Verify subscriber identity, establish encryption
- **Registration & Mobility:** Attach, detach, tracking area updates, handovers
- **Session Management:** Establish/modify/release PDU sessions or bearers
- **QoS Negotiation:** Agree on quality parameters for data flows
- **Paging:** Locate idle UEs for incoming data/calls
- **Policy & Charging:** Apply operator rules to sessions

Key principle: The control plane “programs” the user plane — it sets up the tunnels, rules, and routing that the user plane then uses to forward data.

2 Control Plane Protocols — MAJOR Reference Table

Protocol	Generation	Layer	Plane	Between	Purpose
MAP (Mobile Application Part)	2G/3G	Application	Control	MSC↔HLR, VLR↔HLR	Subscriber data, auth, SMS routing, HO
ISUP (ISDN User Part)	2G/3G	Application	Control	MSC↔MSC, MSC↔PSTN	Circuit-switched call setup/release
RANAP (RAN Application Protocol)	3G	Application	Control	RNC↔CN (MSC/SGSN)	RAN-Core signaling (Iu interface)
S1AP (S1 Application Protocol)	4G	Application	Control	eNB↔MME	RAN-Core signaling (S1-MME)
NGAP (NG Application Protocol)	5G	Application	Control	gNB↔AMF	RAN-Core signaling (N2)
GTP-C v1	3G	Tunnel Control	Control	SGSN↔GGSN	PDP context create/modify/delete
GTP-C v2	4G/5G	Tunnel Control	Control	MME↔S-GW↔P-GW	Session create/modify/delete bearer
Diameter	4G	AAA/Policy	Control	MME↔HSS, PCRF↔P-GW	Auth (S6a), policy (Gx), charging (Gy)

Protocol	Generation	Layer	Plane	Between	Purpose
HTTP/2 SBI	5G	Service-Based	Control	NF↔NF (all 5GC NFs)	Service discovery, registration, session mgmt
NAS EMM	4G	Non-Access Stratum	Control	UE↔MME	Attach, TAU, authentication, security
NAS ESM	4G	Non-Access Stratum	Control	UE↔MME (→P-GW)	Bearer setup/modify/release
NAS 5GMM	5G	Non-Access Stratum	Control	UE↔AMF	Registration, authentication, service request
NAS 5GSM	5G	Non-Access Stratum	Control	UE↔SMF (via AMF)	PDU session establish/modify/release
RRC (LTE)	4G	Access Stratum	Control	UE↔eNB	Connection setup, meas config, HO, SIBs
RRC (NR)	5G	Access Stratum	Control	UE↔gNB	Same + BWP config, beam mgmt, CHO
SCTP	4G/5G	Transport	Control	eNB↔MME, gNB↔AMF	Reliable signaling transport (multi-stream)
SIP (Session Initiation Protocol)	4G/5G (IMS)	Application	Control	UE↔P-CSCF↔S-CSCF	VoLTE/VoNR call setup, IMS registration
PCF	5G (4G CUPS)	Control	Control	SMF↔UPF	User plane rule programming (N4)

Protocol Stack Relationships

APPLICATION PROTOCOLS		
MAP RANAP S1AP NGAP GTP-C Diameter HTTP/2 SBI		
TRANSPORT PROTOCOLS		
SCTP	TCP	UDP
IP		
L2 / L1 (Ethernet, etc.)		

Figure 24.1 How the signalling protocols stack on transport protocols

3 NAS (Non-Access Stratum)

NAS protocols operate **between the UE and the core network** — they pass transparently through the RAN (the eNB/gNB does not interpret NAS messages, only relays them).

3.1 LTE NAS (TS 24.301)

LTE NAS has two sub-layers:

EMM (EPS Mobility Management):

Procedure	Messages	Purpose
Attach	Attach Request/Accept/Complete	Register UE to network
Detach	Detach Request/Accept	Deregister UE
TAU	TAU Request/Accept	Update Tracking Area
Authentication	Auth Request/Response	Mutual authentication (AKA)
Security Mode	SMC Command/Complete	Activate NAS ciphering/integrity
Service Request	Service Request/Accept	Transition IDLE→CONNECTED
Identity	Identity Request/Response	Request IMSI if no GUTI

ESM (EPS Session Management):

Procedure	Messages	Purpose
PDN Connectivity	PDN Conn Request/Activate Default Bearer	Establish PDN connection
Bearer Resource Alloc	Bearer Resource Alloc Req	UE-initiated dedicated bearer
Activate Dedicated Bearer	Act Ded Bearer Req/Accept	Network-initiated QoS bearer
Modify Bearer	Modify Bearer Req/Accept	Change bearer QoS
Deactivate Bearer	Deact Bearer Req/Accept	Release bearer

3.2 5G NAS (TS 24.501)

5G NAS also has two sub-layers but with enhanced capabilities:

5GMM (5G Mobility Management):

Procedure	Messages	Purpose
Registration	Registration Req/Accept/Complete	Initial, mobility, periodic reg
Deregistration	Dereg Request/Accept	UE or network initiated
Authentication	Auth Request/Response/Failure	5G-AKA or EAP-AKA'
Security Mode	SMC Command/Complete	NAS security activation
Service Request	Service Request/Accept	Resume from IDLE/INACTIVE
Configuration Update	Config Update Command/Complete	GUTI reallocation, NSSAI
Notification	Notification/Response	Network triggers (SOR, UPU)

5GSM (5G Session Management):

Procedure	Messages	Purpose
PDU Session Establishment	PDU Session Est Req/Accept	Create PDU session
PDU Session Modification	PDU Session Mod Req/Accept	Modify QoS flows
PDU Session Release	PDU Session Rel Req/Accept	Tear down session

Key 5G NAS enhancements vs. 4G:

- Registration replaces Attach + TAU (unified procedure)
- Supports network slicing (requested NSSAI in registration)
- PDU session types: IPv4, IPv6, IPv4v6, Ethernet, Unstructured
- NAS messages routed via AMF to SMF for session management

3.3 NAS Transparency Through RAN

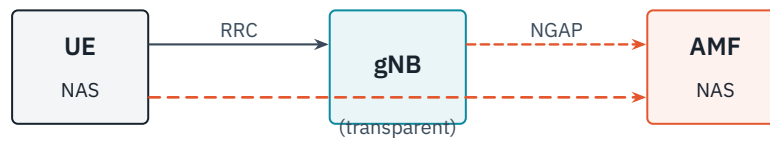


Figure 24.2 NAS passes transparently through the gNB

- NAS messages are carried inside RRC messages on the radio interface
- RRC carries them inside NGAP (5G) or S1AP (4G) to the core
- The RAN node does NOT decrypt or interpret NAS content
- Separate NAS security context (NAS ciphering/integrity) from AS security (RRC/UP)

4 RRC (Radio Resource Control)

RRC operates **between the UE and the base station** (Access Stratum) and controls the radio connection.

4.1 LTE RRC (TS 36.331)

Key Procedures:

Procedure	Messages	Purpose
Connection Setup	RRC Conn Setup Req/Setup/Complete	Establish SRB1
Connection Reconfiguration	RRC Conn Reconfig	Add/modify bearers, handover
Connection Release	RRC Conn Release	Release radio resources
Measurement Configuration	(in Reconfig)	Configure meas events (A1-A6, B1-B2)
Handover	RRC Conn Reconfig (mobilityCtrlInfo)	Execute handover
Security Mode	Security Mode Command/Complete	Activate AS ciphering/integrity
System Information	MIB, SIB1-SIB19	Broadcast cell parameters

LTE RRC States:



Figure 24.3 LTE RRC states

4.2 NR RRC (TS 38.331)

NR RRC includes all LTE RRC functions **plus**:

Feature	Description
BWP Configuration	Configure Bandwidth Parts (active BWP switching)
Beam Management	SSB beam indication, beam failure recovery, L1-RSRP reporting
Conditional Handover (CHO)	Pre-configure HO conditions — execute when met (no HO command delay)
Cell Group Config	MCG + SCG for dual connectivity
MeasConfig enhancements	NR events (similar + inter-RAT NR↔LTE), L3 filtering per beam
SIB enhancements	On-demand SIBs, SIB for RedCap, NTN, sidelink

4.3 RRC States in 5G (Three States)

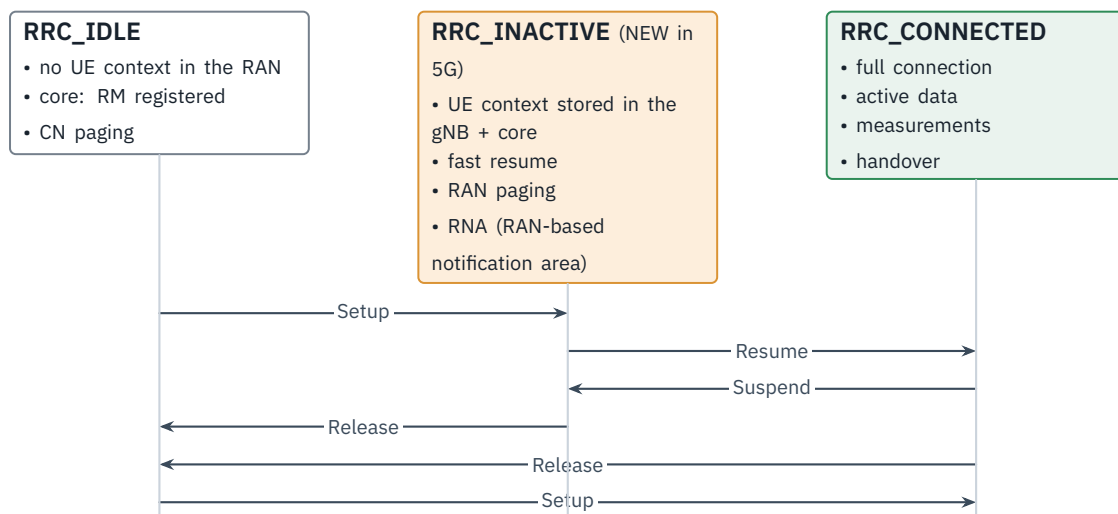


Figure 24.4 The three NR RRC states and their transitions

RRC_INACTIVE benefits:

- UE context preserved in Last Serving gNB → fast resume (~10 ms vs ~50 ms)
- No full connection setup needed — just RRC Resume
- RNA (RAN-based Notification Area) for mobility without core notification
- Ideal for IoT devices with infrequent small data

5 NGAP (NG Application Protocol) – TS 38.413

NGAP is the **5G equivalent of S1AP** (LTE) – it runs between the **gNB and AMF** over the **N2 interface**.

5.1 Transport

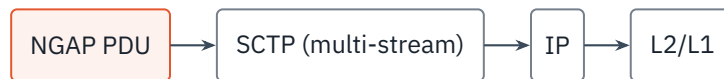


Figure 24.5 NGAP transport stack

- SCTP provides reliable, in-order delivery with multiple streams
- Stream 0: non-UE-associated signaling (setup, reset)
- Other streams: UE-associated signaling (one stream per UE-group)

5.2 Key Procedures

Category	Procedure	Description
Interface Mgmt	NG Setup	gNB registers with AMF (capabilities, served cells)
	AMF Configuration Update	AMF informs gNB of changes
	NG Reset	Error recovery
UE Context	Initial Context Setup	AMF→gNB: create UE context, security, PDU sessions
	UE Context Release	Release UE resources at gNB
	UE Context Modification	Modify existing context
Mobility	Handover Required	gNB→AMF: request inter-gNB HO
	Handover Request	AMF→target gNB: prepare HO
	Handover Notify	Target gNB confirms HO complete
	Path Switch Request	Intra-AMF, X2-based HO notification
NAS Transport	Initial UE Message	First NAS message from UE (gNB→AMF)
	Downlink NAS Transport	AMF→gNB→UE: carry NAS message
	Uplink NAS Transport	UE→gNB→AMF: carry NAS message
Paging	Paging	AMF→gNB: page UE in TA cells
PDU Session	PDU Session Resource Setup	Establish user plane resources
	PDU Session Resource Modify	Modify QoS/resources
	PDU Session Resource Release	Release UP resources

5.3 NGAP vs. S1AP Comparison

Feature	S1AP (4G)	NGAP (5G)
Interface	S1-MME	N2
Nodes	eNB ↔ MME	gNB ↔ AMF
Transport	SCTP	SCTP
Session concept	E-RAB (bearer)	PDU Session + QoS Flows
Slice support	No	Yes (S-NSSAI in procedures)
UE states	IDLE, CONNECTED	IDLE, INACTIVE, CONNECTED
Multi-connectivity	No	Yes (secondary RAT)

6 GTP-C (GTP Control Plane)

6.1 GTP-C v1 (3G – TS 29.060)

Interface	Between	Purpose
Gn/Gp	SGSN ↔ GGSN	PDP Context management

Key messages:

- Create PDP Context Request/Response
- Update PDP Context Request/Response
- Delete PDP Context Request/Response

6.2 GTP-C v2 (4G – TS 29.274)

GTPv2-C is a complete redesign with improved efficiency:

Interface	Between	Purpose
S11	MME ↔ S-GW	Session/bearer management
S5/S8	S-GW ↔ P-GW	Session/bearer management
S10	MME ↔ MME	Inter-MME handover
S3	MME ↔ SGSN	3G↔4G mobility

Key GTPv2-C Messages:

Message	Direction	Purpose
Create Session Request/ Response	MME→S-GW→P-GW	Establish PDN connection
Modify Bearer Request/ Response	MME→S-GW	Update after HO (new eNB TEID)
Delete Session Request/ Response	MME→S-GW→P-GW	Release PDN connection
Create Bearer Request/Response	P-GW→S-GW→MME	Dedicated bearer setup
Delete Bearer Request/Response	P-GW→S-GW→MME	Dedicated bearer release
Release Access Bearers	MME→S-GW	UE goes IDLE (release S1-U)
Downlink Data Notification	S-GW→MME	DL data arrived for idle UE

GTPv2-C Header:

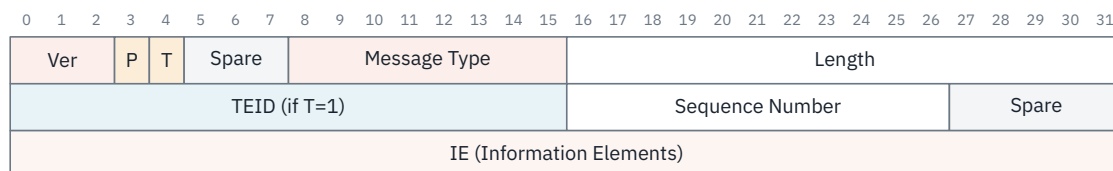


Figure 24.6 GTPv2-C header

6.3 5G: From GTP-C to PCF + HTTP/2

In 5G, the traditional GTP-C role is split:

- **SMF ↔ UPF**: Uses **PCF** (N4) — not GTP-C
- **Inter-NF signaling**: Uses **HTTP/2 SBI** — not GTP-C
- GTPv2-C remains for interworking with 4G (N26 interface: AMF↔MME)

7 HTTP/2 SBI (Service-Based Interface)

7.1 Concept

5G Core replaces point-to-point interfaces with a **Service-Based Architecture (SBA)**:

- Each NF exposes **services** via RESTful APIs
- Communication uses **HTTP/2** over **TLS 1.3** over **TCP**
- Data format: **JSON** (OpenAPI 3.0 specifications)

7.2 Key NF Services

NF	Service	Operations	Purpose
AMF	Namf_Communication	N1N2MessageTransfer, UEContextTransfer	NAS relay, context
SMF	Nsmf_PDUSession	Create, Update, Release	PDU session lifecycle
UDM	Nudm_SubscriberDataMgmt	Get, Subscribe, Unsubscribe	Subscriber data
AUSF	Nausf_UEAuthentication	Authenticate	5G-AKA, EAP-AKA'
PCF	Npcf_SMPolicyControl	Create, Update, Delete	QoS/charging policy
NRF	Nnrf_NFDiscovery	Discover	Find NF instances
NSSF	Nnssf_NSSelection	Get	Network slice selection
NEF	Nnef_EventExposure	Subscribe, Unsubscribe	Event reporting to AF

7.3 Communication Patterns

Pattern	Description	Example
Request-Response	Synchronous API call	SMF queries UDM for subscription
Subscribe-Notify	Asynchronous event	PCF subscribes to AMF for UE location
Service Discovery	Query NRF for NF endpoint	AMF discovers SMF via NRF

7.4 SBI Protocol Stack

NF Service (e.g., Nsmf)
HTTP/2 (multiplexed streams)
TLS 1.3 (mTLS between NFs)
TCP
IP

Figure 24.7 SBI protocol stack

8 Signaling Protocol Evolution: SS7 → Diameter → HTTP/2

8.1 Timeline

Era	Technology	Signaling Stack	Characteristics
1G/2G (1980s-90s)	GSM	SS7 (MAP, ISUP, SCCP, MTP)	Circuit-switched, TDM-based, binary encoding
3G (2000s)	UMTS	SS7 + RANAP + GTP-C v1	IP transport added (SIGTRAN), still SS7 app layer
4G (2010s)	LTE	Diameter + S1AP + GTP-C v2	All-IP, AVP-based encoding, point-to-point
5G (2020s)	NR	HTTP/2 SBI + NGAP + PCF	Cloud-native, RESTful, service-mesh, JSON

8.2 Key Evolutionary Drivers

SS7 (1980s)	Diameter (2010s)	HTTP/2 SBI (2020s)
TDM transport	IP transport	IP transport
Binary encoding	AVP binary encoding	JSON text encoding
Fixed nodes	Fixed nodes	Microservices (NFs)
Point-to-point	Point-to-point	Service mesh
Centralised (STP)	Diameter agents (DRA)	Service discovery (NRF)
Rigid interfaces	Defined interfaces	API-driven, extensible
Hard to scale	Moderately scalable	Cloud-native, auto-scale
Telecom-only	Telecom-specific	IT/cloud standard

8.3 Signaling Evolution Timeline Diagram

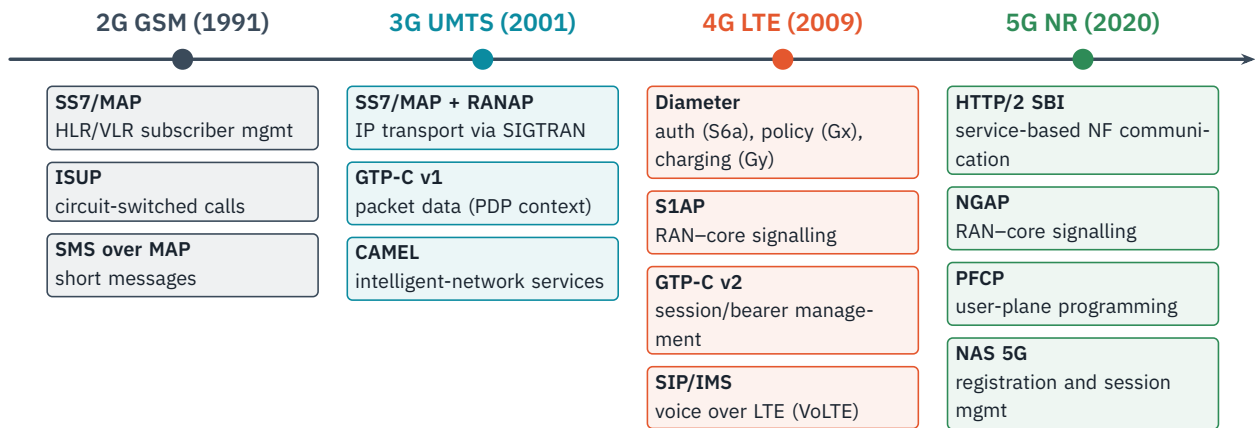


Figure 24.8 Evolution of mobile signalling protocols

8.4 Protocol Comparison: Auth Flow Example

2G/3G (SS7/MAP):

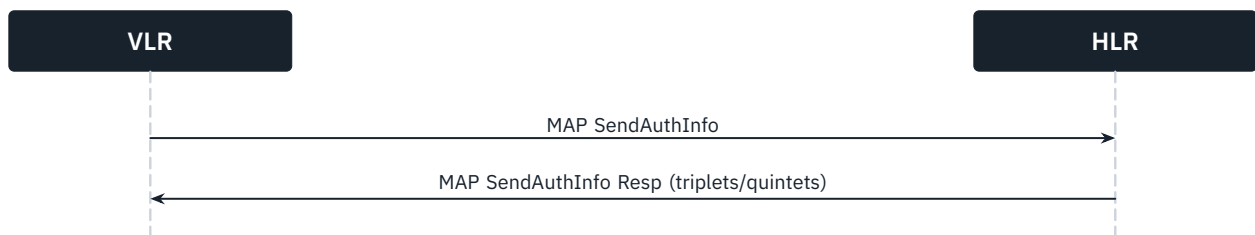


Figure 24.9 2G/3G authentication over SS7/MAP

4G (Diameter):

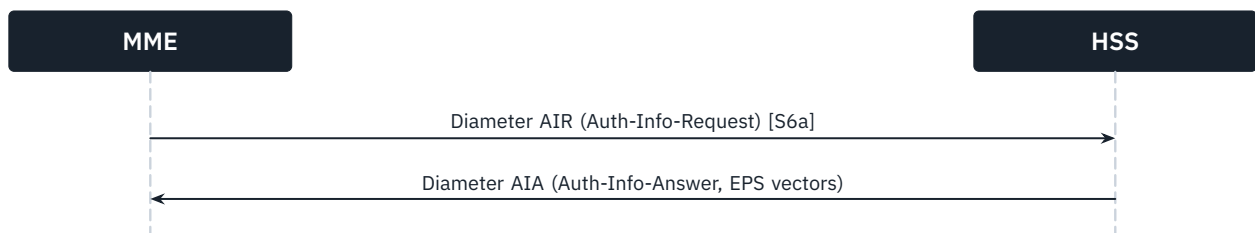


Figure 24.10 4G authentication over Diameter (S6a)

5G (HTTP/2):

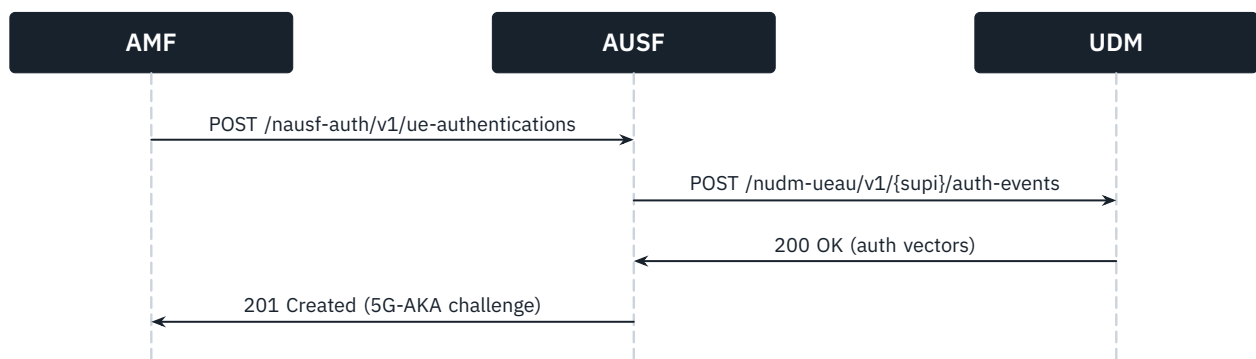


Figure 24.11 5G authentication over HTTP/2 SBI

Complete Control Plane Architecture Diagram (5G)

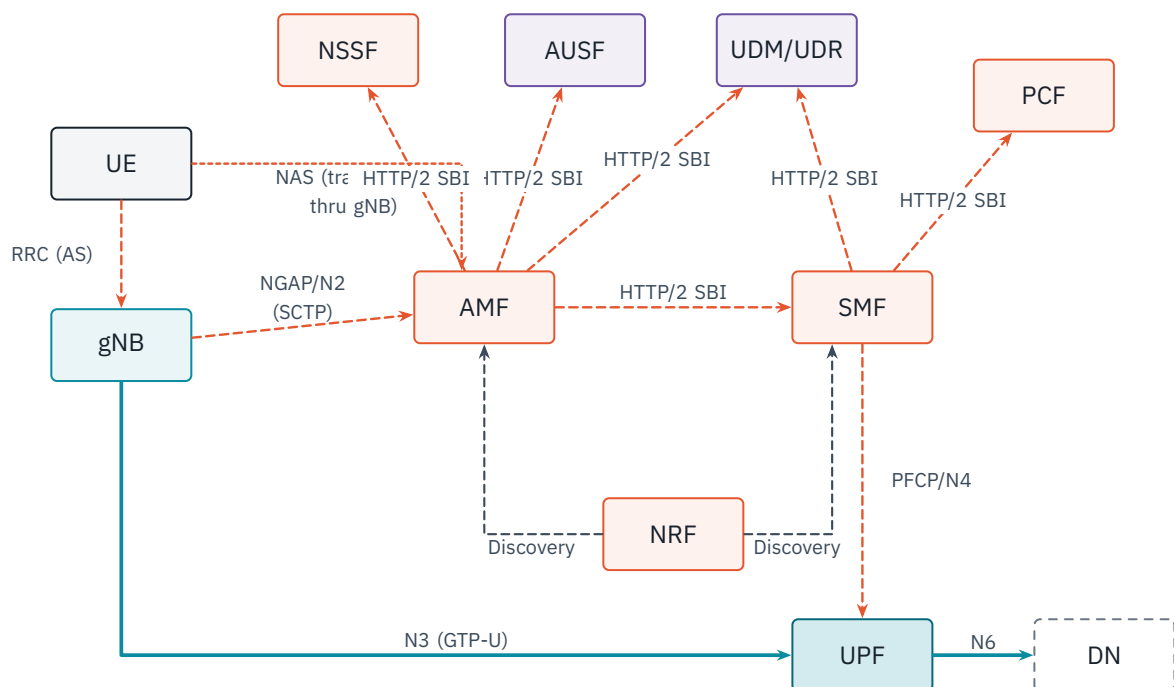


Figure 24.12 Complete 5G control-plane architecture

Summary: Control Plane vs. User Plane

Aspect	Control Plane	User Plane
Carries	Signaling messages	User data (IP, voice, video)
Purpose	Setup, manage, release connections	Transport data end-to-end
Latency tolerance	Moderate (100s of ms OK)	Critical (low latency required)
Volume	Low (small messages)	High (bulk data transfer)
Protocols	NAS, RRC, NGAP, HTTP/2, Diameter	GTP-U, SDAP, PDCP
5G separation	AMF, SMF, AUSF, etc.	UPF (controlled by SMF via PFCP)
Scaling	Scale per signaling load	Scale per throughput demand
State	Stateful (tracks UE context)	Stateless forwarding (rules from CP)

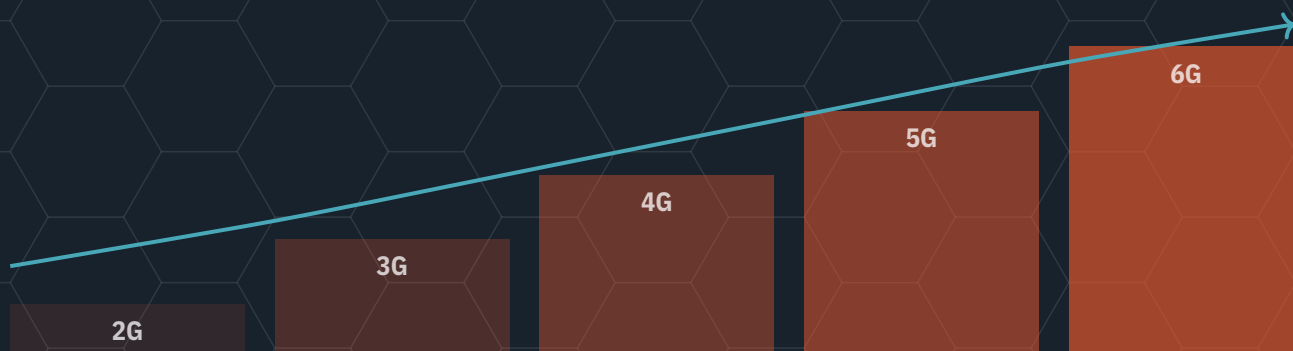
Key Takeaways

1. **Control plane = signaling brain:** Sets up tunnels, authenticates users, manages mobility
2. **NAS is end-to-end (UE↔Core)** — transparent to RAN; separate security context
3. **RRC manages the radio** — connection setup, measurements, handover commands
4. **5G adds RRC_INACTIVE** state for fast resume and IoT efficiency
5. **NGAP replaced S1AP** — supports slicing, QoS flows, and three RRC states
6. **GTP-C evolved:** v1 (3G) → v2 (4G) → replaced by PFCP+HTTP/2 in 5G
7. **HTTP/2 SBI is revolutionary** — cloud-native, RESTful, auto-scalable NF communication
8. **Signaling evolution:** SS7 (binary, TDM) → Diameter (binary, IP) → HTTP/2 (JSON, cloud-native)

Network Evolution — Grand Synthesis

تحويل شبكة — تركيب جامع

25



In This Module

1	Introduction	485
2	Master Comparison Table	485
3	Generation-by-Generation Analysis	487
4	Timeline Diagram	491
5	Architecture Evolution Diagram	492
6	Data Rate Evolution Graph (Exponential Growth)	494
7	Key Lessons Learned from Each Transition	495
8	Summary of Evolutionary Patterns	498
9	Exam Preparation Quick Reference	499
10	Conclusion	500

1 Introduction

This module provides a **comprehensive synthesis** of mobile network evolution from 1G analog systems through 5G and beyond. It serves as a master reference connecting all previous modules, showing how each generation solved specific problems while creating new challenges that motivated the next evolution.

The evolution follows a clear trajectory:

- **Voice → Data → Everything**
- **Circuit → Packet → All-IP → Service-Based**
- **Hierarchical → Flat → Cloud-Native**
- **One-size-fits-all → Network Slicing → AI-Native**

2 Master Comparison Table

Parameter	GSM (2G)	GPRS (2.5G)	EDGE (2.75G)	UMTS (3G)
3GPP Release	Rel-96/97	Rel-97/98	Rel-99	Rel-99/4
Year	1991	2000	2003	2001
Radio Access	TDMA/FDMA	TDMA/FDMA	TDMA/FDMA (8PSK)	WCDMA
Core Network	NSS (MSC/HLR/VLR)	NSS + SGSN/GGSN	NSS + SGSN/GGSN	CS + PS Core
Switching Type	Circuit-Switched	CS + Packet-Switched	CS + PS	CS + PS
Peak DL Rate	9.6 kbps	171.2 kbps	473.6 kbps	2 Mbps
Typical Latency	500 ms	300-600 ms	200-400 ms	100-200 ms
Mobility Mechanism	Handover (hard)	Cell reselection + HO	Cell reselection + HO	Soft handover
Signaling Protocol	MAP/BSSAP	MAP/BSSGP	MAP/BSSGP	RANAP/MAP
User Plane Protocol	PCM (64 kbps)	LLC/SNDCP/GTP-U	LLC/SNDCP/GTP-U	GTP-U/PDCP
Control Plane Protocol	SS7/ISUP/MAP	SS7 + GTP-C	SS7 + GTP-C	SS7 + GTP-C
Main Innovation	Digital voice, roaming	Always-on packet data	Higher modulation (8PSK)	Wideband data (5 MHz)
Main Limitation	No data, low capacity	Slow data, shared TDMA	Still TDMA-limited	High latency, complex RNC
Subscribers (approx.)	1B (2003)	1.5B (2005)	2B (2006)	2B (2007)

Parameter	HSPA (3.5G)	LTE (4G)	LTE-Advanced (4G+)	5G NSA
3GPP Release	Rel-5/6/7	Rel-8/9	Rel-10/11/12	Rel-15
Year	2005-2008	2009	2011-2014	2018
Radio Access	WCDMA + AMC	OFDMA/SC-FDMA	OFDMA + CA + MIMO	NR + LTE
Core Network	CS + PS Core	EPC (MME/SGW/PGW)	EPC Enhanced	EPC + NGC partial
Switching Type	CS + PS (HSPA=PS)	All-IP (Packet)	All-IP	All-IP
Peak DL Rate	42 Mbps (HSPA+)	300 Mbps	3 Gbps	2-4 Gbps
Typical Latency	50-100 ms	10-30 ms	10-20 ms	4-10 ms
Mobility Mechanism	Soft/Hard HO	X2 handover (hard)	X2/S1 handover + CoMP	Dual connectivity
Signaling Protocol	RANAP/MAP	S1AP/GTP-C	S1AP/GTP-C v2	X2AP + S1AP + NGAP
User Plane Protocol	GTP-U/PDCP/MAC	GTP-U/PDCP/RLC	GTP-U/PDCP/RLC	GTP-U (dual pipe)
Control Plane Protocol	SS7 + GTP-C	Diameter/GTP-C	Diameter/GTP-C	GTP-C + NGAP
Main Innovation	Fast scheduling, AMC	All-IP, flat arch, OFDM	Carrier Aggregation, 8×8 MIMO	NR radio with LTE anchor
Main Limitation	Still CS voice, complex	No native IoT, limited MIMO	Spectrum fragmentation	Depends on 4G core
Subscribers (approx.)	3B (2012)	4B (2018)	5B (2020)	1B (2022)

Parameter	5G SA	5G Core (SBA)	6G (Future)
3GPP Release	Rel-15/16	Rel-15/16/17	Rel-20+
Year	2020	2020-2022	~2030
Radio Access	NR (standalone)	NR	Sub-THz, RIS, AI
Core Network	5GC (AMF/SMF/UPF)	5GC SBA (NFs)	6G Core (AI-native)
Switching Type	All-IP	All-IP + SBA	All-IP + Compute
Peak DL Rate	20 Gbps	20 Gbps	1 Tbps
Typical Latency	1-4 ms	1-4 ms	<0.1 ms
Mobility Mechanism	Xn handover	Xn + SBI mobility	Predictive AI HO
Signaling Protocol	NGAP	HTTP/2 + SBI	AI-native signaling
User Plane Protocol	GTP-U/SDAP	GTP-U/SDAP	Semantic comms
Control Plane Protocol	NGAP/NAS 5G	SBI (HTTP/2)	Intent-based
Main Innovation	Independent 5G, slicing	Service-Based Architecture	AI-native, sensing
Main Limitation	Coverage gaps (mmWave)	Complexity, energy	Undefined standards
Subscribers (approx.)	500M (2023)	1B+ (2024)	TBD (~2030+)

3 Generation-by-Generation Analysis

3.1 GSM (2G) – Digital Voice Revolution (1991)

Problem Solved: GSM replaced analog 1G systems (AMPS, TACS) with digital transmission, providing encryption, international roaming via SIM cards, and efficient spectrum use through TDMA. For the first time, a subscriber could travel across Europe and make calls with the same phone and number. The GSM MAP protocol enabled global mobility management, and the hierarchical NSS architecture (MSC/VLR/HLR/AuC) became the template for mobile networks.

What it couldn't do: GSM was fundamentally circuit-switched — a dedicated timeslot was allocated for the entire call duration regardless of actual data flow. Data services (CSD) were limited to 9.6 kbps, making internet access impractical. As the internet exploded in the late 1990s, operators needed packet-switched data without replacing their entire GSM infrastructure.

3.2 GPRS (2.5G) – Packet Data Overlay (2000)

Problem Solved: GPRS introduced packet-switching as an **overlay** on the existing GSM radio and core network. By adding SGSN and GGSN nodes, operators could offer “always-on” internet connectivity where users shared radio resources and were charged by data volume rather than connection time. GPRS introduced the PDP Context concept, IP addressing for mobile devices, and GTP tunneling — concepts that persist through 5G.

What it couldn't do: GPRS reused GSM's TDMA radio interface, sharing timeslots that were still primarily allocated for voice. Theoretical peak of 171.2 kbps was rarely achieved (typical: 30-50 kbps). Latency

remained high (300-600 ms) due to the TDMA frame structure. The radio interface was the bottleneck — you couldn't fix it without changing the air interface technology.

3.3 EDGE (2.75G) — Enhanced Data on GSM (2003)

Problem Solved: EDGE introduced 8PSK modulation (vs. GMSK in GSM/GPRS), tripling the bits per symbol from 1 to 3. With adaptive modulation and coding (9 MCS schemes), EDGE could deliver up to 473.6 kbps theoretical peak. It was a pure software/transceiver upgrade requiring no new infrastructure, giving operators a cost-effective capacity boost while they planned 3G deployments.

What it couldn't do: EDGE was still constrained by the 200 kHz GSM carrier bandwidth and TDMA structure. Real-world speeds of 100-200 kbps couldn't support video streaming or rich web applications. The fundamental limitation was the narrow-band radio interface — achieving true mobile broadband required a completely new wideband radio technology.

3.4 UMTS (3G) — Wideband Mobile Broadband (2001)

Problem Solved: UMTS introduced WCDMA with 5 MHz carriers, providing up to 2 Mbps peak data rates. The spread-spectrum approach offered better spectral efficiency, soft handover for improved cell-edge performance, and native support for simultaneous voice and data. The UTRAN architecture (Node B + RNC) provided centralized radio resource management, and the combined CS+PS core supported both circuit voice and packet data.

What it couldn't do: The RNC was a bottleneck and single point of failure. Latency remained high (100-200 ms round-trip) due to the complex protocol stack. The dedicated channel approach was inefficient for bursty data traffic. Voice still used circuit-switching, maintaining dual-domain complexity. The fixed spreading factor limited adaptation to varying channel conditions.

3.5 HSPA (3.5G) — High-Speed Packet Access (2005-2008)

Problem Solved: HSPA (HSDPA + HSUPA) introduced shared channels, fast scheduling at the Node B (2ms TTI), Adaptive Modulation and Coding (AMC), and HARQ — moving intelligence from the RNC to the base station. HSDPA delivered up to 14.4 Mbps (Rel-5), later HSPA+ reached 42 Mbps with 64-QAM and MIMO. This made mobile video streaming and app stores viable, launching the smartphone era.

What it couldn't do: HSPA was still built on WCDMA's 5 MHz carrier, limiting peak rates. Voice remained circuit-switched (except VoIP experiments). The RNC still existed, adding latency and cost. The architecture couldn't scale efficiently — each RNC managed hundreds of Node Bs, creating capacity planning challenges. Operators needed a clean-slate design.

3.6 LTE (4G) — All-IP Flat Architecture (2009)

Problem Solved: LTE made a revolutionary break: all-IP from day one (no circuit-switching), OFDMA for downlink (eliminating inter-cell interference issues), flat architecture (removed the RNC, eNodeB connects directly to EPC), and the Evolved Packet Core (MME/S-GW/P-GW separation). With 20 MHz bandwidth and 2×2 MIMO, LTE delivered 150-300 Mbps peak. VoLTE replaced circuit-switched voice. The X2 interface enabled direct inter-eNodeB handovers without core involvement.

What it couldn't do: LTE lacked native support for massive IoT (NB-IoT came later in Rel-13), couldn't achieve sub-millisecond latency for industrial applications, had limited MIMO (max 8 layers), and the

EPC's GTP-based point-to-point interfaces couldn't scale for the cloud-native era. Network slicing wasn't possible with monolithic EPC nodes.

3.7 LTE-Advanced (4G+) — Carrier Aggregation Era (2011-2014)

Problem Solved: LTE-Advanced introduced Carrier Aggregation (combining up to 5×20 MHz = 100 MHz), 8×8 MIMO (8 layers DL), enhanced inter-cell interference coordination (eICIC/FeICIC), relay nodes, and Coordinated Multi-Point (CoMP). Peak rates reached 3 Gbps (Rel-12). Heterogeneous Networks (HetNets) with small cells addressed capacity in dense areas. LAA (Licensed Assisted Access) added unlicensed spectrum.

What it couldn't do: LTE-A was an evolution, not revolution — it couldn't fundamentally change the frame structure for ultra-low latency, couldn't use mmWave spectrum, and the EPC remained monolithic. Supporting diverse verticals (automotive, industry, healthcare) with one network configuration was impossible. The industry needed a new radio AND a new core.

3.8 5G NSA — Non-Standalone (2018)

Problem Solved: 5G NSA (Option 3/3a/3x) deployed NR (New Radio) using the existing LTE core (EPC) as an anchor. This allowed operators to launch 5G services quickly without building an entirely new core network. Dual Connectivity (EN-DC) combined LTE and NR carriers, delivering multi-Gbps throughput using mmWave and sub-6 GHz NR spectrum while maintaining LTE for control plane and voice (VoLTE).

What it couldn't do: 5G NSA couldn't deliver the full 5G promise: no network slicing (EPC doesn't support it), no ultra-reliable low-latency (URLLC requires 5GC), no Service-Based Architecture, and the LTE anchor added latency. It was a "5G radio with 4G brain" — a necessary stepping stone, not the destination.

3.9 5G SA — Standalone (2020)

Problem Solved: 5G SA deployed NR with the 5G Core (5GC), eliminating LTE dependency. This enabled network slicing (isolated virtual networks per use case), URLLC with 1ms latency, native support for massive IoT (mMTC), edge computing (MEC) integration, and the Service-Based Architecture. The AMF/SMF/UPF separation allowed independent scaling, and the SBI (HTTP/2, JSON) enabled cloud-native deployments.

What it couldn't do: mmWave coverage remains limited (short range, blockage-sensitive), energy consumption is significantly higher than 4G, the complexity of managing network slices across operators is unsolved, and true deterministic latency for industrial automation requires further enhancements. Migration from NSA to SA requires significant investment with uncertain near-term ROI.

3.10 5G Core — Service-Based Architecture (2020-2022)

Problem Solved: The 5G Core replaced point-to-point interfaces with a Service-Based Architecture where Network Functions (NFs) expose services via RESTful APIs (HTTP/2). NRF enables service discovery, NSSF manages slice selection, PCF provides policy, and UDM unifies subscriber data. Cloud-native principles (microservices, stateless design, containerization) enable elastic scaling, multi-vendor interoperability, and rapid feature deployment.

What it couldn't do: SBA complexity challenges smaller operators. Inter-PLMN roaming with SBA (SEPP) is still maturing. Energy efficiency of distributed cloud infrastructure is a concern. True end-

to-end automation (zero-touch provisioning) requires AI/ML integration that's still evolving. The architecture is ready for 6G evolution but needs AI-native capabilities built in from the ground up.

3.11 6G (Future) — AI-Native Networks (~2030)

Problem Solved (Expected): 6G aims to deliver 1 Tbps peak rates using sub-THz frequencies (100 GHz–3 THz), sub-0.1 ms latency, integrated sensing and communication (ISAC), AI-native air interface and network management, Reconfigurable Intelligent Surfaces (RIS) for programmable propagation, non-terrestrial networks (NTN) integration for global coverage, and semantic/goal-oriented communications that transmit meaning rather than bits.

What it couldn't do (Anticipated challenges): Sub-THz propagation is extremely limited (meters to tens of meters), requiring ultra-dense deployments. Energy efficiency at these frequencies is poor. Standardization of AI-native protocols is unprecedented. Privacy concerns with pervasive sensing. Regulatory frameworks for sub-THz spectrum allocation are undefined. Commercial viability for many proposed features remains unproven.

4 Timeline Diagram

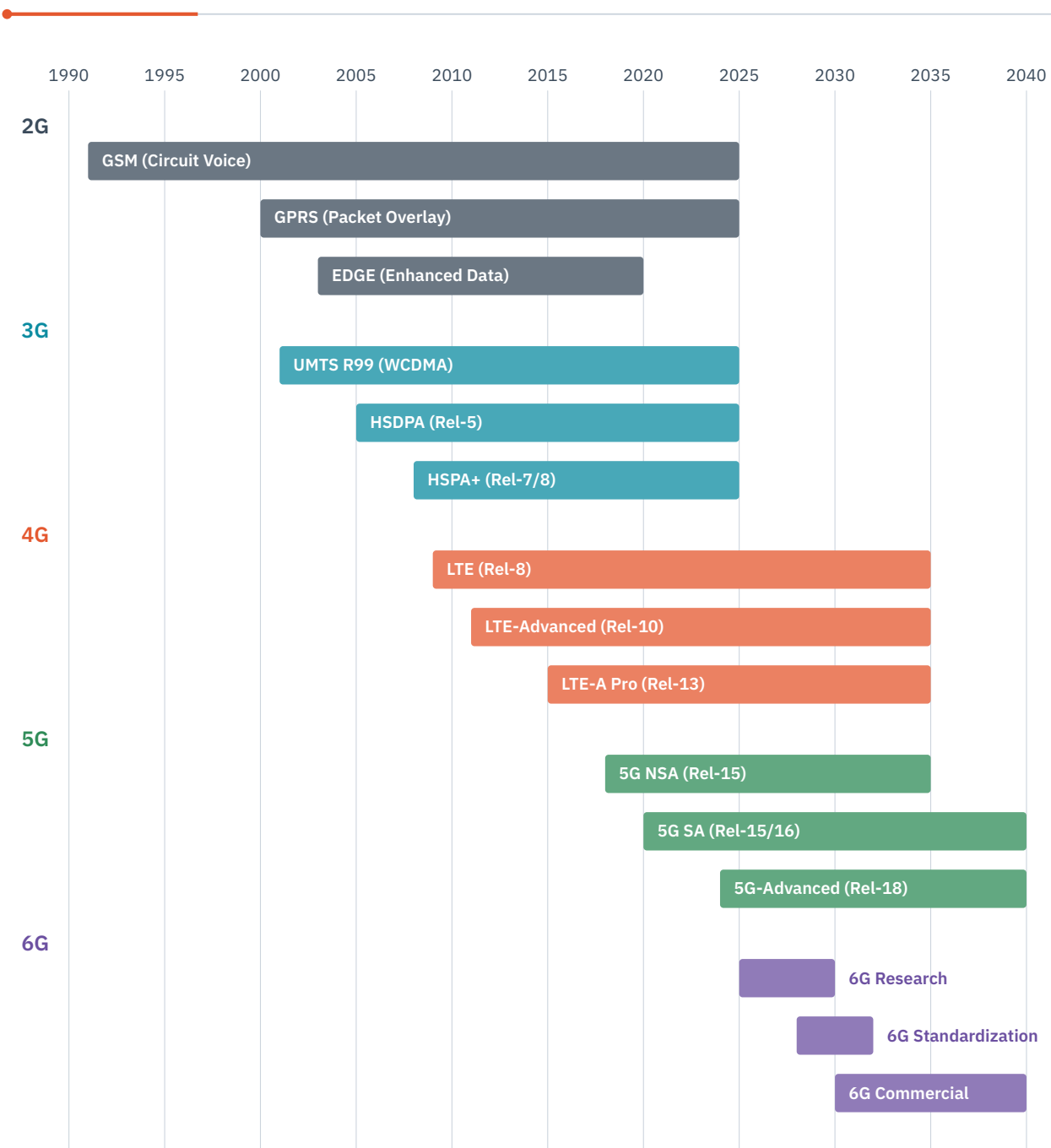


Figure 25.1 Mobile network generations on a timeline

5 Architecture Evolution Diagram

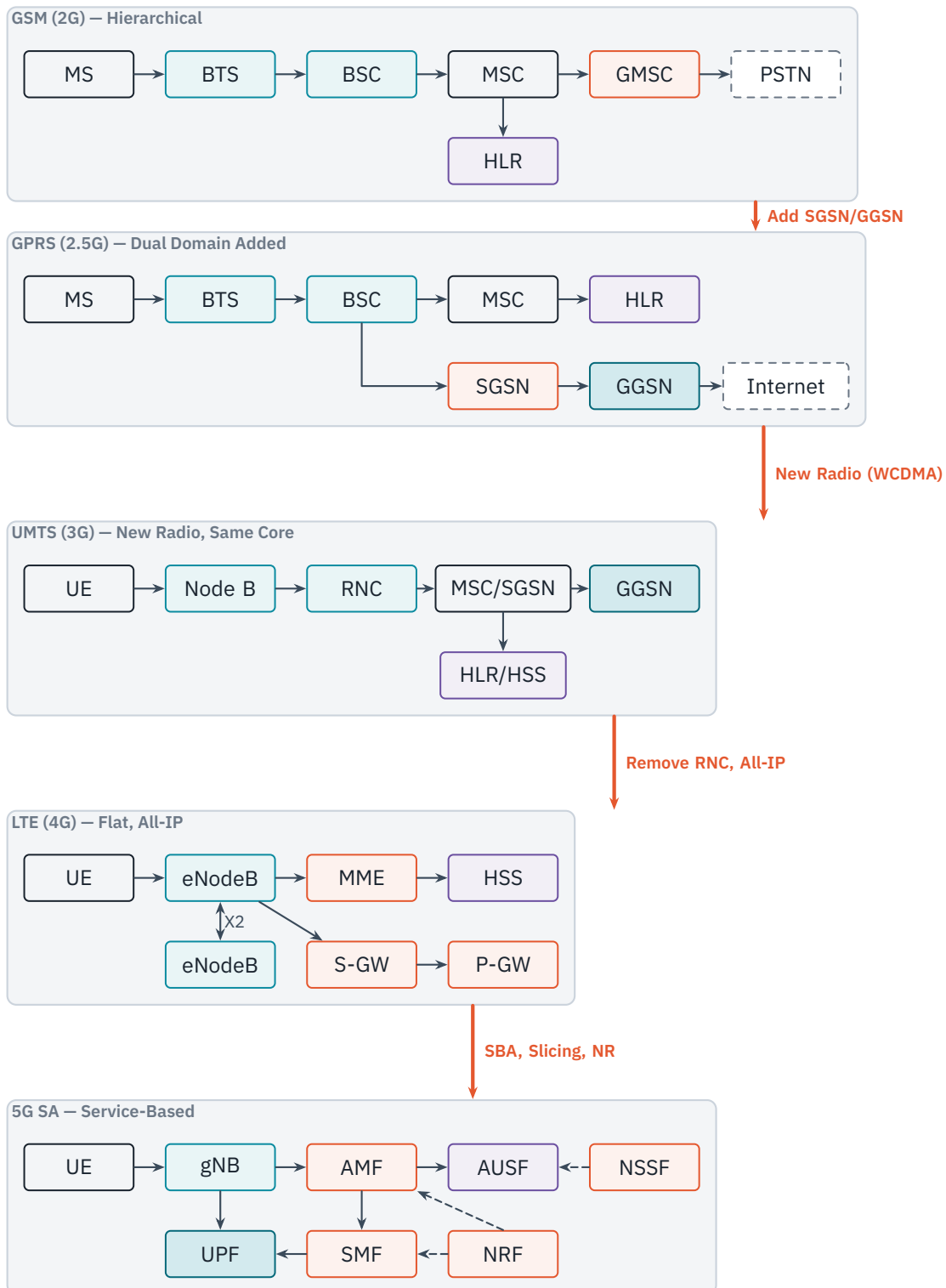


Figure 25.2 Architecture evolution from GSM to the 5G service-based core

Architecture Element Evolution Map

2G Element	3G Equivalent	4G Equivalent	5G Equivalent	Evolution Pattern
BTS	Node B	eNodeB	gNB	More intelligence pushed to edge
BSC	RNC	— (removed)	—	Eliminated (functions in eNB/gNB)
MSC	MSC Server	MME	AMF	Control plane separated from user plane
HLR	HSS	HSS	UDM/UDR	Unified data management
GMSC	GMSC	—	—	Eliminated (all-IP, no PSTN gateway needed)
VLR	VLR	MME (integrated)	AMF (stateless)	From stateful to stateless
SGSN	SGSN	S-GW (partial)	SMF+UPF	Split into control + user plane
GGSN	GGSN	P-GW	UPF	Pure user plane forwarding
—	—	PCRF	PCF	Policy and charging
—	—	—	NRF	Service discovery (new in 5G)
—	—	—	NSSF	Slice selection (new in 5G)

6 Data Rate Evolution Graph (Exponential Growth)

peak downlink data rate

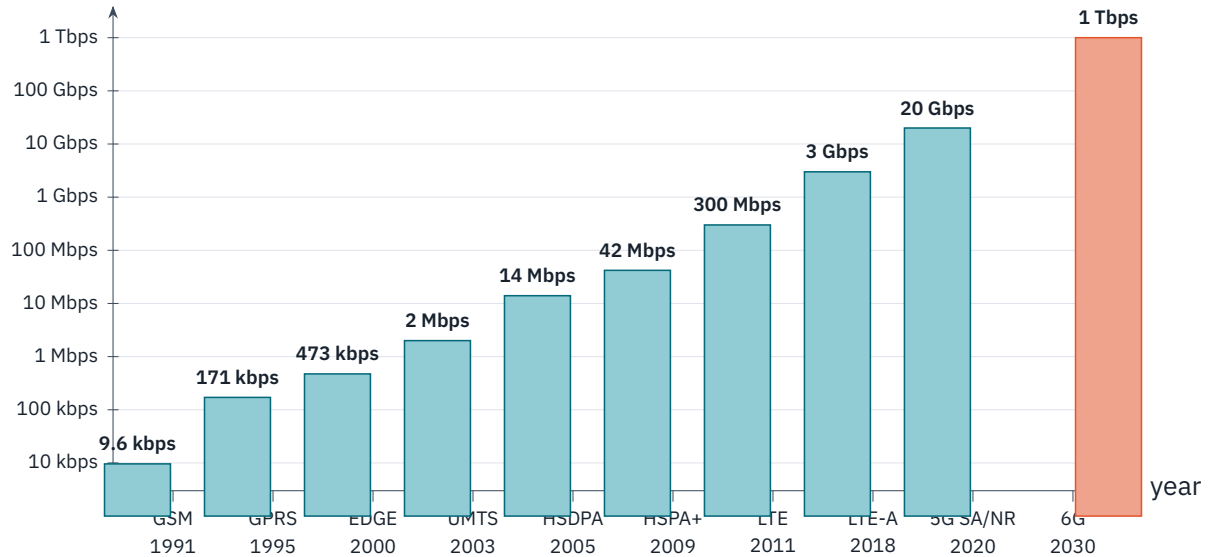


Figure 25.3 Peak downlink data rate of each generation (logarithmic scale)

Growth pattern: about $10\times$ every 5 years ($1000\times$ per decade).

Step	Improvement	Peak rate
GSM → GPRS	$18\times$	9.6 kbps → 171 kbps
GPRS → EDGE	$2.8\times$	171 kbps → 473 kbps
EDGE → UMTS	$4.2\times$	473 kbps → 2 Mbps
UMTS → HSPA	$21\times$	2 Mbps → 42 Mbps
HSPA → LTE	$7\times$	42 Mbps → 300 Mbps
LTE → LTE-A	$10\times$	300 Mbps → 3 Gbps
LTE-A → 5G	$6.7\times$	3 Gbps → 20 Gbps
5G → 6G	$50\times$	20 Gbps → 1 Tbps

7 Key Lessons Learned from Each Transition

7.1 GSM → GPRS: Overlay Packet on Circuit

Aspect	Lesson
Strategy	You don't need to replace everything — overlay new capability on existing infrastructure
Architecture	Added SGSN/GGSN alongside MSC without disrupting voice service
Risk Management	Incremental deployment: operators could add packet capability node by node
Trade-off	Shared radio resources meant neither voice nor data was optimal
Legacy Impact	GTP tunneling protocol invented here persists through 5G
Key Principle	"Evolution over revolution when the installed base is massive"

7.2 GPRS → UMTS: New Radio, Keep Core

Aspect	Lesson
Strategy	When the radio is the bottleneck, change the radio but keep the proven core
Architecture	WCDMA replaced TDMA, but SGSN/GGSN/HLR remained largely unchanged
Risk Management	Operators could run 2G and 3G simultaneously with shared core
Trade-off	New spectrum required (2100 MHz band) — massive licensing costs
Legacy Impact	RNC architecture created a new bottleneck that took a decade to remove
Key Principle	"Change one major subsystem at a time to manage risk"

7.3 UMTS → LTE: Flatten, All-IP, Remove RNC

Aspect	Lesson
Strategy	Sometimes you must break backward compatibility for fundamental improvement
Architecture	Eliminated RNC (functions distributed to eNodeB), all-IP (eliminated circuit-switching)
Risk Management	VoLTE took years to mature — interim CSFB to 2G/3G for voice
Trade-off	Clean-slate design required parallel networks for many years
Legacy Impact	Flat architecture proved far more scalable and cost-effective
Key Principle	“Flatten the hierarchy — push intelligence to the edge, simplify the core”

7.4 LTE → LTE-Advanced: Carrier Aggregation, More MIMO

Aspect	Lesson
Strategy	Combine existing resources more cleverly before seeking new ones
Architecture	CA bonds fragmented spectrum; MIMO multiplies spatial capacity
Risk Management	Backward compatible — LTE devices work alongside LTE-A
Trade-off	Complexity in scheduler and UE increases significantly
Legacy Impact	Established that spectrum aggregation is key to capacity scaling
Key Principle	“Maximize what you have through intelligent combination before building new”

7.5 LTE → 5G: New Radio + New Core, Slicing, SBA

Aspect	Lesson
Strategy	When requirements diverge (eMBB vs. URLLC vs. mMTC), one configuration cannot serve all
Architecture	NR provides flexible numerology; 5GC SBA enables per-slice customization
Risk Management	NSA deployment (Option 3x) as stepping stone before SA
Trade-off	Enormous complexity; operators must justify ROI for diverse verticals
Legacy Impact	SBA and cloud-native principles prepare for continuous software evolution
Key Principle	“Design for diversity — network slicing lets one physical network serve many purposes”

7.6 5G → 6G: AI-Native, Sensing, Sub-THz

Aspect	Lesson
Strategy	The network itself becomes intelligent — AI is not an add-on but a fundamental design principle
Architecture	Digital twin of the network, predictive resource management, semantic communications
Risk Management	Research phase allows exploration of feasibility before standardization
Trade-off	Sub-THz range is extremely limited; cost of ultra-dense deployment is unknown
Legacy Impact	May redefine what “communication” means (sensing + computing + communicating)
Key Principle	“Converge communication, computation, and sensing into a unified intelligent fabric”

8 Summary of Evolutionary Patterns

Pattern 1: Alternating Radio and Core Innovation

Generation	2G	2.5G	3G	3.5G	4G	4G+	5G
Focus	Radio	Core	Radio	Radio	Both	Radio	Both
Key innovation	TDMA	SGSN	WCDMA	HSPA	OFDMA	CA	NR + SBA

Pattern 2: Complexity Cycle

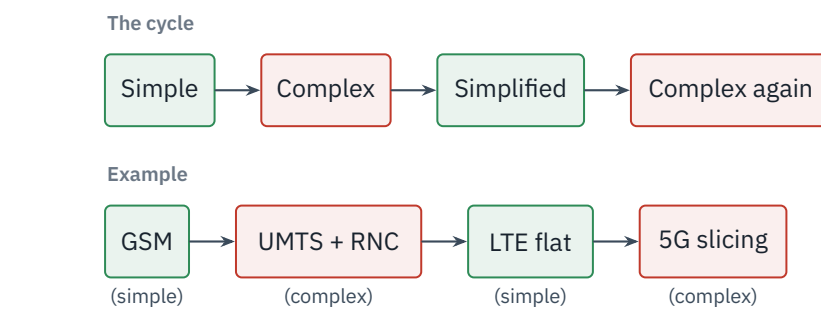


Figure 25.4 The complexity cycle of mobile network architectures

Pattern 3: Control/User Plane Separation Timeline

Generation	Control/user-plane arrangement	Result
2G	MSC handles both	Inefficient
3G	MSC-Server + MGW split (Rel-4)	Partial CUPS
4G	MME (CP) + S-GW/P-GW (UP)	Architectural CUPS
5G	AMF/SMF (CP) + UPF (UP)	Complete CUPS

Pattern 4: Where Intelligence Resides

Generation	Where the intelligence resides
2G	BSC/MSC (centralised)
3G	RNC (centralised)
4G	eNodeB (distributed to the edge)
5G	gNB + edge computing (distributed + edge cloud)
6G	Everywhere (AI at every layer)

9 Exam Preparation Quick Reference

Must-Know Facts per Generation

Gen	Remember This
GSM	TDMA, 200 kHz carrier, 8 timeslots, circuit-switched, MAP/SS7, HLR/VLR/MSC
GPRS	Added SGSN+GGSN, PDP Context, GTP tunnel, packet overlay on CS
EDGE	8PSK modulation (3 bits/symbol vs 1), 9 MCS schemes, software upgrade
UMTS	WCDMA, 5 MHz, spreading codes, RNC, soft handover, Rake receiver
HSPA	Shared channel, 2ms TTI, AMC, HARQ, fast scheduling at Node B
LTE	OFDMA, flat (no RNC), all-IP, EPC (MME/SGW/PGW), X2 interface
LTE-A	CA (5×20 MHz), 8×8 MIMO, CoMP, HetNets, Relay
5G NSA	EN-DC, LTE anchor, NR secondary, Option 3x, EPC core
5G SA	AMF/SMF/UPF, NGAP, network slicing, URLLC, mMTC
5GC	SBA, HTTP/2, NRF discovery, NSSF, stateless NFs, cloud-native
6G	Sub-THz, RIS, AI-native, ISAC, semantic comms, 1 Tbps

Key Protocol Stack Evolution

2G	3G	4G	5G
App	App	App	App
TCP/IP	TCP/IP	TCP/IP	TCP/IP
SNDCP/LLC	PDCP	PDCP	SDAP/PDCP
RLC	RLC	RLC	RLC
MAC	MAC	MAC	MAC
GSM PHY (TDMA)	WCDMA PHY (CDMA)	OFDMA PHY	NR PHY (flex. num.)

Figure 25.5 User-plane protocol stack of each generation

Critical Comparison: 4G EPC vs 5G Core

Feature	4G EPC	5G Core
Interface type	Point-to-point (GTP-C, Diameter)	Service-Based (HTTP/2 REST)
Discovery	Static configuration	NRF dynamic discovery
Scaling	Vertical (bigger boxes)	Horizontal (more instances)
Deployment	Appliance-based	Cloud-native (containers, K8s)
Slicing	Not supported	Native (NSSF, per-slice NFs)
State	Stateful nodes	Stateless NFs + external state (UDSF)
Mobility anchor	GTP tunnel (S-GW)	UPF (can be distributed)
Policy	PCRF (Diameter)	PCF (SBI)
Subscriber data	HSS (Diameter)	UDM/UDR (SBI)
Authentication	MME + HSS (EAP-AKA)	AUSF + UDM (5G-AKA, EAP)

10 Conclusion

The evolution from GSM to 6G represents one of the most remarkable engineering journeys in history — from 9.6 kbps voice calls to terabit-per-second intelligent networks in four decades. Each generation solved the previous generation’s limitations while introducing new complexity that demanded the next evolution.

The three constants across all generations:

1. **Demand always exceeds supply** — users always find ways to consume available capacity
2. **Simplification follows complexity** — every overly complex architecture is eventually flattened
3. **The boundary between network and computing blurs** — from dumb pipes to intelligent platforms

Understanding this evolution is not just historical knowledge — it teaches us how to evaluate future technologies, predict architectural decisions, and design systems that can gracefully evolve.

Module 25 — Network Evolution Grand Synthesis

Advanced Communication Networks Course

Last updated: August 2026

Future 6G Technologies for Advanced Communication Networks

فناوری‌های آینده 6G برای شبکه‌های ارتباطی پیشرفته



In This Module

▲ IMPORTANT DISCLAIMER	503
Overview: The Road to 6G	503
1 AI-Native Networks / AI-RAN	504
2 Extremely Large MIMO (XL-MIMO)	506
3 Cell-Free Massive MIMO	508
4 Integrated Sensing and Communication (ISAC / JCS)	510
5 Reconfigurable Intelligent Surfaces (RIS)	512
6 Sub-THz / THz Communications	515
7 Non-Terrestrial Networks (NTN)	517
8 Semantic Communications	519
9 Digital Twins for Networks	522
10 Advanced Positioning (cm-Level)	524
Technology Maturity Matrix	527
Projected Timeline: 3GPP Releases and 6G Technologies	528
Summary: 5G vs 6G Paradigm Shifts	529
Key Takeaways	529
References and Further Reading	530

⚠ IMPORTANT DISCLAIMER

This module discusses technologies at various stages of maturity. Many concepts presented here are speculative, under early research, or in preliminary standardization study phases. The 6G standard does not yet exist — ITU IMT-2030 framework is under development, and 3GPP has not begun formal 6G specification work (expected ~2025-2028). Information is based on published research, 3GPP study items, and industry roadmaps as of 2024-2025.

■ Maturity Classification Used in This Module

Label	Meaning
STANDARDIZED	Specified in 3GPP Technical Specifications, deployable
STUDY ITEM	Formally studied in 3GPP (TR documents), may become work items
CANDIDATE	Strong industry consensus, likely to enter standardization
RESEARCH	Active academic/industry research, no formal standardization yet
SPECULATIVE	Early-stage concepts, feasibility not fully demonstrated

Overview: The Road to 6G

■ IMT-2030 Vision (ITU-R)

The ITU-R Working Party 5D has outlined the IMT-2030 framework with these usage scenarios:

- **Immersive Communication** (evolution of eMBB)
- **Hyper Reliable and Low-Latency Communication** (evolution of URLLC)
- **Massive Communication** (evolution of mMTC)
- **Ubiquitous Connectivity** (new)
- **Integrated AI and Communication** (new)
- **Integrated Sensing and Communication** (new)

5G Capabilities vs 6G Targets (IMT-2030)

KPI	5G (IMT-2020)	6G Target (IMT-2030)	Improvement
Peak data rate	20 Gbps	200+ Gbps	10×
User experienced rate	100 Mbps	1-10 Gbps	10-100×
Latency (user plane)	1 ms	0.1 ms	10×
Reliability	99.999%	99.99999%	100×
Connection density	10^6 devices/km ²	10^7 - 10^8 devices/km ²	10-100×
Mobility	500 km/h	1000+ km/h	2×
Spectrum efficiency	1× (baseline)	3-5×	3-5×
Energy efficiency	1× (baseline)	10-100×	10-100×
Positioning accuracy	~1 m (R16)	1-10 cm	10-100×
AI integration	External/OTT	Native/intrinsic	Paradigm shift
Sensing capability	Not supported	Integrated	New capability

1 AI-Native Networks / AI-RAN

■ Status: ● STUDY ITEM (Release 18)

■ What

AI-Native Networks integrate artificial intelligence as a fundamental design principle rather than an add-on optimization layer. AI/ML algorithms are embedded directly into the radio access network (AI-RAN) at every protocol layer.

■ Why / Problem Addressed

- Traditional model-based algorithms (MMSE, LMMSE) reach performance limits in complex channels
- Manual network optimization cannot scale to ultra-dense heterogeneous deployments
- 5G NR parameter space is enormous (beamforming, scheduling, power control)
- Need for self-optimizing, self-healing networks

■ AI Integration Across Protocol Layers

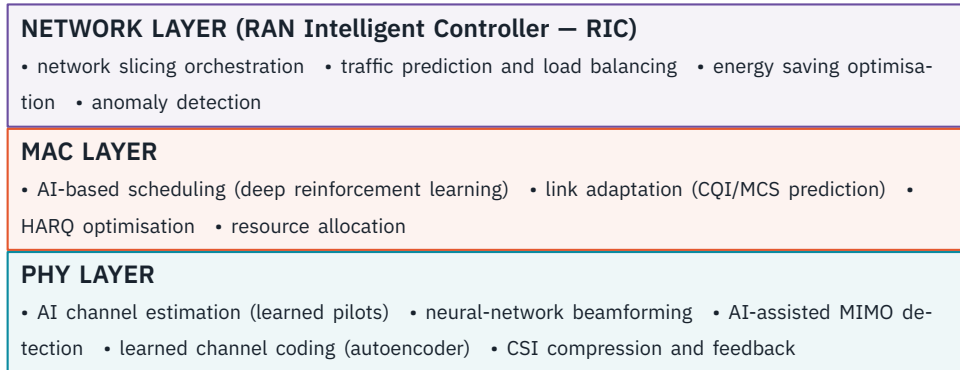


Figure 26.1 AI across the protocol layers

■ 3GPP Progress

- **TR 38.843:** Study on AI/ML for NR air interface (Release 18)
- **Use cases studied:**
 - CSI feedback enhancement (AI-based compression)
 - Beam management (AI-based beam prediction)
 - Positioning enhancement (AI/ML-assisted)
- **TR 38.844:** Study on AI/ML for NG-RAN (network level)
- **Key finding:** Performance gains demonstrated but deployment complexity is high

■ Benefits

- 10-30% throughput improvement in complex scenarios
- Reduced CSI feedback overhead (up to 50% compression)
- Faster beam acquisition (reduced beam sweeping)
- Self-optimizing network behavior

■ Challenges

- **Training data:** Requires massive, representative datasets
- **Generalization:** Models may not transfer across environments
- **Computational cost:** Inference latency vs. real-time requirements
- **Standardization:** How to specify AI models in a standard?
- **Interoperability:** Models from different vendors must coexist
- **Explainability:** Black-box decisions in safety-critical scenarios

■ Relation to 5G

- 5G O-RAN architecture provides RIC framework (xApps/rApps)
- R18 studies lay groundwork; actual specifications expected R19-R20
- Evolution from “AI-assisted” (5G) to “AI-native” (6G)

■ Open Questions

- How to handle model updates over-the-air?
- What is the right split between model-based and data-driven?
- How to ensure robustness against adversarial attacks?
- Standardize the model or just the interface?

2 ■ Extremely Large MIMO (XL-MIMO)

■ Status: ● RESEARCH

■ What

XL-MIMO scales antenna arrays to thousands or tens of thousands of elements, creating electrically very large apertures. Unlike conventional massive MIMO (64-256 elements), XL-MIMO operates partly in the **near-field** regime, fundamentally changing propagation characteristics.

■ Why / Problem Addressed

- Massive MIMO gains saturate with conventional array sizes
- Need for extreme spatial multiplexing (100+ simultaneous streams)
- Higher frequencies require larger arrays for equivalent beamforming gain
- Support for extremely high data rates in 6G

■ Key Physics: Near-Field vs Far-Field

Far-field boundary (Fraunhofer distance):

$$d_F = \frac{2D^2}{\lambda}$$

Example: a $D = 2$ m array at 30 GHz ($\lambda = 10$ mm) gives $d_F = \frac{2 \times 4}{0.01} = 800$ m, so **most users are in the NEAR FIELD of large arrays!**

Near-field characteristics:

- Spherical wavefronts (not planar)
- Beam focusing (not just steering) — energy concentrated at a point in space
- Distance-dependent beamforming

- Spatial non-stationarity: not all array elements “see” the same channel

■ Spatial Non-Stationarity

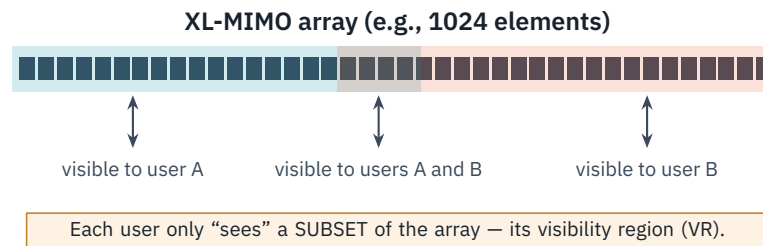


Figure 26.2 Spatial non-stationarity: each user sees only part of the array

■ Benefits

- Beam focusing enables spatial division even for users at same angle
- Near-field MIMO provides additional degrees of freedom
- Extreme spectral efficiency through massive spatial multiplexing
- Inherent interference suppression

■ Challenges

- Channel estimation complexity: $O(N^2)$ pilot overhead without structure exploitation
- Hardware: cost, power, interconnect for thousands of RF chains
- Signal processing: centralized processing infeasible → distributed architectures
- Channel modeling: 3GPP models assume far-field; new models needed
- Calibration: mutual coupling, array imperfections at scale

■ Relation to 5G

- Extension of 5G massive MIMO (R15-R17: up to 256 elements)
- 5G codebook-based and Type-II CSI approaches won't scale
- Requires new channel models beyond 3GPP TR 38.901

■ Open Questions

- Optimal array geometry (planar, cylindrical, conformal)?
- How to exploit near-field properties in standard protocols?
- Hybrid analog/digital architectures at this scale?
- Energy-efficient implementations?

3 Cell-Free Massive MIMO

■ **Status:** ● RESEARCH (5G-Advanced precursors exist)

■ What

Cell-Free Massive MIMO eliminates the concept of cells. A large number of distributed Access Points (APs), connected via fronthaul to Central Processing Units (CPUs), coherently serve all User Equipments (UEs) in the area without cell boundaries.

■ Why / Problem Addressed

- Cell-edge problem: users at cell boundaries suffer from interference and low SINR
- Uneven QoS distribution in cellular networks (10th-percentile vs median)
- Inter-cell interference is a fundamental limitation of cellular architecture
- Need for uniform, ubiquitous coverage

■ Architecture

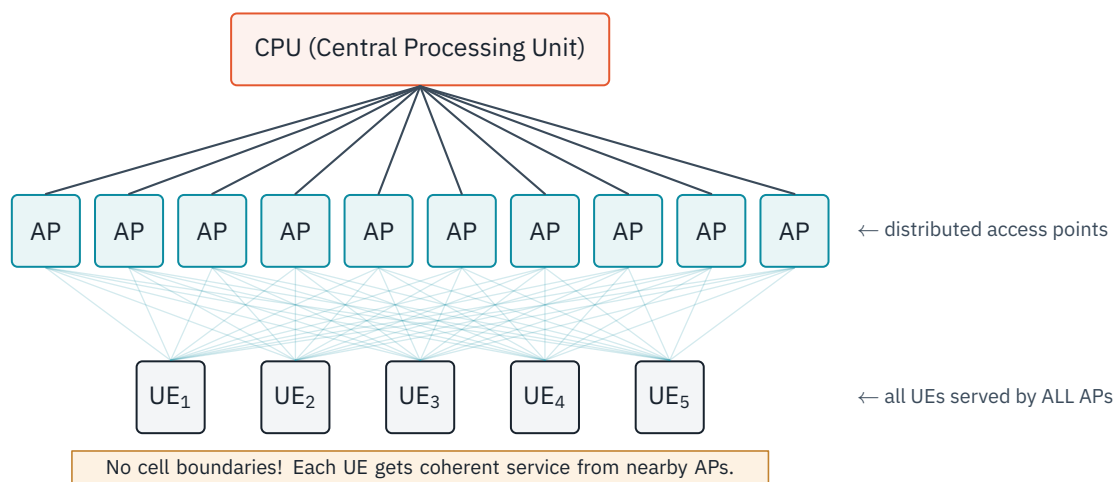


Figure 26.3 Cell-free massive MIMO: all APs serve all UEs

■ Key Principles

1. **All APs serve all UEs** (in principle; practical systems use AP selection)
2. **Coherent joint transmission/reception** across distributed APs
3. **No handover** — UE is always served by the “best” cluster of APs
4. **User-centric clustering** — each UE has its own virtual cell

Benefits

- 95th-percentile throughput improvement: $5\text{-}10\times$ over cellular
- Elimination of cell-edge effect
- Macro-diversity: robust against blockage and shadowing
- Natural interference management through coordination
- Uniform coverage quality

Challenges

- **Fronthaul requirements:** Massive data exchange between APs and CPU
- **Synchronization:** All APs must be phase-synchronized for coherent transmission
- **Scalability:** Computational complexity grows with number of APs $\times$ UEs
- **Channel estimation:** Pilot contamination across large areas
- **Practical deployment:** Cost of dense AP infrastructure

Scalable Implementations

Level	Processing	Fronthaul	Performance
Level 1	Fully centralized MMSE	Very high	Maximum
Level 2	Local MMSE + central combining	High	Near-optimal
Level 3	Local MF + central LSFD	Moderate	Good
Level 4	Local MF, no cooperation	Low	Baseline

Relation to 5G

- 5G CoMP (Coordinated MultiPoint) is a limited precursor
- 5G-Advanced (R18): Multi-TRP enhancements move toward cell-free concepts
- C-RAN architecture provides infrastructure basis
- O-RAN fronthaul interfaces (F1, eCPRI) enable distributed processing

Open Questions

- Optimal AP density and placement?
- How to handle mobility efficiently?
- Power control across distributed APs?
- Economic viability vs. traditional deployment?

4 Integrated Sensing and Communication (ISAC / JCS)

■ **Status:** ● **STUDY ITEM (Release 19)**

■ What

ISAC (also called Joint Communication and Sensing, JCS) uses a shared waveform, shared hardware, and shared spectrum to simultaneously perform wireless communication and radar-like sensing (detection, ranging, velocity estimation, imaging).

■ Why / Problem Addressed

- Separate radar and communication systems waste spectrum and hardware
- Autonomous vehicles, drones, industrial robots need both connectivity and environmental awareness
- Spectrum scarcity drives dual-use of allocated bands
- 5G NR waveform (OFDM) is already suitable for sensing with modifications

■ Sensing Capabilities

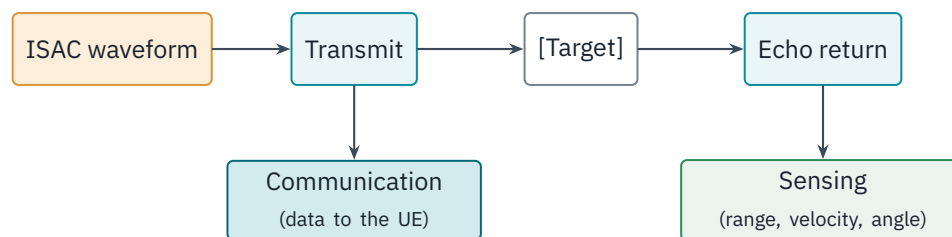


Figure 26.4 Integrated sensing and communication with one waveform

Sensing parameters extracted:

- **Range:** $R = c\tau/2$ (from the delay τ)
- **Velocity:** $v = \lambda f_d/2$ (from the Doppler shift f_d)
- **Angle:** θ from array processing (AoA/AoD)

OFDM Radar Processing

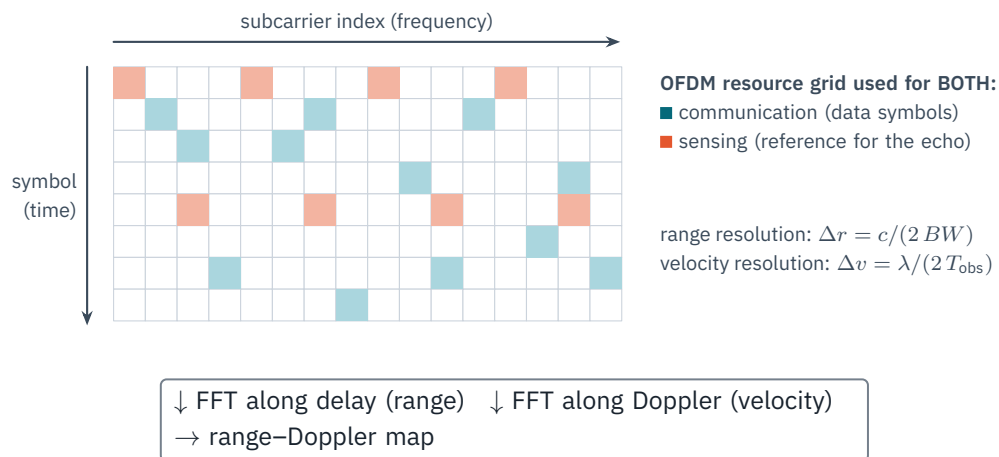


Figure 26.5 OFDM radar processing on the resource grid

3GPP Progress

- **R19 Study Item:** “Study on Integrated Sensing and Communication” (approved 2024)
- Key study areas:
 - Channel model for sensing (bistatic, monostatic)
 - Waveform design and resource allocation
 - Sensing-assisted communication (e.g., beam prediction from sensing)
 - Privacy and regulatory aspects of network-based sensing
- Use cases: gesture recognition, object tracking, intrusion detection, weather monitoring

Benefits

- Spectrum efficiency: dual use of same resources
- Hardware reuse: shared antenna arrays and RF chains
- Synergy: sensing assists communication (predictive beamforming) and vice versa
- New services: network as a sensor (NaaS — Network as a Sensor)

Challenges

- **Waveform tradeoff:** Optimal for communication $\neq$ optimal for sensing
- **Self-interference:** Monostatic sensing requires full-duplex capability
- **Clutter:** Distinguish targets from environment
- **Privacy:** Network can sense people without their devices
- **Regulation:** Sensing in licensed communication bands needs new frameworks
- **Performance metrics:** How to jointly optimize communication rate and sensing accuracy?

■ Relation to 5G

- 5G NR positioning (R16/R17) is a limited precursor
- 5G OFDM waveform is sensing-capable with proper processing
- R19 study prepares for R20 work items
- 5G-Advanced adds framework; 6G makes it native

■ Open Questions

- Monostatic vs bistatic vs networked sensing — which architecture?
- How to allocate resources between sensing and communication optimally?
- What sensing performance is achievable with communication waveforms?
- Regulatory framework for passive sensing of people?

5 Reconfigurable Intelligent Surfaces (RIS)

■ Status: ● STUDY ITEM (Release 18-19)

■ What

Reconfigurable Intelligent Surfaces (RIS) are planar structures composed of many sub-wavelength elements that can be electronically controlled to reflect (or refract) incoming electromagnetic waves in desired directions. They reshape the wireless propagation environment without generating new signals.

■ Why / Problem Addressed

- mmWave/sub-THz signals suffer from severe blockage
- Traditional relays require full RF chains (expensive, power-hungry)
- NLoS coverage gaps in dense urban environments
- Need for energy-efficient coverage extension

Operating Principle

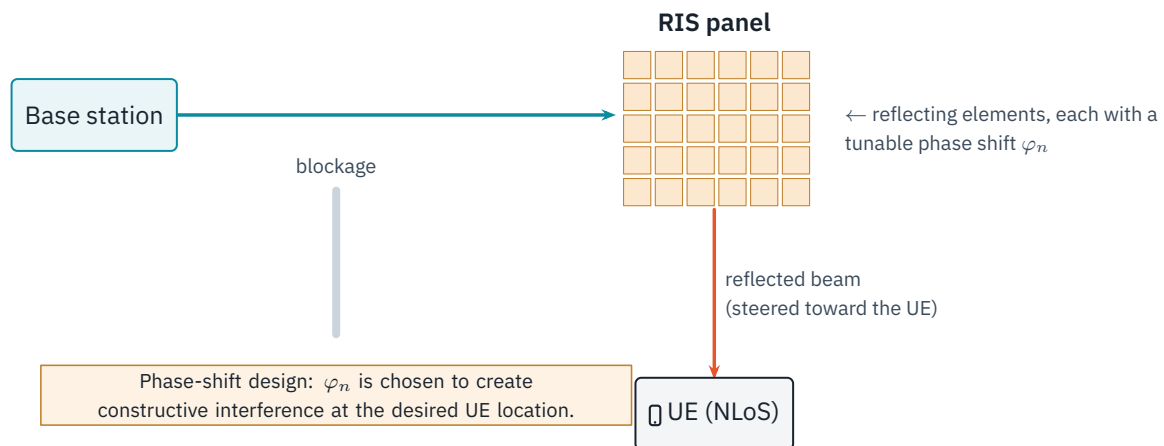


Figure 26.6 A reconfigurable intelligent surface steers the beam to a blocked UE

Types of RIS

Type	Characteristics	Gain	Complexity
Passive RIS	Only phase shift, no amplification	Moderate	Low
Active RIS	Phase shift + amplification	High	Medium
STAR-RIS	Simultaneous transmit and reflect	High	High
Hybrid RIS	Some active, some passive elements	Configurable	Medium

Path Loss Model

For a passive RIS with N elements:

$$P_{\text{Rx}} \propto N^2 \times L_{\text{BS} \rightarrow \text{RIS}} \times L_{\text{RIS} \rightarrow \text{UE}}$$

- “Product-distance” path loss (multiplicative, not additive in dB)
- → the RIS must be placed near the BS or near the UE for best performance
- → the SNR gain scales as N^2 (coherent combining)

3GPP Progress

- **TR 38.857:** Study on NR network-controlled repeaters (R18) — includes smart repeater concepts
- **R19:** Further study on RIS-assisted NR
- Study areas: channel modeling with RIS, deployment scenarios, control signaling

■ Benefits

- Nearly passive operation (only control circuit needs power)
- Low cost per element (printed circuit technology)
- Flexible deployment (walls, ceilings, lamp posts)
- Coverage extension without backhaul
- Green communication: minimal energy consumption

■ Challenges

- **Channel estimation:** Must estimate BS→RIS and RIS→UE channels separately
- **Control overhead:** Configuring hundreds/thousands of elements in real-time
- **Narrowband response:** Phase shifts are typically frequency-flat → wideband limitation
- **Deployment:** Optimal placement, orientation, density
- **Multiplicative path loss:** Double path loss can limit practical gains
- **Hardware impairments:** Discrete phase shifts, element coupling

■ Relation to 5G

- 5G network-controlled repeaters (R18) are a precursor
- 5G mmWave deployments highlight the blockage problem RIS addresses
- R18-R19 studies inform potential R20 specifications
- May complement (not replace) small cells and relays

■ Open Questions

- How many elements needed for meaningful gain in practice?
- Who controls the RIS — operator, third party, autonomous?
- How to handle mobility (real-time reconfiguration speed)?
- Passive vs active: which wins in cost-benefit analysis?
- Can RIS work effectively for broadband (wideband) signals?

6 Sub-THz / THz Communications

■ Status: ● RESEARCH

■ What

Sub-THz (100-300 GHz) and THz (300 GHz - 3 THz) communications exploit vast unused spectrum for extreme data rates. Channels of 10+ GHz bandwidth become available, potentially enabling 100+ Gbps per link.

■ Why / Problem Addressed

- mmWave bandwidth (400 MHz - 2 GHz) insufficient for 6G peak rate targets (200+ Gbps)
- Spectrum below 100 GHz is increasingly congested
- Short-range applications (data kiosks, intra-device communication) need extreme throughput
- Backhaul/fronthaul for dense networks needs wireless alternatives to fiber

■ Frequency Bands of Interest

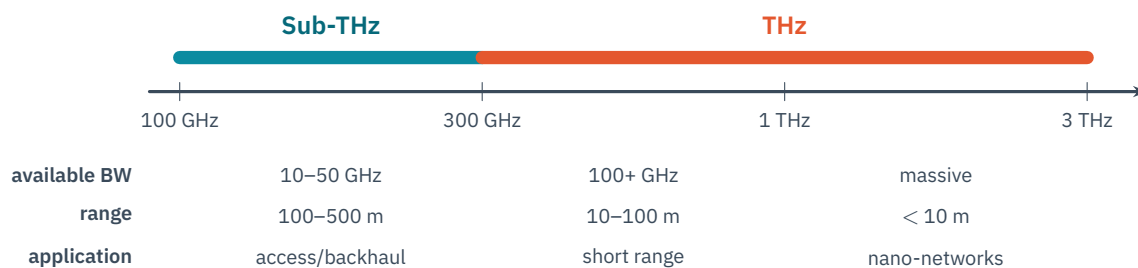


Figure 26.7 Sub-THz and THz bands: bandwidth, range and applications

Key atmospheric absorption windows:

- 130–175 GHz (moderate absorption)
- 200–320 GHz (multiple windows between absorption lines)
- water-vapour absorption lines at 183 and 325 GHz

■ Link Budget Challenge

Free-space path loss at 300 GHz versus 30 GHz (same distance, 100 m):

$$\text{FSPL} = 20 \log_{10} \left(\frac{4\pi d}{\lambda} \right)$$

- at 30 GHz: FSPL = 91.5 dB
- at 300 GHz: FSPL = 111.5 dB (+20 dB!)
- molecular absorption adds a further 1–10 dB/100 m (frequency-dependent)

Compensation requires very high-gain antennas (pencil beams), large arrays (small $\lambda \rightarrow$ many elements in a small area) and short distances.

■ Key Technologies Enabling THz

1. **III-V semiconductors** (InP, GaN): Power amplifiers up to 300 GHz
2. **Silicon-based (CMOS/SiGe)**: Lower cost, lower power, up to ~200 GHz
3. **Photonic approaches**: Optical-to-THz conversion
4. **Graphene-based devices**: Theoretical potential for THz electronics
5. **Massive arrays**: 1000+ elements possible due to small wavelength

■ Benefits

- Enormous available bandwidth (10-100+ GHz channels)
- Very small wavelength enables massive arrays in compact form factor
- Inherent security (highly directional, short range)
- High-resolution sensing capability (sub-mm resolution)

■ Challenges

- **Severe path loss**: Requires LoS and very short range
- **Atmospheric absorption**: Frequency-selective fading from molecular resonances
- **Hardware**: Limited output power, high noise figure at THz frequencies
- **Phase noise**: Oscillator stability at hundreds of GHz
- **Blockage**: Even a hand can block a THz link completely
- **Cost**: Exotic semiconductor processes

■ Relation to 5G

- 5G FR2 (24-52 GHz) demonstrated viability of high-frequency mobile access
- Lessons from 5G mmWave beam management applicable to sub-THz
- WRC-23 identified bands above 100 GHz for study (not yet allocated for mobile)
- 6G may initially target 100-300 GHz (sub-THz) before moving higher

■ Open Questions

- Is mobile access viable above 100 GHz, or only fixed/short-range?
- What is the maximum practical range for useful communication?
- How to handle beam tracking at THz beam widths ($<1^\circ$)?
- Can silicon-based solutions achieve sufficient power at these frequencies?

7 Non-Terrestrial Networks (NTN)

■ **Status:** ● **STANDARDIZED (basic R17)** / ● **STUDY (advanced features)**

■ What

Non-Terrestrial Networks integrate satellite and aerial platforms into the 3GPP network architecture. NR and IoT-NTN enable direct communication between standard UEs (including smartphones) and LEO/MEO/GEO satellites or High-Altitude Platform Systems (HAPS).

■ Why / Problem Addressed

- ~90% of Earth's surface lacks terrestrial mobile coverage
- Maritime, aviation, and remote areas have no connectivity
- Disaster resilience requires non-terrestrial backup
- Global IoT connectivity (agriculture, logistics, environmental monitoring)

■ NTN Platform Types

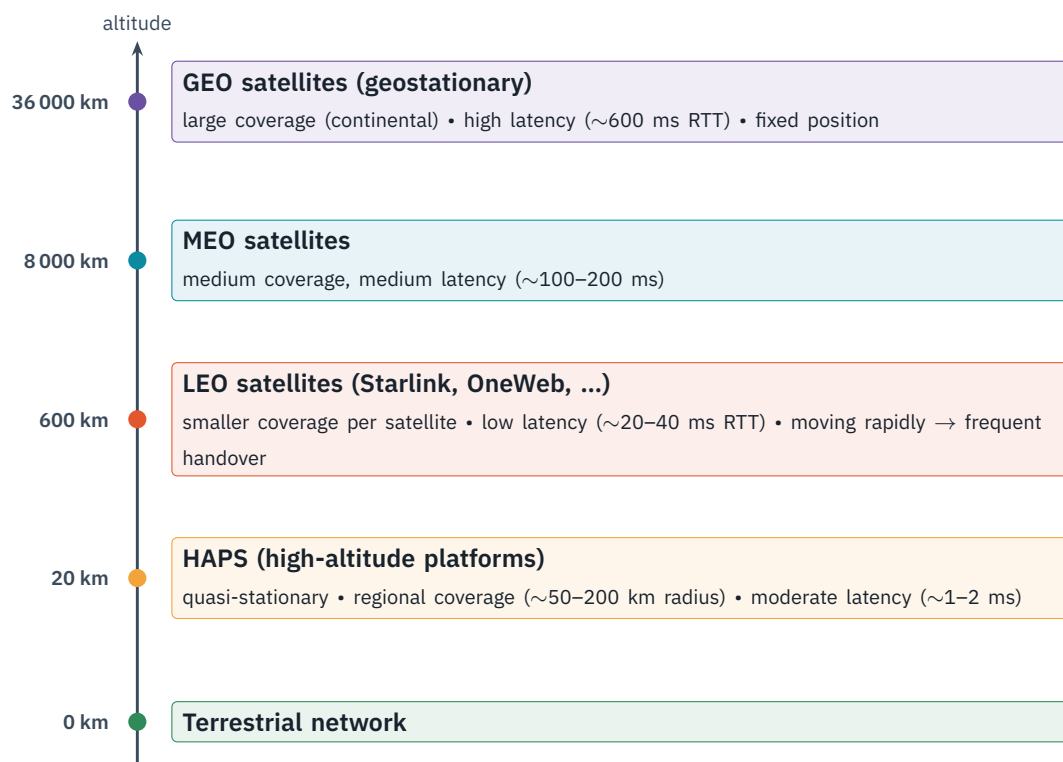


Figure 26.8 Non-terrestrial network platforms by altitude

3GPP Standardization Progress

Release	Feature	Status
R15-R16	Study items (TR 38.811, 38.821)	Completed
R17	Basic NR-NTN (transparent payload, GEO/LEO)	STANDARDIZED
R17	IoT-NTN (NB-IoT/eMTC over satellite)	STANDARDIZED
R18	NTN enhancements (mobility, coverage)	STANDARDIZED
R18	NR-NTN on FR2 (above 10 GHz)	Study
R19	Regenerative payloads, store-and-forward	Study
R19	Direct-to-cell (smartphone satellite)	Study/WI

Key Technical Challenges Solved in R17

1. **Timing advance:** Large propagation delay (up to 540 ms for GEO) → pre-compensation using GNSS at UE
2. **Doppler shift:** LEO at 7.5 km/s → up to ± 24 ppm at S-band → pre-compensation
3. **Large cells:** Satellite beams cover 100s of km → timing advance differential within cell
4. **HARQ:** Disabled or modified due to long RTT (feedback arrives too late)

Architecture Options

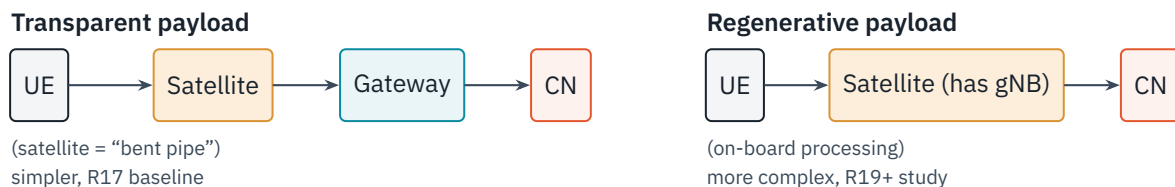


Figure 26.9 Transparent versus regenerative satellite payload

Benefits

- Ubiquitous global coverage
- Disaster resilience and emergency communication
- Direct-to-cell: unmodified smartphones connect to satellites
- IoT coverage in remote areas (agriculture, shipping, pipelines)
- Network redundancy and multi-connectivity

Challenges

- **Link budget:** Very long distances require large antenna gains
- **Spectrum coordination:** Coexistence with terrestrial networks
- **Handover:** LEO satellites move rapidly (4-8 min visibility window)
- **Interference:** Satellite beams overlap with terrestrial cells

- **Capacity:** Limited per-user throughput from satellite
- **Constellation cost:** Thousands of satellites needed for global LEO coverage

Relation to 5G

- NTN is PART of 5G-Advanced (R17-R18 specifications exist)
- Direct-to-cell (smartphone-satellite) commercially launching (T-Mobile/SpaceX, AST SpaceMobile)
- 6G targets tighter integration: seamless terrestrial/NTN handover, multi-layer NTN

Open Questions

- How to achieve seamless handover between LEO satellites?
- Regenerative vs transparent payload — which wins economically?
- How much capacity can satellite provide per user?
- Integration with terrestrial for load balancing in dense areas?

8 Semantic Communications

Status: ● RESEARCH / ● SPECULATIVE

What

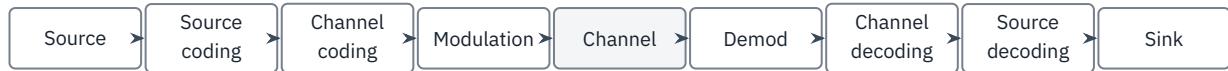
Semantic Communications transmit the **meaning** (semantics) of information rather than exact bit sequences. Instead of ensuring every bit is correctly received, the system ensures the intended meaning or task outcome is preserved, potentially requiring far fewer transmitted bits.

Why / Problem Addressed

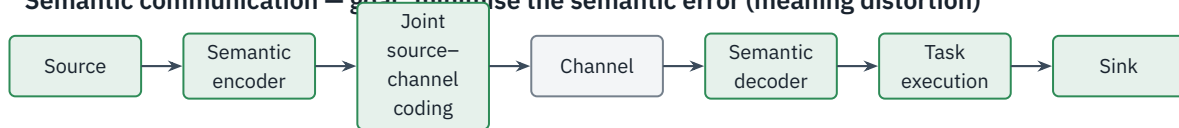
- Shannon's theory optimizes bit transmission, not information utility
- Most transmitted data has redundancy that classical coding doesn't exploit
- Video call: 90%+ of pixels don't change between frames
- Machine-to-machine communication doesn't need human-readable formats
- Approaching Shannon capacity limits → need new paradigm for further gains

Conceptual Framework

Classical communication — goal: minimise the BER



Semantic communication — goal: minimise the semantic error (meaning distortion)



Example (image transmission): classical — transmit all pixels, recover all pixels; semantic — extract key features, transmit a compact representation, reconstruct the image or classify directly.

Figure 26.10 Classical versus semantic communication

Key Technologies

1. Deep Joint Source-Channel Coding (DeepJSCC)

- Single neural network replaces separate source and channel coding
- Graceful degradation (no cliff effect)
- Adapts to channel quality continuously

2. Knowledge Graphs / Shared Knowledge Base

- Transmitter and receiver share background knowledge
- Only novel/unexpected information needs transmission
- Dramatic compression for known contexts

3. Task-Oriented Communication

- Optimize for downstream task (classification, control) not reconstruction
- Transmit only task-relevant features
- Example: Autonomous driving → transmit object positions, not full video

Potential Bandwidth Savings

Application	Classical	Semantic	Reduction
Text (translation task)	Full sentence bits	Intent + entities	10-100×
Image (classification)	All pixels (JPEG)	Feature vector	100-1000×
Video (surveillance)	Full frames (H.265)	Events + changes	10-50×
Control (robotics)	Sensor data stream	Action commands	50-500×

Note: These are theoretical/experimental estimates, not proven at scale

Benefits

- Extreme compression beyond Shannon limits for specific tasks
- Graceful degradation under poor channel conditions
- Natural integration with AI at endpoints
- Efficient machine-to-machine communication
- Reduced latency (fewer bits to transmit)

Challenges

- **No mathematical framework:** Shannon theory doesn't cover semantics
- **Shared knowledge assumption:** Tx and Rx must agree on semantic model
- **Generalization:** Different tasks need different semantic encoders
- **Security:** Semantic attacks (adversarial examples that preserve bits but corrupt meaning)
- **Standardization:** How to specify semantic protocols?
- **Backward compatibility:** Cannot coexist with classical systems easily
- **Evaluation metrics:** What replaces BER/BLER for semantic systems?

Relation to 5G

- 5G has no semantic communication features
- Conceptually orthogonal to current standards
- May first appear in application layer (not radio standard)
- Could eventually change fundamental PHY design in 6G+

Open Questions

- Is a universal semantic language possible, or always task-specific?
- How to handle unknown/novel semantic content?
- Can semantic and classical communication coexist on same channel?
- Who defines the semantic model — standards body, AI training?
- Timeline: Is this 6G or 7G technology?

9 Digital Twins for Networks

■ Status: ● RESEARCH

■ What

A Network Digital Twin is a real-time, high-fidelity virtual replica of a physical network that continuously synchronizes with the live network. It enables what-if analysis, predictive optimization, and autonomous network management by simulating changes before deploying them.

■ Why / Problem Addressed

- Network optimization is reactive (fix after failure)
- Testing changes on live networks is risky
- 5G/6G network complexity exceeds human ability to manage
- Need for predictive maintenance and proactive optimization
- Training AI/ML models requires safe environments

■ Architecture

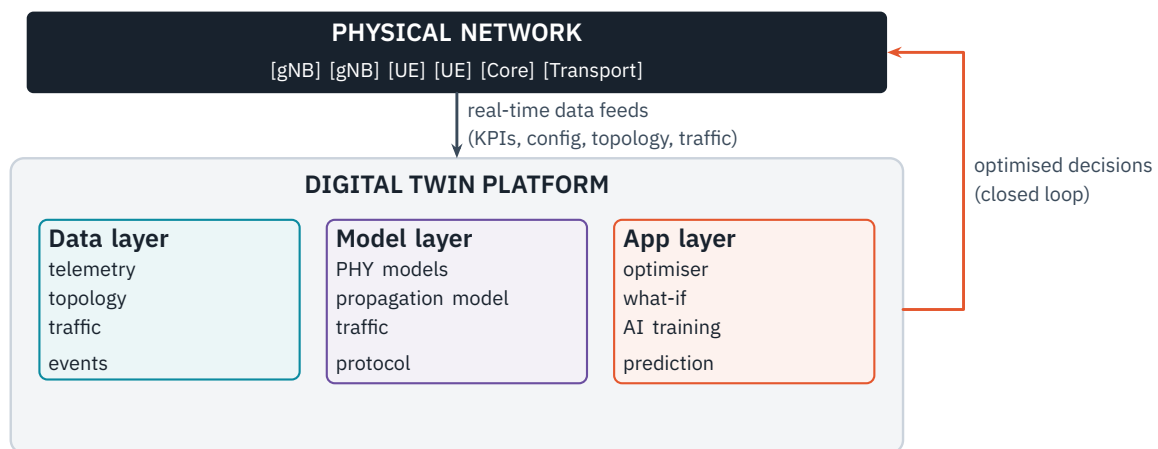


Figure 26.11 Network digital twin in a closed loop

■ Use Cases

1. **Predictive maintenance:** Detect degradation before failure
2. **Configuration optimization:** Test parameter changes in simulation
3. **Capacity planning:** Simulate future growth scenarios
4. **AI/ML training:** Generate synthetic training data safely
5. **Anomaly detection:** Compare twin prediction vs reality
6. **Disaster recovery:** Pre-plan failover configurations
7. **Network slicing:** Simulate SLA compliance before provisioning

I Benefits

- Risk-free experimentation on virtual network
- Continuous optimization without service disruption
- Faster AI model development and validation
- Reduced OPEX through automation and prediction
- Holistic network visibility

I Challenges

- **Fidelity:** Twin must accurately represent physical network behavior
- **Real-time synchronization:** Data collection and model update latency
- **Scale:** Modeling millions of devices in real-time is computationally expensive
- **Validation:** How to verify the twin matches reality?
- **Data requirements:** Enormous telemetry data streams
- **Standardization:** No agreed data model or interface standard yet

I Relation to 5G

- 5G O-RAN provides data interfaces (E2, O1, O2) that feed twins
- 5G network automation (SON) is a simpler precursor
- Some vendors offer “digital twin” products today (mostly offline simulation)
- 6G targets real-time, closed-loop digital twins

I Open Questions

- What level of fidelity is “good enough” for different use cases?
- How to handle model uncertainty and validation?
- Centralized vs distributed twin architecture?
- Standardized data models and APIs?

10 Advanced Positioning (cm-Level)

■ Status: ● STUDY ITEM (Release 18 positioning enhancements)

■ What

6G targets centimeter-level (1-10 cm) positioning accuracy using joint communication and positioning techniques, advanced signal processing, and AI/ML assistance. This goes far beyond GPS (~1-5 m) and 5G R16 positioning (~1 m outdoor).

■ Why / Problem Addressed

- Indoor positioning remains unsolved (GPS unavailable)
- Industrial automation requires mm-cm precision
- Autonomous systems (drones, robots, vehicles) need precise localization
- AR/XR applications require 6DoF tracking at cm level
- Asset tracking in warehouses, hospitals, factories

■ Positioning Techniques Evolution

Release	Accuracy	Techniques
R16 (5G baseline)	~3–10 m	DL-TDoA, UL-TDoA, multi-RTT
R17 (enhancement)	~0.5–3 m	Improved DL-PRS, NR positioning
R18 (study)	~0.2–1 m	Carrier phase, sidelink, AI/ML
R19+ / 6G target	~0.01–0.1 m	cm-level, 6DoF

■ Key Technologies for cm-Level Accuracy

■ Carrier-Phase Positioning

- **Standard ranging** uses the signal envelope: resolution $\sim c/BW$; for $BW = 100$ MHz the resolution is about 3 m (before processing gain).
- **Carrier-phase ranging** uses the carrier wavelength λ : at 3.5 GHz, $\lambda = 8.6$ cm (sub-cm potential); at 28 GHz, $\lambda = 1.07$ cm (mm potential).
- **Challenge:** integer-ambiguity resolution. The measured phase is

$$\varphi = \frac{2\pi}{\lambda} d + 2\pi N + \text{noise}$$

and N must be resolved to obtain the absolute distance.

■ Multi-RTT with Large Bandwidth

- Round-Trip-Time measurement with 400 MHz - 2 GHz bandwidth

- Sub-nanosecond timing resolution
- Multiple base stations for triangulation

■ AI/ML-Assisted Positioning

- **Fingerprinting:** ML models map channel features to position
- **NLOS identification:** AI classifies LoS vs NLoS paths
- **Sensor fusion:** Combine radio with IMU, camera, LiDAR
- **Channel charting:** Unsupervised learning of radio geometry

■ Joint Communication and Positioning

- Same waveform serves dual purpose
- PRS (Positioning Reference Signals) integrated with data transmission
- ISAC synergy: sensing capabilities enhance positioning

■ Accuracy vs Technique Comparison

Technique	Accuracy	Pros	Cons
DL-TDoA (R16)	3-10 m	Standard, UE-based	Clock sync needed
Multi-RTT (R16)	1-5 m	No sync needed	Multiple gNBs
DL-AoD (R16)	5-15 m	Single gNB possible	Antenna cal needed
Enhanced (R17)	0.5-3 m	Improved PRS	Bandwidth limited
Carrier-phase (R18)	1-10 cm	Very high accuracy	Ambiguity resolution
AI-assisted (R18+)	0.3-3 m	Works in NLOS	Training data needed
6G target	1-10 cm	Native, real-time	Full system redesign

■ 3GPP Progress

- **R16:** Basic NR positioning (TS 38.305, 38.455)
- **R17:** Positioning enhancements (improved accuracy, reduced latency)
- **R18:** Study on AI/ML for positioning, carrier-phase, sidelink positioning
- **R18:** Integrity and reliability requirements for positioning
- **R19:** Further accuracy improvements, ISAC-based positioning

I Benefits

- Ubiquitous cm-level positioning (indoor and outdoor)
- No additional infrastructure (reuses communication network)
- 6DoF tracking (position + orientation) for XR/AR
- Enables new applications: digital twins, autonomous navigation

I Challenges

- **Carrier-phase ambiguity:** Integer ambiguity resolution is hard in mobile scenarios
- **Multipath:** Reflections corrupt ranging measurements
- **NLOS:** Dominant in indoor environments
- **Synchronization:** Network timing accuracy limits achievable positioning
- **UE complexity:** Processing requirements for high-accuracy positioning
- **Power consumption:** Continuous positioning drains mobile battery

I Relation to 5G

- 5G NR positioning (R16-R17) provides the foundation
- Each release incrementally improves accuracy
- 6G integrates positioning as a native service (not an add-on)
- Carrier-phase and AI techniques studied in R18 for potential R19 specification

I Open Questions

- Can carrier-phase work reliably in mobile environments?
- How to handle integer ambiguity in real-time with mobility?
- What infrastructure density is needed for cm-level indoor coverage?
- How to balance positioning accuracy vs communication performance?

Technology Maturity Matrix

#	Technology	TRL*	3GPP Status	Standardization Timeline
1	AI-Native Networks / AI-RAN	4-5	Study Item (R18 TR 38.843)	R19-R20 Work Items
2	XL-MIMO	2-3	Not yet studied	R21+ (6G)
3	Cell-Free Massive MIMO	3-4	Not yet studied (CoMP precursor)	R20-R21
4	ISAC	3-4	Study Item (R19)	R20 Work Items
5	RIS	3-4	Study Item (R18-19 TR 38.857)	R20 Work Items
6	Sub-THz Communications	2-3	Not yet studied	R21+ (6G)
7	NTN (basic)	7-8	Standardized (R17)	Deployed
7	NTN (advanced)	4-5	Study/WI (R18-19)	R19-R20
8	Semantic Communications	1-2	Not studied	R22+ (post-6G?)
9	Digital Twins for Networks	3-4	Not yet studied	R20-R21
10	Advanced Positioning (cm)	4-5	Study Item (R18)	R19-R20

*TRL = Technology Readiness Level (1 = basic research, 9 = deployed)

TRL Scale Reference

TRL	Meaning
1	Basic principles observed
2	Technology concept formulated
3	Experimental proof of concept
4	Technology validated in the lab
5	Technology validated in a relevant environment
6	Technology demonstrated in a relevant environment
7	System prototype demonstrated in an operational environment
8	System complete and qualified
9	System proven in an operational environment (deployed)

Projected Timeline: 3GPP Releases and 6G Technologies

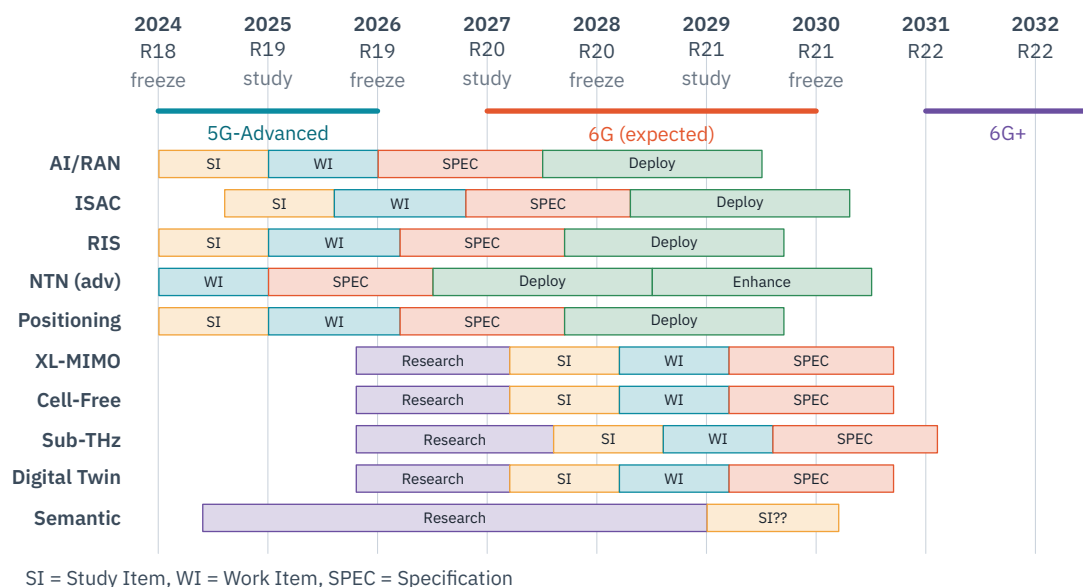


Figure 26.12 Projected timeline of 3GPP releases and 6G technologies

What to Expect per Release

Release	Timeline	Key 6G-Related Features
R19 (5G-Adv)	2025-2026	ISAC study, advanced NTN, positioning enhancements, AI/RAN WI
R20 (early 6G?)	2027-2028	ISAC specification, RIS specification, cell-free studies, sub-THz studies
R21 (6G)	2029-2030	XL-MIMO, sub-THz, semantic study, full AI-native, digital twin
R22 (6G+)	2031-2032	Semantic communication, advanced THz, autonomous networks

Summary: 5G vs 6G Paradigm Shifts

Aspect	5G	6G
AI role	Optimization add-on	Native design principle
Spectrum	Sub-6 + mmWave (FR1+FR2)	+ Sub-THz (FR3?)
Architecture	Cell-based	Cell-free + NTN
Coverage	Terrestrial	Terrestrial + space + aerial
Sensing	Not supported	Integrated (ISAC)
Positioning	~1m accuracy	~1 cm accuracy
Environment control	Passive (adapt to channel)	Active (RIS shapes channel)
Communication goal	Transmit bits reliably	Transmit meaning efficiently
Network management	Human-configured	Autonomous (digital twin + AI)
Antenna scale	64-256 elements	1000-10000+ elements
Peak rate	20 Gbps	200+ Gbps
Latency	1 ms	0.1 ms

Key Takeaways

1. **Not all “6G technologies” are equal** — NTN is already standardized, while semantic communication remains speculative. Always check maturity level.
2. **6G is evolutionary AND revolutionary** — Some features (AI-RAN, ISAC, RIS) evolve from 5G-Advanced. Others (semantic, THz, cell-free) represent paradigm shifts.
3. **3GPP drives standardization** — Technologies must pass: Research → Study Item → Work Item → Technical Specification. This process takes 3-5 years per technology.
4. **Timing: 6G specification expected ~2028-2030**, with first deployments ~2030-2032.
5. **IMT-2030 framework** sets targets but actual specifications may differ from initial visions (as happened with IMT-2020/5G).
6. **Economic viability** ultimately determines adoption — technically superior solutions may lose to “good enough” practical alternatives.

References and Further Reading

■ 3GPP Documents

- TR 38.843: Study on AI/ML for NR air interface (R18)
- TR 38.857: Study on NR network-controlled repeaters (R18)
- TR 38.811: Study on NR to support NTN (R15)
- TR 38.821: Solutions for NR to support NTN (R16)
- TS 38.305: Stage 2 functional specification for positioning (R16+)

■ ITU-R

- IMT-2030 Framework Recommendation (ITU-R M.2160)
- Report ITU-R M.2516: Future technology trends for IMT systems toward 2030+

■ Key Research Papers

- Björnson et al., “Scalable Cell-Free Massive MIMO Systems,” IEEE Trans. Commun., 2020
- Di Renzo et al., “Smart Radio Environments Empowered by RIS,” JSAC, 2020
- Rappaport et al., “Wireless Communications and Applications Above 100 GHz,” IEEE Access, 2019
- Qin et al., “Semantic Communications: Principles and Challenges,” arXiv, 2021
- Liu et al., “Integrated Sensing and Communications,” IEEE JSAC, 2022
- Amiri et al., “Extremely Large MIMO,” IEEE Commun. Mag., 2024

■ Industry Roadmaps

- Next G Alliance (ATIS): 6G Roadmap
- Hexa-X / Hexa-X-II (EU 6G Flagship Project)
- Samsung 6G Vision (2020, updated 2023)
- Nokia Bell Labs 6G Vision
- NGMN Alliance: 6G Drivers and Vision

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Note: This is a living document. Technology maturity assessments will change as research progresses and 3GPP work advances.